WORKING RESEARCH MONOGRAPH

From Kohn–Sham Operators to Effective-Mass Models

Alignment, Reduction, and Evidence for Semiconductor Hamiltonians

Eugene Joseph M. Ragasa

Department of Physics, De La Salle University
2401 Taft Avenue, Manila 0922, Philippines

Status: working research monograph with completed synthetic verification and a proposed material program

Not an institutionally registered dissertation, submitted manuscript, or material validation

Abstract

This monograph develops a controlled framework for investigating when a first-principles Kohn–Sham operator for pristine or substitutionally doped silicon can be replaced by a reduced lattice or continuum Hamiltonian. The proposed reduction path proceeds through retained Bloch subspaces, localized Wannier representations, tight-binding model classes, aligned impurity operators, and effective-mass descriptions.

The central requirement is that operators be compared only after their state spaces, retained subspaces, bases, gauges, geometries, spin conventions, units, and energy references have been explicitly identified. The framework therefore distinguishes projection, representation change, alignment, truncation, and model fitting; separates spectral agreement from aligned-operator agreement; and tracks parent-model, numerical, and model-reduction errors independently. Nested reduced-operator classes and structured learning are developed as proposed diagnostics for determining which physical constraints and model complexities are supported by the parent representation. Spectral fitting, direct Hamiltonian regression, and structural restrictions of particular tight-binding classes are treated as established precedents; the proposed contribution is the narrower contract-dependent decision based on a common witness or a valid positive separation certificate.

A coordinated proof and evidence program records finite-dimensional operator identities, alignment and reduction obligations, analytical bounds, mechanization status, software verification, numerical verification, scientific validation, and uncertainty quantification. At the present evidence boundary, the project supports a represented-operator software foundation, bounded tutorial and bootstrap observations, and bounded synthetic numerical verification. Appendix M inventories the preliminary silicon records and a separate external same-frame Wannier kinetic prototype without promoting either to an accepted production material result. The periodic benchmarks cover one- and two-dimensional scalar, composite-band, topology, hopping, and localization diagnostics. A canonical one-dimensional rank-two calculation additionally retains a common spectral/operator witness and a finite-family separation certificate at two declared threshold pairs. The expanded two-dimensional fixed-protocol study retains nonconverged trajectories and unsupported mesh-convergence and repeated-basin criteria as negative evidence. Separate spin-space and one-dimensional defect benchmarks verify compatibility, alignment, independent extraction routes, a finite-rank spectral oracle, and continuum refinement for their declared synthetic systems.

Only one trial-projection family was executed, so the closed periodic benchmark does not establish projection-family robustness, global optimization, or general Wannier convergence. Two-dimensional defect extraction is active as a controlled synthetic task but has no calculated result; material transfer remains proposed and inactive. The project does not yet support an accepted production bulk parent, material Wannier operator, first-principles impurity operator, effective-mass crossover, scientific validation for silicon, or uncertainty quantification.

Preface

Status working research monograph with completed synthetic numerical verification and a proposed material program. No material validation is claimed.

This research monograph is a long-form working account of the `ksdft2effmass` research program. It is not an institutionally registered dissertation or a submitted manuscript. It combines accepted synthetic software and numerical verification with proposed production and material work. Its purpose is to develop one coherent narrative while preserving the narrower scopes of articles, conference contributions, software records, and computational evidence.

Research assistance and responsibility

Large language models assisted with drafting, code development, analysis, and editorial work. Their outputs were treated as provisional and evaluated under the same source, verification, provenance, and evidentiary requirements as other contributions. Scientific claims in this monograph rely on identified literature and retained calculation or proof records rather than on the use of an AI system. Responsibility for the selection, interpretation, and reporting of the material remains with the author.

Audience

Although this research interest began with the development of a pedagogical problem, I recognize the breadth of the resulting work. This manuscript serves partly as a laboratory notebook and partly as a structured set of research notes. It is intended to help advanced physics undergraduates and graduate students understand the quantum-mechanical, solid-state, semiconductor, computational, and mathematical concepts involved in the program.

The manuscript therefore includes a range of exercises developed to move a reader from reading, through note-taking and computation, and ultimately toward research. These exercises do not all carry the same evidentiary status. A pedagogical derivation, controlled numerical experiment, proposed computational workflow, and retained scientific calculation must remain distinguishable even when they appear in one continuous narrative.

How to read the monograph

The main narrative is ordered by increasing conceptual load rather than by the historical order in which the research notes were written. Part I begins with periodic geometry and centered finite differences. These lectures require mainly linear algebra and multivariable calculus, use concrete cells and stencils, and establish the coordinate vocabulary needed later. Part II asks what makes a reduced model adequate and introduces the comparison and evidence rules used throughout the book. Part III develops the pristine bulk-silicon representation program, from the first-principles parent through retained and localized representations to bulk reduced-model selection. Part IV develops the phosphorus and boron branches, aligned impurity extraction, lattice and continuum model classes, and transferability questions. Part V collects the mathematical and proof program needed to support the preceding scientific claims. The concluding chapter then restates the dependency-ordered research sequence.

The order is pedagogical, not an identity assigned to a source file. Chapter source filenames are semantic and carry no numeric prefix; the composition root generates chapter numbers from the current reading order. This allows a lecture to move when its prerequisites change without renaming it or making durable links encode a temporary table of contents.

The appendices form a reference and analytical workbench. They contain detailed notation, operator geometry, controlled finite-dimensional examples, reduction route comparisons, completed one- and two-dimensional periodic benchmarks, completed spin-space and one-dimensional defect benchmarks, a proposed two-dimensional defect

program, and a dedicated envelope-theory appendix extracting the Luttinger–Kohn, Burt, and Ermoneit equations for the project. They support the main argument while preserving the distinctions among synthetic verification, literature analysis, proposed work, and material validation.

Status and provenance

The monograph explains the research program and reports evidence through identified sources and retained records. It is a living document: its narrative, appendices, and controlled calculations are revised in place as the library gains new capabilities and new evidence changes the supported account. Statements about future work are proposals rather than results. A typeset formula is not by itself a proof, a proof sketch is not a completed proof, a successful software test is not scientific validation, and a generated figure is not a calculated result unless its input, method, software environment, and provenance are identified. When a calculation, source, or proof record changes the supported interpretation, the narrative must be updated rather than treated as independent authority.

Version changes that affect reported calculations or conclusions are stated in the manuscript. The applicable Git commit records the corresponding editorial and calculation change, while retained SHA-256 manifests identify the exact artifacts used by that revision. This commit-and-digest record is the monograph's version boundary; a parallel hierarchy of versioned manuscript or calculation directories is not required.

Objects kept distinct

The discussion uses four related but noninterchangeable levels:

Physical model. The chosen description of the material system and its physical assumptions, including the Kohn–Sham parent problem and the substitutional-impurity branches.

Mathematical operator. A map acting between identified state spaces, with a stated domain and codomain when these are material to the argument.

Finite representation. A matrix obtained only after selecting finite subspaces, bases, ordering conventions, units, geometry, gauge, and energy reference.

Software artifact. An implementation, serialized record, input file, workflow, or test that constructs or evaluates a representation.

Equality or subtraction at one level does not automatically define equality or subtraction at another. In particular, pristine and doped matrices cannot be subtracted meaningfully until their represented spaces and conventions have been aligned.

Evidence vocabulary

The monograph uses explicit status labels:

Calculated result. Produced by an identified calculation with retained provenance.

Literature value. Taken from an identified and verified source.

Expected behavior. A prediction or expectation that has not yet been observed in the project.

Illustrative example. Explanatory material that is not research evidence.

Synthetic test data. Constructed data used to verify software behavior, not output from a physical calculation.

Placeholder. Material awaiting authoritative content or verification.

Proposed work. Planned activity that has not been completed.

Chapter-level status statements apply to the prose that follows unless a more specific label overrides them. Results chapters will identify their retained sources rather than relying on narrative confidence or visual plausibility.

Error categories

Three error sources remain separate throughout the monograph. Parent-model error concerns the difference between the selected physical parent and an independent physical reference such as experiment or a higher-level theory. Numerical or discretization error concerns approximation of the stated parent mathematics by finite cutoffs, meshes, cells, stencils, and solver tolerances. Model-reduction error concerns the loss incurred when replacing an aligned parent representation by a reduced model class. These categories will not be combined unless a later owning specification establishes compatible definitions and a valid mathematical relationship among them.

Program scope

The initial material system is silicon in the diamond structure. The declared impurity branches are substitutional phosphorus and substitutional boron in silicon. Quantum ESPRESSO remains responsible for electronic-structure calculations, and Wannier90 remains responsible for Wannier localization. The software in this repository begins from their retained outputs and addresses representation, alignment, operator extraction, model reduction, evaluation, and provenance.

The immediate bulk pilot and the later impurity program are related but not identical scopes. Bulk development does not establish dopant results. Likewise, a neutral periodic dopant supercell is not silently reinterpreted as an isolated or ionized impurity. Each such interpretation requires its own declared branch and evidence.

Prospective citation note—medium priority
Check the periodic-supercell versus isolated- or ionized-impurity warning against a standard defect-physics source. Candidate: Freysoldt et al. (2014), bibliography key `freysoldt2014` [32] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

Relationship to shorter publications

Articles and presentations may select a narrower question, reorganize the exposition, and adopt venue-specific terminology. Each publication must still identify the evidence it uses and preserve the corresponding status, provenance, and limitations. This monograph remains the broader working account rather than evidence that any shorter publication has been reviewed or accepted.

Contents

Preface **i**

I Computational Foundations **1**

1 Computational Quantities: Units, Coordinates, and Metadata **2**

1.1 Learning goals . . . 2
1.2 From numbers to quantities . . . 3
1.3 Dimensions, units, and reference values . . . 3
1.4 Coordinates are part of the data . . . 4
1.5 Indexing, axis meaning, and ordering . . . 4
1.6 Metadata needed by represented scientific data . . . 5
1.7 A small transparent record . . . 6
1.8 Compatibility before arithmetic . . . 7
1.9 Exercises . . . 7
1.10 Evidence boundary . . . 8

2 Vectors, Bases, Matrices, and Finite Representations **9**

2.1 Learning goals . . . 9
2.2 Scalars, vectors, and vector spaces . . . 10
2.2.1 Examples of vectors . . . 10
2.3 Linear combinations, span, and independence . . . 12
2.4 Bases, dimension, and coordinate columns . . . 12
2.4.1 A coordinate example . . . 13
2.5 Inner products, lengths, and orthogonality . . . 14
2.6 Linear maps . . . 15
2.7 Matrices represent linear maps in chosen bases . . . 15
2.7.1 Reading a matrix by columns . . . 16
2.8 Identity, adjoint, and unitary maps . . . 16
2.9 Changing coordinates without changing the object . . . 17
2.9.1 A rotated-coordinate example . . . 18
2.10 What changes and what does not . . . 19
2.11 Finite representations and their contracts . . . 19

2.12 A two-basis calculation . . . 20
2.13 Common mistakes . . . 22
2.14 From representations to eigensystems . . . 23
2.15 Exercises . . . 23
2.16 Evidence boundary . . . 24

3 Hermitian Eigensystems and Spectral Subspaces 25
3.1 Learning goals . . . 25
3.2 Eigenpairs are invariant-line statements . . . 26
3.3 Why Hermiticity changes the problem . . . 26
3.4 From individual eigenpairs to a spectral decomposition . . . 27
3.5 The characteristic equation . . . 28
3.5.1 A complete two-dimensional example . . . 29
3.6 The Rayleigh quotient and extreme eigenvalues . . . 30
3.7 Degeneracy, eigenspaces, and projectors . . . 30
3.8 Phase, ordering, and subspace freedom . . . 31
3.9 Computing and checking a Hermitian eigensystem . . . 32
3.10 From eigenspaces to retained spaces . . . 34
3.11 Common mistakes . . . 34
3.12 Exercises . . . 35
3.13 Evidence boundary . . . 35

4 Quantum States, Operators, and Hamiltonians 36
4.1 Learning goals . . . 36
4.2 State spaces and Dirac notation . . . 36
4.3 Linear operators and domains . . . 37
4.4 Hamiltonians and additive decompositions . . . 38
4.5 Position-space representation . . . 38
4.6 The one-dimensional particle in a box . . . 39
4.6.1 Solving the stationary equation by a characteristic equation 39
4.6.2 Why the characteristic equation turns calculus into algebra 41
4.6.3 Normalization and the eigenfunction basis . . . 42
4.6.4 Born's interpretation of the eigenfunction . . . 43
4.6.5 The Dirichlet Laplacian on the closed interval $[a, b]$ . . . 44
4.6.6 Discretizing the Dirichlet Laplacian as an operator . . . 45
4.6.7 Eigenpairs of the discretized operator . . . 46
4.7 Finite Hamiltonian representations . . . 47
4.8 What changes in an approximation . . . 49
4.9 Exercises . . . 49
4.10 Evidence boundary . . . 50

5 Particle in a Box in Two and Three Dimensions **51**
5.1 Learning goals . . . 51
5.2 From an interval to a rectangular box . . . 52
5.3 What a Kronecker product actually does . . . 52
5.3.1 A literal block example . . . 53
5.3.2 Kronecker products of vectors . . . 53
5.3.3 Identity factors mean “leave this coordinate alone” . . . 54
5.3.4 Kronecker product versus Kronecker sum . . . 54
5.4 The two-dimensional box . . . 55
5.4.1 Degeneracy in a square box . . . 56
5.5 Finite representation of the two-dimensional box . . . 56
5.5.1 A 2×2 interior grid . . . 57
5.6 The three-dimensional box . . . 57
5.7 Sparse computational construction . . . 58
5.8 What changes outside a rectangular Cartesian box . . . 60
5.9 Common mistakes . . . 60
5.10 Exercises . . . 61
5.11 Evidence boundary . . . 61

6 Periodic Geometry: Intrinsic Cells and Embedded Planes **62**
6.1 Learning goals . . . 62
6.2 Why geometry comes before a periodic discretization . . . 63
6.3 Primitive vectors as a coordinate map . . . 64
6.3.1 Three coordinate languages . . . 64
6.4 The induced metric . . . 65
6.4.1 Orthogonal square example . . . 66
6.4.2 Skew-cell example . . . 66
6.4.3 A fully worked sixty-degree cell . . . 66
6.5 Dual reciprocal vectors . . . 68
6.5.1 Deriving the rectangular formula at the board . . . 68
6.6 The one-dimensional reciprocal quotient . . . 69
6.6.1 Bloch sectors and fiber operators . . . 70
6.7 Projection and coordinate recovery . . . 71
6.7.1 A projection thought experiment . . . 71
6.8 An intrinsic frame that preserves the embedding . . . 72
6.9 An embedded plane is not a slab . . . 72
6.9.1 A language test . . . 73
6.10 Maintained computational representation . . . 73
6.11 How to record a geometry laboratory . . . 74
6.12 End-of-lecture synthesis . . . 74

6.13 Exercises . . . 75
6.14 Evidence boundary . . . 75

7 Bloch-Periodic Finite Differences on Nonorthogonal Cells 76
7.1 Learning goals . . . 76
7.2 Why a familiar stencil needs a full lecture . . . 77
7.3 From Cartesian derivatives to fractional derivatives . . . 77
7.3.1 The chain rule one index at a time . . . 78
7.4 Endpoint-excluded periodic grids . . . 79
7.5 Centered pure derivatives . . . 79
7.6 Centered mixed derivatives . . . 79
7.6.1 Reading the stencil for a sixty-degree cell . . . 80
7.7 Bloch-periodic seams . . . 80
7.7.1 Walking one corner across the seam . . . 81
7.7.2 Hermiticity check . . . 81
7.8 From the Laplacian to a Hamiltonian . . . 82
7.9 A one-dimensional cosine-fiber bridge . . . 82
7.9.1 Cell-periodic plane waves . . . 82
7.9.2 A grid with a Bloch seam . . . 83
7.9.3 What can be compared directly . . . 83
7.10 What the maintained implementation owns . . . 84
7.11 A Fourier-mode diagnostic . . . 86
7.12 A staged computational laboratory . . . 87
7.12.1 Stage 1: orthogonal null mode . . . 87
7.12.2 Stage 2: one-dimensional symbol inside a tensor grid . . . 87
7.12.3 Stage 3: skew intrinsic cell . . . 88
7.12.4 Stage 4: ambient rotation . . . 88
7.12.5 Stage 5: seam and corner phases . . . 88
7.12.6 Stage 6: refinement without overclaiming . . . 88
7.12.7 What a laboratory report should say . . . 88
7.13 Common mistakes . . . 89
7.14 Exercises . . . 89
7.15 Evidence boundary and next use . . . 90

8 Projection and Retained Spaces 91
8.1 Learning goals . . . 91
8.2 Orthogonal projection . . . 92
8.3 Compression to a retained space . . . 92
8.4 Spectral retention . . . 93
8.4.1 The embedded residual and its energy reference . . . 94

8.5 Isolated bands as fiberwise retained spaces . . . 94
8.6 Exact retained operator versus effective model . . . 95
8.7 Spectral projection versus eliminated-space downfolding . . . 96
8.8 Why spectral projection need not preserve finite range . . . 96
8.8.1 A two-band calculation . . . 97
8.9 From exact retention to model-class expressivity . . . 98
8.10 Operations that must not be collapsed . . . 98
8.11 A finite projector calculation . . . 99
8.12 Questions to record before retention . . . 100
8.13 Exercises . . . 101
8.14 Evidence boundary . . . 101

9 From an Isolated Band to a Finite-Range Lattice Hamiltonian 102
9.1 Learning goals . . . 102
9.2 A one-dimensional periodic parent . . . 103
9.2.1 Reduced momentum and natural scales . . . 104
9.3 Plane-wave representation of a Bloch fiber . . . 104
9.4 A finite-difference representation of the same parent . . . 105
9.4.1 A finite represented reference is not the continuum limit 106
9.5 What it means to retain an isolated band . . . 106
9.6 The complete centered reciprocal mesh . . . 107
9.7 From sampled energies to complete hoppings . . . 107
9.7.1 Hermiticity and modular opposites . . . 108
9.7.2 A four-point transform by hand . . . 108
9.8 Finite-range truncation . . . 108
9.9 Parseval's identity for the truncation residual . . . 109
9.10 The direct least-squares route . . . 109
9.10.1 Rank, identifiability, and conditioning . . . 110
9.11 Training and withheld reciprocal coordinates . . . 111
9.12 Band-shape and structural diagnostics . . . 112
9.13 A transparent computational example . . . 113
9.14 A staged isolated-band laboratory . . . 115
9.15 Common mistakes . . . 116
9.16 Exercises . . . 117
9.17 Evidence boundary and next use . . . 117

10 Composite Bloch Frames, Gauge Alignment, and Block Locality 118
10.1 Learning goals . . . 118
10.2 A finite matrix-valued periodic parent . . . 119
10.3 Composite retention and the external gap . . . 120

10.4 Frames, gauges, and projected Hamiltonians . . . 120
10.5 The Bloch–Wannier bridge . . . 121
10.6 Comparing retained frames . . . 122
10.7 Neighborwise polar transport . . . 122
10.7.1 Sewing and closure holonomy . . . 123
10.8 A controlled momentum-dependent gauge attack . . . 124
10.9 Pointwise and globally constrained alignment . . . 124
10.9.1 Pointwise unitary Procrustes alignment . . . 124
10.9.2 Uniqueness, numerical rank, and deterministic output . 126
10.9.3 One global unitary . . . 126
10.9.4 Reproducible method record . . . 127
10.10 Complete block hoppings and gauge-dependent locality . . . 128
10.11 Training and withheld roles . . . 129
10.12 A diagnostic ledger . . . 130
10.13 A transparent computational example . . . 130
10.14 Common mistakes . . . 137
10.15 Exercises . . . 138
10.16 Evidence boundary and next use . . . 139

11 Verification and Evidence Vocabulary 140
11.1 Learning goals . . . 140
11.2 Claims precede evidence . . . 141
11.3 Evidence classes . . . 141
11.4 Examples that share an array but not a claim . . . 142
11.5 Errors from different operations . . . 143
11.6 Finite-setting observations . . . 143
11.7 Finite references and exact language . . . 144
11.8 Transform training data and withheld evaluation . . . 144
11.9 Independent reconstruction is not producer replay . . . 145
11.10 Negative evidence and stop conditions . . . 145
11.11 Reproducibility does not promote evidence . . . 146
11.12 A compact evidence ledger . . . 146
11.13 Exercises . . . 146
11.14 Next use and evidence boundary . . . 147

II Criteria for a Good Reduced Model 148

12 What Makes a Reduced Model Adequate? 149
12.1 Project motivation and significance . . . 149

12.2 Exact retention and model-class approximation . . . 152
12.3 Controlled diagnostic examples . . . 153
12.4 From the particle in a box to electronic-structure reduction . . . 153
12.5 From periodic potentials to impurity operators . . . 153
12.6 Central hypothesis . . . 154
12.7 Relation to Hilbert's sixth problem . . . 156
12.8 Intellectual contribution . . . 156

13 Comparing Operators Responsibly 158
13.1 Gauge constraints in operator comparison . . . 158
13.2 Spectra and operator information . . . 159
13.3 Alignment between retained spaces . . . 159
13.4 Energy references . . . 160
13.5 Represented differences . . . 160
13.6 Hermiticity and residuals . . . 161
13.7 Approximation and evidence boundaries . . . 161

14 Admissible Sets and Compatibility Certificates 162
14.1 From one error to simultaneous requirements . . . 163
14.1.1 Example 1: intervals on a line . . . 163
14.2 The candidate family and its contracts . . . 163
14.2.1 A rank-two shift-and-splitting family . . . 164
14.3 Spectral and operator losses . . . 165
14.3.1 Spectral loss . . . 165
14.3.2 Operator loss . . . 166
14.4 Alignment variables are not model parameters . . . 167
14.5 Admissible sets and their intersection . . . 168
14.6 Why quadratic losses produce ellipses . . . 169
14.6.1 Example 2: a witness and a separation distance in two dimensions . . . 170
14.7 Witnesses and separation certificates . . . 171
14.7.1 The parameter metric is part of the claim . . . 172
14.8 Training, withheld evaluation, and locality . . . 173
14.9 Threshold sensitivity . . . 173
14.10 The controlled M3 role . . . 174
14.11 Common mistakes . . . 174
14.12 Exercises . . . 175
14.13 Chapter summary . . . 176

15 Evidence for Model Adequacy 177

15.1 Verification in the reduction program . . . 177
15.1.1 Software verification . . . 177
15.1.2 Numerical verification . . . 178
15.1.3 Compatibility witnesses and separation certificates . . . 178
15.1.4 Completed controlled benchmark inventory . . . 179
15.1.5 Finite-setting convergence . . . 180
15.1.6 Derived quantities and error amplification . . . 180
15.2 Validation and uncertainty . . . 181
15.2.1 Scientific validation . . . 181
15.2.2 Uncertainty quantification . . . 181
15.3 Model development and provenance . . . 182
15.3.1 Training, validation, and leakage . . . 182
15.3.2 Provenance and reproducibility . . . 183
15.3.3 Current execution-status example . . . 183
15.4 Failure and claim discipline . . . 184
15.4.1 Failure and negative evidence . . . 184
15.4.2 Claim-to-evidence discipline . . . 184

III The Bulk-Silicon Representation Program 185

16 Definition of the Bulk-Silicon Program 186
16.1 The parent description . . . 186
16.2 Pristine silicon host . . . 187
16.3 From parent models to reduced models . . . 187
16.4 Bulk research questions . . . 188
16.5 Interpretive limits . . . 188

17 Construction of the First-Principles Bulk Parent 189
17.1 Declared bulk branch . . . 189
17.2 External-program boundary . . . 189
17.3 Pseudopotential identity . . . 190
17.4 Controlled convergence . . . 191
17.4.1 Plane-wave and charge-density cutoffs . . . 191
17.4.2 Brillouin-zone sampling . . . 191
17.5 Equilibrium lattice procedure . . . 192
17.6 Target parent quantities . . . 192
17.7 Provenance record . . . 193
17.8 Tutorial and production evidence . . . 193

18 Retained and Localized Bulk Representations 194
18.1 Representation construction . . . 194
18.1.1 Bloch representation . . . 194
18.1.2 Target subspace . . . 195
18.1.3 Projection and disentanglement . . . 195
18.1.4 Wannier transformation . . . 196
18.1.5 Localization . . . 197
18.2 Hamiltonian reconstruction and validation . . . 197
18.2.1 Reciprocal- and real-space Hamiltonians . . . 197
18.2.2 Validation of a candidate Wannier representation . . . 198
18.3 Alignment and comparison . . . 198
18.3.1 Why independent constructions require alignment . . . 198
18.3.2 Subspace diagnostics . . . 199
18.3.3 Transport and comparison . . . 199
18.3.4 Energy and gauge sensitivity . . . 200

19 Bulk Reduced-Model Classes 201
19.1 Bulk reduction program . . . 201
19.1.1 Reduction as a constrained problem . . . 201
19.1.2 Bulk orthogonal sp^3s^* hierarchy . . . 202
19.1.3 Two bulk reduction routes . . . 202
19.1.4 Training and withheld validation . . . 203
19.1.5 Aligned operator objective . . . 203
19.1.6 Spectral and operator admissibility . . . 203
19.2 Path dependence and error accounting . . . 204
19.2.1 Path dependence and noncommutativity . . . 204
19.2.2 Error accounting . . . 205

20 Selecting the Bulk-Silicon Representation 206
20.1 The unresolved bulk question . . . 206
20.2 Near-term bulk program . . . 206

IV The Doped-Silicon Representation Program 208

21 Doped-Silicon Parent Systems and Compatibility 209
21.1 Compatible host and dopant branches . . . 209
21.2 Substitutional phosphorus . . . 209
21.3 Substitutional boron . . . 210
21.4 Dopant research questions . . . 211

21.5 Interpretive limits 211

22 Extraction of the Impurity Operator 212
22.1 Purpose and terminology 212
22.2 Matched periodic parent problems 213
22.3 Retained spaces and finite representations 214
22.4 Compatibility gate 216
22.5 Scalar energy-reference alignment 217
22.6 Common-space alignment and signed extraction 219
22.7 Localized matrix interpretation 221
22.8 Locality and finite-extent diagnostics 222
22.9 Supercell convergence and periodic-image contamination 224
22.10 Verification residuals and evidence classes 225
22.11 Dopant-specific interpretation 227

23 Lattice Impurity Model Classes 228
23.1 Nested impurity model classes 228

24 Continuum Effective-Mass Reduction 229
24.1 Continuum effective-mass reduction 229

25 Structured Learning as a Model-Class Diagnostic 231
25.1 The aligned target precedes learning 231
25.2 Structured parameter prediction 232
25.3 Expressiveness through nested model classes 233
25.4 A learned reference for the continuum crossover 234
25.5 Learning the residual beyond a continuum model 235
25.6 Composite objectives and acceptance 235
25.7 Joint learning of identification and reduction 236
25.8 A staged computational progression 237
25.9 Scientific question and evidence boundary 239

26 Dopant Transferability and Validation Program 240
26.1 Phosphorus as the first impurity demonstration 240
26.2 Boron as a transferability branch 240
26.3 Methodological implications 241
26.3.1 Structured learning as a later diagnostic 241
26.3.2 Composition and path dependence 241
26.3.3 Negative results as supported outcomes 241

27 Current Evidence Boundary 243

27.1 Meaning of “result” in this chapter . . . 243
27.2 Accepted represented-operator software foundation . . . 244
27.3 Accepted legacy tutorial observations . . . 244
27.4 Direct PBE bootstrap observations . . . 245
27.5 Exploratory silicon Wannier decomposition . . . 246
27.6 Controlled periodic-reduction benchmarks . . . 247
27.7 Canonical constrained-admissible-set result . . . 248
27.8 Controlled impurity-extraction benchmarks . . . 249
27.9 Current scientific result boundary . . . 249
27.10 Present limitations . . . 250
27.10.1 Parent calculation . . . 250
27.10.2 Representation . . . 251
27.10.3 Bulk model reduction . . . 251
27.10.4 Impurity branches . . . 251
27.10.5 Continuum reduction . . . 251
27.11 Negative findings . . . 252
27.12 Result table contract . . . 252
27.13 Figure contract . . . 253
27.14 Path to populated scientific results . . . 253

V Mathematical and Proof Program 254

28 Proof Architecture 255
28.1 Why a separate proof program is needed . . . 255
28.2 Proof-package graph . . . 256
28.3 Status vocabulary . . . 256

29 Finite-Dimensional Operator Foundations 258
29.1 Finite-dimensional retained frames . . . 258
29.2 Identification pullback and aligned subtraction . . . 259
29.3 Invariant norms and gauge-dependent locality . . . 260
29.4 TB-anchored identification . . . 261

30 Mathematical Structure of Reduction and Continuum Claims 262
30.1 Excluded-space reduction . . . 262
30.2 Crossover and operator-to-observable bounds . . . 263
30.3 Continuum fitting and identifiability . . . 264
30.4 Compatibility and certified incompatibility . . . 264
30.5 Reduction-path commutativity . . . 265

31 Mechanization Status and Open Obligations 267
31.1 Mechanization status . . . 267
31.2 Novelty and evidence boundary . . . 268

32 Conclusions and Research Sequence 269
32.1 Present conclusions . . . 269
32.2 Outputs and perspective . . . 270
32.2.1 Scholarly outputs . . . 270
32.2.2 Final perspective . . . 271

Appendices 272

A Notation, Conventions, and Status Vocabulary 273
A.1 Physical and mathematical objects . . . 273
A.2 Periodic and Wannier notation . . . 274
A.3 Frames, gauges, and alignment . . . 275
A.4 Reduction notation . . . 276
A.5 Excluded-space and continuum notation . . . 277
A.6 Norms and residuals . . . 277
A.7 Proof status . . . 278
A.8 Scientific and evidentiary status . . . 278
A.9 Units and coordinate conventions . . . 279

B Hilbert–Schmidt Geometry and the Frobenius Norm 280
B.1 Hilbert–Schmidt inner product . . . 280
B.2 Common projection and reduced components . . . 281
B.3 Aligned pristine–doped operators . . . 282
B.4 Domains and the particle in a box . . . 283
B.5 Infinite-dimensional qualification . . . 283

C Operator Spaces, Compression, and Alignment 285
C.1 Scope . . . 285
C.2 Operators and their domains . . . 285
C.3 Spectral restriction . . . 286
C.4 Compression and downfolding . . . 287
C.5 Common compression and additive structure . . . 287
C.6 Hilbert–Schmidt geometry . . . 288
C.7 Alignment across retained spaces . . . 289

D Particle-in-a-Box Residuals and Identifiability 290

D.1 Purpose of the example . . . 290
D.2 Continuum Hamiltonian . . . 291
D.3 Finite-difference realization . . . 291
D.4 Spectral reconstruction and retention . . . 292
D.5 Three inequivalent residuals . . . 293
D.5.1 Consistently compressed physical potential . . . 293
D.5.2 Projected Hamiltonian minus unprojected kinetic operator 293
D.5.3 Difference between discrete boundary realizations . . . 294
D.6 Residual identifiability . . . 294
D.7 Retained numerical resolution . . . 295

E Two Finite Harmonic-Oscillator Constructions: A Frobenius Comparison 297
E.1 Question addressed by the experiment . . . 297
E.2 Exact retained ladder Hamiltonian . . . 298
E.3 Harmonic potential inside a Dirichlet box . . . 299
E.4 Map from ladder states to a spatial representation . . . 299
E.5 Finite-box Hamiltonian in ladder coordinates . . . 300
E.6 Comparing different spatial discretizations . . . 301
E.7 Convergence questions . . . 302
E.8 Calculated graphics and retained result . . . 303
E.9 Interpretation boundary . . . 303

F Direct and Wannier-Mediated Tight-Binding Reduction in Silicon 305
F.1 Purpose . . . 305
F.2 Part I: Mathematical issues . . . 306
F.2.1 Parent Bloch fibers and retained subspaces . . . 306
F.2.2 From the retained KS operator to a Wannier representation 306
F.2.3 Tight-binding model class . . . 307
F.2.4 Direct KS-DFT-to-TB route . . . 308
F.2.5 Wannier-mediated route . . . 308
F.2.6 Reciprocal-space and real-space Frobenius residuals . . . 309
F.2.7 The three pairwise relations . . . 309
F.2.8 Path dependence . . . 310
F.2.9 Error accounting . . . 310
F.3 Part II: Proposed numerical experiment . . . 311
F.3.1 Common parent and frozen comparison data . . . 311
F.3.2 Stage A: establish the retained parent target . . . 311
F.3.3 Stage B: construct and diagnose the Wannier representation 312
F.3.4 Stage C: direct spectral TB reduction . . . 312
F.3.5 Stage D: Wannier-mediated operator reduction . . . 313

F.3.6 Stage E: compare the routes . . . 313
F.3.7 Proposed comparison tables . . . 314
F.4 Plot placeholders . . . 314
F.5 Interpretation boundary . . . 315

G Periodic-Reduction Experiments 316
G.1 Introduction . . . 317
G.2 Continuum parent and exact structure . . . 318
G.3 Numerical representations of the parent . . . 319
G.3.1 Plane-wave Galerkin representation . . . 319
G.3.2 Finite-difference Bloch-boundary representation . . . 320
G.3.3 Comparison in common coordinates . . . 320
G.4 Retained subspaces and localization . . . 321
G.4.1 Isolation and subspace diagnostics . . . 321
G.4.2 Isolated-band parallel transport . . . 321
G.4.3 Composite-band transport . . . 322
G.4.4 Wannier90 interface . . . 323
G.5 Finite-range lattice reduction . . . 323
G.5.1 Complete hopping representation . . . 323
G.5.2 Finite hopping range . . . 324
G.5.3 Direct and mediated routes . . . 324
G.6 Numerical protocol . . . 325
G.7 Results . . . 326
G.7.1 Parent representation verification . . . 326
G.7.2 Isolated lowest band . . . 327
G.7.3 Sensitivity and breakdown of scalar-band reduction . . . 328
G.7.4 Composite-band remedy . . . 328
G.7.5 Wannier90 localization . . . 329
G.8 Discussion . . . 330
G.8.1 Representation agreement requires an explicit map . . . 330
G.8.2 Band labels fail before composite subspaces do . . . 330
G.8.3 Localization controls truncation, not the complete spectrum . . . 330
G.8.4 Route equivalence has precise hypotheses . . . 331
G.8.5 Implications for multiband reductions . . . 331
G.9 Canonical constrained admissible-set extension . . . 331
G.10 Limitations and scope . . . 333
G.11 Conclusions . . . 334
G.12 Computational reproducibility . . . 335

H Two-Dimensional Wannier Reduction: From Separable to Coupled Potentials **342**
H.1 Purpose . . . 342
H.2 Continuum parent family . . . 343
H.3 Separable reference limit . . . 344
H.4 Plane-wave Bloch representation . . . 345
H.5 Real-space Bloch representation . . . 345
H.6 Symmetry and degeneracy diagnostics . . . 346
H.7 Effective-mass tensor . . . 347
H.8 Direct two-dimensional gauge construction . . . 347
H.9 Topology and the localization boundary . . . 348
H.10 Wannier90 localization in two dimensions . . . 348
H.11 Two-dimensional Wannier Hamiltonian . . . 350
H.12 Two-dimensional tight-binding hierarchy . . . 350
H.13 Direct and mediated reduction routes . . . 351
H.14 Staged method and execution status . . . 352
H.14.1 Stage A: reproduce the separable product structure . . . 352
H.14.2 Stage B: converge the two parent representations . . . 352
H.14.3 Stage C: construct direct two-dimensional gauges . . . 352
H.14.4 Stage D: compare with Wannier90 . . . 353
H.14.5 Stage E: verify topology and symmetry . . . 353
H.14.6 Stage F: truncate by two-dimensional hopping shell . . . 353
H.14.7 Stage G: compare direct and mediated reductions . . . 353
H.14.8 Stage H: turn on nonseparable coupling . . . 353
H.15 Error accounting . . . 354
H.16 Calculated result . . . 354
H.16.1 Rank-three composite extension . . . 355
H.16.2 Retained failure and corrected Wannier90 comparison . 357
H.16.3 Three distinct topological controls . . . 360
H.16.4 Bounded non-DFT Wannier90 sensitivity study . . . 362
H.16.5 Pilot deterministic multi-initialization study . . . 362
H.16.6 Expanded fixed-embedding and continuation study . . . 367
H.16.7 Separate topological phase sweeps . . . 368
H.17 Interpretation boundary . . . 368

I Controlled Impurity-Extraction Benchmarks **372**
I.1 Physical objective . . . 372
I.2 Representation hierarchy . . . 373
I.3 Pristine silicon reference . . . 373
I.4 Aligned impurity operator . . . 374

I.5 Non-identifiability of an effective potential . . . 375
I.6 Admissible model classes . . . 376
I.7 Continuum effective-mass limit . . . 377
I.8 Dopant branches and validation . . . 378
I.9 Controlled benchmark program . . . 379
I.9.1 Warmup 1: Spinless, spin-degenerate, collinear, and spinor spaces . . . 380
I.9.2 Warmup 2: One-dimensional matched pristine and defect supercells . . . 383
I.10 Benchmark conclusion and two-dimensional handoff . . . 397

J Two-Dimensional Defect Extraction and Material-Transfer Program 398
J.1 Purpose and prerequisite evidence . . . 398
J.2 Two-dimensional specialization of the extraction contract . . . 399
J.3 Completed Stage A null and folding control . . . 400
J.4 Completed Stage B scalar-onsite and symmetry control . . . 400
J.5 Matched pristine and defected-system supercells . . . 402
J.6 Energy-reference, gauge, ordering, and geometry stress tests . . . 403
J.7 Supercell-size and shape convergence . . . 404
J.8 Recovery of known model classes . . . 405
J.9 Smoothness-to-continuum crossover . . . 406
J.10 Operator error versus bound-state error . . . 407
J.11 Null, covariance, and stopping conditions . . . 408
J.12 Required warmup order . . . 409
J.13 Remaining warmup plot placeholder . . . 410
J.14 Research questions . . . 410

K Envelope Theory 411
K.1 Why these papers matter to the present problem . . . 411
K.2 The periodic parent and the adapted basis . . . 412
K.3 The gentle-potential approximation . . . 412
K.4 Removing interband coupling . . . 413
K.5 The simple-band envelope equation . . . 414
K.6 Band minima away from the zone center and multiple valleys . . . 414
K.7 Degenerate bands and coupled envelopes . . . 415
K.8 Spin–orbit coupling and the valence manifold . . . 415
K.9 Magnetic fields . . . 416
K.10 Burt's exact envelope-function representation . . . 416
K.11 Exact equations before reduction . . . 417
K.12 Global slow variation rather than a gentle structure . . . 418

K.13 Dominant envelopes and effective-mass reduction 418
K.14 Operator ordering and whole-matrix Hermiticity 419
K.15 Interfaces, boundary conditions, and current 420
K.16 Luttinger–Kohn and Wannier–Slater envelopes 421
K.17 Out-of-zone solutions and representation completeness 421
K.18 Observables are not determined by the dominant envelope alone . . 422
K.19 Ermoneit's exact multivalley specialization 423
K.19.1 The project–multiply–project potential kernel 424
K.19.2 Energy-reference invariance 424
K.19.3 Intervalley coupling and valley splitting 425
K.19.4 Spectrally filtered local approximation 425
K.20 Mapping the papers onto the project reduction chain 426
K.21 Implications for phosphorus and boron in silicon 428
K.22 Error and evidence interpretation 428
K.23 Summary of what the papers establish 429
K.24 Questions carried forward 430

L Candidate Contemporary Literature for Later Review 431
L.1 Quantum ESPRESSO 432
L.2 Wannier90 432
L.3 PseudoDojo 432
L.4 Defect supercells and charged-defect alignment 433
L.5 Machine-learned tight-binding and DFT Hamiltonians 433
L.6 Verification, validation, and uncertainty quantification 434
L.7 Effective mass and envelope theory 434
L.8 Kohn–Sham and fundamental gaps 435
L.9 Review disposition 435

M Preliminary Silicon Work: Executions, Finite-Setting Evidence, and an Exploratory Wannier Decomposition 436
M.1 Purpose and evidence boundary 436
M.2 Source records 437
M.3 Legacy tutorial baseline 438
M.4 QE 7.5 smoke comparison 439
M.5 PBE/PseudoDojo finite-setting matrix 439
M.5.1 What the finite-setting data do and do not show 440
M.6 Runtime warning diagnosis and remaining block 441
M.7 Exploratory same-frame Wannier kinetic decomposition 441
M.7.1 Same-frame construction 442
M.7.2 Reported finite-representation diagnostics 443

M.7.3 Local rank-three expansion at Γ 443
M.7.4 Finite-mesh block truncation 444
M.8 What has not yet been established 444

Index **458**

PART I

Computational Foundations

CHAPTER 1

Computational Quantities: Units, Coordinates, and Metadata

Status pedagogical foundation. The examples use illustrative synthetic values and establish no material result, numerical convergence claim, scientific validation, or uncertainty quantification.

1.1 Learning goals

After working through this chapter, a reader should be able to

1. distinguish a mathematical number from a computational quantity;
2. attach units, coordinate meaning, indexing conventions, and provenance to numerical values;
3. explain why shape and data type do not determine scientific meaning;
4. convert units and coordinates without silently changing the represented quantity; and
5. identify defining metadata that must accompany a retained array.

This monograph begins with data rather than with a Hamiltonian. Every later vector, matrix, grid, and operator is stored as numbers, and those numbers are usable only when their interpretation is explicit.

1.2 From numbers to quantities

A floating-point value such as 5.43 is a number. It becomes a computational quantity only after its role and unit are supplied. A lattice length might be recorded schematically as

$$q = (5.43,\ \text{angstrom}). \tag{1.1}$$

A conversion changes the numerical value and the unit together while preserving the physical quantity. Changing only one member of the pair changes the meaning.

The same rule applies to arrays. An array does not announce whether it contains positions, reciprocal vectors, energies, expansion coefficients, or matrix elements. Its shape and data type are necessary software facts, but they are not a scientific interpretation.

Definition 1.1 (Computational quantity)**.** A computational quantity is a numerical value or array together with the metadata required to interpret it for a declared operation. Depending on the operation, that metadata can include units, coordinate system, axis meanings, ordering, normalization, reference value, geometry, and provenance. ■

The phrase "required to interpret" is operation dependent. Adding two energies requires compatible energy units and references. Taking a dot product of Cartesian displacement vectors requires compatible coordinates and length units. Comparing represented Hamiltonians requires substantially more, as later chapters show.

1.3 Dimensions, units, and reference values

A physical dimension describes the kind of quantity, such as length, inverse length, or energy. A unit selects a numerical scale within that dimension. Angstrom and bohr are different length units; electronvolt and hartree are different energy units. Unit compatibility does not imply equality of numerical values, so conversion must be explicit.

Some quantities also depend on a reference. If an energy is reported relative to a chosen zero E_0, then

$$\overline{E} = E - E_0. \tag{1.2}$$

Two arrays expressed in electronvolts can still be incompatible when their energy zeros differ. A unit conversion cannot repair a reference mismatch.

A useful dimensional audit asks three questions before arithmetic:

1. Do the operands represent the same physical dimension?
2. Are their units identical, or is an explicit conversion required?

3. Does either quantity depend on an origin, zero, normalization, or other reference convention?

Dimensionless does not mean convention free. Fractional coordinates, reduced Bloch twists, phases, and normalized coefficients can all be dimensionless while still requiring a basis, interval, or phase convention.

1.4 Coordinates are part of the data

The tuple $(1, 0)$ can denote a Cartesian vector, fractional lattice coordinates, grid indices, or coefficients in an abstract basis. These are not interchangeable meanings. A coordinate record should identify at least

- the space in which the coordinates live;
- the ordered axes or basis vectors;
- whether components are Cartesian, fractional, reciprocal, or abstract;
- the component unit, if any;
- the origin or reference point when relevant; and
- the vector-axis and array-order convention.

A coordinate transformation changes the numerical components while preserving the represented object only when the transformation rule is applied consistently. If A stores basis vectors as columns and $\mathbf{s}$ stores fractional coordinates, then

$$\mathbf{r} = A\mathbf{s}. \tag{1.3}$$

Transposing A without changing the convention is not a harmless formatting operation. Chapter 6 develops this map for periodic cells after the required linear algebra has been introduced.

1.5 Indexing, axis meaning, and ordering

Computational arrays have both mathematical axes and storage conventions. For a table $X \in \mathbb{R}^{N \times d}$, rows might enumerate points and columns might enumerate Cartesian

components. Another program may store the transpose. Both layouts can be valid, but an unlabeled shape does not distinguish them.

Index origins also matter. Mathematical derivations often number basis vectors from 1 to N, whereas Python arrays use indices from 0 to $N - 1$. A translation between these conventions should occur at a named boundary, not implicitly inside a scientific argument.

For tensor-product grids, a multi-index must also be mapped to a linear index. C ordering makes the final axis vary fastest; Fortran ordering makes the first axis vary fastest. Two matrices assembled with different flattening conventions can have identical shapes and spectra while assigning entries to different physical grid points.

1.6 Metadata needed by represented scientific data

The interpretation of a finite represented operator can depend on

- the identity and dimension of the represented state space;
- basis identity, orbital convention, and ordering;
- geometry, lattice vectors, site labels, and boundary convention;
- units and normalization;
- scalar energy reference;
- spin representation;
- gauge, phase, or other coordinate conventions; and
- provenance linking the record to its inputs and construction.

Not every elementary array needs every field. The rule is instead that the record must carry enough information for its promised operations to fail closed. If subtraction requires a common energy reference, an absent reference is not permission to assume one. If a decoder promises a column-vector convention, a row-vector document is not an alternative spelling of the same record.

It is useful to distinguish defining data from derived data. A primitive basis can define a metric, reciprocal basis, and projector. Serializing all of them as independent authorities creates the possibility of internal disagreement. Retaining the defining data and deterministically recomputing derived data often produces a stronger record.

1.7 A small transparent record

The following example keeps an energy array, its unit, and its reference in one visible record. It uses a plain dictionary only for pedagogy; maintained project interfaces use typed immutable records and explicit validation.

Python interlude—converting an energy quantity without losing its reference

```
import numpy as np

# These synthetic levels are measured relative to one declared
    energy zero.
levels_ev = np.array([0.0, 0.12, 0.31], dtype=np.float64)
record_ev = {
    "values": levels_ev,
    "unit": "eV",
    "energy_reference": "illustrative band edge",
    "axis_meaning": "state index in ascending eigensystem order",
}

# Convert the values and unit together; preserve the non-numeric
    metadata.
ev_per_hartree = 27.211386245988
record_hartree = {
    **record_ev,
    "values": record_ev["values"] / ev_per_hartree,
    "unit": "hartree",
}

# Recover the original numerical representation as an independent
    check.
np.testing.assert_allclose(
    record_hartree["values"] * ev_per_hartree,
    record_ev["values"],
)
assert record_hartree["energy_reference"] == record_ev["
    energy_reference"]
```

The example verifies one conversion identity for synthetic values. It does not establish a project serialization contract or any physical energy level.

Notebook to do Create `examples/tutorials/research-monograph/foundations/computational_quantities.ipynb` with explicit imports, explanatory inline comments, unit conversion, coordinate annotation, malformed-metadata cases, and no hidden state. Keep it unexecuted in the repository and verify it through a separate bounded test route.

1.8 Compatibility before arithmetic

A robust computational sequence is

$$\begin{aligned}\text{decode} &\longrightarrow \text{validate each record} \longrightarrow \text{check pairwise compatibility} \\ &\longrightarrow \text{perform arithmetic} \longrightarrow \text{retain the result context.}\end{aligned} \tag{1.4}$$

Validation of each operand and compatibility between operands answer different questions. Two individually valid energy arrays can still use different units, references, bases, or state orderings.

Automatic inference should be limited to conventions owned by a documented contract. Inferring a unit from the magnitude of an array, an axis meaning from its length, or a physical correspondence from matching labels replaces an explicit scientific decision with a heuristic.

1.9 Exercises

1. List the metadata needed to add two Cartesian force vectors and explain which fields are unnecessary for that operation.
2. Give two different scientific interpretations of a complex 4×4 array that have the same shape and data type.
3. Describe why two energy arrays with identical units but different scalar zeros cannot be subtracted as a physical perturbation without alignment.
4. Write a small record for three fractional coordinates stored by rows. Include dimension, axis meaning, origin convention, and provenance.
5. Classify the direct basis, Gram metric, and reciprocal basis as defining or derived data under the convention developed in Chapter 6.

1.10 Evidence boundary

Metadata completeness is relative to a declared operation and contract. A successful decode or round trip establishes bounded software behavior; it does not prove that the recorded metadata are truthful, that the source calculation was adequate, or that two physical systems should be compared. The next two chapters add finite coordinate representations and then Hermitian eigensystems while preserving this distinction between numbers and represented meaning.

CHAPTER 2

Vectors, Bases, Matrices, and Finite Representations

Status pedagogical finite-dimensional foundation. The worked examples are illustrative and synthetic. They establish no material result, convergence claim, scientific validation, or uncertainty quantification.

2.1 Learning goals

After working through this chapter, a reader should be able to

1. distinguish an abstract vector from a column of its coordinates;
2. explain span, linear independence, basis, dimension, and orthonormality;
3. distinguish a linear map from the matrix representing it;
4. read the columns of a matrix as the images of basis vectors;
5. derive coordinate and matrix transformation rules under a change of orthonormal basis;
6. distinguish coordinate-dependent entries from basis-invariant properties; and
7. state the mathematical and physical metadata required by a finite representation.

Chapter 1 established that a bare number is not yet a complete computational quantity. The same lesson now applies to arrays. A column of numbers becomes the representation of a vector only after its basis and component meanings are known. A square array becomes the representation of an operator only after its state space, basis, units, and conventions are declared.

This chapter assumes only familiar arithmetic with real and complex numbers. The notation is introduced as it is used. The aim is not abstract algebra for its own sake. It is to

prevent later calculations from silently subtracting, rotating, diagonalizing, or comparing arrays that represent different physical objects.

Notation convention. An abstract vector is written in italic type, such as $v \in \mathcal{V}$. Its finite coordinate column is bold, such as $\mathbf{c} = [v]_{\mathcal{B}} \in \mathbb{C}^N$. Bold symbols such as $\mathbf{r}$, $\mathbf{k}$, and $\mathbf{a}_i$ are also used for physical spatial and reciprocal vectors in declared Cartesian coordinates. Matrices use capital letters such as A, H, and U. Quantum states will later use ket notation, such as $|\psi\rangle$. We do not use arrow accents, because they do not distinguish an abstract state from a finite coordinate column and become awkward for complex and indexed quantities.

2.2 Scalars, vectors, and vector spaces

A *scalar* is an ordinary number used to multiply another mathematical object. In this monograph scalars are usually complex numbers, $\alpha \in \mathbb{C}$, with real numbers as the special case $\alpha \in \mathbb{R}$.

In elementary mechanics, a vector is often pictured as an arrow with a length and direction. That picture works well for displacement, velocity, and force, but it does not cover every vector used in physics. A wavefunction, for example, is a vector even though it is a function rather than an arrow. The common mathematical structure is a vector space: its elements can be added to one another and multiplied by scalars while remaining in the same space.

Definition 2.1 (Vector space)**.** A vector space $\mathcal{V}$ is a collection of objects called vectors for which vector addition and scalar multiplication are defined. If $u, v \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha u + \beta v \in \mathcal{V}. \tag{2.1}$$

These operations obey the familiar distributive and associative rules of ordinary coordinate vectors. ■

2.2.1 Examples of vectors

The definition is deliberately broad. Its meaning is easier to see through examples.

1. **A spatial displacement.** After Cartesian coordinates are declared, a displacement can be represented by

$$\mathbf{r} = \begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix} \in \mathbb{R}^3. \tag{2.2}$$

Vector addition combines displacements, and scalar multiplication changes their magnitude and possibly their orientation.

2. **Orbital amplitudes.** In a finite model with N ordered orbitals, the amplitude column
$$\mathbf{c} = \begin{bmatrix} c_1 & \cdots & c_N \end{bmatrix}^{\mathsf{T}} \in \mathbb{C}^N \tag{2.3}$$
is a vector. Its entries acquire physical meaning only after the orbital ordering and normalization convention are declared.

3. **Samples of a function.** Values stored at ordered grid points can form
$$\mathbf{f} = \begin{bmatrix} f(x_1) & \cdots & f(x_N) \end{bmatrix}^{\mathsf{T}}. \tag{2.4}$$
This finite vector represents sampled information about f; it is not the continuous function itself.

4. **A polynomial.** Polynomials of degree at most q form a vector space because sums and scalar multiples remain polynomials of degree at most q. For example,
$$p(x) = a_0 + a_1 x + \cdots + a_q x^q \tag{2.5}$$
is an abstract vector in that polynomial space, while $\begin{bmatrix} a_0 & \cdots & a_q \end{bmatrix}^{\mathsf{T}}$ is its coordinate column in the ordered basis $(1, x, \ldots, x^q)$.

5. **A wavefunction.** A wavefunction $\psi(x)$ is a vector in a function space. It can be represented approximately by grid samples or by coefficients in a finite basis, but neither finite column is literally identical to the continuous function.

The examples illustrate two related uses of the word *vector*. A spatial displacement, polynomial, or wavefunction is an element of its declared vector space. A bold finite column is an element of $\mathbb{R}^N$ or $\mathbb{C}^N$ and may also serve as the coordinate representation of a vector in another space. The basis or sampling map supplies that connection.

The symbol $0_{\mathcal{V}}$ denotes the zero vector in $\mathcal{V}$. It is the vector satisfying
$$v + 0_{\mathcal{V}} = v \tag{2.6}$$
for every $v \in \mathcal{V}$. Which object plays the role of $0_{\mathcal{V}}$ depends on the space: it may be the zero displacement, the identically zero function, or a column whose entries all vanish.

2.3 Linear combinations, span, and independence

Given vectors $v_1, \dots, v_M$, an expression

$$c_1 v_1 + c_2 v_2 + \cdots + c_M v_M \tag{2.7}$$

is a *linear combination*. The scalars c_j are its coefficients.

Definition 2.2 (Span)**.** The span of $v_1, \dots, v_M$ is the set of all their linear combinations:

$$\operatorname{span}\{v_1, \dots, v_M\} = \left\{ \sum_{j=1}^{M} c_j v_j \;\middle|\; c_j \in \mathbb{C} \right\}. \tag{2.8}$$

■

For two nonparallel arrows in an ordinary plane, the span is the whole plane. For two parallel arrows, the span is only one line. Span answers the question, "Which vectors can these vectors build?"

Definition 2.3 (Linear independence)**.** The vectors $v_1, \dots, v_M$ are linearly independent when

$$\sum_{j=1}^{M} c_j v_j = 0_{\mathcal{V}} \tag{2.9}$$

implies $c_1 = \cdots = c_M = 0$. If a nontrivial choice of coefficients gives the zero vector, the vectors are linearly dependent. ■

Linear dependence means that at least one listed vector is redundant. For example, if $v_2 = 2v_1$, then

$$2v_1 - v_2 = 0_{\mathcal{V}}, \tag{2.10}$$

so v_1 and v_2 are dependent. They describe only one independent direction even though two vectors were written down.

2.4 Bases, dimension, and coordinate columns

Definition 2.4 (Basis)**.** A basis of a vector space $\mathcal{V}$ is a linearly independent set of vectors that spans $\mathcal{V}$. ■

Both conditions matter. Spanning ensures that every vector in the space can be constructed. Independence ensures that the construction is unique.

Suppose

$$\mathcal{B} = (b_1, b_2, \ldots, b_N) \tag{2.11}$$

is an ordered basis of a finite-dimensional space $\mathcal{V}$. Every $v \in \mathcal{V}$ can be written uniquely as

$$v = \sum_{i=1}^{N} c_i b_i. \tag{2.12}$$

The number N is the *dimension* of the space. The ordered column

$$[v]_{\mathcal{B}} = \mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix} \in \mathbb{C}^N \tag{2.13}$$

is the coordinate representation of v in $\mathcal{B}$. Here $\mathbb{C}^N$ denotes the set of columns containing N complex entries.

The brackets $[v]_{\mathcal{B}}$ mean "the coordinates of v in basis $\mathcal{B}$." They are useful whenever confusing the vector with its coordinates would be dangerous.

2.4.1 A coordinate example

Let b_1 point east and b_2 point north, with both having unit length. The displacement

$$v = 3b_1 + b_2 \tag{2.14}$$

has coordinate column

$$[v]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}. \tag{2.15}$$

The column does not intrinsically mean "three units east and one unit north." That meaning comes from the declared basis. In a rotated basis the same physical displacement has different coordinates.

Ordering is also part of the declaration. If the same basis vectors are listed as (b_2, b_1), then the coordinate column becomes $\begin{bmatrix} 1 & 3 \end{bmatrix}^{\mathsf{T}}$. The physical displacement did not change; the coordinate convention did.

In software, indices commonly begin at zero even when mathematical labels begin at one. That implementation convention must not silently change which basis state a stored component represents.

2.5 Inner products, lengths, and orthogonality

Vector addition and scalar multiplication tell us how to combine vectors, but they do not yet tell us how long a vector is or whether two vectors are perpendicular. In ordinary three-dimensional geometry, that information comes from the dot product. An inner product generalizes the dot product to complex coordinate vectors, functions, and the other vector spaces used in physics.

Definition 2.5 (Complex inner product)**.** An inner product $\langle u, v\rangle$ assigns a complex scalar to two vectors. We use the convention that it is conjugate-linear in its first argument and linear in its second. If $\mathbf{u} = [u]_{\mathcal{B}}$ and $\mathbf{v} = [v]_{\mathcal{B}}$ are coordinate columns in an orthonormal basis, then

$$\langle u, v\rangle = \mathbf{u}^{\dagger}\mathbf{v} = \sum_{i=1}^{N} u_i^{*} v_i, \tag{2.16}$$

where * denotes complex conjugation and $\dagger$ denotes conjugate transpose. ■

The conjugation is essential. It ensures

$$\langle v, v\rangle = \sum_{i=1}^{N} |v_i|^2 \geq 0, \tag{2.17}$$

with equality only for the zero vector.

The norm induced by the inner product is

$$\|v\| = \sqrt{\langle v, v\rangle}. \tag{2.18}$$

A vector is *normalized* when $\|v\| = 1$. Two vectors are orthogonal when

$$\langle u, v\rangle = 0. \tag{2.19}$$

A basis is *orthonormal* when

$$\langle b_i, b_j\rangle = \delta_{ij}, \tag{2.20}$$

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise.

For an orthonormal basis, Equation (2.12) can be inverted without solving a linear system:

$$c_i = \langle b_i, v\rangle. \tag{2.21}$$

Taking the inner product with b_i extracts the component along that basis vector.

For the remainder of this chapter, all coordinate bases are orthonormal. That assumption is what permits ordinary coordinate dot products and the simple coefficient formula

above. Nonorthogonal bases require additional information about basis lengths and angles. Chapter 6 derives that information through the Gram matrix when it becomes necessary for nonorthogonal periodic cells.

2.6 Linear maps

Definition 2.6 (Linear map)**.** A map $A : \mathcal{V} \to \mathcal{W}$ is linear when

$$A(\alpha u + \beta v) = \alpha A(u) + \beta A(v) \tag{2.22}$$

for every $u, v \in \mathcal{V}$ and all scalars α, β. ■

The space $\mathcal{V}$ is the domain and $\mathcal{W}$ is the codomain. These spaces are part of the definition of the map. Two formulas that produce the same numbers but act between different declared spaces are not the same represented map.

Linearity means that knowing the action of A on a basis determines its action on every vector. From

$$v = \sum_{j=1}^{N} c_j b_j \tag{2.23}$$

we obtain

$$A(v) = \sum_{j=1}^{N} c_j A(b_j). \tag{2.24}$$

This observation explains why matrix columns have the meaning they do.

2.7 Matrices represent linear maps in chosen bases

Let

$$\mathcal{B} = (b_1, \dots, b_N) \tag{2.25}$$

be an ordered orthonormal basis of $\mathcal{V}$, and let

$$\mathcal{D} = (d_1, \dots, d_M) \tag{2.26}$$

be an ordered orthonormal basis of $\mathcal{W}$. The matrix representing $A : \mathcal{V} \to \mathcal{W}$ has entries

$$[A]_{\mathcal{D} \leftarrow \mathcal{B}, ij} = \langle d_i, A(b_j) \rangle. \tag{2.27}$$

It has M rows and N columns.

The arrow in the subscript records the coordinate direction: input coordinates are taken in $\mathcal{B}$, and output coordinates are returned in $\mathcal{D}$. Column j contains all coordinates of $A(b_j)$:

$$A(b_j) = \sum_{i=1}^{M} [A]_{\mathcal{D}\leftarrow\mathcal{B},ij} d_i. \tag{2.28}$$

Therefore, if $\mathbf{x} = [v]_{\mathcal{B}}$, then

$$[A(v)]_{\mathcal{D}} = [A]_{\mathcal{D}\leftarrow\mathcal{B}}\mathbf{x}. \tag{2.29}$$

2.7.1 Reading a matrix by columns

Suppose a map on a two-dimensional space satisfies

$$A(b_1) = 2b_1 + b_2, \qquad A(b_2) = -b_1 + 3b_2. \tag{2.30}$$

Its matrix in $\mathcal{B} = (b_1, b_2)$ is

$$[A]_{\mathcal{B}} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}. \tag{2.31}$$

The first column is the coordinate column of $A(b_1)$; the second is the coordinate column of $A(b_2)$.

For

$$v = 4b_1 + 2b_2, \qquad [v]_{\mathcal{B}} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \tag{2.32}$$

matrix multiplication gives

$$[A(v)]_{\mathcal{B}} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \end{bmatrix}. \tag{2.33}$$

The multiplication combines the two matrix columns with the same coefficients used to build v, exactly as Equation (2.24) requires.

A square matrix represents a map from a finite space to itself. A rectangular matrix represents a map between spaces of different dimensions or between different selected subspaces. Shape alone does not identify what either axis means.

2.8 Identity, adjoint, and unitary maps

The identity map leaves every vector unchanged:

$$I(v) = v. \tag{2.34}$$

In any basis its matrix is the identity matrix I_N, with ones on the diagonal and zeros elsewhere.

For a matrix A, the conjugate transpose $A^\dagger$ is obtained by transposing the array and complex-conjugating every entry. It represents the adjoint map when the domain and codomain bases are orthonormal. The adjoint is characterized abstractly by

$$\langle w, Av \rangle = \langle A^\dagger w, v \rangle. \tag{2.35}$$

For real matrices, conjugate transpose reduces to ordinary transpose.

Definition 2.7 (Unitary matrix)**.** A square complex matrix U is unitary when

$$U^\dagger U = U U^\dagger = I_N. \tag{2.36}$$

Equivalently, $U^{-1} = U^\dagger$. ■

The columns of a unitary matrix are an orthonormal basis of $\mathbb{C}^N$. Unitary matrices preserve coordinate inner products and Euclidean norms:

$$(U\mathbf{u})^\dagger (U\mathbf{v}) = \mathbf{u}^\dagger \mathbf{v}, \qquad \|U\mathbf{v}\|_2 = \|\mathbf{v}\|_2. \tag{2.37}$$

Real unitary matrices are called orthogonal matrices.

Chapter 3 will define Hermitian matrices and show why unitary basis matrices are central to their diagonalization.

2.9 Changing coordinates without changing the object

Let

$$\mathcal{B} = (b_1, \dots, b_N) \tag{2.38}$$

and

$$\mathcal{B}' = (b'_1, \dots, b'_N) \tag{2.39}$$

be two ordered orthonormal bases of the same space. Store the new basis vectors as columns in the old basis:

$$b'_j = \sum_{i=1}^{N} b_i U_{ij}. \tag{2.40}$$

The matrix U is unitary because both bases are orthonormal.

Let $\mathbf{c} = [v]_{\mathcal{B}}$ and $\mathbf{c}' = [v]_{\mathcal{B}'}$. Expanding the same abstract vector in both bases gives

$$v = \sum_i c_i b_i = \sum_j c'_j b'_j. \tag{2.41}$$

Substitute Equation (2.40) into the second sum. The old coordinate column then satisfies

$$\mathbf{c} = U\mathbf{c}'. \tag{2.42}$$

Multiplication by $U^\dagger = U^{-1}$ gives

$$\mathbf{c}' = U^\dagger \mathbf{c}. \tag{2.43}$$

The vector did not move. Only its coordinate description changed. This is a *passive* change of basis. It should not be confused with actively applying a physical rotation or symmetry operation to the vector.

If $A : \mathcal{V} \to \mathcal{V}$, then its matrix in the new basis is

$$[A]_{\mathcal{B}'} = U^\dagger [A]_{\mathcal{B}} U. \tag{2.44}$$

The factor on the right converts new input coordinates to old coordinates. The factor on the left converts the old output coordinates back to new coordinates.

For a map between two different spaces, the domain and codomain can have different basis changes. If $U_{\mathcal{V}}$ changes the domain basis and $U_{\mathcal{W}}$ changes the codomain basis, then

$$[A]' = U_{\mathcal{W}}^\dagger [A] U_{\mathcal{V}}. \tag{2.45}$$

This more general rule matters whenever a rectangular projection, embedding, or comparison map is represented.

2.9.1 A rotated-coordinate example

Define

$$b_1' = \frac{b_1 + b_2}{\sqrt{2}}, \qquad b_2' = \frac{-b_1 + b_2}{\sqrt{2}}. \tag{2.46}$$

Then

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}. \tag{2.47}$$

For the earlier vector $v = 3b_1 + b_2$,

$$[v]_{\mathcal{B}'} = U^{\mathsf{T}} [v]_{\mathcal{B}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ -\sqrt{2} \end{bmatrix}. \tag{2.48}$$

The columns $\begin{bmatrix} 3 & 1 \end{bmatrix}^{\mathsf{T}}$ and $\begin{bmatrix} 2\sqrt{2} & -\sqrt{2} \end{bmatrix}^{\mathsf{T}}$ look different, but they represent the same vector in different bases.

2.10 What changes and what does not

Under the unitary similarity transformation in Equation (2.44), individual matrix entries generally change. The following quantities do not:

- dimension and rank;
- trace and determinant;
- eigenvalues, including multiplicities; and
- unitarily invariant norms, including the Frobenius norm.

These are basis invariants. In contrast, a particular diagonal entry, a named row, or a componentwise residual depends on the basis unless additional structure identifies those coordinates.

A diagonal matrix is not intrinsically diagonal in every basis. It states that the selected coordinate axes are adapted to the map. After a general basis change, the same map may have off-diagonal entries. Conversely, making a matrix look different by changing coordinates does not by itself create a different physical operator.

Basis invariance also has limits. Two arrays with equal eigenvalues are not automatically representations of the same physical operator. Their state spaces, basis meanings, energy references, and other physical metadata must still be identified. Chapter 3 develops this distinction for spectra and eigenspaces.

2.11 Finite representations and their contracts

Definition 2.8 (Finite representation)**.** A finite representation is a finite array together with the mathematical and physical declarations needed to interpret its indices and permitted operations. ■

A useful representation contract answers at least four questions.

1. **What is represented?** Name the vector, linear map, bilinear form, or operator and its source model.
2. **On which finite spaces?** State the domain, codomain, dimensions, and any selected subspaces.
3. **In which coordinates?** State ordered bases, axis meanings, index ordering, normalization, and gauge conventions.

4. **With which quantities?** State units, physical references, geometry, and provenance required by the promised operations.

A finite matrix can arise in several inequivalent ways:

- by declaring a model with finitely many degrees of freedom;
- by restricting an existing map to a finite invariant subspace;
- by projecting onto a selected subspace;
- by sampling a continuous kernel or function; or
- by discretizing a differential operator.

These constructions are not synonyms. A 100×100 matrix does not reveal whether it is an exact finite model, a projection, or a grid approximation. That distinction belongs to the representation contract.

Sparse and dense arrays may represent the same finite map, but their computational contracts differ. Sparse structure should be preserved whenever the mathematics and downstream operations allow it. Converting a sparse operator to a dense array merely to test symmetry or Hermiticity can turn a local diagnostic into an unnecessary memory-intensive operation.

2.12 A two-basis calculation

We first perform the complete calculation symbolically. Let a vector and a linear map have the following representations in the original basis $\mathcal{B}$:

$$\mathbf{c} = [v]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad A_{\mathcal{B}} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}. \tag{2.49}$$

Use the rotated basis whose columns in the original coordinates form

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad U^{\mathsf{T}} U = I_2. \tag{2.50}$$

Because this example is real, $U^{\dagger} = U^{\mathsf{T}}$. The coordinates of the same vector in the new basis are

$$\mathbf{c}' = U^{\mathsf{T}} \mathbf{c} = \begin{bmatrix} 2\sqrt{2} \\ -\sqrt{2} \end{bmatrix}. \tag{2.51}$$

The matrix representing the same linear map in the new basis is

$$A_{\mathcal{B}'} = U^{\mathsf{T}} A_{\mathcal{B}} U \tag{2.52}$$

$$= \begin{bmatrix} 5/2 & -1/2 \\ 3/2 & 5/2 \end{bmatrix}. \tag{2.53}$$

Neither $\mathbf{c}' = \mathbf{c}$ nor $A_{\mathcal{B}'} = A_{\mathcal{B}}$. That is expected: the coordinate descriptions changed.

We can confirm that the represented action did not change. In the original basis,

$$[A(v)]_{\mathcal{B}} = A_{\mathcal{B}}\mathbf{c} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}. \tag{2.54}$$

In the new basis,

$$[A(v)]_{\mathcal{B}'} = A_{\mathcal{B}'}\mathbf{c}' \tag{2.55}$$

$$= \begin{bmatrix} 11/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \tag{2.56}$$

$$= U^{\mathsf{T}}[A(v)]_{\mathcal{B}}. \tag{2.57}$$

The two output columns differ because they use different bases, but the final equality shows that they are coordinates of the same output vector.

The code below repeats these steps numerically. It prints the transformed objects and small round-trip errors so that a reader can inspect what happened. It does not use a testing framework.

Python interlude—representing one linear map in two bases

```
import numpy as np

# Coordinates of one abstract vector in the original ordered basis.
c = np.array([3.0, 1.0])

# Columns of U are the new orthonormal basis vectors in old
    coordinates.
U = np.array([[1.0, -1.0], [1.0, 1.0]]) / np.sqrt(2.0)

# Represent one synthetic linear map in the original basis.
A = np.array([[2.0, -1.0], [1.0, 3.0]])

# Describe the same vector and map in the new basis.
c_prime = U.T @ c
A_prime = U.T @ A @ U
```

```
# Compute the map's action in both coordinate systems.
Av_old = A @ c
Av_new = A_prime @ c_prime

print("coordinates in the old basis:", c)
print("coordinates in the new basis:", c_prime)
print("matrix in the old basis:\n", A)
print("matrix in the new basis:\n", A_prime)
print("output coordinates in the old basis:", Av_old)
print("output coordinates in the new basis:", Av_new)

# Values near zero show agreement up to floating-point roundoff.
print("basis orthonormality error:", np.linalg.norm(U.T @ U - np.
    eye(2)))
print("vector round-trip error:", np.linalg.norm(U @ c_prime - c))
print("matrix round-trip error:", np.linalg.norm(U @ A_prime @ U.T -
     A))
print("action consistency error:", np.linalg.norm(Av_new - U.T @
    Av_old))
```

This example demonstrates coordinate transformation rules on one synthetic vector and one synthetic 2×2 matrix. The displayed errors are floating-point consistency diagnostics, not physical uncertainties. The example does not identify a physical state space, operator, Hamiltonian, or material system.

Notebook to do Create `examples/tutorials/research-monograph/foundations/finite_representations.ipynb` with explicit imports, explanatory inline comments, span and independence examples, ordered-basis metadata, rectangular maps, orthonormal basis changes, and sparse-preserving invariant checks. Keep it unexecuted in the repository.

2.13 Common mistakes

- Treating an abstract vector as though it were identical to one of its coordinate columns.
- Calling a set a basis after checking only that it spans, or only that it is independent.
- Forgetting that basis ordering determines coordinate ordering.

- Reading matrix rows as the images of input basis vectors; those images are stored in columns under the convention used here.
- Applying U where a passive coordinate change requires $U^\dagger$.
- Comparing entries of two matrices before identifying their domain, codomain, and bases.
- Assuming equal shape or equal eigenvalues establishes equal represented meaning.
- Densifying a sparse representation merely to run a diagnostic.

2.14 From representations to eigensystems

A matrix may admit a basis in which its action becomes diagonal. Finding and interpreting that basis is not merely another array conversion: existence, orthogonality, multiplicity, and numerical stability depend on the class of matrix being studied. Chapter 3 develops those questions for the real-symmetric and complex-Hermitian matrices used throughout the monograph.

2.15 Exercises

1. Decide whether each of the following is a vector, a coordinate representation, or either depending on context: a velocity arrow, a list of grid values, a polynomial, and a NumPy array.
2. Determine whether $\{(1, 0), (2, 0)\}$ spans $\mathbb{R}^2$, and determine whether it is linearly independent.
3. Show that the coordinate coefficients in an orthonormal basis satisfy $c_i = \langle b_i, v \rangle$.
4. Derive Equation (2.43) from Equation (2.40).
5. Derive Equation (2.44) by following an input column from new coordinates to old coordinates, applying the old matrix, and converting the result back.
6. Give an example of a rectangular matrix and identify its domain basis, codomain basis, row meaning, and column meaning.
7. Construct two matrices representing the same map in different orthonormal bases. Identify entries that change and quantities that remain invariant.

8. Give one example each of a finite matrix produced by restriction, projection, sampling, and discretization.

2.16 Evidence boundary

Exact coordinate round trips and invariant checks can verify implementation of a finite representation. They do not establish that the represented space is physically appropriate, that a discretization has converged, that metadata are truthful, or that the matrix models a real material. Those are separate claims requiring model, provenance, numerical, and validation evidence.

CHAPTER 3

Hermitian Eigensystems and Spectral Subspaces

Status pedagogical finite-dimensional foundation. All matrices and numerical examples are synthetic. This chapter establishes no material result, discretization-convergence claim, scientific validation, or uncertainty quantification.

3.1 Learning goals

After working through this chapter, a reader should be able to

1. distinguish an eigenpair, spectrum, eigenspace, assembled eigensystem, and spectral decomposition;
2. explain why Hermitian matrices have real eigenvalues and orthogonal eigenspaces;
3. derive a small characteristic equation without mistaking it for a production numerical algorithm;
4. interpret degeneracy through invariant subspaces and spectral projectors rather than arbitrary eigenvector columns;
5. use the Rayleigh quotient to characterize extreme eigenvalues; and
6. verify a computed Hermitian eigensystem with residual, orthogonality, and reconstruction diagnostics.

Chapter 2 distinguished a linear map from its matrix in a selected basis. We now ask when a basis can be chosen so that a matrix acts by separately scaling its basis vectors. That question is central to the monograph: finite Hamiltonians are diagonalized, low-energy spaces are retained, Bloch bands are assembled from fiber eigenvalues, and composite frames are compared through their subspaces rather than through arbitrary raw columns.

3.2 Eigenpairs are invariant-line statements

Let

$$H \in \mathbb{C}^{N\times N}. \tag{3.1}$$

An *eigenpair* consists of a scalar $\lambda \in \mathbb{C}$ and a nonzero vector $\mathbf{v} \in \mathbb{C}^N$ satisfying

$$H\mathbf{v} = \lambda\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}. \tag{3.2}$$

The line spanned by $\mathbf{v}$ is invariant under H: applying H keeps the result on that line. Saying that the "direction does not change" is less precise because a negative real eigenvalue reverses an oriented vector, and a complex eigenvalue changes its phase.

Several related terms should not be conflated.

Eigenvalue. One scalar λ satisfying Equation (3.2) for at least one nonzero vector.

Eigenvector. One nonzero vector associated with that eigenvalue.

Eigenspace. The complete null space $\ker(H - \lambda I_N)$, including the zero vector.

Spectrum. The collection of eigenvalues, counted with a declared multiplicity convention when needed.

Assembled eigensystem. Eigenvalues and selected eigenvector columns, together with their ordering and normalization conventions.

Spectral decomposition. A representation of the matrix in terms of its eigenvalues and invariant spectral projectors.

A general square matrix need not have an orthonormal eigenbasis and may even be defective. The remainder of this chapter concerns Hermitian matrices, for which a much stronger theorem is available.

3.3 Why Hermiticity changes the problem

A complex matrix is Hermitian when

$$H = H^\dagger, \tag{3.3}$$

where $\dagger$ denotes conjugate transpose. A real symmetric matrix, $H = H^\mathsf{T}$, is the real-valued special case.

Suppose $(\lambda, \mathbf{v})$ is an eigenpair of a Hermitian matrix. Using an inner product that is conjugate-linear in its first argument,

$$\langle \mathbf{v}, H\mathbf{v} \rangle = \lambda \langle \mathbf{v}, \mathbf{v} \rangle. \tag{3.4}$$

Hermiticity also implies that $\langle \mathbf{v}, H\mathbf{v} \rangle$ is real. Because $\langle \mathbf{v}, \mathbf{v} \rangle > 0$, the eigenvalue λ must be real.

Now let $\mathbf{v}_m$ and $\mathbf{v}_n$ have distinct eigenvalues. Then

$$\lambda_n \langle \mathbf{v}_m, \mathbf{v}_n \rangle = \langle \mathbf{v}_m, H\mathbf{v}_n \rangle, \tag{3.5}$$

$$\lambda_m \langle \mathbf{v}_m, \mathbf{v}_n \rangle = \langle H\mathbf{v}_m, \mathbf{v}_n \rangle. \tag{3.6}$$

The two right-hand sides are equal by Hermiticity, so

$$(\lambda_n - \lambda_m) \langle \mathbf{v}_m, \mathbf{v}_n \rangle = 0. \tag{3.7}$$

Distinct eigenvalues therefore have orthogonal eigenspaces.

The finite-dimensional spectral theorem adds the crucial completeness result: every Hermitian $N \times N$ matrix has N orthonormal eigenvectors when multiplicity is counted [51]. The reality and orthogonality arguments above explain important parts of the theorem, but they do not by themselves prove the existence of a complete eigenbasis.

3.4 From individual eigenpairs to a spectral decomposition

Choose normalized eigenvectors and place them in columns,

$$V = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_N \end{bmatrix}, \qquad V^\dagger V = V V^\dagger = I_N. \tag{3.8}$$

Place the correspondingly ordered eigenvalues on a diagonal matrix,

$$\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_N). \tag{3.9}$$

The individual equations $H\mathbf{v}_n = \lambda_n \mathbf{v}_n$ then become

$$HV = V\Lambda. \tag{3.10}$$

Multiplying by $V^\dagger$ gives the equivalent forms

$$H = V\Lambda V^\dagger, \qquad V^\dagger H V = \Lambda. \tag{3.11}$$

The first reconstructs the matrix in the original basis. The second represents the same linear map in the eigenvector basis. Diagonalization is therefore a basis transformation with a basis whose existence follows from Hermiticity.

Define the rank-one projector

$$P_n = \mathbf{v}_n \mathbf{v}_n^\dagger. \tag{3.12}$$

Then

$$I_N = \sum_{n=1}^{N} P_n, \qquad H = \sum_{n=1}^{N} \lambda_n P_n. \tag{3.13}$$

For any vector

$$\mathbf{x} = \sum_{n=1}^{N} c_n \mathbf{v}_n, \qquad c_n = \mathbf{v}_n^\dagger \mathbf{x}, \tag{3.14}$$

the action of H is

$$H\mathbf{x} = \sum_{n=1}^{N} \lambda_n c_n \mathbf{v}_n. \tag{3.15}$$

Thus the eigenbasis resolves a coupled matrix action into independent scalar actions. In a degenerate eigenspace, however, the individual rank-one projectors depend on the selected basis; their sum over the complete degenerate space does not.

3.5 The characteristic equation

Move the right-hand side of Equation (3.2) to the left:

$$(H - \lambda I_N)\mathbf{v} = \mathbf{0}. \tag{3.16}$$

A nonzero solution exists only when $H - \lambda I_N$ is singular. Therefore

$$\det(H - \lambda I_N) = 0. \tag{3.17}$$

This is the characteristic equation. Its left-hand side is a degree-N polynomial in λ, and its roots are the eigenvalues.

For a real $N \times N$ matrix, $|\det A|$ is its N-dimensional volume-scaling factor. For a complex matrix viewed as a map of a $2N$-dimensional real space, the corresponding real-volume factor is $|\det A|^2$. In either scalar field, $\det A = 0$ means that the map collapses at least one nonzero direction and is not invertible. Applied to $A = H - \lambda I_N$, that collapsed direction is an eigenvector. The determinant therefore detects the eigenvalue; the null space identifies its eigenspace.

The characteristic polynomial is basis invariant. If $H' = U^\dagger H U$ for a unitary U, then

$$\det(H' - \lambda I_N) = \det\big(U^\dagger (H - \lambda I_N) U\big) \tag{3.18}$$
$$= \det(U^\dagger)\det(H - \lambda I_N)\det(U) \tag{3.19}$$
$$= \det(H - \lambda I_N). \tag{3.20}$$

The entries change under the basis transformation, while the roots remain the same.

3.5.1 A complete two-dimensional example

Consider

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. \tag{3.21}$$

The characteristic equation is

$$0 = \det \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} \tag{3.22}$$
$$= (2-\lambda)^2 - 1 \tag{3.23}$$
$$= (\lambda - 1)(\lambda - 3). \tag{3.24}$$

The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 3$. Solving the two null space problems and normalizing gives

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \tag{3.25}$$

Consequently,

$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}, \tag{3.26}$$

and direct multiplication verifies

$$HV = V\Lambda, \qquad H = V\Lambda V^{\mathsf{T}}. \tag{3.27}$$

The transpose appears because this example is real. The complex-Hermitian formula uses $V^\dagger$.

For any matrix, with eigenvalues counted according to algebraic multiplicity,

$$\det H = \prod_{n=1}^{N} \lambda_n, \qquad \operatorname{tr} H = \sum_{n=1}^{N} \lambda_n. \tag{3.28}$$

These identities are useful checks, but they do not replace eigenvectors or spectral projectors.

Equation (3.17) is invaluable for small hand calculations and for understanding singularity and multiplicity. It is not the usual stable numerical route for a large matrix. Production Hermitian eigensolvers use matrix factorizations or iterative subspace methods rather than evaluating determinants and finding roots of an explicitly formed characteristic polynomial [40].

3.6 The Rayleigh quotient and extreme eigenvalues

For any nonzero vector $\mathbf{x}$, define

$$\rho_H(\mathbf{x}) = \frac{\mathbf{x}^\dagger H \mathbf{x}}{\mathbf{x}^\dagger \mathbf{x}}. \tag{3.29}$$

This quotient is real when H is Hermitian. Expand $\mathbf{x} = \sum_n c_n \mathbf{v}_n$. The spectral decomposition gives

$$\rho_H(\mathbf{x}) = \frac{\sum_n |c_n|^2 \lambda_n}{\sum_n |c_n|^2}. \tag{3.30}$$

The quotient is therefore a weighted average of eigenvalues. If they are ordered as $\lambda_1 \leq \cdots \leq \lambda_N$, then

$$\lambda_1 \leq \rho_H(\mathbf{x}) \leq \lambda_N. \tag{3.31}$$

Moreover,

$$\lambda_1 = \min_{\mathbf{x} \neq \mathbf{0}} \rho_H(\mathbf{x}), \qquad \lambda_N = \max_{\mathbf{x} \neq \mathbf{0}} \rho_H(\mathbf{x}). \tag{3.32}$$

The minimizing and maximizing vectors lie in the corresponding eigenspaces. Intermediate eigenvalues admit related constrained variational characterizations [51]. This is the finite-dimensional foundation of methods that seek low-energy states without computing a complete spectrum.

3.7 Degeneracy, eigenspaces, and projectors

For an eigenvalue λ, the eigenspace is

$$\mathcal{E}_\lambda = \ker(H - \lambda I_N). \tag{3.33}$$

For a Hermitian matrix, the dimension of this space equals the algebraic multiplicity of λ. A repeated eigenvalue is called degenerate when this multiplicity exceeds one.

Let

$$W_\lambda = \begin{bmatrix} \mathbf{w}_1 & \cdots & \mathbf{w}_g \end{bmatrix} \tag{3.34}$$

contain any orthonormal basis of a g-dimensional eigenspace. Its spectral projector is

$$P_\lambda = W_\lambda W_\lambda^\dagger. \tag{3.35}$$

If $Q \in U(g)$, then $W'_\lambda = W_\lambda Q$ is another orthonormal basis of the same eigenspace, but

$$W'_\lambda W'^\dagger_\lambda = W_\lambda Q Q^\dagger W_\lambda^\dagger = P_\lambda. \tag{3.36}$$

The basis columns are not unique; the projector is.

As a concrete example, consider

$$H = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}. \tag{3.37}$$

The normalized vector

$$\mathbf{v}_+ = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \tag{3.38}$$

has eigenvalue 4. Every vector orthogonal to $\mathbf{v}_+$, equivalently every $\mathbf{x}$ satisfying $x_1 + x_2 + x_3 = 0$, has eigenvalue 1. Thus

$$\mathcal{E}_1 = \left\{ \mathbf{x} \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0 \right\}, \qquad \dim \mathcal{E}_1 = 2, \tag{3.39}$$

and the projector onto that plane is

$$P_1 = I_3 - \mathbf{v}_+ \mathbf{v}_+^{\mathsf{T}}. \tag{3.40}$$

No particular pair of axes inside the plane is preferred by H. Different software runs or algorithms may return different valid orthonormal columns spanning it.

Exact degeneracy must be distinguished from numerical *near-degeneracy*. Two computed eigenvalues that differ by a small amount are not automatically mathematically equal. A calculation must declare the dimensional scale and tolerance used to group them, and its conclusions should remain stable under that declared rule.

3.8 Phase, ordering, and subspace freedom

Even a normalized nondegenerate eigenvector is not unique. For any real θ,

$$\mathbf{v}'_n = e^{i\theta} \mathbf{v}_n \tag{3.41}$$

is an equally valid eigenvector. In a real calculation this includes the sign change $\mathbf{v}_n \mapsto -\mathbf{v}_n$. Eigenvalue sorting is also an output convention rather than a theorem. Inside a degenerate

eigenspace, the freedom expands from one phase per column to an arbitrary unitary mixing of all columns in the block.

Raw component equality is therefore not a universal comparison rule for independently computed eigenvectors. Suitable comparisons include eigenvalue residuals, spectral projectors, and invariant-subspace diagnostics. Later chapters refine these ideas for momentum-dependent Bloch frames, where choices made independently at neighboring reciprocal points affect smoothness and real-space locality.

The spectrum is basis invariant but does not completely identify an operator. Isospectral matrices can attach the same eigenvalues to different physical sites, orbitals, spin channels, or spatial regions. Spectral agreement is valuable evidence, but it supplies no physical alignment map by itself.

3.9 Computing and checking a Hermitian eigensystem

For a dense Hermitian matrix, use a Hermitian eigensolver such as `numpy.linalg.eigh`, not a general nonsymmetric routine such as `numpy.linalg.eig`. A Hermitian solver exploits the matrix structure and returns real eigenvalues and orthonormal eigenvectors up to floating-point error. For a large sparse matrix, computing only a requested portion of the spectrum is often more appropriate than constructing a complete dense factorization [40].

A returned pair should be checked through the residual

$$\mathbf{r}_n = H\mathbf{v}_n - \lambda_n \mathbf{v}_n. \tag{3.42}$$

For a normalized eigenvector and an energy-valued H, $\|\mathbf{r}_n\|_2$ has units of energy. Its tolerance must therefore be stated with a scale rather than treated as a dimensionless universal constant. For several returned columns, define

$$R = HV - V\Lambda. \tag{3.43}$$

Then three distinct checks are

$$\|R\|_{\mathrm{F}} \quad \text{eigenpair residual}, \tag{3.44}$$

$$\|V^\dagger V - I\|_{\mathrm{F}} \quad \text{orthonormality defect}, \tag{3.45}$$

$$\|H - V\Lambda V^\dagger\|_{\mathrm{F}} \quad \text{full reconstruction defect}. \tag{3.46}$$

The reconstruction check applies only when V contains a complete basis. It should not be manufactured by densifying a large sparse operator. Partial sparse results can be checked with sparse matrix–block products, column orthogonality, and projectors on the returned subspace.

Python interlude—checking eigenpairs and a degenerate subspace

```
import numpy as np

# This dense matrix is synthetic and has eigenvalues 1, 1, and 4.
H = np.array(
    [[2.0, 1.0, 1.0], [1.0, 2.0, 1.0], [1.0, 1.0, 2.0]],
    dtype=np.float64,
)
np.testing.assert_allclose(H, H.T, rtol=0.0, atol=0.0)

# eigh uses Hermitian structure and returns eigenvectors as columns.

eigenvalues, eigenvectors = np.linalg.eigh(H)
Lambda = np.diag(eigenvalues)

residual = H @ eigenvectors - eigenvectors @ Lambda
orthogonality_defect = eigenvectors.T @ eigenvectors - np.eye(3)
reconstruction = eigenvectors @ Lambda @ eigenvectors.T

np.testing.assert_allclose(residual, 0.0, atol=1e-14)
np.testing.assert_allclose(orthogonality_defect, 0.0, atol=1e-14)
np.testing.assert_allclose(reconstruction, H, atol=1e-14)
np.testing.assert_allclose(eigenvalues, [1.0, 1.0, 4.0], atol=1e
    -14)

# Rotate the two-dimensional degenerate eigenspace without changing
    it.
W = eigenvectors[:, :2]
angle = 0.37
Q = np.array(
    [[np.cos(angle), -np.sin(angle)], [np.sin(angle), np.cos(angle)
    ]],
    dtype=np.float64,
)
W_rotated = W @ Q

np.testing.assert_allclose(H @ W_rotated, W_rotated, atol=1e-14)
np.testing.assert_allclose(W_rotated @ W_rotated.T, W @ W.T, atol=1
    e-14)
```

The successful checks establish finite linear-algebra behavior for this synthetic matrix. They do not establish that a discretized Hamiltonian has converged or that any physical material is represented.

Notebook to do Create `examples/tutorials/research-monograph/foundations/hermitian_eigensystems.ipynb` with explicit imports, explanatory inline comments, characteristic-polynomial hand checks, Hermitian solver diagnostics, phase changes, degenerate rotations, projector comparisons, and sparse-preserving partial-eigensystem checks. Keep it unexecuted in the repository.

3.10 From eigenspaces to retained spaces

This chapter treated eigenspaces selected by exact eigenvalues of one finite Hermitian matrix. Chapter 4 next supplies the quantum meaning of states, observables, and Hamiltonian eigenproblems. Chapter 8 later asks a different question: which eigenvectors or spectral subspaces should be retained across a family of represented operators, and what information is lost by that selection? Diagonalization, projection, gauge choice, and truncation remain separate operations.

3.11 Common mistakes

- Treating $\det(H - \lambda I) = 0$ as the preferred large-scale numerical eigensolver.
- Saying an eigenvector is unique without accounting for normalization, phase, ordering, or degenerate rotations.
- Comparing raw eigenvector components from independent calculations without first aligning their coordinate spaces and gauge freedoms.
- Calling a small difference between floating-point eigenvalues an exact degeneracy without a declared scale and grouping tolerance.
- Using a full reconstruction check for a partial eigensystem.
- Densifying a sparse matrix merely to perform a diagnostic.
- Treating equality of spectra as equality of represented operators.

3.12 Exercises

1. Prove directly that an eigenvalue of a Hermitian matrix is real.
2. Derive the characteristic equation and normalized eigenvectors for $\begin{bmatrix} 3 & i \\ -i & 3 \end{bmatrix}$.
3. Use Equation (3.30) to prove the bounds on the Rayleigh quotient.
4. Construct two different orthonormal bases of the plane $x_1 + x_2 + x_3 = 0$ and verify that they produce the same projector.
5. Explain why two independently returned eigenvectors can represent the same non-degenerate eigendirection while failing componentwise equality.
6. State which diagnostics remain available when only the lowest $q < N$ eigenpairs of a sparse Hermitian matrix have been computed.
7. Construct two isospectral 3×3 Hermitian matrices with visibly different entries. State what additional alignment information is needed before their difference can be interpreted physically.

3.13 Evidence boundary

Residual, orthogonality, projector, and reconstruction checks can verify a finite eigensystem calculation within declared tolerances. They do not establish that the parent operator is physically appropriate, that a numerical discretization has converged, that a retained subspace is adequate for a later observable, or that an eigensystem models a real material. Those are separate numerical and scientific claims.

CHAPTER 4

Quantum States, Operators, and Hamiltonians

Status pedagogical quantum-mechanical foundation. The finite example is illustrative and establishes no material Hamiltonian, calculated scientific result, validation, or uncertainty quantification.

4.1 Learning goals

After working through this chapter, a reader should be able to

1. distinguish an abstract quantum state from its coordinates;
2. state the domain-qualified definition of a linear operator;
3. interpret expectation values and Hamiltonian eigenproblems;
4. distinguish an operator identity from a matrix identity;
5. solve the one-dimensional particle-in-a-box eigenproblem from its boundary conditions;
6. apply Born's interpretation to position and energy measurements; and
7. translate that continuum problem into a finite representation without identifying the finite matrix with the abstract operator itself.

4.2 State spaces and Dirac notation

A one-particle quantum model begins by declaring a Hilbert space $\mathcal{H}$. A quantum state is represented by a ket $|\psi\rangle \in \mathcal{H}$. Its corresponding bra $\langle\psi|$ is the linear functional associated

with the ket by the Hilbert-space inner product. For $|\phi\rangle, |\psi\rangle \in \mathcal{H}$,

$$\langle\phi|\psi\rangle, \qquad \|\psi\| = \sqrt{\langle\psi|\psi\rangle}. \tag{4.1}$$

A normalized state satisfies $\langle\psi|\psi\rangle = 1$.

If $\{|\phi_a\rangle\}$ is an orthonormal basis, then

$$|\psi\rangle = \sum_a c_a|\phi_a\rangle, \qquad c_a = \langle\phi_a|\psi\rangle. \tag{4.2}$$

The ket is the abstract state; the coefficients c_a are its coordinates in the selected basis. Changing the basis changes the coordinates without, by itself, changing the state. This is the quantum-mechanical use of the finite-representation distinction developed in Chapter 2.

4.3 Linear operators and domains

Definition 4.1 (Linear operator)**.** Let $\mathcal{H}$ be a Hilbert space of quantum states. A linear operator $\hat{A}$ is a map

$$\hat{A} : \mathcal{D}(\hat{A}) \subseteq \mathcal{H} \longrightarrow \mathcal{H}, \tag{4.3}$$

with a domain $\mathcal{D}(\hat{A})$ on which its action is defined. ■

The domain is part of the operator. Writing $\hat{A}|\psi\rangle$ is meaningful only when $|\psi\rangle \in \mathcal{D}(\hat{A})$. Differential expressions that look identical can define different operators when supplied with different domains or boundary conditions.

For a normalized state in the domain of a self-adjoint observable $\hat{A}$, the expectation value is

$$\langle\hat{A}\rangle_\psi = \langle\psi|\hat{A}|\psi\rangle. \tag{4.4}$$

Self-adjointness is an operator property, not merely a visual symmetry of one array. In a finite orthonormal representation it implies a Hermitian matrix, but that finite test does not reconstruct unrecorded domain or boundary information.

4.4 Hamiltonians and additive decompositions

Definition 4.2 (Hamiltonian)**.** A Hamiltonian $\hat{H}$ is the operator that generates the dynamics of the selected physical model. For the elementary nonrelativistic decomposition used here,

$$\hat{H} = \hat{T} + \hat{V}, \tag{4.5}$$

where $\hat{T}$ is the kinetic-energy operator and $\hat{V}$ is the potential or interaction operator. ■

For every state in the common domain,

$$\hat{H}|\psi\rangle = \big(\hat{T} + \hat{V}\big)|\psi\rangle. \tag{4.6}$$

This is an equality of operator actions on states. It is not yet a finite matrix equality, because no finite space or basis has been selected.

The stationary eigenproblem is

$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle. \tag{4.7}$$

Energy eigenvalues are important outputs, but they do not alone identify the state correspondence or the action of $\hat{H}$ on arbitrary superpositions. A list of energies is therefore not a complete Hamiltonian representation.

4.5 Position-space representation

Let $\langle \mathbf{r}|$ be the generalized position bra and define the wavefunction by $\psi(\mathbf{r}) = \langle \mathbf{r}|\psi\rangle$. The position-space representation of the abstract operator equation is

$$\langle \mathbf{r}|\hat{H}|\psi\rangle = \langle \mathbf{r}|\hat{T}|\psi\rangle + \langle \mathbf{r}|\hat{V}|\psi\rangle. \tag{4.8}$$

For the nonrelativistic kinetic operator and a local scalar potential, this becomes

$$\langle \mathbf{r}|\hat{H}|\psi\rangle = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}). \tag{4.9}$$

The generalized coordinate bra $\langle \mathbf{r}|$ is not a finite-rank projector. Coordinate representation, subspace projection, basis selection, and model reduction are distinct operations.

A local multiplication operator in one representation need not remain diagonal or local after projection or basis transformation. “Kinetic” and “potential” describe roles in the selected model; they do not authorize subtraction of matrices assembled in incompatible spaces.

4.6 The one-dimensional particle in a box

The one-dimensional particle in a box turns the preceding definitions into one complete eigenproblem. Consider a particle of mass m confined between two endpoints $a < b$. The interior and its length are

$$\Omega = (a, b), \qquad L = b - a > 0. \tag{4.10}$$

Its state space is $L^2(a, b)$, the space of square-integrable complex wavefunctions on the interval. The inner product is

$$\langle \phi | \psi \rangle = \int_a^b \phi(x)^* \psi(x) \, \mathrm{d}x. \tag{4.11}$$

A normalized state satisfies

$$\int_a^b |\psi(x)|^2 \, \mathrm{d}x = 1. \tag{4.12}$$

Inside the box, the potential is zero. Confinement is imposed at the closed interval's endpoints through the Dirichlet boundary conditions

$$\psi(a) = 0, \qquad \psi(b) = 0. \tag{4.13}$$

The Hamiltonian is therefore the kinetic operator

$$\hat{H}_{\text{box}} = -\frac{\hbar^2}{2m} \frac{\mathrm{d}^2}{\mathrm{d}x^2} \tag{4.14}$$

on a domain of sufficiently regular wavefunctions that satisfy those boundary conditions. The boundary conditions are part of the operator. Reusing the same second-derivative expression with periodic or Neumann boundary conditions would define a different Hamiltonian.

The phrase "infinite square well" is common because one may imagine an exterior potential that becomes infinitely large. In the operator used here, there is no finite interior wall matrix to add. The confinement has already entered through the interval and the domain. Appendix D develops this distinction formally and uses it for a separate retained-space residual diagnostic [9, 1].

4.6.1 Solving the stationary equation by a characteristic equation

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{\mathrm{d}^2 \psi}{\mathrm{d}x^2} = E\psi. \tag{4.15}$$

For $E > 0$, define

$$k^2 = \frac{2mE}{\hbar^2}. \tag{4.16}$$

The differential equation becomes

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + k^2\psi = 0. \tag{4.17}$$

Use the exponential trial function

$$\psi(x) = e^{rx}, \tag{4.18}$$

where r is initially an unknown number. Differentiation gives

$$\frac{\mathrm{d}\psi}{\mathrm{d}x} = re^{rx}, \qquad \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = r^2e^{rx}. \tag{4.19}$$

Substitution into the differential equation yields

$$r^2e^{rx} + k^2e^{rx} = 0. \tag{4.20}$$

The exponential never vanishes, so divide by e^{rx}:

$$r^2 + k^2 = 0. \tag{4.21}$$

This polynomial is the characteristic equation of the constant-coefficient ordinary differential equation. Its roots are

$$r_+ = ik, \qquad r_- = -ik. \tag{4.22}$$

Therefore the complex exponential form of the general solution is

$$\psi(x) = C_+e^{ikx} + C_-e^{-ikx}. \tag{4.23}$$

Euler's identities give the equivalent real-function form

$$\psi(x) = A\sin(kx) + B\cos(kx). \tag{4.24}$$

For the interval $[a, b]$, it is convenient to measure distance from the left endpoint. The same two-dimensional solution space can be written as

$$\psi(x) = A\sin\big(k(x-a)\big) + B\cos\big(k(x-a)\big). \tag{4.25}$$

Apply the left boundary condition:

$$\psi(a) = B = 0. \tag{4.26}$$

The surviving solution is

$$\psi(x) = A\sin\big(k(x-a)\big). \tag{4.27}$$

At the right endpoint,

$$\psi(b) = A\sin\big(k(b-a)\big) = A\sin(kL) = 0. \tag{4.28}$$

A nonzero eigenfunction has $A \neq 0$, so

$$\sin(kL) = 0, \qquad kL = n\pi, \qquad n = 1, 2, 3, \ldots. \tag{4.29}$$

Thus the permitted wave numbers are

$$k_n = \frac{n\pi}{L}. \tag{4.30}$$

Substitution into $E = \hbar^2 k^2/(2m)$ gives

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots. \tag{4.31}$$

There is no $n = 0$ state: it would produce the zero function, which cannot be normalized.

4.6.2 Why the characteristic equation turns calculus into algebra

Let $\mathrm{D} = \mathrm{d}/\mathrm{d}x$. A constant-coefficient linear differential equation can be written as a polynomial in the differentiation operator:

$$P(\mathrm{D})\psi = 0, \tag{4.32}$$

where, for example,

$$P(\mathrm{D}) = \mathrm{D}^2 + k^2 \tag{4.33}$$

for the box interior. Exponentials are special because differentiation acts on them by scalar multiplication:

$$\mathrm{D}^j e^{rx} = r^j e^{rx}. \tag{4.34}$$

Consequently,

$$P(\mathrm{D})e^{rx} = P(r)e^{rx}. \tag{4.35}$$

The trial function converts the differential operator D into the algebraic number r, and converts the operator polynomial $P(\mathrm{D})$ into the ordinary polynomial $P(r)$. The differential equation is satisfied when

$$P(r) = 0. \tag{4.36}$$

We have not discarded the calculus. We have chosen functions on which the derivative has the simple eigenvalue action

$$\mathrm{D}e^{rx} = re^{rx}. \tag{4.37}$$

That eigenfunction property is the bridge from the differential equation to the algebraic characteristic equation.

This construction parallels the matrix characteristic equation from Chapter 3. For a matrix, one seeks numbers λ for which $H - \lambda I$ loses invertibility. For a constant-coefficient differential equation, one seeks numbers r for which $P(r) = 0$, making e^{rx} a solution of $P(\mathrm{D})\psi = 0$. Both methods turn an operator question into an algebraic root problem, although the operators and root variables have different meanings.

The method has a boundary. If differential-equation coefficients depend on x, then substituting e^{rx} generally does not reduce the complete problem to one polynomial in r. Constant coefficients are what make the simple characteristic equation possible here.

4.6.3 Normalization and the eigenfunction basis

Before normalization, the nth eigenfunction is

$$\psi_n(x) = A_n \sin\left(\frac{n\pi(x-a)}{L}\right). \tag{4.38}$$

Using

$$\int_a^b \sin^2\left(\frac{n\pi(x-a)}{L}\right)\,\mathrm{d}x = \frac{L}{2}, \tag{4.39}$$

the normalization condition gives $|A_n| = \sqrt{2/L}$. Choosing a real positive phase convention,

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi(x-a)}{L}\right). \tag{4.40}$$

The eigenfunctions obey

$$\int_a^b \psi_m(x)^* \psi_n(x)\,\mathrm{d}x = \delta_{mn}. \tag{4.41}$$

They form an orthonormal basis of $L^2(a,b)$. Consequently, a general state can be expanded as

$$\psi(x) = \sum_{n=1}^{\infty} c_n \psi_n(x), \qquad c_n = \int_a^b \psi_n(x)^* \psi(x)\,\mathrm{d}x. \tag{4.42}$$

This is the infinite-dimensional analogue of assembling matrix eigenvectors as basis columns in Chapter 3. Applying the Hamiltonian scales each basis component by its

energy:

$$\hat{H}_{\text{box}}\psi = \sum_{n=1}^{\infty} c_n E_n \psi_n. \tag{4.43}$$

Each one-dimensional energy level is nondegenerate: apart from normalization and phase, the two boundary conditions leave one eigenfunction for each positive integer n.

4.6.4 Born's interpretation of the eigenfunction

The eigenfunction $\psi_n(x)$ is a probability amplitude, not a probability by itself. Born's interpretation assigns the position probability density

$$\rho_n(x) = |\psi_n(x)|^2 = \frac{2}{L}\sin^2\!\left(\frac{n\pi(x-a)}{L}\right). \tag{4.44}$$

For any subinterval $[c,d] \subseteq [a,b]$, the probability that an ideal position measurement returns a value in that subinterval is

$$\Pr(x \in [c,d] \mid \psi_n) = \int_c^d |\psi_n(x)|^2 \,\mathrm{d}x. \tag{4.45}$$

Normalization ensures

$$\Pr(x \in [a,b] \mid \psi_n) = \int_a^b |\psi_n(x)|^2 \,\mathrm{d}x = 1. \tag{4.46}$$

The wavefunction has physical dimension length$^{-1/2}$ in one spatial dimension, so $|\psi_n|^2$ has dimension length^{-1}, as a position density must.

The density vanishes at the walls and at every interior node of the sine function. For the nth eigenfunction, the nodes occur at

$$x_j = a + \frac{jL}{n}, \qquad j = 0, 1, \ldots, n. \tag{4.47}$$

There are $n-1$ interior nodes. A maximum of $|\psi_n(x)|^2$ identifies a position region with larger measurement probability; it is not a classical trajectory or a claim that the particle follows the plotted curve.

Multiplying an eigenfunction by a constant phase does not alter any position probability:

$$\left|e^{i\alpha}\psi_n(x)\right|^2 = |\psi_n(x)|^2. \tag{4.48}$$

This makes the phase freedom of an individual eigenvector physically visible: the coordinate representation changes, while the position distribution does not.

Born's rule also applies to the energy basis. For a normalized general state

$$\psi(x) = \sum_{n=1}^{\infty} c_n \psi_n(x), \qquad \sum_{n=1}^{\infty} |c_n|^2 = 1, \tag{4.49}$$

an ideal energy measurement returns E_n with probability

$$\Pr(E_n \mid \psi) = |c_n|^2 = |\langle \psi_n | \psi \rangle|^2 . \tag{4.50}$$

A state may therefore have a continuous position distribution and, at the same time, a discrete distribution over energy outcomes. The chosen observable determines which eigenbasis supplies the alternatives.

For a single stationary energy eigenstate, time evolution contributes only a phase:

$$\psi_n(x,t) = e^{-iE_n t/\hbar} \psi_n(x). \tag{4.51}$$

Its position density is time independent because

$$|\psi_n(x,t)|^2 = |\psi_n(x)|^2. \tag{4.52}$$

A superposition generally contains relative phases $e^{-i(E_n - E_m)t/\hbar}$, so its position density can change with time. A full development of measurement and time evolution requires the broader postulates of quantum mechanics; the statements here provide the interpretation needed for the box eigensystem [108].

4.6.5 The Dirichlet Laplacian on the closed interval $[a, b]$

In one Cartesian dimension, the scalar Laplacian is the second derivative:

$$\Delta = \frac{\mathrm{d}^2}{\mathrm{d}x^2}. \tag{4.53}$$

To specify an operator on the interval, the differential expression must be combined with boundary conditions. The Dirichlet Laplacian uses

$$\psi(a) = \psi(b) = 0. \tag{4.54}$$

The positive operator used in the kinetic energy is the negative Dirichlet Laplacian,

$$\hat{L}_D = -\Delta_D = -\frac{\mathrm{d}^2}{\mathrm{d}x^2}. \tag{4.55}$$

In formal domain notation,

$$\mathcal{D}(\hat{L}_D) = H^2(a,b) \cap H_0^1(a,b). \tag{4.56}$$

Here the Sobolev-space notation records square-integrable derivatives and zero boundary traces. The essential point for the present construction is that the endpoints and Dirichlet conditions belong to the operator definition.

The continuum box Hamiltonian factors into a physical coefficient and this geometric operator:

$$\hat{H}_{\text{box}} = \frac{\hbar^2}{2m}\hat{L}_D. \tag{4.57}$$

The eigenvalues of $\hat{L}_D$ are

$$\mu_n = \left(\frac{n\pi}{L}\right)^2, \tag{4.58}$$

and the physical energies are

$$E_n = \frac{\hbar^2}{2m}\mu_n. \tag{4.59}$$

This separates the interval geometry and boundary condition from the mass and Planck-constant scaling.

4.6.6 Discretizing the Dirichlet Laplacian as an operator

Choose N interior grid points on the closed interval $[a, b]$:

$$x_j = a + jh, \qquad h = \frac{b-a}{N+1} = \frac{L}{N+1}, \qquad j = 1, \ldots, N. \tag{4.60}$$

The endpoints are

$$x_0 = a, \qquad x_{N+1} = b. \tag{4.61}$$

Dirichlet data fix

$$\psi(x_0) = \psi(x_{N+1}) = 0, \tag{4.62}$$

so those two values are not independent components of the numerical state vector. Define the interior sample vector

$$\boldsymbol{\psi}_h = \begin{bmatrix}\psi(x_1) & \psi(x_2) & \cdots & \psi(x_N)\end{bmatrix}^{\mathsf{T}} \in \mathbb{C}^N. \tag{4.63}$$

At an interior point, the centered second-difference formula is

$$\psi''(x_j) \approx \frac{\psi(x_{j+1}) - 2\psi(x_j) + \psi(x_{j-1})}{h^2}. \tag{4.64}$$

Therefore the discrete second-derivative operator is

$$D_{2,h} = \frac{1}{h^2}\begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \ddots & \vdots \\ 0 & 1 & -2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & -2 \end{bmatrix}. \tag{4.65}$$

The finite representation of the positive operator $-\Delta_D$ is

$$L_{D,h} = -D_{2,h} = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix} \in \mathbb{R}^{N\times N}. \tag{4.66}$$

Finally, the discretized box Hamiltonian is

$$H_h = \frac{\hbar^2}{2m} L_{D,h}. \tag{4.67}$$

The three symbols $D_{2,h}$, $L_{D,h}$, and H_h represent different objects: the second derivative, the positive operator representing the negative Laplacian, and the kinetic Hamiltonian. Their signs and units must not be collapsed.

4.6.7 Eigenpairs of the discretized operator

The discrete Dirichlet eigenvectors have components

$$v_n(j) = \sqrt{\frac{2}{N+1}}\,\sin\!\left(\frac{jn\pi}{N+1}\right), \qquad j,n = 1,\dots,N. \tag{4.68}$$

To see why, set

$$\theta_n = \frac{n\pi}{N+1} \tag{4.69}$$

and temporarily ignore normalization. The jth component of $L_{D,h}\mathbf{v}_n$ is

$$(L_{D,h}\mathbf{v}_n)_j = \frac{1}{h^2}\left(2\sin(j\theta_n) - \sin((j-1)\theta_n) - \sin((j+1)\theta_n)\right) \tag{4.70}$$

$$= \frac{2-2\cos\theta_n}{h^2}\sin(j\theta_n) \tag{4.71}$$

$$= \frac{4}{h^2}\sin^2\!\left(\frac{\theta_n}{2}\right)\sin(j\theta_n). \tag{4.72}$$

The endpoint conditions are built into the sine sequence:

$$v_n(0) = 0, \qquad v_n(N+1) = \sin(n\pi) = 0. \tag{4.73}$$

Hence

$$L_{D,h}\mathbf{v}_n = \mu_{n,h}\mathbf{v}_n, \tag{4.74}$$

with discrete Laplacian eigenvalues

$$\mu_{n,h} = \frac{4}{h^2}\sin^2\left(\frac{n\pi}{2(N+1)}\right). \tag{4.75}$$

The Hamiltonian eigenvalues are therefore

$$\varepsilon_n = \frac{\hbar^2}{2m}\mu_{n,h} = \frac{2\hbar^2}{mh^2}\sin^2\left(\frac{n\pi}{2(N+1)}\right). \tag{4.76}$$

The N discrete eigenvectors form an orthonormal basis of $\mathbb{R}^N$. The infinitely many continuum sine functions form an orthonormal basis of $L^2(a,b)$. For a fixed mode n, their normalization conventions are related by the quadrature factor

$$v_n(j) = \sqrt{h}\,\psi_n(x_j). \tag{4.77}$$

For each fixed n, $\mu_{n,h} \to \mu_n$ and $\varepsilon_n \to E_n$ as $h \to 0$. This statement must not be strengthened into uniform convergence of every resolvable mode as N grows; Appendix D examines that distinction with retained calculated evidence.

4.7 Finite Hamiltonian representations

Let $\mathcal{S}$ be a finite state space with ordered orthonormal basis $\mathcal{B} = \{|b_i\rangle\}_{i=0}^{N-1}$. Once a Hamiltonian acting on $\mathcal{S}$ has been defined, its represented matrix is

$$H_{ij}^{(\mathcal{B})} = \langle b_i|\hat{H}_{\mathcal{S}}|b_j\rangle. \tag{4.78}$$

If the decomposition has been transported consistently to the same space and basis, then

$$H^{(\mathcal{B})} = T^{(\mathcal{B})} + V^{(\mathcal{B})}. \tag{4.79}$$

Without the common-space and common-basis qualification, this array equation has no justified operator interpretation.

Python interlude—diagonalizing the one-dimensional box matrix

```
import numpy as np

# Use dimensionless pedagogical units and declare both interval
    endpoints.
point_count = 8
```

```
left_endpoint = -0.5
right_endpoint = 1.5
length = right_endpoint - left_endpoint
mass = 1.0
hbar = 1.0
spacing = length / (point_count + 1)

# Assemble the centered Dirichlet Hamiltonian on interior grid
    points.
diagonal = 2.0 * np.ones(point_count, dtype=np.float64)
off_diagonal = -np.ones(point_count - 1, dtype=np.float64)
kinetic_core = (
    np.diag(diagonal)
    + np.diag(off_diagonal, k=1)
    + np.diag(off_diagonal, k=-1)
)
H_box = hbar**2 * kinetic_core / (2.0 * mass * spacing**2)

# eigh returns ascending energies and normalized eigenvectors as
    columns.
energies, states = np.linalg.eigh(H_box)
mode_indices = np.arange(1, point_count + 1, dtype=np.float64)
expected_discrete = (
    2.0
    * hbar**2
    / (mass * spacing**2)
    * np.sin(mode_indices * np.pi / (2.0 * (point_count + 1))) ** 2
)
np.testing.assert_allclose(energies, expected_discrete)

# Sample the shifted continuum sine basis with its quadrature
    factor.
grid = left_endpoint + spacing * mode_indices
sampled_sines = np.sqrt(spacing) * np.sqrt(2.0 / length) * np.sin(
    np.outer(grid - left_endpoint, mode_indices * np.pi / length)
)

# Eigenvector signs are arbitrary, so compare absolute basis
    overlaps.
overlaps = np.abs(states.conjugate().T @ sampled_sines)
```

```
np.testing.assert_allclose(overlaps, np.eye(point_count), atol=1.0e
    -12)
```

The arrays are synthetic teaching data in declared dimensionless units. The checks establish exact identities of this finite matrix and sampled sine basis; they do not establish a material energy or a general convergence result.

Notebook to do Create `examples/tutorials/research-monograph/foundations/quantum_operators.ipynb` with explicit imports, explanatory inline comments, normalized states, expectation values, the one-dimensional particle-in-a-box derivation and finite matrix, and a clear separation between the continuum operator and its finite representation. Keep it unexecuted in the repository.

4.8 What changes in an approximation

Several distinct operations can produce a smaller or simpler Hamiltonian: selecting a different physical model, projecting onto a retained subspace, changing basis, discretizing a differential expression, truncating matrix entries, or fitting parameters in an admissible model class. A correct account names the operation rather than calling every output an "effective Hamiltonian."

Chapter 5 next extends the box to two and three dimensions and explains how Kronecker products represent separate coordinate actions. Chapter 6 then supplies the geometry needed by periodic quantum representations, before later chapters introduce Bloch-periodic finite differences, retained spaces, isolated-band Fourier reduction, and evidence vocabulary.

4.9 Exercises

1. Explain why an operator domain is not recoverable from the entries of a finite matrix alone.

2. Show that the expectation value is unchanged under a consistent unitary basis transformation of both the state coordinates and operator matrix.

3. Starting from the stationary Schrödinger equation, derive the permitted wave numbers and energies of the one-dimensional box.

4. Normalize the shifted sine eigenfunctions and explain why they form a basis of $L^2(a, b)$.

5. Use Born's rule to compute the probability of finding the particle in a chosen subinterval $[c, d] \subseteq [a, b]$, and distinguish this position probability from the probability of an energy outcome.

6. Derive $D_{2,h}$, $L_{D,h}$, and H_h, explaining their signs, units, and represented meanings.

7. Derive the discrete sine eigenvectors and distinguish their finite basis of $\mathbb{R}^N$ from the infinite continuum basis.

8. Give an example in which $V(\mathbf{r})$ is local in position space but has off-diagonal entries in another basis.

9. State the compatibility requirements needed before adding separately supplied finite kinetic and potential matrices.

10. Distinguish a list of eigenvalues, an eigensystem, a represented Hamiltonian, and an abstract Hamiltonian.

4.10 Evidence boundary

A Hermiticity check, normalized eigenvectors, and a small residual can verify bounded properties of a finite calculation. They do not establish that the state space, domain, boundary conditions, Hamiltonian terms, or physical model are adequate. Those choices must be declared before stronger evidence can be interpreted.

CHAPTER 5

Particle in a Box in Two and Three Dimensions

Status pedagogical extension of the one-dimensional box. The chapter derives separable continuum solutions and sparse finite representations. Its examples are synthetic mathematics, not material calculations, scientific validation, or uncertainty quantification.

5.1 Learning goals

After working through this chapter, a reader should be able to

1. construct two- and three-dimensional box eigenfunctions from one-dimensional eigenfunctions;
2. explain what a Kronecker product does to matrices, vectors, dimensions, and tensor-product coordinates;
3. distinguish a Kronecker product from a Kronecker sum;
4. assemble two- and three-dimensional finite-difference Hamiltonians with a declared flattening order;
5. derive product eigenvectors and additive energies; and
6. explain symmetry-induced degeneracy in square and cubic boxes.

Chapter 4 solved the one-dimensional box. The higher-dimensional rectangular box is the cleanest setting in which to see how separate coordinate directions combine. The continuum construction uses tensor products of state spaces. Its finite matrix representation uses Kronecker products.

5.2 From an interval to a rectangular box

A two-dimensional rectangular box occupies

$$\Omega_2 = (0, L_x) \times (0, L_y). \tag{5.1}$$

A three-dimensional rectangular box occupies

$$\Omega_3 = (0, L_x) \times (0, L_y) \times (0, L_z). \tag{5.2}$$

The wavefunction vanishes whenever any coordinate reaches a boundary. In two dimensions,

$$\psi(0, y) = \psi(L_x, y) = 0, \qquad \psi(x, 0) = \psi(x, L_y) = 0. \tag{5.3}$$

The corresponding three-dimensional statement also imposes zero values on the two faces normal to the z direction.

Inside either box the potential is zero. The Hamiltonian is the Dirichlet realization of the kinetic operator

$$\hat{H}_d = -\frac{\hbar^2}{2m}\nabla_d^2, \tag{5.4}$$

where $d = 2$ or 3. Cartesian derivatives separate:

$$\nabla_2^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}, \tag{5.5}$$

$$\nabla_3^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}. \tag{5.6}$$

Each term changes one coordinate while leaving the others untouched. That simple sentence is exactly what the later identity factors in the Kronecker representation will encode.

5.3 What a Kronecker product actually does

Let

$$A \in \mathbb{R}^{m\times n}, \qquad B \in \mathbb{R}^{p\times q}. \tag{5.7}$$

The Kronecker product $A \otimes B$ is the block matrix

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix} \in \mathbb{R}^{mp\times nq}. \tag{5.8}$$

Every scalar entry a_{ij} of A is replaced by a complete scaled copy $a_{ij}B$. Thus the Kronecker product is not elementwise multiplication, and it is not ordinary matrix multiplication. It constructs one larger linear map from two maps acting on separate coordinate factors.

5.3.1 A literal block example

Take

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix}. \tag{5.9}$$

Then

$$A \otimes B = \begin{bmatrix} 1B & 2B \\ 3B & 4B \end{bmatrix} \tag{5.10}$$

$$= \begin{bmatrix} 0 & 5 & 0 & 10 \\ 6 & 7 & 12 & 14 \\ 0 & 15 & 0 & 20 \\ 18 & 21 & 24 & 28 \end{bmatrix}. \tag{5.11}$$

Both inputs are 2×2, but their Kronecker product is 4×4. The block structure retains which factor supplied each action.

5.3.2 Kronecker products of vectors

For column vectors

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, \tag{5.12}$$

the Kronecker product is

$$\mathbf{a} \otimes \mathbf{b} = \begin{bmatrix} a_1\mathbf{b} \\ a_2\mathbf{b} \end{bmatrix} = \begin{bmatrix} a_1 b_1 \\ a_1 b_2 \\ a_1 b_3 \\ a_2 b_1 \\ a_2 b_2 \\ a_2 b_3 \end{bmatrix} \in \mathbb{R}^6. \tag{5.13}$$

The product lists every pair of components in a specific order. Reversing the factors generally changes that order:

$$\mathbf{a} \otimes \mathbf{b} \neq \mathbf{b} \otimes \mathbf{a} \tag{5.14}$$

as coordinate columns, even though a permutation matrix relates them.

The central action identity is

$$(A \otimes B)(\mathbf{x} \otimes \mathbf{y}) = (A\mathbf{x}) \otimes (B\mathbf{y}), \tag{5.15}$$

whenever the dimensions are compatible. The larger matrix applies A to the first factor and B to the second factor. This identity is why Kronecker products are natural for separable multidimensional operators.

More generally, compatible products obey the mixed-product rule

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD). \tag{5.16}$$

They also obey

$$(A \otimes B)^{\mathsf{T}} = A^{\mathsf{T}} \otimes B^{\mathsf{T}}, \tag{5.17}$$

and, for complex matrices,

$$(A \otimes B)^{\dagger} = A^{\dagger} \otimes B^{\dagger}. \tag{5.18}$$

Consequently, the Kronecker product of Hermitian matrices is Hermitian.

5.3.3 Identity factors mean "leave this coordinate alone"

Let A_x act on an N_x-component x-coordinate vector, and let I_y be the $N_y \times N_y$ identity. Then

$$I_y \otimes A_x \tag{5.19}$$

contains N_y copies of A_x on its block diagonal. It applies the same x-direction action independently at every y index. The identity factor means that the y coordinate is carried through unchanged.

Conversely,

$$A_y \otimes I_x \tag{5.20}$$

uses A_y to couple different y slices while I_x preserves the x index inside each slice. These statements depend on the declared flattening order, which we now make explicit.

Suppose samples are stored in an array Ψ_{j_y, j_x} and flattened in C order, so that x varies fastest:

$$\mathrm{vec}_C(\Psi) = \begin{bmatrix} \Psi_{1,1} & \cdots & \Psi_{1,N_x} & \Psi_{2,1} & \cdots & \Psi_{N_y,N_x} \end{bmatrix}^{\mathsf{T}}. \tag{5.21}$$

With this ordering, $I_y \otimes A_x$ acts along x, and $A_y \otimes I_x$ acts along y. A different flattening convention changes the displayed matrices and requires a corresponding permutation. The physical operator does not choose an array ordering for us.

5.3.4 Kronecker product versus Kronecker sum

For square matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{p \times p}$, one common convention defines the Kronecker sum by

$$A \oplus B = I_p \otimes A + B \otimes I_n. \tag{5.22}$$

The symbol $\oplus$ here does not mean a direct sum of block-diagonal spaces. It names the particular operator that applies A in one factor plus B in the other. Because conventions can reverse factor order, a computational record should state the expanded formula rather than rely on the symbol alone.

A separable Hamiltonian is represented by a Kronecker *sum*, not by the single product $H_y \otimes H_x$. The sum encodes

$$\text{energy from the } x \text{ direction} + \text{energy from the } y \text{ direction}. \tag{5.23}$$

5.4 The two-dimensional box

The two-dimensional Hamiltonian is

$$\hat{H}_2 = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right). \tag{5.24}$$

Seek a separated eigenfunction

$$\psi(x, y) = X(x)Y(y). \tag{5.25}$$

Substitution into $\hat{H}_2\psi = E\psi$ gives

$$-\frac{\hbar^2}{2m}\left(X''(x)Y(y) + X(x)Y''(y)\right) = EX(x)Y(y). \tag{5.26}$$

Away from zeros of the factors, divide by $X(x)Y(y)$:

$$-\frac{\hbar^2}{2m}\frac{X''(x)}{X(x)} - \frac{\hbar^2}{2m}\frac{Y''(y)}{Y(y)} = E. \tag{5.27}$$

The first term depends only on x, the second only on y, and their sum is constant. Assign one-dimensional energies $E_{n_x}^{(x)}$ and $E_{n_y}^{(y)}$. Each factor obeys the one-dimensional box equation and its own boundary conditions.

The normalized product eigenfunctions are

$$\psi_{n_x,n_y}(x, y) = \frac{2}{\sqrt{L_x L_y}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right), \tag{5.28}$$

where

$$n_x, n_y = 1, 2, 3, \ldots. \tag{5.29}$$

Their energies add:

$$E_{n_x,n_y} = \frac{\hbar^2 \pi^2}{2m}\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2}\right). \tag{5.30}$$

The product states form an orthonormal basis of $L^2(0, L_x) \otimes L^2(0, L_y)$, naturally identified with $L^2(\Omega_2)$.

5.4.1 Degeneracy in a square box

If $L_x = L_y = L$, interchanging n_x and n_y leaves the energy unchanged:

$$E_{n_x,n_y} = \frac{\hbar^2\pi^2}{2mL^2}\left(n_x^2 + n_y^2\right) = E_{n_y,n_x}. \tag{5.31}$$

For $n_x \neq n_y$, the two product states are linearly independent, so this exchange produces at least a two-dimensional degenerate eigenspace. For example,

$$E_{1,2} = E_{2,1}. \tag{5.32}$$

The Hamiltonian fixes the eigenspace spanned by $\psi_{1,2}$ and $\psi_{2,1}$, but any orthonormal rotation inside that space is also a valid eigenbasis. A rectangular box with $L_x \neq L_y$ generally breaks this exchange degeneracy, though accidental coincidences may still occur for particular length ratios and quantum numbers.

5.5 Finite representation of the two-dimensional box

Let

$$H_x \in \mathbb{R}^{N_x \times N_x}, \qquad H_y \in \mathbb{R}^{N_y \times N_y} \tag{5.33}$$

be the one-dimensional Dirichlet finite-difference Hamiltonians from Chapter 4. Flatten the grid with x varying fastest. The two-dimensional represented Hamiltonian is

$$H_{2,h} = I_{N_y} \otimes H_x + H_y \otimes I_{N_x} \in \mathbb{R}^{N_xN_y \times N_xN_y}. \tag{5.34}$$

The first term applies H_x independently within every y slice. The second term couples the slices according to H_y while preserving each x index.

Suppose

$$H_x\mathbf{v}_{n_x}^{(x)} = \varepsilon_{n_x}^{(x)}\mathbf{v}_{n_x}^{(x)}, \qquad H_y\mathbf{v}_{n_y}^{(y)} = \varepsilon_{n_y}^{(y)}\mathbf{v}_{n_y}^{(y)}. \tag{5.35}$$

Then the product vector

$$\mathbf{v}_{n_y}^{(y)} \otimes \mathbf{v}_{n_x}^{(x)} \in \mathbb{R}^{N_xN_y} \tag{5.36}$$

is an eigenvector of $H_{2,h}$. Apply each term separately:

$$(I_{N_y} \otimes H_x)(\mathbf{v}_{n_y}^{(y)} \otimes \mathbf{v}_{n_x}^{(x)}) = \mathbf{v}_{n_y}^{(y)} \otimes (\varepsilon_{n_x}^{(x)}\mathbf{v}_{n_x}^{(x)}), \tag{5.37}$$

$$(H_y \otimes I_{N_x})(\mathbf{v}_{n_y}^{(y)} \otimes \mathbf{v}_{n_x}^{(x)}) = (\varepsilon_{n_y}^{(y)}\mathbf{v}_{n_y}^{(y)}) \otimes \mathbf{v}_{n_x}^{(x)}. \tag{5.38}$$

Adding the results gives

$$H_{2,h}(\mathbf{v}_{n_y}^{(y)} \otimes \mathbf{v}_{n_x}^{(x)}) = (\varepsilon_{n_x}^{(x)} + \varepsilon_{n_y}^{(y)})(\mathbf{v}_{n_y}^{(y)} \otimes \mathbf{v}_{n_x}^{(x)}). \tag{5.39}$$

Thus product eigenvectors accompany additive energies in the discrete problem just as they do in the continuum problem.

5.5.1 A 2×2 interior grid

For equal grid spacings, suppress the common kinetic prefactor and use the one-dimensional core

$$K = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}. \tag{5.40}$$

With two interior points along each axis,

$$K_2 = I_2 \otimes K + K \otimes I_2 \tag{5.41}$$

$$= \begin{bmatrix} 4 & -1 & -1 & 0 \\ -1 & 4 & 0 & -1 \\ -1 & 0 & 4 & -1 \\ 0 & -1 & -1 & 4 \end{bmatrix}. \tag{5.42}$$

The four rows correspond, in one declared C-order convention, to

$$(y_1, x_1),\ (y_1, x_2),\ (y_2, x_1),\ (y_2, x_2). \tag{5.43}$$

Entries from $I_2 \otimes K$ connect points with the same y index and neighboring x indices. Entries from $K \otimes I_2$ connect points with the same x index and neighboring y indices. The zero corner entries show that this axial Laplacian does not directly connect diagonal neighbors.

5.6 The three-dimensional box

The normalized continuum eigenfunctions are

$$\psi_{n_x,n_y,n_z}(x, y, z) = \sqrt{\frac{8}{L_x L_y L_z}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right) \sin\left(\frac{n_z \pi z}{L_z}\right), \tag{5.44}$$

with energies

$$E_{n_x,n_y,n_z} = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}\right). \tag{5.45}$$

For a cube, $L_x = L_y = L_z$, permutations of (n_x, n_y, n_z) have the same energy. The states associated with $(1, 1, 2)$, $(1, 2, 1)$, and $(2, 1, 1)$, for example, span a three-dimensional degenerate eigenspace.

Let $H_z \in \mathbb{R}^{N_z \times N_z}$ be the third one-dimensional finite-difference Hamiltonian. Flatten with x fastest, then y, then z. The three-dimensional Kronecker sum is

$$H_{3,h} = I_{N_z} \otimes I_{N_y} \otimes H_x \tag{5.46}$$

$$+ I_{N_z} \otimes H_y \otimes I_{N_x} \tag{5.47}$$

$$+ H_z \otimes I_{N_y} \otimes I_{N_x}. \tag{5.48}$$

It has shape

$$H_{3,h} \in \mathbb{R}^{N_x N_y N_z \times N_x N_y N_z}. \tag{5.49}$$

The associated product eigenvectors and eigenvalues are

$$\mathbf{v}^{(z)}_{n_z} \otimes \mathbf{v}^{(y)}_{n_y} \otimes \mathbf{v}^{(x)}_{n_x}, \qquad \varepsilon^{(x)}_{n_x} + \varepsilon^{(y)}_{n_y} + \varepsilon^{(z)}_{n_z}. \tag{5.50}$$

Each identity factor means that the corresponding coordinate index is preserved while another one-dimensional Hamiltonian acts.

5.7 Sparse computational construction

A multidimensional finite-difference Hamiltonian is sparse. Each grid point couples only to itself and to a bounded set of neighbors. The Kronecker construction should preserve that sparsity rather than form a dense $(N_x N_y N_z) \times (N_x N_y N_z)$ array.

Python interlude—assembling sparse two- and three-dimensional boxes

```
import numpy as np
from scipy.sparse import diags, eye, kron


def dirichlet_kinetic(point_count: int, length: float):
    # Use interior points; boundary values are fixed to zero and
    not stored.
    spacing = length / (point_count + 1)
    off_diagonal = -np.ones(point_count - 1, dtype=np.float64)
    diagonal = 2.0 * np.ones(point_count, dtype=np.float64)

    # Set hbar = mass = 1 for this dimensionless pedagogical
    example.
    core = diags(
        (off_diagonal, diagonal, off_diagonal),
        offsets=(-1, 0, 1),
        format="csr",
    )
    return core / (2.0 * spacing**2)


# Declare each one-dimensional factor and its identity explicitly.
Nx, Ny, Nz = 4, 3, 2
```

```
Hx = dirichlet_kinetic(Nx, length=1.0)
Hy = dirichlet_kinetic(Ny, length=1.5)
Hz = dirichlet_kinetic(Nz, length=2.0)
Ix = eye(Nx, format="csr")
Iy = eye(Ny, format="csr")
Iz = eye(Nz, format="csr")

# C-order flattening makes x the fastest coordinate.
H2 = kron(Iy, Hx, format="csr") + kron(Hy, Ix, format="csr")
H3 = (
    kron(kron(Iz, Iy, format="csr"), Hx, format="csr")
    + kron(kron(Iz, Hy, format="csr"), Ix, format="csr")
    + kron(kron(Hz, Iy, format="csr"), Ix, format="csr")
)

# Check shapes without converting either sparse operator to a dense
    array.
assert H2.shape == (Nx * Ny, Nx * Ny)
assert H3.shape == (Nx * Ny * Nz, Nx * Ny * Nz)

# Preserve sparsity in the Hermiticity diagnostic.
residual = H3 - H3.conjugate().transpose()
residual.eliminate_zeros()
assert residual.nnz == 0
print("2D shape and nonzeros:", H2.shape, H2.nnz)
print("3D shape and nonzeros:", H3.shape, H3.nnz)
```

The example constructs synthetic dimensionless operators. It checks assembly shape and exact sparse Hermiticity only. It does not establish refinement, continuum convergence, or a physical material result.

Notebook to do Create `examples/tutorials/research-monograph/foundations/particle_in_box_dimensions.ipynb` with explicit imports, explanatory inline comments, hand-checked Kronecker products, declared flattening order, product-state eigenpairs, square/cubic degeneracies, and sparse-preserving two- and three-dimensional diagnostics. Keep it unexecuted in the repository.

5.8 What changes outside a rectangular Cartesian box

The simple Kronecker sum depends on separability. It works here because

- the domain is a Cartesian product of intervals;
- the mass is a constant scalar;
- the boundary conditions separate by coordinate;
- the potential is zero, or more generally is a sum of one-coordinate terms; and
- the coordinate metric is diagonal and constant.

A coupled potential $V(x, y)$, a nonrectangular boundary, a spatially varying mass, or nonorthogonal coordinates can destroy the simple sum of independent one-dimensional operators. Chapter 6 introduces the metric needed for nonorthogonal cells, and Chapter 7 shows why mixed derivatives then add diagonal-neighbor stencil terms that are absent from the rectangular box matrix above.

5.9 Common mistakes

1. Writing $H_y \otimes H_x$ when the separable Hamiltonian requires the Kronecker sum $I_y \otimes H_x + H_y \otimes I_x$.
2. Omitting identity factors and therefore changing the represented space.
3. Reversing Kronecker factors without changing the flattening convention.
4. Treating $\mathbf{v}_y \otimes \mathbf{v}_x$ and $\mathbf{v}_x \otimes \mathbf{v}_y$ as identical coordinate columns.
5. Assuming every repeated numerical eigenvalue is an exact physical degeneracy without a tolerance and symmetry argument.
6. Densifying a sparse multidimensional Hamiltonian merely to inspect or test it.
7. Calling a finite-difference matrix the continuum Hamiltonian rather than a finite representation of a declared discretization.

5.10 Exercises

1. Compute $A \otimes B$ by blocks for two chosen 2×2 matrices, and verify its action on one vector product.
2. Prove $(A \otimes B)(\mathbf{x} \otimes \mathbf{y}) = (A\mathbf{x}) \otimes (B\mathbf{y})$ directly from components.
3. Starting from one-dimensional box eigenfunctions, verify the normalization of $\psi_{n_x,n_y}(x, y)$.
4. Derive the additive two-dimensional energy from separation of variables.
5. Reconstruct the displayed 4×4 matrix $I_2 \otimes K + K \otimes I_2$ one term at a time.
6. For a cubic box, list the degeneracy due to permutations of $(1, 2, 3)$, and distinguish those states from possible accidental degeneracies involving different triples.
7. Show that the three-dimensional product vector is an eigenvector of $H_{3,h}$ and that its eigenvalue is the sum of the three one-dimensional eigenvalues.
8. Describe the permutation needed if a code changes from x-fastest to z-fastest flattening.

5.11 Evidence boundary

The continuum product formulas are exact for the stated rectangular Dirichlet model. The finite Kronecker sums are exact identities for the chosen one-dimensional matrices and ordering. Neither statement establishes numerical convergence at an untested grid, adequacy for a nonseparable model, or relevance to a semiconductor material. Those stronger claims require separate numerical and scientific evidence.

CHAPTER 6

Periodic Geometry: Intrinsic Cells and Embedded Planes

Status pedagogical synthesis and verified software-capability description. The derivations are illustrative mathematics. The maintained implementation has software-verification evidence, but this chapter reports no new calculated material result, numerical convergence claim, scientific validation, or uncertainty quantification.

6.1 Learning goals

After working through this chapter, a reader should be able to

1. distinguish fractional, intrinsic Cartesian, and ambient Cartesian coordinates;
2. construct the Gram metric of a nonorthogonal primitive basis;
3. derive the dual reciprocal basis for both square and rectangular direct bases;
4. interpret a one-dimensional Brillouin zone as a reciprocal quotient and distinguish physical from reduced momentum;
5. define a Bloch fiber as a fixed-quasimomentum sector and distinguish its quasiperiodic and cell-periodic representations;
6. explain why an intrinsic two-dimensional cell and a two-dimensional plane embedded in three dimensions are different records even when they have the same metric;
7. construct the orthogonal projector onto an embedded plane; and
8. identify which quantities define the geometry and which should instead be reconstructed as derived quantities.

The required background is elementary linear algebra: matrix multiplication, inner products, rank, inverses of positive-definite matrices, and orthogonal projection. The chapter prepares the finite-difference construction in Chapter 7 and the controlled periodic example in Appendix H.

6.2 Why geometry comes before a periodic discretization

Chapter 4 introduced Hamiltonians abstractly. Before a periodic derivative can be discretized, a reciprocal vector can be named, or two represented operators can be compared, we must additionally know which coordinates and geometry those objects use. A matrix of numbers is not yet a lattice. It becomes a lattice representation only when we say what its rows and columns mean and what units its entries carry.

Keep a sheet of ordinary graph paper in mind. On the graph paper, a point can be located by counting horizontal and vertical grid lines. Those counts play the role of fractional coordinates. Now imagine stretching the sheet more in one direction than the other and shearing it so that the grid lines meet at an oblique angle. The counting coordinates did not change: one still moves by "one cell" when one count increases by one. Physical distances did change. The matrix A records that change from counting coordinates to Cartesian vectors.

Now lift the same sheet, without bending it, and rotate it in a three-dimensional room. Distances measured on the sheet do not change, but the sheet has acquired an orientation relative to the room. This is the distinction between intrinsic geometry and embedding data. The Gram metric records the distances and angles available to an observer confined to the sheet. The ambient basis, frame, and projector additionally record how the sheet sits in the room.

That picture is useful because the software boundary follows the same logic. An intrinsic scalar differential operator needs the metric. A later model that couples the plane to a laboratory-axis electric field also needs the ambient orientation. If we discard the orientation when constructing the intrinsic operator, we may not be able to reconstruct the coupling correctly. If we pass all ambient data into every scalar stencil routine, on the other hand, we make a simple intrinsic algorithm depend on information it does not mathematically need. The adapter introduced later in this lecture preserves both sides of the boundary.

A recurring discipline of these notes is therefore

$$\text{coordinates} \longrightarrow \text{geometry} \longrightarrow \text{represented operator} \longrightarrow \text{physical model}. \tag{6.1}$$

The arrow is not permission to collapse the objects on either side of it. Each step adds meaning and assumptions.

6.3 Primitive vectors as a coordinate map

Let a d-dimensional periodic cell have ordered primitive vectors $\mathbf{a}_1, \ldots, \mathbf{a}_d$. Store those vectors as the columns of a matrix

$$A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_d \end{bmatrix}. \tag{6.2}$$

A fractional coordinate $\mathbf{s} \in \mathbb{R}^d$ maps to Cartesian position through

$$\mathbf{r} = A\mathbf{s}. \tag{6.3}$$

The entries of $\mathbf{s}$ are dimensionless. The columns of A carry the length unit. Consequently, $\mathbf{r}$ carries that same length unit.

Two cases must be distinguished.

Intrinsic two-dimensional basis. $A \in \mathbb{R}^{2\times 2}$ maps two fractional coordinates into an intrinsic Cartesian plane.

Two-dimensional plane embedded in three dimensions. $A \in \mathbb{R}^{3\times 2}$ maps the same number of fractional coordinates into ambient three-space.

Both cases describe two periodic directions because both matrices have two columns. The number of rows instead records the ambient Cartesian dimension. A 3×2 basis therefore does not define a third periodic direction.

6.3.1 Three coordinate languages

During a calculation we will move repeatedly among three related coordinate languages. It is worth naming them before doing any algebra.

Fractional coordinates s. These say how many primitive vectors are used. Their components are dimensionless, and integer translations identify equivalent points in a periodic cell.

Intrinsic Cartesian coordinates x. These are ordinary orthonormal coordinates inside the plane. They carry a length unit. Choosing them requires an in-plane orthonormal frame.

Ambient Cartesian coordinates r. These are coordinates in the surrounding space. For an embedded plane, $\mathbf{r}$ has three components even though only two independent coordinates are periodic.

The same physical in-plane point can consequently be written in several ways:

$$\mathbf{r} = A\mathbf{s} = Q\mathbf{x}, \qquad \mathbf{x} = Q^{\mathsf{T}} A\mathbf{s}. \tag{6.4}$$

Nothing physical changes when we translate the point faithfully between these languages. What changes is the representation and therefore the matrices that act on coordinate arrays.

A good notebook habit is to annotate every vector at first use. For example, write "**s**: fractional, dimensionless, two components" and "**r**: ambient Cartesian, angstrom, three components." Such notes may look excessive in a two-line exercise. They become invaluable when a retained artifact is read months later or when a two-dimensional model is passed through a three-dimensional file interface.

Remark 6.1 (Columns, not rows)**.** The convention in Equation (6.2) is part of the representation. Transposing A without changing the formulas changes the coordinate map. A retained record or file should state the vector-axis convention rather than relying on visual inspection of a rectangular array.

6.4 The induced metric

The Cartesian squared length of a displacement is

$$\|\mathrm{d}\mathbf{r}\|_2^2 = (A\,\mathrm{d}\mathbf{s})^{\mathsf{T}}(A\,\mathrm{d}\mathbf{s}) \tag{6.5}$$

$$= \mathrm{d}\mathbf{s}^{\mathsf{T}} G\,\mathrm{d}\mathbf{s}, \tag{6.6}$$

where

$$G = A^{\mathsf{T}} A. \tag{6.7}$$

The matrix G is the Gram metric of the ordered primitive vectors. It contains all intrinsic lengths and angles:

$$G_{ij} = \mathbf{a}_i \cdot \mathbf{a}_j. \tag{6.8}$$

If the columns of A are linearly independent, then G is symmetric positive definite and G^{-1} exists.

This observation explains why an intrinsic 2×2 basis and an embedded 3×2 basis can define the same scalar periodic geometry. If $A_{\mathrm{int}}^{\mathsf{T}} A_{\mathrm{int}} = A_{\mathrm{emb}}^{\mathsf{T}} A_{\mathrm{emb}}$, they assign the same length to every fractional displacement. The embedded basis nevertheless carries extra orientation information that the metric alone cannot recover.

6.4.1 Orthogonal square example

For

$$A = \begin{bmatrix} a & 0 \\ 0 & a \\ 0 & 0 \end{bmatrix}, \tag{6.9}$$

we obtain

$$G = a^2 I_2, \qquad G^{-1} = a^{-2} I_2. \tag{6.10}$$

Fractional coordinate lines are orthogonal and have equal physical scale. This is the geometry used by the original square-lattice controlled model in Appendix H.

6.4.2 Skew-cell example

Let

$$\mathbf{a}_1 = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{a}_2 = a \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix}, \qquad 0 < \theta < \pi. \tag{6.11}$$

Then

$$G = a^2 \begin{bmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{bmatrix}. \tag{6.12}$$

The off-diagonal entries are geometric: they record that the coordinate axes are not orthogonal. They do not, by themselves, describe an anisotropic material response.

6.4.3 A fully worked sixty-degree cell

Let us pause over one cell long enough to make the abstractions concrete. Take

$$\mathbf{a}_1 = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{a}_2 = a \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \\ 0 \end{bmatrix}. \tag{6.13}$$

The two primitive vectors have length a and meet at sixty degrees. Their Gram metric is

$$G = a^2 \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}, \qquad G^{-1} = \frac{1}{a^2} \begin{bmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{bmatrix}. \tag{6.14}$$

Notice the negative off-diagonal entries of the inverse metric. They do not mean that a squared length can be negative. Positive definiteness concerns the whole quadratic form $\mathrm{d}\mathbf{s}^{\mathsf{T}} G\, \mathrm{d}\mathbf{s}$, not the sign of each matrix entry.

Consider three fractional displacements:

$$\begin{aligned} \mathrm{d}\mathbf{s}_1 &= (1, 0)^{\mathsf{T}}, \\ \mathrm{d}\mathbf{s}_2 &= (0, 1)^{\mathsf{T}}, \\ \mathrm{d}\mathbf{s}_3 &= (1, -1)^{\mathsf{T}}. \end{aligned} \tag{6.15}$$

Equation (6.6) assigns all three the same physical length a. The third result is a useful check: in a sixty-degree rhombus, the difference between the two primitive vectors is also an edge of the associated equilateral triangle.

Now rotate both vectors in ambient three-space. The individual Cartesian components change, but every entry of G remains the same. An intrinsic scalar calculation should not change merely because the flat sample was rotated on a laboratory table. A coupling to a fixed laboratory direction may change, which is exactly why we retain the ambient representation separately.

The following small program turns the board derivation into explicit arrays. The imports are shown because a reader should always be able to identify which package owns each numerical operation. Inline comments explain why a line is present rather than merely restating its syntax.

Python interlude—checking an embedded periodic plane

```
import numpy as np

# Store primitive vectors as columns of one ambient 3-by-2 basis.
a = 2.0
A = a * np.array(
    [[1.0, 0.5], [0.0, np.sqrt(3.0) / 2.0], [0.0, 0.0]],
    dtype=np.float64,
)

# Derive intrinsic geometry from the defining basis.
G = A.T @ A
G_inverse = np.linalg.inv(G)
B = 2.0 * np.pi * A @ G_inverse
P = A @ G_inverse @ A.T

# Measure duality and orthogonal-projector identities independently.

duality_error = np.max(np.abs(A.T @ B - 2.0 * np.pi * np.eye(2)))
symmetry_error = np.max(np.abs(P.T - P))
idempotence_error = np.max(np.abs(P @ P - P))
```

```
print("Gram metric:\n", G)
print("projector rank:", np.linalg.matrix_rank(P))
print("maximum duality error:", duality_error)
print("maximum projector symmetry error:", symmetry_error)
print("maximum projector idempotence error:", idempotence_error)
```

This code is pedagogical synthetic data, not a material calculation.

Notebook to do Create `examples/tutorials/research-monograph/foundations/periodic_geometry.ipynb` with explicit imports, explanatory inline comments, physical units, transparent NumPy checks, and exercises using the maintained PhysKit adapter and strict codec. Keep the notebook unexecuted in the repository and verify it through a separate bounded test route.

6.5 Dual reciprocal vectors

For a full-column-rank $A \in \mathbb{R}^{m\times d}$, with $m \geq d$, define

$$B = 2\pi A \left(A^{\mathsf{T}} A\right)^{-1} = 2\pi A G^{-1}. \tag{6.16}$$

Its columns $\mathbf{b}_j$ obey

$$A^{\mathsf{T}} B = 2\pi I_d, \qquad \mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}. \tag{6.17}$$

When A is square and invertible, Equation (6.16) reduces to the familiar expression

$$B = 2\pi A^{-\mathsf{T}}. \tag{6.18}$$

The rectangular formula is therefore not a different reciprocal-lattice convention. It is the full-column-rank extension of the same duality condition.

6.5.1 Deriving the rectangular formula at the board

The defining requirement is not “take an inverse.” The defining requirement is the duality relation

$$A^{\mathsf{T}} B = 2\pi I_d. \tag{6.19}$$

For a rectangular A, an ordinary inverse does not exist. We therefore ask for columns of B that lie in the same plane as the columns of A. Any such candidate has the form $B = AC$

for some $d \times d$ coefficient matrix C. Substitution gives

$$A^{\mathsf{T}} A C = G C = 2\pi I_d. \tag{6.20}$$

Full column rank makes G invertible, so

$$C = 2\pi G^{-1} \tag{6.21}$$

and hence $B = 2\pi A G^{-1}$. This short derivation tells us more than a memorized formula. It tells us why the dual vectors remain in the tangent plane and why rank loss makes the construction fail.

For the sixty-degree cell in Equation (6.13), the result is

$$B = \frac{2\pi}{a} \begin{bmatrix} 1 & 0 \\ -1/\sqrt{3} & 2/\sqrt{3} \\ 0 & 0 \end{bmatrix}. \tag{6.22}$$

A quick multiplication by A^{T} recovers $2\pi I_2$. Performing that multiplication is preferable to trusting the visual appearance of the vectors.

Because A carries length, G carries length squared, G^{-1} carries inverse length squared, and B carries inverse length. These unit checks often reveal a transposition or scaling mistake before a numerical calculation is run. In lecture, it is useful to say the units aloud while moving through the product: “length times inverse length squared gives inverse length.”

6.6 The one-dimensional reciprocal quotient

The general dual-basis construction becomes especially transparent in one dimension. Let the direct primitive vector have length $a > 0$. Its reciprocal primitive vector has magnitude

$$G = \frac{2\pi}{a}, \tag{6.23}$$

because $aG = 2\pi$. A physical wave vector k can be written in the dimensionless reduced coordinate

$$\kappa = \frac{k}{G}. \tag{6.24}$$

Wave vectors whose reduced coordinates differ by an integer occupy the same point of the reciprocal quotient. Thus the first Brillouin zone is not fundamentally a closed interval with two distinct endpoints. It is a circle of reciprocal classes that can be represented, for example, by

$$-\frac{1}{2} \le \kappa < \frac{1}{2}. \tag{6.25}$$

The endpoint $1/2$ is excluded because it is equivalent to $-1/2$ after one reciprocal translation. Including both may be useful when displaying a closed plotting path, but it duplicates one point when constructing a complete periodic mesh.

6.6.1 Bloch sectors and fiber operators

The reciprocal quotient labels more than plotting coordinates. Suppose that an operator $\widehat{H}$ on the infinite periodic system commutes with translation by a. Bloch theory resolves its state space into sectors labelled by a reciprocal class $[k]$ [55, 112]. In the sector labelled by k, a wavefunction obeys the quasiperiodic condition

$$\psi_k(x+a) = e^{ika}\psi_k(x). \tag{6.26}$$

Changing k by an integer multiple of G gives the same reciprocal class and therefore does not create a new physical sector.

A *Bloch fiber space* is the state space belonging to one fixed reciprocal class $[k]$. The corresponding *Bloch-fiber operator* is the action of $\widehat{H}$ restricted to that sector. This is an operator and state-space statement, not yet a choice of numerical basis or finite matrix.

There are two common, unitarily related ways to represent the same fiber. One may retain the original wavefunction on a primitive cell and impose Equation (6.26). Alternatively, write

$$\psi_k(x) = e^{ikx}u_k(x), \qquad u_k(x+a) = u_k(x), \tag{6.27}$$

and work in the fixed cell-periodic space $\mathcal{H}_{\text{per}} = L^2_{\text{per}}([0,a])$. In this representation, the fiber operator is

$$\widehat{H}(k) = e^{-ikx}\widehat{H}e^{ikx} \quad \text{on } \mathcal{H}_{\text{per}}, \tag{6.28}$$

with an operator domain carrying periodic boundary conditions. For example, if

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x), \qquad V(x+a) = V(x), \tag{6.29}$$

then

$$\widehat{H}(k) = \frac{\hbar^2}{2m}\left(-i\frac{\mathrm{d}}{\mathrm{d}x} + k\right)^2 + V(x). \tag{6.30}$$

The k dependence therefore belongs to the represented derivative and its domain convention; it is not an additional potential.

At the operator level, the Floquet–Bloch transformation expresses this sector decomposition as a direct integral [63],

$$\widehat{H} \simeq \int_{[-G/2,G/2)}^{\oplus} \widehat{H}(k)\,\frac{a\,\mathrm{d}k}{2\pi}. \tag{6.31}$$

The symbol $\simeq$ denotes unitary equivalence, and $\oplus$ indicates that there is one fiber space at each reciprocal class. Solving one fiber produces energies and states at one k; sampling many fibers produces a band representation. A plane-wave cutoff or finite-difference grid then produces a finite representation of each fiber. Such a matrix approximates or represents the declared fiber operator under a stated discretization; it is not the continuum fiber operator itself.

For a particle of mass m, the reciprocal vector also defines the energy scale

$$E_G = \frac{\hbar^2 G^2}{2m}. \tag{6.32}$$

An energy written as E/E_G and a potential amplitude written as V_0/E_G are dimensionless only after a, G, m, and the scaling convention have been declared. Chapter 9 uses these coordinates and scales to derive a complete one-dimensional Bloch-to-lattice reduction.

6.7 Projection and coordinate recovery

The left pseudoinverse of a full-column-rank embedded basis is

$$A^+ = G^{-1} A^{\mathsf{T}}. \tag{6.33}$$

For an in-plane point $\mathbf{r} = A\mathbf{s}$,

$$A^+ \mathbf{r} = \mathbf{s}. \tag{6.34}$$

For a general ambient vector, $A^+\mathbf{r}$ returns the fractional coordinates of its orthogonal in-plane projection. The associated ambient projector is

$$P = AA^+ = AG^{-1}A^{\mathsf{T}}. \tag{6.35}$$

It satisfies

$$P^{\mathsf{T}} = P, \qquad P^2 = P, \qquad \mathrm{Ran}\, P = \mathrm{Ran}\, A. \tag{6.36}$$

An out-of-plane component is removed by this operation. That removal must be explicit: it is not evidence that the ambient vector was originally confined to the plane.

6.7.1 A projection thought experiment

Suppose the plane is the ordinary xy plane and an ambient vector is $\mathbf{v} = (v_x, v_y, v_z)^{\mathsf{T}}$. The projector returns $(v_x, v_y, 0)^{\mathsf{T}}$. No information inside the projected vector tells us whether v_z was originally zero or was discarded. This is why a projected result should retain provenance linking it to its ambient source and to the projection action.

The same issue is easy to miss for a tilted plane because none of the ambient components need vanish. The equation $P\mathbf{v} = \mathbf{v}$ is the appropriate test for whether $\mathbf{v}$ already lies in the plane. Merely observing three nonzero components says nothing either way.

The distinction has a physical counterpart. An ambient electric field may have both tangent and normal components. A strictly two-dimensional model can use the tangent component only after a modeling decision has said what happens to the normal component. Linear algebra can perform the projection, but it cannot make that physical decision for us.

6.8 An intrinsic frame that preserves the embedding

Let $Q \in \mathbb{R}^{3\times 2}$ contain an ordered orthonormal frame for the plane, so that

$$Q^{\mathsf{T}}Q = I_2, \qquad QQ^{\mathsf{T}} = P. \tag{6.37}$$

The coordinates of the primitive vectors in that frame form the intrinsic basis

$$C = Q^{\mathsf{T}}A \in \mathbb{R}^{2\times 2}. \tag{6.38}$$

The intrinsic basis has exactly the same Gram metric:

$$C^{\mathsf{T}}C = A^{\mathsf{T}}QQ^{\mathsf{T}}A = A^{\mathsf{T}}A = G. \tag{6.39}$$

This is the key adapter identity. Algorithms that depend only on the intrinsic metric can operate on C, while the original A, Q, and P retain the ambient orientation needed by later composition.

If the ambient basis is rigidly rotated to RA with $R^{\mathsf{T}}R = I_3$, then

$$(RA)^{\mathsf{T}}(RA) = G. \tag{6.40}$$

A geometric scalar Laplacian is therefore invariant under ambient rigid rotation. Coupling to an ambient vector field need not be invariant unless the field is rotated consistently.

6.9 An embedded plane is not a slab

A flat embedded plane and a slab answer different questions.

3×2 **embedded plane.** Two periodic coordinates; no third direct primitive vector; intrinsic differential operators act only in the plane.

3×3 **slab or bulk cell.** Three direct primitive vectors. Even if one direction contains vacuum or is sampled sparsely, it remains part of the declared three-dimensional cell and boundary-condition model.

This distinction is also separate from the auxiliary third direction used by some historical Wannier90 interfaces in Appendix H. The phrase "embedding" there can refer to satisfying a three-dimensional interface convention. The present chapter instead concerns the geometric map from an intrinsically two-dimensional lattice into ambient Cartesian three-space.

6.9.1 A language test

Whenever the word "two-dimensional" appears, ask two questions.

1. How many independent coordinates does the state depend on?
2. In how many Cartesian dimensions is the geometry represented?

For an intrinsic plane the answers are two and two. For an embedded plane they are two and three. For a three-dimensional slab model they are three and three, even if the third direction is physically long or contains vacuum. This language test prevents a data-shape observation from silently becoming a physical claim.

6.10 Maintained computational representation

The maintained PhysKit dependency represents the defining basis with `EmbeddedDirectLattice2D`. An explicit adapter constructs the intrinsic basis, induced metric, inverse metric, ambient dual reciprocal basis, orthonormal frame, and tangent-plane projector. The adapter preserves the original A rather than silently replacing it by C.

The corresponding strict JSON codec serializes only the defining 3×2 basis, its unit, a schema version, a representation discriminator, and the column-vector convention. It deliberately does not serialize G, G^{-1}, B, Q, or P. Those quantities are recomputed and checked after decoding. This design prevents a retained document from carrying mutually inconsistent defining and derived geometry.

The accepted dependency revision for the software-capability description in this draft is `949d106bc98975a18ad86d6fa84092ef0c68298e`. A consuming-project smoke test constructs a physical 3×2 basis, performs the strict JSON round trip, adapts it to the intrinsic metric, and constructs a Hermitian two-dimensional Bloch-periodic scalar Laplacian. That test establishes bounded software composition only.

6.11 How to record a geometry laboratory

A reproducible geometry exercise should retain more than the final metric. Before computing, write down the following items:

1. the ordered primitive vectors and whether they are stored as rows or columns;
2. the ambient and intrinsic dimensions;
3. the length unit of the direct basis;
4. the fractional-coordinate origin and cell convention;
5. the intended interpretation: intrinsic plane, embedded plane, slab, or bulk cell; and
6. the invariants that will be checked independently.

After computing, retain at least the defining basis and observations of the checks

$$G = A^{\mathsf{T}}A, \qquad A^{\mathsf{T}}B = 2\pi I, \qquad P^{\mathsf{T}} = P, \qquad P^2 = P. \tag{6.41}$$

Derived arrays may be useful in a report, but they should not be allowed to compete with the defining basis as independent authorities. If a serialized metric and a serialized basis disagree, the document has not preserved one coherent geometry.

For a classroom calculation, print a few small matrices and inspect them. For a maintained research artifact, use typed records, strict decoding, explicit units, and a manifest. The scale changes; the conceptual obligation does not.

6.12 End-of-lecture synthesis

The central object of this lecture is the map $\mathbf{r} = A\mathbf{s}$. From it we obtained four families of derived information:

1. $G = A^{\mathsf{T}}A$ describes intrinsic lengths and angles;
2. $B = 2\pi AG^{-1}$ supplies reciprocal vectors dual to the direct basis and thereby defines reduced reciprocal coordinates;
3. $A^{+} = G^{-1}A^{\mathsf{T}}$ recovers fractional coordinates of the in-plane projection; and
4. $P = AG^{-1}A^{\mathsf{T}}$ records the plane inside the ambient space.

The formulas are short, but the distinctions are durable. Fractional coordinates are not Cartesian coordinates. Intrinsic geometry is not ambient orientation. A flat embedded plane is not a slab. A geometric off-diagonal metric entry is not a material effective-mass tensor. These distinctions are the vocabulary needed for the next lecture, where the metric becomes a differential operator.

6.13 Exercises

1. For the skew basis in Equation (6.11), compute G^{-1} and show explicitly that it diverges as $\theta \to 0$ or $\theta \to \pi$. Interpret the result as loss of column rank rather than as a material singularity.

2. Starting from Equation (6.16), prove Equation (6.17) without assuming that A is square.

3. Show that the projector in Equation (6.35) is unchanged when A is replaced by AS for any invertible 2×2 matrix S. Explain what does and does not remain the same about the periodic cell.

4. Choose two nonorthogonal vectors in three dimensions, construct A, G, B, and P, and check the duality and projector identities numerically.

5. Explain why retaining only G is sufficient for a scalar intrinsic Laplacian but insufficient for coupling an in-plane model to a declared ambient electric-field direction.

6. Design one malformed JSON case for each of the following failures: wrong vector-axis declaration, duplicate field, nonfinite number, numeric string, incompatible unit, and rank-deficient basis. State which boundary should reject each case.

6.14 Evidence boundary

The identities in this chapter are geometry and linear algebra. Passing the associated software tests establishes implementation behavior for the tested contracts. It does not establish convergence of a finite-difference model, adequacy of a kinetic Hamiltonian, a material lattice, a slab approximation, or a semiconductor effective-mass tensor. Those questions require the separate operators, calculations, and evidence developed later in the monograph.

CHAPTER 7

Bloch-Periodic Finite Differences on Nonorthogonal Cells

Status pedagogical derivation and verified software-capability description. The maintained operator is a centered second-order geometric scalar Laplacian. This chapter reports no new M4 calculation, material Hamiltonian, scientific validation, or uncertainty quantification.

7.1 Learning goals

After working through this chapter, a reader should be able to

1. derive the Cartesian scalar Laplacian in fractional coordinates;
2. distinguish metric-induced mixed derivatives from physical effective-mass anisotropy;
3. construct pure and mixed centered second-order stencils;
4. apply Bloch phases when a stencil crosses one or several periodic seams;
5. construct plane-wave and finite-difference representations of a one-dimensional periodic cosine fiber;
6. distinguish the geometric Laplacian from kinetic-energy and potential operators; and
7. design software and numerical checks without turning an expected trend into a scientific conclusion.

Chapter 6 supplies the direct basis A, Gram metric $G = A^{\mathsf{T}}A$, inverse metric G^{-1}, and explicit route from an ambient 3×2 basis to an intrinsic 2×2 basis. Standard finite-difference background may be found in Strikwerda [118]; the evidence classifications used here follow Chapter 11.

7.2 Why a familiar stencil needs a full lecture

Most readers have seen the three-point approximation to a second derivative. It is therefore reasonable to ask why we should spend an entire lecture on a Laplacian. The answer is that the one-dimensional stencil is only the local algebraic ingredient. A periodic multidimensional operator also commits us to a coordinate system, a metric, a tensor ordering, a boundary phase convention, a matrix ordering, a physical unit, and a sign convention. Each choice is simple when isolated. Their composition is where silent errors appear.

On a square grid with zero twist, several mistakes can hide. If one treats fractional coordinates as Cartesian coordinates, the square unit cell may make the error invisible. If one omits mixed derivatives, an orthogonal metric makes the omitted coefficients zero anyway. If one mishandles simultaneous seam crossings, an axial stencil never tests the corner case. A nonorthogonal cell with nonzero twist is therefore not needless generality: it exposes assumptions that an easy special case allows us to forget.

The lecture will proceed in the same order one would use at a blackboard. We first derive the continuous operator in reduced coordinates. We then replace each derivative by a centered stencil. Only after the interior stencil is clear do we discuss periodic seams. Finally, we attach units and explain how another model layer turns a geometric Laplacian into a Hamiltonian. This order keeps the source of each coefficient visible.

A useful question to ask at every stage is: "What has been added since the previous line?" Moving from G^{-1} to Δ adds differential structure. Moving from Δ to Δ_h adds a discretization. Moving from an interior stencil to a Bloch matrix adds a boundary representation. Moving from Δ_h to H_h adds physical scaling and a potential. None of these steps is merely a change of notation.

7.3 From Cartesian derivatives to fractional derivatives

Let

$$\mathbf{r} = A\mathbf{s}, \qquad G = A^{\mathsf{T}}A, \tag{7.1}$$

where the primitive basis is constant over the cell. For a scalar function $f(\mathbf{r}) = \widetilde{f}(\mathbf{s})$, the gradient transformation gives a constant-coefficient Laplacian

$$\nabla_{\mathbf{r}}^2 f = \sum_{i,j=1}^{d} g^{ij} \frac{\partial^2 \widetilde{f}}{\partial s_i \partial s_j}, \qquad [g^{ij}] = G^{-1}. \tag{7.2}$$

Since G^{-1} is symmetric, the same expression can be written

$$\nabla_{\mathbf{r}}^2 = \sum_{i=1}^{d} g^{ii} \partial_{s_i}^2 + 2 \sum_{i<j} g^{ij} \partial_{s_i} \partial_{s_j}. \tag{7.3}$$

For an orthogonal cell, $g^{ij} = 0$ when $i \neq j$, and the mixed derivatives vanish. For a skew cell, the mixed terms are required to represent the ordinary isotropic Cartesian Laplacian. Omitting them changes the mathematical operator.

7.3.1 The chain rule one index at a time

The compact expression in Equation (7.2) is easy to quote and just as easy to misremember. We therefore derive it. Let $\nabla_{\mathbf{s}}\widetilde{f}$ be the column of fractional derivatives. A small fractional displacement changes the function by

$$d\widetilde{f} = \left(\nabla_{\mathbf{s}}\widetilde{f}\right)^{\mathsf{T}} \mathrm{d}\mathbf{s}. \tag{7.4}$$

The same change, described in the tangent Cartesian plane, is

$$df = \left(\nabla_{\parallel} f\right)^{\mathsf{T}} \mathrm{d}\mathbf{r} = \left(\nabla_{\parallel} f\right)^{\mathsf{T}} A\, \mathrm{d}\mathbf{s}. \tag{7.5}$$

Equality for every d$\mathbf{s}$ requires

$$A^{\mathsf{T}} \nabla_{\parallel} f = \nabla_{\mathbf{s}}\widetilde{f}. \tag{7.6}$$

The in-plane solution is

$$\nabla_{\parallel} f = AG^{-1}\nabla_{\mathbf{s}}\widetilde{f}. \tag{7.7}$$

Substitution confirms the result because $A^{\mathsf{T}}AG^{-1} = I_d$.

The basis is constant, so differentiating once more does not produce derivatives of A or G. Taking the tangent divergence leaves one inverse-metric coefficient for each pair of fractional derivatives, giving Equation (7.2). For a curvilinear or spatially varying basis, this shortcut would be invalid; metric determinants and coefficient derivatives would enter. The present implementation is explicitly a constant primitive-cell construction.

This derivation is also valid for an embedded plane because $AG^{-1}\nabla_{\mathbf{s}}\widetilde{f}$ lies in the column space of A. We are not taking a three-dimensional Laplacian of a function whose normal dependence has been left unspecified. We are taking the intrinsic scalar Laplacian of a flat plane represented in ambient coordinates.

Remark 7.1 (Geometry is not constitutive anisotropy)**.** Off-diagonal entries of G^{-1} arise because the selected coordinate axes are skew. An anisotropic effective-mass tensor describes a physical constitutive model. Although both can produce cross derivatives after a coordinate choice, they belong to different layers and must not be identified merely because their matrices have similar shapes.

7.4 Endpoint-excluded periodic grids

For axis i, select N_i points

$$s_{i,n_i} = \frac{n_i}{N_i}, \qquad n_i = 0, 1, \ldots, N_i - 1, \qquad h_i = \frac{1}{N_i}. \tag{7.8}$$

The endpoint $s_i = 1$ is excluded because it is equivalent to $s_i = 0$ up to the boundary condition. Including both endpoints would duplicate one physical location and complicate the seam rule.

A tensor-product grid also needs a declared linear ordering. The maintained implementation uses C order: the last fractional axis varies fastest. Ordering is not a cosmetic implementation detail because it determines which matrix row and column correspond to each grid point.

7.5 Centered pure derivatives

The centered second derivative along fractional axis i is

$$(D_{ii} f)_{\mathbf{n}} = \frac{f_{\mathbf{n}+\mathbf{e}_i} - 2f_{\mathbf{n}} + f_{\mathbf{n}-\mathbf{e}_i}}{h_i^2}. \tag{7.9}$$

For a smooth function away from representation boundaries, this is a second-order approximation to $\partial_{s_i}^2 f$. The corresponding axial matrix entries are real before seam phases are applied.

The pure-derivative contribution to the discrete metric Laplacian is

$$\sum_i g^{ii} D_{ii}. \tag{7.10}$$

Because G^{-1} is positive definite, each g^{ii} is positive. With the sign convention in Equation (7.9), the periodic discrete Laplacian has nonpositive eigenvalues.

7.6 Centered mixed derivatives

For $i \neq j$, the centered mixed derivative uses four corner points:

$$(D_{ij} f)_{\mathbf{n}} = \frac{f_{\mathbf{n}+\mathbf{e}_i+\mathbf{e}_j} - f_{\mathbf{n}+\mathbf{e}_i-\mathbf{e}_j} - f_{\mathbf{n}-\mathbf{e}_i+\mathbf{e}_j} + f_{\mathbf{n}-\mathbf{e}_i-\mathbf{e}_j}}{4h_i h_j}. \tag{7.11}$$

The complete centered operator is

$$\Delta_h = \sum_i g^{ii} D_{ii} + 2 \sum_{i<j} g^{ij} D_{ij}. \tag{7.12}$$

For each pair $i < j$, the factor of two comes from combining the (i, j) and (j, i) terms in Equation (7.2).

In two dimensions, the pure stencil touches the center and four axial neighbors; a nonzero mixed coefficient adds four diagonal neighbors. In three dimensions, three axis pairs contribute corresponding corner displacements. The represented operator remains sparse.

7.6.1 Reading the stencil for a sixty-degree cell

Return to the sixty-degree cell from Equation (6.13). Its inverse metric is

$$G^{-1} = \frac{1}{a^2} \begin{bmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{bmatrix}. \tag{7.13}$$

The continuous Laplacian is therefore

$$\nabla^2 = \frac{4}{3a^2} \left(\partial_{s_1}^2 + \partial_{s_2}^2 - \partial_{s_1} \partial_{s_2} \right). \tag{7.14}$$

The minus sign on the mixed derivative is not a negative kinetic energy and not an instability. It is the coordinate correction required by the oblique axes. The full quadratic symbol remains nonnegative for $-\nabla^2$ because the inverse metric is positive definite.

Suppose, for simplicity, that $h_1 = h_2 = h$. The center receives $-16/(3a^2h^2)$. Each of the four axial neighbors receives $4/(3a^2h^2)$. The four corner coefficients alternate in sign: same-sign corner displacements receive $-1/(3a^2h^2)$, while opposite-sign corner displacements receive $+1/(3a^2h^2)$. Adding every coefficient in a row gives zero before a nonzero boundary twist is applied. That row-sum check is the discrete statement that a constant periodic function has zero Laplacian.

It is worth tracing the signs directly from Equation (7.11). The mixed derivative has positive same-sign corners and negative opposite-sign corners. Multiplication by the negative g^{12} reverses those signs. This two-step explanation is safer than memorizing a nine-point stencil whose signs depend on the selected cell.

7.7 Bloch-periodic seams

Let the reduced twist $\boldsymbol{\kappa}$ be measured in turns along the direct primitive axes. The maintained quotient-seam convention is

$$\psi(\mathbf{s} + \mathbf{q}) = \exp(2\pi i \, \boldsymbol{\kappa} \cdot \mathbf{q}) \, \psi(\mathbf{s}), \qquad \mathbf{q} \in \mathbb{Z}^d. \tag{7.15}$$

Suppose a stencil displacement carries a raw target outside the stored grid. Reduce it as

$$\mathbf{n} + \boldsymbol{\delta} = \overline{\mathbf{n}} + N\mathbf{q}, \tag{7.16}$$

where $\overline{\mathbf{n}}$ lies in the stored index range and $\mathbf{q}$ counts signed primitive-cell crossings. The matrix contribution receives the phase

$$\exp(2\pi i\, \boldsymbol{\kappa} \cdot \mathbf{q}) = \prod_a \exp(2\pi i \kappa_a q_a)\,. \tag{7.17}$$

The product form matters for a mixed-derivative corner that crosses two seams at once. Applying only one axis phase produces the wrong operator. Integer shifts of the twist define the same quotient-seam phases, but the complete unreduced lift and its quotient reduction can still be retained as provenance.

7.7.1 Walking one corner across the seam

Take a 5×4 grid. The stored indices are $n_1 \in \{0, 1, 2, 3, 4\}$ and $n_2 \in \{0, 1, 2, 3\}$. Begin at the upper-right stored point $(4, 3)$ and apply the corner displacement $(+1, +1)$. The raw target is $(5, 4)$. Componentwise Euclidean division gives the wrapped point $(0, 0)$ and the quotient

$$\mathbf{q} = (1, 1). \tag{7.18}$$

The associated matrix entry receives

$$\exp[2\pi i(\kappa_1 + \kappa_2)]\,. \tag{7.19}$$

Both seam crossings matter.

Now begin at $(4, 0)$ and apply $(+1, -1)$. The wrapped point is $(0, 3)$, but the quotient is $(1, -1)$. Its phase is

$$\exp[2\pi i(\kappa_1 - \kappa_2)]\,. \tag{7.20}$$

The negative quotient records that the second coordinate crossed the lower seam. Using an unsigned "did wrap" flag would lose this distinction.

Finally consider the reverse matrix connection. It carries the opposite quotient and therefore the complex-conjugate phase. Paired with the real symmetric interior coefficient, this is the local reason the assembled matrix can be Hermitian. The global Hermiticity test should still be performed; the hand argument tells us what the test is checking.

7.7.2 Hermiticity check

For a real symmetric metric and a consistent seam convention, the represented matrix should satisfy

$$L_h^\dagger = L_h. \tag{7.21}$$

This is an important software invariant. It is not, by itself, a convergence or scientific-validation result. Many incorrect Hermitian matrices can still be assembled.

7.8 From the Laplacian to a Hamiltonian

The geometric result constructed above represents ∇^2. For an isotropic scalar mass m, a familiar kinetic operator is

$$\hat{T} = -\frac{\hbar^2}{2m}\nabla^2. \tag{7.22}$$

A finite Hamiltonian may then have the form

$$H_h = -\alpha L_h + V_h, \qquad \alpha = \frac{\hbar^2}{2m}, \tag{7.23}$$

where V_h is a separately defined potential representation. The basis, units, sampling convention, and energy reference of both terms must agree before addition.

An anisotropic effective-mass model instead involves a tensor such as

$$\hat{T}_M = -\frac{\hbar^2}{2}\nabla\cdot M^{-1}\nabla, \tag{7.24}$$

subject to the model's ordering and regularity assumptions. That tensor is not supplied by the scalar geometric Laplacian. It requires a separate physical contract and, for spatially varying coefficients, a separate discretization analysis.

7.9 A one-dimensional cosine-fiber bridge

Before returning to the maintained multidimensional constructor, it is useful to specialize the preceding rules to the one-dimensional parent used later in the isolated-band reduction. A Bloch fiber was defined in Equation (6.28) as the restriction of the periodic parent to one fixed quasimomentum sector, represented here on cell-periodic functions. Let

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V_0\cos(Gx), \qquad G = \frac{2\pi}{a}. \tag{7.25}$$

There are two common one-cell representations.

7.9.1 Cell-periodic plane waves

Write $\psi_k(x) = e^{ikx}u_k(x)$ with $u_k(x+a) = u_k(x)$. In the periodic basis

$$\phi_n(x) = a^{-1/2}e^{inGx}, \qquad n \in \mathbb{Z}, \tag{7.26}$$

the cell-periodic fiber has entries

$$H^{\mathrm{PW}}_{nm}(k) = \frac{\hbar^2(k+nG)^2}{2m}\delta_{nm} + \frac{V_0}{2}\left(\delta_{n,m+1} + \delta_{n,m-1}\right). \tag{7.27}$$

The kinetic term is diagonal because each basis function is a derivative eigenfunction. The cosine contains only the harmonics $+G$ and $-G$, so it connects reciprocal indices separated by one. This reciprocal-index sparsity is a statement about the potential's Fourier content, not yet a short-range real-space hopping model.

A cutoff P retains $n = -P, \ldots, P$. The resulting matrix dimension is $2P + 1$, and changing P changes the represented space. Low eigenvalues may stabilize rapidly while states near the reciprocal cutoff remain poorly represented.

7.9.2 A grid with a Bloch seam

Alternatively, store the original wavefunction on

$$x_j = \frac{ja}{N_x}, \qquad j = 0, \ldots, N_x - 1, \tag{7.28}$$

and impose $\psi(x + a) = e^{ika}\psi(x)$. If

$$c_h = \frac{\hbar^2}{2m(a/N_x)^2}, \tag{7.29}$$

the centered kinetic matrix has diagonal entries $2c_h$, adjacent entries $-c_h$, and seam entries

$$T_{0,N_x-1}(k) = -c_h e^{-ika}, \qquad T_{N_x-1,0}(k) = -c_h e^{ika}. \tag{7.30}$$

The potential is diagonal with values $V_0 \cos(Gx_j)$. The two seam entries are complex conjugates, which is the one-dimensional instance of the quotient rule in Equation (7.17).

7.9.3 What can be compared directly

The two finite matrices approximate the same declared continuum fibers but use different bases and generally different dimensions. Their ordered low eigenvalues can be compared at common k values. The matrices themselves cannot be subtracted merely because both are called Hamiltonians. An operator residual requires an explicit map into common coordinates.

A high plane-wave cutoff can serve as a declared finite represented reference for a bounded refinement study. Such a reference is not the exact continuum spectrum. Plane-wave cutoff error and finite-difference grid error also remain separate channels even when both are measured against that reference. Chapter 9 continues from these fiber eigensystems to an isolated scalar band and its finite hopping representation.

7.10 What the maintained implementation owns

The pinned PhysKit implementation owns

- endpoint-excluded tensor grids in two or three dimensions;
- intrinsic nonorthogonal direct metrics;
- centered second-order pure and mixed derivatives;
- quotient-seam Bloch phases, including simultaneous multi-axis wraps;
- C-order sparse assembly;
- explicit physical or unitless basis units; and
- an immutable canonical complex CSR representation correlated with its request and twist context.

For an ambient 3×2 plane, the explicit adapter from Chapter 6 supplies the intrinsic metric while retaining the ambient basis and maps. The Laplacian constructor does not need to pretend that the rectangular basis is square.

The implementation intentionally does not own kinetic-energy scaling, effective-mass tensors, potential sampling, material parameters, model selection, campaign thresholds, or scientific acceptance. Those belong to the consumer that constructs a Hamiltonian and states the question being tested.

The next program deliberately uses the maintained public objects rather than reproducing their implementation. The example remains small enough that every construction step is visible.

Python interlude—constructing a twisted scalar Laplacian

```
import numpy as np
from projectkoios.physkit.periodic.finite_difference.grid import (
    PeriodicFiniteDifferenceGrid,
)
from projectkoios.physkit.periodic.finite_difference.laplacian
    import (
    BlochPeriodicLaplacianConstructionRequest,
    BlochPeriodicLaplacianConstructor,
)
from projectkoios.physkit.periodic.lattice.boundary_phases import (
    BoundaryTwistLift,
```

```
)
from projectkoios.physkit.periodic.lattice.embedding import (
    EmbeddedDirectLattice2D,
    EmbeddedDirectLattice2DAdaptationRequest,
    EmbeddedDirectLattice2DAdapter,
)
from projectkoios.physkit.periodic.lattice.finite_domain import (
    FinitePeriodicDomain,
    LatticeDimension,
)
from projectkoios.physkit.units import MatrixQuantity, PhysicalUnit

# Define a flat two-dimensional lattice in ambient three-space.
unit = PhysicalUnit("angstrom")
ambient_basis = np.array(
    [[1.0, 0.25], [0.0, 1.2], [0.5, -0.1]],
    dtype=np.float64,
)
embedded = EmbeddedDirectLattice2D(
    ambient_basis=MatrixQuantity(ambient_basis, unit)
)

# Adapt ambient geometry to the intrinsic metric used by the
    stencil.
adaptation = EmbeddedDirectLattice2DAdapter().action(
    request=EmbeddedDirectLattice2DAdaptationRequest(
        embedded_lattice=embedded
    )
)
grid = PeriodicFiniteDifferenceGrid(
    metric=adaptation.intrinsic_metric,
    domain=FinitePeriodicDomain(LatticeDimension.TWO, (4, 5)),
    direct_basis_unit=unit,
)

# The twist is measured in turns along the two primitive axes.
request = BlochPeriodicLaplacianConstructionRequest(
    grid=grid,
    twist=BoundaryTwistLift(
        LatticeDimension.TWO,
        (0.125, -0.25),
```

```
    ),
)
result = BlochPeriodicLaplacianConstructor().action(request=request
    )
L = result.represented_laplacian.to_csr()

# Preserve sparsity while checking the exact construction invariant.

hermitian_residual = L - L.conjugate().transpose()
assert hermitian_residual.nnz == 0
print("shape:", L.shape)
print("unit:", result.represented_laplacian.unit)
```

Notebook to do Create `examples/tutorials/research-monograph/foundations/bloch_periodic_finite_differences.ipynb` with explicit imports, explanatory inline comments, a zero-twist null-mode check, and a sparse-preserving nonzero-twist Hermiticity check. The notebook must not densify the represented Laplacian and must not imply kinetic scaling or material physics.

7.11 A Fourier-mode diagnostic

A periodic constant-coefficient stencil should be understood in Fourier space as well as in matrix form. Consider a reduced-coordinate mode

$$f_{\boldsymbol{\nu}}(\mathbf{s}) = \exp(2\pi i\, \boldsymbol{\nu} \cdot \mathbf{s})\,, \qquad \boldsymbol{\nu} = \mathbf{m} + \boldsymbol{\kappa}, \tag{7.31}$$

where $\mathbf{m}$ is an integer mode index and $\boldsymbol{\kappa}$ supplies the Bloch twist. Translating by one primitive cell multiplies this mode by the phase in Equation (7.15).

Applying the pure stencil gives the eigenvalue

$$\widehat{D}_{ii}(\boldsymbol{\nu}) = -\frac{4}{h_i^2}\sin^2(\pi \nu_i h_i)\,. \tag{7.32}$$

Applying the mixed stencil gives

$$\widehat{D}_{ij}(\boldsymbol{\nu}) = -\frac{\sin(2\pi \nu_i h_i)\sin(2\pi \nu_j h_j)}{h_i h_j}. \tag{7.33}$$

The full discrete symbol follows by inserting these expressions into Equation (7.12).

This is an especially useful numerical oracle because it is derived without assembling the sparse matrix row by row. On a small grid, one can construct the mode vector, apply the

matrix, and compare the result with the analytical stencil symbol. A discrepancy can reveal a wrong flattening order, a missing mixed term, or an incorrect seam phase. Agreement remains bounded numerical verification of this operator; it does not validate any material model.

In the small-spacing limit, $\sin(\pi\nu_i h_i) \sim \pi\nu_i h_i$, so the pure symbol approaches $-(2\pi\nu_i)^2$. The mixed symbol similarly approaches $-(2\pi\nu_i)(2\pi\nu_j)$. This recovers the continuous metric symbol and makes the expected second-order truncation behavior visible.

7.12 A staged computational laboratory

A useful student exercise should separate contracts rather than beginning with a large material calculation. The point of the laboratory is not to obtain one impressive matrix. It is to make each layer fail in an interpretable way before the layers are composed.

For every stage, record the request, the expected invariant, the observed quantity, the acceptance rule, and the interpretation. If a stage fails, stop and diagnose it rather than carrying a suspect operator into the next stage. This is slower than writing the final Hamiltonian immediately and much faster than debugging an unexplained band-structure discrepancy at the end.

7.12.1 Stage 1: orthogonal null mode

Use zero twist and an orthogonal unit cell. Construct the discrete scalar Laplacian and verify that the constant vector is a null vector up to the declared numerical tolerance. Check matrix shape, ordering, units, and Hermiticity separately.

7.12.2 Stage 2: one-dimensional symbol inside a tensor grid

Choose a function that varies along only one fractional axis. Compare the observed eigenvalue with the centered-difference symbol

$$\lambda_h(\xi) = -\frac{4}{h^2}\sin^2\left(\frac{\xi h}{2}\right). \tag{7.34}$$

This isolates the pure derivative before introducing mixed terms.

7.12.3 Stage 3: skew intrinsic cell

Choose a full-rank skew 2×2 basis. Compare the represented action on a fractional Fourier mode with the metric-weighted discrete symbol. Repeat with the off-diagonal metric entry artificially omitted and explain why the latter operator is not the same Cartesian Laplacian.

7.12.4 Stage 4: ambient rotation

Embed the same plane with a 3×2 basis, rotate it rigidly in ambient three-space, and adapt both bases to the intrinsic route. The geometric scalar Laplacians should agree because the induced metrics agree. The stored ambient frames and projectors should rotate rather than disappear.

7.12.5 Stage 5: seam and corner phases

Use a nonzero two-component twist and inspect a mixed-stencil point that wraps both axes. Verify that its phase equals the product in Equation (7.17). Then check Hermiticity of the full matrix.

7.12.6 Stage 6: refinement without overclaiming

Apply the operator to a smooth manufactured Bloch mode over a sequence of grids. Estimate the observed order over the tested sequence. Report this as numerical verification of the discretization for the manufactured case, not as material validation or uncertainty quantification.

7.12.7 What a laboratory report should say

A strong report does not merely state that the code "worked." It says, for example, that the matrix shape and unit matched the declared grid contract; that the maximum absolute Hermiticity residual was evaluated; that an independently derived Fourier symbol agreed over the tested modes; and that a manufactured smooth mode displayed a stated observed refinement rate over a listed grid sequence. It also says what those observations do not establish.

If physical units are used, the report should show the dimensional path. The basis has length, the inverse metric has inverse length squared, fractional spacings are dimensionless, and the represented Laplacian has inverse length squared. Kinetic scaling then converts that unit to energy. A bare floating array cannot communicate this path on its own.

The report should also retain negative observations. If a coarse grid breaks an expected

asymptotic trend, that point is not improved by deletion after the fact. It may reveal that the sequence has not entered the asymptotic regime. The appropriate conclusion is bounded to the tested sequence.

7.13 Common mistakes

1. Treating fractional coordinates as Cartesian coordinates when the cell is skew.
2. Using only diagonal entries of G^{-1} for a nonorthogonal basis.
3. Confusing a 3×2 embedded plane with a 3×3 slab.
4. Applying a seam phase for only one axis when a corner wraps several axes.
5. Adding a potential matrix whose ordering or units differ from the Laplacian representation.
6. Calling L_h a kinetic-energy matrix before supplying the sign, coefficient, and mass model.
7. Interpreting Hermiticity or a passing software test as convergence.
8. Interpreting convergence to the stated synthetic operator as evidence that the operator models a real material adequately.

7.14 Exercises

1. Derive Equation (7.2) from the chain rule for a constant full-rank basis.
2. Derive the four-corner stencil in Equation (7.11) by composing two centered first differences.
3. For the skew basis in Equation (6.11), write the complete two-dimensional discrete stencil coefficients in terms of a, θ, h_1, and h_2.
4. Show that the zero-twist constant mode is annihilated by every pure and mixed stencil contribution.
5. Determine the seam quotient $\mathbf{q}$ and phase for a corner displacement from $(N_1 - 1, N_2 - 1)$ to (N_1, N_2).

6. Explain why a scalar potential sampled at grid points is diagonal in this real-space representation but need not be diagonal after transport to a plane-wave representation.

7. Propose independent oracles for shape/order, Hermiticity, the orthogonal reduction, the skew-cell symbol, second-order refinement, and unit propagation. Classify each as software or numerical verification.

8. State the additional physical information required to replace Equation (7.22) by an anisotropic multivalley silicon effective-mass operator.

7.15 Evidence boundary and next use

The software dependency and consuming-project smoke test establish that the accepted public records compose through the embedded-plane, intrinsic-metric, strict-serialization, and scalar-Laplacian route. PhysKit's own synthetic tests cover a broader numerical surface. Neither set of tests is a new calculation for Appendix H.

The planned M4 calculation will be the first monograph-owned opportunity to use the maintained nonorthogonal parent route in a prospectively frozen controlled comparison. Until its protocol, retained result, independent reconstruction, and manifest pass, this chapter remains training material and software capability documentation rather than calculated numerical evidence.

CHAPTER 8

Projection and Retained Spaces

Status pedagogical foundation for finite retained-space constructions. The examples are illustrative mathematics and synthetic numerical checks, not validated reductions of silicon.

8.1 Learning goals

After working through this chapter, a reader should be able to

1. construct an orthogonal projector from an orthonormal frame;
2. distinguish a retained subspace from a basis used to represent it;
3. define a compressed operator on the retained space;
4. interpret an isolated Bloch band as a fiberwise retained space without confusing its scalar dispersion with a gauge or localized orbital;
5. calculate the discarded-sector residual of an exact spectral projection and explain its energy-reference dependence;
6. distinguish exact spectral projection from Feshbach–Löwdin elimination and Schrieffer–Wolff block decoupling;
7. explain why projection can preserve selected eigenpairs without preserving exact finite hopping range;
8. distinguish exact retention from an approximate effective-model class; and
9. separate projection, disentanglement, basis transformation, localization, and truncation.

The preceding chapters supplied finite representations, Hermitian eigensystems and spectral projectors, quantum operators, periodic geometry, and a concrete discretization.

We can now state precisely what it means to retain only part of a state space. Chapter 3 treated an exact eigenspace of one matrix; this chapter adds selection rules, compression, and fiberwise retained families.

8.2 Orthogonal projection

Definition 8.1 (Orthogonal projection)**.** Let $\mathcal{S} \subseteq \mathcal{H}$ be a closed subspace. Its orthogonal projector $\hat{P}_{\mathcal{S}}$ is the operator satisfying

$$\begin{aligned} \hat{P}_{\mathcal{S}}^2 &= \hat{P}_{\mathcal{S}}, \quad \text{(idempotence)}, \\ \hat{P}_{\mathcal{S}}^\dagger &= \hat{P}_{\mathcal{S}}, \quad \text{(self-adjointness)}, \\ \operatorname{Ran} \hat{P}_{\mathcal{S}} &= \mathcal{S}, \quad \text{(range condition)}. \end{aligned} \tag{8.1}$$

■

For a finite orthonormal basis $\{|\phi_a\rangle\}_{a=1}^N$ of $\mathcal{S}$,

$$\hat{P}_{\mathcal{S}} = \sum_{a=1}^{N} |\phi_a\rangle\langle\phi_a|. \tag{8.2}$$

Each term retains the component along $|\phi_a\rangle$, while the sum removes every component orthogonal to $\mathcal{S}$.

If the orthonormal frame is stored as columns of $V \in \mathbb{C}^{D\times N}$, then

$$V^\dagger V = I_N, \qquad P = VV^\dagger. \tag{8.3}$$

Replacing V by VG for a unitary $G \in U(N)$ changes the frame inside $\mathcal{S}$ but not the projector:

$$(VG)(VG)^\dagger = VV^\dagger. \tag{8.4}$$

The subspace and a selected basis of that subspace are therefore different records.

8.3 Compression to a retained space

Let $\widehat{H}$ act on its stated domain in a parent state space $\mathcal{H}$. When the required domain conditions hold, the compressed operator on $\mathcal{S}$ is

$$\widehat{H}_{\mathcal{S}} = \hat{P}_{\mathcal{S}} \widehat{H} \hat{P}_{\mathcal{S}}\Big|_{\mathcal{S}} : \mathcal{S} \longrightarrow \mathcal{S}. \tag{8.5}$$

In the frame V, its finite matrix is

$$H_{\mathcal{S}} = V^{\dagger} H V \tag{8.6}$$

when H is a compatible ambient finite representation.

Selecting $\mathcal{S}$ is a model or representation decision. Writing the coordinates of $\widehat{H}_{\mathcal{S}}$ in a basis is a separate operation. Neither operation is synonymous with Wannier localization or with truncation of matrix entries.

Consistently compressing an additive decomposition gives

$$\hat{P}_{\mathcal{S}} \hat{H} \hat{P}_{\mathcal{S}} = \hat{P}_{\mathcal{S}} \hat{T} \hat{P}_{\mathcal{S}} + \hat{P}_{\mathcal{S}} \hat{V} \hat{P}_{\mathcal{S}}. \tag{8.7}$$

In general, one may not compress $\hat{H}$ while leaving $\hat{T}$ in an uncompressed parent representation. Projection may also change mathematical form: a potential local in the original coordinate representation need not be representable by multiplication by a scalar function on the retained space.

8.4 Spectral retention

Assume for this pedagogical case that the relevant spectrum is discrete and that the eigenstates form a complete orthonormal basis:

$$\hat{H}|\psi_i\rangle = E_i|\psi_i\rangle, \qquad \hat{H} = \sum_i E_i |\psi_i\rangle\langle\psi_i|. \tag{8.8}$$

For an energy cutoff E_{cut}, define

$$\mathcal{H}^{(P)} = \text{span}\left\{|\psi_i\rangle \mid E_i \le E_{\text{cut}}\right\}, \qquad \hat{P} = \sum_{E_i \le E_{\text{cut}}} |\psi_i\rangle\langle\psi_i|. \tag{8.9}$$

The retained Hamiltonian is

$$\hat{H}^{(P)} = \hat{P}\hat{H}\hat{P}\Big|_{\mathcal{H}^{(P)}} = \sum_{E_i \le E_{\text{cut}}} E_i |\psi_i\rangle\langle\psi_i|. \tag{8.10}$$

It exactly reproduces every retained eigenpair but does not specify the action of $\hat{H}$ outside $\mathcal{H}^{(P)}$. It is an exact restriction of the parent Hamiltonian, not yet an approximate effective model.

This construction is pedagogical. A periodic crystal uses Bloch-fiber spaces, and entangled bands may require construction of a smooth retained family rather than selection by one global energy cutoff.

8.4.1 The embedded residual and its energy reference

Let

$$\hat{Q} = \hat{I} - \hat{P}. \tag{8.11}$$

Because the projector above is built from eigenvectors of $\hat{H}$, it is an invariant spectral projector:

$$[\hat{P}, \hat{H}] = 0, \qquad \hat{P}\hat{H}\hat{Q} = \hat{Q}\hat{H}\hat{P} = 0. \tag{8.12}$$

If the retained operator is embedded back into the parent space as $\hat{P}\hat{H}\hat{P}$, then

$$\hat{H} - \hat{P}\hat{H}\hat{P} = \hat{Q}\hat{H}\hat{Q} = \sum_{E_i > E_{\text{cut}}} E_i |\psi_i\rangle\langle\psi_i|. \tag{8.13}$$

For a finite D-dimensional representation this gives

$$\|H - PHP\|_{\text{F}}^2 = \sum_{E_i > E_{\text{cut}}} |E_i|^2. \tag{8.14}$$

This is an exact discarded-sector identity, not an approximation estimate for a new model class.

The displayed Frobenius residual depends on the energy zero. Replacing $\hat{H}$ by $\hat{H} + c\hat{I}$ shifts the retained operator on Ran $\hat{P}$ by $c\hat{I}_P$, as expected, but it also changes the full-space embedded residual by $c\hat{Q}$. The reason is simple: $\hat{P}\hat{H}\hat{P}$, viewed on the parent space, acts as zero on the discarded sector. That zero action is an embedding convention. It is not a claim that the discarded parent energies are physically zero. Energy references must therefore be fixed before comparing embedded projection residuals.

8.5 Isolated bands as fiberwise retained spaces

For a periodic parent, retention is performed over a family of Bloch fibers. Let

$$\widehat{H}(k)|u_{nk}\rangle = E_n(k)|u_{nk}\rangle. \tag{8.15}$$

A nondegenerate band is isolated over a reciprocal domain when its eigenvalue remains separated from the excluded spectrum throughout that domain. The associated rank-one projector is

$$\widehat{P}_n(k) = |u_{nk}\rangle\langle u_{nk}|. \tag{8.16}$$

Changing the phase by $|u_{nk}\rangle \mapsto e^{i\theta(k)}|u_{nk}\rangle$ changes the frame but leaves $\widehat{P}_n(k)$ unchanged. For an isolated group $\mathcal{B}$ of M bands, the corresponding higher-rank projector is

$$\widehat{P}_{\mathcal{B}}(k) = \sum_{n \in \mathcal{B}} |u_{nk}\rangle\langle u_{nk}|. \tag{8.17}$$

A unitary change of the M-column frame changes its coordinates but not this projector.

Regularity of the projector family is what connects spectral selection to spatial localization. Kohn established the relation between analytic Bloch waves and localized Wannier functions in one dimension [57]; des Cloizeaux studied analytic and asymptotic properties of crystalline band projection operators [20]; Marzari and Vanderbilt formulated maximally localized generalized Wannier functions for composite bands [73]; and Brouder et al. related exponentially localized composite Wannier functions to the topology of insulating Bloch bundles [13]. These results reinforce a distinction already visible in the finite calculation: selecting a subspace is not the same operation as selecting a smooth, periodic, localized frame inside that subspace.

The retained Hamiltonian in this one-dimensional fiber space is the scalar matrix

$$H_n^{(P)}(k) = \langle u_{nk}|\widehat{H}(k)|u_{nk}\rangle = E_n(k). \tag{8.18}$$

This explains both the usefulness and the limitation of scalar isolated-band reduction. The dispersion $E_n(k)$ completely specifies the Hamiltonian's action inside each rank-one eigenspace. It does not specify a phase-connected family of eigenvectors, a Wannier function, a localization center, or a real-space probability density.

On a finite reciprocal mesh, a positive minimum sampled adjacent gap is a conditioning diagnostic. It does not prove that the gap remains positive between sample points. Near a degeneracy, energy ordering can exchange state identity even when the ordered eigenvalue table remains easy to compute. A composite retained projector may then remain meaningful after the individual scalar labels cease to be stable.

Chapter 9 starts from the scalar retained operator in Equation (8.18) and derives its finite hopping representation. Chapter 10 then retains a higher-rank frame and makes gauge transport, alignment, the Bloch–Wannier bridge, and block locality explicit. Neither Fourier transformation is an additional projection or, by itself, a localization calculation.

8.6 Exact retained operator versus effective model

Reserve $\hat{H}_{\text{eff}}(\boldsymbol{\theta})$ for an operator belonging to a prescribed effective-model class with parameters $\boldsymbol{\theta}$. Then

$$\underbrace{\hat{H}^{(P)}}_{\text{exact retained operator}} \qquad \text{and} \qquad \underbrace{\hat{H}_{\text{eff}}(\boldsymbol{\theta})}_{\text{approximate model on an identified retained space}} \tag{8.19}$$

are different constructions.

The first is fixed after the parent, retained space, and compression rule are fixed. The second additionally selects a model class and usually a fitting or reduction criterion. Agreement of selected eigenvalues does not by itself show that the effective operator reproduces the retained operator in the coordinates needed by later perturbations.

8.7 Spectral projection versus eliminated-space downfolding

An exact spectral projector is a special projector: it commutes with the parent Hamiltonian. A projector chosen from atomic orbitals, spatial basis functions, or another proposed model space need not. To see the consequence, let $\hat{Q} = \hat{I} - \hat{P}$ and write the parent Hamiltonian in block form,

$$\hat{H} = \begin{pmatrix} \hat{P}\hat{H}\hat{P} & \hat{P}\hat{H}\hat{Q} \\ \hat{Q}\hat{H}\hat{P} & \hat{Q}\hat{H}\hat{Q} \end{pmatrix}. \tag{8.20}$$

For an eigenstate $|\psi\rangle$, define $|\psi_P\rangle = \hat{P}|\psi\rangle$ and $|\psi_Q\rangle = \hat{Q}|\psi\rangle$. The Q-sector equation is

$$(E - \hat{Q}\hat{H}\hat{Q})|\psi_Q\rangle = \hat{Q}\hat{H}\hat{P}|\psi_P\rangle. \tag{8.21}$$

Whenever the displayed inverse exists,

$$|\psi_Q\rangle = (E - \hat{Q}\hat{H}\hat{Q})^{-1}\hat{Q}\hat{H}\hat{P}|\psi_P\rangle. \tag{8.22}$$

Substitution into the P-sector equation gives the energy-dependent Feshbach–Löwdin effective operator

$$\hat{H}_{\text{down}}(E) = \hat{P}\hat{H}\hat{P} + \hat{P}\hat{H}\hat{Q}(E - \hat{Q}\hat{H}\hat{Q})^{-1}\hat{Q}\hat{H}\hat{P}. \tag{8.23}$$

Projection-operator elimination in this form is associated with Feshbach and Löwdin [28, 67]. The second term records excursions through the eliminated sector. For an exact spectral projector, $\hat{P}\hat{H}\hat{Q} = 0$, so the term vanishes and the downfolded operator reduces to the exact compression.

A Schrieffer–Wolff construction asks a related but different question: it seeks a unitary change of basis that block diagonalizes or perturbatively decouples selected sectors [12]. Thus three statements must not be exchanged: an eigenspace was retained exactly; an orbital sector was eliminated through a resolvent; or two sectors were approximately decoupled by a unitary transformation.

8.8 Why spectral projection need not preserve finite range

Suppose that a periodic parent matrix is exactly finite range in a declared orbital basis,

$$H(\mathbf{k}) = \sum_{\mathbf{R}\in\mathcal{S}_0} e^{i\mathbf{k}\cdot\mathbf{R}} T_{\mathbf{R}}, \qquad |\mathcal{S}_0| < \infty. \tag{8.24}$$

The embedded spectral projection is

$$H_P(\mathbf{k}) = P(\mathbf{k})H(\mathbf{k})P(\mathbf{k}). \tag{8.25}$$

Even when $H(\mathbf{k})$ is a finite Fourier polynomial, $P(\mathbf{k})$ generally is not. Consequently the projected coefficients

$$T_{\mathbf{R}}^{(P)} = \frac{1}{|\mathrm{BZ}|}\int_{\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{R}} H_P(\mathbf{k})\,\mathrm{d}\mathbf{k} \tag{8.26}$$

need not vanish outside any finite translation set. Classical work on crystalline projection operators and modern matrix-decay analysis explain why gapped projectors can nevertheless be spatially localized or rapidly decaying [20, 4]. Exact finite support is a stronger property than decay.

8.8.1 A two-band calculation

Consider the nearest-neighbor two-orbital Bloch matrix

$$H(k) = \begin{pmatrix} \Delta & t(1+e^{-ik}) \\ t(1+e^{ik}) & -\Delta \end{pmatrix}, \qquad t \in \mathbb{R}, \qquad \Delta \neq 0. \tag{8.27}$$

Direct multiplication gives

$$H(k)^2 = d(k)^2 I, \qquad d(k) = \sqrt{\Delta^2 + 4t^2\cos^2(k/2)}, \tag{8.28}$$

so the two energies are $E_{\pm}(k) = \pm d(k)$. The lower-band projector is

$$P_-(k) = \frac{1}{2}\left(I - \frac{H(k)}{d(k)}\right). \tag{8.29}$$

Because $H(k)P_-(k) = -d(k)P_-(k)$, its embedded projected Hamiltonian is

$$H_-(k) = P_-(k)H(k)P_-(k) = -d(k)P_-(k) = \frac{1}{2}\big(H(k) - d(k)I\big). \tag{8.30}$$

For generic $t \neq 0$, the square root $d(k)$ is not a finite Fourier polynomial. One quick way to see this is to suppose that it had a largest nonzero Fourier index $m > 0$. Its square would then have a nonzero coefficient at index $2m$, whereas

$$d(k)^2 = \Delta^2 + 2t^2 + 2t^2\cos k \tag{8.31}$$

contains only indices 0 and ± 1. Taking $m = 0$ would make $d(k)$ constant, which it is not when $t \neq 0$. Therefore both the embedded matrix $H_-(k)$ and the scalar lower-band operator $-d(k)$ generally have nonzero Fourier coefficients beyond nearest neighbors.

Nothing failed: the lower eigenpair is retained exactly. What failed is a different proposition—that spectral projection preserves the parent's exact finite hopping support. For a composite retained space, a momentum-dependent unitary frame change can redistribute the matrix-valued Fourier blocks further, as Chapter 10 demonstrates. Finite-range representability must therefore be tested in the declared frame and model class; it does not follow from spectral exactness.

8.9 From exact retention to model-class expressivity

The preceding calculation separates three maps that answer different questions:

$$\text{spectral projection:} \quad H \longmapsto PHP, \tag{8.32}$$

$$\text{hopping truncation:} \quad T_{\mathbf{R}} \longmapsto \mathbf{1}_{\mathbf{R}\in\mathcal{S}} T_{\mathbf{R}}, \tag{8.33}$$

$$\text{structural restriction:} \quad H \longmapsto H_j(\boldsymbol{\theta}) \in \mathcal{M}_j. \tag{8.34}$$

The first selects an invariant state space, the second enforces a translation support, and the third enforces a candidate Hamiltonian family. They preserve different information and need not commute.

An exact spectral projection can therefore supply a controlled target for a model-class test, but it does not answer that test. The later adequacy question asks whether one candidate $H_j(\boldsymbol{\theta})$, after the declared alignment, satisfies both the spectral and represented-operator criteria. That requires the common witness or globally valid separation argument developed in Chapter 14; exact retention alone supplies neither conclusion.

8.10 Operations that must not be collapsed

Projection. Applies an idempotent map onto a declared subspace.

Spectral selection. Chooses eigenvectors according to a declared spectral rule in a setting where that rule is meaningful.

Disentanglement. Constructs a retained subspace from a larger window when target bands are not globally isolated.

Downfolding. Eliminates a complementary sector and may produce an energy-dependent, nonlocal effective operator on the model space.

Basis transformation. Changes coordinates inside an already identified space without changing the represented abstract operator.

Localization. Selects or optimizes a basis or gauge according to a locality criterion; it is not automatically a reduction in rank.

Truncation. Removes matrix elements, basis functions, hopping blocks, or other represented content according to a separate rule.

Model fitting. Selects parameters in a declared admissible class using an objective and data split.

Several operations may appear in one workflow, but each introduces different assumptions and error sources. Calling all of them "projection" prevents those errors from being evaluated separately.

8.11 A finite projector calculation

Python interlude—compressing a synthetic Hermitian operator

```
import numpy as np

# Define a synthetic ambient Hermitian operator on a four-
    coordinate space.
H = np.array(
    [
        [0.0, 0.2, 0.0, 0.0],
        [0.2, 1.0, 0.1, 0.0],
        [0.0, 0.1, 2.0, 0.3],
        [0.0, 0.0, 0.3, 3.0],
    ],
    dtype=np.complex128,
)

# Use two orthonormal columns as the retained frame.
V = np.eye(4, dtype=np.complex128)[:, :2]
P = V @ V.conjugate().T
H_retained = V.conjugate().T @ H @ V

# Report projector and compression defects explicitly.
idempotence_defect = np.linalg.norm(P @ P - P, ord="fro")
self_adjoint_defect = np.linalg.norm(P.conjugate().T - P, ord="fro"
    )
```

```
compression_defect = np.linalg.norm(
    H_retained - V.conjugate().T @ P @ H @ P @ V,
    ord="fro",
)

print(f"idempotence defect: {idempotence_defect:.3e}")
print(f"self-adjointness defect: {self_adjoint_defect:.3e}")
print(f"compression defect: {compression_defect:.3e}")
```

The example verifies exact identities for synthetic arrays. It does not justify this retained frame for a physical problem.

8.12 Questions to record before retention

A retained-space calculation should state

1. the parent state space and operator;
2. the rule that defines the retained space;
3. the retained rank and any conditioning requirement;
4. the projector or frame, including basis and gauge conventions;
5. the treatment of degeneracies or entangled bands;
6. the operator terms compressed through the same map;
7. whether the construction is an invariant spectral restriction, an eliminated-space downfolding, or another reduction;
8. the distinction between exact finite support, spatial decay, and a chosen finite-range model class;
9. the relation between the exact retained operator and any fitted model; and
10. the evidence and tolerance used to assess each approximation.

8.13 Exercises

1. Prove that $P = VV^\dagger$ is an orthogonal projector when $V^\dagger V = I$.
2. Show that V and VG define the same projector for unitary G, and derive the corresponding covariance of $V^\dagger HV$.
3. Construct a local diagonal matrix V in an ambient basis and show that its compression can be dense in another retained frame.
4. Explain why deleting small entries from H is not the same operation as projecting onto a retained subspace.
5. State why a spectral cutoff can become ambiguous near a degeneracy or for entangled bands.
6. Derive the Feshbach–Löwdin operator by solving the Q-sector equation, and show why its correction vanishes for an exact spectral projector.
7. Verify $H(k)^2 = d(k)^2 I$ for the two-band example and explain why the lower-band dispersion cannot terminate at nearest-neighbor range.

8.14 Evidence boundary

Projector identities and exact recovery of retained eigenpairs can establish mathematical or software properties of a declared construction. They do not show that the retained space preserves the structures needed by a later continuum model or that a fitted effective Hamiltonian is scientifically adequate. Chapter 11 supplies the vocabulary needed to state those stronger claims without conflation.

CHAPTER 9

From an Isolated Band to a Finite-Range Lattice Hamiltonian

Status pedagogical derivation and synthetic computational example. This chapter explains the mathematical chain exercised by the controlled one-dimensional single-band calculation. The maintained numerical successor is reported in Appendix G; this chapter itself adds no material calculation, scientific validation, or uncertainty quantification.

9.1 Learning goals

After working through this chapter, a reader should be able to

1. construct a one-dimensional Bloch-fiber matrix in plane-wave and finite-difference representations;
2. distinguish a continuum parent, a finite represented reference, and a sampled band dispersion;
3. explain what is retained when one isolated scalar band is selected;
4. derive the finite reciprocal-to-cell Fourier transform and its inverse;
5. interpret complete and finite-range hopping coefficients without confusing truncation with projection or localization;
6. derive the Parseval relation for the stated transform convention;
7. formulate direct hopping recovery as a least-squares problem and state the conditions under which it equals Fourier truncation;
8. keep training and withheld reciprocal samples in distinct roles; and
9. interpret Hermiticity, bandwidth, curvature, rank, conditioning, and imaginary residuals as bounded diagnostics.

Chapter 6 introduced direct and reciprocal geometry. Chapter 7 constructed Bloch-periodic finite representations, and Chapter 8 distinguished a retained space from coordinates and later truncations. We now join those ideas in one complete but deliberately one-dimensional reduction.

The sequence is

$$\begin{aligned} &\text{periodic parent} \longrightarrow \text{Bloch fibers} \longrightarrow \text{isolated band } E(k), \\ &E(k) \longrightarrow \text{complete hoppings } t_R \longrightarrow \text{finite-range model.} \end{aligned} \tag{9.1}$$

Every arrow changes the mathematical object. The purpose of this chapter is to make those changes visible rather than hiding them behind one plotting routine.

9.2 A one-dimensional periodic parent

Let the direct-lattice period be $a > 0$ and define

$$G = \frac{2\pi}{a}. \tag{9.2}$$

Translation by a in real space and translation by G in reciprocal space supply the fundamental periodicities. Consider the continuum operator

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V_0 \cos(Gx) \quad \text{on } L^2(\mathbb{R}). \tag{9.3}$$

The cosine is a synthetic periodic potential. It is useful because its Fourier content is transparent; it is not a material potential.

A Bloch state can be written

$$\psi_k(x) = e^{ikx} u_k(x), \qquad u_k(x + a) = u_k(x). \tag{9.4}$$

Substitution into Equation (9.3) gives the cell-periodic fiber operator

$$\widehat{H}(k) = \frac{\hbar^2}{2m}\left(-i\frac{\mathrm{d}}{\mathrm{d}x} + k\right)^2 + V_0 \cos(Gx). \tag{9.5}$$

The same physics can instead be represented by the original differential expression on one cell with Bloch boundary conditions. These conventions are unitarily related, but their matrices are not interchangeable without the corresponding transformation.

9.2.1 Reduced momentum and natural scales

Define the reduced momentum, reciprocal energy scale, and dimensionless potential strength by

$$\kappa = \frac{k}{G}, \qquad E_G = \frac{\hbar^2 G^2}{2m}, \qquad \lambda = \frac{V_0}{E_G}. \tag{9.6}$$

If $y = Gx$, division by E_G turns the fiber operator into

$$\widehat{h}(\kappa) = \left(-i\frac{\mathrm{d}}{\mathrm{d}y} + \kappa\right)^2 + \lambda \cos y. \tag{9.7}$$

Thus a statement such as $a = 2\pi$, $G = 1$, and $E_G = 1$ is a dimensionless representation convention. It does not erase the physical meanings from which the scales were constructed.

The first Brillouin zone may be represented by the half-open interval

$$-\frac{1}{2} \le \kappa < \frac{1}{2}. \tag{9.8}$$

The two endpoints $-1/2$ and $1/2$ differ by one reciprocal-lattice translation and represent the same boundary point in the reciprocal quotient. A table may include both endpoints for a symmetry check, but a complete discrete Fourier mesh should not duplicate them.

9.3 Plane-wave representation of a Bloch fiber

Because u_k is periodic, expand it in the orthonormal cell basis

$$\phi_n(x) = \frac{1}{\sqrt{a}} e^{inGx}, \qquad n \in \mathbb{Z}. \tag{9.9}$$

Acting on one basis function gives

$$\left(-i\frac{\mathrm{d}}{\mathrm{d}x} + k\right)\phi_n = (k + nG)\phi_n. \tag{9.10}$$

The kinetic term is therefore diagonal. The potential identity

$$V_0 \cos(Gx) = \frac{V_0}{2} e^{iGx} + \frac{V_0}{2} e^{-iGx} \tag{9.11}$$

shows that multiplication by the cosine changes the reciprocal index by exactly one. Consequently,

$$H^{\mathrm{PW}}_{nm}(k) = \frac{\hbar^2 (k + nG)^2}{2m}\delta_{nm} + \frac{V_0}{2}\left(\delta_{n,m+1} + \delta_{n,m-1}\right). \tag{9.12}$$

In dimensionless form this becomes

$$h^{\mathrm{PW}}_{nm}(\kappa) = (n+\kappa)^2\delta_{nm} + \frac{\lambda}{2}\left(\delta_{n,m+1} + \delta_{n,m-1}\right). \tag{9.13}$$

A cutoff P retains $n = -P, \ldots, P$ and gives a $(2P+1) \times (2P+1)$ Hermitian matrix. Increasing P changes the represented state space. A cutoff sequence can therefore supply finite-representation refinement evidence for selected low eigenvalues; it does not make any one finite matrix identical to the continuum operator.

The nearest-neighbor pattern in reciprocal index deserves careful language. It comes from the single cosine harmonic. It is not a nearest-neighbor real-space tight-binding approximation. Reciprocal-index coupling and real-space hopping are different representations introduced at different stages.

9.4 A finite-difference representation of the same parent

A second representation uses the endpoint-excluded grid

$$x_j = j\Delta x, \qquad \Delta x = \frac{a}{N_x}, \qquad j = 0, \ldots, N_x - 1. \tag{9.14}$$

Represent the original differential expression on one cell and impose

$$\psi(x+a) = e^{ika}\psi(x). \tag{9.15}$$

For the centered second derivative, define

$$c_h = \frac{\hbar^2}{2m\Delta x^2}. \tag{9.16}$$

The kinetic matrix has diagonal entries $2c_h$, adjacent entries $-c_h$, and seam entries

$$T_{0,N_x-1}(k) = -c_h e^{-ika}, \qquad T_{N_x-1,0}(k) = -c_h e^{ika}. \tag{9.17}$$

The two seam entries are complex conjugates. Together with the real diagonal potential samples $V_0\cos(Gx_j)$, this gives a Hermitian finite matrix.

The plane-wave and finite-difference matrices have different bases and usually different dimensions. Their eigenvalues can be compared as approximations to the same declared spectral quantities. The matrices themselves cannot be subtracted until an explicit common-space map has been supplied. Spectral refinement and represented-operator refinement are therefore different questions.

9.4.1 A finite represented reference is not the continuum limit

Suppose a high plane-wave cutoff P_{ref} is declared as the reference for a finite study. For a candidate cutoff P, one may report

$$\epsilon_P = \max_{\kappa \in \mathcal{K}_{\mathrm{parent}}} \max_{1 \le n \le n_{\mathrm{cmp}}} \left| E^{\mathrm{PW}}_{n,P}(\kappa) - E^{\mathrm{PW}}_{n,P_{\mathrm{ref}}}(\kappa) \right| . \tag{9.18}$$

This is an error relative to the stated finite reference. It is not automatically $|E_{n,P} - E_{n,\infty}|$. Once discrepancies reach floating-point and eigensolver scales, they need not decrease monotonically. The accurate report is the observed finite discrepancy over the tested sequence, unless a separate argument supplies a continuum bound.

A finite-difference sequence can be compared with the same represented reference, but its grid and stencil error remains a separate channel from the plane-wave cutoff error. Agreement of the two sequences is useful precisely because their discretization mechanisms differ.

9.5 What it means to retain an isolated band

Let $E_n(\kappa)$ be ordered fiber eigenvalues. A band is isolated over a declared reciprocal domain $\mathcal{K}$ when it remains separated from the excluded spectrum. For an interior band, a sampled isolation diagnostic is

$$\gamma_n^{\mathrm{sampled}} = \min_{\kappa \in \mathcal{K}} \min \left\{ E_n(\kappa) - E_{n-1}(\kappa), E_{n+1}(\kappa) - E_n(\kappa) \right\} . \tag{9.19}$$

For the lowest band, only the upper gap is present. A positive value on a finite mesh establishes sampled separation, not a theorem that the gap is positive at every unsampled point.

At each momentum, four objects must remain distinct:

1. the scalar eigenvalue $E_n(\kappa)$;
2. a normalized eigenvector, whose phase is not unique;
3. the one-dimensional eigenspace or its projector; and
4. a choice of phases over momentum, called a band gauge.

For a scalar isolated band, the represented Hamiltonian within the eigenspace is just the 1×1 value $E_n(\kappa)$. Fourier transformation of these scalar values therefore does not require selecting eigenvector phases. It constructs a scalar lattice representation of the sampled dispersion. It does not, by itself, construct Wannier functions, calculate a localization gauge, or retain a real-space probability density. Those additional objects require eigenvectors and overlap information [57, 72].

9.6 The complete centered reciprocal mesh

Let N be even. A complete centered half-open mesh is

$$\kappa_j = -\frac{1}{2} + \frac{j}{N}, \qquad j = 0, \ldots, N-1. \tag{9.20}$$

It contains $-1/2$ and excludes the equivalent endpoint $1/2$. One convenient set of cell representatives is

$$\mathcal{R}_N = \left\{ -\frac{N}{2}, -\frac{N}{2}+1, \ldots, \frac{N}{2}-1 \right\}. \tag{9.21}$$

The labels in $\mathcal{R}_N$ are representatives of residue classes modulo N. Thus R and $R + Nq$ produce the same phase at every mesh point. In particular, the even-mesh endpoint representative $-N/2$ is its own opposite modulo N, because $N/2$ belongs to the same residue class even though it is not separately stored.

For $R, S \in \mathcal{R}_N$, finite Fourier orthogonality gives

$$\sum_{j=0}^{N-1} e^{2\pi i (R-S)\kappa_j} = N\, \delta^{(N)}_{R,S}, \tag{9.22}$$

where $\delta^{(N)}_{R,S}$ equals one when R and S are the same residue class modulo N and zero otherwise. This identity is the algebraic reason that the complete transform is invertible.

9.7 From sampled energies to complete hoppings

Let $E_j = E(\kappa_j)$ be one real scalar band sampled on the complete mesh. Define

$$t_R = \frac{1}{N} \sum_{j=0}^{N-1} e^{-2\pi i R \kappa_j} E_j, \qquad R \in \mathcal{R}_N. \tag{9.23}$$

Multiplying by the inverse phases and using Equation (9.22) gives

$$E_j = \sum_{R \in \mathcal{R}_N} e^{2\pi i R \kappa_j} t_R. \tag{9.24}$$

The normalization and signs in these two equations form one convention. Moving the factor $1/N$ or reversing both signs is possible, but every producer, interpolator, and verifier must then change consistently [124].

The coefficient t_0 is the cell-diagonal or onsite scalar term. A coefficient t_R with $R \neq 0$ couples cells separated by R direct-lattice periods in the represented lattice model. The finite mesh produces exactly N independent coefficients. Calling this the complete hopping model means complete relative to that finite transform, not infinite real-space exactness.

9.7.1 Hermiticity and modular opposites

A scalar lattice Hamiltonian is Hermitian when

$$t_{\overline{-R}} = t_R^*, \tag{9.25}$$

where $\overline{-R}$ is the selected representative of the opposite residue class modulo N. If the sampled dispersion is real, the complete discrete transform satisfies this relation up to numerical error. If it is also even in κ, the hoppings are real and satisfy $t_{\overline{-R}} = t_R$.

The modular qualifier matters at the representative boundary. A verifier that searches only for the literal integer $-R$ can incorrectly report the Nyquist representative as unpaired. The comparison belongs to the finite periodic index set, not to an unbounded integer list.

9.7.2 A four-point transform by hand

Consider the already finite scalar model

$$E(\kappa) = \varepsilon_0 + 2t\cos(2\pi\kappa). \tag{9.26}$$

For $N = 4$, the mesh is $\{-1/2, -1/4, 0, 1/4\}$ and the sampled values are

$$\varepsilon_0 - 2t, \quad \varepsilon_0, \quad \varepsilon_0 + 2t, \quad \varepsilon_0. \tag{9.27}$$

Substitution into Equation (9.23) gives

$$t_0 = \varepsilon_0, \qquad t_{-1} = t_1 = t, \qquad t_{-2} = 0. \tag{9.28}$$

The inverse transform recovers all four samples. This example makes the meaning of "range one" literal: only the onsite and adjacent-cell coefficients are nonzero. A general sampled band need not terminate at any finite range.

9.8 Finite-range truncation

For an integer $R_{\max} < N/2$, retain the symmetric representative set

$$\mathcal{R}_{R_{\max}} = \{R \in \mathcal{R}_N : |R| \le R_{\max}\}. \tag{9.29}$$

The Fourier-mediated finite-range model is

$$E^{\mathrm{med}}_{R_{\max}}(\kappa) = \sum_{R \in \mathcal{R}_{R_{\max}}} e^{2\pi i R\kappa} t_R. \tag{9.30}$$

This operation removes hopping coefficients. It does not reduce the dimension of the already selected scalar band, select a new eigenspace, or choose a Wannier gauge. It is a model-class approximation performed after band retention.

The omitted-coefficient norm

$$\eta_{R_{\max}} = \left(\sum_{R \notin \mathcal{R}_{R_{\max}}} |t_R|^2 \right)^{1/2} \tag{9.31}$$

is a representation diagnostic with energy units. It is not a probability, an energy fraction, or a material uncertainty.

9.9 Parseval's identity for the truncation residual

On the training mesh define

$$r_j = E^{\mathrm{med}}_{R_{\max}}(\kappa_j) - E_j = - \sum_{R \notin \mathcal{R}_{R_{\max}}} e^{2\pi i R \kappa_j} t_R. \tag{9.32}$$

Then

$$\sum_{j=0}^{N-1} |r_j|^2 = \sum_{j} \sum_{R \notin \mathcal{R}_{R_{\max}}} \sum_{S \notin \mathcal{R}_{R_{\max}}} e^{2\pi i (R-S)\kappa_j} t_R t_S^* \tag{9.33}$$

$$= N \sum_{R \notin \mathcal{R}_{R_{\max}}} |t_R|^2. \tag{9.34}$$

The second line follows from finite Fourier orthogonality. This is an exact identity for the stated complete mesh, representatives, and normalization.

Parseval equates a squared residual summed over the training mesh with a squared omitted-coefficient norm. It does not state that the maximum error is the same quantity. It also does not determine the error at a new reciprocal coordinate, where finite Fourier interpolation rather than discrete orthogonality controls the comparison.

9.10 The direct least-squares route

Instead of first calculating all N coefficients, one can fit a chosen finite-range class directly. Let the columns be indexed by $R \in \mathcal{R}_{R_{\max}}$ and define

$$A_{jR} = e^{2\pi i R \kappa_j}. \tag{9.35}$$

For nonnegative declared weights w_j, the direct route solves

$$\min_{\boldsymbol{\tau}} \sum_{j=0}^{N-1} w_j \left| E_j - \sum_R A_{jR}\tau_R \right|^2 . \tag{9.36}$$

The unknown vector $\boldsymbol{\tau}$ contains the directly fitted hoppings. A stable numerical implementation normally uses a QR or singular-value method rather than explicitly forming inverse normal equations [40].

With the complete uniform mesh, equal weights, and distinct representatives, Fourier orthogonality gives

$$A^\dagger A = NI. \tag{9.37}$$

The least-squares solution is therefore

$$\boldsymbol{\tau} = \frac{1}{N} A^\dagger \mathbf{E}, \tag{9.38}$$

which selects the same coefficients as the complete transform and then discards the unrepresented columns. Under these hypotheses, direct fitting and Fourier-mediated truncation are the same orthogonal projection.

This equivalence is conditional, not a general property of two workflows. It can be lost by changing

- the reciprocal coordinates or Fourier signs;
- the weights;
- the fitting domain;
- the retained representatives;
- Hermiticity or parameter-sharing constraints;
- the fitted observable; or
- , for a multiband operator, the aligned frame.

When one of these changes, a nonzero route defect may correctly report two different approximation problems rather than a software failure.

9.10.1 Rank, identifiability, and conditioning

The design rank is the dimension of the coefficient combinations visible on the training coordinates. If

$$\operatorname{rank}(A) < |\mathcal{R}_{R_{\max}}|, \tag{9.39}$$

then multiple coefficient vectors produce the same fitted samples and the individual hoppings are not identified by that design. Full column rank gives a unique least-squares minimizer, but uniqueness does not guarantee numerical stability.

If $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest nonzero singular values, the two-norm condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}. \tag{9.40}$$

A large condition number warns that small perturbations in samples may produce large coefficient changes. On the matched complete uniform design above, the columns are orthogonal and the condition number is one. That favorable value is a property of this design, not of every hopping fit.

9.11 Training and withheld reciprocal coordinates

The complete training mesh has a special algebraic role: it defines the finite Fourier transform. Reconstructing those same N values with all N coefficients verifies the transform pair, but it is interpolation of the data that defined the coefficients. It is not evidence about unsampled coordinates.

A separate uniform withheld mesh can be staggered so that it shares no training coordinates. For training extent N, withheld extent M, and

$$\eta = \frac{1}{N+1}, \tag{9.41}$$

define

$$\kappa_i^{\mathrm{W}} = -\frac{1}{2} + \frac{i+\eta}{M}, \qquad i = 0, \ldots, M-1. \tag{9.42}$$

To see why the meshes are disjoint, suppose a training and withheld coordinate were equal. Cancelling $-1/2$ would give

$$\frac{j}{N} = \frac{i + 1/(N+1)}{M}. \tag{9.43}$$

Multiplication by $MN(N+1)$ would require

$$jM(N+1) = iN(N+1) + N. \tag{9.44}$$

The left side is divisible by $N+1$, whereas the right side has remainder N and is not. The assumed equality is impossible.

The withheld values are evaluated only after the coefficients, tested ranges, and tolerances have been frozen. They may reveal interpolation behavior between training points. They must not update the fit, choose a range, alter a threshold, or become a new training

set while retaining the label “withheld.” In this synthetic setting, withheld evaluation is still numerical evidence relative to the selected parent representation; it is not scientific validation.

9.12 Band-shape and structural diagnostics

A diagnostic has meaning only after its domain and coordinate are declared. On a finite comparison set $\mathcal{K}_{\text{cmp}}$, the sampled bandwidth is

$$W_{\text{sampled}} = \max_{\kappa \in \mathcal{K}_{\text{cmp}}} E(\kappa) - \min_{\kappa \in \mathcal{K}_{\text{cmp}}} E(\kappa). \tag{9.45}$$

It need not equal the exact continuous bandwidth unless the extrema are known to lie in the sampled set.

For the trigonometric polynomial

$$E_R(\kappa) = \sum_R e^{2\pi i R\kappa} t_R, \tag{9.46}$$

differentiation is analytical:

$$\frac{\mathrm{d}^2 E_R}{\mathrm{d}\kappa^2}(0) = -(2\pi)^2 \sum_R R^2 t_R. \tag{9.47}$$

Because $\kappa = k/G$,

$$\frac{\mathrm{d}^2 E_R}{\mathrm{d}k^2} = \frac{1}{G^2} \frac{\mathrm{d}^2 E_R}{\mathrm{d}\kappa^2}. \tag{9.48}$$

Thus reduced-coordinate curvature and physical-wave-vector curvature have different units. Neither becomes a material effective mass until the extremum, physical scales, band identity, and effective-mass convention are supplied.

Numerical interpolation may also produce small imaginary components even when the target band is real. A scalar diagnostic can report

$$\epsilon_{\text{imag}} = \max \left\{ \max_{\kappa \in \mathcal{K}_{\text{cmp}}} |\operatorname{Im} E_R(\kappa)|, \left| \operatorname{Im} \frac{\mathrm{d}^2 E_R}{\mathrm{d}\kappa^2}(0) \right| \right\}. \tag{9.49}$$

This checks consistency with a real scalar observable. It does not replace the coefficient-level Hermiticity check in Equation (9.25).

Maximum error and root-mean-square error must also be named separately. For residuals r_i,

$$\epsilon_{\max} = \max_i |r_i|, \qquad \epsilon_{\text{RMS}} = \left(\frac{1}{M} \sum_i |r_i|^2 \right)^{1/2}. \tag{9.50}$$

They answer different questions and generally have different values. A report must not transfer a number from one metric to the other because both are called “band error.”

9.13 A transparent computational example

The following synthetic example implements the transform, truncation, direct fit, and withheld comparison directly. It is intentionally smaller than a research calculation so that every array has an evident meaning.

Python interlude—transforming a sampled scalar band into hoppings

```
import numpy as np

# Use an even complete mesh on the half-open reduced Brillouin zone.

N = 8
training_kappa = -0.5 + np.arange(N, dtype=np.float64) / N
representatives = np.arange(-N // 2, N // 2, dtype=np.int64)

# Define a synthetic scalar band with range-two Fourier content.
epsilon_0 = 0.7
t_1 = -0.12
t_2 = 0.015
training_energy = (
    epsilon_0
    + 2.0 * t_1 * np.cos(2.0 * np.pi * training_kappa)
    + 2.0 * t_2 * np.cos(4.0 * np.pi * training_kappa)
)

# Apply the declared forward transform one representative at a time.

hoppings = np.array(
    [
        np.mean(
            training_energy
            * np.exp(-2j * np.pi * R * training_kappa)
        )
        for R in representatives
    ],
    dtype=np.complex128,
)

# The complete inverse transform must reproduce every training
    value.
```

```
complete_design = np.exp(
    2j * np.pi * np.outer(training_kappa, representatives)
)
reconstructed = complete_design @ hoppings
np.testing.assert_allclose(reconstructed.real, training_energy,
    atol=1e-14)
np.testing.assert_allclose(reconstructed.imag, 0.0, atol=1e-14)

# Retain the symmetric range-one model and evaluate its training
    residual.
range_mask = np.abs(representatives) <= 1
range_representatives = representatives[range_mask]
mediated_hoppings = hoppings[range_mask]
range_design = np.exp(
    2j * np.pi * np.outer(training_kappa, range_representatives)
)
mediated_training = range_design @ mediated_hoppings

# Equal-weight direct least squares on the same complete mesh is
    equivalent.
direct_hoppings, _, rank, singular_values = np.linalg.lstsq(
    range_design,
    training_energy.astype(np.complex128),
    rcond=None,
)
np.testing.assert_allclose(direct_hoppings, mediated_hoppings, atol
    =1e-14)
assert rank == range_representatives.size
condition_number = singular_values[0] / singular_values[-1]

# Use a frozen staggered mesh only for evaluation, never for the
    fit.
M = 17
offset = 1.0 / (N + 1)
withheld_kappa = -0.5 + (np.arange(M) + offset) / M
withheld_target = (
    epsilon_0
    + 2.0 * t_1 * np.cos(2.0 * np.pi * withheld_kappa)
    + 2.0 * t_2 * np.cos(4.0 * np.pi * withheld_kappa)
)
withheld_design = np.exp(
```

```
    2j * np.pi * np.outer(withheld_kappa, range_representatives)
)
withheld_candidate = withheld_design @ mediated_hoppings
withheld_maximum_error = np.max(
    np.abs(withheld_candidate.real - withheld_target)
)

print("direct design rank:", rank)
print("direct design condition number:", condition_number)
print("withheld maximum error:", withheld_maximum_error)
```

The example's range-one error is intentional because the target contains a range-two term. Exact complete-mesh reconstruction checks the transform. The nonzero withheld error checks the deliberately truncated candidate. Neither observation is a material result.

Notebook to do Create `examples/tutorials/research-monograph/foundations/isolated_band_fourier_reduction.ipynb` only after this lecture contract is stable. Include explicit imports, explanatory inline comments, a hand-checkable transform, modular Hermiticity, Parseval, equal-weight route equivalence, a deliberately changed-weight route, and disjoint withheld evaluation. Keep all examples synthetic and do not create or execute the notebook as part of the present manuscript change.

9.14 A staged isolated-band laboratory

A useful laboratory should preserve the sequence of objects rather than jumping directly from a potential to a final error curve.

1. Declare a, G, E_G, λ, the reciprocal interval, and the energy zero.
2. Construct plane-wave fiber matrices and verify their Hermiticity and reciprocal-index coupling pattern.
3. Construct the independent finite-difference fibers, including the conjugate seam phases.
4. Compare a frozen set of low eigenvalues against a declared finite reference while retaining the two discretization channels separately.
5. Check the sampled isolation gap before assigning a scalar band role.

6. Freeze the complete training mesh, representative convention, tested ranges, withheld rule, metrics, and numerical tolerances.
7. Calculate the complete transform and verify inverse reconstruction and modular hopping Hermiticity.
8. For each range, retain the omitted norm, training and withheld errors, Parseval residual, direct-fit rank and conditioning, route defects, and band-shape diagnostics.
9. Interpret numerical-consistency tolerances separately from any later model-selection or scientific-acceptance threshold.

If one stage fails, the workflow should stop or retain the failure explicitly. For example, a poorly isolated band is not repaired by making the hopping range larger, and a rank-deficient direct design is not repaired by reporting only its training residual.

9.15 Common mistakes

1. Including both reciprocal-zone endpoints in a complete Fourier mesh.
2. Calling the reciprocal-index tridiagonal plane-wave matrix a real-space nearest-neighbor model.
3. Treating a high finite cutoff as the exact continuum spectrum.
4. Assuming a positive sampled gap proves isolation between sample points.
5. Claiming that scalar energy samples construct a Wannier gauge or localized orbital.
6. Ignoring modular opposites at the even-mesh Nyquist representative.
7. Calling hopping truncation a projection onto fewer quantum states.
8. Applying Parseval's training-mesh identity directly to a withheld maximum error.
9. Assuming direct fitting and mediated truncation agree after changing weights, coordinates, constraints, or model classes.
10. Using withheld points to choose the hopping range and then continuing to call them withheld.
11. Reporting reduced-coordinate curvature as a physical effective mass.
12. Treating a binary64 numerical tolerance as scientific uncertainty.

9.16 Exercises

1. Starting from Equation (9.5), derive every term in Equation (9.12).
2. Construct the $P = 1$ plane-wave matrix explicitly for arbitrary κ and λ, and verify its Hermiticity.
3. Derive the two finite-difference seam entries in Equation (9.17) from the Bloch boundary condition.
4. Prove Equation (9.22) by summing a finite geometric series.
5. Complete every sum in the four-point example and explain why the representative -2 is its own modular opposite.
6. Derive Equation (9.34) without invoking an infinite Fourier-series theorem.
7. Use Equation (9.37) to derive the direct-fit solution in Equation (9.38).
8. Change one weight in the Python example and measure the resulting direct-versus-mediated coefficient defect.
9. Prove the withheld-mesh disjointness argument for arbitrary positive integers N and M.
10. Convert a reduced curvature to physical-k curvature and list the additional assumptions required to call it an effective mass.

9.17 Evidence boundary and next use

The derivations establish finite-transform identities under their stated conventions. The Python example checks those identities for synthetic data. Neither establishes that a particular continuum calculation is converged, that a finite hopping range is adequate for an application, or that a physical material is described correctly.

Appendix G applies this chain to retained controlled calculations. Its numerical outcomes, historical localization channels, and provenance remain owned by their calculation packages. The present chapter supplies the explanatory mathematics and does not inherit their evidence status. Chapter 10 next replaces the scalar retained space by a composite frame and derives the missing Bloch–Wannier, transport, alignment, and block-locality relations. The final Part I chapter then explains how to classify claims, references, observations, and dispositions.

CHAPTER 10

Composite Bloch Frames, Gauge Alignment, and Block Locality

Status pedagogical derivation and synthetic computational example. This chapter explains the mathematical chain exercised by a controlled one-dimensional rank-two alignment calculation, but it reports no M2 numerical outcome, material calculation, localization result, topological classification, scientific validation, or uncertainty quantification.

10.1 Learning goals

After working through this chapter, a reader should be able to

1. distinguish a matrix-valued periodic parent from a continuum parent and from a reduced represented operator;
2. define a composite retained space by its frame and projector while allowing internal band crossings;
3. separate an external spectral gap from neighboring-frame overlap conditioning;
4. derive the covariance of a projected Hamiltonian under a momentum-dependent unitary gauge;
5. connect a Bloch frame, a Wannier basis, and matrix-valued hopping blocks without confusing that connection with localization;
6. construct neighborwise polar transport and interpret reciprocal sewing and closure holonomy;
7. derive pointwise and globally constrained unitary Procrustes alignment;
8. explain why complete spectra can be gauge invariant while finite-range locality is gauge dependent; and

9. interpret the Frobenius, singular-value, gap, Hermiticity, and spectral diagnostics used in a bounded multiband calculation.

Chapter 8 introduced frames, projectors, and exact retained operators. Chapter 9 then considered a rank-one retained Hamiltonian, for which the represented operator at each reciprocal coordinate is the scalar energy $E_n(k)$. A composite retained space has rank greater than one. Its Hamiltonian is a matrix, and that matrix cannot be discussed independently of the frame in which it is written.

The new sequence is

$$\begin{aligned} \text{periodic parent} &\longrightarrow \text{composite space} \longrightarrow \text{frame path}, \\ \text{frame path} &\longrightarrow \text{transport and alignment} \longrightarrow H_F(\kappa), \\ H_F(\kappa) &\longrightarrow \text{block hoppings} \longrightarrow \text{finite range}. \end{aligned} \tag{10.1}$$

Each arrow carries information that a table of eigenvalues alone does not contain.

10.2 A finite matrix-valued periodic parent

Let $G > 0$ be one reciprocal period and let $\kappa = k/G \in [-1/2, 1/2)$. Consider the finite Bloch matrix

$$H_{\mathrm{p}}(\kappa) = \sum_{R \in \mathcal{R}_{\mathrm{p}}} e^{2\pi i R\kappa} T_R, \tag{10.2}$$

where each $T_R \in \mathbb{C}^{D\times D}$ acts on a declared D-dimensional orbital coordinate space. The parent is Hermitian at every κ when

$$T_{-R} = T_R^\dagger \tag{10.3}$$

for every represented displacement, including $T_0 = T_0^\dagger$. Indeed,

$$H_{\mathrm{p}}(\kappa)^\dagger = \sum_R e^{-2\pi i R\kappa} T_R^\dagger = \sum_R e^{-2\pi i R\kappa} T_{-R} = H_{\mathrm{p}}(\kappa). \tag{10.4}$$

This parent differs from the continuum cosine Hamiltonian used in Chapter 9. Equation (10.2) is already an exact finite represented model. Evaluating its finite sum does not introduce a plane-wave cutoff or finite-difference stencil error. It may be a useful controlled parent, but it is not thereby a continuum Hamiltonian or a material model.

The symbols R and T_R already resemble lattice hoppings. They describe the *ambient parent* coordinate space. The reduced blocks constructed later act in a different, rank-M retained space. Equal notation or equal range does not identify those two spaces.

10.3 Composite retention and the external gap

At each reciprocal coordinate, diagonalize the parent:

$$H_{\mathrm{p}}(\kappa)u_n(\kappa) = E_n(\kappa)u_n(\kappa), \qquad E_1(\kappa) \le \cdots \le E_D(\kappa). \tag{10.5}$$

Suppose the lowest $M < D$ states are retained. Collect an orthonormal basis for their span in the frame matrix

$$F(\kappa) = \begin{bmatrix} u_1(\kappa) & \cdots & u_M(\kappa) \end{bmatrix}, \qquad F(\kappa)^\dagger F(\kappa) = I_M. \tag{10.6}$$

The corresponding ambient-space projector is

$$P(\kappa) = F(\kappa)F(\kappa)^\dagger. \tag{10.7}$$

For this lowest-M selection, the direct external gap is

$$\gamma_{\mathrm{ext}}(\kappa) = E_{M+1}(\kappa) - E_M(\kappa). \tag{10.8}$$

A sampled calculation may retain

$$\gamma_{\mathrm{ext}}^{\mathrm{sampled}} = \min_{\kappa \in \mathcal{K}_{\mathrm{declared}}} \gamma_{\mathrm{ext}}(\kappa). \tag{10.9}$$

A positive sampled value supports separation on the declared coordinates. It does not prove a positive gap between those coordinates.

The retained bands may cross one another internally. Such an internal degeneracy can make individual eigenvectors or energy-ordered labels unstable without destroying the M-dimensional projector. This is a principal reason to retain a composite space: the projector can remain smooth and meaningful even when the individual columns returned by an eigensolver are not.

An energy gap and a neighboring-frame overlap diagnose different things. The gap separates retained and excluded eigenvalues at one coordinate. An overlap singular value, introduced below, measures how well two sampled retained spaces see one another. Neither quantity can replace the other.

10.4 Frames, gauges, and projected Hamiltonians

For any $U(\kappa) \in U(M)$, define another frame

$$F'(\kappa) = F(\kappa)U(\kappa). \tag{10.10}$$

Its projector is unchanged:

$$F'F'^{\dagger} = FUU^{\dagger}F^{\dagger} = FF^{\dagger} = P. \tag{10.11}$$

The multiplication order matters: the frame is $D \times M$, whereas U is $M \times M$.

The parent compressed into the frame is

$$H_F(\kappa) = F(\kappa)^{\dagger} H_{\mathrm{p}}(\kappa) F(\kappa) \in \mathbb{C}^{M \times M}. \tag{10.12}$$

Under Equation (10.10),

$$H_{F'}(\kappa) = U(\kappa)^{\dagger} H_F(\kappa) U(\kappa). \tag{10.13}$$

Thus the two matrices generally differ entry by entry even though they represent the same retained abstract operator. Their eigenvalues agree. Subtracting them before alignment measures a coordinate change, not a physical perturbation.

If the columns of F are the individual eigenvectors, then H_F is diagonal. A smooth composite gauge may mix those eigenvectors, in which case H_F need not be diagonal. Diagonality and smoothness are different coordinate properties.

10.5 The Bloch–Wannier bridge

Let $\{|\psi_{mk}\rangle\}_{m=1}^{M}$ be a retained Bloch frame on a uniform N_k-point mesh. For a chosen unitary gauge $U(k)$, define

$$|w_{nR}\rangle = \frac{1}{\sqrt{N_k}} \sum_{k} e^{-ikRa} \sum_{m=1}^{M} |\psi_{mk}\rangle U_{mn}(k), \tag{10.14}$$

where R is an integer cell displacement. This is the finite Bloch–Wannier transformation [129, 73, 72]. It constructs a basis only after a frame and its boundary convention have been chosen.

Let $H_U(k) = U(k)^{\dagger} H_F(k) U(k)$. Orthogonality between different Bloch coordinates gives

$$\langle w_{m0}|\widehat{H}|w_{nR}\rangle = \frac{1}{N_k} \sum_{k} e^{-ikRa} [H_U(k)]_{mn}. \tag{10.15}$$

The inverse finite transform is

$$H_U(k) = \sum_{R} e^{ikRa} H_U(R), \qquad [H_U(R)]_{mn} = \langle w_{m0}|\widehat{H}|w_{nR}\rangle. \tag{10.16}$$

Equations (10.15) and (10.16) are the missing bridge between a Bloch-frame Hamiltonian and block hoppings.

For one isolated eigenband, $M = 1$ and $H_F(k) = E_n(k)$. The phase $U(k)$ cancels from $U(k)^* E_n(k) U(k)$, so the complete scalar hopping coefficients depend only on the dispersion. The Wannier function itself still depends on the phase. For $M > 1$, a momentum-dependent $U(k)$ generally changes the matrix $H_U(k)$ and therefore its individual real-space blocks.

The transform of Hamiltonian matrices is not a localization calculation. It does not by itself calculate Wannier functions, centers, spreads, probability densities, or a minimizer of a localization functional. Those quantities require eigenvectors, overlaps, and an explicit localization construction.

10.6 Comparing retained frames

Let F_{r} and F_{c} be reference and candidate frames in the same ambient coordinates. Three comparisons must remain distinct.

The raw frame defect

$$d_{\text{frame}} = \|F_{\mathrm{c}} - F_{\mathrm{r}}\|_{\mathrm{F}} \tag{10.17}$$

is gauge dependent. The projector defect

$$d_{\text{proj}} = \|F_{\mathrm{c}} F_{\mathrm{c}}^{\dagger} - F_{\mathrm{r}} F_{\mathrm{r}}^{\dagger}\|_{\mathrm{F}} \tag{10.18}$$

is invariant under independent unitary rotations of either frame. It vanishes exactly when the two orthogonal projectors agree.

Finally, the singular values of

$$S = F_{\mathrm{c}}^{\dagger} F_{\mathrm{r}} \tag{10.19}$$

are the cosines of the principal angles between the retained spaces. Values near one indicate compatible subspaces; values near zero expose directions that cannot be stably identified by a unitary rotation. Equal rank alone does not guarantee favorable singular values.

The Frobenius norm used here is

$$\|A\|_{\mathrm{F}} = \left(\sum_{i,j} |A_{ij}|^2 \right)^{1/2} = \sqrt{\mathrm{tr}(A^{\dagger} A)}. \tag{10.20}$$

A frame or projector defect is dimensionless. A Hamiltonian-matrix defect has energy units. The same norm formula does not erase that dimensional difference.

10.7 Neighborwise polar transport

Raw eigensolver frames at neighboring points can contain arbitrary phases and unitary mixtures. Suppose a transported frame F_j has already been chosen and G_{j+1} is the next

raw frame. Form

$$S_j = F_j^\dagger G_{j+1}. \tag{10.21}$$

Let its singular-value decomposition be

$$S_j = L_j \Sigma_j R_j^\dagger. \tag{10.22}$$

Then choose

$$Q_j = R_j L_j^\dagger, \qquad F_{j+1} = G_{j+1} Q_j. \tag{10.23}$$

The new overlap is

$$F_j^\dagger F_{j+1} = L_j \Sigma_j L_j^\dagger, \tag{10.24}$$

which is Hermitian positive semidefinite. This is the unitary polar or orthogonal-Procrustes step that makes the next frame as close as possible to the previous one in the declared Frobenius metric [104, 46].

The minimum singular value over neighbor and boundary overlaps is a conditioning diagnostic. A frozen threshold can make insufficient overlap a fail-closed stop condition. If an overlap is rank deficient, the positive-overlap condition does not determine a unique unitary action on its null directions: a complete SVD still returns matrices, but their null-space columns can depend on the backend. Neighborwise transport therefore requires the same rank and nonuniqueness policy as the Procrustes problem derived below. The threshold is not a universal physical constant, and passing it does not prove continuous smoothness between mesh points.

10.7.1 Sewing and closure holonomy

The first and last stored mesh points are not duplicate endpoints. To close the reciprocal loop, the first frame must be mapped across the zone boundary by a declared ambient sewing map S_G. After neighborwise transport, form

$$C = F_{N_k-1}^\dagger S_G F_0. \tag{10.25}$$

The unitary polar factor of C is the discrete closure holonomy. If its eigenvalues are $e^{i\phi_1}, \ldots, e^{i\phi_M}$, the phases are defined only modulo 2π and their ordering can change at degeneracies or branch cuts. They must therefore be treated as a phase set with an explicit branch convention [72].

A periodic orbital-coordinate parent may use $S_G = I$ because its ambient basis is exactly periodic. Other representations can require a nontrivial sewing map. For example, reciprocal-index plane-wave coordinates relabel across $k \mapsto k + G$. Identity sewing must never be assumed merely because the Hamiltonian spectrum is periodic.

One can distribute a matrix root of the closure holonomy along the path to produce a closed smooth representative. This is a gauge construction. Its closure eigenphases are

Wilson-loop-like data, but a single finite loop does not by itself establish a polarization, a topological invariant, or an infinite-mesh localization theorem.

10.8 A controlled momentum-dependent gauge attack

For a rank-two retained space, a transparent periodic rotation is

$$A(\kappa) = e^{-i\theta(\kappa)\sigma_y} = \begin{pmatrix} \cos\theta(\kappa) & -\sin\theta(\kappa) \\ \sin\theta(\kappa) & \cos\theta(\kappa) \end{pmatrix}, \tag{10.26}$$

with

$$\theta(\kappa) = \theta_0 + \sum_{m=1}^{M_a} a_m \sin(2\pi m\kappa). \tag{10.27}$$

Integer harmonics make the attack periodic. Define

$$F_{\mathrm{a}}(\kappa) = F_{\mathrm{r}}(\kappa)A(\kappa). \tag{10.28}$$

Then

$$P_{\mathrm{a}} = P_{\mathrm{r}}, \qquad H_{\mathrm{a}} = A^{\dagger}H_{\mathrm{r}}A. \tag{10.29}$$

The frame and matrix entries can change substantially while the projector and pointwise spectrum remain unchanged.

A finite Fourier series for θ does not imply that the entries of $A(\kappa)$ have the same finite Fourier range: the nonlinear sine and cosine of $\theta(\kappa)$ can generate longer Fourier tails. Consequently a smooth, known gauge transformation can change the apparent range of the projected Hamiltonian without changing its complete spectral content.

This attack is a controlled positive test. Recovering it does not show that an unknown alignment between independent physical calculations has been found.

10.9 Pointwise and globally constrained alignment

10.9.1 Pointwise unitary Procrustes alignment

At each reciprocal coordinate, solve

$$\min_{U(\kappa)\in U(M)} \|F_{\mathrm{c}}(\kappa)U(\kappa) - F_{\mathrm{r}}(\kappa)\|_{\mathrm{F}}. \tag{10.30}$$

The orientation is part of the method: the returned matrix right-multiplies the candidate frame. If $S = F_{\mathrm{c}}^{\dagger} F_{\mathrm{r}}$, the squared objective is

$$J(U) = \|F_{\mathrm{c}} U - F_{\mathrm{r}}\|_{\mathrm{F}}^{2} \tag{10.31}$$

$$= 2M - 2\,\mathrm{Re}\,\mathrm{tr}(U^{\dagger} S), \tag{10.32}$$

because both frames are isometries. Let

$$S = L \Sigma V^{\dagger}, \qquad \Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_M) \tag{10.33}$$

be a singular-value decomposition. Maximizing the trace term gives

$$U_{\star} = L V^{\dagger}, \qquad J_{\min} = 2M - 2 \sum_{m=1}^{M} \sigma_m. \tag{10.34}$$

Schönemann gives the corresponding real orthogonal least-squares formulation [104]. For the complex unitary result, set Higham's $A = F_{\mathrm{r}}$ and $B = F_{\mathrm{c}}$, so that $B^{\dagger} A = S$. Equations (2.1)–(2.3) of Ref. [46] construct the unitary polar factor from the SVD, and Higham's Theorem 2.2, specifically inequality (2.4), proves that this factor minimizes the Frobenius residual over all unitary matrices. Section 6.1 of the same article identifies the real specialization as the orthogonal Procrustes problem. Since the σ_m are the cosines of the principal angles θ_m, the minimum also has the geometric form

$$J_{\min} = 2 \sum_{m=1}^{M} (1 - \cos \theta_m). \tag{10.35}$$

For comparison, the squared projector defect is $2 \sum_m (1 - \cos^2 \theta_m)$. Procrustes alignment can choose coordinates within each retained space, but it cannot change either projector or erase a genuine subspace mismatch.

The same orientation determines the represented-operator transformation. If $H_{\mathrm{c}} = F_{\mathrm{c}}^{\dagger} H_{\mathrm{p}} F_{\mathrm{c}}$, then the operator in the aligned frame $F_{\mathrm{c}} U_{\star}$ is

$$H_{\mathrm{c}\to r} = U_{\star}^{\dagger} H_{\mathrm{c}} U_{\star}. \tag{10.36}$$

Using the opposite overlap convention reverses this transformation. A reported alignment must therefore state the overlap order, which frame receives the right action, and which similarity transform is applied to its operator.

For the constructed attack $F_{\mathrm{c}} = F_{\mathrm{r}} A$, the overlap is the full-rank unitary $A^{\dagger}$, so pointwise alignment uniquely recovers $A^{\dagger}$ up to floating-point error. Exact recovery in that setting is expected mathematics, not evidence for a general optimizer or for unknown physical correspondence.

10.9.2 Uniqueness, numerical rank, and deterministic output

In exact arithmetic, $U_\star$ is unique if and only if S has full rank. The full-rank direction is Higham's Theorem 1.1: the unitary polar factor is uniquely determined when the matrix has full column rank [46]. Repeated positive singular values may make individual singular vectors nonunique, but they do not make the polar factor $LV^\dagger$ nonunique when S is nonsingular. The relevant instability is a small minimum singular value, not repetition by itself.

If $r = \operatorname{rank} S < M$, every optimizer can be written, for a compatible complete SVD, as

$$U_W = L \begin{pmatrix} I_r & 0 \\ 0 & W \end{pmatrix} V^\dagger, \qquad W \in U(M-r). \tag{10.37}$$

The objective fixes the partial polar map on the rank-r support and leaves an arbitrary unitary identification between the two null spaces. A numerical implementation must replace exact rank with a declared cutoff. Returning $LV^\dagger$ from a complete backend SVD without further policy silently uses backend-selected null-space bases; it is an optimizer, but not a portable deterministic scientific identification.

A reusable method must therefore choose one of two contracts: reject numerical rank deficiency, or define a versioned deterministic completion. One possible coordinate-dependent completion projects the standard coordinate vectors into each null space in fixed index order, reorthogonalizes them with a declared acceptance tolerance, fixes each phase by a declared first-pivot rule, and pairs the resulting bases in order. Such a completion is deterministic in the stated coordinates, not basis invariant. The present manuscript does not select that policy and makes no claim for a rank-deficient returned map.

For every admitted problem, retain the overlap orientation, singular values, applied rank cutoff, numerical rank, selected unitary, objective residual, unitarity defect, and backend identity. A finite condition number should be reported only for numerical full rank; near the cutoff, sensitivity must remain visible rather than being hidden by the returned unitary.

Pointwise freedom is a large admissible family: every reciprocal coordinate receives its own unitary. It can be inappropriate when the intended physical comparison permits only a smaller parameterized transformation.

10.9.3 One global unitary

A deliberately restricted comparison uses one $U \in U(M)$ for the entire path:

$$\min_{U \in U(M)} \sum_j \|F_{\mathrm{c}}(\kappa_j)U - F_{\mathrm{r}}(\kappa_j)\|_{\mathrm{F}}^2. \tag{10.38}$$

Let

$$B = \sum_j F_{\mathrm{c}}(\kappa_j)^\dagger F_{\mathrm{r}}(\kappa_j) = L_B \Sigma_B V_B^\dagger. \tag{10.39}$$

The objective reduces to the same trace maximization, now with aggregate cross-covariance B. For N_k isometric frame pairs,

$$U_{\text{global}} = L_B V_B^\dagger, \qquad J_{\text{global,min}} = 2N_k M - 2\,\mathrm{tr}\,\Sigma_B. \tag{10.40}$$

Equal normalization by N_k changes the reported objective scale but not its minimizer. Nonuniform weights instead define a different aggregate and must be part of the frozen protocol.

The global minimizer is unique exactly when B is full rank. It can be rank deficient through cancellation even when every pointwise overlap is nonsingular. Repeated nonzero singular values alone do not destroy uniqueness. Consequently the global channel requires its own singular values, numerical-rank decision, residual, and rank-deficiency policy rather than inheriting the pointwise conditioning record.

One constant unitary generally cannot undo a genuinely momentum-dependent attack. A nonzero global residual establishes a limitation of this restricted family for the declared data. It does not establish failure of every nonidentity alignment family. Enlarging the admissible family is a new model class and requires a separately frozen calculation.

This distinction later becomes part of an adequacy decision. A model parameter and an alignment variable play different roles: the first selects a candidate Hamiltonian, while the second identifies its coordinates with the target within a declared family. Chapter 14 defines operator-admissible parameter sets by minimizing only over that frozen alignment family. Failure for one global unitary, a finite list of rotations, or a pointwise family therefore produces three different claims.

10.9.4 Reproducible method record

A Procrustes result is interpretable only when its record binds the complete method, not merely the returned matrix. Before solving, establish common ambient coordinates, equal retained rank, compatible sewing, and the admissible alignment family. Then retain

1. the ordered reference and candidate identities and sample-point order;
2. the right-action and overlap convention;
3. the equal or explicitly weighted Frobenius objective and its normalization;
4. all cross-covariance singular values, the rank rule, and the applied cutoff;
5. the selected transformation, residual, unitarity defect, and whether uniqueness was established;
6. the backend and deterministic-completion policy, if any; and

7. the independently reconstructed aligned frames and transformed operators.

The independent check should evaluate the objective and unitary identities from the original frames. Repeating the same SVD call is parity evidence, not an independent oracle. None of these numerical checks establishes that the chosen alignment family is physically appropriate.

10.10 Complete block hoppings and gauge-dependent locality

On the complete centered mesh, transform the projected matrices according to

$$H_F(R) = \frac{1}{N_k} \sum_{j=0}^{N_k-1} e^{-2\pi i R\kappa_j} H_F(\kappa_j). \tag{10.41}$$

The inverse is

$$H_F(\kappa_j) = \sum_{R\in\mathcal{R}_{N_k}} e^{2\pi i R\kappa_j} H_F(R). \tag{10.42}$$

Hermiticity requires the modular relation

$$H_F(\overline{-R}) = H_F(R)^\dagger, \tag{10.43}$$

including the self-opposite even-mesh Nyquist representative.

A symmetric range-$R_{\max}$ model retains $|R| \le R_{\max}$. Its omitted-block norm is

$$\eta_F(R_{\max}) = \left(\sum_{|R|>R_{\max}} \|H_F(R)\|_{\mathrm{F}}^2 \right)^{1/2}. \tag{10.44}$$

This quantity has energy units. It measures represented block content outside the selected range; it is not a probability or uncertainty.

For a complete transform, reference, attacked, and correctly realigned gauges have the same pointwise eigenvalues. After truncation they need not. The momentum-dependent gauge redistributes matrix content among the R blocks, so it changes both η_F and the spectral error of a finite-range model. This is why localization or gauge selection matters to short-range approximation even though it does not change the complete retained spectrum.

The maximum spectral error on a declared set $\mathcal{K}$ is

$$\epsilon_{\text{spec,max}} = \max_{\kappa\in\mathcal{K}} \max_{1\le n\le M} \left| E_n^{\text{truncated}}(\kappa) - E_n^{\text{target}}(\kappa) \right|. \tag{10.45}$$

It is an eigenvalue metric. It does not measure entrywise or aligned-operator agreement.

When the complete and truncated matrices use the same retained coordinates, the complementary rank-normalized Hamiltonian residuals on $\mathcal{K}$ are

$$\epsilon_{H,F,\max} = \max_{\kappa\in\mathcal{K}} \frac{\|H^{\text{truncated}}(\kappa) - H^{\text{target}}(\kappa)\|_{\text{F}}}{\sqrt{M}}, \tag{10.46}$$

$$\epsilon_{H,F,\text{rms}} = \left[\frac{1}{M|\mathcal{K}|}\sum_{\kappa\in\mathcal{K}}\|H^{\text{truncated}}(\kappa) - H^{\text{target}}(\kappa)\|_{\text{F}}^2\right]^{1/2}. \tag{10.47}$$

Here M is the retained rank. These quantities have energy units; division by a separately declared positive energy scale produces dimensionless residuals. They are not defined until state spaces, bases, gauges, energy references, units, and reciprocal coordinates agree. The maximum spectral discrepancy obeys the pointwise perturbation bound supplied by the unnormalized Frobenius residual, but need not determine it.

For a complete uniform transform, when the target is the untruncated finite interpolant and $\mathcal{K}$ supplies exact quadrature for the squared finite series, Parseval makes $\epsilon^2_{H,F,\text{rms}}$ equal the omitted-block power divided by M. This identity does not extend automatically to interpolation-plus-truncation residuals, nonuniform weights, incomplete transforms, or a separately removed cyclic Nyquist term.

10.11 Training and withheld roles

The complete training mesh determines raw eigenspaces, transported frames, alignment maps, projected matrices, Fourier blocks, and every finite-range candidate. A staggered disjoint mesh can then evaluate spectra of those frozen candidates between training coordinates.

In a bounded composite-space protocol, withheld coordinates may also enter a prospectively declared sampled external-gap gate. That use must be named: the withheld values then contribute to a frozen prerequisite and to final spectral evaluation, even though they do not alter frames, coefficients, ranges, or tolerances. They must not be used after inspection to redesign the attack or select which range is reported while retaining an untouched-withheld label.

No withheld frame is needed merely to compare the eigenvalues of an interpolated block Hamiltonian with the parent spectrum. Consequently such a comparison does not establish withheld frame, projector, or operator-matrix agreement.

10.12 A diagnostic ledger

Each tolerance belongs to one row and one predicate. A dimensionless frame tolerance cannot silently become an energy tolerance. A numerical consistency tolerance is not a physical uncertainty or scientific acceptance threshold.

10.13 A transparent computational example

The following synthetic example constructs a rank-two retained frame, applies a known periodic gauge attack, recovers it pointwise, and compares complete and truncated block representations. It is deliberately small and contains no material parameters.

Python interlude—aligning a synthetic composite Bloch frame

```
import numpy as np
from scipy.linalg import schur

# Define a Hermitian four-orbital parent with cell ranges -1, 0,
    and +1.
onsite = np.array(
    [
        [-1.20, 0.06, 0.00, 0.00],
        [0.06, -0.35, 0.00, 0.00],
        [0.00, 0.00, 1.00, 0.04],
        [0.00, 0.00, 0.04, 1.75],
    ],
    dtype=np.complex128,
)
positive = np.array(
    [
        [0.15, 0.00, 0.12, 0.00],
        [0.00, -0.10, 0.00, 0.10],
        [0.00, 0.00, 0.05, 0.00],
        [0.00, 0.00, 0.00, -0.04],
    ],
    dtype=np.complex128,
)
parent_R = np.array([-1, 0, 1], dtype=np.int64)
parent_blocks = np.array(
```

Table 10.1: Distinct diagnostics in a bounded composite-frame calculation.

Quantity	Compared objects	Unit	Interpretation
External gap	retained and excluded eigenvalues	energy	sampled spectral separation
Minimum overlap singular value	neighboring or closure frames	unitless	transport conditioning and rank gate
Procrustes record	overlap spectrum, rank cutoff, optimum residual, and unitary	unitless	reproducible alignment method and numerical checks
Closure eigenphases	unitary closure holonomy	radians	branch-dependent Wilson-loop-like phase set
Closure and unitarity defects	corrected boundary frames and returned unitaries	unitless	numerical sewing checks
Frame Frobenius defect	two coordinate frames	unitless	gauge-dependent basis difference
Projector Frobenius defect	two retained subspaces	unitless	gauge-invariant subspace difference
Projected-operator defect	two matrices in declared coordinates	energy	coordinate-sensitive operator difference
Block Hermiticity defect	$H(-R)$ and $H(R)^{\dagger}$	energy	represented structural consistency
Omitted-block norm	discarded hopping blocks	energy	range-truncation diagnostic
Hamiltonian truncation error	complete and truncated matrices in common coordinates	energy	sampled operator discrepancy
Maximum spectral error	ordered eigenvalues on a mesh	energy	worst sampled band discrepancy
Reconstruction, Parseval, and Weyl defects	redundant matrix, coefficient, and eigenvalue relations	declared tolerance units	numerical consistency only

```
    [positive.conjugate().T, onsite, positive],
    dtype=np.complex128,
)

# Evaluate the exact finite parent on a complete centered
    reciprocal mesh.
N = 16
kappa = -0.5 + np.arange(N, dtype=np.float64) / N
parent_phases = np.exp(2j * np.pi * np.outer(kappa, parent_R))
parent = np.einsum("kr,rij->kij", parent_phases, parent_blocks)

# Retain the lowest two eigenvectors at every reciprocal coordinate.

raw_frames = []
retained_energies = []
for matrix in parent:
    values, vectors = np.linalg.eigh(matrix)
    retained_energies.append(values[:2])
    raw_frames.append(vectors[:, :2])
raw_frames = np.asarray(raw_frames, dtype=np.complex128)
retained_energies = np.asarray(retained_energies, dtype=np.float64)

# Apply neighborwise polar transport to remove arbitrary
    eigensolver gauges.
transported = [raw_frames[0].copy()]
neighbor_singular_values = []
for index in range(N - 1):
    overlap = transported[index].conjugate().T @ raw_frames[index +
     1]
    left, singular_values, right_h = np.linalg.svd(overlap)
    neighbor_singular_values.extend(singular_values)
    rotation = right_h.conjugate().T @ left.conjugate().T
    transported.append(raw_frames[index + 1] @ rotation)

# Close the half-open reciprocal loop with identity ambient sewing.
closure = transported[-1].conjugate().T @ transported[0]
left, closure_singular_values, right_h = np.linalg.svd(closure)
neighbor_singular_values.extend(closure_singular_values)
unitary_closure = left @ right_h
triangular, eigenvectors = schur(unitary_closure, output="complex")
closure_phases = np.angle(np.diag(triangular))
```

```
order = np.argsort(closure_phases)
closure_phases = closure_phases[order]
eigenvectors = eigenvectors[:, order]

# Distribute the closure holonomy over the represented loop.
for index in range(N):
    fraction = index / N
    root = (
        eigenvectors
        @ np.diag(np.exp(1j * closure_phases * fraction))
        @ eigenvectors.conjugate().T
    )
    transported[index] = transported[index] @ root
reference = np.asarray(transported, dtype=np.complex128)

# Apply a known periodic, momentum-dependent rank-two gauge
    rotation.
theta = 0.30 + 0.65 * np.sin(2.0 * np.pi * kappa)
attacks = np.array(
    [
        [[np.cos(value), -np.sin(value)],
         [np.sin(value), np.cos(value)]]
        for value in theta
    ],
    dtype=np.complex128,
)
attacked = np.einsum("kdi,kij->kdj", reference, attacks)

# Projectors are invariant even though the frame matrices differ.
reference_projectors = np.einsum(
    "kdi,kei->kde", reference, reference.conjugate()
)
attacked_projectors = np.einsum(
    "kdi,kei->kde", attacked, attacked.conjugate()
)
np.testing.assert_allclose(attacked_projectors,
    reference_projectors, atol=1e-13)

# Recover one unitary Procrustes rotation independently at each
    mesh point.
```

```
# The cutoff is an illustrative numerical control, not a universal
    threshold.
procrustes_rank_cutoff = 1.0e-12
aligned = []
recovered_rotations = []
pointwise_overlap_singular_values = []
for reference_frame, candidate_frame in zip(reference, attacked,
    strict=True):
    overlap = candidate_frame.conjugate().T @ reference_frame
    left, singular_values, right_h = np.linalg.svd(overlap)
    if singular_values[-1] <= procrustes_rank_cutoff:
        raise RuntimeError("pointwise Procrustes overlap is rank
    deficient")
    pointwise_overlap_singular_values.extend(singular_values)
    rotation = left @ right_h
    recovered_rotations.append(rotation)
    aligned.append(candidate_frame @ rotation)
aligned = np.asarray(aligned, dtype=np.complex128)
recovered_rotations = np.asarray(recovered_rotations, dtype=np.
    complex128)
pointwise_frame_defect = max(
    np.linalg.norm(candidate - target)
    for candidate, target in zip(aligned, reference, strict=True)
)
pointwise_unitarity_defect = max(
    np.linalg.norm(rotation.conjugate().T @ rotation - np.eye(2))
    for rotation in recovered_rotations
)
np.testing.assert_allclose(aligned, reference, atol=1e-13)
np.testing.assert_allclose(
    recovered_rotations,
    attacks.conjugate().transpose(0, 2, 1),
    atol=1e-13,
)

# Fit the distinct restricted family containing one unitary for the
    whole path.
aggregate = sum(
    candidate.conjugate().T @ target
    for candidate, target in zip(attacked, reference, strict=True)
)
```

```
left, global_singular_values, right_h = np.linalg.svd(aggregate)
if global_singular_values[-1] <= procrustes_rank_cutoff:
    raise RuntimeError("global Procrustes aggregate is rank
    deficient")
global_rotation = left @ right_h
globally_aligned = np.asarray(
    [frame @ global_rotation for frame in attacked],
    dtype=np.complex128,
)
global_frame_defect = max(
    np.linalg.norm(candidate - target)
    for candidate, target in zip(globally_aligned, reference,
    strict=True)
)

# Compress the same parent into the frame choices.
def project(frames):
    return np.asarray(
        [
            frame.conjugate().T @ matrix @ frame
            for matrix, frame in zip(parent, frames, strict=True)
        ],
        dtype=np.complex128,
    )

reference_operator = project(reference)
attacked_operator = project(attacked)
aligned_operator = project(aligned)
global_operator = project(globally_aligned)

# All complete projected operators have the retained parent
    eigenvalues.
for operator in (
    reference_operator,
    attacked_operator,
    aligned_operator,
    global_operator,
):
    values = np.linalg.eigvalsh(operator)
    np.testing.assert_allclose(values, retained_energies, atol=1e
    -13)
```

```
# Transform each matrix path into complete block hoppings.
cell_R = np.arange(-N // 2, N // 2, dtype=np.int64)
def blocks_from_samples(samples):
    return np.asarray(
        [
            np.mean(
                samples
                * np.exp(-2j * np.pi * R * kappa)[:, None, None],
                axis=0,
            )
            for R in cell_R
        ],
        dtype=np.complex128,
    )

reference_blocks = blocks_from_samples(reference_operator)
attacked_blocks = blocks_from_samples(attacked_operator)
aligned_blocks = blocks_from_samples(aligned_operator)
np.testing.assert_allclose(aligned_blocks, reference_blocks, atol=1
    e-13)

# A fixed finite range can retain different amounts in different
    gauges.
range_mask = np.abs(cell_R) <= 1
reference_omitted = np.linalg.norm(reference_blocks[~range_mask])
attacked_omitted = np.linalg.norm(attacked_blocks[~range_mask])

print("minimum neighbor/closure singular value:", min(
    neighbor_singular_values))
print("minimum pointwise Procrustes singular value:",
      min(pointwise_overlap_singular_values))
print("pointwise frame defect:", pointwise_frame_defect)
print("pointwise unitarity defect:", pointwise_unitarity_defect)
print("global Procrustes singular values:", global_singular_values)
print("one-global-unitary frame defect:", global_frame_defect)
print("reference omitted-block norm:", reference_omitted)
print("attacked omitted-block norm:", attacked_omitted)
```

The example establishes finite-array identities for a synthetic parent. It does not establish continuous-gap protection, optimal localization, or a physical alignment. The

default matrix norm used by `numpy.linalg.norm` is the Frobenius norm in these calls; production interfaces should name that choice rather than rely on an unstated default.

Notebook to do Create `examples/tutorials/research-monograph/foundations/composite_band_gauge_alignment.ipynb` only after this lecture contract is stable. Include frame/projector invariance, neighbor singular values, polar transport, reciprocal sewing, closure phases, pointwise and global Procrustes alignment, block Hermiticity, gauge-dependent truncation, and disjoint withheld spectral evaluation. Keep the example synthetic and do not create or execute the notebook as part of the present manuscript change.

10.14 Common mistakes

1. Tracking individual energy-ordered eigenvectors through an internal degeneracy instead of retaining the composite projector.
2. Treating an external energy gap as a substitute for overlap singular values, or vice versa.
3. Comparing frame matrices before accounting for unitary gauge freedom.
4. Calling a large unaligned matrix defect a physical operator error.
5. Assuming periodic eigenvalues imply identity reciprocal sewing in every basis.
6. Comparing closure eigenphases as ordinary real numbers without branch and permutation handling.
7. Describing exact recovery of a constructed pointwise attack as a general alignment solution.
8. Treating the complete-SVD product as a deterministic scientific map on rank-deficient overlaps without declaring a null-space policy.
9. Confusing repeated positive singular values with rank-deficient Procrustes nonuniqueness.
10. Concluding from failure of one global unitary that every nonidentity alignment family fails.
11. Assuming a finite harmonic series for the attack angle gives a finite-range attacked Hamiltonian.

12. Calling a smooth transported gauge maximally localized without evaluating a localization functional.
13. Inferring Wannier centers, spreads, or densities from Hamiltonian blocks alone.
14. Assuming gauge invariance of the complete spectrum survives an arbitrary finite-range truncation.
15. Treating an omitted-block Frobenius norm as a probability or physical uncertainty.
16. Using withheld spectra to redesign the gauge or range while preserving an untouched-evaluation claim.

10.15 Exercises

1. Starting from Equation (10.2), prove Equation (10.4).
2. Verify the dimensions of every factor in Equations (10.7) and (10.12).
3. Prove projector invariance and projected-operator covariance under $F \mapsto FU$.
4. Derive Equation (10.15) from the finite Wannier transformation and Bloch orthogonality.
5. Show explicitly why the gauge cancels from the complete scalar isolated-band Hamiltonian but not from a general composite matrix.
6. Derive the positive overlap in Equation (10.24).
7. Prove the pointwise Procrustes solution by maximizing $\mathrm{Re}\,\mathrm{tr}(U^\dagger F_{\mathrm{c}}^\dagger F_{\mathrm{r}})$ and derive the minimum in Equation (10.34).
8. Derive the rank-deficient optimizer family in Equation (10.37) and explain why a complete backend SVD does not select a portable map between the null spaces.
9. Derive the aggregate matrix and minimum in Equations (10.39) and (10.40).
10. Construct an attack that is constant in κ and explain why the pointwise and global alignment families then contain the same inverse.
11. Compare a frame defect, projector defect, and spectral defect for two frames related by a nontrivial unitary.

12. Show that symmetric block truncation preserves Equation (10.43) when the complete representative set is paired modulo N_k.

13. Explain what additional evidence would be required to interpret closure eigenphases as Wannier centers or topological data.

10.16 Evidence boundary and next use

The derivations in this chapter establish finite-dimensional identities under stated frame, mesh, sewing, Fourier, and norm conventions. The Python example checks those identities for synthetic arrays. Its constructed pointwise overlaps are full rank; its illustrative cutoff neither qualifies a general numerical-rank rule nor a rank-deficient completion, and the example does not define deterministic degenerate closure coordinates. Neither the derivation nor the example establishes that an unknown physical alignment has been recovered, that a chosen gauge is maximally localized, that a hopping range is adequate for an application, or that a material is represented correctly.

A controlled M2-like calculation can use this chain to test a known gauge attack, distinguish pointwise from globally constrained alignment, and expose gauge-dependent finite-range locality. Such a calculation remains a bounded positive and negative control. Its retained result and independent verifier own any numerical outcomes; this chapter does not import them.

Part II uses these foundations to ask whether an alignment or reduced model is adequate for its declared purpose. Appendix G applies composite transport, closure, block hopping, and localization concepts to retained controlled calculations. Later material chapters add projection windows, disentanglement, geometry, orbital identity, spin, and energy-reference requirements that the present synthetic construction deliberately omits.

CHAPTER 11

Verification and Evidence Vocabulary

Status foundational evidence vocabulary. This chapter classifies claims and records; it introduces no new numerical result, scientific validation, or uncertainty-quantification result.

11.1 Learning goals

After working through this chapter, a reader should be able to

1. write a bounded claim before selecting evidence;
2. distinguish mathematical proof, software verification, numerical verification, scientific validation, and uncertainty quantification;
3. distinguish an oracle, metric, tolerance, observation, and disposition;
4. distinguish a finite represented reference from an exact limit;
5. separate transform training samples from withheld evaluation and independent reconstruction;
6. keep parent-model, numerical, and model-reduction errors separate; and
7. state what a passing check does not establish.

Scientific software can run successfully, reproduce its own output, and still answer the wrong mathematical or physical question. Evidence vocabulary is therefore a computational prerequisite rather than an afterword.

11.2 Claims precede evidence

An evidence record begins with a bounded claim. The claim identifies the system, quantity, representation, parameter domain, metric, tolerance, and intended use. Only then can an oracle and acceptance rule be selected. A recurring record shape is

$$(\text{reference}, \text{candidate}, \text{metric}, \text{tolerance}, \text{observation}, \text{disposition}). \tag{11.1}$$

The record is incomplete unless the reference and candidate carry the metadata required to make the metric meaningful.

These terms have distinct roles.

Claim. The bounded statement the evidence is intended to support.

Reference or oracle. The independently justified source of expected behavior for the stated check.

Candidate. The object or behavior under examination.

Metric. The declared comparison quantity.

Tolerance or acceptance rule. The prospectively stated decision rule.

Observation. The value or categorical outcome actually obtained.

Disposition. The conclusion permitted by the rule and evidence class.

A threshold-free diagnostic can report an observation without accepting or rejecting it. Conversely, a tolerance without a named metric and scale is not a complete acceptance rule.

11.3 Evidence classes

Mathematical proof. Establishes a theorem from explicitly stated assumptions by a recorded derivation or checked formal encoding. Proof does not establish that the assumptions describe the physical system, that a numerical implementation is correct, or that a bound is useful at project scale.

Software verification. Establishes that software follows a documented contract. Examples include shape and type invariants, exact serialization round trips, basis-order checks, failure behavior, and declared symmetry or Hermiticity tests.

Numerical verification. Establishes agreement between a numerical implementation and independently derived mathematics, or controlled convergence toward the stated mathematical problem. Examples include analytic limiting cases, manufactured solutions, independent norm calculations, and stencil-refinement studies.

Scientific validation. Evaluates adequacy against independent trusted physical, experimental, or scientific reference evidence for a declared use. It addresses whether the selected model represents relevant physical behavior, not merely whether its code is internally consistent.

Uncertainty quantification. Identifies relevant uncertainty sources and characterizes or propagates them to declared outputs. Sensitivity alone is not automatically a probabilistic uncertainty statement.

No class implies another. A proved implication does not validate its physical assumptions. A passing software regression does not establish numerical convergence. Agreement with a selected Kohn–Sham parent does not establish agreement with experiment. A validation comparison does not quantify every uncertainty source.

Prospective citation note—high priority
Check the evidence-class taxonomy and the claim that the classes do not imply one another against an authoritative verification, validation, and uncertainty-quantification source. Candidate: National Research Council (2012), provisional key `nationalResearchCouncil2012` [82]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

11.4 Examples that share an array but not a claim

A Hermitian matrix may participate in several different evidence records.

- Comparing H with $H^\dagger$ can verify a declared software invariant.
- Comparing H with an independently derived small-matrix oracle can numerically verify an implementation.
- Comparing an observable derived from H with independent physical data can contribute to scientific validation for a declared use.

The array does not carry an evidence class intrinsically. The class follows from the claim, reference, construction, and interpretation.

Similarly, successful process completion is an observation about execution. It is not automatically evidence that inputs were scientifically accepted, a solver converged to the desired state, a discretization converged, or the model is valid.

11.5 Errors from different operations

Projection error, discretization error, fitting error, truncation error, and parent-model error arise from different operations. They should be combined only when an owning mathematical contract defines a compatible relationship. The broad categories used throughout this monograph are

Parent-model error. Discrepancy attributable to the selected physical parent and its assumptions relative to the intended physical use.

Numerical or discretization error. Discrepancy introduced while approximating the stated mathematical problem at finite settings.

Model-reduction error. Discrepancy introduced when replacing the accepted parent representation by a reduced class.

A reduced model can reproduce an unconverged parent accurately. A converged numerical solution can solve an inadequate parent model precisely. Agreement with experiment can arise through cancellation between parent and reduction errors. Evidence records must preserve these possibilities rather than report one undifferentiated "model error."

11.6 Finite-setting observations

Suppose a quantity q_j is computed at two successive finite settings and satisfies

$$|q_{j+1} - q_j| \leq \tau. \tag{11.2}$$

This establishes stability over those tested settings under the recorded controls. It does not by itself bound

$$|q_j - q_\infty|. \tag{11.3}$$

An infinite-setting bound requires additional structure, such as a justified asymptotic error model, monotonicity result, variational bound, or sufficient higher-resolution evidence.

"Stable over the tested sequence" is therefore a stronger description than "converged" when only the finite contrast is known.

Derived quantities can amplify errors. A central second-difference estimate of curvature divides an energy combination by a squared step. Pointwise energy agreement therefore need not control effective mass. Conditioning and cancellation must be evaluated for the derived claim, not inferred from a visually smooth parent plot.

11.7 Finite references and exact language

A numerical study may compare candidates with a declared high-resolution finite representation. If q_h is the candidate and $q_{h_{\mathrm{ref}}}$ is that reference, the observed quantity

$$\epsilon_{h,\mathrm{ref}} = |q_h - q_{h_{\mathrm{ref}}}| \tag{11.4}$$

is a finite-reference discrepancy. It is not automatically an error bound to the continuum quantity q. The latter would require information about $|q_{h_{\mathrm{ref}}} - q|$ or a theorem that otherwise controls the limit.

This distinction changes the permitted language. A decreasing sequence of finite-reference discrepancies can be described as observed refinement over the tested settings. Calling the final value an exact continuum error would state more than the comparison measured. Small nonmonotone variations near floating-point or iterative-solver scales should also be retained rather than removed to manufacture a monotone convergence plot.

11.8 Transform training data and withheld evaluation

Training data can play a mathematical role beyond statistical parameter estimation. In a complete finite Fourier transform, the training mesh defines all transform coefficients. Exact reconstruction on that same mesh verifies the forward–inverse transform pair, but it is interpolation of the defining samples. It does not test behavior between them.

A withheld mesh can evaluate the frozen interpolant at coordinates that were not used to calculate coefficients. To preserve that role, the withheld values must not alter

- the fitted coefficients;
- the candidate model class or hopping range;
- the stopping rule;
- the metric or tolerance; or

- the selection of which outcomes are reported.

If withheld observations motivate redesign, the original observation remains part of the evidence history and a new untouched evaluation design is needed for a later confirmatory claim. In a synthetic numerical study, disjoint withheld evaluation remains numerical verification relative to the selected parent. The word "withheld" does not promote it to scientific validation.

11.9 Independent reconstruction is not producer replay

A producer may verify internal invariants before returning a result. That is valuable, but importing the producer and asking it to generate the same object again can repeat the same mistake. Stronger numerical-consistency evidence reconstructs decisive quantities from frozen inputs and retained outputs by a separately maintained route.

Independence is bounded rather than absolute. A verifier may share the same language runtime, numerical libraries, or mathematical assumptions. Its record should therefore state what it reconstructs independently and what remains shared. A useful verifier also checks identities and roles—for example, coordinate inventories, transform representatives, tolerances, and the separation of training and withheld samples—rather than comparing only one headline scalar.

Agreement within a prospectively frozen numerical tolerance establishes consistency for the reconstructed finite protocol. It does not select a model for a scientific use, quantify physical uncertainty, or validate the parent model. Chapter 9 supplies a concrete transform, truncation, and route-comparison setting to which these rules apply.

11.10 Negative evidence and stop conditions

A failed acceptance criterion is retained rather than weakened. It may indicate an implementation defect, inadequate resolution, incompatible representation, insufficient model-class expressiveness, or mismatch between a model and its intended use. These diagnoses have different remedies.

A missing prerequisite can also be a valid stop condition. If units, geometry, basis ordering, energy reference, or alignment are incompatible, direct matrix subtraction should not proceed to a residual. A fail-closed compatibility finding is evidence about the attempted operation, not an inconvenience to be removed by guessing metadata.

A well-executed negative result is bounded by the tested domain. Failure of one frozen model hierarchy does not prove failure of every possible model, and one nonconverged

numerical sequence does not prove that convergence is impossible.

11.11 Reproducibility does not promote evidence

A result is reproducible only relative to retained identities and declared conditions. A useful record generally includes immutable inputs, code and dependency versions, physical and numerical settings, commands, artifact checksums, metric definitions, structured outcomes, warnings, and locations of externally retained large artifacts.

Reproduction preserves the class of the reproduced claim. Reproducing a pedagogical calculation reproduces pedagogical behavior; reproducing a synthetic verification case reproduces bounded verification evidence; reproducing a production calculation reproduces that calculation under its recorded conditions. None is promoted automatically to scientific validation.

11.12 A compact evidence ledger

For each computational stage, a ledger can record

1. the claim identifier and bounded statement;
2. identities of reference and candidate records;
3. compatibility checks and stop conditions;
4. metric, unit, normalization, and tolerance;
5. observed value and pass, fail, or diagnostic-only disposition;
6. evidence class and limitations; and
7. provenance for reproduction.

This structure makes limitations part of the result rather than prose added only after a surprising outcome.

11.13 Exercises

1. Classify a JSON round trip, a manufactured finite-difference solution, and comparison with an experimental band edge by evidence class.

2. Write a bounded claim for a Hermiticity check, including metric, unit, and tolerance. State three conclusions it does not support.
3. Explain why two stable successive settings do not establish a bound to an infinite-setting limit.
4. Give one example of parent-model, numerical, and model-reduction error in a single reduction workflow.
5. Design a negative-evidence record for incompatible energy references that stops before matrix subtraction.

11.14 Next use and evidence boundary

Part II applies this vocabulary to model adequacy, aligned operator comparison, and the retained benchmark inventory. The vocabulary itself does not establish that any project calculation has passed. Evidence status remains owned by the applicable proof package, test record, calculation record, scientific review, or accepted disposition.

PART II

Criteria for a Good Reduced Model

CHAPTER 12

What Makes a Reduced Model Adequate?

Status conceptual and illustrative framework. The central hypothesis, continuum criteria, and intellectual contributions are proposed work; the diagnostic examples are not research evidence or validated reductions of silicon.

12.1 Project motivation and significance

Part I established the computational prerequisites: quantities and metadata, finite representations, quantum operators, separable box Hamiltonians and Kronecker constructions, periodic geometry, discretization, retained spaces, isolated-band Fourier reduction, and evidence vocabulary. This chapter now asks the research-specific question that motivates their composition: when does a reduced semiconductor Hamiltonian remain adequate for its declared use?[1]

Chapter 8 defined orthogonal compression and distinguished an exact retained operator from an approximate effective model. One nearby electronic-structure term requires additional care.

The projector augmented-wave (PAW) method provides an instructive contrast to this orthogonal compression. In PAW, a linear transformation $\hat{\mathcal{T}}$ maps a smooth auxiliary Kohn–Sham state to an all-electron state,

$$|\psi_n\rangle = \hat{\mathcal{T}}|\widetilde{\psi}_n\rangle. \tag{12.1}$$

The auxiliary state consequently satisfies the generalized eigenproblem

$$\hat{\mathcal{T}}^\dagger \hat{H} \hat{\mathcal{T}} |\widetilde{\psi}_n\rangle = \varepsilon_n \hat{\mathcal{T}}^\dagger \hat{\mathcal{T}} |\widetilde{\psi}_n\rangle. \tag{12.2}$$

rather than an ordinary orthonormal eigenproblem [98]. The localized PAW projector functions recover expansion coefficients inside augmentation regions; despite their name,

[1] For quantum mechanics, see Shankar [108]. For solid-state physics, see Kittel [55] and Simon [112]. Sze and Lee [119] provide a device-oriented introduction to semiconductor physics, while Yu and Cardona [133] provide a graduate-level treatment.

they are not the global orthogonal projector $\hat{P}_{\mathcal{S}}$ defined in Chapter 8. Rostgaard's tutorial also shows how the transformation must be carried into expectation values and Wannier matrix elements. PAW therefore illustrates why a state transformation, an induced overlap metric, and an orthogonal subspace projection must not be silently identified.

The active silicon parent uses approved norm-conserving pseudopotentials rather than PAW datasets. This comparison is explanatory and does not alter that scientific setting.

Reading note Read Blöchl's original PAW paper [8] before using PAW transformations, projector functions, or augmentation corrections in a scientific claim. Rostgaard [98] is a useful self-contained tutorial, but the original paper is the primary source for the method.

Semiconductor physics is commonly taught through two quantum-mechanical descriptions that are physically related but mathematically disconnected. Students first solve a microscopic periodic Hamiltonian,

$$\hat{H}_{\text{micro}} = -\frac{\hbar^2}{2m_0}\nabla^2 + V_{\text{per}}(\mathbf{r}), \tag{12.3}$$

and obtain Bloch states and energy bands. They are later introduced to effective-mass theory through a continuum envelope Hamiltonian. Written relative to the selected band-edge energy, it is

$$\hat{H}_{\text{EMT}} = -\frac{\hbar^2}{2}\nabla \cdot \mathbf{m}^{*-1}\nabla + V_{\text{ext}}(\mathbf{r}). \tag{12.4}$$

For a smooth, nondegenerate band near an extremum $\mathbf{k}_0$, the usual connection expands the dispersion as

$$E_n(\mathbf{k}_0 + \mathbf{q}) = E_n(\mathbf{k}_0) + \frac{\hbar^2}{2}\mathbf{q}^{\mathsf{T}}\mathbf{m}_n^{*-1}\mathbf{q} + O\big(\|\mathbf{q}\|^3\big)\,. \tag{12.5}$$

After subtracting the constant $E_n(\mathbf{k}_0)\hat{I}$ and replacing $\mathbf{q}$ by $-i\nabla$, this argument explains the form of the effective kinetic operator. It does not construct the envelope-function state space, define the map from microscopic Bloch states to continuum states, or quantify the approximations introduced by band truncation, gauge selection, valley decomposition, and spatial coarse-graining.

The Luttinger–Kohn derivation, its assumptions, its multivalley and degenerate- band extensions, and its relation to the present first-principles reduction problem are developed in Appendix K. The detailed summary and remaining source-review note are kept there rather than duplicated in this chapter [70].

Effective-mass theory is therefore often presented as a second quantum mechanics rather than as a controlled reduction of the first. The effective-mass tensor is obtained from the curvature of a Bloch band, but the corresponding state-space reduction is usually left implicit.

This project began from a practical attempt to close that pedagogical gap. The initial objective was to construct an explicit periodic potential that students could discretize and diagonalize, yet which remained sufficiently expressive to produce an indirect band gap. Such a model would provide an executable bridge between elementary periodic quantum mechanics and the band structure of an indirect-gap semiconductor.

The most direct strategy appears elementary: calculate Kohn–Sham density-functional-theory Bloch eigenpairs, reconstruct a retained Hamiltonian, subtract the known kinetic operator, and interpret the remainder as an effective periodic potential.

Let $\hat{H}_{\mathrm{KS}}(\mathbf{k})$ be a Kohn–Sham Bloch-fiber operator with cell-periodic eigenstates $|u_{n\mathbf{k}}\rangle$. For a retained set of bands $\mathcal{B}$, define

$$\hat{P}(\mathbf{k}) = \sum_{n\in\mathcal{B}} |u_{n\mathbf{k}}\rangle\langle u_{n\mathbf{k}}| \tag{12.6}$$

and

$$\hat{H}^{(P)}_{\mathrm{KS}}(\mathbf{k}) = \hat{P}(\mathbf{k})\hat{H}_{\mathrm{KS}}(\mathbf{k})\hat{P}(\mathbf{k})\Big|_{\mathcal{H}^{(P)}(\mathbf{k})} = \sum_{n\in\mathcal{B}} \varepsilon_n(\mathbf{k})|u_{n\mathbf{k}}\rangle\langle u_{n\mathbf{k}}|, \tag{12.7}$$

where

$$\mathcal{H}^{(P)}(\mathbf{k}) = \mathrm{Ran}\,\hat{P}(\mathbf{k}). \tag{12.8}$$

For entangled bands, constructing a smooth retained family may require projection or disentanglement rather than selection by a single global energy cutoff.

The apparently obvious construction is

$$\hat{V}^{(P)}_{\mathrm{eff}}(\mathbf{k}) \stackrel{?}{=} \hat{H}^{(P)}_{\mathrm{KS}}(\mathbf{k}) - \hat{T}^{(P)}(\mathbf{k}), \tag{12.9}$$

where

$$\hat{T}^{(P)}(\mathbf{k}) = \hat{P}(\mathbf{k})\hat{T}(\mathbf{k})\hat{P}(\mathbf{k})\Big|_{\mathcal{H}^{(P)}(\mathbf{k})}. \tag{12.10}$$

This subtraction is mathematically valid on the retained fiber, but its result is a projected operator, not necessarily a unique local scalar potential. Projection may induce nonlocality, while nonlocal pseudopotentials, orbital structure, spin–orbit interactions, and band truncation may introduce components that cannot be represented by a simple function $V(\mathbf{r})$. Chapter 8 gives an explicit nearest-neighbor two-band example whose exact lower-band projection has non-finite-range Fourier coefficients. Bloch gauge freedom further changes the matrix coordinates of a composite retained operator without changing its abstract action or spectrum.

Prospective citation note—medium priority
Check the entangled-band disentanglement and momentum-dependent Wannier-gauge statements against Souza–Marzari–Vanderbilt (2001) and Marzari et al. (2012), bibliography keys `souza2001` [115] and `marzari2012` [72] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The desired teaching potential must therefore be obtained by reducing the operator residual into an explicitly chosen model class:

$$\hat{V}_{\mathrm{teach}} = \arg\min_{\hat{V}\in\mathcal{V}_{\mathrm{adm}}} \sum_{\mathbf{k}\in\mathcal{K}} \left\| \hat{H}_{\mathrm{KS}}^{(P)}(\mathbf{k}) - \hat{T}^{(P)}(\mathbf{k}) - \hat{P}(\mathbf{k})\hat{V}(\mathbf{k})\hat{P}(\mathbf{k}) \Big|_{\mathcal{H}^{(P)}(\mathbf{k})} \right\|_*^2 . \tag{12.11}$$

Here, $\mathcal{K}$ is the sampled reciprocal-space region and $\mathcal{V}_{\mathrm{adm}}$ may represent local periodic potentials, truncated Fourier expansions, finite real-space stencils, or constrained tight-binding operator classes. The norm $\|\cdot\|_*$ and any reciprocal-space weights must be chosen to measure operator error consistently with the physical observables and continuum limit of interest. This is a proposed model-class problem, not a presently accepted fitting criterion.

The foundational construction in Chapter 8 separates exact operator retention from model-class approximation. The diagnostic discussion below shows why all components of an additive Hamiltonian must be reduced through the same map before the chapter returns to pristine–doped comparison, continuum admissibility, and the broader research program.

12.2 Exact retention and model-class approximation

Chapter 8 constructed the exact retained operator for a pedagogical discrete spectrum and distinguished it from an approximate effective-model class. The present program applies that distinction fiberwise: the retained Kohn–Sham operator is not itself the teaching potential, tight-binding hierarchy, or continuum Hamiltonian later fitted to it. The applicable reduction must name its admissible class, objective, retained structures, and error criterion.

This framing does not claim that fitting band energies, fitting Hamiltonian matrix elements, or identifying restrictions of a tight-binding class is new. Chapter 14 introduces representative precedents alongside each corresponding concept and states the narrower proposed contribution: a contract-dependent decision about whether one model satisfies both spectral and aligned-operator criteria, supported by a common witness or a valid separation certificate.

12.3 Controlled diagnostic examples

Simple finite-dimensional examples expose the distinction between matching a spectrum and identifying an operator. Appendix D develops the particle-in-a-box residual and its identifiability limits in detail, while Appendix E compares two finite harmonic-oscillator constructions. These examples are analytical and numerical diagnostics; they are not evidence for the silicon reduction program.

12.4 From the particle in a box to electronic-structure reduction

The particle-in-a-box example is elementary, but the obstruction it reveals is general. In a common parent state space, the decomposition

$$\mathbf{H} = \mathbf{T} + \mathbf{V} \tag{12.12}$$

is meaningful because all three matrices act on the same discrete state space and are represented in the same basis. Reduction preserves this additive decomposition only when every term is reduced through the same map:

$$\mathcal{R}_{P_M}(\mathbf{H} - \mathbf{T}) = \mathcal{R}_{P_M}(\mathbf{H}) - \mathcal{R}_{P_M}(\mathbf{T}). \tag{12.13}$$

In general,

$$\mathcal{R}_{P_M}(\mathbf{H} - \mathbf{T}) \neq \mathcal{R}_{P_M}(\mathbf{H}) - \mathbf{T}. \tag{12.14}$$

The distinction becomes more consequential in electronic-structure calculations. A Kohn–Sham Hamiltonian, a Wannier Hamiltonian, a tight-binding Hamiltonian, and an effective-mass Hamiltonian generally do not begin as matrices acting on an identical numerical state space. They may differ in dimension, basis, gauge, spatial resolution, boundary conditions, geometry, units, and energy reference. Their matrices therefore cannot be physically compared merely because they are all called Hamiltonians.

The central problem of this work is consequently not the mechanical subtraction of two matrices. It is the construction and validation of the state-space identifications that make such a subtraction mathematically defined and physically interpretable.

12.5 From periodic potentials to impurity operators

The same structural problem becomes more consequential when the model is extended from bulk bands to semiconductor impurities. Given pristine and doped Hamiltonians,

the natural proposal is

$$\hat{V}_{\text{imp}} \stackrel{?}{=} \hat{H}_d - \hat{H}_b. \tag{12.15}$$

This expression is not yet an operator identity because independently constructed pristine and doped calculations may use different retained subspaces, Bloch or Wannier gauges, basis orderings, geometries, spin structures, energy references, and numerical dimensions. Their matrices cannot be interpreted as representations in common physical coordinates without an explicit identification.

Let $s \in \{b, d\}$ denote matched bulk and doped branches, with retained operators

$$\hat{H}_s^{(P)} = \hat{P}_s \hat{H}_s \hat{P}_s \Big|_{\mathcal{H}_s^{(P)}}, \qquad \mathcal{H}_s^{(P)} = \text{Ran}\, \hat{P}_s. \tag{12.16}$$

If the retained ranks agree, a meaningful comparison requires an identification

$$\hat{U}_d : \mathcal{H}_b^{(P)} \longrightarrow \mathcal{H}_d^{(P)}. \tag{12.17}$$

Let c_b and c_d be the scalar reference energies selected by the declared alignment procedure, and define

$$\overline{\hat{H}}_s^{(P)} = \hat{H}_s^{(P)} - c_s \hat{I}_s. \tag{12.18}$$

For a unitary identification $\hat{U}_d$, the aligned impurity operator on the bulk retained space is

$$\Delta \hat{H}_d^{(P)} = \hat{U}_d^\dagger \overline{\hat{H}}_d^{(P)} \hat{U}_d - \overline{\hat{H}}_b^{(P)}. \tag{12.19}$$

Unequal retained ranks require a separately defined partial identification or further model reduction; they do not admit this unitary formula unchanged.

Equal-dimensional spaces admit many abstract unitary maps. Most are not physically admissible: an arbitrary unitary can scramble localization, orbital character, crystal momentum, valley content, symmetry, and spin while leaving the spectrum unchanged. Gauge alignment is therefore necessary to define the subtraction, but it is not sufficient to select a physical impurity operator. The identification must also respect the structures needed by the intended reduced theory.

12.6 Central hypothesis

The central hypothesis proposed by this project is:

> *The intended continuum limit is not merely a downstream approximation to an already constructed atomistic model. It constrains, from the beginning, the admissible retained spaces, gauges, identification maps, and reduced operator classes.*

For a semiconductor representation to support an effective-mass interpretation, its atomistic-to-continuum identification must preserve or control the structures used by the continuum theory. These include

- real-space localization and lattice translation;
- Bloch wave vector and valley correspondence;
- orbital, symmetry, and spin content;
- smoothness across the relevant reciprocal-space region;
- common bulk behavior away from an impurity;
- boundary conditions and normalization; and
- separation of slowly varying envelopes from lattice-periodic components.

Let

$$\epsilon = \frac{a}{L} \tag{12.20}$$

denote the ratio of lattice scale a to an envelope or impurity scale L. For a specified family of retained atomistic operators $\hat{H}_\epsilon^{(P)}$ and appropriately normalized identification maps

$$\hat{J}_\epsilon : \mathcal{H}_\epsilon^{(P)} \longrightarrow \mathcal{H}_{\text{EMT}}, \tag{12.21}$$

one possible continuum criterion is norm-resolvent convergence,

$$\left\| \hat{J}_\epsilon \left(\hat{H}_\epsilon^{(P)} - z \right)^{-1} \hat{J}_\epsilon^\dagger - \left(\hat{H}_{\text{EMT}} - z \right)^{-1} \right\| \longrightarrow 0, \qquad \epsilon \longrightarrow 0, \tag{12.22}$$

for z in an identified common resolvent set. This is a candidate analytical form, not an established result of the project. A later specification may instead require a weaker, explicitly defined combination of spectral, subspace, operator, and observable convergence criteria.

Prospective citation note—low priority
Check the mathematical meaning and implications of norm-resolvent convergence with an operator-theory source. Candidate: Teschl (2014), bibliography key `teschl2014` [121] (already present). The project's choice of this criterion remains explicitly provisional.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The continuum limit thereby becomes an upstream admissibility condition. A gauge, basis, identification, or reduced operator class is acceptable only if it preserves, or approximates with controlled error, the microscopic structures needed to recover the declared envelope theory.

12.7 Relation to Hilbert's sixth problem

This research is situated within the broad mathematical tradition associated with Hilbert's sixth problem: the rigorous formulation of physical theories and the justification of limiting relations between microscopic and continuum descriptions. It does not claim to solve Hilbert's sixth problem in its general form. Instead, it addresses a bounded and experimentally consequential instance: the reduction of microscopic electronic-structure operators to continuum envelope models for indirect-gap semiconductors and their impurities.

Prospective citation note—medium priority
Check this characterization against Hilbert's published problem statement rather than an uncited historical paraphrase. Candidate: Hilbert, "Mathematical Problems" (1902), provisional key `hilbert1902` [47]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The relevant limiting process is not only a passage between different length scales. It is a passage between different state spaces, operator algebras, coordinate systems, and model classes:

$$\begin{aligned} \text{KS–DFT} &\longrightarrow \text{retained Bloch operator} \\ &\longrightarrow \text{localized atomistic operator} \\ &\longrightarrow \text{effective-mass continuum operator}. \end{aligned} \tag{12.23}$$

Each arrow requires an explicit map, a statement of preserved structure, and an error criterion. Without these elements, agreement between selected band energies does not establish that two models represent the same reduced physics.

The effective-mass limit is therefore both an endpoint and a constraint. The program asks not only whether an atomistic calculation can reproduce a continuum observable, but which atomistic representations and reductions are capable of supporting the continuum theory at all.

12.8 Intellectual contribution

The proposed program aims to develop a gauge-equivariant, continuum-constrained framework for semiconductor operator reduction. Its intended contributions are

1. reconstruction of retained operators from Kohn–Sham Bloch eigenpairs;
2. separation of operator reconstruction from model-class approximation;

3. quantitative projection onto interpretable periodic-potential, stencil, and tight-binding classes;
4. explicit alignment of pristine and doped retained state spaces;
5. construction of impurity operators whose abstract action is invariant under compatible coordinate changes after alignment;
6. formulation of continuum-compatibility criteria using operator, subspace, spectral, and observable errors; and
7. an executable pedagogical chain connecting periodic potentials, indirect band structure, impurity operators, and effective-mass theory.

The conceptual distinction is concise:

> *Gauge alignment makes an operator comparison mathematically defined; continuum compatibility makes the resulting reduced model physically admissible.*

What began as an attempt to construct a teachable periodic potential therefore exposes a broader mathematical problem. Obtaining the potential requires reconstructing an operator rather than merely fitting a spectrum. Introducing an impurity requires comparing independently constructed operators. Comparing those operators requires identifying their state spaces and gauges. Selecting physically admissible identifications and reduced models requires a declared continuum limit.

This chain provides both the scientific rationale and the educational significance of the proposed research. It converts the normally implicit transition from Bloch theory to effective-mass theory into an explicit, testable, and executable sequence of operator reductions while keeping every approximation and evidence claim explicit. Chapter 13 next develops the common-space operator comparisons needed by this framework. Chapter 14 then turns the phrase “satisfies several criteria” into a precise intersection problem and separates common-witness evidence from certified separation. The later program chapters specialize these ideas to the silicon host and its substitutional phosphorus and boron branches.

CHAPTER 13

Comparing Operators Responsibly

Status expository mathematical framework; no numerical or scientific result is claimed.

Part I fixed the distinctions among quantities, finite representations, operators, and retained spaces. In particular, Chapter 10 derived frame and projector diagnostics, polar transport, pointwise and globally constrained Procrustes alignment, and gauge-dependent block locality in a controlled setting. This chapter applies those foundations to alignment, subtraction, and residual analysis. It does not prescribe a particular alignment algorithm or reduced model class; those choices belong to the applicable methodological specifications.

13.1 Gauge constraints in operator comparison

Chapter 2 established the unitary basis rule $H^{(\mathcal{B}')} = U^\dagger H^{(\mathcal{B})} U$. Its periodic specialization carries additional structure.

For an isolated retained Bloch subspace, momentum-dependent unitary freedom similarly changes the Wannier basis generated from that subspace. Localization selects or optimizes a gauge; it does not remove the need to record the selected frame. Equal band energies do not determine Wannier functions or provide a physically distinguished alignment between two calculations.

Gauge covariance of the complete represented operator does not imply gauge invariance after real-space truncation. Two gauges can represent the same pointwise spectrum and projector while producing different hopping decay and therefore different finite-range residuals. Appendix G demonstrates this distinction with controlled smooth and rough composite-band frames.

Prospective citation note—medium priority
Check the specialization from finite-dimensional basis changes to Bloch/Wannier gauge and localization against Marzari et al. (2012) and Souza–Marzari–Vanderbilt (2001), bibliography keys `marzari2012` [72] and `souza2001` [115] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

13.2 Spectra and operator information

The spectrum of a finite Hermitian matrix is invariant under unitary conjugation. Spectral comparison is consequently valuable but incomplete for operator comparison. Even when two Hermitian matrices are isospectral, the spectrum alone does not identify which basis vectors, sites, orbitals, spin channels, or spatial regions correspond. It supplies no distinguished physical alignment map.

Conversely, a reduced model may approximate selected eigenvalues while retaining different onsite terms or couplings in the coordinates used to represent a later perturbation. A spectral acceptance rule and an aligned operator acceptance rule are therefore separate constraints. Direct regression of Hamiltonian and overlap matrix blocks in a declared atomic-orbital representation is also established methodology [137]; it does not remove the need to identify that representation before comparison. The relationship between spectral and operator criteria must be investigated rather than assumed. Chapter 14 makes that investigation explicit: each threshold defines a parameter-space admissible set, and adequacy requires one common candidate rather than two unrelated minimizers.

13.3 Alignment between retained spaces

Let $\mathcal{S}_{\mathrm{r}}$ and $\mathcal{S}_{\mathrm{c}}$ denote reference and candidate retained spaces of equal finite dimension. An alignment is an explicitly defined isometric identification

$$A : \mathcal{S}_{\mathrm{r}} \longrightarrow \mathcal{S}_{\mathrm{c}}, \qquad A^{\dagger}A = I_{\mathcal{S}_{\mathrm{r}}}. \tag{13.1}$$

When the spaces have equal dimension and A is onto, A is unitary. The candidate operator pulled back to the reference space is

$$\widehat{H}_{\mathrm{c}\to r} = A^{\dagger}\widehat{H}_{\mathrm{c}}A : \mathcal{S}_{\mathrm{r}} \longrightarrow \mathcal{S}_{\mathrm{r}}. \tag{13.2}$$

The direction of A, its coordinate convention, and the information used to construct it must be stated. Orbital labels, matrix dimension, or similar spectra do not substitute for this map.

If no justified alignment exists, coordinate-wise subtraction is undefined for the intended physical comparison. One may instead use a declared invariant comparison, but the invariant and the claim it supports must be named explicitly.

13.4 Energy references

An energy-valued operator is unchanged in its eigenvectors but shifted in its spectrum when a scalar multiple of the identity is added. For a recorded energy zero E_0, define

$$\overline{H} = H - E_0 I. \tag{13.3}$$

Reference and candidate operators must use a common physical energy convention before their difference is interpreted. A valence-band maximum used to plot bulk bands and an electrostatic estimator used to align pristine and doped supercells are not automatically the same convention.

The physical specification requires a common scalar energy zero before bulk–dopant subtraction, while the operational estimator and its uncertainty belong to a later alignment stage. Until that estimator is supplied, the requirement is fixed but the corresponding impurity difference is not yet a calculated project result.

Prospective citation note—high priority
Check the distinction between plotting references and electrostatic defect-supercell alignment against Freysoldt et al. (2009, 2014), bibliography keys `freysoldt2009` [33] and `freysoldt2014` [32] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

13.5 Represented differences

After compatibility and alignment have been established, define the signed represented difference by an explicit operand order. For example,

$$\Delta H = \overline{H}_{\mathrm{c}\to r} - \overline{H}_{\mathrm{r}}. \tag{13.4}$$

Reversing the operands reverses the sign. A difference matrix is not intrinsically an impurity operator, perturbation, approximation error, or observable. That interpretation follows only from the identities of the two operators, their common-space construction, and the declared scientific use.

For pristine and doped systems, compatibility includes at least the represented state spaces, matrix dimension, basis ordering, geometry and site mapping, units, energy reference, spin convention, and gauge alignment. An unresolved item is a stop condition for direct matrix subtraction.

13.6 Hermiticity and residuals

A represented Hamiltonian is expected to be Hermitian within a stated numerical criterion. One possible absolute residual is

$$\varepsilon_{\mathrm{H}} = \max_{i,j} \left| H_{ij} - H_{ji}^{*} \right| . \tag{13.5}$$

The residual, its units, any normalization scale, and its acceptance tolerance are distinct quantities. Passing a Hermiticity tolerance verifies only that declared numerical criterion; it does not validate the physical model or the metadata used to interpret the matrix.

For a compatible difference ΔH, examples of absolute residual measures include

$$\varepsilon_{\max} = \max_{i,j} |\Delta H_{ij}|, \tag{13.6}$$

$$\varepsilon_{\mathrm{F}} = \|\Delta H\|_{\mathrm{F}}, \tag{13.7}$$

$$\varepsilon_{2} = \|\Delta H\|_{2}. \tag{13.8}$$

A normalized residual must identify its reference scale and define behavior when that scale vanishes. Numerical implementations must also handle nonfinite values and overflow explicitly rather than silently reporting an infinite or undefined metric.

13.7 Approximation and evidence boundaries

Chapter 11 separates parent-model, numerical, projection, fitting, and truncation errors and defines the evidence classes used here. An aligned residual establishes only its declared common-space comparison. It does not by itself validate the physical model or combine distinct error sources.

Chapter 14 next combines a normalized spectral loss and a compatible aligned-operator loss without conflating them. It distinguishes an explicit common witness from a failed search and from a certified positive separation. Chapter 28 later records the corresponding proof obligations and their current analytical and mechanization status.

CHAPTER 14

Admissible Sets and Compatibility Certificates

Status foundational model-adequacy framework with synthetic worked examples; no material acceptance decision, physical threshold calibration, or uncertainty-quantification result is claimed.

A reduced Hamiltonian is rarely judged by one number. A candidate may reproduce selected energy bands while failing to reproduce the represented operator, its real-space locality, or an observable needed by the intended application. The opposite can also occur: a candidate may be close in one operator norm while missing a small but important spectral feature. This chapter develops a precise way to ask whether one candidate satisfies two declared criteria at the same time.

The central objects are *admissible sets*. Each criterion selects the parameters that satisfy its threshold. A common acceptable candidate exists when the corresponding sets intersect. An explicit point in the intersection is a *common witness*. If the sets do not intersect, a valid positive lower bound on their distance is a *separation certificate*. A failed numerical search is neither one of these.

After this chapter, the reader should be able to

- distinguish model parameters from alignment variables and numerical controls;
- define normalized spectral and operator losses on declared data;
- distinguish established spectral fitting and operator regression from the proposed simultaneous witness-or-certificate question;
- construct thresholded admissible sets;
- distinguish a witness, a failed search, and a separation certificate;
- recognize ellipsoidal sublevel sets of positive-definite quadratic losses;
- keep training, withheld evaluation, and locality diagnostics in their proper roles; and

- state exactly what a bounded synthetic compatibility result does and does not establish.

14.1 From one error to simultaneous requirements

Suppose that a reduced model is controlled by a parameter $\boldsymbol{\theta}$ in a declared domain Θ. A spectral comparison produces a nonnegative loss $L_E(\boldsymbol{\theta})$, while a represented- operator comparison produces a nonnegative loss $L_H(\boldsymbol{\theta})$. Two thresholds, τ_E and τ_H, state the largest losses accepted for the present purpose.

It is not enough to minimize the losses separately. The minimizer of L_E may be a different model from the minimizer of L_H. The scientific question is whether there is one parameter value for which

$$L_E(\boldsymbol{\theta}) \le \tau_E \quad \text{and} \quad L_H(\boldsymbol{\theta}) \le \tau_H. \tag{14.1}$$

The word "and" is the essential point. Reporting one spectrally successful model and a different operator-successful model does not exhibit a single adequate model.

14.1.1 Example 1: intervals on a line

Before considering Hamiltonians, take one real parameter x and define

$$L_E(x) = |x|, \qquad L_H(x) = |x - 1|. \tag{14.2}$$

With equal thresholds $\tau_E = \tau_H = 0.6$, the two admissible sets are

$$\mathcal{A}_E = [-0.6, 0.6], \qquad \mathcal{A}_H = [0.4, 1.6]. \tag{14.3}$$

Their intersection is $[0.4, 0.6]$. The point $x = 0.5$ is a common witness. Neither $x = 0$, which minimizes L_E, nor $x = 1$, which minimizes L_H, is needed to prove compatibility.

If both thresholds are instead 0.4, then

$$\mathcal{A}_E = [-0.4, 0.4], \qquad \mathcal{A}_H = [0.6, 1.4]. \tag{14.4}$$

The sets are separated by 0.2. This positive number is stronger evidence than checking a finite list of points and failing to find an intersection.

14.2 The candidate family and its contracts

Let

$$\mathfrak{M} = \{H_{\boldsymbol{\theta}}(k) : \boldsymbol{\theta} \in \Theta\} \tag{14.5}$$

be a family of finite Hermitian Bloch matrices. A complete definition of this family states

- the meaning, units, and allowed range of every component of $\boldsymbol{\theta}$;
- the reciprocal coordinates k on which the matrices are evaluated;
- the retained rank, basis, gauge, energy reference, and energy unit;
- the parent or target data to which the candidates are compared; and
- the distinction between fitting data, withheld data, and purely diagnostic data.

A bounded domain is not merely a programming convenience. It is part of the model class. Enlarging Θ asks a different question and may change both compatibility and separation.

The matrix structure is likewise part of the scientific question, not merely an implementation choice. Stiles showed how the independent matrix elements allowed by a declared crystal structure can be enumerated before a tight- binding fit [117]. Such enumeration determines which coordinates are available to the fit; it does not prove that the resulting family can reproduce every desired property. In a specified spin–orbit $sp^3d^5s^*$ empirical tight-binding class, for example, Boykin, Klimeck, and Oyafuso derived exact relations among selected band gaps and effective masses [11]. Those relations are intrinsic restrictions of that model class. This is stronger than observing that one optimization run failed to locate a satisfactory parameter set.

14.2.1 A rank-two shift-and-splitting family

A useful controlled family begins with a rank-two reference matrix $H_{\mathrm{ref}}(k)$. Define its mean energy by

$$\overline{E}(k) = \frac{1}{2}\,\mathrm{tr}\,H_{\mathrm{ref}}(k) \tag{14.6}$$

and its traceless part by

$$H_0(k) = H_{\mathrm{ref}}(k) - \overline{E}(k) I_2. \tag{14.7}$$

Given a positive energy scale E_0, consider

$$H_{s,q}(k) = \overline{E}(k) I_2 + s E_0 I_2 + q H_0(k). \tag{14.8}$$

The dimensionless parameter s shifts both levels equally. The dimensionless parameter q rescales their splitting about the mean. The parameter column is

$$\boldsymbol{\theta} = \begin{bmatrix} s \\ q \end{bmatrix}, \qquad \boldsymbol{\theta} \in \Theta. \tag{14.9}$$

Equation (14.8) is a deliberately restricted model family. It does not represent every rank-two Hamiltonian. A compatibility or separation result obtained from it is therefore conditional on this family and its bounds.

When the domain requires $q > 0$, the ordered eigenvalues obey

$$E_n^{\mathrm{cand}}(k; s, q) = \overline{E}(k) + sE_0 + q\left(E_n^{\mathrm{ref}}(k) - \overline{E}(k)\right). \tag{14.10}$$

Each spectral residual is therefore affine in (s, q). For any fixed global alignment, the entries of the aligned candidate matrix are also affine in (s, q). Squaring and summing these residuals produces the quadratic losses derived below. The positivity restriction on q is important: allowing its sign to change can reverse the ordering of the two eigenvalues and invalidate this single ordered formula.

14.3 Spectral and operator losses

The two criteria use the same declared sampling contract but compare different objects. Let $k_1, \ldots, k_{N_k}$ be a frozen training mesh, let the retained rank be r, and assign each sample a finite positive dimensionless weight w_j. Write

$$W = \sum_{j=1}^{N_k} w_j. \tag{14.11}$$

14.3.1 Spectral loss

Suppose the target energies are $E_n^{\mathrm{tar}}(k_j)$, ordered according to a declared convention.

Definition 14.1 (Spectral RMS error)**.** The energy-valued ordered spectral root-mean-square error of the candidate specified by $\boldsymbol{\theta}$ is

$$R_E(\boldsymbol{\theta}) = \left[\frac{1}{rW}\sum_{j=1}^{N_k} w_j \sum_{n=1}^{r} \left|E_n^{\mathrm{cand}}(k_j; \boldsymbol{\theta}) - E_n^{\mathrm{tar}}(k_j)\right|^2\right]^{1/2}. \tag{14.12}$$

It has the same energy unit as the candidate and target eigenvalues. ■

Definition 14.2 (Normalized spectral loss)**.** Given a finite positive energy scale E_0,

the normalized spectral loss is

$$L_E(\boldsymbol{\theta}) = \frac{R_E(\boldsymbol{\theta})}{E_0}. \tag{14.13}$$

It is dimensionless. ■

Uniform weights recover the factor $1/(N_k r)$. Another normalization defines another loss and must be named rather than silently substituted. Neither the weights nor E_0 are automatically uncertainty estimates.

Objectives built from energy residuals have a long history. Starrost and collaborators used a genetic algorithm to fit an empirical tight-binding Hamiltonian to energies at selected high-symmetry wave vectors [116]. More recently, Wang and collaborators trained adjustable tight-binding matrix elements with an eigenvalue mean-squared-error objective against first-principles band energies [128]. These are precedents for spectral supervision, although their sampling, weights, and normalizations need not equal those in Equation (14.13).

The ordered-band formula also has a matching premise. Away from crossings, an energy ordering may be sufficient. At crossings or inside nearly degenerate clusters, the comparison must state how bands or clusters are identified. A sorting rule, overlap-based tracking rule, projector comparison, or another declared convention changes the loss contract and cannot be inferred from the energy arrays alone.

14.3.2 Operator loss

The corresponding operator comparison requires both matrices to act in an identified retained space. Let $H^{\mathrm{tar}}(k_j)$ be the target matrices in that space, and let C be an allowed global unitary alignment.

Definition 14.3 (Represented-operator RMS residual)**.** The energy-valued represented-operator root-mean-square residual of the candidate specified by $\boldsymbol{\theta}$ under alignment C is

$$R_H(\boldsymbol{\theta}, C) = \left[\frac{1}{rW}\sum_{j=1}^{N_k} w_j \left\| C^\dagger H_{\boldsymbol{\theta}}(k_j) C - H^{\mathrm{tar}}(k_j)\right\|_{\mathrm{F}}^2\right]^{1/2}. \tag{14.14}$$

It has the same energy unit as the candidate and target matrices. ■

Definition 14.4 (Normalized operator loss)**.** Given the same finite positive energy scale E_0, the normalized operator loss under alignment C is

$$L_H(\boldsymbol{\theta}, C) = \frac{R_H(\boldsymbol{\theta}, C)}{E_0}. \tag{14.15}$$

It is dimensionless. ■

The Frobenius norm and its sample aggregation are meaningful only after the state spaces, bases, gauges, energy references, units, reciprocal coordinates, sample identities, and weights have been declared. The scientific loss definition chooses the samples and weights; represented-operator mechanics validate compatibility, form differences, and apply the declared aggregation. Chapter 13 develops those prerequisites.

Direct matrix supervision addresses a different object from spectral supervision. Zhang and collaborators learned Hamiltonian and overlap blocks in a declared atomic-orbital representation using equivariant atomic-cluster- expansion models [137]. Schwade and collaborators instead combined an approximate tight-binding model with learned environment-dependent corrections and evaluated the resulting electronic-property predictions at larger scales [105]. The targets and representations are method-specific, but both examples reinforce why a matrix-level objective must name its representation rather than inherit meaning from its eigenvalues.

Some fitting studies augment energy residuals with band gaps, effective masses, or other observables. Such an objective must state each observable's definition, weight, and scale separately. A gap is derived from selected eigenvalues, an effective mass requires a derivative or local fit, and an optical transition strength additionally depends on states and another operator. These terms should not be called "purely spectral" merely because the final reported quantity is a scalar. Klimeck and collaborators provide a concrete mixed-target example: they fitted silicon nearest- and second-neighbor sp^3s^* models to band-edge and effective-mass targets, reported a successful second-neighbor parameterization, and reported that their search found no satisfactory global nearest-neighbor fit [56]. That last statement is evidence about the completed search, not by itself an infeasibility proof.

Two matrices can have identical eigenvalues and a nonzero operator residual. Equation (14.13) therefore cannot replace Equation (14.15). The two losses ask different questions.

14.4 Alignment variables are not model parameters

A composite retained space permits unitary changes of frame. The alignment matrix C is consequently an additional comparison variable, not necessarily a physical model

parameter. Let $\mathcal{U}$ be the declared family of allowed alignments. Assume that $\mathcal{U}$ is finite, or that it is compact and the loss depends continuously on C, so that the best permitted value is attained. Define

$$L_H^{\min}(\boldsymbol{\theta}) = \min_{C \in \mathcal{U}} L_H(\boldsymbol{\theta}, C). \tag{14.16}$$

Without an attainment condition, "minimum" must be replaced by "infimum," and a boundary value equal to that infimum need not supply an actual alignment witness. The family $\mathcal{U}$ is part of the contract. It may be one fixed identity, a finite list of global rotations, a continuous symmetry family, or some other physically justified set. These choices are not interchangeable.

Chapter 10 distinguished pointwise Procrustes alignment from one globally constrained unitary. A pointwise map $C(k_j)$ may recover a known gauge attack exactly at every sample, while one constant C may not. Failure within one finite family of constant rotations does not prove failure for every nonidentity alignment, every smooth gauge, or every physically admissible map. Enlarging $\mathcal{U}$ changes the model-comparison problem.

If $\mathcal{U} = \{C_\alpha : \alpha \in \mathcal{F}\}$ is a finite family, then Equation (14.16) is evaluated by explicit enumeration. Such enumeration is deterministic and complete for $\mathcal{F}$; it is not a continuous alignment optimizer.

14.5 Admissible sets and their intersection

Define the parameter-space spectral-admissible set by

$$\mathcal{A}_E(\tau_E) = \{\boldsymbol{\theta} \in \Theta : L_E(\boldsymbol{\theta}) \le \tau_E\} \tag{14.17}$$

and the operator-admissible set by

$$\mathcal{A}_H(\tau_H) = \left\{\boldsymbol{\theta} \in \Theta : L_H^{\min}(\boldsymbol{\theta}) \le \tau_H\right\}. \tag{14.18}$$

For a finite alignment family,

$$\mathcal{A}_H(\tau_H) = \bigcup_{\alpha \in \mathcal{F}} \{\boldsymbol{\theta} \in \Theta : L_H(\boldsymbol{\theta}, C_\alpha) \le \tau_H\}. \tag{14.19}$$

Thus the operator-admissible set need not be one connected region even when each fixed-alignment loss has a simple shape.

The family is compatible at the declared thresholds when

$$\mathcal{A}_E(\tau_E) \cap \mathcal{A}_H(\tau_H) \neq \varnothing. \tag{14.20}$$

Any retained $\boldsymbol{\theta}_{\mathrm{w}}$ satisfying both inequalities is a common witness. A complete operator witness also records an alignment $C_{\mathrm{w}} \in \mathcal{U}$ that attains the accepted operator loss. Substitution into independently evaluated losses proves membership directly.

A witness proves existence but does not establish uniqueness, parameter identifiability, robustness, or physical validity. It also says nothing about parameters outside the frozen family.

This intersection is the contribution boundary. Spectral fitting, direct Hamiltonian regression, and analytic restrictions on particular tight-binding classes are established ideas. The question introduced here is narrower: whether one candidate in a frozen family satisfies independently declared spectral and aligned-operator criteria. The answer must be supported by a common witness or, when the sets do not intersect, by a valid positive separation certificate under the declared family, alignment, losses, metric, and thresholds.

14.6 Why quadratic losses produce ellipses

The geometry becomes especially transparent when a residual depends linearly on two parameters. Suppose

$$r_\ell(\boldsymbol{\theta}) = \mathbf{a}_\ell^{\mathsf{T}}\boldsymbol{\theta} - b_\ell, \qquad \ell = 1, \dots, M, \tag{14.21}$$

and define

$$L^2(\boldsymbol{\theta}) = \frac{1}{M}\sum_{\ell=1}^{M} r_\ell(\boldsymbol{\theta})^2. \tag{14.22}$$

Expanding the squares gives

$$L^2(\boldsymbol{\theta}) = \boldsymbol{\theta}^{\mathsf{T}} Q \boldsymbol{\theta} + 2\mathbf{g}^{\mathsf{T}}\boldsymbol{\theta} + d, \tag{14.23}$$

where

$$Q = \frac{1}{M}\sum_{\ell=1}^{M} \mathbf{a}_\ell \mathbf{a}_\ell^{\mathsf{T}}, \qquad \mathbf{g} = -\frac{1}{M}\sum_{\ell=1}^{M} b_\ell \mathbf{a}_\ell, \qquad d = \frac{1}{M}\sum_{\ell=1}^{M} b_\ell^2. \tag{14.24}$$

If Q is positive definite, the center is

$$\mathbf{c} = -Q^{-1}\mathbf{g}. \tag{14.25}$$

Completing the square yields

$$L^2(\boldsymbol{\theta}) = L^2_{\min} + (\boldsymbol{\theta} - \mathbf{c})^{\mathsf{T}} Q (\boldsymbol{\theta} - \mathbf{c}), \tag{14.26}$$

with

$$L^2_{\min} = d - \mathbf{g}^{\mathsf{T}} Q^{-1} \mathbf{g}. \tag{14.27}$$

Consequently, the unconstrained threshold set $L \le \tau$ is empty if $\tau^2 < L^2_{\min}$, is the single point $\mathbf{c}$ if $\tau^2 = L^2_{\min}$, and is a filled ellipse if $\tau^2 > L^2_{\min}$. Intersecting that ellipse with a

rectangular parameter domain may clip its boundary. If Q is only positive semidefinite, flat directions require separate treatment; the positive-definite formulas must not be used without checking their premise.

For a finite alignment family, each fixed-alignment operator loss may generate one ellipse. Equation (14.19) then becomes a union of clipped ellipses.

14.6.1 Example 2: a witness and a separation distance in two dimensions

Let $\boldsymbol{\theta} = (s, q)^{\mathsf{T}}$ and consider the circular quadratic losses

$$L_E^2(s, q) = s^2 + (q - 0.9)^2, \tag{14.28}$$

$$L_H^2(s, q) = (s - 0.2)^2 + (q - 0.5)^2. \tag{14.29}$$

Their centers are

$$\mathbf{c}_E = \begin{bmatrix} 0 \\ 0.9 \end{bmatrix}, \qquad \mathbf{c}_H = \begin{bmatrix} 0.2 \\ 0.5 \end{bmatrix}, \tag{14.30}$$

and the center distance is

$$d_c = \|\mathbf{c}_E - \mathbf{c}_H\|_2 = \sqrt{0.2} \approx 0.447214. \tag{14.31}$$

With equal thresholds 0.3, the midpoint

$$\boldsymbol{\theta}_{\mathrm{w}} = \frac{\mathbf{c}_E + \mathbf{c}_H}{2} = \begin{bmatrix} 0.1 \\ 0.7 \end{bmatrix} \tag{14.32}$$

has both losses equal to $d_c/2 \approx 0.223607$. It is therefore a common witness.

With equal thresholds 0.2, the two closed disks are disjoint. Their exact Euclidean separation is

$$\delta^* = d_c - 0.2 - 0.2 \approx 0.047214. \tag{14.33}$$

This exact expression is available because the sets are ordinary disks. General ellipses and unions require more care.

Python interlude—checking the two-dimensional witness and separation

```
import numpy as np

# These centers and radii come from the symbolic calculation above.
center_e = np.array([0.0, 0.9])
center_h = np.array([0.2, 0.5])
```

```
center_distance = np.linalg.norm(center_e - center_h)

# At radius 0.3, the midpoint is an explicit common witness.
witness = 0.5 * (center_e + center_h)
loss_e = np.linalg.norm(witness - center_e)
loss_h = np.linalg.norm(witness - center_h)

# At radius 0.2, the disk-to-disk distance is positive.
separation = max(0.0, center_distance - 0.2 - 0.2)

print("center distance:", center_distance)
print("candidate witness:", witness)
print("spectral loss at witness:", loss_e)
print("operator loss at witness:", loss_h)
print("witness passes radius 0.3:", loss_e <= 0.3 and loss_h <=
    0.3)
print("separation at radius 0.2:", separation)
```

14.7 Witnesses and separation certificates

A common witness is easy to verify: evaluate both losses at the same retained parameter and compare them with the frozen thresholds. Proving that no witness exists is harder.

The underlying implications are standard metric-space facts: a common point forces zero set distance, any feasible pair bounds the defining infimum from above, and a valid positive lower bound excludes intersection [100]. Treating threshold-consistent parameters as a feasible set has direct precedent in bounded-error and set-membership identification [75]. The combined wording *common witness or certified positive separation* is used here as a descriptive synthesis for this contract-defined comparison, not as the name of an established theory.

Let d_Θ be a declared metric on the parameter domain. Define

$$\delta^* = \inf_{\substack{\boldsymbol{\theta}_E \in \mathcal{A}_E \\ \boldsymbol{\theta}_H \in \mathcal{A}_H}} d_\Theta(\boldsymbol{\theta}_E, \boldsymbol{\theta}_H). \tag{14.34}$$

If either admissible set is empty, that fact should be reported separately rather than hidden inside an arbitrary distance convention. If both are nonempty, a certified lower bound

$$0 < \delta_{\text{lower}} \leq \delta^* \tag{14.35}$$

proves separation. A displayed pair of feasible boundary points supplies an upper bound

$$\delta^* \le d_\Theta(\boldsymbol{\theta}_E^{\mathrm{b}}, \boldsymbol{\theta}_H^{\mathrm{b}}) = \delta_{\mathrm{upper}}. \tag{14.36}$$

Equal lower and upper bounds identify the distance exactly, up to the declared numerical tolerance. If the protocol declares a separation resolution $\rho > 0$, the disposition is certified separated only when the valid lower bound exceeds ρ. A positive bound no larger than ρ remains unresolved at that declared resolution.

For Euclidean coordinates, one useful certificate projects both sets onto a unit direction $\mathbf{e}$. If every spectral-admissible point satisfies

$$\mathbf{e}^{\mathsf{T}}\boldsymbol{\theta}_E \ge a \tag{14.37}$$

while every operator-admissible point satisfies

$$\mathbf{e}^{\mathsf{T}}\boldsymbol{\theta}_H \le b, \tag{14.38}$$

then

$$\delta^* \ge \max(0, a - b). \tag{14.39}$$

Both inequalities must hold over the complete applicable sets, including every alignment branch and every effect of clipping by Θ.

For the ellipse in Equation (14.26), its unclipped extreme in direction $\mathbf{e}$ is

$$\mathbf{e}^{\mathsf{T}}\mathbf{c} \pm \sqrt{(\tau^2 - L_{\mathrm{min}}^2)\mathbf{e}^{\mathsf{T}}Q^{-1}\mathbf{e}}. \tag{14.40}$$

This formula follows by transforming the ellipse to a unit ball and applying the Cauchy–Schwarz inequality. It is valid only when the ellipse is nonempty, Q is positive definite, and the relevant extreme is not changed by the parameter-domain boundary. Those conditions are premises of the certificate, not implementation details.

A local optimizer, a dense plot, or a finite grid may suggest separation but does not establish the global lower bound in Equation (14.34). The nearest-neighbor search reported by Klimeck and collaborators above is an instructive example of this evidence boundary: "no satisfactory fit found" does not become a theorem that the admissible set is empty. Validated global-search methods can sometimes provide such bounds, but their applicability and assumptions must be stated explicitly [83, 77].

14.7.1 The parameter metric is part of the claim

Euclidean distance is not invariant under arbitrary reparameterization. If one coordinate is replaced by 100 times itself, the numerical distance changes unless the metric changes too. A separation result must therefore state

- the canonical parameter coordinates;
- their units or dimensionless normalization;
- the metric and any coordinate weights; and
- the resolution below which separation is treated as unresolved.

A positive distance in a synthetic parameter metric is not automatically a physical energy, probability, or uncertainty interval.

14.8 Training, withheld evaluation, and locality

Training data define the losses whose sublevel sets enter the compatibility decision. Withheld data evaluate selected parameters after that decision. If a withheld mesh is consulted while choosing thresholds, changing the family, or selecting the witness, then it is no longer purely withheld for that decision.

For selected $\boldsymbol{\theta}$, one may report

$$L_E^{\text{train}}(\boldsymbol{\theta}), \qquad L_H^{\text{train}}(\boldsymbol{\theta}), \tag{14.41}$$

$$L_E^{\text{withheld}}(\boldsymbol{\theta}), \qquad L_H^{\text{withheld}}(\boldsymbol{\theta}). \tag{14.42}$$

The withheld values test interpolation or generalization to the frozen withheld coordinates. They do not retroactively alter the training-set admissible sets unless the protocol explicitly defines a second acceptance stage.

A reciprocal operator may also be transformed to hopping blocks and truncated at several real-space ranges. The omitted-block norm then measures locality in the chosen gauge and convention. Unless a locality threshold appears in the formal definition of admissibility, this quantity is a diagnostic rather than a third acceptance condition. Reporting it beside the two losses must not silently turn it into one.

14.9 Threshold sensitivity

Compatibility is relative to (τ_E, τ_H). Increasing a threshold can enlarge its admissible set; decreasing it can shrink the set or make it empty. A threshold-sensitivity calculation asks how the disposition changes as one or both thresholds vary.

Such an analysis is useful for identifying

- whether a conclusion is far from or close to a transition;

- which criterion controls the transition;
- whether a reported positive separation is numerically resolved; and
- whether small changes in a proposed threshold change the disposition.

It does not calibrate a physically meaningful threshold by itself. If the sensitivity sweep was designed after the confirmatory result was inspected, it must be labelled post hoc. It may characterize the retained geometry without becoming a prospectively frozen prediction.

14.10 The controlled M3 role

The maintained one-dimensional M3 calculation instantiates the framework above with a synthetic rank-two parent inherited from the controlled multiband calculation, the two parameters (s, q) from Equation (14.8), a bounded rectangular domain, and a frozen finite family of global real rotations. It uses uniform sample weights as the special case $w_j = 1$, constructs spectral and operator quadratic losses on the training mesh, evaluates selected points on a disjoint withheld mesh, and records hopping-locality diagnostics at selected ranges.

This role is deliberately narrow. M3 is a controlled example of how to retain one common witness or a positive analytic separation certificate within one frozen family. It is not a general continuous alignment solver, a statement about every rank-two Hamiltonian, a localization calculation, a material calculation, or a calibration of physical acceptance tolerances. The retained outcomes and their provenance belong in Section G.9 and the owning calculation package, not in this foundational derivation. Figure G.5 there shows the actual compatible and separated M3 regions; the circles in Example 2 above remain a simpler pedagogical model.

14.11 Common mistakes

1. **Presenting the loss as the contribution.** Weighted spectral fitting, effective-mass targets, and direct Hamiltonian regression all have precedents. The present contribution is the bounded simultaneous witness-or-certificate question.
2. **Comparing two separate minimizers.** The spectral optimum and operator optimum need not define one common candidate.
3. **Calling a failed grid search a proof.** A grid leaves untested points between samples and supplies no global lower bound by itself.

4. **Optimizing over an undeclared gauge family.** Changing the alignment family changes the admissible set.

5. **Using incompatible represented operators.** A Frobenius norm cannot repair mismatched spaces, bases, units, or energy references.

6. **Choosing thresholds from the desired answer.** Thresholds are acceptance inputs, not knobs adjusted until an intersection appears or disappears.

7. **Treating withheld values as training evidence.** Withheld data have a distinct evaluation role.

8. **Promoting a locality diagnostic to a criterion.** That requires a declared locality loss and threshold.

9. **Generalizing beyond the frozen family.** Separation inside one bounded family does not establish universal incompatibility.

14.12 Exercises

1. For Example 1, determine the intersection for equal thresholds 0.5. Is there one witness or more than one?

2. Starting from Equation (14.23), carry out the completion of the square and verify Equation (14.26).

3. Let $Q = \operatorname{diag}(4, 1)$, $\mathbf{c} = \mathbf{0}$, and $L_{\min} = 0$. Sketch the sets for $\tau = 1$ and $\tau = 1/2$ and explain which parameter is more tightly constrained.

4. Add a second operator-alignment branch centered at $(-0.3, 0.4)$ to Example 2. Draw the union of the two operator disks and determine which branch is closer to the spectral center.

5. Explain why multiplying the first parameter coordinate by 100 changes an unweighted Euclidean separation. Propose a dimensionless metric that preserves the intended relative scales.

6. Construct an example in which no sampled grid point satisfies both criteria but an unsampled common witness exists.

7. State which additional contract would be required to make an omitted-hopping norm part of the admissibility decision rather than a diagnostic.

14.13 Chapter summary

Spectral fitting, direct Hamiltonian regression, and analytic restrictions on particular tight-binding classes predate this framework. The role of the framework is to make a narrower simultaneous-adequacy question explicit. A model family is compatible with two declared criteria only when the same parameter lies in both thresholded admissible sets. A common witness proves that such a parameter exists. A positive, globally valid lower bound proves separation; failure of a finite search does not. Quadratic losses make this logic visible through ellipses and finite unions of ellipses, but their analytic certificates remain conditional on positive curvature, domain geometry, and the declared alignment family. Training data define the decision, withheld data evaluate selected candidates, and locality remains a diagnostic unless it is explicitly promoted to a thresholded criterion. These distinctions allow later bulk and impurity chapters to use the language of adequacy without turning a bounded synthetic calculation into a material claim.

CHAPTER 15

Evidence for Model Adequacy

Status evidence framework and retained-status interpretation; no new scientific validation or uncertainty-quantification result is claimed.

Chapter 11 defined bounded claims, oracles, metrics, tolerances, dispositions, and the five evidence classes used throughout the monograph. This chapter applies that vocabulary to the reduction program's verification inventory, convergence questions, validation targets, uncertainty sources, provenance, and negative results.

15.1 Verification in the reduction program

15.1.1 Software verification

Software-verification evidence uses public contracts, schemas, exact language semantics, or otherwise independently stated behavior as its oracle. For the present program, relevant subjects include

- construction of represented-operator records and their intrinsic invariants;
- basis, geometry, energy-reference, and spin metadata;
- Hermiticity and symmetry analyzers;
- compatibility checks that precede arithmetic;
- signed difference and residual conventions;
- serializer field vocabularies and exact promised round trips;
- tight-binding limiting cases and Fourier conventions;
- alignment recovery under known synthetic unitary transformations;
- continuum-embedding behavior for synthetic envelopes; and

- external-tool lifecycle, stream capture, and provenance manifests.

Tests should use the supported public surface and an oracle that does not simply reproduce the production algorithm. Exact contracts use exact acceptance; numerical contracts use explicit tolerances justified by the operation and scale. Helpers and fixtures remain distinguishable from evidence-bearing cases.

A round trip establishes only the promised wire representation. A compatibility result establishes only that the represented prerequisites checked by that operation agree. Neither proves that recorded provenance is truthful or that the physical model is adequate.

15.1.2 Numerical verification

Numerical verification asks whether the implementation approximates the stated mathematics. Controlled cases are especially important before first-principles artifacts are available. Examples include

- exact recovery of an operator after a known unitary gauge change;
- principal angles for analytically constructed subspaces;
- residual norms derived independently for small matrices;
- Hermitian tight-binding operators in zero-hopping or symmetry-limited cases;
- Fourier transforms checked against finite hand-constructed lattices;
- effective-mass stencils applied to an analytic quadratic dispersion;
- continuum solvers checked against solvable potentials where the model assumptions match; and
- embedding maps checked with synthetic normalized envelopes.

Synthetic data are labelled as such. Exact recovery on a controlled example is strong evidence about the corresponding mathematical implementation, but it is not a first-principles calculation or physical validation.

15.1.3 Compatibility witnesses and separation certificates

Chapter 14 defines compatibility as the intersection of thresholded parameter sets. The evidence supporting an intersection differs logically from the evidence supporting separation.

A retained parameter is a *common witness* only when independently evaluated losses show that the same parameter satisfies every declared criterion. It proves existence within the frozen family and thresholds. It does not prove uniqueness, stability, transferability, or scientific validation.

A *separation certificate* requires a lower bound that applies to the complete admissible sets, including every applicable alignment branch and parameter-domain boundary. Two numerically optimized points supply only an upper bound on the distance between sets. A failed grid or local search supplies neither a common witness nor a positive global lower bound. When neither form of evidence is available, the correct disposition is unresolved.

The certificate's premises are part of its evidence. For a quadratic axis-reduction argument these include the retained quadratic coefficients, positive curvature, symmetry, zero or otherwise accounted-for cross terms, the location of the centers, and the absence or explicit treatment of domain clipping. A verifier that recomputes a final scalar without authenticating these premises has not independently reconstructed the full certificate.

Threshold sensitivity has a separate role. A prospectively frozen threshold defines the original decision; a later sweep may reveal how close that decision lies to a transition. The latter is post-hoc sensitivity evidence unless it was itself frozen in advance. It must not be presented as physical calibration of the threshold or as uncertainty quantification.

15.1.4 Completed controlled benchmark inventory

Appendix G supplies a calculated example of this evidence class. It compares independently refined plane-wave and finite-difference parents, checks selected energies against Mathieu characteristic values, and compares direct and Wannier90 localization using the same spectral and overlap data. Discretization, subspace conditioning, gauge, localization, serialization, and hopping truncation remain separate error sources. Nonconverged localization attempts are retained as diagnostic evidence rather than relabelled as successful calculations.

Section G.9 separately records the canonical M3 constrained-admissible-set extension. M3 supplies one retained common witness and one analytic positive separation certificate for two frozen threshold pairs in a bounded synthetic rank-two family. Its public and standalone verifiers reconstruct the role, loss, threshold, finite-alignment, and certificate contracts. The associated threshold-sensitivity package is post-hoc exploratory evidence and does not calibrate a physical acceptance threshold.

Appendix H extends the controlled periodic benchmark to two dimensions. It verifies separable and coupled scalar parents, composite-band projection with the corrected polar gauge, mass-tensor recovery, real-space hopping reconstruction, common-estimator localization comparisons, and distinct QWZ, Hofstadter, and Haldane topology controls. Mesh, cutoff, embedding, hopping truncation, serialization, topology, optimizer-basin,

and continuation effects remain separate evidence axes.

The bounded localization study is complete as negative numerical evidence. Across the 256 frozen start/interface trajectories, the final effective endpoints comprise 196 native convergences and 60 iteration-limit nonconvergences. The frozen mesh-convergence and repeated-basin criteria are not supported. The censored convergence-time regression is exploratory and descriptive. Only one frozen trial-projection family was executed, so projection-family sensitivity is outside the scope of the closed benchmark. These results establish neither global optimization nor general Wannier convergence.

Appendix I separately verifies spin-space contracts and one-dimensional pristine–defect extraction, including blind alignment, an independent extraction route, a finite-rank spectral oracle, and continuum refinement. Those accepted synthetic records do not establish a two-dimensional defect result or material validation. The proposed two-dimensional extension is therefore kept outside the completed evidence inventory in Appendix J.

15.1.5 Finite-setting convergence

Chapter 11 explains why stability over a finite tested sequence does not by itself bound an infinite-setting limit. In the present program, cutoff and mesh errors may additionally interact. For an accepted candidate pair (E_*, K_*) and higher guards (E_+, K_+), a finite two-factor contrast is

$$I_q = \left| q(E_+, K_+) - q(E_+, K_*) - q(E_*, K_+) + q(E_*, K_*) \right| . \tag{15.1}$$

A small value detects limited interaction in that tested rectangle; it does not prove global separability of cutoff and sampling errors.

15.1.6 Derived quantities and error amplification

A pointwise energy tolerance need not control a derivative. For a central second-difference estimate,

$$E''(k_0) \approx \frac{E(k_0 + h) - 2E(k_0) + E(k_0 - h)}{h^2}, \tag{15.2}$$

pointwise energy perturbations are amplified by a scale proportional to h^{-2}. Effective-mass verification must therefore vary the stencil or fit window, report conditioning, identify the selected valley and axes, and check stability under at least one smaller scale. A converged band plot is not an effective-mass convergence study.

The same principle applies to subtraction of two large aligned operators, normalization by a small reference scale, and composition of successive reduction maps. Numerical cancellation, conditioning, and nonfinite intermediates must be evaluated at the level of the derived quantity.

15.2 Validation and uncertainty

15.2.1 Scientific validation

Scientific validation compares a declared model output with independent physical evidence appropriate to the intended use. Potential bulk-silicon references include experimental structural or band-edge measurements and higher-level theoretical results, but each comparison must match temperature, geometry, relativistic treatment, observable definition, and uncertainty as far as the claim requires.

The PBE Kohn–Sham parent is not fitted silently to the experimental silicon gap. Its disagreement with experiment belongs to parent-model assessment. A Wannier or tight-binding model is first evaluated against that selected parent to measure reduction fidelity. If the reduced model is then compared with experiment, the account must not attribute the parent's discrepancy to the reduction or claim cancellation as validation without a declared rationale.

Prospective citation note—high priority
Check the silicon-specific PBE gap statement with a settings-relevant PBE benchmark and an experimental or evaluated silicon reference. Candidate sources: Paier et al. (2006), provisional key `paier2006` [86], and Yu–Cardona (2010), `yuCardona2010` [133]. Perdew–Levy and Sham–Schluter, keys `perdewLevy1983` [90] and `shamSchluter1983` [107], address why the exact Kohn–Sham and physical gaps require care.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

For dopants, validation references must match the charge-state interpretation, periodic or isolated setting, spin and SOC branch, band-edge reference, and observable definition. A neutral periodic P:Si calculation cannot be validated as though it were directly the potential of an isolated ionized donor.

Prospective citation note—high priority
Check this charge-state and boundary-condition matching rule against defect-supercell literature. Candidate: Freysoldt et al. (2014), bibliography key `freysoldt2014` [32] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

15.2.2 Uncertainty quantification

Relevant uncertainty sources can include

- parent-model choice and external-reference uncertainty;
- cutoff, mesh, cell-size, and solver discretization;
- lattice-fit and valley-location uncertainty;
- effective-mass fit-window and stencil sensitivity;
- Wannier windows, projections, gauge, and localization choices;
- alignment and scalar energy-reference uncertainty;
- parameter non-identifiability and optimization variability;
- model-class and real-space truncation sensitivity; and
- continuum discretization, boundary, and embedding uncertainty.

These sources are not combined automatically. A valid propagation statement requires compatible quantities, an explicit dependence model, and treatment of correlation where material. Until then, the sources are reported separately as sensitivity or bounded numerical evidence.

15.3 Model development and provenance

15.3.1 Training, validation, and leakage

Model fitting uses only the frozen training set. Withheld points, band-edge quantities, operator blocks, or dopant observables designated for validation must not influence parameter selection, model-class expansion, stopping rules, or tolerance adjustment. If a validation failure motivates a new model class, that class and a new untouched validation design must be recorded explicitly; the original result remains part of the evidence history.

Prospective citation note—medium priority
Check the validation-leakage rule and the need for a new untouched validation design after model redesign against model-selection literature. Candidate: Cawley and Talbot (2010), provisional key `cawley2010` [18]; the candidate record is present for this editorial note.
Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

Visual agreement can support diagnosis but is not an acceptance rule. Plots must expose the data identity, axes, units, reference, and residual definition, while machine-readable results retain the exact values used for the decision.

15.3.2 Provenance and reproducibility

A result is reproducible only relative to retained identities and declared conditions. The minimum record depends on the operation but generally includes

- immutable input and source identities;
- code revision and dependency or executable versions;
- physical and numerical settings;
- exact commands and environment information;
- artifact manifests and checksums;
- metric and tolerance definitions;
- structured outcomes and warnings; and
- locations of externally retained large artifacts.

Reproduction does not promote the evidence class. Reproducing a tutorial calculation reproduces tutorial behavior; reproducing a synthetic test reproduces software or numerical verification; reproducing a production calculation reproduces that calculation under its recorded conditions.

Prospective citation note—medium priority
Check the minimum reproducibility record against established computational-research guidance. Candidate: Sandve et al. (2013), provisional key `sandve2013` [101]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

15.3.3 Current execution-status example

The repository retains a direct bulk-silicon bootstrap campaign in which nine SCF and nine linked NSCF Quantum ESPRESSO invocations exited successfully and reported completion. The owning disposition classifies these observations as bootstrap scientific-harness development evidence. They are not a canonical scientific workflow record and do not accept a final cutoff, mesh, lattice parameter, gap, mass, infinite-basis result, or scientific validation claim.

This example demonstrates why process completion and evidentiary acceptance are separate. The observations may inform integration and future workflow design without being discarded or promoted into a production result.

15.4 Failure and claim discipline

15.4.1 Failure and negative evidence

A failed acceptance criterion is retained rather than weakened. The diagnosis may identify an implementation defect, inadequate numerical resolution, incompatible representation, insufficient model-class expressiveness, or mismatch between a model and its intended physical use. These dispositions have different remedies.

Similarly, a well-executed negative result can be scientifically useful: no model in a frozen hierarchy may satisfy both spectral and operator criteria; no finite tested crossover radius may satisfy continuum criteria; two reduction paths may fail to commute within tolerance. The evidence must report the tested domain and must not generalize beyond it.

15.4.2 Claim-to-evidence discipline

Later results are included only when each statement can point to an applicable retained record. Software verification supports contract claims. Numerical verification supports stated mathematical or convergence claims. Scientific validation supports adequacy for a declared physical use. UQ supports only the uncertainties it actually identifies and propagates.

This discipline is not administrative overhead added after computation. It is how the project prevents a passing test, a successful executable, a smooth plot, or an aligned matrix from being mistaken for a stronger scientific conclusion.

PART III

The Bulk-Silicon Representation Program

CHAPTER 16

Definition of the Bulk-Silicon Program

Status expository synthesis of the accepted bulk-silicon physical specification; production outcomes remain proposed work.

This chapter defines the physical parent, pristine host, and representation sequence for the bulk-silicon program. Dopant-specific branches are introduced separately in Part III.

16.1 The parent description

The program asks when a first-principles Kohn–Sham description of silicon and substitutional dopants can be replaced by a smaller model without losing the operator information needed for its declared use. The question is not whether one model is universally "correct." It is whether a stated reduced-model class preserves stated features of a stated parent problem within separately declared errors.

The accepted physical conventions are owned by `specification/ksdft2Effmass.physical-specification.v1.md`. This chapter organizes those conventions as a research narrative and introduces no new physical branch.

The parent is a self-consistent Kohn–Sham density-functional calculation in the Born–Oppenheimer fixed-ion framework [50, 59]. Its output is a one-particle Kohn–Sham operator and associated finite numerical representations. It is not the complete interacting many-body Hamiltonian, and its eigenvalues are not identified here with the complete many-body excitation spectrum.

Prospective citation note—high priority
Check the distinction between Kohn–Sham eigenvalue gaps and physical fundamental or quasiparticle gaps against Perdew–Levy (1983) and Sham–Schluter (1983), bibliography keys `perdewLevy1983` [90] and `shamSchluter1983` [107] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

For the version-one program, exchange and correlation are described with the PBE generalized-gradient approximation [89], and the electron–ion interaction is represented by compatible optimized norm-conserving Vanderbilt pseudopotentials from the approved PseudoDojo family [44, 106]. These choices define the parent against which later reductions are measured. In particular, disagreement between this parent and experiment is a parent-model question, not automatically an error introduced by tight-binding or effective-mass reduction.

16.2 Pristine silicon host

The host is crystalline silicon in the diamond structure: an fcc Bravais lattice with a two-atom basis. Primitive-cell and conventional-cell coordinates may both be useful, but every retained representation must identify which lattice vectors, axes, basis sites, and reciprocal-coordinate convention it uses.

The immediate bulk pilot is scalar relativistic, non-spin-polarized, and does not include spin–orbit coupling. Its production lattice constant is to be obtained from a zero-pressure PBE bulk relaxation using the selected silicon pseudopotential. Until that calculation and its convergence record exist, no numerical lattice constant is a calculated result of this project.

The pilot targets a band-edge subspace sufficient for an orthogonal sp^3s^* reduction, including the valence and low-conduction states needed to describe the indirect Kohn–Sham gap and electron effective masses. The exact band count, projection choices, and Wannier windows belong to later methodological records. They are not fixed by the phrase "band-edge subspace" alone.

The declared bulk observables are the indirect Kohn–Sham gap, the conduction-valley position, longitudinal and transverse electron effective masses, and withheld-point band errors. These quantities serve distinct roles: they characterize the selected parent, test numerical convergence, and later evaluate reduced representations. Passing one role does not imply passing the others.

16.3 From parent models to reduced models

The program studies a sequence of representations rather than one direct leap from density-functional theory to continuum theory:

1. a converged Kohn–Sham parent problem;

2. a retained projected or Wannier subspace;

3. an aligned finite operator representation;
4. a constrained lattice model, initially an orthogonal sp^3s^* hierarchy for bulk silicon;
5. an extracted and reduced impurity operator; and
6. where justified, a continuum effective-mass description.

Projection, localization, alignment, parameter fitting, truncation, and continuum approximation are different operations. Their errors and assumptions must therefore remain identifiable. Wannier localization alone is not a low-rank approximation, and a good spectral fit alone does not establish agreement of aligned operator matrix elements.

16.4 Bulk research questions

The bulk program asks which finite subspace preserves the declared silicon features, whether a prescribed tight-binding class can satisfy both spectral and aligned-operator criteria, and which tested class is the least complex one that does so. These are proposed questions rather than completed findings.

16.5 Interpretive limits

A finite periodic supercell is not an isolated impurity. A Kohn–Sham band gap is not an exact quasiparticle gap. Spectral agreement is not operator equality. Matrix subtraction is not physically meaningful before common-space and energy alignment. Software execution and passing tests establish neither numerical convergence nor scientific validation unless the corresponding evidence and acceptance rules are present.

Prospective citation note—medium priority
Check the isolated-impurity limitation for a finite periodic supercell against defect-physics literature. Candidate: Freysoldt et al. (2014), bibliography key `freysoldt2014` [32] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

These limits are part of the scientific problem rather than editorial caveats: they determine which comparisons are well posed and which conclusions the resulting hierarchy can support.

CHAPTER 17

Construction of the First-Principles Bulk Parent

Status accepted model and numerical protocols are described; no production calculation or converged value is claimed.

This chapter describes how the pristine-silicon Kohn–Sham parent is to be constructed and recorded. The physical branch is owned by `specification/ksdft2Effmass.physical-specification.v1.md`; the numerical protocol is owned by `specification/ksdft2Effmass.numerical-specification.v1.md`. The present account does not replace either specification.

17.1 Declared bulk branch

The immediate parent problem is pristine silicon in the diamond structure, represented by a two-atom primitive cell for primary bulk calculations. The accepted branch uses PBE, a scalar-relativistic PseudoDojo optimized norm-conserving Vanderbilt silicon pseudopotential, no spin–orbit coupling, and no spin polarization. The production lattice parameter is an output of a zero-pressure PBE optimization using the selected pseudopotential; it is not inserted from experiment by convention.

This branch defines a Kohn–Sham reference. Its known physical limitations and any comparison with experiment are parent-model issues. They remain distinct from convergence of the numerical representation and from later errors caused by projection or model reduction.

17.2 External-program boundary

Quantum ESPRESSO is responsible for self-consistent, non-self-consistent, band-path, and lattice-optimization calculations [39]. The project does not reimplement plane-wave

Kohn–Sham DFT. For the declared workflow, `pw.x` is the relevant executable for the bulk electronic-structure stages. Wannier construction and the QE–Wannier interface are addressed in Chapter 18; they remain responsibilities of Wannier90 and `pw2wannier90.x`, respectively.

Prospective citation note—low priority
Check the stated current Quantum ESPRESSO capabilities and executable boundary against the 2017 suite paper in addition to the existing 2009 paper. Candidate: Giannozzi et al. (2017), provisional key `giannozzi2017` [37]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The software in this repository begins at explicit integration and provenance boundaries. It may prepare or validate declared inputs, record execution identity, parse retained outputs, and construct downstream representations. A successful process exit does not establish convergence, correctness of the physical branch, or scientific acceptance.

No production executable is invoked in developing this monograph. Production QE and Wannier90 execution is a separately authorized activity with declared inputs, resources, outputs, and retention policy.

17.3 Pseudopotential identity

The numerical specification records the approved silicon pseudopotential by source table, release, filename, relativistic form, valence count, cutoff hints, and SHA-256 hashes of both compressed and decompressed files. This identity is part of the physical and numerical parent. A filename alone is insufficient, and a later download is not assumed identical merely because its URL or element label matches.

The PseudoDojo hints are starting information for convergence, not accepted production cutoffs. They are expressed in Hartree, whereas Quantum ESPRESSO uses Rydberg for `ecutwfc`; the conversion and the units of every recorded setting must therefore be explicit. Phosphorus and boron pseudopotentials are not selected by the bulk protocol and belong to their later specialization gates.

Prospective citation note—medium priority
Verify the Hartree cutoff hints and Quantum ESPRESSO `ecutwfc` Rydberg convention against the PseudoDojo source and a versioned official `pw.x` input reference. Candidate keys: `vansetten2018` [106] and provisional `quantumEspressoPwInput` [96]; verify and

record the documentation version and access date.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

17.4 Controlled convergence

Numerical convergence is evaluated by changing one controlled variable at a time while holding the physical branch and the remaining numerical settings fixed. A parameter is accepted only when the declared target observables meet the applicable stability criteria and the retained record identifies both accepted and rejected candidates.

17.4.1 Plane-wave and charge-density cutoffs

The cutoff study begins from a monotone sequence that brackets and extends the approved pseudopotential hints. For the norm-conserving silicon potential, the first charge-density cutoff trial is tied explicitly to the wavefunction cutoff; a separate ratio refinement is required if charge-density sensitivity controls the result. The smallest pair meeting every declared tolerance is selected, with a higher guard setting retained when feasible.

A total-energy plateau alone is not sufficient when a stricter setting is required by the indirect gap, valley position, effective masses, stress, or another declared parent observable. Conversely, a higher cutoff is not adopted merely because it is more expensive or appears more conservative.

17.4.2 Brillouin-zone sampling

Four sampling purposes remain separate:

SCF mesh. A uniform Monkhorst–Pack sequence [76] converges the charge density and total energy of the primitive bulk cell.

Band path. An explicitly recorded high-symmetry path reports the band structure and supplies declared withheld points.

Valley sampling. A local one-dimensional refinement locates the conduction minimum along the selected Γ–X-type direction.

Effective-mass sampling. Local stencils around the selected conduction minimum estimate longitudinal and transverse curvature.

A dense path is not a substitute for an SCF integration mesh. The valley and mass calculations use the accepted self-consistent density but require their own coordinates, refinement intervals, fit or finite-difference rules, and stability checks.

17.5 Equilibrium lattice procedure

The production lattice parameter is determined from a symmetric set of compressed and expanded primitive cells. Every point uses identical numerical settings and retains energy and stress conventions. An equation of state or a stated local polynomial is fit near the minimum, with included points, excluded points, residuals, and uncertainty recorded. Refinement continues according to the accepted numerical criterion, followed by a calculation or constrained relaxation that checks the selected minimum.

The resulting lattice parameter is then frozen for compatible bulk and dopant lattice vectors. An experimental lattice parameter may be evaluated as an external validation branch, but it does not silently replace the internally consistent PBE parent geometry.

17.6 Target parent quantities

The bulk parent record eventually supplies

- the equilibrium lattice parameter and its numerical uncertainty;
- the indirect Kohn–Sham gap relative to the declared band-edge convention;
- the conduction-valley position in an explicitly stated reciprocal or path coordinate;
- longitudinal and transverse electron effective masses, including fit axes and stencil stability; and
- selected band energies at points withheld from later fitting.

These are future calculated results. The accepted numerical specification fixes how stability is judged but deliberately contains no converged cutoff, mesh, lattice parameter, gap, valley coordinate, or mass. Values cannot be promoted into this chapter until their calculation identity and retained provenance are available.

17.7 Provenance record

A reproducible parent calculation requires more than its input text. The compact record must identify, as applicable,

- repository revision and immutable input identities;
- executable path, version, and build or environment provenance;
- pseudopotential source, release, checksums, and recorded license;
- physical branch, cell vectors, positions, units, and symmetry choices;
- cutoff, mesh, occupation, solver, and convergence settings;
- MPI and OpenMP configuration and the exact command;
- expected and observed output inventory; and
- checksums and locations for externally retained artifacts.

Large wavefunctions, densities, restart state, scratch directories, and dense native outputs remain outside Git. Compact sanitized inputs, manifests, checksums, software identities, and calculation-status summaries may be retained when authorized.

17.8 Tutorial and production evidence

The repository also defines an inactive Quantum ESPRESSO tutorial campaign. Those exercises are intended to learn executable behavior, file boundaries, stream capture, failure modes, and provenance mechanics. Even if executed later, tutorial observations remain software-workflow evidence. They do not become the production silicon parent, establish numerical convergence, or validate the scientific model.

This distinction allows operational learning to proceed without contaminating the evidentiary status of the research calculation. The parent model becomes a calculated project result only through its own authorized execution, convergence record, and scientific interpretation.

CHAPTER 18

Retained and Localized Bulk Representations

Status technical framework and planned diagnostics; no accepted Wannier operator or alignment result is claimed. Selected foundational references are included, while a broader literature review remains future editorial work.

The Kohn–Sham parent is naturally described by Bloch states, whereas local lattice models and substitutional perturbations are naturally discussed in real space. Moving between these descriptions requires several operations that must not be collapsed into the single word "Wannierization." This chapter separates subspace selection, projection, disentanglement, gauge selection, localization, basis transformation, real-space truncation, and alignment.

18.1 Representation construction

18.1.1 Bloch representation

For a periodic parent problem, Bloch states may be written

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}), \tag{18.1}$$

where $u_{n\mathbf{k}}$ has the periodicity of the crystal lattice, n is a band label, and $\mathbf{k}$ is a crystal momentum expressed in a declared reciprocal coordinate convention. The eigenvalue relation at fixed $\mathbf{k}$ is

$$\widehat{H}(\mathbf{k})|\psi_{n\mathbf{k}}\rangle = \varepsilon_{n\mathbf{k}}|\psi_{n\mathbf{k}}\rangle. \tag{18.2}$$

Band labels alone do not provide a globally smooth frame. Degeneracies, crossings, and entanglement with excluded bands make the retained subspace, not an informal list of band indices, the stable object to identify. The controlled calculation in Appendix G shows this boundary explicitly: energy-ordered scalar labels become poorly conditioned as adjacent gaps close, while a grouped subspace can remain transportable.

18.1.2 Target subspace

At each sampled **k**, let $\mathcal{S}(\mathbf{k})$ be an M-dimensional retained subspace with orthogonal projector

$$P(\mathbf{k}) = \sum_{m=1}^{M} |\psi_{m\mathbf{k}}\rangle\langle\psi_{m\mathbf{k}}| \tag{18.3}$$

for any orthonormal frame spanning that subspace. The immediate bulk target must contain the valence and low-conduction information needed for the declared sp^3s^* reduction, indirect gap, conduction valley, and electron masses. The exact rank, energy windows, excluded states, and interpolation domain remain outputs of the Stage 03 target-subspace task.

A target subspace specification must say which parent calculation it consumes, which **k** grid it uses, how states are selected or optimized, and over which energies and reciprocal-space region fidelity is required. A visually plausible band plot does not uniquely define this object.

18.1.3 Projection and disentanglement

Initial orbital projections provide trial directions from which a retained frame may be constructed. Projection is an overlap operation between trial orbitals and parent states; it is not by itself a proof that the desired subspace has been selected.

When the desired bands are entangled with other states, disentanglement selects an M-dimensional subspace from a larger outer space [115], often subject to inner and outer energy-window constraints. This changes the selected subspace and is therefore conceptually distinct from a unitary rotation within an already fixed M-dimensional subspace. The following operations must remain separate:

Projection. Measures or constructs overlap with specified trial functions.

Disentanglement. Selects a retained subspace from a larger candidate space.

Gauge transformation. Changes the orthonormal frame within the same retained subspace.

Localization. Chooses a frame according to a real-space localization objective.

Truncation. Discards selected matrix elements or real-space blocks after representation.

Only the last of these directly removes represented couplings. Wannierization alone is therefore not described as a low-rank approximation.

18.1.4 Wannier transformation

Let

$$\{|\psi_{m\mathbf{k}}\rangle\}_{m=1}^{M} \tag{18.4}$$

be an orthonormal Bloch frame spanning the retained subspace on a uniform grid of $N_{\mathbf{k}}$ points. A corresponding set of Wannier functions [129] is

$$|w_{n\mathbf{R}}\rangle = \frac{1}{\sqrt{N_{\mathbf{k}}}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \sum_{m=1}^{M} |\psi_{m\mathbf{k}}\rangle U_{mn}(\mathbf{k}), \tag{18.5}$$

where $\mathbf{R}$ is a Bravais-lattice vector and $U(\mathbf{k}) \in U(M)$ specifies the selected frame.

For a single isolated band, $U(\mathbf{k})$ reduces to a momentum-dependent phase. For a composite subspace it mixes the retained states. Replacing $U(\mathbf{k})$ by another unitary frame changes the individual Wannier functions while preserving the subspace projector. The Wannier basis is therefore gauge dependent even though the retained subspace is gauge invariant.

Prospective citation note—medium priority
Check the composite-band unitary-frame construction and projector invariance against Wannier (1937), Marzari–Vanderbilt (1997), and the modern review. Candidate keys: `wannier1937` [129] and `marzari2012` [72], plus provisional `marzari1997` [73]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The overlap quantities supplied to a Wannier construction must correspond to the parent representation actually used. For a PAW parent, the transformation from smooth auxiliary states to all-electron states contributes augmentation terms to the relevant matrix elements; Rostgaard gives an explicit derivation for Wannier overlaps [98]. The present norm-conserving branch does not inherit PAW corrections, but this example demonstrates why interface matrix elements cannot be transferred between pseudopotential or augmented-wave formalisms without an explicit contract.

Prospective citation note—low priority
Check the PAW augmentation qualification for Wannier overlaps directly against Rostgaard's derivation and Blochl's primary PAW paper, bibliography keys `rostgaard2009` [98] and `blochl1994paw` [8] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

18.1.5 Localization

Spatial localization depends on the regularity of the selected Bloch frame over the Brillouin zone [57]. Maximally localized Wannier functions choose a gauge by minimizing a declared spread functional [72]. The spread and its decomposition are diagnostics of that construction, not standalone measures of operator fidelity.

Prospective citation note—medium priority
Check the regularity/localization claim in more than the one-dimensional setting. Candidate: Brouder et al. (2007), provisional key `brouder2007` [13], alongside `kohn1959` [57].

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

A small spread does not guarantee correct band interpolation, correct orbital character, symmetry compatibility, or alignment with a separately generated basis.

The construction record must retain at least the initial projections, disentanglement windows when applicable, localization settings, convergence history, Wannier centers, spreads, and Wannier90 software identities [91]. Multiple candidate constructions should consume the same accepted QE–Wannier interface artifacts when they are intended to isolate sensitivity to projection or localization choices.

18.2 Hamiltonian reconstruction and validation

18.2.1 Reciprocal- and real-space Hamiltonians

In the selected Bloch frame, let $H^{(\mathrm{W})}(\mathbf{k})$ denote the retained Hamiltonian matrix. Its real-space blocks are

$$H^{(\mathrm{W})}_{mn}(\mathbf{R}) = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} H^{(\mathrm{W})}_{mn}(\mathbf{k}), \tag{18.6}$$

subject to the precise Fourier and degeneracy conventions used by the producing software. The inverse transform reconstructs an interpolated reciprocal-space matrix,

$$H^{(\mathrm{W})}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} H^{(\mathrm{W})}(\mathbf{R}), \tag{18.7}$$

when all represented blocks and the matching convention are retained.

Real-space decay can be diagnosed with quantities such as $\|H^{(\mathrm{W})}(\mathbf{R})\|_{\mathrm{F}}$. Discarding distant blocks is a separate approximation whose cutoff, boundary convention, and induced spectral and operator residuals must be reported. The existence of localized functions does not authorize an unrecorded hopping truncation. Localization selects the coordinates in

which a short-range approximation is attempted; it does not change the complete represented spectrum. Appendix G shows that smooth and rough gauges can preserve the complete operator while producing substantially different finite-range errors.

18.2.2 Validation of a candidate Wannier representation

Stage 03 declares four complementary diagnostic families:

- interpolation error over a declared domain, including withheld points;
- band-edge errors for the bulk quantities used downstream;
- Wannier centers and spreads, including unresolved symmetry or localization pathologies; and
- decay of the real-space Hamiltonian blocks.

These diagnostics answer different questions. Interpolation tests whether the represented subspace reproduces selected parent eigenvalues. Centers and spreads characterize the chosen localized frame. Real-space decay informs later truncation choices. None alone establishes a physically aligned basis for subtracting an independently generated operator.

Acceptance of a bulk Wannier operator requires reproducible subspace and gauge construction, a declared interpolation domain, and explicit truncation status. No such accepted operator is asserted in this chapter.

18.3 Alignment and comparison

18.3.1 Why independent constructions require alignment

Pristine and doped calculations, or two independent pristine constructions, may produce localized orbitals with different phases, orderings, centers, mixtures, and subspaces. Even identical software settings do not force identical gauges. Directly subtracting their raw matrices can therefore measure coordinate choice rather than a physical perturbation.

Before an identification map is formed, the comparison records must establish compatible

- cell and supercell geometry;
- atomic-site and lattice-translation mapping;
- orbital identities and ordering;

- retained dimensions and spin conventions;
- units and Fourier conventions; and
- scalar energy references.

For doped systems, the matched pristine comparison must use the same underlying supercell convention. Structural relaxation is part of the physical impurity branch, but it also makes the geometry map nontrivial and therefore explicit.

18.3.2 Subspace diagnostics

Let B_{r} and B_{c} contain orthonormal basis vectors for reference and candidate spaces in a common ambient representation. Their overlap matrix is

$$S = B_{\mathrm{r}}^{\dagger} B_{\mathrm{c}}. \tag{18.8}$$

The singular values σ_j of S define principal angles [7]

$$\theta_j = \arccos \sigma_j, \tag{18.9}$$

within the applicable numerical range. These quantities diagnose subspace agreement before coordinate alignment. Rank loss or large principal angles cannot be repaired merely by relabelling orbitals. No universal singular-value or neighboring-overlap cutoff is assumed: these diagnostics must be interpreted with external gaps, reciprocal-mesh refinement, and the intended downstream use.

When the spaces are sufficiently compatible, an orthogonal-Procrustes or polar construction can supply a unitary coordinate identification [104, 46]. Schönemann supplies the primary real orthogonal least-squares formulation, while Higham's polar-factor best-approximation result supports the complex unitary form used here. The exact map, its direction, numerical conditioning, and residual diagnostics must be stored. This method is a planned Stage 05 operation, not an alignment result already obtained by the project.

18.3.3 Transport and comparison

For an alignment $A : \mathcal{S}_{\mathrm{r}} \to \mathcal{S}_{\mathrm{c}}$, the candidate operator is transported to the reference space as

$$H_{\mathrm{c}\to r} = A^{\dagger} H_{\mathrm{c}} A. \tag{18.10}$$

Only after energy-reference compatibility and the remaining metadata checks may a signed difference be formed. The difference convention must identify which operator is candidate, reference, pristine, or doped rather than relying on an ambiguous symbol ΔH.

The alignment protocol is to be tested first on synthetic, controlled gauge transformations where the expected map and transformed operator are known. First-principles sensitivity tests then vary windows, projections, and localization choices while preserving the relevant comparison controls. Exact recovery in a synthetic test verifies software and mathematics; it does not by itself validate a physical alignment between independent calculations.

18.3.4 Energy and gauge sensitivity

Alignment uncertainty is not exhausted by the unitary map. The eventual record must report principal angles, overlap residuals, gauge-equivariance error, and sensitivity to the construction choices used to produce both spaces. The operational scalar energy-alignment estimator and its uncertainty must also be frozen before interpreting a pristine–doped difference as an impurity operator.

If rank compatibility, geometry correspondence, spin convention, energy zero, or gauge validity remains unresolved, the comparison stops. Reporting that boundary is preferable to constructing a numerically convenient subtraction whose physical meaning is unidentified.

CHAPTER 19

Bulk Reduced-Model Classes

Status proposed bulk-reduction framework; no fitted or selected bulk tight-binding model is claimed.

Reduction replaces an accepted parent representation by a member of a more restricted model class for a declared purpose. It is not merely a change of basis.

19.1 Bulk reduction program

19.1.1 Reduction as a constrained problem

Let H_{p} be a compatible finite representation of a parent operator and let

$$\mathfrak{M}_j = \{H_j(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta_j\} \tag{19.1}$$

denote a model class indexed by j. The parameter domain Θ_j includes all symmetry, range, orbital, and sign conventions imposed by that class. A reduction operation must state

- the parent operator and its retained representation;
- the candidate model class and parameter constraints;
- the fitting or projection map;
- the training data and objective;
- the withheld validation data;
- the metrics and tolerances associated with the intended use; and
- the provenance of the resulting parameters and operator.

The least complex model is meaningful only relative to a nested or otherwise ordered family and a complete acceptance rule. Fewer parameters do not make a model acceptable when one required criterion fails.

19.1.2 Bulk orthogonal sp^3s^* hierarchy

The initial bulk lattice model is an orthogonal Slater–Koster sp^3s^* model for diamond-structure silicon [113, 127]. Each silicon site carries a prescribed orbital set, and symmetry relates hopping matrix elements to a smaller parameter vector. The executable operator must define site, orbital, spin, neighbor-shell, lattice, Fourier, and energy-zero conventions before any parameter value is interpreted.

The first model class uses the approved nearest-neighbor structure. If that class cannot satisfy the declared criteria, later classes may introduce second-neighbor or selected symmetry-allowed corrections in a fixed sequence. The sequence must be frozen before the compatibility search so that model selection is not changed after inspecting the outcomes.

Hermiticity, crystal symmetries, and controlled limiting cases are software-verification requirements for every model class. Passing them shows that the implementation follows its mathematical contract; it does not show that the class is expressive enough for the silicon parent.

19.1.3 Two bulk reduction routes

Stage 04 compares two paths that share the same accepted Kohn–Sham parent:

$$\text{direct path:} \quad \text{DFT} \longrightarrow \text{TB}, \tag{19.2}$$

$$\text{operator-mediated path:} \quad \text{DFT} \longrightarrow \text{Wannier} \longrightarrow \text{TB}. \tag{19.3}$$

The direct path fits selected spectral and band-edge information. The mediated path first constructs a validated Wannier operator and then aligns the Wannier and tight-binding orbital spaces so that represented operator residuals can be defined.

The paths cannot be compared merely because both produced parameter vectors. Their result records must identify the same accepted parent dataset and source manifest and must carry compatible physical, numerical, pseudopotential, workflow, energy, representation, artifact, and validation metadata. This provenance-compatible join prevents path differences from being attributed to reduction when the branches actually consumed different parents.

In the special case of an unconstrained finite Fourier class, direct least-squares fitting and truncation of the complete localized operator are the same orthogonal projection only when they use the same complete uniform mesh, equal weights, aligned frame, Fourier convention, and retained model class. Nonuniform weights, restricted fitting domains, parameter-sharing constraints, or different gauges define different reduction problems and need not produce the same coefficients. Appendix G verifies the equality under matched conditions and measures its loss when the objective is changed one condition at a time.

19.1.4 Training and withheld validation

Training data determine or constrain $\boldsymbol{\theta}$; withheld data test the resulting model after fitting. The two sets must be frozen and disjoint. For the bulk pilot, validation includes withheld band energies, the indirect gap, conduction-valley position, and longitudinal and transverse electron effective masses. Operator metrics are added only where a common aligned representation exists.

A weighted spectral objective may be written abstractly as

$$\mathcal{L}_E(\boldsymbol{\theta}) = \sum_{q \in \mathcal{T}_E} w_q \, \ell_q(H_j(\boldsymbol{\theta}), H_{\mathrm{p}}) \,, \tag{19.4}$$

where $\mathcal{T}_E$ is the frozen spectral training set, w_q are declared weights, and every loss term ℓ_q defines its units and normalization. The objective value on training data is not a withheld validation result.

19.1.5 Aligned operator objective

Let $H_{\mathrm{W}}(\mathbf{R})$ be the retained real-space Wannier blocks and $H_{\mathrm{TB}}(\mathbf{R}; \boldsymbol{\theta})$ the corresponding tight-binding blocks. For a declared set $\mathcal{U}$ of symmetry-compatible alignment maps, one possible operator objective is

$$\mathcal{L}_H(C, \boldsymbol{\theta}) = \sum_{\mathbf{R}} \omega_{\mathbf{R}} \left\| H_{\mathrm{W}}(\mathbf{R}) - C H_{\mathrm{TB}}(\mathbf{R}; \boldsymbol{\theta}) C^{\dagger} \right\|_{\mathrm{F}}^2 , \qquad C \in \mathcal{U}. \tag{19.5}$$

The direction of C, the retained $\mathbf{R}$ blocks, shell weights $\omega_{\mathbf{R}}$, normalization, units, and treatment of omitted blocks must be fixed. Optimizing over C does not waive the subspace, geometry, orbital, spin, or energy-reference compatibility requirements established in Chapter 18.

The residual can be decomposed by onsite and hopping contribution, orbital block, symmetry channel, and neighbor shell. Such a decomposition may identify where a model class lacks expressiveness, but it remains a reduction diagnostic relative to the selected parent representation.

19.1.6 Spectral and operator admissibility

Chapter 14 owns the general definitions, quadratic geometry, and witness-versus-certificate logic. Here those definitions are specialized to the bulk hierarchy. That chapter's $\mathcal{A}_{E,j}$ and $\mathcal{A}_{H,j}$ are subsets of the parameter domain Θ_j; the fraktur symbols below denote their images in the Hamiltonian class $\mathfrak{M}_j$.

For a model class $\mathfrak{M}_j$, define the spectral-admissible image

$$\mathfrak{A}_{E,j}(\tau_E) = \{ H_j(\boldsymbol{\theta}) \in \mathfrak{M}_j : \mathcal{L}_E(\boldsymbol{\theta}) \le \tau_E \} \tag{19.6}$$

and the operator-admissible set

$$\mathfrak{A}_{H,j}(\tau_H) = \left\{ H_j(\boldsymbol{\theta}) \in \mathfrak{M}_j : \min_{C \in \mathcal{U}} \mathcal{L}_H(C, \boldsymbol{\theta}) \leq \tau_H \right\}. \tag{19.7}$$

The tolerances τ_E and τ_H are proposed acceptance inputs, not values selected by this chapter. They must be fixed before the compatibility result is interpreted.

A common acceptable Hamiltonian exists when

$$\mathfrak{A}_{E,j}(\tau_E) \cap \mathfrak{A}_{H,j}(\tau_H) \neq \varnothing. \tag{19.8}$$

One explicit Hamiltonian in this intersection is a common witness. Separately optimized spectral and operator models are not a common witness unless they are the same represented candidate.

If the intersection is empty, a declared dimensionless metric d_{TB} on common canonical model coordinates may define the set separation

$$\delta_j^* = \inf_{\substack{H_E \in \mathfrak{A}_{E,j} \\ H_H \in \mathfrak{A}_{H,j}}} d_{\mathrm{TB}}(H_E, H_H). \tag{19.9}$$

A numerical optimizer returning two parameter vectors does not by itself prove that this infimum has been found: their distance is an upper bound, not a positive global lower bound. A separation claim requires a certificate covering the full frozen domain and alignment family. Optimization stability, parameter identifiability, class sensitivity, and the definition of the metric remain part of the evidence.

The bulk model-selection rule is the lowest-order tested class satisfying all prespecified criteria. If no tested class passes, the correct outcome is that no accepted reduced class was found in the tested hierarchy; complexity is not expanded after the fact merely to force a positive result.

19.2 Path dependence and error accounting

19.2.1 Path dependence and noncommutativity

Reduction and impurity extraction need not commute. The program therefore compares

$$\text{extract then reduce}, \tag{19.10}$$

$$\text{reduce parent operators then extract}. \tag{19.11}$$

After aligning their outputs, a declared norm measures the defect between the two paths. Similar *route defects* compare direct and Wannier-mediated bulk reductions and test gauge

equivariance. A route defect is interpretable only when its comparison domain, weights, model class, gauge, and normalization are part of the definition.

A nonzero defect is not automatically an implementation bug. It may arise from projection, fitting, truncation, incompatible approximation domains, or sensitivity to alignment. The final interpretation requires precisely defined paths, compatible outputs, a stated norm, a validation domain, and evidence that separates numerical artifacts from structural noncommutativity.

19.2.2 Error accounting

Every reduction result remains relative to the chosen Kohn–Sham parent. Parent-model error concerns that parent's relationship to independent physical references. Numerical error concerns approximation of the parent calculation and the reduction algorithms. Model-reduction error concerns replacement by a restricted class. Impurity finite-size error and continuum discretization may require still more specific components.

No scalar "total error" is reported until an owning contract defines how its components relate and whether their units, norms, correlations, and validation domains are compatible. The immediate purpose of the hierarchy is instead to make each loss visible and to determine which scientific statements survive a particular sequence of reductions.

CHAPTER 20

Selecting the Bulk-Silicon Representation

Status decision framework and proposed work; no bulk representation or reduced model has crossed its acceptance boundary.

20.1 The unresolved bulk question

The first unresolved scientific question is whether one member of a prescribed bulk tight-binding hierarchy can satisfy both spectral and aligned-operator criteria relative to the same accepted Kohn–Sham parent. The answer may take one of three forms:

1. the first tested class contains a common admissible Hamiltonian;
2. a later, more expressive class is the first to contain one; or
3. no tested class satisfies the joint criteria.

None of these outcomes is assumed. If no common model is found, the set separation and its orbital-, symmetry-, or shell-resolved contributions may be more informative than forcing an expanded fit. Such a result would identify the information that the tested class cannot preserve.

20.2 Near-term bulk program

The next scientific sequence remains within pristine silicon:

1. complete the canonical convergence boundary for the declared PBE/PseudoDojo parent;
2. determine and accept the internally consistent equilibrium lattice reference;

3. construct the production SCF parent and purpose-specific band-path, valley, and mass children;
4. freeze the target subspace, QE–Wannier interface, and validated Wannier operator;
5. verify the alignment machinery first on controlled transformations and then on independent bulk constructions; and
6. compare direct and Wannier-mediated reductions on frozen training and withheld datasets from the same parent.

This is a scientific dependency sequence, not execution authorization. Each protected calculation still requires its applicable activation, preflight, resource report, and human authorization.

PART IV

The Doped-Silicon Representation Program

CHAPTER 21

Doped-Silicon Parent Systems and Compatibility

Status expository synthesis of the accepted phosphorus and boron branches; no production dopant calculation or impurity operator is claimed.

The dopant program begins only after the bulk representation and its alignment conventions have been identified. It treats phosphorus and boron as distinct physical branches rather than interchangeable instances of an abstract defect.

21.1 Compatible host and dopant branches

Each doped calculation uses a periodic supercell built from the frozen host lattice vectors, with one substitutional dopant and a matched pristine comparison supercell. In the primary dopant branches, internal atomic coordinates are relaxed while the approved bulk lattice vectors remain fixed; unrelaxed structures are retained only as diagnostics. Geometry and site correspondence must be identified before pristine and doped operators are compared.

The immediate non-spin-polarized, non-SOC bulk pilot does not by itself supply every host reference required by the dopant program. A phosphorus comparison requires a compatible bulk reference with documented spin and SOC conventions. Final boron acceptor and continuum claims require a matching fully relativistic, noncollinear SOC bulk reference. Operators from these different spin spaces are not directly subtractable without an explicit identification.

21.2 Substitutional phosphorus

The phosphorus branch replaces one silicon atom at a diamond-lattice site by phosphorus in a periodic supercell. Its primary version-one charge state is neutral P_{Si}^{0}. With the ac-

cepted pseudopotential valences, this system has one additional valence electron relative to the matched pristine silicon supercell and therefore an odd valence-electron count. The primary branch is consequently scalar relativistic, collinearly spin polarized, and non-SOC.

The neutral supercell contains both the donor-core perturbation relative to silicon and the additional electron that occupies and screens that perturbation. It must not be relabelled silently as an isolated ionized donor. A charged P_{Si}^{+} calculation, or an extraction intended to isolate an ionized-core potential, requires a separately specified controlled branch.

The eventual phosphorus questions concern donor energies relative to a compatible conduction-band reference, valley composition, localization or envelope diagnostics, and the spatial structure of an aligned impurity operator. Each finite supercell represents a periodic dopant array. An isolated-donor interpretation requires supercell-size evidence rather than the presence of a single substituted atom in the simulation cell.

Prospective citation note—medium priority
Check the periodic-dopant-array and dilute-limit warning against defect-supercell literature. Candidate: Freysoldt et al. (2014), bibliography key `freysoldt2014` [32] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

21.3 Substitutional boron

The boron branch replaces one silicon atom by boron and uses neutral B_{Si}^{0} as its primary version-one charge state. The accepted valence convention gives one fewer valence electron than in the matched pristine supercell and again produces an odd electron count.

For final acceptor and continuum claims, the physical specification requires a fully relativistic, noncollinear spinor calculation with spin–orbit coupling. A scalar-relativistic non-SOC calculation may be retained only as an explicitly labelled method-development branch. This distinction prevents an early simplification from being presented as the final physical treatment of the spin–orbit-coupled valence manifold.

Prospective citation note—medium priority
Check the requirement to preserve the spin–orbit-coupled valence manifold in an acceptor continuum treatment against the shallow-acceptor literature. Candidate: Baldereschi–Lipari (1973), provisional key `baldereschiLipari1973` [2]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The eventual boron questions concern acceptor energies relative to a compatible valence-band reference, valence-subspace fidelity, localization or envelope diagnostics, and the radial structure of the aligned impurity operator. Charged B_{Si}^{-} calculations are deferred to a separate controlled branch.

21.4 Dopant research questions

The dopant program asks which identifications are required before pristine and doped operators can be compared, which nested lattice classes preserve the aligned perturbation, and whether a continuum effective-mass description is adequate for a declared donor or acceptor use. Answers require retained calculations and separately declared acceptance rules.

21.5 Interpretive limits

A finite periodic supercell is not an isolated impurity. A Kohn–Sham band gap is not an exact quasiparticle gap. Spectral agreement is not operator equality. Matrix subtraction is not physically meaningful before common-space and energy alignment. Software execution and passing tests establish neither numerical convergence nor scientific validation unless the corresponding evidence and acceptance rules are present.

These limits are part of the scientific problem rather than editorial caveats: they determine which comparisons are well posed and which conclusions the resulting hierarchy can support.

Prospective citation note—high priority

Check the Kohn–Sham versus quasiparticle-gap warning against Perdew–Levy (1983) and Sham–Schluter (1983), bibliography keys `perdewLevy1983` [90] and `shamSchluter1983` [107] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

CHAPTER 22

Extraction of the Impurity Operator

Status proposed aligned-extraction contract; no phosphorus or boron impurity operator is claimed.

22.1 Purpose and terminology

This chapter defines the method by which a pristine host Hamiltonian and a doped Hamiltonian may be compared. The central rule is that an impurity operator is not obtained by subtracting two arrays merely because they have the same shape. The two operators must first be represented on identified finite spaces, placed on a common scalar energy reference, and related by an explicit coordinate map. Chapter 13 gives the general operator-comparison framework; this chapter specializes it to substitutional dopants.

Definition 22.1 (Host, doped system, defect, and dopant)**.** The *host* or *pristine* system is the declared periodic reference material without the substitution. The *doped system* is the matched periodic supercell in which one host atom is replaced by a dopant species. A *defect* is the resulting change relative to the host under the declared structural, charge, spin, and relativistic branch. A *dopant* is the chemical species producing that substitution. In this monograph the dopant label is

$$d \in \{\mathrm{P}, \mathrm{B}\}, \tag{22.1}$$

where

d the chemical dopant label.

P substitutional phosphorus in silicon.

B substitutional boron in silicon.

where P denotes substitutional phosphorus in silicon and B denotes substitutional boron in silicon. ■

Definition 22.2 (Perturbation and impurity operator)**.** A *perturbation operator* is a signed operator change relative to a declared parent operator. An *impurity operator* is the perturbation obtained from pristine and doped operators only after their state spaces, coordinates, geometry, spin structure, units, and scalar energy references have been made compatible. It is therefore an aligned represented difference, not a synonym for the raw dopant Hamiltonian and not automatically a scalar potential. ■

Definition 22.3 (Onsite potential and finite-extent operator perturbation)**.** In an ordered localized basis, an *onsite scalar potential* is diagonal in site and internal basis labels. A *finite-extent operator perturbation* may also contain orbital-dependent onsite blocks or off-diagonal bond terms that change hopping. Thus every onsite scalar potential is an operator perturbation, but a perturbation containing bond, orbital-mixing, or spin-mixing terms is not described as a scalar potential. Finite extent is a separate claim relative to a declared spatial partition and tolerance; it does not follow merely because a represented matrix has finite dimension. ■

Here and below, *alignment* means the construction of an explicit map between two represented state spaces. *Gauge* means an admissible unitary change of coordinates within one retained space. *Energy-reference alignment* means removal of a scalar offset so that energy-valued operators use the same zero. *Compatibility* means that all prerequisites for the intended operation have been checked. These operations are distinct even when a practical workflow performs them consecutively.

22.2 Matched periodic parent problems

Let the supercell be specified by positive integers N_1, N_2, N_3 and primitive lattice vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. Its lattice vectors are

$$\mathbf{L}_\mu = N_\mu \mathbf{a}_\mu, \qquad \mu \in \{1, 2, 3\}. \tag{22.2}$$

where

$\mathbf{L}_\mu$ the supercell lattice vector in direction μ.
N_μ the positive number of primitive repeats in direction μ.
$\mathbf{a}_\mu$ the primitive lattice vector in direction μ.
μ a lattice-direction index in $\{1, 2, 3\}$.

Here μ labels a lattice direction, N_μ is the number of primitive repeats in that direction,

and $\mathbf{L}_\mu$ is the corresponding supercell vector. The collective label $\mathbf{L}$ denotes the supercell shape and lattice vectors, not a single vector.

At supercell crystal momentum $\mathbf{K}$, let

$$\widehat{H}_{b,\mathbf{L}}(\mathbf{K}) \quad \text{and} \quad \widehat{H}_{d,\mathbf{L}}(\mathbf{K}) \tag{22.3}$$

where

$\widehat{H}_{b,\mathbf{L}}(\mathbf{K})$ the abstract pristine Hamiltonian at supercell momentum $\mathbf{K}$.

$\widehat{H}_{d,\mathbf{L}}(\mathbf{K})$ the abstract doped Hamiltonian at the same momentum.

b **and** d the pristine branch and the doped branch associated with the selected chemical dopant, respectively.

$\mathbf{L}$ the complete supercell geometry.

$\mathbf{K}$ the crystal momentum in the reduced supercell Brillouin zone.

denote the pristine and doped Hamiltonian operators. The subscript b means bulk or pristine, while d denotes the doped branch associated with the selected chemical dopant $d \in \{\mathrm{P}, \mathrm{B}\}$. The label $\mathbf{L}$ identifies the supercell, and $\mathbf{K}$ belongs to the reduced supercell Brillouin zone. The operators must belong to matched physical branches: exchange–correlation approximation, host pseudopotential, lattice vectors, boundary conditions, relativistic treatment, and applicable spin convention must agree. A doped supercell remains a periodic array of dopants at finite $\mathbf{L}$; it is not an isolated impurity.

The doped internal coordinates may differ from the pristine coordinates when local atomic relaxation is included. That relaxation is part of the selected physical defect branch, but it makes the site correspondence nontrivial. A *site map* is the declared correspondence between pristine sites and the host-derived sites of the doped supercell. The substituted site and every unmatched or symmetry-related site must be identified rather than inferred from matrix position.

22.3 Retained spaces and finite representations

Let

$$\mathcal{S}_{b,\mathbf{L}}(\mathbf{K}) \subseteq \mathcal{H}_{b,\mathbf{L}}(\mathbf{K}), \qquad \mathcal{S}_{d,\mathbf{L}}(\mathbf{K}) \subseteq \mathcal{H}_{d,\mathbf{L}}(\mathbf{K}) \tag{22.4}$$

where

$\mathcal{S}_{b,\mathbf{L}}(\mathbf{K})$ the finite pristine retained subspace.

$\mathcal{S}_{d,\mathbf{L}}(\mathbf{K})$ the finite doped retained subspace.

$\mathcal{H}_{b,\mathbf{L}}(\mathbf{K})$ the ambient pristine one-particle space.

$\mathcal{H}_{d,\mathbf{L}}(\mathbf{K})$ the ambient doped one-particle space.

$\subseteq$ inclusion of a retained subspace in its ambient space.

be the pristine and doped retained subspaces. The symbols $\mathcal{H}_b$ and $\mathcal{H}_d$ denote the ambient one-particle spaces, while $\mathcal{S}_b$ and $\mathcal{S}_d$ denote the finite subspaces selected for comparison. For direct unitary alignment, their dimensions must agree:

$$\dim \mathcal{S}_{b,\mathbf{L}}(\mathbf{K}) = \dim \mathcal{S}_{d,\mathbf{L}}(\mathbf{K}) = M. \tag{22.5}$$

where

dim the dimension of a finite retained space.

$\mathcal{S}_{b,\mathbf{L}}(\mathbf{K})$ **and** $\mathcal{S}_{d,\mathbf{L}}(\mathbf{K})$ the pristine and doped retained spaces.

M their common finite represented dimension.

The integer M is the represented dimension of one supercell fiber. Equal M is necessary but not sufficient for compatibility.

Choose ordered orthonormal bases

$$\mathcal{B}_b = \{|b_j\rangle\}_{j=1}^{M}, \qquad \mathcal{B}_d = \{|d_i\rangle\}_{i=1}^{M}. \tag{22.6}$$

where

$\mathcal{B}_b$ **and** $\mathcal{B}_d$ the ordered pristine and doped orthonormal bases.

$|b_j\rangle$ **and** $|d_i\rangle$ their pristine and doped basis vectors.

i, j basis-coordinate indices running from 1 through M.

M the common number of basis vectors.

The vectors $|b_j\rangle$ and $|d_i\rangle$ are basis states in the pristine and doped retained spaces; i and j are coordinate indices. The represented matrices are

$$[H_b]_{ij} = \langle b_i|\widehat{H}_{b,\mathbf{L}}(\mathbf{K})|b_j\rangle, \tag{22.7}$$

$$[H_d]_{ij} = \langle d_i|\widehat{H}_{d,\mathbf{L}}(\mathbf{K})|d_j\rangle. \tag{22.8}$$

where

$[H_b]_{ij}$ **and** $[H_d]_{ij}$ matrix entries of the pristine and doped represented Hamiltonians.

$\widehat{H}_{b,\mathbf{L}}(\mathbf{K})$ **and** $\widehat{H}_{d,\mathbf{L}}(\mathbf{K})$ the corresponding abstract operators.

$|b_i\rangle, |b_j\rangle, |d_i\rangle, |d_j\rangle$ ordered basis vectors used to take matrix elements.

$\langle\cdot|\cdot\rangle$ the Hilbert-space matrix-element pairing.

i, j row and column indices; b, d, $\mathbf{L}$, $\mathbf{K}$ retain the meanings defined above.

Thus $H_b, H_d \in \mathbb{C}^{M\times M}$ are finite matrix representations, not the abstract operators themselves. Their interpretation additionally requires the basis identities, site and orbital ordering, geometry, Fourier convention, energy unit, scalar energy zero, spin representation, and gauge.

22.4 Compatibility gate

Before alignment or subtraction, the comparison must establish all of the following:

1. matched physical parent branches and boundary conditions;
2. matched supercell shape and a declared geometry and site map;
3. compatible retained spaces and represented dimensions;
4. declared orbital, spin, and basis ordering;
5. common energy units and Fourier conventions;
6. a justified scalar energy-reference procedure; and
7. an explicit alignment direction and admissible gauge class.

A failed item is a stop condition. The method does not repair incompatible spaces by padding matrices, discarding unexplained states, sorting eigenvalues, or fitting an arbitrary unitary solely to reduce the final difference.

When both retained frames are available in one ambient coordinate system, let $B_b, B_d \in \mathbb{C}^{N_{\text{amb}} \times M}$ contain their orthonormal basis vectors as columns. The integer N_{amb} is the dimension of the common ambient discretization. The overlap matrix is

$$C = B_d^\dagger B_b \in \mathbb{C}^{M \times M}. \tag{22.9}$$

where

C the overlap matrix from pristine coordinates to doped-frame overlaps.

B_b **and** B_d matrices whose columns are the pristine and doped orthonormal frame vectors.

$\dagger$ the conjugate transpose.

$\mathbb{C}^{M \times M}$ the space of complex M-by-M matrices.

M the common retained dimension.

The matrix C measures overlap between doped and pristine retained frames; it is not a Hamiltonian. If

$$C = W \Sigma V^\dagger \tag{22.10}$$

where

C the retained-frame overlap matrix.

W **and** V unitary left and right singular-vector matrices.

Σ the diagonal matrix of nonnegative singular values.

$\dagger$ the conjugate transpose.

is a singular-value decomposition, then W and V are unitary matrices and $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_M)$ contains singular values $0 \le \sigma_j \le 1$. The principal angles are

$$\theta_j = \arccos \sigma_j, \qquad j \in \{1, \ldots, M\}. \tag{22.11}$$

where

θ_j the jth principal angle between retained spaces.

σ_j the jth singular value of the overlap matrix.

arccos the inverse cosine on $[0, 1]$.

j an index in $\{1, \ldots, M\}$; M is the retained dimension.

Each θ_j diagnoses one direction of subspace agreement. A singular value near zero signals an unresolved or nearly orthogonal direction; it cannot be fixed by relabelling the basis.

For compatible full-rank frames, the unitary polar factor

$$U_{\mathrm{pol}} = WV^\dagger \tag{22.12}$$

where

U_{pol} the unitary polar factor mapping pristine coordinates to doped coordinates under the overlap convention.

W **and** V the singular-vector matrices of C.

$\dagger$ the conjugate transpose.

is the nearest unitary factor associated with C in the standard orthogonal-Procrustes construction. Here U_{pol} maps pristine coordinate vectors to doped coordinate vectors under the stated overlap convention. It is one possible alignment ingredient, not a universal physical answer. A project alignment may additionally constrain site permutations, orbital anchors, spin frames, crystal symmetries, smoothness in $\mathbf{K}$, and periodicity across the reciprocal mesh. The selected map, its direction, conditioning, singular values, residuals, and non-uniqueness must be retained. No universal singular-value or principal-angle cutoff is asserted here.

22.5 Scalar energy-reference alignment

Let $E_{0,b}$ and $E_{0,d}$ be the scalar energy zeros assigned to the pristine and doped representations. Define

$$\overline{H}_b = H_b - E_{0,b} I_M, \qquad \overline{H}_d = H_d - E_{0,d} I_M, \tag{22.13}$$

where

$\overline{H}_b$ **and** $\overline{H}_d$ pristine and doped matrices after scalar energy alignment.
H_b **and** H_d the source represented Hamiltonian matrices.
$E_{0,b}$ **and** $E_{0,d}$ their declared scalar energy zeros.
I_M the identity matrix on the M-dimensional represented space.
b **and** d pristine and doped branches; the overline denotes energy alignment.

where I_M is the $M \times M$ identity matrix and the overline denotes an energy-aligned representation. Both $E_{0,b}$ and $E_{0,d}$ have the same energy unit as the matrix entries. Only their difference affects the extracted operator, but both source conventions must be recorded.

One general diagnostic for a scalar offset uses a declared reference-sector projector P_{ref}. Suppose a provisional coordinate alignment U has been supplied and define

$$X = H_d - U H_b U^\dagger. \tag{22.14}$$

where

X the provisional uncorrected difference in doped coordinates.
H_d **and** H_b doped and pristine represented Hamiltonians.
U a supplied provisional map from pristine to doped coordinates.
$\dagger$ the conjugate transpose.

The matrix X is the uncorrected doped-space difference. If $P_{\text{ref}} = P_{\text{ref}}^\dagger = P_{\text{ref}}^2$ projects onto a sector assumed not to contain the localized defect signal, the least-squares scalar offset on that sector is

$$c_{\text{ref}} = \frac{\operatorname{Re}\operatorname{Tr}(P_{\text{ref}} X P_{\text{ref}})}{\operatorname{rank}(P_{\text{ref}})}. \tag{22.15}$$

where

c_{ref} the least-squares scalar energy offset estimated on the reference sector.
P_{ref} the orthogonal projector onto that declared sector.
X the provisional uncorrected doped-space difference.
Tr **and** Re the matrix trace and real-part operations.
$\operatorname{rank}(P_{\text{ref}})$ the number of retained reference directions.

Here Tr is the matrix trace, Re takes the real part, and $\operatorname{rank}(P_{\text{ref}})$ is the number of retained reference directions. This formula explains a class of exterior-anchor estimators; it is not yet the frozen material estimator. It is undefined when the reference rank is zero and can be biased when the defect perturbation extends into the reference sector. The production estimator, spatial sampling region, convergence test, and uncertainty remain required inputs to the dopant program.

22.6 Common-space alignment and signed extraction

Let

$$U_d : \mathcal{S}_{b,\mathbf{L}}(\mathbf{K}) \longrightarrow \mathcal{S}_{d,\mathbf{L}}(\mathbf{K}) \tag{22.16}$$

where

U_d the selected coordinate identification from pristine to doped retained space for dopant d.

$\mathcal{S}_{b,\mathbf{L}}(\mathbf{K})$ the pristine retained space.

$\mathcal{S}_{d,\mathbf{L}}(\mathbf{K})$ the doped retained space.

L and K the supercell specification and supercell crystal momentum.

be the selected identification from pristine to doped retained space. For an equal-rank comparison, U_d is required to be unitary:

$$U_d^\dagger U_d = I_b, \qquad U_d U_d^\dagger = I_d, \tag{22.17}$$

where

U_d the equal-rank pristine-to-doped unitary identification.

$U_d^\dagger$ its adjoint and inverse.

I_b **and** I_d identity operators on pristine and doped retained spaces.

where I_b and I_d are the identity operators on the pristine and doped retained spaces. The dagger denotes the adjoint. If only a common sector is identified, U_d may instead be a declared partial isometry; then no full-space impurity operator is inferred on the unidentified complement.

The pristine operator pushed into doped coordinates is

$$\overline{H}_{b\to d} = U_d \overline{H}_b U_d^\dagger. \tag{22.18}$$

where

$\overline{H}_{b\to d}$ the energy-aligned pristine Hamiltonian represented in doped coordinates.

$\overline{H}_b$ the energy-aligned pristine matrix in its original coordinates.

U_d **and** $U_d^\dagger$ the pristine-to-doped map and its adjoint.

$b \to d$ the direction of coordinate transport.

The arrow $b \to d$ states the direction of coordinate transport. The signed impurity operator represented in doped coordinates is

$$\boxed{\Delta H_{b\to d} = \overline{H}_d - U_d \overline{H}_b U_d^\dagger}. \tag{22.19}$$

where

$\Delta H_{b\to d}$ the signed impurity matrix in doped coordinates.
$\overline{H}_d$ the energy-aligned doped Hamiltonian.
$\overline{H}_b$ the energy-aligned pristine Hamiltonian.
U_d **and** $U_d^\dagger$ the pristine-to-doped alignment and its adjoint.
$b \to d$ the common doped-coordinate comparison direction.

The first operand is the doped operator, so a positive onsite entry denotes an increase relative to the aligned host convention.

Alternatively, pull the doped operator into pristine coordinates:

$$\overline{H}_{d\to b} = U_d^\dagger \overline{H}_d U_d, \tag{22.20}$$

where

$\overline{H}_{d\to b}$ the energy-aligned doped Hamiltonian represented in pristine coordinates.
$\overline{H}_d$ the energy-aligned doped matrix in its original coordinates.
$U_d^\dagger$ **and** U_d the pullback map and its inverse orientation.
$d \to b$ the direction of coordinate transport.

and define

$$\boxed{\Delta H_{d\to b} = U_d^\dagger \overline{H}_d U_d - \overline{H}_b}. \tag{22.21}$$

where

$\Delta H_{d\to b}$ the signed impurity matrix in pristine coordinates.
$\overline{H}_d$ **and** $\overline{H}_b$ the aligned doped and pristine Hamiltonians.
$U_d^\dagger$ **and** U_d the maps that pull the doped operator into pristine coordinates.
$d \to b$ the common pristine-coordinate comparison direction.

If U_d is unitary, the two differences describe the same aligned perturbation in different coordinates:

$$\Delta H_{b\to d} = U_d \Delta H_{d\to b} U_d^\dagger. \tag{22.22}$$

where

$\Delta H_{b\to d}$ **and** $\Delta H_{d\to b}$ the same aligned perturbation represented in doped and pristine coordinates.
U_d **and** $U_d^\dagger$ the unitary coordinate identification and its adjoint.

This equality explains why the alignment direction must be recorded. The two matrices need not be entrywise equal, although unitary-invariant norms and spectra agree. The extraction operation itself performs no site mapping, gauge inference, basis rotation, geometry transformation, or energy-zero estimation; those are prerequisites represented by U_d, the overlines, and the compatibility record.

22.7 Localized matrix interpretation

After a localized basis has been justified, write one basis index as a pair (i, α), where i labels a lattice site and α labels an internal degree of freedom such as an orbital or spin component. The matrix elements of the aligned impurity operator are

$$[\Delta H]_{i\alpha,j\beta} = \langle i, \alpha | \Delta \widehat{H} | j, \beta \rangle. \tag{22.23}$$

where

$[\Delta H]_{i\alpha,j\beta}$ one represented matrix element of the aligned impurity operator.
$\Delta \widehat{H}$ the corresponding abstract perturbation operator.
i, j **and** α, β lattice-site labels and internal orbital or spin labels, respectively.
$|i, \alpha\rangle$ **and** $|j, \beta\rangle$ localized basis states.

The indices j and β label the ket-side site and internal state. Terms with $i = j$ are onsite blocks. Terms with $i \neq j$ are bond or hopping changes. Terms with $\alpha \neq \beta$ mix internal components and may describe orbital or spin mixing, depending on the basis.

A scalar onsite perturbation has the form

$$[\Delta H_{\text{pot}}]_{i\alpha,j\beta} = V_i \, \delta_{ij} \delta_{\alpha\beta}, \tag{22.24}$$

where

$[\Delta H_{\text{pot}}]_{i\alpha,j\beta}$ one matrix entry of a scalar onsite perturbation.
V_i the energy-valued scalar onsite change at site i.
i, j **and** α, β site labels and internal labels, respectively.
δ_{ij} **and** $\delta_{\alpha\beta}$ Kronecker deltas enforcing equality of their indices.

where V_i is an energy-valued onsite coefficient and δ_{ij} and $\delta_{\alpha\beta}$ are Kronecker deltas. This operator is diagonal in both site and internal labels and is proportional to the identity within each onsite internal space.

A Hermitian bond perturbation over an authored bond set may be written

$$\Delta H_{\text{bond}} = \sum_{(i\alpha,j\beta)\in\mathcal{E}} \left(t_{i\alpha,j\beta} |i, \alpha\rangle\langle j, \beta| + t^*_{i\alpha,j\beta} |j, \beta\rangle\langle i, \alpha| \right). \tag{22.25}$$

where

ΔH_{bond} the Hermitian bond-change operator.
$\mathcal{E}$ a finite set containing one orientation of each modified bond between distinct sites $i \neq j$, with no reverse-pair duplicate.

$t_{i\alpha,j\beta}$ the complex energy-valued directed bond change.

$t^*_{i\alpha,j\beta}$ its complex conjugate.

$|i,\alpha\rangle\langle j,\beta|$ **and** $|j,\beta\rangle\langle i,\alpha|$ mutually adjoint rank-one maps between the two localized basis states.

Here $\mathcal{E}$ contains one chosen orientation of each modified bond between distinct sites, and it does not also contain the reverse pair. The coefficient $t_{i\alpha,j\beta}$ is a complex energy-valued bond change, and the second term is its Hermitian conjugate. This convention prevents the same Hermitian bond contribution from being counted twice. Such a perturbation modifies hopping and must not be relabelled as a scalar potential. Its spatial extent must be established by the declared support or locality criterion below; the finiteness of $\mathcal{E}$ in one finite matrix is not sufficient.

22.8 Locality and finite-extent diagnostics

Assign site i the integer primitive-cell coordinate $\mathbf{n}_i = (n_{i1}, n_{i2}, n_{i3})$, with $n_{i\mu} \in \{0, \dots, N_\mu - 1\}$. Multiple intracell sites may share the same $\mathbf{n}_i$; any intracell site label remains part of i. Let o denote the declared defect-origin site. The minimum-image cell-index separation in direction μ is

$$\delta_\mu(i,o) = \min\left(|n_{i\mu} - n_{o\mu}|, N_\mu - |n_{i\mu} - n_{o\mu}|\right). \tag{22.26}$$

where

$\delta_\mu(i,o)$ the minimum-image primitive-cell-index separation in lattice direction μ.

i **and** o the evaluated site and declared defect-origin site.

$n_{i\mu}$ **and** $n_{o\mu}$ their integer coordinates in direction μ.

N_μ the periodic primitive-cell count in direction μ.

$|\cdot|$ **and** min absolute value and selection of the shorter periodic displacement.

The minimum-image Chebyshev cell-shell distance is

$$d_\infty^{\mathrm{MI}}(i,o) = \max_\mu \delta_\mu(i,o). \tag{22.27}$$

where

$d_\infty^{\mathrm{MI}}(i,o)$ the minimum-image Chebyshev cell-shell distance between site i and origin o.

$\delta_\mu(i,o)$ the minimum-image cell-index separation in direction μ.

$\max_\mu$ the maximum over lattice directions.

The minimum implements periodic wrapping, while the maximum assigns a primitive-cell shell. This metric places every intracell site in the same shell as its cell and is therefore not a geometric interatomic distance. A physical- distance partition must instead declare the intracell positions, lattice metric, periodic-image rule, and units. Other metrics may be used only when declared; they define different spatial partitions.

For a nonnegative integer radius r, define the core projector

$$P_r = \sum_{i:\, d_\infty^{\mathrm{MI}}(i,o)\le r} \sum_{\alpha} |i,\alpha\rangle\langle i,\alpha|, \qquad Q_r = I - P_r. \tag{22.28}$$

where

P_r the orthogonal projector onto all localized basis states within cell-shell radius r of origin o.

Q_r the complementary exterior projector.

r a nonnegative integer shell radius.

i **and** α a site and its internal basis-state label.

$d_\infty^{\mathrm{MI}}(i,o)$ the declared minimum-image cell-shell distance from site i to origin o.

$|i,\alpha\rangle\langle i,\alpha|$ the rank-one projector onto one localized basis state.

I the identity on the full represented localized-basis space.

The projector P_r selects all internal basis states on sites within the core; Q_r selects its exterior. The partition-resolved Frobenius norms are

$$\varepsilon_{\mathrm{core}}(r) = \|P_r \Delta H P_r\|_{\mathrm{F}}, \tag{22.29}$$

$$\varepsilon_{\mathrm{ext}}(r) = \|Q_r \Delta H Q_r\|_{\mathrm{F}}, \tag{22.30}$$

$$\varepsilon_{\mathrm{cross}}(r) = \left(\|P_r \Delta H Q_r\|_{\mathrm{F}}^2 + \|Q_r \Delta H P_r\|_{\mathrm{F}}^2\right)^{1/2}. \tag{22.31}$$

where

$\varepsilon_{\mathrm{core}}(r)$ the Frobenius norm of the core–core block.

$\varepsilon_{\mathrm{ext}}(r)$ the Frobenius norm of the exterior–exterior block.

$\varepsilon_{\mathrm{cross}}(r)$ the combined Frobenius norm of both core–exterior blocks.

P_r **and** Q_r the core and exterior projectors at radius r.

ΔH the aligned represented impurity operator.

$\|\cdot\|_{\mathrm{F}}$ the Frobenius matrix norm.

The core norm measures perturbation wholly inside the core. The exterior norm measures perturbation wholly outside it. The cross norm measures couplings that cross the core boundary. All three have energy units.

For shell $s \geq 0$, define $P_{-1} = 0$ and

$$S_s = P_s - P_{s-1}, \qquad \eta_s = \|S_s \Delta H\|_{\mathrm{F}}. \tag{22.32}$$

where

S_s the projector onto lattice shell s.
P_s **and** P_{s-1} cumulative core projectors through shells s and $s-1$; $P_{-1} = 0$.
η_s the row-resolved Frobenius norm on shell s.
ΔH the aligned represented impurity operator.
$\|\cdot\|_{\mathrm{F}}$ the Frobenius matrix norm.
s a nonnegative integer shell index.

The projector S_s selects rows associated with sites at shell s, and η_s is the row-resolved shell norm. Because a general unitary can mix sites, η_s and the partitioned norms are meaningful only in the declared localized gauge or under a restricted site-local gauge class.

A finite-extent acceptance rule at radius r takes explicit nonnegative energy-valued tolerances τ_{ext} and τ_{cross} and requires

$$\varepsilon_{\mathrm{ext}}(r) \leq \tau_{\mathrm{ext}}, \qquad \varepsilon_{\mathrm{cross}}(r) \leq \tau_{\mathrm{cross}}. \tag{22.33}$$

where

$\varepsilon_{\mathrm{ext}}(r)$ the exterior–exterior residual at radius r.
$\varepsilon_{\mathrm{cross}}(r)$ the core–exterior residual at radius r.
τ_{ext} **and** τ_{cross} authored nonnegative energy-valued acceptance tolerances.
r the declared nonnegative core radius.

The tolerances, radius, metric, basis, and gauge are part of the claim. Passing this finite represented criterion does not prove compact support in a continuum or infinite crystal.

22.9 Supercell convergence and periodic-image contamination

Let $\Delta H_{\mathbf{L}}$ and $\Delta H_{\mathbf{L}'}$ be aligned impurity operators from two supercells. They cannot be subtracted until a common local space has been identified. Let

$$J_{\mathbf{L}\to c} : \mathcal{S}_{\mathbf{L}} \longrightarrow \mathcal{S}_c, \qquad J_{\mathbf{L}'\to c} : \mathcal{S}_{\mathbf{L}'} \longrightarrow \mathcal{S}_c \tag{22.34}$$

where

$J_{\mathbf{L}\to c}$ **and** $J_{\mathbf{L}'\to c}$ declared maps from two supercell spaces into one common comparison space.

$\mathcal{S}_{\mathbf{L}}$ **and** $\mathcal{S}_{\mathbf{L}'}$ represented spaces for supercells $\mathbf{L}$ and $\mathbf{L}'$.

$\mathcal{S}_c$ the common local comparison space.

$\mathbf{L}\to c$ **and** $\mathbf{L}'\to c$ map directions, not arithmetic operations.

be declared restriction or embedding maps into a common comparison space $\mathcal{S}_c$. The core stability residual is

$$\varepsilon_{\text{size}}(r;\mathbf{L},\mathbf{L}') = \left\| P_r \left(J_{\mathbf{L}\to c}\Delta H_{\mathbf{L}} J_{\mathbf{L}\to c}^{\dagger} - J_{\mathbf{L}'\to c}\Delta H_{\mathbf{L}'} J_{\mathbf{L}'\to c}^{\dagger} \right) P_r \right\|_{\mathrm{F}}. \tag{22.35}$$

where

$\varepsilon_{\text{size}}(r;\mathbf{L},\mathbf{L}')$ the core Frobenius discrepancy between two supercell extractions.

$\Delta H_{\mathbf{L}}$ **and** $\Delta H_{\mathbf{L}'}$ aligned impurity matrices extracted from supercells $\mathbf{L}$ and $\mathbf{L}'$.

$J_{\mathbf{L}\to c}$ **and** $J_{\mathbf{L}'\to c}$ their maps into the common comparison space; daggers denote adjoints.

P_r the common core projector at radius r.

$\|\cdot\|_{\mathrm{F}}$ the Frobenius norm.

The maps J define which sites, orbitals, and spin components are compared; $\mathcal{S}_c$ is their common represented space. The same radius r and core projector P_r must refer to the same physical neighborhood in both representations.

A convergence study must vary both supercell size and shape. Increasing size changes dopant-image separation. Changing shape probes anisotropic image coupling and exposes conclusions that depend accidentally on a square or cubic sequence. Defect-derived eigenvalues must also be tracked over supercell momentum $\mathbf{K}$: their bandwidth is evidence of periodic-image coupling and must not be collapsed into one nominal isolated bound-state energy.

22.10 Verification residuals and evidence classes

For a synthetic control with a known planted perturbation $\Delta H_\star$ and extracted result ΔH_{ext}, define the recovery error

$$R = \Delta H_{\text{ext}} - \Delta H_\star. \tag{22.36}$$

where

R the extraction-recovery error matrix.

ΔH_{ext} the perturbation returned by the extraction method.

$\Delta H_\star$ the independently authored planted perturbation.

$\star$ an authored-reference marker, not complex conjugation in this equation.

The symbol $\star$ denotes the authored reference, not an optimized value. Two absolute recovery measures are

$$\varepsilon_{\max} = \max_{p,q} |R_{pq}|, \qquad \varepsilon_{\mathrm{F}} = \|R\|_{\mathrm{F}}, \tag{22.37}$$

where

$\varepsilon_{\max}$ the largest absolute entry of the recovery matrix.

ε_{F} the Frobenius norm of the recovery matrix.

R_{pq} the recovery-matrix entry at represented indices p, q.

p, q indices over the represented basis.

$|\cdot|$**,** max**, and** $\|\cdot\|_{\mathrm{F}}$ absolute value, maximum, and Frobenius norm.

where p and q range over represented basis indices. The maximum-entry norm locates the largest coordinate discrepancy; the Frobenius norm aggregates all entries. These residuals verify the declared finite representation and extraction route when compared with authored tolerances. They do not establish that the planted model is physically adequate for silicon.

The error sources remain separate:

- *parent-model error* concerns the Kohn–Sham approximation, exchange–correlation functional, pseudopotentials, and physical branch;
- *numerical error* concerns discretization, self-consistency, reciprocal meshes, Wannier convergence, and finite supercells;
- *alignment error* concerns site, subspace, gauge, spin, and scalar-energy identification; and
- *reduction error* concerns approximation of the aligned impurity operator by a lattice or continuum model class.

A successful software test establishes only its software contract. Numerical verification requires an independent mathematical or computational route. Scientific validation requires comparison with independent physical evidence for a declared use. Sensitivity scans are not uncertainty quantification unless uncertain inputs and their propagation have been defined.

22.11 Dopant-specific interpretation

For phosphorus, the primary branch is neutral spin-polarized P_{Si}^{0}. It contains both the extra electron and the positively charged donor core relative to silicon. Therefore Eq. (22.19) yields the aligned perturbation of the complete selected neutral Kohn–Sham branch. It must not be silently relabelled as the potential of an ionized donor core.

For boron, the final acceptor branch must retain the accepted noncollinear spinor and spin–orbit-coupled valence structure. Its pristine comparison must use the same relativistic and spin convention. A scalar-relativistic non-SOC boron calculation may support method development only; it is not the final acceptor impurity operator.

For both dopants, the matrix must be extracted over a supercell sequence. Stability of local blocks, gauge and alignment sensitivity, exterior and cross-coupling norms, defect-band dispersion, and shape dependence are required before an isolated-impurity interpretation is considered. One finite-cell matrix difference is an impurity-operator candidate, not an isolated-impurity result.

Appendix I records the controlled one-dimensional extraction, alignment, model-class, and continuum exercises. Appendix J records the finite-periodic two-dimensional representation, extraction, locality, symmetry, and adverse controls. Those controlled calculations verify bounded synthetic contracts. They do not constitute phosphorus or boron extraction, material validation, transferability evidence, or uncertainty quantification.

CHAPTER 23

Lattice Impurity Model Classes

Status proposed nested impurity-model hierarchy; no minimal model has been selected.

23.1 Nested impurity model classes

Stage 08 proposes a hierarchy that can distinguish scalar, orbital-dependent, onsite, hopping, and range-restricted components. Let $\mathfrak{I}_{0,d} \subseteq \mathfrak{I}_{1,d} \subseteq \cdots$ denote the frozen nested family for dopant d. Each candidate is obtained by a declared projection or fit of the accepted atomistic impurity operator.

A model level is evaluated with a vector of complementary errors,

$$\boldsymbol{\varepsilon}_{m,d} = \left(\varepsilon_{H,m,d}, \varepsilon_{\Pi,m,d}, \Delta E_{b,m,d}, 1 - F_{m,d}\right), \tag{23.1}$$

where the components represent, respectively, an operator error, a retained subspace or projector error, a bound-state energy error, and loss of a declared fidelity measure. Each component still requires its exact definition, units, normalization, and tolerance in the owning specification.

The selected impurity model is the least complex member satisfying every prespecified tolerance. If no reduced member passes, the full atomistic operator remains the accepted model. Failure to reduce is therefore a valid scientific outcome rather than a software failure.

CHAPTER 24

Continuum Effective-Mass Reduction

Status proposed continuum-reduction framework; no finite crossover or validated donor or acceptor continuum model is claimed.

24.1 Continuum effective-mass reduction

The continuum stage introduces host band-edge tensors and channel data, a multivalley donor solver for phosphorus, and a multiband acceptor solver for boron. Screened-Coulomb and central-cell model families supply candidate continuum impurity terms. These are mathematical models with their own state spaces; their matrices cannot be subtracted directly from lattice operators.

Prospective citation note—high priority
Check the selected multivalley donor, multiband acceptor, screened-Coulomb, and central-cell model lineages against Luttinger–Kohn (1955), Kohn–Luttinger donor theory (1955), and Baldereschi–Lipari (1973). Candidate keys are `luttingerKohn1955` [70], `kohnLuttinger1955donor` [58], and `baldereschiLipari1973` [2]; the latter two records are provisional.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

A continuum-to-Wannier embedding is therefore required. Let

$$J : \mathcal{H}_{\mathrm{cont}} \longrightarrow \mathcal{S}_{\mathrm{W}} \tag{24.1}$$

be the declared embedding over its validation domain. A continuum operator H_{cont} is represented in the atomistic comparison space through an explicitly defined construction such as

$$H_{\mathrm{cont}\to W} = J H_{\mathrm{cont}} J^{\dagger}, \tag{24.2}$$

with normalization, channel ordering, spatial discretization, boundary conditions, and any non-isometric behavior of J stated. Synthetic-envelope tests verify the implementation

before first-principles claims are attempted.

The proposed crossover radius $r_{c,d}$ is the smallest tested radius beyond which both exterior and cross-coupling criteria satisfy their declared tolerances. If no tested radius passes, the result is recorded as $r_{c,d} = +\infty$ over the tested domain. A finite crossover is not assumed in advance.

CHAPTER 25

Structured Learning as a Model-Class Diagnostic

Status proposed learning framework. No trained model, learned impurity operator, model-class error sequence, or atomistic-to-continuum crossover is claimed.

Machine learning enters this program only after the first-principles and representation problems have been separated. Kohn–Sham DFT supplies a parent one-particle operator. Projection, retention, energy alignment, and an admissible identification map then place the pristine and doped operators in a common represented space. Learning does not replace any of these steps. Its proposed role is to test how much structure a reduced impurity model requires in order to approximate the aligned parent operator.

Prospective citation note—low priority
Check the statement that Kohn–Sham DFT supplies the parent one-particle operator against the primary Kohn–Sham paper, bibliography key `kohn1965` [59] (already present).
Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

This viewpoint makes the machine-learning question subordinate to a physical model-selection question. The objective is not simply to fit Kohn–Sham bands or to emit an unconstrained matrix. It is to compare physically defined operator classes and determine which additional degrees of freedom materially reduce a declared discrepancy.

25.1 The aligned target precedes learning

At wavevector $\mathbf{k}$, let P denote the declared retained-space projector, and let $H_b^{(P)}(\mathbf{k})$ and $H_d^{(P)}(\mathbf{k})$ be finite representations of the pristine and doped Kohn–Sham operators in their retained spaces. The superscript (P) denotes retention by this projector; the dopant phosphorus is written P when it appears as a species label. Suppose that all prerequisite

geometry, spin, unit, and scalar-energy conventions have been made compatible, and let

$$U_d(\mathbf{k}) : \mathcal{H}_b^{(P)}(\mathbf{k}) \longrightarrow \mathcal{H}_d^{(P)}(\mathbf{k}) \tag{25.1}$$

be an accepted unitary identification. In pristine coordinates, the signed aligned impurity target is

$$\Delta H_{d\to b}^{(P)}(\mathbf{k}) = U_d^\dagger(\mathbf{k})\, \overline{H}_d^{(P)}(\mathbf{k})\, U_d(\mathbf{k}) - \overline{H}_b^{(P)}(\mathbf{k}), \tag{25.2}$$

where the overbar denotes the recorded common scalar-energy convention. This is one represented target that a reduced model may be asked to reproduce. Its meaning remains conditional on the retained subspaces and identification validated in Chapter 18.

A direct supervised objective for a parameterized candidate $\Delta H_\theta(\mathbf{k})$ is

$$\mathcal{L}_H(\theta) = \sum_{\mathbf{k}\in\mathcal{K}_{\text{train}}} \omega_{\mathbf{k}} \left\| \Delta H_{d\to b}^{(P)}(\mathbf{k}) - \Delta H_\theta(\mathbf{k}) \right\|_{\text{F}}^2. \tag{25.3}$$

The sampled set, weights, normalization, units, Fourier convention, and withheld domain must all be declared. Equation (25.3) is a legitimate regression objective, but an unrestricted network that emits an arbitrary $M \times M$ array would abandon most of the structure established by the reduction framework.

25.2 Structured parameter prediction

The preferred architecture separates learned parameter prediction from a deterministic Hamiltonian constructor:

$$x \xrightarrow{f_\phi} \boldsymbol{\theta}(x) = \{t_i, \epsilon_i, V_i, \ldots\} \xrightarrow{\mathcal{C}} H_\phi(x). \tag{25.4}$$

Here x denotes declared physical descriptors, f_ϕ is the learned map, and $\mathcal{C}$ constructs an operator in a prescribed model class. Depending on the class, the constructor can enforce by construction

- Hermiticity and the declared spin structure;
- lattice, site, and orbital ordering;
- crystal or defect symmetries;
- prescribed hopping topology and finite interaction range;
- locality or a specified form of nonlocality; and
- scalar, orbital-dependent, or central-cell potential structure.

The descriptors and constructor are part of the scientific model, not neutral software details. Their invariance, equivariance, boundary, and completeness properties must therefore be stated. A network cannot repair a descriptor map that identifies physically distinct environments or a constructor that omits an essential operator component.

Abstractly, the reduction problem is

$$\inf_{H\in\mathfrak{M}} d\Big(\Delta H^{(P)}_{d\to b}, H\Big)\,, \tag{25.5}$$

where $\mathfrak{M}$ is a physically defined reduced-operator class and d is a discrepancy on compatible representations. Neural optimization is one possible way to search a class; it does not define the class by itself and does not prove that the infimum has been found.

25.3 Expressiveness through nested model classes

After embedding all candidates in one comparison space, one may prescribe a nested hierarchy such as

$$\mathfrak{M}_{\text{EMT}} \subseteq \mathfrak{M}_{\text{scalar}} \subseteq \mathfrak{M}_{\text{orbital}} \subseteq \mathfrak{M}_{\text{finite-range}} \subseteq \mathfrak{M}_{\text{NN}}. \tag{25.6}$$

This nesting is a contract to be demonstrated, not inferred from the class names. For example, the EMT operator requires an explicit continuum-to-atomistic embedding before it can be a member of a lattice comparison class, and a structured neural constructor contains the earlier classes only if its parameterization can reproduce them exactly.

For a nonzero target, define the best normalized represented-operator error

$$\epsilon(\mathfrak{M}) = \inf_{H\in\mathfrak{M}} \left(\frac{\sum_{\mathbf{k}\in\mathcal{K}} \omega_{\mathbf{k}} \left\| \Delta H^{(P)}_{d\to b}(\mathbf{k}) - H(\mathbf{k}) \right\|_{\mathrm{F}}^2}{\sum_{\mathbf{k}\in\mathcal{K}} \omega_{\mathbf{k}} \left\| \Delta H^{(P)}_{d\to b}(\mathbf{k}) \right\|_{\mathrm{F}}^2} \right)^{1/2}. \tag{25.7}$$

A zero-target case requires an absolute scale rather than this normalization. For genuinely nested classes, the exact infima obey

$$\epsilon_1 \geq \epsilon_2 \geq \cdots \geq \epsilon_J. \tag{25.8}$$

Numerically obtained training errors need not obey or attain this sequence because optimization, sampling, regularization, and finite precision introduce additional error. Approximation error, estimation or generalization error, and optimization error must therefore be reported separately whenever the evidence supports that decomposition.

The useful scientific information lies in the changes between adjacent classes. Consider, for example, the scalar onsite hypothesis

$$[\Delta H]_{i\alpha,j\beta} \approx V_i\, \delta_{ij}\delta_{\alpha\beta}. \tag{25.9}$$

If allowing orbital dependence yields no material improvement under controlled optimization and withheld evaluation, then the tested evidence does not require orbital dependence at that scale. If allowing off-site blocks yields a large, reproducible reduction, nonlocality is implicated as an important correction within the tested representation. Neither observation is automatically a universal physical conclusion: both remain conditional on the target, embedding, metric, domain, and tolerances.

A range hierarchy may impose

$$[\Delta H]_{i\alpha,j\beta} = 0 \quad \text{when} \quad |\mathbf{R}_i - \mathbf{R}_j| > R \tag{25.10}$$

and evaluate $\epsilon(R)$. This tests the represented spatial range of the impurity operator. The attribution of a nonlocal matrix element to a radius, its periodic-image treatment, and the tested supercell sequence must be fixed before $\epsilon(R)$ is interpreted physically.

25.4 A learned reference for the continuum crossover

The continuum question can be phrased as the emergence, over a declared exterior domain, of a smaller effective-mass operator class. A screened Coulomb donor term has the asymptotic form

$$V_{\text{EMT}}(r) \sim -\frac{e^2}{4\pi\epsilon_0\epsilon_r r}, \tag{25.11}$$

but the physical specification cautions that the primary neutral P:Si supercell perturbation must not be silently relabelled as an ionized donor-core potential. The charge-state interpretation and screening content must be established before this form is used as a target.

Prospective citation note—high priority
Check the screened-Coulomb donor asymptote and its silicon multivalley context against Kohn–Luttinger donor theory (1955), provisional key `kohnLuttinger1955donor` [58]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

Let $P_{>R}$ denote the declared projector onto an exterior atomistic region, and let $H^*_{\mathfrak{M}}$ be a best or controlled approximate fit in class $\mathfrak{M}$. One class-relative exterior discrepancy is

$$\eta_{\mathfrak{M}}(R) = \left\| P_{>R} \left(\Delta H^{(P)}_{d\to b} - H^*_{\mathfrak{M}} \right) P_{>R} \right\|, \tag{25.12}$$

with cross-boundary blocks assessed separately as required by the crossover contract in Chapter 19. A highly expressive structured model can serve as a reference class. If its

validated exterior error is much smaller than that of EMT near the impurity but offers no material advantage beyond a radius, that pattern can contribute to an operational crossover assessment.

Agreement between two fitted errors is not sufficient by itself. Both models must satisfy the prespecified absolute tolerances; optimization and sampling errors must be controlled; and the exterior and cross-coupling criteria must hold for every tested larger radius. A learned reference therefore does not establish EMT by assumption. It provides a more expressive comparator within a larger body of crossover evidence.

25.5 Learning the residual beyond a continuum model

A particularly interpretable construction assigns the known continuum model as a baseline and learns only a correction:

$$\Delta H_{d\to b}^{(P)} = \Delta H_{\mathrm{EMT}\to\mathrm{ret}} + \delta H_\phi + E_\phi, \tag{25.13}$$

where $\Delta H_{\mathrm{EMT}\to\mathrm{ret}}$ denotes the continuum operator after a validated embedding into the retained atomistic space, δH_ϕ is a constrained atomistic correction, and E_ϕ is the remaining represented residual. One proposed objective is

$$\mathcal{L}_{\mathrm{residual}}(\phi) = \left\| \Delta H_{d\to b}^{(P)} - \Delta H_{\mathrm{EMT}\to\mathrm{ret}} - \delta H_\phi \right\|_{\mathrm{F}}^2 + \lambda\, \mathcal{R}(\delta H_\phi), \tag{25.14}$$

where $\mathcal{R}$ is a declared complexity or locality penalty. The value of λ changes the fitted model and must be selected without using withheld acceptance data.

This formulation asks which smallest admissible correction to effective-mass theory is required relative to the aligned Kohn–Sham target. The correction can be decomposed by onsite contribution, orbital dependence, off-diagonal orbital mixing, hopping range, symmetry channel, and radial decay. Such decompositions are more interpretable than a validation score alone, provided that their gauge and localization dependence is also reported.

25.6 Composite objectives and acceptance

Operator fidelity can be combined with other declared reduction quantities:

$$\mathcal{L} = \lambda_H \mathcal{L}_H + \lambda_{\mathrm{spec}} \mathcal{L}_{\mathrm{spec}} + \lambda_{\Pi} \mathcal{L}_{\Pi} + \lambda_{\mathrm{loc}} \mathcal{L}_{\mathrm{loc}} + \lambda_{\mathrm{EMT}} \mathcal{L}_{\mathrm{EMT}}. \tag{25.15}$$

For example, a normalized operator term may be

$$\mathcal{L}_H = \frac{\sum_{\mathbf{k}} \omega_{\mathbf{k}} \left\| H_{\mathrm{target}}(\mathbf{k}) - H_\phi(\mathbf{k}) \right\|_{\mathrm{F}}^2}{\sum_{\mathbf{k}} \omega_{\mathbf{k}} \left\| H_{\mathrm{target}}(\mathbf{k}) \right\|_{\mathrm{F}}^2}, \tag{25.16}$$

a spectral term may be

$$\mathcal{L}_{\text{spec}} = \sum_{n,\mathbf{k}} w_{n\mathbf{k}} \left(E^{\text{target}}_{n\mathbf{k}} - E^{\phi}_{n\mathbf{k}} \right)^2 , \tag{25.17}$$

and a locality penalty in real space may be

$$\mathcal{L}_{\text{loc}} = \sum_{\mathbf{R}} w(\mathbf{R}) \, \| H_\phi(\mathbf{R}) \|_{\text{F}}^2 , \tag{25.18}$$

where $w(\mathbf{R})$ increases with separation.

A weighted sum is an optimization device, not automatically an acceptance rule. Its weights can hide a failed criterion behind success on another. Final model selection should therefore retain the separate metrics and apply the prespecified admissible-set criteria developed in Chapter 19. Symmetry, Hermiticity, and other exact requirements should be enforced by the constructor where possible rather than encouraged only by soft penalties.

25.7 Joint learning of identification and reduction

A more ambitious problem optimizes both the coordinate identification and the reduced operator:

$$\inf_{\substack{U \in \mathcal{G} \\ H \in \mathfrak{M}}} d\Big(U^\dagger \overline{H}_d^{(P)} U - \overline{H}_b^{(P)}, H \Big) , \tag{25.19}$$

where $\mathcal{G}$ is an admissible identification class. A learned unitary may be parameterized through an anti-Hermitian generator,

$$A_\psi^\dagger(\mathbf{k}) = -A_\psi(\mathbf{k}), \tag{25.20}$$

$$U_\psi(\mathbf{k}) = \exp A_\psi(\mathbf{k}), \tag{25.21}$$

so that unitarity holds by construction. The joint sampled objective would then compare

$$U_\psi^\dagger(\mathbf{k}) \overline{H}_d^{(P)}(\mathbf{k}) U_\psi(\mathbf{k}) - \overline{H}_b^{(P)}(\mathbf{k}) \tag{25.22}$$

with $\Delta H_\phi(\mathbf{k})$.

This problem exposes the search for both coordinates and a simple operator in those coordinates, but it is generally less identifiable than either problem alone. A sufficiently expressive reduced model can absorb an incorrect identification, while a flexible identification can make an operator appear artificially simple. Low training loss would not establish physical correspondence. Any future joint method therefore requires independent overlap, locality, symmetry, smoothness, and gauge-equivariance constraints together with controlled synthetic recovery tests.

The safer progression and the staged escalation of model expressiveness are summarized in Figure 25.1. Joint optimization over U and H remains a later proposed investigation, not the first learning experiment.

25.8 A staged computational progression

The proposed progression introduces flexibility only when a simpler level is shown to be inadequate under its declared tests.

Level 1: linear physical model. Use fixed stencils or a linear Slater–Koster or tight-binding parameterization and solve by linear least squares when the model permits it.

> **Prospective citation note—medium priority**
> Check the named Slater–Koster parameterization against the primary source, bibliography key `slater1954` [113] (already present).
>
> *Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.*

Level 2: nonlinear differentiable physical model. Retain a deterministic Hamiltonian constructor while optimizing nonlinear physical parameters.

Level 3: learned parameter map. Introduce a neural map only when admissible parameters must depend nonlinearly on a declared local environment.

Level 4: learned residual operator. Fit the constrained correction beyond simpler accepted physical classes and decompose the residual by physical operator content.

Level 5: learned identification. Only after fixed-identification behavior is understood, investigate optimization over an admissible gauge or identification manifold.

Failure can be informative. Success at Level 1 would show that the tested target requires no learned parameter map for the declared use. A gain at Level 2 would show the value of nonlinear parameter dependence within the same constructor. A reproducible gain from a structured learned model would identify missing expressiveness in the simpler tested classes. Failure of an expressive model under fixed constraints may indicate incompatible constraints, inadequate descriptors, insufficient data, or optimization failure; these alternatives must be distinguished before a scientific interpretation is assigned.

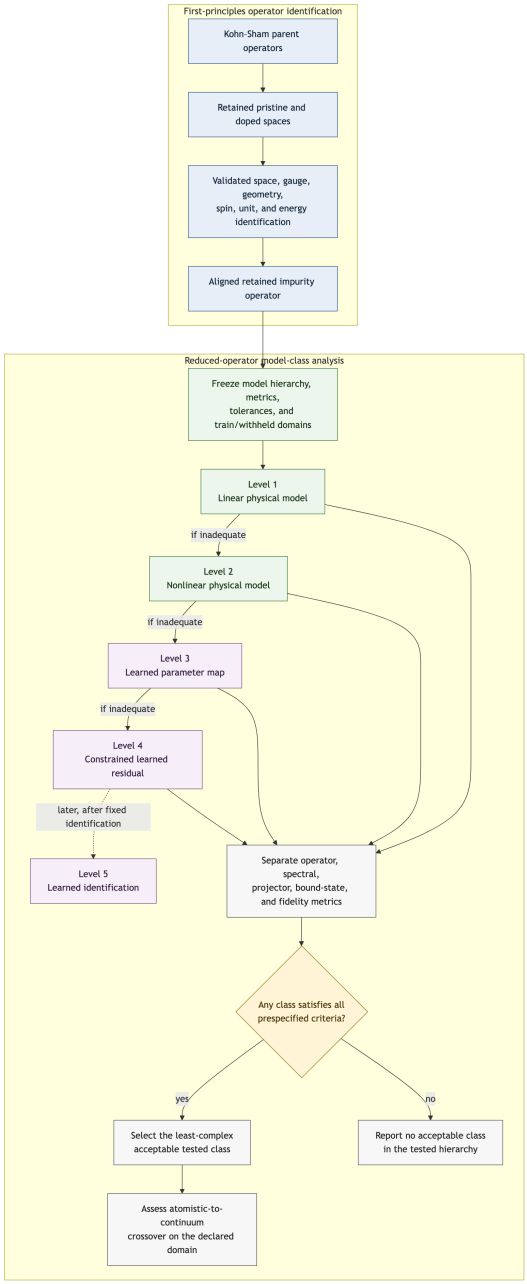


Figure 25.1: Proposed operator-identification and structured-learning process. Solid arrows show the fixed-identification progression and escalation through increasingly expressive model classes when a simpler tested class is inadequate. The dotted arrow marks learned identification as deferred work. The editable Mermaid source is retained in `figures/structured-learning-process.mmd`.

25.9 Scientific question and evidence boundary

The proposed central question is

> What is the lowest-complexity operator class in which the aligned retained Kohn–Sham impurity problem is representable within prespecified tolerances for a declared use?

For each candidate class $\mathfrak{M}$, let $\mathfrak{A}_{\mathfrak{M}}$ be the subset satisfying every prespecified operator, spectral, projector, bound-state, fidelity, and exact structural criterion applicable to the declared use. One formal expression is

$$\mathfrak{M}^{*} \in \arg \min_{\substack{\mathfrak{M}\in\mathfrak{H} \\ \mathfrak{A}_{\mathfrak{M}}\neq\varnothing}} \operatorname{complexity}(\mathfrak{M}). \tag{25.23}$$

The condition $\epsilon(\mathfrak{M}) \leq \epsilon_{\text{target}}$ may define the operator-error component of $\mathfrak{A}_{\mathfrak{M}}$, but it cannot replace the other required criteria. The candidate hierarchy $\mathfrak{H}$, complexity ordering, metrics, tolerances, training protocol, and withheld validation domain are frozen before selection. If no tested class passes, no acceptable class has been identified within the tested hierarchy.

A constrained neural model is thus an expressive comparator, not the central physical model and not an infallible oracle. It can help test whether a simpler physical class has exhausted its representational capacity only when optimization adequacy and generalization have independent evidence. The complete proposed pipeline is the process shown in Figure 25.1.

This framing makes the learning component useful because it is accountable to the operator-identification framework. It asks not whether a flexible model can fit a matrix, but which physically interpretable restrictions can be retained without losing the parent information required for the intended reduction.

CHAPTER 26

Dopant Transferability and Validation Program

Status proposed phosphorus demonstration and boron transferability program; no production impurity reduction or validation result is claimed.

26.1 Phosphorus as the first impurity demonstration

After the bulk representation and alignment gates pass, phosphorus is the first complete impurity path. The branch requires a compatible P pseudopotential, neutral spin-polarized P_{Si}^{0} supercells, controlled internal relaxation, matched pristine comparison cells, and supercell-size evidence. The aligned difference is then decomposed into nested scalar, orbital, onsite, hopping, and range-restricted model classes.

The objective is not to assume that a screened Coulomb potential plus one central-cell parameter is sufficient. It is to determine which operator and bound-state information each tested class preserves. A continuum donor model becomes a first-principles reduction only after its host parameters, embedding, exterior error, cross-coupling error, and bound-state comparison are tied to the accepted atomistic parent.

26.2 Boron as a transferability branch

Boron tests whether the methods transfer to a physically different band-edge problem. The final acceptor branch requires a fully relativistic noncollinear spinor treatment with SOC and a compatible valence subspace. The phosphorus basis or scalar model class cannot be transferred unchanged merely for convenience.

Comparing the P and B hierarchies may reveal which reduction features are common to substitutional impurities and which are specific to conduction- or valence-manifold structure. Such a comparison becomes eligible only after each branch has its own accepted

atomistic operator and minimal-model evidence.

26.3 Methodological implications

26.3.1 Structured learning as a later diagnostic

After an aligned impurity operator exists, structured learning may provide an expressive comparison class for the reduction hierarchy. The proposed network does not solve the Kohn–Sham parent problem and does not replace the validated identification map. It predicts parameters of a deterministic Hamiltonian constructor or a constrained residual beyond a simpler physical model.

The relevant result would be the sequence of attainable errors across frozen scalar, orbital, finite-range, continuum, and learned classes, together with separate optimization and withheld-evaluation evidence. A learned model that improves on effective-mass theory near the impurity but offers no material advantage in an accepted exterior domain could contribute to the crossover analysis. Such a pattern is proposed work, not a result of the present draft. Joint learning of the gauge and reduced operator is deferred until the fixed-identification problem is understood, because low joint training loss would not by itself establish a physically meaningful correspondence.

26.3.2 Composition and path dependence

A completed hierarchy still requires cross-path assessment. Direct and Wannier-mediated bulk fits may disagree. Extracting an impurity before reduction may differ from reducing pristine and doped parents before subtraction. Successive lattice and continuum reductions may amplify earlier errors.

These noncommuting diagrams are not expected to vanish by definition. Their paths, aligned outputs, norms, and validation domains must be measured. The final compositional claim may be that a diagram commutes within tolerance over a declared domain, that it fails by a quantified defect, or that available metadata are insufficient to compare the paths.

26.3.3 Negative results as supported outcomes

The research program is designed to retain negative findings. An unacceptable nearest-neighbor model, an unstable alignment, a nonconverged impurity sequence, an empty admissible-set intersection, or an infinite crossover radius over the tested domain can all be valid outcomes. Their value depends on preserving the frozen model hierarchy, rejected

candidates, tested range, metrics, and limitations.

A negative finding is not generalized beyond its tested classes and domain. A failed software contract is corrected; a failed numerical tolerance is refined or reported; a physically inadequate model class is not repaired by weakening its acceptance rule after the result is known.

CHAPTER 27

Current Evidence Boundary

Status retained evidence inventory and result-reporting framework. No production bulk, impurity, tight-binding, or continuum result is accepted.

This chapter records what the project can presently support and, equally importantly, what it cannot. Current status is derived from the owning task, calculation, and evidence records associated with this draft. Because the monograph is a living document, this inventory is rewritten when new library capabilities and calculations change the evidence boundary. The manuscript states consequential version changes; the applicable Git commit records the revision, and retained SHA-256 manifests identify its calculation artifacts.

27.1 Meaning of "result" in this chapter

A result is reported with its evidence class and provenance. The categories used here are

- proved or mechanized mathematical result with declared assumptions and proof status;
- verified software behavior;
- verified numerical behavior on controlled mathematics;
- tutorial or bootstrap execution observation;
- calculated physical result with retained provenance;
- scientific validation result; and
- uncertainty-quantification result.

These categories are not ordered stages through which an array advances automatically. A proof establishes only its implication from its assumptions, and the same artifact may support one bounded software claim while remaining unusable for a physical claim.

27.2 Accepted represented-operator software foundation

Evidence class software verification, with declared controlled numerical-verification cases.

The proof registry separately records nine frozen finite-dimensional PRF−05 contracts. Only projector invariance under unitary frame rotation, PRF−05.01, has been checked by one pinned Lean backend. The other Lean targets and every Isabelle and Rocq target remain unencoded; no contract is cross-checked and all physics proof packages remain proposed. These facts are mechanization-status results, not physical validation.

The accepted Python operator-record foundation provides

- finite represented-operator storage with explicit state-space, basis, geometry, energy-reference, provenance, and matrix metadata;
- Hermiticity analysis in a fixed representation;
- deterministic version-one JSON serialization with a public schema and golden fixtures;
- exact compatibility auditing for represented metadata;
- signed subtraction for records that are already compatible;
- maximum-entry, Frobenius, and spectral residual analysis; and
- a composed comparison workflow for differencing and residual analysis.

This is a software capability result, not an accepted scientific operator. The foundation does not align bases or gauges, convert units, align scalar energy zeros, transform geometries, establish physical equivalence, or decide that a generic difference is an impurity operator. It has not supplied scientific validation or uncertainty quantification, and it does not by itself pass the later bulk or alignment gates.

27.3 Accepted legacy tutorial observations

Evidence class tutorial execution, artifact provenance, and bounded software-extraction behavior.

Four legacy silicon tutorial tasks are retained as human-accepted bounded evidence: an SCF tutorial reproduction, artifact inventory, periodic-record extraction, and a matching

tutorial band calculation. The band execution consumed an isolated identity-verified copy of the accepted tutorial SCF state and retained 28 ordered **k** points with eight bands at each point.

Against the bundled legacy printed reference, 166 of 224 eigenvalues matched at printed precision and the largest printed difference was approximately 0.0001 eV. No numerical comparison tolerance or pass/fail rule was accepted, so this observation is not promoted into numerical verification. An IEEE floating-point exception report remains unresolved and unclassified.

The tutorial parent uses a legacy LDA pseudopotential and tutorial numerical settings. Its energy reference, basis identity, retained subspace, and gauge are insufficient for the production representation and reduction questions. These records therefore establish neither a production silicon band structure nor a band gap, valley position, effective mass, Wannier target, or tight-binding reference.

27.4 Direct PBE bootstrap observations

Evidence class provisional calculated and finite-setting numerical-verification evidence with downstream execution blocked.

A separately retained PBE/PseudoDojo bootstrap campaign executed nine SCF and nine linked NSCF Quantum ESPRESSO invocations. Every invocation exited with status zero and emitted `JOB DONE.`; no retry was performed. A later read-only audit matched the compact boundary identities, all 18 staged inputs, 41 retained text-output and receipt identities, all nine parsed case records, and all eight reconstructed adjacent-setting comparisons. Exact input, executable, and pseudopotential identities and compact finite-setting records are retained by the owning convergence design and disposition.

Prospective citation note—high priority
Attach the standard scholarly sources for PBE, PseudoDojo, and Quantum ESPRESSO—keys `perdew1996` [89], `vansetten2018` [106], and `giannozzi2009` [39] (already present)—only after checking their local scope. These sources identify methods and software; they do not substantiate the repository-specific execution count or outcomes, which require durable internal provenance.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The finite-setting disposition provisionally selects a 48-Ry wavefunction cutoff, a 192-Ry charge-density cutoff, and a shifted $8 \times 8 \times 8$ SCF mesh by the frozen two-sided adjacent-

setting rules. This is not an infinite-basis estimate, a cutoff–mesh interaction bound, or production-parent acceptance. The provisional geometry has not passed the zero-pressure lattice gate, and no indirect gap, valley coordinate, or effective mass is accepted.

The recurring four-class IEEE report has been narrowed but not closed. A later non-MPI OpenBLAS diagnostic classified one matching first operation below `ZHEGVX` as an IEEE divide-by-zero exception, but its operands, observable effect, underflow source, and every later flag origin remain unresolved. Downstream protected execution remains blocked and unauthorized. Scientific-harness reproduction is later comparison work rather than a prerequisite for using the audited finite-setting observations. Appendix M records the complete bounded synthesis.

27.5 Exploratory silicon Wannier decomposition

Evidence class external read-only content-bound software and numerical prototype with a bounded branch-retained publication extraction; not production evidence.

A separate local analysis consumed an externally retained QE 7.5–Wannier90 3.1.0 silicon source run without executing either calculator. It reports a rank-four same-frame construction of the represented Hamiltonian, canonical plane-wave kinetic operator, and non-kinetic remainder on a $4\times4\times4$ mesh, together with Fourier, Hermiticity, interpolation, derivative, and finite-mesh truncation diagnostics. The complete analysis package remains in a separate development worktree; a bounded subset is retained under `calculations/ICMSEP2026/conference/paper_1/silicon-wannier-kinetic-preview/`. The analysis did not consume a pre-existing authenticated production-lineage manifest and used content-bound working-tree source over a base commit that alone is insufficient to reconstruct the implementation.

The source run's four original files match the official Wannier90 3.1.0 `examples/example11` silicon tutorial by SHA-256. The executed `silicon.win` adds only two bound output-artifact requests and changes no tutorial numerical setting. That tutorial uses a legacy ultrasoft PBE pseudopotential and is not the frozen PseudoDojo production branch. The reported values are therefore tutorial-scale prototype observations only. The tutorial's visually convincing total-band interpolation does not certify separate locality or short-range compressibility of the represented kinetic and non-kinetic components; retaining 57 of 64 blocks still permits an 84 meV maximum kinetic eigenvalue error. These observations do not establish an accepted material Wannier operator, mesh convergence, a short-range model, or effective masses. Appendix M states the construction, selected numerical observations, and exclusions.

27.6 Controlled periodic-reduction benchmarks

Evidence class human-accepted calculated illustrative numerical verification.

Appendix G reports a controlled one-dimensional cosine-lattice reduction. Its maintained nominal single-band evidence is the publication-independent family `ksdft2effmass.campaign.periodic-reduction.periodic1d.single-band-fourier-reduction`, immutable version `...v1`, run `...v1.run-20261014T184707Z`, and evidence package `sha256:6e8d5db988df97d4b8cf7843b3baceb16e2eb70d2b03c729497a82251fca6377`, bound by the monograph's `experiment-references.json`. Independent reconstruction passes, and matched historical M1 channels have zero retained binary64 defect. The successor additionally retains positive weights, construction/evaluation roles, sampled external gaps, per-range rank and conditioning, strict schemas, and exact predecessor provenance. Its sampled gaps do not certify continuum isolation.

The broader historical benchmark independently refines plane-wave and finite-difference representations toward the same low-mode parent; selected energies agree with Mathieu characteristic values; and scalar- and composite-band constructions verify parallel transport, alignment, hopping reconstruction, and finite-range truncation. A separate Wannier90 calculation uses the same eigenvalue and overlap interface and reproduces the aligned represented operators after convergence.

The benchmark also identifies bounded failure modes. Scalar energy-ordered labels become poorly conditioned near adjacent-gap closure, rough gauges degrade finite-range locality without changing the complete spectrum, and direct and mediated reductions cease to agree when their weights or fitting domains differ. These are numerical-verification results for the synthetic cosine parent. They are not a production silicon calculation, scientific validation, uncertainty quantification, or evidence of transferability to silicon. At the evidence boundary represented by this draft, the calculated records pass their declared verification checks and the bounded one-dimensional periodic-reduction task has received final human acceptance under these stated limitations.

Appendix H extends the controlled periodic program to a two-dimensional synthetic parent. The accepted evidence covers separable and coupled scalar operators, a corrected composite-band polar gauge, mass-tensor recovery, real-space hopping reconstruction, common-estimator localization comparisons, distinct QWZ, Hofstadter, and Haldane topology controls, and bounded mesh, cutoff, embedding, and hopping sensitivities. Native Berry-link spreads and common finite-supercell spread estimates remain separate quantities.

The expanded optimizer study retains 256 frozen start/interface trajectories. Their final effective endpoints comprise 196 native convergences and 60 iteration-limit nonconvergences. The frozen mesh-convergence and repeated-basin criteria are not supported,

and the density-aware symmetry analysis leaves every frozen basin as a singleton. The right-censored convergence-time regression is exploratory and descriptive. Only one trial-projection family was executed; projection-family sensitivity is explicitly excluded from the closed benchmark.

This is human-accepted negative numerical evidence for one frozen synthetic protocol. It does not establish global optimization, general Wannier convergence, uncertainty quantification, a DFT interface, or material validation.

27.7 Canonical constrained-admissible-set result

Evidence class calculated synthetic numerical verification with an independently reconstructed common witness and finite-family separation certificate.

The canonical M3 package composes the retained rank-two multiband baseline with a bounded two-parameter shift-and-splitting family and nine frozen global real rotations. On its 64-point training mesh, the dimensionless RMS threshold pair $(\tau_E, \tau_H) = (0.03, 0.33)$ retains the common witness $(s, q) = (0, 1)$. For $(\tau_E, \tau_H) = (0.03, 0.31)$, independently reconstructed analytic lower and upper bounds coincide at the Euclidean parameter separation 0.0994674010272163, above the frozen resolution 0.05. A disjoint 257-point staggered mesh supplies withheld diagnostics and does not alter the witness or certificate. Section G.9 records the calculation in context.

The public and standalone verification routes now bind evaluation roles, positive definition tolerances, the finite alignment family, and certificate quantities. The separately retained threshold-sensitivity reanalysis also checks source-manifest correlation and the axis-aligned positive-quadratic premises on which its exact reduction depends. That sensitivity analysis is post hoc, not part of the prospectively frozen confirmatory decision.

This evidence establishes the two stated outcomes only within the frozen synthetic family, parameter metric, alignment components, training mesh, and thresholds. It establishes neither a general nonconvex alignment solution nor material compatibility, physical threshold calibration, scientific validation, transferability to silicon, or uncertainty quantification.

27.8 Controlled impurity-extraction benchmarks

Evidence class human-accepted calculated synthetic software and numerical verification, plus a human-accepted bounded literature synthesis.

Appendix I reports a spin-space diagnostic, a matched one-dimensional pristine–defect extraction, and four follow-ons that separate blind alignment, independent extraction routes, a finite-rank resolvent oracle, and continuum refinement. The numerical records verify their declared finite-model contracts and preserve failed compatibility controls, finite-volume effects, operator residuals, spectral errors, projector errors, and continuum axes as separate quantities. They do not supply a production silicon operator or a validated material model.

The accompanying primary-literature comparison narrows the contribution boundary. Finite-rank impurity methods, weak localization, band-edge homogenization, explicit discrete–continuum identifications, periodic-volume effects, multivalley and central-cell donor models, Wannier defect operators, and gauge-aligned model stitching all have established or contemporary precedents [62, 110, 87, 48, 26, 81, 61, 58, 36, 5, 21, 66, 69]. Two relevant 2026 records are explicitly preprints: an onsite-only substitutional-defect screening protocol and a clean/defect Wannier workflow using assignment, unitary Procrustes alignment, and a nontrivial overlap metric [52, 109].

The bounded contribution is therefore a reproducible integration of these concerns into one fail-closed synthetic diagnostic chain, not a new component theorem, a priority claim, or scientific validation. The review's negative search outcome is conditional on its declared concepts, queries, interfaces, dates, and stopping rule; it is not evidence that no unreviewed precedent exists.

Appendix J separately records the two-dimensional defect and material-transfer sequence. The controlled synthetic defect task is active and consumes the accepted periodic parent and spin-space and one-dimensional extraction contracts, but it contains no completed two-dimensional defect calculation. Material transfer remains proposed and inactive.

27.9 Current scientific result boundary

At the status represented by this draft, the project has not accepted

- a converged production bulk-silicon Kohn–Sham parent;
- a production equilibrium lattice parameter;

- a calculated production indirect gap or conduction-valley position;
- longitudinal or transverse production electron effective masses;
- a validated bulk-silicon Wannier operator;
- a fitted or selected sp^3s^* tight-binding model;
- a frozen basis- and gauge-alignment protocol;
- a calculated two-dimensional defect-extraction or material-transfer result;
- a phosphorus or boron first-principles impurity operator;
- a minimal impurity lattice model;
- a donor or acceptor continuum crossover radius;
- scientific validation for a declared physical use; or
- propagated uncertainty for the proposed reduction hierarchy.

Accordingly, this chapter contains no production band plot, fitted parameter table, impurity profile, binding energy, or crossover curve. Adding an expected shape or literature value to fill visual space would obscure the evidentiary boundary.

27.10 Present limitations

27.10.1 Parent calculation

The physical and numerical protocols are accepted, but the production parent has not crossed its convergence and acceptance boundary. Finite candidate settings and successful process completion cannot substitute for that gate. The selected PBE parent will also retain parent-model limitations relative to experiment even after numerical convergence.

Prospective citation note—medium priority
Narrow this broad statement to a named observable before citing it. If the intended claim is the electronic gap, check Perdew–Levy (1983) and Sham–Schluter (1983), bibliography keys `perdewLevy1983` [90] and `shamSchluter1983` [107] (already present). Do not use those sources to imply a universal PBE limitation.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

27.10.2 Representation

For production silicon, the target Wannier rank, projections, energy windows, uniform interface grid, localization outcome, and interpolation domain have not been accepted. The accepted synthetic benchmarks do not select or validate those material settings. The alignment map between independently generated spaces is likewise unavailable. Operator subtraction beyond already compatible synthetic or software records would therefore lack required meaning.

27.10.3 Bulk model reduction

The executable sp^3s^* hierarchy, frozen training and withheld sets, alignment group, shell weights, objective normalizations, operator tolerance, and complete model-selection rule remain downstream work. Proposed manuscript tolerances are not treated as accepted computational evidence merely because they appear in a draft.

27.10.4 Impurity branches

Phosphorus and boron specialization gates have not supplied their production pseudopotentials, supercell sequences, doped Wannier representations, or alignment records. Periodic-image convergence is unavailable. The boron final branch additionally requires compatible SOC and spinor treatment, while the phosphorus neutral branch requires its open-shell and charge-state interpretation to remain explicit.

27.10.5 Continuum reduction

Verified continuum solver infrastructure would not by itself establish a first-principles reduction. Host parameters, accepted atomistic impurity operators, continuum-to-Wannier embedding, spatial error criteria, and bound-state validation are all required before a crossover statement can be made. Established finite-rank identities, band-edge homogenization results, discrete-to-continuum convergence theorems, and semiconductor effective-mass models provide methodological precedents, but citations to those results do not establish that this project's represented impurity operator satisfies their hypotheses or has reached a material continuum crossover.

27.11 Negative findings

The failed periodic-2D localization criteria are accepted negative numerical evidence for their frozen synthetic protocol. They are not a scientific negative finding about silicon, defects, or the proposed material model hierarchy.

No scientific negative finding has yet been accepted for the model hierarchy. In particular, the project has not shown that

- nearest-neighbor sp^3s^* is incompatible with the joint criteria;
- a larger tight-binding class is required;
- no reduced impurity model passes;
- no finite continuum crossover exists; or
- two declared reduction paths fail to commute beyond tolerance.

These are possible future outcomes, not conclusions. When a frozen tested domain produces such an outcome, the result should state the domain, metric, tolerance, optimization or convergence evidence, and unresolved alternatives.

27.12 Result table contract

Each future numerical table should identify

- the calculated quantity and units;
- immutable parent, child, and analysis identities;
- representation, basis, gauge, geometry, spin, and energy reference where applicable;
- numerical settings and tested refinement domain;
- uncertainty or observed sensitivity without false precision;
- comparison reference and evidence class;
- metric, tolerance, and pass/fail rule; and
- limitations on interpretation.

Training results and withheld results appear in separate columns or tables. Parent-model, numerical, and model-reduction errors remain separate quantities. A missing entry is marked unavailable rather than estimated from an unrelated source.

27.13 Figure contract

A future figure must be a deterministic view of retained data. Its caption and associated record identify the data source, processing code, axes, units, reference zero, plotted subset, and evidentiary status. Expected diagrams, synthetic regressions, tutorial bands, and production results use visibly distinct labels.

Band plots state the reciprocal-coordinate path and energy reference. Operator plots state the basis, alignment, block or shell definition, norm, and normalization. Convergence plots show rejected settings and guard points rather than only the selected candidate. Residual plots distinguish training from withheld data.

27.14 Path to populated scientific results

The first populated production sequence requires an accepted bulk convergence record, equilibrium lattice reference, and production SCF parent. Purpose-built band-path and local-valley children then provide the bulk observables. A separately approved uniform NSCF and QE–Wannier interface support candidate Wannier operators, followed by interpolation and alignment evidence. Tight- binding, impurity, and continuum result sections become eligible only after their corresponding parent artifacts exist.

Until those records are available, the absence of production numbers in this chapter is an accurate result boundary rather than an incomplete editorial placeholder.

PART V

Mathematical and Proof Program

CHAPTER 28

Proof Architecture

Status proof contracts and proof strategies are presented with explicit status; no proposed physics proof package is represented as complete or reviewed.

28.1 Why a separate proof program is needed

The computational hierarchy contains claims of several different kinds. Some are elementary finite-dimensional identities, such as covariance under a unitary coordinate change. Some require analytical hypotheses, such as a spectral gap, compact parameter domain, resolvent bound, scale separation, or asymptotic locality. Others are numerical or scientific statements that cannot be established by proof alone.

A successful calculation may suggest a theorem but does not prove it. A formal proof may establish an implication from assumptions while leaving the physical adequacy of those assumptions unvalidated. The proof program therefore records

- state spaces, domains, codomains, and group actions;
- assumptions needed by each result;
- analytical dependencies among results;
- mechanized theorem contracts and backend status;
- nonclaims and failure boundaries; and
- numerical or scientific evidence that remains separate from proof.

The proof workspace is subordinate to the versioned physical and numerical specifications. It cannot repair an unresolved scientific convention by choosing a mathematically convenient one.

28.2 Proof-package graph

The physical proof program is divided into six packages:

PRF–00: Foundations. State spaces, Bloch-fiber correspondence, and representation and reduction maps.

PRF–05: Mechanized operator lemmas. Nine frozen finite-dimensional contracts targeted independently in Lean, Isabelle, and Rocq.

PRF–10: Operator alignment. TB-anchored identification, gauge actions, and aligned pristine–doped subtraction.

PRF–20: Reduction. Gauge-constrained locality, excluded-space reduction, atomistic-to-continuum consistency, and path commutativity.

PRF–30: Bounds. Crossover, operator-to-observable, and continuum-fitting statements.

PRF–40: Compatibility. Spectral/operator admissibility, model-class distance, and certified incompatibility.

Prospective citation note—medium priority

Add scholarly system references for Lean 4, mathlib, Isabelle/HOL, and Rocq only after checking the sources and current naming. Provisional keys: `demouraUllrich2021` [79], `mathlib2020` [122], `nipkowPaulsonWenzel2002` [85], and `bertotCasteran2004` [6]. These sources identify the systems; they cannot substantiate repository-specific proof status.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

Their dependency structure is a partial order rather than a single pipeline: foundations support every later package; mechanized elementary lemmas support applicable alignment, reduction, and compatibility claims; alignment and reduction support the proposed bounds; and compatibility also depends on explicit model classes and admissible gauges.

The separate PRF–90 package concerns agentic-workflow properties such as authorization safety and replay determinism. It is intentionally outside the physics proof chain. A workflow theorem would not establish a physical or numerical theorem.

28.3 Status vocabulary

The proof registry distinguishes `proposed`, `in development`, `blocked`, `complete`, `unreviewed`, `reviewed`, `numerically checked`, and `retired`. Mathematical review,

numerical verification, scientific validation, and human publication acceptance remain distinct.

A frozen mechanization contract fixes the current theorem target; it does not mean that the theorem has been encoded or proved. A backend status of `checked` means that the identified checker accepted the identified encoding under the pinned toolchain. Cross-checking additionally requires all three independent backends and semantic conformance review.

The derivations below explain selected contracts. They do not modify the owning registry status.

CHAPTER 29

Finite-Dimensional Operator Foundations

Status proof contracts and proof strategies are presented with explicit status; no proposed physics proof package is represented as complete or reviewed.

29.1 Finite-dimensional retained frames

Let $V \in \mathbb{C}^{D\times M}$ have orthonormal columns,

$$V^\dagger V = I_M, \tag{29.1}$$

and let $G \in \mathbb{C}^{M\times M}$ be unitary. The columns of V represent an ordered orthonormal frame for an M-dimensional retained subspace of an ambient D-dimensional space.

Proposition 29.1 (PRF-05.01: projector invariance)**.** *For $P = VV^\dagger$ and $V' = VG$, the rotated frame induces the same ambient orthogonal projector:*

$$V'V'^\dagger = P. \tag{29.2}$$

Proof. Using $(VG)^\dagger = G^\dagger V^\dagger$ and $GG^\dagger = I_M$,

$$(VG)(VG)^\dagger = VGG^\dagger V^\dagger = VV^\dagger. \tag{29.3}$$

The condition $V^\dagger V = I_M$ is needed to interpret $VV^\dagger$ as the orthogonal projector, although the displayed equality uses only the unitarity of G. □

This is the sole frozen contract currently encoded and checked in Lean. The pinned trial uses Lean `4` and mathlib `v4.33.0`; the project source contains no `sorry` or `admit`. The check establishes acceptance of that encoding, not physical admissibility of the frame, smoothness in $\mathbf{k}$, or conformance with Isabelle or Rocq.

Prospective citation note—medium priority
Check and add system references for Lean 4, mathlib, Isabelle/HOL, and Rocq. Provisional keys: `demouraUllrich2021` [79], `mathlib2020` [122], `nipkowPaulsonWenzel2002` [85], and `bertotCasteran2004` [6]. The claim that this repository contains a checked encoding and no `sorry` or `admit` still requires durable internal evidence rather than scholarly citations.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

Proposition 29.2 (PRF-05.02: covariance of compression)**.** *For an ambient operator $A \in \mathbb{C}^{D\times D}$, define*

$$H = V^\dagger A V, \qquad H' = (VG)^\dagger A (VG). \tag{29.4}$$

Then

$$H' = G^\dagger H G. \tag{29.5}$$

Proof. Associativity and the adjoint product rule give

$$(VG)^\dagger A (VG) = G^\dagger (V^\dagger A V) G. \tag{29.6}$$

No Hermiticity assumption on A is needed for this coordinate identity. □

The contract does not assert that A preserves the retained subspace or that the compressed spectrum equals the full ambient spectrum.

29.2 Identification pullback and aligned subtraction

Let $H_b \in \mathbb{C}^{M_b\times M_b}$ and $H_d \in \mathbb{C}^{M_d\times M_d}$ represent pristine and doped retained operators. Let

$$U : \mathbb{C}^{M_b} \longrightarrow \mathbb{C}^{M_d} \tag{29.7}$$

be an identification map. Under independent unitary frame changes G_b and G_d, define

$$H_b' = G_b^\dagger H_b G_b, \qquad H_d' = G_d^\dagger H_d G_d, \qquad U' = G_d^\dagger U G_b. \tag{29.8}$$

Proposition 29.3 (PRF-05.03: pullback covariance)**.** *The doped operator pulled into pristine coordinates transforms as*

$$U'^\dagger H_d' U' = G_b^\dagger (U^\dagger H_d U) G_b. \tag{29.9}$$

Proof. Insert the primed definitions and cancel $G_d G_d^\dagger = I_{M_d}$ on both sides of H_d:

$$U'^\dagger H'_d U' = G_b^\dagger U^\dagger G_d G_d^\dagger H_d G_d G_d^\dagger U G_b \tag{29.10}$$

$$= G_b^\dagger U^\dagger H_d U G_b. \tag{29.11}$$

Unitarity of U is not required for this covariance identity. □

Two common-space differences must be distinguished:

$$\Delta H_{d\to b} = U^\dagger H_d U - H_b, \qquad \Delta H_{d\to b} : \mathbb{C}^{M_b} \to \mathbb{C}^{M_b}, \tag{29.12}$$

$$\Delta H_{b\to d} = H_d - U H_b U^\dagger, \qquad \Delta H_{b\to d} : \mathbb{C}^{M_d} \to \mathbb{C}^{M_d}. \tag{29.13}$$

Proposition 29.4 (PRF-05.04: aligned-difference covariance and equivalence)**.** *Each difference transforms covariantly in its own common space. If U is a unitary identification, then*

$$\Delta H_{b\to d} = U \Delta H_{d\to b} U^\dagger, \qquad \Delta H_{d\to b} = U^\dagger \Delta H_{b\to d} U. \tag{29.14}$$

Proof. Covariance follows by applying Proposition 29.3 to the pullback difference and its pushforward analogue to the doped-space difference. For unitary U,

$$U \Delta H_{d\to b} U^\dagger = U(U^\dagger H_d U - H_b) U^\dagger \tag{29.15}$$

$$= H_d - U H_b U^\dagger = \Delta H_{b\to d}. \tag{29.16}$$

The inverse relation follows similarly. □

This algebra does not establish that a physically adequate U exists, that it is smooth or symmetry compatible, or that the scalar energy zeros of H_b and H_d agree. Those remain prerequisites before either difference is called an impurity operator.

29.3 Invariant norms and gauge-dependent locality

For a unitary G, the Frobenius and induced Euclidean operator norms satisfy

$$\|G^\dagger A G\|_{\mathrm{F}} = \|A\|_{\mathrm{F}}, \qquad \|G^\dagger A G\|_2 = \|A\|_2. \tag{29.17}$$

These are the frozen PRF-05.05a and PRF-05.05b targets. They support gauge-invariant residuals only when the compared paths transform conformantly, as required by PRF-05.06.

Spatial blocks and shell decompositions require a narrower statement. A general unitary rotation can mix orbitals associated with different sites or cells and therefore change

which entries are classified as local or exterior. The frozen PRF−05.07 target is a two-point counterexample showing that gauge transformation and real-space truncation need not commute. The PRF−05.08 target instead restricts the gauge to declared lattice-local constant rotations under which shell Frobenius diagnostics remain invariant.

Thus a global unitarily invariant norm and a spatially resolved residual have different admissible gauge groups. A proof about the first cannot be reused for the second without the additional locality assumptions.

29.4 TB-anchored identification

Suppose $\widetilde{V}_b(\mathbf{k})$ and $\widetilde{V}_d(\mathbf{k})$ are orthonormal frames of equal rank after both have been related to a declared tight-binding anchor. The candidate fiberwise map is

$$U_d(\mathbf{k}) = \widetilde{V}_d(\mathbf{k})\widetilde{V}_b(\mathbf{k})^\dagger. \tag{29.18}$$

On the corresponding retained spaces, this formula gives an isometric unitary identification under the stated orthonormality and equal-rank assumptions.

The substantial PRF−10.01 obligations are not the elementary product identity. They concern existence of suitable anchored frames, rank and conditioning, global smoothness and periodicity, crystal symmetry, and physical adequacy. PRF−10.02 then owns equivariance of the construction, while PRF−10.03 owns the common-space subtraction and its covariance. All three remain proposed.

CHAPTER 30

Mathematical Structure of Reduction and Continuum Claims

Status proof contracts and proof strategies are presented with explicit status; no proposed physics proof package is represented as complete or reviewed.

30.1 Excluded-space reduction

Let $P + Q = I$ be complementary orthogonal projectors and write the block components of a self-adjoint operator H as $H_{PP} = PHP$, $H_{PQ} = PHQ$, and so on. When a real energy E belongs to the resolvent set of H_{QQ}, eliminating the Q component gives the energy-dependent Feshbach effective operator [28, 29]

$$H_{\text{eff}}(E) = H_{PP} + H_{PQ}(E - H_{QQ})^{-1}H_{QP}. \tag{30.1}$$

If

$$\Delta_Q = \operatorname{dist}(E, \sigma(H_{QQ})) > 0, \tag{30.2}$$

then the self-adjoint resolvent estimate suggests the proposed bound

$$\left\| H_{PQ}(E - H_{QQ})^{-1}H_{QP} \right\| \leq \frac{\|H_{PQ}\|\,\|H_{QP}\|}{\Delta_Q}. \tag{30.3}$$

For $H_{QP} = H_{PQ}^{\dagger}$ this becomes $\|H_{PQ}\|^2/\Delta_Q$.

The PRF−20 . 02 obligation must keep this real off-spectrum setting separate from open-system or analytically continued resonance formulations, where non-Hermiticity and widths require additional theory. No small correction is inferred without both a coupling estimate and resolvent distance.

30.2 Crossover and operator-to-observable bounds

A continuum comparison may contain several mathematically distinct limiting operations. A continuum-discretization limit refines the numerical mesh for a fixed continuum operator; a discrete-to-continuum limit changes the represented lattice scale and requires an explicit identification between changing state spaces [81]; a band-edge homogenization limit weakens and widens a defect relative to the crystal period while deriving an envelope operator [48, 26]; and an infinite-volume limit enlarges the domain while holding local physics fixed. Changing the defect profile or switching between fixed-peak and fixed-integrated normalization is a model-family change unless the scaling is part of the declared limit. Convergence on a path that varies several of these quantities therefore does not identify which approximation produced the observed agreement.

The sources also differ in operator class and spectral claim. Weak-binding results concern existence below a threshold or in periodic gaps [110, 87]; homogenization results concern modes bifurcating from a band edge under weak/slow scaling; and finite-periodic-volume formulas concern image shifts under separate range and tail assumptions [61]. None implies that a finite represented multiorbital operator satisfies another source's hypotheses. Numerical agreement of one energy or projector therefore cannot substitute for the state-space map, operator residual, and scaling assumptions of the applicable result.

Let $D_d = \Delta H_d - V_{\mathrm{cont},d}$ be an aligned atomistic-minus-continuum discrepancy in a common space, and let $P_{>R}$ project onto a declared exterior region. Define

$$\eta_d(R) = \|P_{>R} D_d P_{>R}\|. \tag{30.4}$$

When the exterior subspaces are nested, compression by a smaller exterior projector cannot increase the operator norm. A crossover criterion may then be defined by

$$r_{c,d}(\tau) = \inf\{R : \eta_d(R) \le \tau\}. \tag{30.5}$$

The PRF-30.01 obligation must distinguish nonemptiness of the set, existence of the infimum, attainment by a tested radius, numerical estimation, and physical interpretation. The assumption $\eta_d(R) \to 0$ as $R \to \infty$ is an asymptotic-locality hypothesis, not a consequence established by general Kohn–Sham DFT.

Prospective citation note—high priority

Check the asymptotic-locality discussion against nearsightedness literature while preserving the statement as an explicit hypothesis. Candidate: Prodan and Kohn (2005), provisional key prodanKohn2005 [92]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or

changing a citation.

For self-adjoint atomistic and reduced operators with perturbation $E = H_{\text{atom}} - H_{\text{red}}$, standard perturbation theory motivates spectral-distance bounds controlled by $\|E\|$. Invariant-subspace rotation can be controlled by a Davis–Kahan-type estimate when the target cluster is separated by a declared spectral gap [24]. The proposed PRF-30.02 result must identify the finite or infinite setting, norm, spectral window, gap, state identification, and degeneracy treatment. A worst-case bound is not assumed sharp, and a small spectral error does not imply a small operator error.

Prospective citation note—medium priority

Check the general spectral-perturbation statement against Kato's standard treatment, provisional key `kato1995` [53], while retaining `davis1970` [24] for the named Davis–Kahan invariant-subspace estimate.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

30.3 Continuum fitting and identifiability

For continuum parameters $\boldsymbol{\theta} \in \Theta$, define a declared exterior objective

$$J_R(\boldsymbol{\theta}) = \|P_{>R}[\Delta H_d - V_{\text{cont}}(\boldsymbol{\theta})]P_{>R}\| \,. \tag{30.6}$$

If Θ is nonempty and compact and J_R is continuous, the extreme-value theorem gives existence of a minimizer. The proposed PRF-30.03 program separates this existence result from uniqueness, conditioning, stability, and physical identifiability. An optimizer output supplies none of those stronger properties automatically.

30.4 Compatibility and certified incompatibility

Let $\mathfrak{A}_E$ and $\mathfrak{A}_H$ be nonempty spectral- and operator-admissible subsets of a declared metric model space. If both sets are compact, their distance

$$\delta^* = \min_{H_E \in \mathfrak{A}_E,\, H_H \in \mathfrak{A}_H} d(H_E, H_H) \tag{30.7}$$

is attained. If the compact sets are disjoint, the distance is positive. This standard existence argument is only part of PRF-40.01: the scientific application must also establish closure or compactness, continuity of the losses, and a reproducible admissible gauge group.

A failed local optimizer does not prove that the sets are disjoint or that $\delta^* > 0$. The proposed PRF-40.03 certification obligation requires a valid global lower bound, for example from analytic estimates, interval methods, branch-and-bound, exhaustive certified reduction, or an applicable validated convex relaxation. PRF-40.04 then asks what can be inferred when a nested model class reduces the residual; a smaller residual does not uniquely identify the physical cause of the improvement.

Prospective citation note—medium priority
Check the certified global-optimization examples, especially interval and branch-and-bound methods, against a validated global optimization source. Candidate: Neumaier (2004), provisional key `neumaier2004` [83]; the candidate record is present for this editorial note.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

30.5 Reduction-path commutativity

For aligned operands in one common space, let $\mathcal{R}$ be a reduction map. The path defect is

$$\varepsilon_{\text{path}} = \left\| \mathcal{R}(H_d - H_b^{\text{aligned}}) - [\mathcal{R}(H_d) - \mathcal{R}(H_b^{\text{aligned}})] \right\| . \tag{30.8}$$

If the same linear map acts on both operands in the same coordinates, exact commutativity follows from linearity. This elementary fact does not apply automatically to nonlinear fitting, separately selected model classes, gauge-dependent truncation, or independently chosen alignment maps. Appendix G supplies a related controlled route comparison: matched direct and mediated Fourier projections agree, whereas altered reciprocal weights or fitting domains produce nonzero route defects. That experiment does not instantiate the impurity-difference expression above; it demonstrates the same need to hold maps, coordinates, and objectives fixed before interpreting path agreement.

The PRF-20.04 program distinguishes gauge equivariance from path commutativity and seeks bounds when exact equality fails. The frozen PRF-05.06 contract addresses gauge invariance of a residual only after the two paths are already conformant.

This conformity requirement is operational rather than merely notational. Localized-basis disorder and electron–defect methods require a common Wannier representation or explicit Bloch–Wannier maps before defect contributions are transported [5, 69]. Model-stitching work aligns orbital pairing, spin orientation, phase, and scalar potential before minimizing overlap-region Hamiltonian differences [66]. A recent preprint similarly uses assignment, a dense unitary Procrustes map, and an induced overlap metric when embedding independently generated clean and defect Wannier models [109]. These constructions

are precedents for explicit identification; they do not prove that an inferred map is unique, well conditioned, or complete on an unobserved sector.

CHAPTER 31

Mechanization Status and Open Obligations

Status proof contracts and proof strategies are presented with explicit status; no proposed physics proof package is represented as complete or reviewed.

31.1 Mechanization status

The current prover-neutral PRF-05 target set contains nine frozen contracts:

Contract	Subject	Current backend evidence
05.01	Projector invariance	Lean checked
05.02	Compression covariance	Unencoded
05.03	Pullback covariance	Unencoded
05.04	Aligned subtraction	Unencoded
05.05a	Frobenius-norm invariance	Unencoded
05.05b	Operator-norm invariance	Unencoded
05.06	Equivariant path residual	Unencoded
05.07	Gauge–truncation counterexample	Unencoded
05.08	Shell-Frobenius invariance	Unencoded

Every Isabelle and Rocq target is unencoded. Eight Lean targets are unencoded. No contract is cross-checked. The package remains proposed, while PRF-05.01 is individually in development because the multi-backend completion and review criteria have not been met.

Prospective citation note—medium priority
Check and add scholarly system references for Lean 4, mathlib, Isabelle/HOL, and Rocq. Provisional keys: demouraUllrich2021 [79], mathlib2020 [122], nipkowPaulsonWenzel2002 [85], and bertotCasteran2004 [6]. The table's checked and unencoded statuses require durable internal provenance; system citations do not

establish those statuses.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

31.2 Novelty and evidence boundary

The finite-dimensional projector, compression, pullback, subtraction, and norm identities are standard linear algebra. The project does not claim those identities as new mathematics. A possible contribution is their application-specific organization into one prover-neutral contract set, independent Lean–Isabelle–Rocq encodings, semantic conformance review, dual common-space impurity extraction, and composition with later locality, continuum, and compatibility arguments. Priority over prior formalization has not been established.

Potentially substantive mathematical results lie in the assumptions and bounds that connect the elementary layer to the physical reduction: existence and regularity of an adequate identification, gauge-constrained locality, quantitative excluded-space control, well-posed crossover criteria, operator-to-observable bounds, certified model-class incompatibility, and atomistic-to-continuum consistency. Every such result remains proposed.

Proof establishes only the stated conclusion from the stated assumptions. Numerical verification must still test implementations and bound usefulness; scientific validation must still test the physical model for a declared use; and human review must still decide whether the assumptions and interpretation are acceptable.

CHAPTER 32

Conclusions and Research Sequence

Status synthesis of the present evidence boundary and proposed work. No completed production reduction or physical validation is claimed.

The monograph proceeds from criteria for a good model, through the bulk-silicon and doped-silicon representation programs, to the mathematical proof program that records their obligations. The current project state does not yet supply a material-specific numerical answer to the central reduction question. It does, however, retain accepted synthetic numerical evidence for periodic reduction, spin-space compatibility, and one-dimensional impurity extraction.

32.1 Present conclusions

Several conclusions are supported at the present evidence level.

First, spectral fitting and represented-operator fitting are different requirements. Eigenvalues do not identify localized orbitals, gauge, site and orbital correspondence, or the matrix elements through which later impurity operators act. A reduced model intended only for spectral interpolation may be adequate for that use while remaining unsuitable as a common coordinate system for perturbations. Spectral fitting, direct Hamiltonian regression, and analytical restrictions on particular tight-binding classes all predate this work; the narrower proposed contribution is the contract-dependent common- witness or certified-separation decision developed in Chapter 14.

Second, pristine and doped matrix subtraction is not defined by equal array shape. The retained state spaces, basis ordering, geometry, units, spin convention, energy zero, and gauge must agree or be related by an explicit identification map. A difference formed before these prerequisites are met has no established impurity-operator meaning.

Third, projection, disentanglement, gauge transformation, Wannier localization, and real-space truncation are separate operations. Their assumptions and losses must remain attributable. In particular, localization does not itself establish a low-rank approximation or authorize an unstated hopping cutoff.

Fourth, parent-model, numerical, and reduction errors answer different questions. The PBE parent's relationship to experiment is not the same as a finite-cutoff error, and neither is the same as the error introduced by an sp^3s^* model class. Combining them without a compatible mathematical contract would hide rather than explain the quality of the reduction.

Fifth, the controlled periodic benchmarks establish a bounded verification chain from one-dimensional plane-wave and finite-difference parents to two-dimensional scalar, composite-band, topology, hopping, and localization diagnostics. The two-dimensional optimizer study is a completed negative result for its frozen synthetic protocol: its mesh-convergence and repeated-basin criteria are unsupported. Because only one trial-projection family was executed, the result establishes neither projection-family robustness, global optimization, nor general Wannier convergence.

Sixth, the accepted spin-space and one-dimensional defect benchmarks verify representation compatibility, folding, known-map and blind alignment, independent extraction routes, finite-rank spectral checks, model-class separation, finite-volume diagnostics, and independently refined continuum comparisons for their declared synthetic systems. They do not establish a two-dimensional defect operator, a silicon impurity model, or material transferability.

Finally, the repository has an accepted represented-operator software foundation, a coordinated proposed proof program, one frozen operator identity checked by a pinned Lean backend, retained tutorial and bootstrap execution observations, and human-accepted synthetic periodic and impurity-extraction benchmarks. Those artifacts support their bounded proof, software, workflow, and numerical-verification claims. They do not supply the converged production parent, material Wannier operator, reduced bulk model, first-principles impurity operator, or continuum result needed to answer the silicon questions.

32.2 Outputs and perspective

32.2.1 Scholarly outputs

The monograph remains the broad account of definitions, methods, provenance, results, and limitations. Article manuscripts extract narrower questions. P01, for example, can focus on bulk model-class expressiveness and the compatibility of spectral and operator criteria without claiming completion of the impurity program. Later articles may address alignment, impurity-operator hierarchies, or continuum crossover only when their evidence exists.

Presentations may use the same retained results at a different explanatory level, but they do not change evidentiary status. A conference figure derived from tutorial data remains

tutorial evidence; a proposed workflow diagram remains proposed work; a production result retains its calculation identity and limitations.

32.2.2 Final perspective

The intended contribution is not a single fitted parameter table. It is a controlled account of when representations can be identified, when reduced operators preserve their declared parent information, and where approximation paths cease to agree. Reaching that account requires the full chain from physical specification through numerical convergence, representation, alignment, reduction, and evidence.

At present, the project has established the framework, bounded software foundation, and a synthetic verification chain, but it has not completed the proof packages or material reduction sequence. The accepted spin-space and one-dimensional defect contracts and the accepted periodic two-dimensional parent are frozen prerequisites for the two-dimensional defect extension in Appendix J. That controlled synthetic task is active but has no calculated result; it should test only genuinely two-dimensional defect structure rather than repeat the completed prerequisites. Material transfer follows only after that separate controlled stage and its own authorization.

The remaining work is therefore stated as work: complete and review the applicable proofs, construct the accepted material parent, measure each loss, retain negative outcomes, and draw conclusions only at the strength supported by the resulting evidence.

Appendices

CHAPTER A

Notation, Conventions, and Status Vocabulary

This appendix is an expository index for the monograph. It does not override the applicable specifications, proof contracts, software schemas, or retained evidence. A symbol used by an owning record retains the definition in that record when it is more specific than the summary below.

A.1 Physical and mathematical objects

Symbol or term	Meaning in this monograph
Physical model	A declared material and physical approximation, including its state variables, branch, and intended interpretation.
$\widehat{H}$	An abstract Hamiltonian operator on an identified state space; not a finite matrix unless a representation has been selected.
H	A finite matrix representation whose space, basis, geometry, units, energy reference, spin convention, and gauge must be identified.
$\mathcal{H}$	An ambient one-particle state space.
$\mathcal{S}$	A finite retained subspace of an ambient state space.
$P_{\mathcal{S}}$	Orthogonal projector from the ambient space onto $\mathcal{S}$.
$\mathcal{B}$	An ordered basis for a represented finite space.
N or M	A finite represented or retained dimension, as locally defined.
$\dagger$	Adjoint or conjugate transpose in the stated finite-dimensional inner-product setting.

A.2 Periodic and Wannier notation

Symbol	Meaning
$\mathbf{r}$	Real-space position coordinate.
$\mathbf{k}$	Crystal momentum in a declared reciprocal-coordinate convention.
$\mathbf{R}$	Bravais-lattice translation.
$\psi_{n\mathbf{k}}$	Bloch state with local band or frame index n.
$u_{n\mathbf{k}}$	Cell-periodic factor of a Bloch state.
$U(\mathbf{k})$	Momentum-dependent unitary frame choice within a retained Bloch subspace; not to be confused with an interspace identification below.
$w_{n\mathbf{R}}$	Wannier basis function generated from a retained Bloch frame.
$H^{(\mathrm{W})}(\mathbf{k})$	Hamiltonian matrix in the selected Wannier frame at $\mathbf{k}$.
$H^{(\mathrm{W})}(\mathbf{R})$	Real-space Wannier Hamiltonian block under the recorded Fourier convention.
$N_{\mathbf{k}}$	Number of sampled wavevectors in the applicable uniform grid.

The same letter U is conventional for both a Wannier gauge and an interspace identification. The local domain and codomain must disambiguate it. Where confusion is likely, the monograph uses G for an in-space gauge rotation and U or A for a map between retained spaces.

A.3 Frames, gauges, and alignment

Symbol	Meaning
V	Rectangular matrix whose orthonormal columns form a retained frame.
G	Unitary coordinate change within one retained space.
$VV^\dagger$	Ambient-space projector associated with an orthonormal frame.
U or A	Explicit identification between two retained spaces; its direction is stated with its definition.
H_b	Pristine or bulk represented Hamiltonian in its declared coordinates.
H_d	Doped represented Hamiltonian for dopant d.
G_b, G_d	Independent unitary coordinate changes in pristine and doped retained spaces.
$\Delta H_{d\to b}$	Doped operator pulled into pristine coordinates and then subtracted: $U^\dagger H_d U - H_b$.
$\Delta H_{b\to d}$	Pristine operator pushed into doped coordinates and then subtracted: $H_d - U H_b U^\dagger$.
$\overline{H}$	Operator after application of the explicitly recorded scalar energy-reference convention.
θ_j	Principal angle between two retained subspaces.

A subscript b denotes bulk or pristine material, while d denotes a dopant branch. When the identity of the dopant matters, $d \in \{\mathrm{P}, \mathrm{B}\}$. Neither notation fixes charge state, spin branch, or supercell size without an accompanying record.

A.4 Reduction notation

Symbol	Meaning
$\mathfrak{M}_j$	Declared reduced-model class indexed by complexity or another frozen ordering.
$\boldsymbol{\theta}$	Parameter vector within a declared model class.
f_ϕ	Learned parameter map with trainable parameters ϕ and declared physical descriptors as inputs.
$\mathcal{C}$	Deterministic constructor mapping admissible parameters to a represented Hamiltonian.
$\epsilon(\mathfrak{M})$	Best declared normalized discrepancy attainable in model class $\mathfrak{M}$; a computed fit need not attain this infimum.
$\mathcal{L}_E$	Spectral training objective with stated data, weights, units, and normalization.
$\mathcal{L}_H$	Aligned represented-operator objective.
τ_E, τ_H	Prespecified spectral and operator tolerances.
$\mathfrak{A}_{E,j}$	Spectral-admissible subset of model class j.
$\mathfrak{A}_{H,j}$	Operator-admissible subset of model class j.
d_{TB}	Declared metric on compatible canonical tight-binding coordinates.
δ_j^*	Minimum or infimum separation between the two admissible sets, according to the established assumptions.
$\mathcal{R}$	Reduction map with an explicitly stated domain and codomain.
$\varepsilon_{\mathrm{path}}$	Norm of the difference between two aligned reduction paths.

The words *projection*, *disentanglement*, *basis transformation*, *localization*, *truncation*, *fitting*, and *reduction* denote different operations. They are not used as synonyms.

A.5 Excluded-space and continuum notation

Symbol	Meaning
P, Q	Complementary orthogonal projectors with $P+Q=I$. A superscript (P) marks restriction to the retained P space; phosphorus is written P when ambiguity is possible.
H_{PP}, H_{PQ}	Blocks such as PHP and PHQ in a declared decomposition.
$H_{\mathrm{eff}}(E)$	Energy-dependent Feshbach effective operator in the retained P space.
Δ_Q	Distance from E to the spectrum of H_{QQ} in the declared self-adjoint resolvent setting.
$V_{\mathrm{cont},d}$	Candidate continuum impurity operator for dopant d.
D_d	Aligned atomistic-minus-continuum discrepancy.
$P_{>R}$	Projector onto a declared exterior region beyond radius R.
$\eta_d(R)$	Exterior residual $\|P_{>R} D_d P_{>R}\|$ in a declared norm.
$r_{c,d}(\tau)$	Tolerance-dependent crossover infimum; not a universal material constant.
J	Continuum-to-atomistic or continuum-to-Wannier embedding, with its normalization and validation domain stated separately.

Prospective citation note—low priority

Check the naming and definition of the energy-dependent Feshbach effective operator against Feshbach (1958), bibliography key `feshbach1958` [28] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

A.6 Norms and residuals

$$\|A\|_{\mathrm{F}} = \left(\sum_{i,j} |A_{ij}|^2 \right)^{1/2}, \tag{A.1}$$

$$\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}, \tag{A.2}$$

$$\|A\|_{\max} = \max_{i,j} |A_{ij}|. \tag{A.3}$$

The Frobenius and induced Euclidean operator norms are invariant under unitary conjugation. Entrywise maxima, shell decompositions, and spatial truncations may have

narrower invariance groups. Every reported residual states whether it is absolute or normalized and gives its units and zero-scale behavior.

A.7 Proof status

`proposed`. Statement, definitions, or strategy exists without a complete established proof.

`in development`. Proof construction is active and immediate prerequisites are sufficiently identified.

`blocked`. A named missing assumption, definition, prerequisite, or human scientific decision prevents progress.

`complete, unreviewed`. A complete derivation is recorded but has not passed independent mathematical review.

`reviewed`. The derivation and declared prerequisites passed independent mathematical review.

`numerically checked`. Separate numerical evidence agrees under declared conditions; this supplements rather than replaces proof.

`retired`. The claim was superseded, merged, or rejected with a recorded disposition.

A mechanization contract may separately be `frozen`, `encoded`, `checked`, or `cross-checked` in each backend. These words do not replace the proof-package status. In particular, the currently checked Lean encoding of `PRF-05.01` does not make the entire `PRF-05` package complete or reviewed.

A.8 Scientific and evidentiary status

Calculated result. Produced by an identified calculation with retained provenance.

Literature value. Taken from an identified and verified source.

Expected behavior. A prediction, not an observation.

Illustrative example. Explanatory material without research-evidence status.

Synthetic test data. Constructed for software or numerical testing.

Placeholder. Incomplete material awaiting authoritative content.

Proposed work. Planned but not completed.

Mathematical proof, software verification, numerical verification, scientific validation, and uncertainty quantification remain distinct evidence types. A claim may require more than one of them, and none is inferred merely from the presence of the others.

A.9 Units and coordinate conventions

Every physical numerical value states its unit. Energy references used for plots, bulk observables, and pristine–doped subtraction are identified separately. Real-space coordinates identify lattice vectors, cell convention, and site mapping. Reciprocal coordinates identify the reciprocal basis and any path parameterization. Spinless, collinear-spin, and non-collinear spinor representations are never mixed implicitly.

The notation in this appendix is intentionally insufficient to authorize a comparison. The represented records and owning specifications must still supply the exact metadata needed by the operation.

CHAPTER B

Hilbert–Schmidt Geometry and the Frobenius Norm

Status expository appendix. This appendix records the operator-space geometry underlying Frobenius-norm comparisons of reduced operators. It does not define a scientific acceptance metric.

B.1 Hilbert–Schmidt inner product

Let V be a finite-dimensional complex Hilbert space, and let $\mathcal{L}(V)$ denote the vector space of linear operators from V to itself. If $\dim V = d$, then $\dim \mathcal{L}(V) = d^2$. The function

$$\langle \cdot, \cdot \rangle_{\mathrm{HS}} : \mathcal{L}(V) \times \mathcal{L}(V) \longrightarrow \mathbb{C} \tag{B.1}$$

defined by

$$\langle A, B \rangle_{\mathrm{HS}} = \mathrm{Tr}\big(A^\dagger B\big) \tag{B.2}$$

is the *Hilbert–Schmidt inner product*, also called the trace inner product [84]. With this convention it is conjugate-linear in its first argument and linear in its second. Its induced norm is

$$\|A\|_{\mathrm{HS}} = \sqrt{\mathrm{Tr}(A^\dagger A)}. \tag{B.3}$$

Let $\mathcal{H}^{(P)}$ be a finite-dimensional retained Hilbert space. The operator space

$$\mathcal{L}\big(\mathcal{H}^{(P)}\big) \tag{B.4}$$

is itself a finite-dimensional Hilbert space under this inner product. If

$$\hat{A}, \hat{B} : \mathcal{H}^{(P)} \longrightarrow \mathcal{H}^{(P)}, \tag{B.5}$$

then their sum and difference are well-defined elements of the same operator space:

$$\hat{A} \pm \hat{B} : \mathcal{H}^{(P)} \longrightarrow \mathcal{H}^{(P)}. \tag{B.6}$$

Given an orthonormal basis of $\mathcal{H}^{(P)}$, let $\mathbf{A}$ and $\mathbf{B}$ be the matrices representing $\hat{A}$ and $\hat{B}$. The Hilbert–Schmidt inner product becomes the Frobenius inner product,

$$\langle \mathbf{A}, \mathbf{B} \rangle_{\mathrm{F}} = \mathrm{Tr}\big(\mathbf{A}^{\dagger}\mathbf{B}\big) , \tag{B.7}$$

and the corresponding norm satisfies

$$\|\mathbf{A}\|_{\mathrm{F}}^{2} = \mathrm{Tr}\big(\mathbf{A}^{\dagger}\mathbf{A}\big) = \sum_{i,j} |A_{ij}|^{2}. \tag{B.8}$$

Thus the Frobenius norm is the matrix-coordinate form of the Hilbert–Schmidt norm when the coordinates are defined by an orthonormal basis.

Operators may be added and subtracted because they belong to a common operator vector space. The legitimacy of an operator subtraction does not follow merely from two matrices having the same dimensions. It follows from the represented operators having compatible domains and codomains and from the matrices using an identified common state space and basis convention.

B.2 Common projection and reduced components

Let $\hat{P}$ be an orthogonal projector and define

$$\mathcal{H}^{(P)} = \mathrm{Ran}\, \hat{P}. \tag{B.9}$$

For possibly unbounded parent operators $\hat{H}$ and $\hat{T}$, assume that the retained range lies in the domains needed to define the compressions below. Set

$$\hat{H}_{\mathrm{eff}} = \hat{P}\hat{H}\hat{P}\Big|_{\mathcal{H}^{(P)}} , \tag{B.10}$$

$$\hat{T}_{\mathrm{eff}} = \hat{P}\hat{T}\hat{P}\Big|_{\mathcal{H}^{(P)}} . \tag{B.11}$$

Both reduced operators belong to $\mathcal{L}\big(\mathcal{H}^{(P)}\big)$. Their difference is therefore a well-defined reduced operator,

$$\hat{V}_{\mathrm{eff}} = \hat{H}_{\mathrm{eff}} - \hat{T}_{\mathrm{eff}}, \tag{B.12}$$

and, by construction,

$$\hat{H}_{\mathrm{eff}} = \hat{T}_{\mathrm{eff}} + \hat{V}_{\mathrm{eff}}. \tag{B.13}$$

If the parent operators satisfy $\hat{H} = \hat{T} + \hat{V}$ on a compatible domain, linearity gives

$$\hat{V}_{\mathrm{eff}} = \hat{P}\hat{V}\hat{P}\Big|_{\mathcal{H}^{(P)}} . \tag{B.14}$$

Even when $\hat{V}$ is a local multiplication operator in the parent position representation, its compression need not act as multiplication by a scalar function within the retained representation. Projection can produce an operator that is nonlocal in the retained coordinates. If the parent Hamiltonian contains nonlocal pseudopotential, spin–orbit, or other operator components, the chosen kinetic–remainder decomposition must identify those components explicitly.

B.3 Aligned pristine–doped operators

The same common-space requirement applies to independently constructed pristine and doped retained operators. Suppose first that the two retained spaces have equal finite dimension and that a physically admissible unitary identification has been constructed,

$$\hat{U}_d : \mathcal{H}_b^{(P)} \longrightarrow \mathcal{H}_d^{(P)}. \tag{B.15}$$

Let c_b and c_d denote scalar energy references selected by the declared alignment procedure, and define

$$\overline{\hat{H}}_s^{(P)} = \hat{H}_s^{(P)} - c_s \hat{I}_s, \qquad s \in \{b, d\}. \tag{B.16}$$

The aligned impurity operator on the bulk retained space is then

$$\Delta\hat{H}_d^{(P)} = \hat{U}_d^{\dagger}\overline{\hat{H}}_d^{(P)}\hat{U}_d - \overline{\hat{H}}_b^{(P)} \in \mathcal{L}\left(\mathcal{H}_b^{(P)}\right). \tag{B.17}$$

Equal matrix dimensions alone would not establish this membership. The identification must also preserve the physical structures required by the comparison, including the applicable gauge, geometry, orbital ordering, spin structure, and energy reference.

The Hilbert–Schmidt discrepancy of the aligned operator is

$$\left\|\Delta\hat{H}_d^{(P)}\right\|_{\mathrm{HS}}^2 = \mathrm{Tr}\left[\left(\Delta\hat{H}_d^{(P)}\right)^{\dagger}\Delta\hat{H}_d^{(P)}\right]. \tag{B.18}$$

In a common orthonormal numerical basis, this is exactly

$$\left\|\Delta\mathbf{H}_d^{(P)}\right\|_{\mathrm{F}}^2 = \sum_{\alpha,\beta}\left|\Delta H_{d,\alpha\beta}^{(P)}\right|^2. \tag{B.19}$$

Frobenius-norm minimization is therefore least-squares approximation in the finite-dimensional Hilbert space of retained operators.

B.4 Domains and the particle in a box

The differential expression

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} \tag{B.20}$$

does not by itself specify a quantum-mechanical operator. An operator is determined by both its action and its domain. The free kinetic operator on the real line and the kinetic operator on a finite interval with Dirichlet boundary conditions are therefore different operators even though they share the same differential expression.

Prospective citation note—medium priority
Check the operator-domain and self-adjoint realization distinction against a standard Schrodinger-operator source. Candidate: Teschl (2014), bibliography key `teschl2014` [121] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

For the particle in a box, let

$$\hat{H}_{\text{box}} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}, \qquad \mathcal{D}(\hat{H}_{\text{box}}) = H^2(0,L) \cap H_0^1(0,L). \tag{B.21}$$

If $\hat{T}$ is this same Dirichlet kinetic operator, then $\hat{H}_{\text{box}} = \hat{T}$ and consistent projection gives

$$\hat{H}_{\text{eff}} - \hat{T}_{\text{eff}} = \hat{0}. \tag{B.22}$$

A nonzero difference can arise if the chosen reference kinetic operator does not carry the same confinement or boundary-domain structure. Such a comparison first requires an explicit common-space identification. The resulting residual can then record the missing confinement or boundary contribution relative to the chosen reference kinetic operator; it must not be identified automatically with a local physical potential. This distinction separates a mathematically valid represented subtraction from its physical interpretation.

B.5 Infinite-dimensional qualification

For an infinite-dimensional Hilbert space $\mathcal{H}$, the space of all bounded linear operators is not a Hilbert space under the Hilbert–Schmidt inner product. One instead uses the Hilbert–Schmidt class

$$\mathcal{L}_2(\mathcal{H}) = \left\{ \hat{A} : \mathrm{Tr}\left(\hat{A}^\dagger \hat{A}\right) < \infty \right\}. \tag{B.23}$$

For $\hat{A} \in \mathcal{L}_2(\mathcal{H})$,

$$\|\hat{A}\|_{\mathrm{HS}}^2 = \mathrm{Tr}\left(\hat{A}^\dagger \hat{A}\right). \tag{B.24}$$

Finite-rank operators are Hilbert–Schmidt. Consequently, a well-defined compression to a finite-dimensional retained space has finite Hilbert–Schmidt norm.

Prospective citation note—medium priority

Check the infinite-dimensional Hilbert–Schmidt class qualification and the finite-rank inclusion against Simon (2005), bibliography key `simon2005` [111] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The Frobenius norm is invariant under a common unitary change of orthonormal coordinates:

$$\left\|\mathbf{U}^\dagger \mathbf{A}\mathbf{U} - \mathbf{U}^\dagger \mathbf{B}\mathbf{U}\right\|_{\mathrm{F}} = \|\mathbf{A} - \mathbf{B}\|_{\mathrm{F}}. \tag{B.25}$$

It is not meaningful to invoke this invariance when only one operand has been transformed or when the two matrices have not first been placed in a common represented space.

The Hilbert–Schmidt geometry supplies a rigorous finite-dimensional setting for operator comparison and Frobenius-norm minimization. It does not by itself make the Frobenius norm the scientifically appropriate loss. The common state space, retained subspace, normalization, alignment, weighting, domains, energy reference, and relation of the norm to the observables of interest must still be specified.

CHAPTER C

Operator Spaces, Compression, and Alignment

C.1 Scope

The replacement of a first-principles Hamiltonian by a finite model is not defined solely by retaining a selected set of eigenvalues. A reduced description must specify the state space on which the reduced operator acts, the map from the parent space to that state space, the coordinate representation used for its matrix elements, and the class of operators admitted as candidate models. These choices determine which algebraic operations are meaningful and which physical interpretations can be assigned to their results.

The distinction is particularly important when the parent Hamiltonian is an unbounded differential operator, while the reduced description is a finite matrix. In the continuum, the operator domain and boundary conditions are part of the Hamiltonian. In the retained representation, those data are no longer written as an explicit domain, but their consequences remain encoded in the matrix. A rigorous reduction must therefore distinguish the continuum operator, the retained operator, and the numerical matrix representing that operator.

C.2 Operators and their domains

Let $\mathcal{H}$ be a complex Hilbert space. An unbounded operator is a map

$$\hat{A} : \mathcal{D}(\hat{A}) \subseteq \mathcal{H} \longrightarrow \mathcal{H}, \tag{C.1}$$

where the domain $\mathcal{D}(\hat{A})$ is part of the definition of $\hat{A}$. Two operators may have the same formal differential action and nevertheless define different physical systems because they act on different domains. For quantum Hamiltonians, the distinction between a symmetric differential expression and a self-adjoint operator is essential: self-adjointness requires equality of both the operator action and the adjoint domain [121, 97, 103].

The sum or difference of two unbounded operators is initially defined on the intersection

of their domains,

$$\mathcal{D}(\hat{A} \pm \hat{B}) = \mathcal{D}(\hat{A}) \cap \mathcal{D}(\hat{B}). \tag{C.2}$$

Consequently, an expression such as

$$\hat{V} = \hat{H} - \hat{T} \tag{C.3}$$

is an operator identity only after the operators and their compatible domains have been specified. This qualification is not an objection to subtraction. It identifies the mathematical data required for the subtraction to refer to a common operator space.

C.3 Spectral restriction

Consider a self-adjoint Hamiltonian $\hat{H}$ with a pure-point spectral sector,

$$\hat{H}|\psi_i\rangle = E_i|\psi_i\rangle, \tag{C.4}$$

where the retained eigenvectors are orthonormal. Let $\mathcal{I}_P$ denote the selected set of spectral indices and define

$$\hat{P} = \sum_{i \in \mathcal{I}_P} |\psi_i\rangle\langle\psi_i|, \tag{C.5}$$

with retained space

$$\mathcal{H}^{(P)} = \text{Ran}\, \hat{P}. \tag{C.6}$$

If an energy cutoff intersects a degenerate eigenspace, the full degenerate eigenspace must be retained if the construction is to remain invariant under unitary rotations within that eigenspace. More generally, the selected range should be an invariant subspace of the parent operator.

The retained Hamiltonian is

$$\hat{H}^{(P)} = \hat{P}\hat{H}\hat{P}\Big|_{\mathcal{H}^{(P)}} : \mathcal{H}^{(P)} \longrightarrow \mathcal{H}^{(P)}. \tag{C.7}$$

It reproduces the action of the parent Hamiltonian exactly on the selected invariant subspace. When $\hat{P}\hat{H}\hat{P}$ is instead regarded as an operator on the full parent space, it acts as zero on $\text{Ran}(\hat{I} - \hat{P})$. That zero action is an embedding convention; it is not the parent Hamiltonian on the discarded sector.

This distinction separates three objects:

$$\hat{H}, \qquad \hat{H}^{(P)}, \qquad \mathbf{H}^{(P)}. \tag{C.8}$$

They denote, respectively, the parent operator, the retained operator, and a matrix representation of the retained operator in a declared basis. Treating them as interchangeable obscures the state-space maps on which later comparisons depend.

C.4 Compression and downfolding

A fixed-subspace compression should not be identified with an effective Hamiltonian obtained by eliminating the complementary sector. With

$$\hat{Q} = \hat{I} - \hat{P}, \tag{C.9}$$

a Feshbach–Löwdin construction gives the energy-dependent operator

$$\hat{H}_{\text{down}}(E) = \hat{P}\hat{H}\hat{P} + \hat{P}\hat{H}\hat{Q}\left(E - \hat{Q}\hat{H}\hat{Q}\right)^{-1}\hat{Q}\hat{H}\hat{P}, \tag{C.10}$$

whenever the reduced resolvent exists. This projection-operator elimination is the Feshbach–Löwdin construction [28, 67, 68]. The second term represents virtual excursions into the eliminated subspace. In general,

$$\hat{H}_{\text{down}}(E) \neq \hat{H}^{(P)}. \tag{C.11}$$

For the exact spectral projector used in the preceding section, $\hat{P}\hat{H}\hat{Q} = 0$, so the correction vanishes. For a projector chosen from an orbital or other proposed model space, the coupling blocks need not vanish. A related Schrieffer–Wolff construction instead seeks a unitary transformation that exactly or perturbatively block diagonalizes the selected sectors; it must not be identified with merely deleting the Q block [12].

The approximation target must therefore be declared explicitly. A model fitted to $\hat{H}^{(P)}$ approximates the parent action on a selected subspace. A model fitted to $\hat{H}_{\text{down}}(E)$ also attempts to reproduce eliminated-space effects and may inherit energy dependence and nonlocality.

C.5 Common compression and additive structure

Suppose that

$$\hat{H} = \hat{T} + \hat{V} \tag{C.12}$$

is valid on a compatible domain, and assume that the finite-dimensional retained space lies within the required domains. Applying the same compression to every term gives

$$\hat{H}^{(P)} = \hat{T}^{(P)} + \hat{V}^{(P)}, \tag{C.13}$$

where

$$\hat{T}^{(P)} = \hat{P}\hat{T}\hat{P}\Big|_{\mathcal{H}^{(P)}}, \qquad \hat{V}^{(P)} = \hat{P}\hat{V}\hat{P}\Big|_{\mathcal{H}^{(P)}}. \tag{C.14}$$

It follows that

$$\hat{V}^{(P)} = \hat{H}^{(P)} - \hat{T}^{(P)}. \tag{C.15}$$

The subtraction is valid because the corresponding addition was performed in the common vector space

$$\mathcal{L}\big(\mathcal{H}^{(P)}\big)\,. \tag{C.16}$$

The physical interpretation of the difference is a separate question. Although $\hat{V}$ may be a local multiplication operator in the parent position representation, $\hat{P}\hat{V}\hat{P}$ is generally nonlocal. If

$$\langle x|\hat{V}|x'\rangle = V(x)\delta(x-x'), \tag{C.17}$$

then

$$V^{(P)}(x,x') = \int \mathrm{d}y\, P(x,y)V(y)P(y,x'), \tag{C.18}$$

which is not generally proportional to $\delta(x-x')$. In periodic problems, even an exactly finite-range parent can acquire nonzero projected Fourier blocks at arbitrarily distant translations because its spectral projector need not be a finite Fourier polynomial. Gaps and regularity may still imply decay, but exact finite support is a stronger condition [20, 4]. Locality must therefore be treated as an admissible model assumption rather than as an automatic consequence of projection.

C.6 Hilbert–Schmidt geometry

The space of all bounded operators on an infinite-dimensional Hilbert space is not a Hilbert space under the trace inner product. The relevant infinite-dimensional space is the Hilbert–Schmidt class

$$\mathfrak{S}_2(\mathcal{H}) = \left\{ \hat{A} \in \mathcal{B}(\mathcal{H}) \;\middle|\; \mathrm{Tr}(\hat{A}^\dagger \hat{A}) < \infty \right\}, \tag{C.19}$$

with

$$\langle \hat{A}, \hat{B}\rangle_{\mathrm{HS}} = \mathrm{Tr}(\hat{A}^\dagger \hat{B}) \tag{C.20}$$

and

$$\|\hat{A}\|_{\mathrm{HS}}^2 = \mathrm{Tr}(\hat{A}^\dagger \hat{A}) \tag{C.21}$$

[111]. Full continuum Hamiltonians are generally unbounded and are not Hilbert–Schmidt operators. The trace inner product should not be applied to them without an additional construction.

Finite-dimensional retained operators avoid this obstruction. In any orthonormal basis of an M-dimensional retained space,

$$\|\hat{A}\|_{\mathrm{HS}}^2 = \sum_{\alpha,\beta=1}^{M} |A_{\alpha\beta}|^2 = \|\mathbf{A}\|_{\mathrm{F}}^2. \tag{C.22}$$

The Frobenius norm is therefore the coordinate representation of the Hilbert–Schmidt norm. Its unitary invariance makes it suitable for operator comparisons after the relevant state spaces have been identified. It does not determine the identification itself.

C.7 Alignment across retained spaces

Let $\hat{H}_b^{(P)}$ and $\hat{H}_d^{(P)}$ denote retained pristine and doped Hamiltonians acting on $\mathcal{H}_b^{(P)}$ and $\mathcal{H}_d^{(P)}$. Equal matrix dimensions do not imply that the two matrices represent operators on the same physical state space. Before subtraction, assign the declared scalar energy zeros $E_{0,b}$ and $E_{0,d}$ and form

$$\overline{\hat{H}}_b^{(P)} = \hat{H}_b^{(P)} - E_{0,b}\hat{I}_b, \tag{C.23}$$

$$\overline{\hat{H}}_d^{(P)} = \hat{H}_d^{(P)} - E_{0,d}\hat{I}_d, \tag{C.24}$$

where $\hat{I}_b$ and $\hat{I}_d$ are the identities on the two retained spaces. A unitary identification

$$\hat{U}_d : \mathcal{H}_b^{(P)} \longrightarrow \mathcal{H}_d^{(P)} \tag{C.25}$$

must then be specified. The signed aligned difference pulled into pristine coordinates is

$$\Delta\hat{H}_{d\to b}^{(P)} = \hat{U}_d^\dagger \overline{\hat{H}}_d^{(P)} \hat{U}_d - \overline{\hat{H}}_b^{(P)}. \tag{C.26}$$

The arrow records the coordinate-transport direction. When both source operators already use one recorded scalar energy zero, the corresponding $E_{0,b}$ and $E_{0,d}$ corrections vanish. Otherwise, omitting them would add an arbitrary identity component to the represented difference. Chapter 22 gives the detailed compatibility, energy-reference, directional-extraction, and locality contract for impurity operators.

Wannier functions provide localized frames for such comparisons, but their construction includes unitary gauge freedom, and an entangled band manifold requires an explicit retained-subspace selection [72, 115]. The alignment record must therefore include the retained subspaces, orbital labels, gauge, geometry, units, spin convention, and energy reference.

The fundamental conclusion is that model reduction is performed on identified operator spaces, not on arrays whose dimensions happen to agree. Once the spaces and maps are fixed, subtraction and Hilbert–Schmidt comparison are mathematically straightforward. The scientific content lies in justifying the projection, the identification, and the admissible model class.

CHAPTER D

Particle-in-a-Box Residuals and Identifiability

Status calculated illustrative finite-dimensional exercise. The retained $N = 8$, $M = 3$ residual result and a grid-refinement series over $N = 8, 16, 32, 64, 128, 256$, together with exact inputs, reproduction and independent verification scripts, software identities, graphics, and checksums, are recorded under `calculations/piab1d/`. The residual identities pass their declared checks. Fixed-index eigenvalue series approach observed order two, including the higher modes $n = 8, 16, 32$. A full-spectrum sweep shows nonuniform eigenvalue dispersion when the mode index scales with grid dimension, while nodal eigenvector overlap remains at binary64 roundoff. A multi-norm sweep records raw and same-grid-normalized Frobenius, spectral, and maximum-entry diagnostics and keeps algebraic eigensolver residuals separate from continuum error. An executable retained-space demonstration verifies two distinct exact decompositions of the same reduced Hamiltonian and compares four nested admissible model classes. This is numerical verification for the stated matrices, modes, grids, and illustrative model classes, not uniform spectral or operator convergence, physical identification of the alternative decomposition, semiconductor evidence, or scientific validation.

D.1 Purpose of the example

The one-dimensional particle in a box provides a minimal setting in which the continuum Hamiltonian, boundary conditions, discrete representation, spectral restriction, and effective-model interpretation can be separated explicitly. The example is useful not because the spectrum is difficult to obtain, but because every state-space map can be written in closed form. It therefore exposes several distinct residual operators that can otherwise be conflated under the notation V_{eff}.

D.2 Continuum Hamiltonian

Let a particle of mass m be confined to the interval

$$\Omega = (0, L). \tag{D.1}$$

The Hamiltonian is the Dirichlet realization of the kinetic-energy differential expression,

$$\hat{H}_{\text{box}} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}, \tag{D.2}$$

with domain

$$\mathcal{D}(\hat{H}_{\text{box}}) = H^2(0, L) \cap H_0^1(0, L). \tag{D.3}$$

Equivalently,

$$\psi(0) = \psi(L) = 0. \tag{D.4}$$

The eigenfunctions and eigenvalues are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \tag{D.5}$$

and

$$E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}, \qquad n = 1, 2, \ldots. \tag{D.6}$$

The interior potential is zero. Confinement is encoded in the domain of the self-adjoint kinetic operator rather than by finite values of a multiplication potential inside the interval [9, 1]. A different boundary condition applied to the same formal differential expression defines a different operator.

D.3 Finite-difference realization

Introduce N interior points

$$x_j = j\Delta x, \qquad \Delta x = \frac{L}{N+1}, \qquad j = 1, \ldots, N. \tag{D.7}$$

The boundary values are fixed to zero and are not independent components of the numerical state vector. With the centered second-order approximation,

$$\left.\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\right|_{x=x_j} \approx \frac{\psi(x_{j+1}) - 2\psi(x_j) + \psi(x_{j-1})}{\Delta x^2}, \tag{D.8}$$

the discrete Hamiltonian is

$$\mathbf{H}_h = \frac{\hbar^2}{2m\Delta x^2} \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}. \tag{D.9}$$

This matrix is the second-order discrete Dirichlet Laplacian multiplied by the kinetic prefactor [118]. Its eigenvalues are

$$\varepsilon_i = \frac{2\hbar^2}{m\Delta x^2} \sin^2\left(\frac{i\pi}{2(N+1)}\right), \qquad i = 1, \dots, N. \tag{D.10}$$

For fixed i,

$$\varepsilon_i \longrightarrow E_i \tag{D.11}$$

as $N \to \infty$. Writing

$$z_i = \frac{i\pi}{2(N+1)}, \tag{D.12}$$

the dispersion error satisfies

$$\frac{\varepsilon_i}{E_i} = \left(\frac{\sin z_i}{z_i}\right)^2, \tag{D.13}$$

and hence

$$\frac{\varepsilon_i - E_i}{E_i} = -\frac{z_i^2}{3} + O(z_i^4). \tag{D.14}$$

The relation between a Euclidean-normalized eigenvector $\mathbf{v}_i$ and samples of a normalized continuum eigenfunction contains the quadrature factor

$$v_i(j) \approx \sqrt{\Delta x}\, \psi_i(x_j), \tag{D.15}$$

up to an arbitrary phase. This factor distinguishes the numerical inner product from the continuum L^2 inner product.

D.4 Spectral reconstruction and retention

Let

$$\mathbf{\Phi} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_N \end{bmatrix}, \tag{D.16}$$

and

$$\mathbf{\Lambda} = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_N). \tag{D.17}$$

The discrete Hamiltonian has the exact spectral reconstruction

$$\mathbf{H}_h = \mathbf{\Phi}\mathbf{\Lambda}\mathbf{\Phi}^\dagger. \tag{D.18}$$

This is an exact identity for the discrete matrix. It is not an equality between the finite-difference matrix and the continuum differential operator.

Retain the lowest M discrete eigenvectors in

$$\mathbf{\Phi}_M = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_M \end{bmatrix}, \tag{D.19}$$

and define

$$\mathbf{P}_M = \mathbf{\Phi}_M \mathbf{\Phi}_M^\dagger. \tag{D.20}$$

The retained Hamiltonian embedded in the full grid space is

$$\mathbf{H}_h^{(P_M)} = \mathbf{P}_M \mathbf{H}_h \mathbf{P}_M = \mathbf{\Phi}_M \mathbf{\Lambda}_M \mathbf{\Phi}_M^\dagger, \tag{D.21}$$

whereas its representation in retained coordinates is

$$\mathbf{H}_{\text{red}} = \mathbf{\Phi}_M^\dagger \mathbf{H}_h \mathbf{\Phi}_M = \mathbf{\Lambda}_M. \tag{D.22}$$

The first matrix is $N \times N$ and has rank M; the second is $M \times M$. They represent the same retained operator in different coordinate spaces. Their dimensions and embeddings must be accounted for before subtraction.

D.5 Three inequivalent residuals

D.5.1 Consistently compressed physical potential

For the particle in the box,

$$\mathbf{H}_h = \mathbf{T}_{\text{D},h}, \tag{D.23}$$

where $\mathbf{T}_{\text{D},h}$ is the discrete Dirichlet kinetic operator. Applying the same spectral compression to both terms gives

$$\mathbf{H}_h^{(P_M)} - \mathbf{T}_{\text{D},h}^{(P_M)} = \mathbf{0}. \tag{D.24}$$

This is the consistently reduced physical potential. It vanishes because the confinement is already included in the Dirichlet realization of the kinetic operator.

D.5.2 Projected Hamiltonian minus unprojected kinetic operator

Consider instead

$$\widetilde{\mathbf{R}}_h = \mathbf{H}_h^{(P_M)} - \mathbf{T}_{\text{D},h}. \tag{D.25}$$

Because $\mathbf{P}_M$ is a spectral projector of $\mathbf{T}_{\mathrm{D},h}$, it commutes with that matrix. With

$$\mathbf{Q}_M = \mathbf{I} - \mathbf{P}_M, \tag{D.26}$$

one obtains

$$\widetilde{\mathbf{R}}_h = -\mathbf{Q}_M \mathbf{T}_{\mathrm{D},h} \mathbf{Q}_M. \tag{D.27}$$

The nonzero residual is therefore the negative discarded kinetic sector. It results from applying the retention map to the Hamiltonian but not to the kinetic operator. It is neither the consistently projected potential nor a boundary operator.

D.5.3 Difference between discrete boundary realizations

Let

$$\mathbf{H}_{\mathrm{box},h}, \mathbf{T}_{\mathrm{ref},h} \in \mathcal{L}(\mathcal{V}_h) \tag{D.28}$$

be two discrete operators acting on an explicitly identified numerical state space $\mathcal{V}_h$. Suppose that $\mathbf{H}_{\mathrm{box},h}$ carries the box boundary closure and $\mathbf{T}_{\mathrm{ref},h}$ carries a different declared closure. Then

$$\mathbf{B}_h = \mathbf{H}_{\mathrm{box},h} - \mathbf{T}_{\mathrm{ref},h} \tag{D.29}$$

is a well-defined finite-dimensional operator. Depending on the discretization, $\mathbf{B}_h$ may be expressed through altered boundary rows, modified stencils, eliminated degrees of freedom, or nonlocal couplings. In this construction the residual encodes boundary structure relative to the selected reference realization.

The identification of the two numerical state spaces is part of the construction. The result should not be described as a representation-independent continuum potential.

D.6 Residual identifiability

The three cases demonstrate that the symbol

$$\mathbf{H}_{\mathrm{eff}} - \mathbf{T} \tag{D.30}$$

does not have a unique physical interpretation. Its meaning depends on which operator was reduced, which reference was selected, and whether both terms were passed through the same map.

More generally, define

$$\mathbf{R}_{T_0} = \mathbf{H}_{\mathrm{red}} - \mathbf{T}_0. \tag{D.31}$$

For any Hermitian matrix $\mathbf{K}$ on the retained space,

$$\mathbf{H}_{\mathrm{red}} = (\mathbf{T}_0 + \mathbf{K}) + (\mathbf{R}_{T_0} - \mathbf{K}). \tag{D.32}$$

The reduced Hamiltonian alone therefore does not identify a unique kinetic–potential decomposition. A potential interpretation requires prior restrictions on the kinetic reference and on the admissible potential class.

Let

$$\mathcal{M}_V = \{\mathbf{V}(\boldsymbol{\theta}) \mid \boldsymbol{\theta} \in \Theta\} \tag{D.33}$$

be a prescribed model class. The best admissible potential is

$$\mathbf{V}^{\star}_{\mathcal{M}} = \arg\min_{\mathbf{V}\in\mathcal{M}_V} \|\mathbf{R}_{T_0} - \mathbf{V}\|_{\mathrm{F}}^2 , \tag{D.34}$$

with unexplained residual

$$\mathbf{E}_{\mathcal{M}} = \mathbf{R}_{T_0} - \mathbf{V}^{\star}_{\mathcal{M}}. \tag{D.35}$$

If arbitrary Hermitian matrices are admitted, the minimization is trivial. The inverse problem becomes scientifically informative only when $\mathcal{M}_V$ is restricted by locality, range, symmetry, differential form, parameter sharing, or transferability.

The particle in a box therefore supplies a controlled demonstration of the central methodological point: operator subtraction is straightforward after a common numerical state space has been established, but the physical category assigned to the result is determined by the reference operator and the admissible model class.

D.7 Retained numerical resolution

The retained illustrative calculation operationalizes this argument rather than using spectral convergence as a substitute for it. On the $N = 8$ full coordinate space with $M = 3$ retained spectral coordinates, independent checks recover the closed-form finite-difference spectrum, the projector and embedding identities, the zero consistently compressed potential, the negative discarded kinetic sector, and the declared two-corner boundary-realization difference.

The convergence studies are retained as supporting numerical verification. For $N = 8, 16, 32, 64, 128, 256$, fixed-index eigenvalue errors approach second order. The full-spectrum sweep shows that this convergence is not uniform when the mode index grows with the grid dimension. Raw and same-grid-normalized Frobenius, spectral, and maximum-entry norms further show that norm choice changes which aspect of a residual is emphasized; no norm assigns the residual a physical category.

The central identifiability statement is also executable. On the same $M = 3$ retained coordinate space, both

$$\mathbf{H}_{\text{red}} = \mathbf{T}_{\text{D},red} + \mathbf{0} \tag{D.36}$$

and

$$\mathbf{H}_{\text{red}} = (\mathbf{T}_{\text{D},red} - \mathbf{K}) + \mathbf{K} \tag{D.37}$$

reconstruct the reduced Hamiltonian to binary64 roundoff for a declared illustrative Hermitian shift $\mathbf{K}$. The second decomposition is not given a physical interpretation. Fitting that shift in Frobenius norm leaves unexplained norms of approximately 1.997, 1.856, 0.566, and 0 for the scalar-identity, retained-basis diagonal, real-symmetric tridiagonal, and arbitrary-real-symmetric model classes, respectively. The zero unrestricted fit is the trivial case predicted above; the differing restricted residuals make the model-class dependence explicit.

The exact protocol, inputs, outputs, graphics, independent verification scripts, software identities, and checksums are retained under `calculations/piab1d/`. These calculated illustrative results resolve the finite example posed here. They do not select a physically preferred alternative kinetic reference or validate a material model.

CHAPTER E

Two Finite Harmonic-Oscillator Constructions: A Frobenius Comparison

Status calculated illustrative numerical experiment. The retained sweep, comparison maps, verifier, report, and graphics are under `calculations/research-monograph/harmonic-oscillator/`. The calculation compares a ladder-operator representation with a harmonic-oscillator potential enclosed by a finite Dirichlet box after both have been placed in the same retained ladder coordinates. The results provide numerical verification for the declared finite construction, not semiconductor validation.

E.1 Question addressed by the experiment

The one-dimensional quantum harmonic oscillator admits the exact ladder-operator representation

$$\hat{H}_{\mathrm{QHO}} = \hbar\omega \left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\hat{I} \right) \tag{E.1}$$

on $L^2(\mathbb{R})$. The same potential can be placed inside a particle in a box and represented on a finite spatial grid. Enlarging the box is expected to reduce the influence of the Dirichlet walls on low-lying oscillator states, while refining the grid is expected to reduce spatial-discretization error.

The central quantity is not a raw difference between a ladder matrix and a grid matrix. Those matrices act in different coordinate spaces and may have different dimensions. Instead, the finite-box Hamiltonian must first be pulled back to a declared finite ladder space. The primary question is then

> How does the Frobenius discrepancy between the pulled-back finite-box operator and the exact retained ladder Hamiltonian behave as the box is enlarged and the spatial discretization is refined?

This construction also permits comparisons between spatial discretization schemes with different grid spacings or dimensions because every represented operator is mapped to the same retained ladder coordinates before subtraction.

E.2 Exact retained ladder Hamiltonian

The oscillator length is

$$\ell = \sqrt{\frac{\hbar}{m\omega}}. \tag{E.2}$$

The normalized number states satisfy

$$\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{E.3}$$

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle, \tag{E.4}$$

and

$$\hat{H}_{\mathrm{QHO}}|n\rangle = E_n|n\rangle, \qquad E_n = \hbar\omega\left(n+\frac{1}{2}\right). \tag{E.5}$$

For a retained dimension K, define the abstract comparison space

$$\mathcal{K}_K = \mathrm{span}\,\{|0\rangle,\ldots,|K-1\rangle\}. \tag{E.6}$$

In the ordered number-state basis, the exact retained Hamiltonian is

$$\mathbf{H}_{\mathrm{lad}}^{(K)} = \hbar\omega\ \mathrm{diag}\left(\frac{1}{2},\frac{3}{2},\ldots,K-\frac{1}{2}\right). \tag{E.7}$$

This is an exact spectral restriction of the real-line oscillator. It is the reference operator for the retained Frobenius comparison.

A finite ladder representation has an algebraic truncation boundary. If $\mathbf{a}_K$ is the compressed annihilation operator, then

$$[\mathbf{a}_K, \mathbf{a}_K^\dagger] = \mathbf{I}_K - K|K-1\rangle\langle K-1|. \tag{E.8}$$

This top-state defect should be recorded when ladder operators themselves are compared, although it does not alter the diagonal reference energies written above.

Prospective citation note—medium priority

Check the exact one-dimensional harmonic- oscillator ladder representation, spectrum, oscillator length, and retained finite-ladder qualification against a standard quantum-mechanics source. Candidate: Shankar (1994), bibliography key `shankar1994` [108] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

E.3 Harmonic potential inside a Dirichlet box

Let $X > 0$ be the box half-width. The finite-box continuum Hamiltonian is

$$\hat{H}_X = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2 x^2 \tag{E.9}$$

on $L^2((-X, X))$, with domain

$$\mathcal{D}(\hat{H}_X) = H^2((-X, X)) \cap H_0^1((-X, X)). \tag{E.10}$$

Thus

$$\psi(-X) = \psi(X) = 0. \tag{E.11}$$

For finite X, this is a different continuum operator from the oscillator on $\mathbb{R}$. The difference is a boundary-domain effect, not a change to the interior multiplication potential.

For a spatial representation labeled by h, let the numerical state space be $\mathcal{V}_{X,h}$ and let

$$\mathbf{H}_{X,h} = \mathbf{T}_{\mathrm{D},X,h} + \mathbf{V}_{\mathrm{QHO},X,h} \in \mathcal{L}(\mathcal{V}_{X,h}). \tag{E.12}$$

Here $\mathbf{T}_{\mathrm{D},X,h}$ represents the kinetic operator with the declared Dirichlet closure, and $\mathbf{V}_{\mathrm{QHO},X,h}$ represents the interior potential $m\omega^2 x^2/2$. For a uniform grid with N interior points,

$$x_j = -X + j\Delta x, \qquad \Delta x = \frac{2X}{N+1}, \qquad j = 1, \ldots, N. \tag{E.13}$$

The comparison below does not require two spatial representations to have the same N or Δx.

E.4 Map from ladder states to a spatial representation

Let

$$\psi_n(x) = \langle x \mid n \rangle \tag{E.14}$$

be the normalized real-line oscillator eigenfunction. For a uniform grid, define the quadrature-scaled sampled vector

$$\mathbf{s}_{n;X,h}(j) = \sqrt{\Delta x}\, \psi_n(x_j). \tag{E.15}$$

Collect the first K sampled states in

$$\mathbf{S}_{X,h}^{(K)} = \begin{bmatrix} \mathbf{s}_{0;X,h} & \cdots & \mathbf{s}_{K-1;X,h} \end{bmatrix}. \tag{E.16}$$

The columns need not be exactly orthonormal after restriction to a finite box and spatial representation. Their Gram matrix is

$$\mathbf{G}_{X,h}^{(K)} = \left(\mathbf{S}_{X,h}^{(K)}\right)^{\dagger} \mathbf{S}_{X,h}^{(K)}. \tag{E.17}$$

Provided that $\mathbf{G}_{X,h}^{(K)}$ is positive definite, define the symmetrically orthonormalized injection

$$\mathbf{J}_{X,h}^{(K)} = \mathbf{S}_{X,h}^{(K)} \left(\mathbf{G}_{X,h}^{(K)}\right)^{-1/2}. \tag{E.18}$$

It satisfies

$$\left(\mathbf{J}_{X,h}^{(K)}\right)^{\dagger} \mathbf{J}_{X,h}^{(K)} = \mathbf{I}_K. \tag{E.19}$$

The columns of $\mathbf{J}_{X,h}^{(K)}$ identify the abstract ladder coordinates with an orthonormal K-dimensional subspace of $\mathcal{V}_{X,h}$.

Prospective citation note—low priority

Check the inverse-square-root symmetric orthonormalization terminology and conclusion against Higham (1986), bibliography key `higham1986` [46] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The deviation

$$\delta G(X, h; K) = \left\| \mathbf{G}_{X,h}^{(K)} - \mathbf{I}_K \right\|_{\mathrm{F}} \tag{E.20}$$

and the condition number of $\mathbf{G}_{X,h}^{(K)}$ should be reported. They measure whether finite-box restriction and spatial representation have made the declared ladder-state identification inaccurate or ill-conditioned. An ill-conditioned Gram matrix is a failure of the selected comparison map, not a small operator discrepancy.

For a nonuniform grid or another spatial scheme, the same construction requires the declared discrete inner product or quadrature matrix. The injection must be isometric with respect to that inner product before a Frobenius comparison is interpreted as a Hilbert–Schmidt comparison.

E.5 Finite-box Hamiltonian in ladder coordinates

Pull the represented finite-box Hamiltonian back to the retained ladder space:

$$\mathbf{A}_{X,h}^{(K)} = \left(\mathbf{J}_{X,h}^{(K)}\right)^{\dagger} \mathbf{H}_{X,h} \mathbf{J}_{X,h}^{(K)}. \tag{E.21}$$

Both

$$\mathbf{A}_{X,h}^{(K)}, \mathbf{H}_{\mathrm{lad}}^{(K)} \in \mathbb{C}^{K\times K} \tag{E.22}$$

now represent operators in the same declared ladder coordinates. Their subtraction is therefore defined.

The primary absolute discrepancy is

$$D_F(X,h;K) = \left\| \mathbf{A}_{X,h}^{(K)} - \mathbf{H}_{\mathrm{lad}}^{(K)} \right\|_{\mathrm{F}}. \tag{E.23}$$

A dimensionless relative discrepancy is

$$\delta_F(X,h;K) = \frac{\left\| \mathbf{A}_{X,h}^{(K)} - \mathbf{H}_{\mathrm{lad}}^{(K)} \right\|_{\mathrm{F}}}{\left\| \mathbf{H}_{\mathrm{lad}}^{(K)} \right\|_{\mathrm{F}}}. \tag{E.24}$$

This is the principal retained output of the experiment. Its interpretation is the finite-box-plus-representation discrepancy on the declared K-state ladder subspace. It is not a norm difference between the full unbounded continuum Hamiltonians.

Because the reference matrix is diagonal, the discrepancy separates as

$$D_F(X,h;K)^2 = D_{\mathrm{diag}}(X,h;K)^2 + D_{\mathrm{off}}(X,h;K)^2, \tag{E.25}$$

$$D_{\mathrm{diag}}(X,h;K)^2 = \sum_{n=0}^{K-1} \left| A_{X,h;nn}^{(K)} - E_n \right|^2, \tag{E.26}$$

$$D_{\mathrm{off}}(X,h;K)^2 = \sum_{\substack{m,n=0 \\ m\neq n}}^{K-1} \left| A_{X,h;mn}^{(K)} \right|^2. \tag{E.27}$$

The diagonal term measures errors in the ladder-coordinate energy entries. The off-diagonal term measures mixing produced by the finite boundary, spatial representation, and finite-sample identification. Reporting only eigenvalue errors would omit this off-diagonal operator information.

E.6 Comparing different spatial discretizations

Let h and h' denote two spatial representations. Their grid matrices may have different dimensions, so the raw expression

$$\mathbf{H}_{X,h} - \mathbf{H}_{X,h'} \tag{E.28}$$

is generally undefined. Each operator is instead pulled back independently:

$$\mathbf{A}^{(K)}_{X,h} = \left(\mathbf{J}^{(K)}_{X,h}\right)^{\dagger} \mathbf{H}_{X,h}\mathbf{J}^{(K)}_{X,h}, \tag{E.29}$$

$$\mathbf{A}^{(K)}_{X,h'} = \left(\mathbf{J}^{(K)}_{X,h'}\right)^{\dagger} \mathbf{H}_{X,h'}\mathbf{J}^{(K)}_{X,h'}. \tag{E.30}$$

The cross-discretization discrepancy is then

$$D_F(h, h'; X, K) = \left\| \mathbf{A}^{(K)}_{X,h} - \mathbf{A}^{(K)}_{X,h'} \right\|_{\mathrm{F}}. \tag{E.31}$$

This remains well-defined when the two grids have different spacings, numbers of points, stencils, or other declared representation details. The ladder space supplies the common coordinates.

The maps $\mathbf{J}^{(K)}_{X,h}$ and $\mathbf{J}^{(K)}_{X,h'}$ are part of the comparison and must be retained with the result. Equal output dimensions do not make the comparison independent of these maps.

E.7 Convergence questions

Use the dimensionless controls

$$b = \frac{X}{\ell}, \qquad \eta = \frac{\Delta x}{\ell} \tag{E.32}$$

for a uniform grid. For fixed K, the controlled limiting question is whether

$$\delta_F(X, h; K) \longrightarrow 0 \tag{E.33}$$

when the box expands and the spatial representation is refined. The retained sweep uses $b = 4, 6, 8$, $\eta = 0.2, 0.1, 0.05$, and $K = 2, 4, 6$. At fixed $b = 8$, the final observed spatial-refinement orders are 1.99875, 1.99792, and 1.99696 for $K = 2$, 4, and 6, respectively. This supports the expected second-order behavior for the declared fixed-box sequence; it does not establish uniform convergence in K.

The two limiting effects must be separated:

1. **Spatial refinement at fixed box.** Hold b fixed and reduce η to estimate representation error for the finite-box operator.

2. **Box expansion at controlled resolution.** Increase b while keeping η fixed or sufficiently refined to estimate the effect of the Dirichlet walls.

3. **Retained-space sensitivity.** Repeat the two studies for several values of K. Higher oscillator states extend farther in space and are expected to require larger boxes and finer resolution.

Holding N fixed while increasing X does not isolate box convergence because $\Delta x = 2X/(N+1)$ then grows and the spatial resolution becomes worse.

A nested study may first estimate the refined-grid limit at each fixed X and then examine the large-box behavior of those estimates. The reverse order may also be reported, but the order of limits and the numerical controls must be explicit.

E.8 Calculated graphics and retained result

The retained `convergence-summary.png` reports the relative Frobenius discrepancy under box expansion and fixed-box spatial refinement, the diagonal and off-diagonal discrepancy components, and the sampled-state Gram deviation. The retained `operator-difference-heatmap.png` displays

$$\frac{\mathbf{A}_{X,h}^{(K)} - \mathbf{H}_{\mathrm{lad}}^{(K)}}{\hbar\omega}$$

for $b = 8$, $\eta = 0.05$, and $K = 6$. Exact comparison-map reconstruction contracts and content identities, compact common-coordinate operators, cross-grid comparisons, and diagnostics are in `result.json`; the figures are visualizations of those retained values, not independent evidence.

At $\eta = 0.05$, expansion from $b = 4$ to $b = 6$ substantially reduces the total discrepancy for $K = 4$ and $K = 6$. The $K = 2$ total discrepancy is instead slightly smaller at $b = 4$ than at $b = 6$ or $b = 8$. This does not establish that the narrower box is independently more accurate: boundary-domain and spatial-discretization contributions can partially cancel in the total matrix difference. A continuous two-parameter surface beyond the retained grid remains proposed work and is not represented as a calculated result.

E.9 Interpretation boundary

The calculated experiment tests represented finite-box harmonic oscillators against the exact retained ladder operator under an explicit state-space map. It keeps boundary-domain effects, spatial-representation error, retained-space sensitivity, and map conditioning distinct. The Frobenius norm is applied only after the two operators have been placed in the same orthonormal ladder coordinates, as described in Appendix B.

The retained verifier establishes numerical agreement with the declared finite mathematics and independent analytic oscillator states under binary64 tolerances. The observed refinement behavior is numerical verification for this illustrative comparison under the declared conditions. It does not validate a semiconductor model, establish a continuum

limit for silicon, or supply evidence for an unperformed first-principles calculation.

CHAPTER F

Direct and Wannier-Mediated Tight-Binding Reduction in Silicon

Status mathematical framework and proposed numerical experiment. No bulk-silicon production calculation, accepted Wannier operator, fitted tight-binding model, convergence result, or route comparison is claimed in this appendix. The physical and numerical parent remain governed by the applicable specifications and by Chapters 16 and 17. The representation, alignment, and reduction requirements remain governed by Chapters 18 and 19.

F.1 Purpose

The bulk-silicon program contains two proposed routes from a retained Kohn–Sham parent to an orthogonal sp^3s^* tight-binding model:

$$\text{direct route:} \qquad \text{KS-DFT} \longrightarrow \text{TB}, \tag{F.1}$$

$$\text{operator-mediated route:} \qquad \text{KS-DFT} \longrightarrow \text{Wannier} \longrightarrow \text{TB}. \tag{F.2}$$

The routes share a parent but do not perform the same mathematical operations. The direct route is initially a constrained spectral fit. The mediated route constructs a retained localized representation and then compares aligned operators. Their outputs cannot be compared responsibly until the state spaces, maps, gauges, energy references, model classes, and fitting objectives have been declared.

This appendix first states the mathematical issues that determine which comparisons are meaningful. It then describes a proposed numerical experiment for the bulk-silicon pilot. The experiment is intended to compare KS-DFT-to-TB, KS-DFT-to-Wannier, and Wannier-to-TB relations without treating them as three instances of one undifferentiated matrix norm.

F.2 Part I: Mathematical issues

F.2.1 Parent Bloch fibers and retained subspaces

Let $\hat{H}_{\mathrm{KS}}(\mathbf{k})$ denote a Kohn–Sham Bloch-fiber operator at crystal momentum $\mathbf{k}$. Let $\mathcal{S}(\mathbf{k})$ be an M-dimensional retained subspace with orthogonal projector $\hat{P}(\mathbf{k})$. The retained parent operator is

$$\hat{H}^{(P)}_{\mathrm{KS}}(\mathbf{k}) = \hat{P}(\mathbf{k})\hat{H}_{\mathrm{KS}}(\mathbf{k})\hat{P}(\mathbf{k})\Big|_{\mathcal{S}(\mathbf{k})}. \tag{F.3}$$

The exact band count, reciprocal-space domain, projection choices, and energy windows are not fixed by this appendix. They must be inherited from the accepted target-subspace specification.

A list of selected eigenvalues does not by itself define $\mathcal{S}(\mathbf{k})$. Degeneracies, crossings, and entanglement make the projector the stable mathematical object. If disentanglement is used, it selects a subspace from a larger candidate space and can therefore introduce a subspace-selection discrepancy. A later unitary gauge transformation within a fixed retained subspace does not change that projector.

The parent differential operator, the retained operator, and any finite matrix representation remain distinct:

$$\hat{H}_{\mathrm{KS}}(\mathbf{k}), \qquad \hat{H}^{(P)}_{\mathrm{KS}}(\mathbf{k}), \qquad \mathbf{H}^{(P)}_{\mathrm{KS}}(\mathbf{k}). \tag{F.4}$$

Only the last object is directly available to a Frobenius-norm calculation, and only after its basis and normalization have been declared.

F.2.2 From the retained KS operator to a Wannier representation

Let

$$\mathbf{J}_{\mathrm{W}}(\mathbf{k}) : \mathbb{C}^M \longrightarrow \mathcal{S}(\mathbf{k}) \tag{F.5}$$

be an isometric frame map for the selected Wannier gauge, so that

$$\mathbf{J}_{\mathrm{W}}(\mathbf{k})^\dagger \mathbf{J}_{\mathrm{W}}(\mathbf{k}) = \mathbf{I}_M. \tag{F.6}$$

The retained Hamiltonian in Wannier-gauge Bloch coordinates is

$$\mathbf{H}_{\mathrm{W}}(\mathbf{k}) = \mathbf{J}_{\mathrm{W}}(\mathbf{k})^\dagger \hat{H}^{(P)}_{\mathrm{KS}}(\mathbf{k})\mathbf{J}_{\mathrm{W}}(\mathbf{k}). \tag{F.7}$$

For a fixed retained subspace and a complete representation, this is a change of coordinates rather than a tight-binding approximation. Localization chooses a gauge within the subspace. It does not by itself discard rank or hopping blocks.

On a uniform grid of $N_{\mathbf{k}}$ points, the corresponding real-space blocks use the declared Fourier convention

$$\mathbf{H}_{\mathrm{W}}(\mathbf{R}) = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \mathbf{H}_{\mathrm{W}}(\mathbf{k}). \tag{F.8}$$

The inverse transform is

$$\mathbf{H}_{\mathrm{W}}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \mathbf{H}_{\mathrm{W}}(\mathbf{R}) \tag{F.9}$$

when all represented blocks and matching conventions are retained. Discarding blocks beyond a chosen range is a separate truncation.

The KS-DFT-to-Wannier relation must therefore distinguish at least:

1. retained-subspace selection or disentanglement;
2. gauge selection and localization;
3. finite reciprocal-grid interpolation;
4. any omitted real-space blocks; and
5. the numerical fidelity of the producing calculation.

A band-interpolation comparison tests selected spectral consequences of these operations. It is not by itself an operator equality in an undeclared ambient space.

Prospective citation note—medium priority
Check the separation among disentanglement, gauge localization, Fourier interpolation, and real-space truncation against Souza–Marzari–Vanderbilt (2001) and Marzari et al. (2012), bibliography keys `souza2001` [115] and `marzari2012` [72] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

F.2.3 Tight-binding model class

Let

$$\mathbf{H}_{\mathrm{TB}}(\mathbf{k};\boldsymbol{\theta}) \in \mathbb{C}^{M\times M} \tag{F.10}$$

be a member of a declared orthogonal sp^3s^* model class. The parameter domain fixes the orbital ordering, site convention, neighbor shells, Slater–Koster structure, crystal symmetries, Fourier sign, spin convention, and energy reference. Changing the permitted shell range or symmetry-allowed terms changes the model class.

A tight-binding model is not defined solely by a parameter vector. Its operator is

$$\mathbf{H}_{\mathrm{TB}}(\mathbf{k};\boldsymbol{\theta}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \mathbf{H}_{\mathrm{TB}}(\mathbf{R};\boldsymbol{\theta}), \tag{F.11}$$

where the retained blocks and Fourier convention are part of the model. A nearest-neighbor model and a second-neighbor model belong to different restricted classes even when their matrices have the same dimension at each $\mathbf{k}$.

Prospective citation note—medium priority
Check the declared orthogonal sp^3s^* silicon hierarchy, Slater–Koster structure, and neighbor-shell restrictions against Slater–Koster (1954) and Vogl et al. (1983), bibliography keys `slater1954` [113] and `vogl1983` [127] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

F.2.4 Direct KS-DFT-to-TB route

The direct route determines a parameter vector $\boldsymbol{\theta}_{\mathrm{dir}}$ from selected parent spectral data. An abstract training objective is

$$\mathcal{L}_{\mathrm{dir}}(\boldsymbol{\theta}) = \sum_{q \in \mathcal{T}_E} w_q \, \ell_q \left(\mathbf{H}_{\mathrm{TB}}(\boldsymbol{\theta}), \hat{H}_{\mathrm{KS}}^{(P)} \right), \tag{F.12}$$

where $\mathcal{T}_E$ is the frozen spectral training set and every loss term specifies its units and normalization.

This route can compare eigenvalues and derived band quantities without first constructing a common orbital basis. It does not automatically supply an operator Frobenius norm between the KS and TB Hamiltonians. Such a norm would require an additional isometric embedding or alignment between the TB orbital space and the retained KS subspace. Equal numbers of bands do not provide that map.

The direct route must therefore report spectral fitting error separately from withheld spectral and band-edge errors. A small training objective does not establish agreement of orbital character, projectors, gauges, or matrix elements.

F.2.5 Wannier-mediated route

The mediated route first constructs a candidate Wannier operator and then aligns the TB orbital coordinates with the Wannier coordinates. Let

$$\mathbf{C} : \mathbb{C}_{\mathrm{TB}}^{M} \longrightarrow \mathbb{C}_{\mathrm{W}}^{M} \tag{F.13}$$

be a symmetry-compatible unitary alignment after geometry, orbital identity, spin, units, and energy reference have been checked. A proposed real-space operator objective is

$$\mathcal{L}_{\mathrm{med}}(\mathbf{C}, \boldsymbol{\theta}) = \sum_{\mathbf{R}} \omega_{\mathbf{R}} \left\| \mathbf{H}_{\mathrm{W}}(\mathbf{R}) - \mathbf{C}\mathbf{H}_{\mathrm{TB}}(\mathbf{R}; \boldsymbol{\theta})\mathbf{C}^{\dagger} \right\|_{\mathrm{F}}^{2}. \tag{F.14}$$

The direction of $\mathbf{C}$, the included blocks, shell weights, normalization, and treatment of omitted blocks must be fixed before the objective is interpreted.

The alignment does not repair incompatible subspaces. Principal angles, overlap singular values, orbital correspondence, and alignment conditioning must be checked before the operator residual is minimized. Optimizing $\mathbf{C}$ over an unrestricted unitary group could hide physically inadmissible orbital mixing; the admissible alignment class must preserve the structures required by the sp^3s^* interpretation.

F.2.6 Reciprocal-space and real-space Frobenius residuals

For a common aligned gauge, define

$$\Delta\mathbf{H}(\mathbf{k}) = \mathbf{H}_{\mathrm{W}}(\mathbf{k}) - \mathbf{C}\mathbf{H}_{\mathrm{TB}}(\mathbf{k};\boldsymbol{\theta})\mathbf{C}^{\dagger} \tag{F.15}$$

and

$$\Delta\mathbf{H}(\mathbf{R}) = \mathbf{H}_{\mathrm{W}}(\mathbf{R}) - \mathbf{C}\mathbf{H}_{\mathrm{TB}}(\mathbf{R};\boldsymbol{\theta})\mathbf{C}^{\dagger}. \tag{F.16}$$

For a complete discrete Fourier pair with the convention above, Parseval's identity gives

$$\sum_{\mathbf{k}} \|\Delta\mathbf{H}(\mathbf{k})\|_{\mathrm{F}}^2 = N_{\mathbf{k}} \sum_{\mathbf{R}} \|\Delta\mathbf{H}(\mathbf{R})\|_{\mathrm{F}}^2 . \tag{F.17}$$

This equivalence does not survive unchanged when nonuniform weights, incomplete $\mathbf{R}$ sets, different reciprocal grids, or shell-dependent objectives are introduced. Those choices define different losses and must be reported as such.

The Frobenius norm is invariant under a common unitary coordinate change:

$$\left\|\mathbf{U}^{\dagger}\Delta\mathbf{H}\mathbf{U}\right\|_{\mathrm{F}} = \|\Delta\mathbf{H}\|_{\mathrm{F}} . \tag{F.18}$$

It is not invariant under independently changing only one operand. Gauge invariance therefore follows only after the two representations and their alignment map are transformed consistently.

F.2.7 The three pairwise relations

The proposed bulk experiment contains three pairwise relations, but they answer different questions:

KS-DFT to Wannier. Does the selected Wannier subspace and interpolation reproduce the declared retained parent information over the training and withheld domains? Relevant diagnostics include principal angles where a common ambient representation exists, interpolation errors, band-edge errors, centers, spreads, and real-space block decay.

KS-DFT to TB. Does the direct TB model reproduce the frozen parent spectral and band-edge targets? Without an additional orbital embedding, this is primarily a spectral comparison rather than an operator Frobenius comparison.

Wannier to TB. Can the declared TB model class approximate the aligned Wannier operator? This is the natural location for blockwise and shell-resolved Frobenius residuals.

These relations should not be collapsed into a single score. Their errors have different meanings, domains, and dependencies.

F.2.8 Path dependence

Let $\boldsymbol{\theta}_{\mathrm{dir}}$ be the direct-route result and $\boldsymbol{\theta}_{\mathrm{med}}$ the mediated-route result. If both final TB operators can be placed in one canonical orbital convention, define

$$\Delta\mathbf{H}_{\mathrm{path}}(\mathbf{k}) = \mathbf{H}_{\mathrm{TB}}(\mathbf{k};\boldsymbol{\theta}_{\mathrm{dir}}) - \mathbf{C}_{\mathrm{path}}\mathbf{H}_{\mathrm{TB}}(\mathbf{k};\boldsymbol{\theta}_{\mathrm{med}})\mathbf{C}_{\mathrm{path}}^{\dagger}, \tag{F.19}$$

where $\mathbf{C}_{\mathrm{path}}$ records any required admissible alignment. A proposed route discrepancy is

$$D_{\mathrm{path}}^{2} = \sum_{\mathbf{k}\in\mathcal{K}_{\mathrm{cmp}}} w_{\mathbf{k}} \left\|\Delta\mathbf{H}_{\mathrm{path}}(\mathbf{k})\right\|_{\mathrm{F}}^{2}. \tag{F.20}$$

A parameter-vector difference is meaningful only if both vectors use the same identifiable parameterization and canonical convention. Distinct parameter vectors can represent the same operator, while similar parameter vectors can produce different operators when conventions differ. The operator discrepancy is therefore primary.

A nonzero path discrepancy is not automatically an error. It may reflect the difference between a spectral objective and an operator objective, sensitivity to the retained Wannier subspace, alignment constraints, model-class limitations, optimization nonuniqueness, or numerical error. These causes must be investigated separately.

F.2.9 Error accounting

The comparison should retain the following error classes separately:

1. **Parent-model error:** limitations of the declared Kohn–Sham physical approximation relative to independent physical evidence.
2. **Parent numerical error:** incomplete convergence of the plane-wave, reciprocal-space, structural, or solver representation.
3. **Subspace error:** sensitivity to retained rank, projections, and disentanglement windows.

4. **Wannier representation error:** interpolation and finite-grid effects for the declared retained subspace.
5. **Alignment sensitivity:** dependence on orbital matching, gauge, energy zero, and the admissible map.
6. **TB model-reduction error:** inability of the prescribed orbital, symmetry, and neighbor-range class to reproduce the aligned parent representation.
7. **Optimization and identifiability error:** uncertainty caused by incomplete minimization or nonunique parameter descriptions.

No scalar total error is defined by this appendix. Combining these components would require an owning rule that establishes compatible units, norms, correlations, and validation domains.

F.3 Part II: Proposed numerical experiment

F.3.1 Common parent and frozen comparison data

Both routes must consume the same accepted pristine-silicon Kohn–Sham parent, including the same physical branch, lattice, pseudopotential identity, converged numerical settings, reciprocal conventions, and energy reference. The parent dataset must be divided before fitting into frozen training and withheld comparison sets.

The bulk pilot targets the valence and low-conduction information needed for the orthogonal sp^3s^* hierarchy, indirect Kohn–Sham gap, conduction valley, and electron effective masses. The retained rank, windows, projection choices, and interpolation domain must be supplied by the applicable accepted records rather than selected retrospectively to improve the comparison.

No production parent is asserted to exist by this appendix. The following stages are a proposed experiment to be performed only after the required parent and its provenance are available.

F.3.2 Stage A: establish the retained parent target

The first stage should record:

1. the sampled $\mathbf{k}$ points and weights;
2. the retained rank and target energy region;

3. the training and withheld band data;
4. the band-edge quantities and their extraction rules;
5. the parent numerical uncertainty relevant to those quantities; and
6. the representation in which subspace overlaps can be evaluated.

This stage supplies the common target for both reduction routes. It does not fit a TB model or select a Wannier gauge.

F.3.3 Stage B: construct and diagnose the Wannier representation

A candidate Wannier90-derived representation should retain its initial projections, inner and outer windows when applicable, localization settings, centers, spreads, convergence history, Fourier conventions, and software identity. Its evaluation should include:

1. agreement with retained parent bands on the construction grid;
2. interpolation errors at withheld $\mathbf{k}$ points;
3. errors in the indirect gap, valley position, and effective masses;
4. principal-angle or projector diagnostics where the common ambient data permit them; and
5. decay of $\|\mathbf{H}_{\mathrm{W}}(\mathbf{R})\|_{\mathrm{F}}$ with lattice vector and neighbor shell.

A candidate that fails the declared retained-subspace or interpolation criteria should not be promoted to the mediated TB comparison merely because its Wannier functions appear localized.

F.3.4 Stage C: direct spectral TB reduction

The direct route should fit the first declared sp^3s^* model class to the frozen spectral training data. The objective, weights, parameter bounds, symmetry constraints, initialization strategy, and optimization stability checks must be fixed and retained.

After fitting, the model should be evaluated on the withheld band points and on the declared indirect gap, valley position, and longitudinal and transverse effective masses. Training and withheld errors must be shown separately. If the nearest-neighbor class fails a prespecified criterion, the experiment may proceed through the already declared hierarchy; the class sequence must not be expanded after inspecting results merely to force acceptance.

F.3.5 Stage D: Wannier-mediated operator reduction

The mediated route should align the declared TB orbital space with the accepted Wannier space and fit or project the same ordered sp^3s^* hierarchy against the retained Wannier blocks. The experiment should record:

1. the direction and admissible class of the alignment map;
2. overlap, principal-angle, and conditioning diagnostics;
3. the scalar energy-reference convention;
4. included and omitted **R** blocks;
5. shell weights and normalization;
6. onsite, hopping, orbital-block, and symmetry-channel residuals; and
7. sensitivity to the permitted neighbor range.

The resulting model must also be evaluated on the same withheld spectral and band-edge quantities used for the direct route. Operator fitting does not waive spectral evaluation, and spectral agreement does not waive the operator residual.

F.3.6 Stage E: compare the routes

The final comparison should place the direct and mediated TB operators in one canonical orbital, lattice, spin, Fourier, unit, and energy-reference convention. It should then report:

1. the route discrepancy D_{path} over a frozen reciprocal comparison set;
2. corresponding real-space block differences where the complete Fourier conventions permit them;
3. differences in withheld bands and band-edge quantities;
4. model-class and neighbor-shell dependence;
5. alignment and optimization sensitivity; and
6. whether one parameter set satisfies both the prespecified spectral and operator criteria.

If no tested model satisfies the declared criteria, the result should be reported as failure of the tested reduction class rather than as failure of the Kohn–Sham or Wannier parent.

F.3.7 Proposed comparison tables

A first table should identify the immutable inputs and conventions shared by both routes: parent identity, lattice, reciprocal coordinates, retained rank, training set, withheld set, spin convention, units, and energy reference.

A second table should report the KS-to-Wannier diagnostics: interpolation errors, band-edge discrepancies, projector or principal-angle information, centers, spreads, and block-decay summaries.

A third table should compare direct and mediated TB models using separate columns for training spectral error, withheld spectral error, indirect gap, valley position, effective masses, aligned operator residual, model class, and neighbor range. Blank or inapplicable cells should remain explicit rather than being filled by incomparable quantities.

F.4 Plot placeholders

1. **Placeholder—route diagram.** Show the direct KS-DFT-to-TB route and the KS-DFT-to-Wannier-to-TB route, labeling subspace selection, gauge choice, alignment, spectral fitting, and operator reduction as distinct operations.

2. **Placeholder—KS-to-Wannier interpolation.** Overlay retained parent and Wannier-interpolated bands on the frozen training and withheld reciprocal-space domains.

3. **Placeholder—Wannier interpolation error.** Plot bandwise or subspace-aware interpolation discrepancies against $\mathbf{k}$, marking the indirect valley and withheld points.

4. **Placeholder—real-space block decay.** Plot $\|\mathbf{H}_{\mathrm{W}}(\mathbf{R})\|_{\mathrm{F}}$ against distance and neighbor shell, with onsite and symmetry-distinct blocks identified.

5. **Placeholder—direct-route spectral comparison.** Overlay the parent and direct-fit TB bands, distinguishing training from withheld points.

6. **Placeholder—mediated-route spectral comparison.** Overlay the parent, Wannier-interpolated, and operator-fit TB bands on the same reciprocal-space domain.

7. **Placeholder—shell-resolved operator residual.** Plot $\|\Delta\mathbf{H}(\mathbf{R})\|_{\mathrm{F}}$ by distance, shell, and selected orbital block.

8. **Placeholder—model-class progression.** Plot spectral and operator discrepancies separately for the prespecified nearest-neighbor and subsequent tested model classes.

9. **Placeholder—route discrepancy.** Plot $\|\Delta\mathbf{H}_{\mathrm{path}}(\mathbf{k})\|_{\mathrm{F}}$ over the frozen reciprocal comparison path or mesh.

10. **Placeholder—band-edge comparison.** Display the indirect gap, valley position, and longitudinal and transverse effective-mass discrepancies for the Wannier, direct-TB, and mediated-TB representations without combining them into one score.

11. **Placeholder—alignment sensitivity.** Show the change in the operator objective under the declared admissible alignment variants, together with principal-angle and conditioning diagnostics.

12. **Placeholder—residual decomposition.** Display onsite, hopping, orbital-block, symmetry-channel, and omitted-range contributions to the mediated operator discrepancy.

Each completed plot must identify the parent dataset, retained subspace, training or withheld status, units, normalization, alignment convention, model class, and evidentiary status. A placeholder is proposed work, not a calculated result.

F.5 Interpretation boundary

The proposed experiment asks whether a declared sp^3s^* hierarchy can preserve the selected bulk-silicon Kohn–Sham information through two different reduction routes. The KS-to-Wannier relation tests retained representation and interpolation. The direct KS-to-TB relation tests a constrained spectral fit. The Wannier-to-TB relation tests an aligned operator reduction. The comparison of their final TB operators tests path dependence.

Agreement between the routes would not establish that the Kohn–Sham parent is an exact description of experimental silicon. Disagreement would not by itself identify a single cause. Scientific interpretation requires the parent, subspace, representation, alignment, model-class, optimization, and withheld errors to remain visible as separate quantities.

CHAPTER G

Periodic-Reduction Experiments: From Bloch Fibers to Finite-Range Lattice Hamiltonians

Status calculated illustrative numerical verification. Independent discretizations, isolated- and composite-band reductions, and a converged Wannier90 comparison establish the numerical behavior of the specified cosine model. The results do not constitute semiconductor validation or evidence of transferability to silicon.

This appendix is the manuscript home for the dimensional periodic-reduction experiment sequence. The maintained family-first layout places the current one-dimensional single-band experiment at `experiments/periodic-reduction/single-band-fourier-reduction/periodic1d/`. Two- and three-dimensional variants will appear beside it only after their scientific contracts, schemas, and evidence boundaries exist. The present appendix therefore reports the completed one-dimensional scalar and composite cases without implying that higher-dimensional successors have already been executed.

Abstract

Reducing a periodic continuum Hamiltonian to a finite-range lattice model requires controlled choices of representation, retained subspace, gauge, and truncation. We examine these choices in a one-dimensional cosine lattice using independently refined plane-wave and finite-difference discretizations. For the isolated lowest band, localized hopping reconstruction reproduces the parent dispersion, withheld errors decrease with hopping range, and direct Fourier fitting agrees with truncation of the localized operator. Stress tests show that scalar-band tracking becomes unreliable near gap closures and for poorly conditioned higher bands. Composite-band transport resolves this label instability, while controlled gauge perturbations demonstrate that finite-range locality is not gauge invariant. An independent Wannier90 localization implementation applied to the same spectral

and overlap data converges with preconditioning and reproduces the direct centers and common-frame operators to the reported numerical accuracy.

G.1 Introduction

Localized lattice Hamiltonians mediate between periodic continuum operators and the reduced models used in solid-state and semiconductor physics. Wannier's original construction introduced localized orbitals for periodic systems, and subsequent work established their analytic localization properties and the modern variational treatment of isolated and composite band manifolds [129, 57, 73, 115, 72]. Contemporary implementations make these constructions available for realistic electronic structure calculations [91].

The passage from a continuum operator to a finite-range model is nevertheless not determined by band energies alone. It also requires a retained state space, a reciprocal-space gauge, a localization convention, and a real-space truncation. Each choice introduces a distinct approximation or coordinate dependence. Comparing final spectra without controlling these intermediate objects can therefore conceal representation errors, subspace mismatch, or compensating fit errors.

The cosine lattice considered here isolates those issues in a problem whose Bloch fibers, Fourier couplings, reciprocal sewing, and hopping transform can be inspected directly. Chapter 9 provides the preceding pedagogical derivation of the one-dimensional plane-wave and finite-difference fibers, centered scalar transform, modular hopping Hermiticity, Parseval identity, direct least-squares route, and staggered withheld mesh. Chapter 10 then derives composite frames and projectors, the Bloch–Wannier bridge, polar transport, reciprocal closure, Procrustes alignment, and gauge-dependent block locality. The present appendix applies those definitions to retained controlled calculations rather than treating the formulas below as their first explanation. It supplies a controlled analogue of two reduction routes:

$$\text{periodic parent} \longrightarrow \text{finite-range lattice model}, \tag{G.1}$$

$$\text{periodic parent} \longrightarrow \text{localized represented operator} \longrightarrow \text{finite-range lattice model}. \tag{G.2}$$

The first route fits a specified Fourier class directly to reciprocal-space data. The second constructs the complete represented hopping operator and then truncates it in real space. Under matched grids, weights, coordinates, and model classes, the two routes are the same orthogonal projection. Controlled changes to those conditions expose when and why they cease to agree.

The study addresses four related questions. First, do independent plane-wave and finite-difference representations converge to the same parent operator after transport into

common coordinates? Second, when is an individual band sufficiently isolated for scalar parallel transport and Wannier construction? Third, can a composite retained subspace replace unstable energy-ordered band labels while preserving a comparable represented operator? Fourth, how do gauge choice and hopping range affect the accuracy of the final lattice model? A separate Wannier90 calculation tests the localization interface and implementation without introducing an independent physical parent. The contribution is not a new localization algorithm; it is a controlled benchmark that places discretization error, subspace conditioning, gauge covariance, finite-range truncation, and route equivalence in one quantitative setting.

G.2 Continuum parent and exact structure

Let the lattice period be a, define $G = 2\pi/a$, and consider

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V_0\cos(Gx) \quad \text{on } L^2(\mathbb{R}). \tag{G.3}$$

The natural reciprocal energy scale is

$$E_G = \frac{\hbar^2 G^2}{2m}, \qquad \lambda = \frac{V_0}{E_G}. \tag{G.4}$$

Bloch states have the form

$$\psi_{nk}(x) = e^{ikx}u_{nk}(x), \qquad u_{nk}(x+a) = u_{nk}(x), \tag{G.5}$$

and the cell-periodic fiber operator is

$$\hat{H}(k) = \frac{1}{2m}\left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}x} + \hbar k\right)^2 + V_0\cos(Gx). \tag{G.6}$$

Equivalently, the original differential expression can act on a single cell with Bloch boundary conditions

$$\psi(a) = e^{ika}\psi(0), \qquad \psi'(a) = e^{ika}\psi'(0). \tag{G.7}$$

These two conventions are unitarily related, but their matrices must not be mixed without an explicit transformation.

The real inversion-symmetric parent obeys

$$E_n(k+G) = E_n(k), \qquad E_n(-k) = E_n(k), \tag{G.8}$$

after consistent band matching. The sign change $V_0 \mapsto -V_0$ translates the potential by $a/2$ and leaves the spectrum invariant. In the weak-potential regime, the Fourier coefficient $V_0/2$ couples crossing free-particle states and gives a leading first-zone-boundary gap of $|V_0|$. These identities provide structural and perturbative checks that are independent of either numerical representation.

With $z = Gx/2$, Eq. (G.3) becomes the Mathieu equation

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}z^2} + [A - 2q\cos(2z)]\,\psi = 0, \qquad A = \frac{4E}{E_G}, \quad q = \frac{2V_0}{E_G}, \tag{G.9}$$

with Floquet condition

$$\psi(z+\pi) = e^{ika}\psi(z), \qquad \nu = \frac{2k}{G} \pmod 2. \tag{G.10}$$

For the chosen Brillouin zone, $-1 \le \nu < 1$; the equivalent representatives $0 \le \nu < 2$ follow by reduction modulo 2. The checks used below are restricted to branches whose indexing is unambiguous: the lowest zone-center energy is $a_0(q)E_G/4$, while the two lowest zone-boundary energies are the sorted values $a_1(q)E_G/4$ and $b_1(q)E_G/4$. These characteristic values provide a reference that does not reuse either matrix discretization; the notation and Floquet convention follow the standard Mathieu-function treatment [74].

G.3 Numerical representations of the parent

G.3.1 Plane-wave Galerkin representation

In the Bloch plane-wave basis

$$\langle x \mid n, k\rangle = \frac{1}{\sqrt{a}} e^{i(k+nG)x}, \qquad n \in \mathbb{Z}, \tag{G.11}$$

the fiber matrix is

$$H^{\mathrm{PW}}_{nm}(k) = \frac{\hbar^2(k+nG)^2}{2m}\delta_{nm} + \frac{V_0}{2}\left(\delta_{n,m+1} + \delta_{n,m-1}\right). \tag{G.12}$$

The nearest-neighbor structure in reciprocal index is an exact consequence of the potential's Fourier content, not a real-space tight-binding approximation. A cutoff P retains $n = -P, \ldots, P$ and defines the Galerkin matrix $\mathbf{H}^{\mathrm{PW}}_P(k) \in \mathbb{C}^{(2P+1)\times(2P+1)}$. At fixed k, the largest represented free-particle energy is $E_G \max_{|n|\le P}(k/G+n)^2$; its supremum over the first Brillouin zone is $(P+1/2)^2 E_G$. Convergence of low bands must be distinguished from the behavior of states near this reciprocal cutoff. This finite reciprocal-basis construction is the one-dimensional specialization of the plane-wave representation used in periodic electronic-structure methods [88].

G.3.2 Finite-difference Bloch-boundary representation

A second representation acts on one unit cell with a real-space finite-difference method of the type used for basis-free electronic-structure calculations [19]; the stencil-order and truncation-error conventions follow standard finite-difference analysis [118]. Specifically,

$$x_j = j\Delta x, \qquad \Delta x = \frac{a}{N}, \qquad j = 0, \ldots, N-1. \tag{G.13}$$

For the centered second-order kinetic stencil, let

$$t_h = \frac{\hbar^2}{2m\Delta x^2}. \tag{G.14}$$

The kinetic matrix has diagonal entries $2t_h$, nearest-neighbor entries $-t_h$, and boundary entries

$$T^{\mathrm{FD}}_{0,N-1}(k) = -t_h e^{-ika}, \qquad T^{\mathrm{FD}}_{N-1,0}(k) = -t_h e^{ika}. \tag{G.15}$$

Together with

$$\mathbf{V}_h = \mathrm{diag}\left(V_0 \cos(Gx_0), \ldots, V_0 \cos(Gx_{N-1})\right), \tag{G.16}$$

this gives

$$\mathbf{H}^{\mathrm{FD}}_h(k) = \mathbf{T}^{\mathrm{FD}}_h(k) + \mathbf{V}_h. \tag{G.17}$$

Hermiticity requires conjugate boundary phases. A failure of this condition is an implementation error rather than a physical band effect.

For a discrete Bloch wave with $q = k + nG$, the represented kinetic energy is

$$\varepsilon^{\mathrm{kin}}_h(q) = \frac{2\hbar^2}{m\Delta x^2}\sin^2\left(\frac{q\Delta x}{2}\right) = \frac{\hbar^2 q^2}{2m}\left[1 - \frac{(q\Delta x)^2}{12} + O\big((q\Delta x)^4\big)\right]. \tag{G.18}$$

The expected second-order error applies to fixed smooth modes, not to modes approaching the grid cutoff.

G.3.3 Comparison in common coordinates

Matrices of different dimensions or in different bases cannot be subtracted directly. Choosing $N = 2P + 1$ gives equal matrix dimensions, after which the discrete Bloch–Fourier matrix

$$F_{jn}(k) = \frac{1}{\sqrt{N}} e^{i(k+nG)x_j}, \qquad n = -P, \ldots, P, \tag{G.19}$$

transports the finite-difference matrix into discrete plane-wave coordinates:

$$\mathbf{A}^{\mathrm{FD}\to\mathrm{PW}}_h(k) = \mathbf{F}(k)^\dagger \mathbf{H}^{\mathrm{FD}}_h(k)\mathbf{F}(k). \tag{G.20}$$

The full common-coordinate discrepancy is

$$D_{\mathrm{rep}}(k;h,P)=\left\|\mathbf{A}_h^{\mathrm{FD}\to\mathrm{PW}}(k)-\mathbf{H}_P^{\mathrm{PW}}(k)\right\|_{\mathrm{F}}. \tag{G.21}$$

Because cutoff modes can dominate this norm, a fixed low-mode sector is also used. If $\mathbf{P}_L$ selects $|n|\le L<P$, then

$$D_{\mathrm{rep}}^{(L)}(k;h,P)=\left\|\mathbf{P}_L\left(\mathbf{A}_h^{\mathrm{FD}\to\mathrm{PW}}(k)-\mathbf{H}_P^{\mathrm{PW}}(k)\right)\mathbf{P}_L\right\|_{\mathrm{F}}. \tag{G.22}$$

The sampled potential may wrap Fourier couplings across the finite cyclic grid. Discrete Fourier indices are identified modulo N, so on a coarse grid the $\pm G$ harmonic can connect the endpoint reciprocal indices through cyclic wraparound. That aliasing is a property of the representation and must not be interpreted as a physical long-range Fourier component. The common-coordinate map and its aliasing behavior are finite Fourier/spectral representation statements in the sense of Ref. [124], not an identification of the two untransformed matrices.

G.4 Retained subspaces and localization

G.4.1 Isolation and subspace diagnostics

Let $E_n(k)$ denote ordered fiber eigenvalues. An individual band is treated as isolated only when its separation from neighboring bands remains above the prescribed threshold over the complete reciprocal mesh. Near close approaches, independent energy sorting is insufficient: it can exchange labels while leaving the combined subspace continuous.

For two retained frames represented in common ambient coordinates, $\mathbf{B}_1(k)$ and $\mathbf{B}_2(k)$, define

$$\mathbf{S}(k)=\mathbf{B}_1(k)^{\dagger}\mathbf{B}_2(k). \tag{G.23}$$

The singular values of $\mathbf{S}(k)$ measure neighboring or cross-representation subspace compatibility. The associated projector discrepancy is

$$D_P(k)=\left\|\mathbf{B}_1(k)\mathbf{B}_1(k)^{\dagger}-\mathbf{B}_2(k)\mathbf{B}_2(k)^{\dagger}\right\|_{\mathrm{F}}. \tag{G.24}$$

Because the Frobenius norm is definite, $D_P(k)=0$ if and only if the two orthogonal projectors are identical. Spectral agreement, subspace agreement, and basis agreement are therefore reported separately.

G.4.2 Isolated-band parallel transport

For normalized cell-periodic states at neighboring mesh points, define

$$m_j=\langle u_{k_j}\mid u_{k_{j+1}}\rangle. \tag{G.25}$$

When $m_j \neq 0$, the phase of $u_{k_{j+1}}$ is chosen so that m_j is real and nonnegative. Sequential transport gives a smooth frame, while reciprocal sewing at the Brillouin-zone boundary supplies a closure holonomy. Distributing that phase over the mesh produces a periodic gauge. A small $|m_j|$ instead signals poor matching, insufficient sampling, or failure of the isolated-band model; phase smoothing cannot repair the loss of a retained subspace. No universal overlap threshold is imposed here. Overlap magnitudes are interpreted together with gap closure and reciprocal-mesh refinement rather than converted into an automatic scalar-to-composite switch.

For a uniform N_k-point mesh and Born–von Karman representatives $R \in \mathcal{R}_{N_k}$, the Wannier states are

$$|w_R\rangle = \frac{1}{\sqrt{N_k}} \sum_k e^{-ikR} |\psi_k\rangle. \tag{G.26}$$

A smooth periodic phase changes the Wannier profile but preserves the center modulo a. A reciprocal winding translates its representative by a lattice vector. The isolated-band projector and scalar dispersion remain invariant under either compatible gauge change [57, 72].

G.4.3 Composite-band transport

For an M-band retained space, the neighboring overlap matrix is

$$\mathbf{M}^{(j)}_{mn} = \langle u_{m,k_j} \mid u_{n,k_{j+1}} \rangle. \tag{G.27}$$

Its singular values distinguish a change of frame from a loss of subspace overlap. The unitary polar factor supplies the orthogonal-Procrustes rotation that maximizes neighboring-frame overlap [104, 46]. Sequential polar transport and reciprocal closure define a smooth composite frame and a Wilson-loop matrix. In this one-dimensional setting, the Wilson eigenphases give Wannier centers modulo lattice translations, consistent with the composite-band framework of Refs. [73, 115]. The underlying identification of a one-dimensional geometric phase with a Wannier center modulo a lattice vector is the modern-polarization relation of King-Smith and Vanderbilt [54]. Here that relation fixes the phase-to-center convention only; the finite-loop calculation is not presented as a material polarization measurement.

A momentum-dependent unitary transformation changes the frame, individual Wannier functions, and real-space matrix blocks but leaves the retained projector invariant. Pointwise operator subtraction is meaningful only after both frames have been transformed into common coordinates. In this study the required alignment is obtained from the unitary polar factor of the frame overlap at each reciprocal point.

G.4.4 Wannier90 interface

Wannier90 provides an independent implementation of localization, not an independent parent calculation. It consumes eigenvalues, neighboring-state overlaps

$$M_{mn}^{(\mathbf{k},\mathbf{b})} = \langle u_{m\mathbf{k}} \mid u_{n,\mathbf{k}+\mathbf{b}} \rangle, \tag{G.28}$$

and trial projections

$$A_{mn}(\mathbf{k}) = \langle \psi_{m\mathbf{k}} \mid g_n \rangle. \tag{G.29}$$

The standard file interface represents these quantities through `.eig`, `.mmn`, `.amn`, and `.win` files. The overlap matrices must use cell-periodic states, the specified inner product, and consistent reciprocal-boundary relabelling; energies alone cannot define the localization problem [115, 91].

The one-dimensional parent is embedded in a $128 \times 1 \times 1$ reciprocal mesh with an auxiliary transverse cell and a unit transverse form factor. Those inactive directions are representation conventions, not physical dimensions of the parent. The comparison retains two bands and two Wannier functions per group, identity projections in the retained eigenbasis, `search_shells = 130`, `num_iter = 5000`, `conv_tol` $= 10^{-12}$, a five-iteration convergence window, and `precond = true`. Direct and Wannier90 outputs are compared only after orbital order, lattice translations, and composite-space unitary freedom have been aligned.

G.5 Finite-range lattice reduction

The transform signs and normalization in this section agree with Equations (9.23) –(9.24). The scalar construction uses only the retained dispersion; the later composite construction additionally requires a declared frame.

G.5.1 Complete hopping representation

For one isolated band, the complete represented hopping coefficients are

$$t(R) = \langle w_0 | \hat{H} | w_R \rangle = \frac{1}{N_k} \sum_k e^{-ikR} E(k), \qquad R \in \mathcal{R}_{N_k}, \tag{G.30}$$

with inverse transform

$$E(k) = \sum_{R \in \mathcal{R}_{N_k}} e^{ikR} t(R). \tag{G.31}$$

For a composite group, the corresponding matrix blocks are

$$\mathbf{H}_{\mathrm{W}}(R) = \frac{1}{N_k} \sum_k e^{-ikR} \mathbf{H}_{\mathrm{W}}(k). \tag{G.32}$$

The complete scalar transform depends only on the represented band dispersion. It is the finite-mesh form of Wannier Hamiltonian interpolation [72, 124]. For the present time-reversal- and inversion-symmetric scalar band, $E(k)$ is real and even, so the exact transform gives real $t(R) = t(-R)$; the residual imaginary components below measure numerical error. Composite blocks additionally depend on the selected frame.

G.5.2 Finite hopping range

For a symmetric retained set $\mathcal{R}_c \subset \mathcal{R}_{N_k}$, define

$$E_{\mathrm{TB}}^{(\mathcal{R}_c)}(k) = \sum_{R \in \mathcal{R}_c} e^{ikR} t(R), \tag{G.33}$$

$$\mathbf{H}_{\mathrm{TB}}^{(\mathcal{R}_c)}(k) = \sum_{R \in \mathcal{R}_c} e^{ikR} \mathbf{H}_{\mathrm{W}}(R). \tag{G.34}$$

Hermiticity requires $t(-R) = t(R)^*$ and $\mathbf{H}_{\mathrm{W}}(-R) = \mathbf{H}_{\mathrm{W}}(R)^\dagger$ under the chosen Born–von Karman representative convention. The scalar residual is

$$\Delta E^{(\mathcal{R}_c)}(k) = \sum_{R \notin \mathcal{R}_c} e^{ikR} t(R). \tag{G.35}$$

On the complete uniform transform, Parseval's identity gives

$$\sum_k \left|\Delta E^{(\mathcal{R}_c)}(k)\right|^2 = N_k \sum_{R \notin \mathcal{R}_c} |t(R)|^2. \tag{G.36}$$

Thus reciprocal residuals and omitted hopping norms are equivalent descriptions only under the stated finite-transform conventions [124]. In particular, decreasing the omitted hopping norm decreases the mean-square reciprocal-space error on the complete training mesh. For composite bands the same construction applies to Frobenius norms of the aligned matrix residuals.

The lattice model approximates a retained Bloch operator or its dispersion. It does not converge directly to the scalar multiplication potential. Recovering a potential from reduced spectral or hopping data is a different inverse problem.

G.5.3 Direct and mediated routes

For an isolated band, a weighted direct Fourier fit minimizes

$$\mathcal{L}_{\mathrm{dir}}(\{t(R)\}) = \sum_{k \in \mathcal{T}} w_k \left| E_{\mathrm{parent}}(k) - \sum_{R \in \mathcal{R}_c} e^{ikR} t(R) \right|^2. \tag{G.37}$$

This is an ordinary complex linear least-squares problem, with rank and conditioning interpreted according to standard matrix-computation conventions [40]. The mediated route first evaluates Eq. (G.30) on the complete mesh and then discards blocks outside $\mathcal{R}_c$. With a complete uniform mesh, equal weights, compatible transforms, and the same unconstrained Hermitian Fourier class, both procedures are the same orthogonal projection. They need not agree for nonuniform weights, restricted training domains, parameter-sharing constraints, different gauges, incompatible grids, or different objective observables.

For aligned final matrix models, define the *route defect*

$$D^2_{\text{path}} = \sum_{k \in \mathcal{K}_{\text{cmp}}} w_k \left\| \mathbf{H}_{\text{TB,dir}}(k) - \mathbf{H}_{\text{TB,med}}(k) \right\|_{\text{F}}^2 . \tag{G.38}$$

The route defect separates a change of reduction route from a change of fitting problem.

G.6 Numerical protocol

The maintained nominal single-band result is now the publication-independent experiment version `ksdft2effmass.campaign.periodic-reduction.periodic1d.single-band-fourier-reduction.v1`, run `...v1.run-20261014T184707Z`, evidence package `sha256:6e8d5db988df97d4b8cf7843b3baceb16e2eb70d2b03c729497a82251fca6377`. The publication binding is retained in `experiment-references.json`. This successor reproduces the matched historical M1 channels with zero retained binary64 defect and adds positive objective weights, explicit construction/evaluation roles, sampled external gaps, rank and conditioning, strict schemas, and independent verification. The historical paper-owned package remains unchanged provenance.

The nominal single-band calculation uses the dimensionless convention $a = 2\pi$, $G = 1$, and $E_G = 1$, with $V_0/E_G = 0.5$. Plane-wave cutoffs $P = 3, 5, 7, 9, 11$ are compared with a $P = 15$ reference. The corresponding Brillouin-zone supremum kinetic cutoffs are obtained from $(P + 1/2)^2 E_G$. The finite-difference study uses $N = 31, 63, 127, 255$ cell points, with exact spacings $\Delta x/a = 1/N$. Parent comparisons are made at $k/G = -0.5, -0.25, 0, 0.25, 0.5$ in the fixed low-mode sector $|n| \leq 3$. The single-band hopping transform uses 64 reciprocal points, ranges $r_c/a = 0, 1, 2, 3, 4, 6, 8$, and an independent 257-point withheld mesh. Bandwidth and zone-boundary gap are obtained directly from the sampled parent spectrum. The parent zone-center curvature is a three-point central difference with reduced-momentum step 10^{-3}, whereas each finite-range hopping curvature is the analytical second derivative of its trigonometric polynomial. Curvature is retained as a model observable; no material effective mass is inferred without the additional band-edge and unit conventions required by effective-mass theory [70].

The sensitivity analysis spans $V_0/E_G = 0, 0.02, 0.1, 0.5, 1, 2, 4$, reciprocal meshes from 8 to 128 points, and the first eight bands. A $P = 21$ plane-wave calculation is the higher-

band reference. Translated, constant-shifted, second-harmonic, inversion-breaking, and three-harmonic real potentials test symmetry and model dependence. Random phase perturbations test gauge covariance, while nonuniform weights and restricted training regions change the direct fitting objective one condition at a time.

The composite calculation uses $P = 15$, a 128-point reciprocal mesh, a 257-point withheld mesh, and band groups 0–1 and 2–3. Hopping ranges extend to $r_c = 12a$. The external-gap threshold is $10^{-8}E_G$. Controlled smooth and rough $U(2)$ transformations test covariance and locality. Equal-weight direct and mediated matrix reductions are compared at $r_c = 4a$.

The analysis reports the following error sources separately:

1. plane-wave cutoff error;
2. finite-difference grid and stencil error;
3. Bloch-boundary and structural residuals;
4. reciprocal sampling and Fourier-transform error;
5. retained-band or retained-subspace conditioning;
6. overlap, reciprocal-sewing, and localization error;
7. gauge and alignment sensitivity;
8. hopping-range truncation error;
9. direct-fit objective and route discrepancy;
10. serialized-Hamiltonian precision; and
11. observable-extraction error.

No combined uncertainty is formed because no mathematical relation among these error classes has been established.

G.7 Results

G.7.1 Parent representation verification

The plane-wave and finite-difference sequences converge independently. Table G.1 reports representative endpoint errors. For the lowest three bands, the plane-wave error relative to $P = 15$ decreases by more than seven orders of magnitude between $P = 3$ and $P = 5$. The

finite-difference sequence exhibits the slower convergence expected from the second-order dispersion error in Eq. (G.18). Selected plane-wave energies agree with independently evaluated Mathieu characteristic values within $1.2 \times 10^{-16} E_G$.

Table G.1: Representative parent-discretization errors in units of E_G. Spectral errors are maxima over the lowest three bands and the sampled momenta. The common-coordinate discrepancy is the Frobenius norm in the fixed $|n| \leq 3$ sector.

Quantity	Coarse	Fine
Plane-wave spectrum	1.21×10^{-5} ($P = 3$)	5.31×10^{-13} ($P = 5$)
Finite-difference spectrum	1.75×10^{-2} ($N = 31$)	2.60×10^{-4} ($N = 255$)
Common-coordinate operator	5.39×10^{-1} ($N = 31$)	8.10×10^{-3} ($N = 255$)

Transport into common $|n| \leq 3$ coordinates is essential: the operator discrepancy decreases systematically only after this identification. Its reduction by a factor of 66.5 between $N = 31$ and $N = 255$ is consistent with the second-order finite-difference kinetic error. The comparison therefore does not rely on subtracting matrices that act in different numerical spaces. Time-reversal symmetry, reciprocal periodicity, and the half-cell translation relating V_0 to $-V_0$ are satisfied at the reported numerical precision.

G.7.2 Isolated lowest band

The lowest band remains well isolated on the 64-point mesh. Its minimum neighboring overlap is 0.999466, and parallel transport closes with holonomy phase π. The representative Wannier center is $-a/2$ modulo a, the normalized spread is $0.04175a^2$, and the quadrature norm agrees with unity to binary64 precision.

The complete hopping transform reconstructs the represented band with maximum mesh error $4.17 \times 10^{-17} E_G$. The disjoint evaluation maximum error decreases from $4.62 \times 10^{-2} E_G$ at $r_c = 0$ to $9.47 \times 10^{-9} E_G$ at $r_c = 8a$. The sampled training and evaluation external gaps are $0.49233922 E_G$ and $0.49233924 E_G$; these finite samples do not certify continuum isolation. No preferred range is selected because the benchmark specifies no application-dependent accuracy threshold.

On the same complete uniform mesh, equal-weight direct fitting and mediated truncation agree within $7.38 \times 10^{-17} E_G$ in coefficient norm and $1.56 \times 10^{-16} E_G$ in sampled maximum route defect. Every fit is full rank and retains its condition number. Figure G.1 summarizes the finite-range error and locality channels.

G.7.3 Sensitivity and breakdown of scalar-band reduction

The sensitivity study distinguishes robust structure from assumptions that hold only in the baseline regime. Translation of the cosine changes the spectrum by less than $8.9 \times 10^{-15} E_G$, and removal of the known constant shift restores the baseline to the same tolerance. Every tested real potential preserves spinless time-reversal energy symmetry within $2.0 \times 10^{-14} E_G$.

Cutoff adequacy depends on the claimed band range. The first eight $P = 15$ bands agree with the $P = 21$ reference within $8.9 \times 10^{-15} E_G$ throughout the amplitude sweep, whereas the corresponding $P = 5$ error reaches $1.25 \times 10^{-2} E_G$ at $V_0/E_G = 4$. A cutoff verified for three low bands is therefore not evidence for higher-band accuracy.

The free limit closes every tested isolation gap. At $V_0/E_G = 0.5$, the minimum adjacent gap falls from $4.92 \times 10^{-1} E_G$ for band 0 to $1.20 \times 10^{-12} E_G$ for band 7. At the fixed range $r_c = 3a$, the associated withheld maximum error rises from $6.01 \times 10^{-5} E_G$ to $2.02 \times 10^{-1} E_G$. Additional Fourier harmonics alter upper-band gaps by orders of magnitude. Energy ordering alone is therefore not a stable higher-band identity.

Random phase perturbations preserve projectors, transported frames, and closure holonomy to binary64 precision. Route equivalence is lost, as expected, when the fitting problem changes: nonuniform reciprocal weights give a coefficient defect of 1.32×10^{-5}, while restricting the training region gives 3.44×10^{-4}, compared with 5.2×10^{-17} for the complete uniform-mesh control. These coefficient route defects quantify the changed fitting problems; they are not implementation residuals. Figure G.2 collects the principal sensitivity results.

G.7.4 Composite-band remedy

Bands 0–1 provide a well-conditioned composite baseline, while bands 2–3 provide a more difficult higher-band case. Table G.2 shows the principal conditioning and reduction results. Both groups satisfy the external-gap threshold, but the higher pair is much less isolated and has a smaller minimum neighboring-overlap singular value.

Table G.2: Composite-band conditioning, centers, and range-$12a$ errors. Center defects compare the converged Wannier90 center set with the direct Wilson center set modulo one lattice period.

Group	Ext. gap (E_G)	Min. σ	Center defect (a)	Range-$12a$ error (E_G)
Bands 0–1	1.14×10^{-1}	0.991696	2.16×10^{-7}	9.32×10^{-4}
Bands 2–3	2.16×10^{-4}	0.708327	2.63×10^{-3}	3.13×10^{-2}

Direct polar transport gives Wilson-loop eigenphases ± 1.98562 radians for bands 0–1

and ± 1.56593 radians for bands 2–3. Under a controlled smooth momentum-dependent $U(2)$ rotation, projector defects remain below 5.1×10^{-16}, Wilson-phase set defects below 2.5×10^{-15}, and represented eigenvalue defects below $7.6 \times 10^{-15} E_G$. Pointwise unitary alignment recovers frames within 2.1×10^{-15} and represented operators within $6.9 \times 10^{-15} E_G$.

The complete block-hopping transforms reconstruct the reciprocal operators within $3.3 \times 10^{-14} E_G$. Smooth polar frames yield monotonically improving finite-range models. At $r_c = 12a$, their withheld maximum errors are $9.32 \times 10^{-4} E_G$ and $3.13 \times 10^{-2} E_G$. A deliberately rough periodic gauge preserves the complete pointwise spectra but leaves errors of $1.99 \times 10^{-1} E_G$ and $2.82 \times 10^{-1} E_G$. Thus gauge covariance of the complete operator does not imply gauge invariance of a finite-range truncation.

At $r_c = 4a$, equal-weight direct matrix Fourier fitting and mediated block truncation agree within 2.5×10^{-15} in coefficient norm when they use the same mesh, aligned frame, and model class. Figure G.3 shows the conditioning, Wilson phases, hopping decay, and finite-range errors.

G.7.5 Wannier90 localization

The Wannier90 interface uses the same eigenvalues, band groups, overlap matrices, reciprocal sewing, and auxiliary embedding as the direct calculation. The strongly anisotropic mesh requires an extended reciprocal-shell search to construct a neighbor list satisfying the three-dimensional B1 condition. With those data unchanged, unpreconditioned calculations remained nonstationary at 500 iterations; increasing only the ceiling to 5000 iterations also failed for the low pair. The spread decreased monotonically, and even the smallest late-stage change remained above the prescribed tolerance. Relaxing that tolerance would therefore have classified an evolving gauge as converged.

Wannier90 documents its real-space preconditioner for slow spread minimization on fine reciprocal grids. Adding only `precond = true` while preserving the parent, interface, objective, 10^{-12} tolerance, five-step window, and 5000-iteration ceiling changes the outcome. The low pair converges in 69 iterations and the higher pair in 4176. The total spreads are 0.337439093 and 3.673238490 cell^2, respectively.

The low-pair centers are $\{+0.316021, -0.316021\}a$ and agree with the direct Wilson set within 2.16×10^{-7} cell. Modulo lattice translations, the higher-pair centers are $\{-0.246592, +0.246600\}a$, with a 2.63×10^{-3}-cell discrepancy from the direct set. The discrepancy is reported without changing tolerances or relabelling orbitals. It occurs in the group with the smaller external gap and poorer overlap conditioning, but this calculation does not establish a unique causal attribution.

The exported gauges are unitary within 1.9×10^{-10} at their printed precision. After pointwise alignment, the represented operators agree with the direct construction within

$4.9 \times 10^{-10} E_G$. At $r_c = 12a$, the Wannier90-gauge errors are $9.32 \times 10^{-4} E_G$ and $3.09 \times 10^{-2} E_G$, close to the direct-polar values in Table G.2. The formatted `hr.dat` files add separate maximum training eigenvalue discrepancies of approximately $1.1 \times 10^{-5} E_G$ and $1.2 \times 10^{-4} E_G$. Figure G.4 summarizes the converged comparison.

G.8 Discussion

G.8.1 Representation agreement requires an explicit map

The parent comparison demonstrates why equality of eigenvalues is not a sufficient operator comparison. The finite-difference and plane-wave matrices act in different numerical spaces until the discrete Fourier map identifies a common sector. Once that map is supplied, the low-mode discrepancy converges as expected and can be attributed to the finite-difference stencil rather than to an undefined subtraction. The same principle governs later pristine–doped and cross-code comparisons: basis and state-space identification precede matrix residuals.

G.8.2 Band labels fail before composite subspaces do

The sensitivity analysis shows a qualitative distinction between an energy-ordered band and a retained subspace. As adjacent gaps collapse, scalar overlaps and fixed-range models deteriorate even though a neighboring group can remain transportable. Composite polar transport does not remove poor conditioning; it makes that conditioning visible through singular values and external gaps while preserving a gauge-covariant operator. This is the relevant analogue of a multiorbital retained manifold.

G.8.3 Localization controls truncation, not the complete spectrum

Smooth and rough gauges produce complete operators with identical pointwise spectra but markedly different block decay. Localization is therefore not a cosmetic post-processing step when the final model is truncated. It selects the coordinates in which a short-range approximation is attempted. The direct and Wannier90 comparisons reinforce the same conclusion: raw hopping blocks cannot be compared across gauges, while aligned complete operators can.

G.8.4 Route equivalence has precise hypotheses

Agreement between direct fitting and mediated truncation is exact only when the two calculations implement the same finite Fourier projection. The controlled weight and training-domain changes break those hypotheses and produce nonzero route defects without indicating a software failure. In a material calculation, the route defect must therefore be interpreted together with the fitting measure, model constraints, gauge, and comparison domain.

G.8.5 Implications for multiband reductions

This calculation verifies the middle of the intended reduction chain: retained Bloch information can be transported into localized coordinates, reconstructed as a complete lattice operator, and truncated with separately measured error. It also supplies explicit breakdown cases for weak isolation, unstable scalar labels, rough gauges, and mismatched fitting objectives. These observations define concrete diagnostics for a subsequent silicon calculation. The present benchmark does not test a Kohn–Sham parent, a three-dimensional crystal, pristine–doped alignment, an impurity operator, multivalley structure, spin–orbit coupling, or a continuum effective-mass reduction.

G.9 Canonical constrained admissible-set extension

The historical calculations above and the canonical M3 package have different identities. M3 is retained under `calculations/ICMSEP2026/conference/paper_1/constrained-admissible-sets/`; it does not rewrite the historical `calculations/research-monograph/periodic-1d/` records. Its evidence class is a calculated and independently reconstructed synthetic numerical result for a frozen finite family. Chapter 14 owns the general definitions and derivations.

M3 composes the retained rank-two multiband baseline with the candidate

$$H_{s,q}(k) = \overline{E}(k)I_2 + sE_0I_2 + q\left(H_{\mathrm{ref}}(k) - \overline{E}(k)I_2\right), \tag{G.39}$$

over

$$-0.25 \le s \le 0.25, \qquad 0.4 \le q \le 1.2. \tag{G.40}$$

The alignment family contains the following nine global real-rotation angles, in radians:

$$-1.2, -0.9, -0.6, -0.3, 0, 0.3, 0.6, 0.9, 1.2. \tag{G.41}$$

The losses use 64 training coordinates, retained rank two, an energy scale of 1 in the model's

dimensionless energy convention, and the normalization

$$\frac{1}{E_0}\sqrt{\frac{\text{sum of squared residuals}}{N_k r}}. \tag{G.42}$$

A staggered 257-point mesh is disjoint from the training coordinates and is used only for withheld evaluation. The locality ranges $0, 1, 2, 3, 4, 6, 8$ produce hopping diagnostics; no locality threshold enters the admissibility decision.

The thresholds are benchmark-design controls rather than inferred physical tolerances. They were selected from the known synthetic model and preparatory analytic training-loss geometry to produce one compatible case and one resolved separated case. This selection was not blind. It used neither the withheld values nor an experimental uncertainty, literature tolerance, or material acceptance criterion. The controls were frozen before the retained confirmatory execution, so the calculation confirms the prespecified synthetic decision logic without turning the thresholds into universal estimates.

For the prospectively frozen dimensionless RMS thresholds

$$(\tau_E, \tau_H) = (0.03, 0.33), \tag{G.43}$$

the retained parameter $(s, q) = (0, 1)$ is a common witness. The selected global rotation is 0.3, the training spectral loss is $7.58984639784215\times10^{-16}$, and the training operator loss is 0.3286318914348874. The withheld values are respectively $7.928256904202344 \times 10^{-16}$ and 0.3286318914348874. These withheld values diagnose the retained candidate; they do not define its witness status.

For the separately frozen dimensionless RMS thresholds

$$(\tau_E, \tau_H) = (0.03, 0.31), \tag{G.44}$$

the analytic certificate uses the spectral boundary point

$$(s_E, q_E) = (3.885780586188048 \times 10^{-16}, 0.9354687051718992) \tag{G.45}$$

and the operator boundary point

$$(s_H, q_H) = (0, 0.8360013041446829). \tag{G.46}$$

Only the 0.3 rotation component is operator-feasible in this case. The lower and upper Euclidean parameter-separation bounds coincide at

$$\delta^* = 0.0994674010272163, \tag{G.47}$$

which exceeds the frozen separation resolution 0.05. The resulting `certified-separated` disposition is conditional on the two-parameter domain, Euclidean parameter

metric, nine-component alignment family, training mesh, and thresholds. Figure G.5 displays the two decision geometries. Each curve is a threshold boundary; its enclosed region is admissible.

The independent standalone reconstruction passes with maximum absolute defect $4.6629367034256575 \times 10^{-15}$ against tolerance 10^{-11}. The source and artifact manifests pass at the audited repository state. The verifier rejects altered evaluation roles, nonpositive definition tolerances, inconsistent source-manifest correlation, and violations of the quadratic premises used by the analytic certificate. It does not import the M3 producer Actions, but it shares NumPy, SciPy, eigensolver behavior, floating-point arithmetic, and the declared scientific conventions with the producer.

The protocol freeze binds the editable configuration, deterministic input builder, derived input, inherited M2 identity, implementation sources, runner, and standalone verifier. Nine retained amendments disclose post-freeze corrections and verification hardening without changing the M3 scientific controls or numerical outcomes. In particular, amendment 6 retracts an earlier incorrect diagnosis of overlap between the training and staggered withheld coordinates, while amendment 9 records the post-result role and positive- tolerance checks added after audit. The SHA-256 records bind retained bytes; they are not an external trusted timestamp and do not independently prove chronology.

A separate retained package under `constrained-admissible-sets-threshold-sensitivity/` is explicitly post hoc. At fixed $\tau_E = 0.03$, the operator set first becomes nonempty at $\tau_H \approx 0.3074197403$. The separation exceeds the declared 0.05 resolution through $\tau_H \approx 0.3137907655$, remains positive but below resolution until the compatibility transition at $\tau_H \approx 0.3192641695$, and is zero thereafter. The original parameter $(0, 1)$ itself becomes operator admissible at $\tau_H \approx 0.3286318914$. These values characterize the frozen synthetic quadratic geometry. They are not prospectively predicted transitions, physical tolerance calibrations, or uncertainty quantification.

M3 therefore demonstrates both forms of evidence needed by the methodology: an explicit common witness for one threshold pair and a positive separation certificate for another. It does not establish incompatibility for unrestricted unitary alignments, another candidate family, a material Hamiltonian, or the silicon reduction program.

G.10 Limitations and scope

The results establish numerical behavior only for the specified cosine parent. Several limitations remain:

- The transverse Wannier90 embedding and identity trial projections are synthetic conventions rather than material-derived orbitals.

- The study contains no reciprocal-mesh convergence analysis of the final Wannier90 centers across alternative projection choices.
- The higher pair is only weakly separated from excluded bands, and its center discrepancy is not assigned to one error source.
- No hopping range is selected for a physical use because no application-specific accuracy threshold is specified.
- The calculations do not establish scientific validation, uncertainty quantification, or transferability to silicon.

The successful convergence of two numerical routes does not promote the cosine model into a semiconductor result. Likewise, the failed intermediate Wannier90 calculations diagnose optimizer behavior for this interface; they do not contradict the direct composite construction.

G.11 Conclusions

The one-dimensional cosine lattice provides a controlled test of the complete Bloch-to-lattice reduction rather than of any one eigensolver or localization algorithm. Independent parent discretizations converge toward a common low-mode operator, and the isolated lowest band admits a rapidly convergent hopping expansion. Direct Fourier fitting and localized-operator truncation agree when they implement the same projection and diverge when their weights or fitting domains differ.

The multiband calculations identify the principal boundary of the scalar construction. Near small adjacent gaps, energy-ordered labels become poorly conditioned before the composite subspace is lost. Composite transport restores a coherent operator description, but finite-range accuracy remains dependent on the gauge in which truncation is performed. The converged Wannier90 comparison reproduces the aligned operators and low-pair centers while preserving a finite higher-pair center discrepancy. The separately identified M3 extension then shows, within its bounded rank-two family, how the same candidate can satisfy spectral and operator thresholds and how a different threshold pair can be separated by an analytic certificate. These results support the use of explicit subspace, alignment, localization, truncation, and admissibility diagnostics in more complex periodic reductions; they do not by themselves establish accuracy for a semiconductor Hamiltonian.

G.12 Computational reproducibility

Inputs, calculation programs, deterministic verification scripts, numerical results, figures, and SHA-256 checksums are available under `calculations/research-monograph/periodic-1d/`. Separate records distinguish the isolated-band calculation, sensitivity analysis, composite-band calculation, unconverged Wannier90 iterations, and converged preconditioned calculation. Large native Wannier90 artifacts remain outside the repository and are identified by file size and SHA-256 digest in the execution metadata.

The public `Periodic1DIsolatedBandCalculationWorkflow` now owns the execution-local nonlocalization calculation channels: plane-wave and finite-difference studies, common-low-mode defects, Mathieu and weak-gap references, symmetry residuals, lowest-band reciprocal samples, the complete hopping pair, finite-range training and withheld metrics, Parseval and direct-fit comparisons, and parent bandwidth, gap, and curvature. Its curvature step is explicit, and its ResultObject keeps these channels separate. The implementation uses NumPy and SciPy [45, 125], reads no retained result, performs no filesystem discovery or external execution, and does not calculate the separately owned gauge-transport or localization channels.

The public `Periodic1DStressResultVerifier` separately reconstructs every retained sensitivity channel from the correlated version-one controls. It assembles plane-wave and periodic finite-difference matrices directly, evaluates centered finite Fourier sums and reciprocal sewing, performs scalar parallel transport under the deterministic phase attack, and solves the complete, restricted-domain, and nonuniform-weight least-squares routes without importing the historical runner or the production construction algorithms. The corresponding `Periodic1DStressVerifiedWorkflow` preserves retained correlation and independent numerical verification as distinct results under one explicit unitless tolerance. This establishes bounded numerical verification of the represented stress campaign only; it does not rerun the historical calculation or establish material validation or uncertainty quantification.

The isolated retained-result verifier independently reconstructs the parent, reciprocal-energy, complete-hopping, finite-range, direct-fit, and parent-observable channels. Transported frames and localization-density samples were not retained, so neighbor overlaps, closure holonomy, center, spread, and density identity remain calculated producer diagnostics rather than independently reconstructed numerical evidence.

The composite retained-result verifier independently reconstructs the retained reciprocal Hamiltonians, complete smooth hopping transform, exact-pair Hermiticity, finite-range training errors, direct-route fits, and array identities. Source frames, projectors, attacked gauges, rough reciprocal matrices, and withheld reciprocal matrices were not retained. The sampled-gap, overlap and Wilson, controlled-gauge, alignment, rough-gauge, and withheld-range channels are consequently calculated producer diagnostics with software and structural checks. Neighbor-overlap singular values are reported as conditioning

diagnostics; no universal singular-value pass threshold is imposed.

The canonical M3 verifier independently reconstructs the inherited multiband baseline, quadratic spectral and operator losses, evaluation roles, common witness, finite alignment components, boundary points, separation bounds, withheld losses, and locality diagnostics without invoking the M3 producer Action. Its standalone counterpart also decodes the frozen documents directly and rejects invalid definition domains. The post-hoc sensitivity verifier correlates the exact M3 result with its package manifest and checks the positive-definite axis-aligned premises before reconstructing the threshold curve. These are independent orchestration and reconstruction routes, not an independent parent Hamiltonian, physical oracle, or material validation.

The compact Wannier90 verifier reconstructs frame, aligned-operator, and finite-range diagnostics from retained unitary matrices. Independent native Wilson reconstruction requires the externally retained overlap and output files identified by size and SHA-256 digest, and verification does not rerun Wannier90 or reproduce its convergence trajectory.

The reports `report.md` and `composite-report.md` provide exact reproduction commands and additional numerical detail. Passing verification is bounded to the explicitly reconstructable channels; it does not constitute material validation or evidence of transferability to silicon.

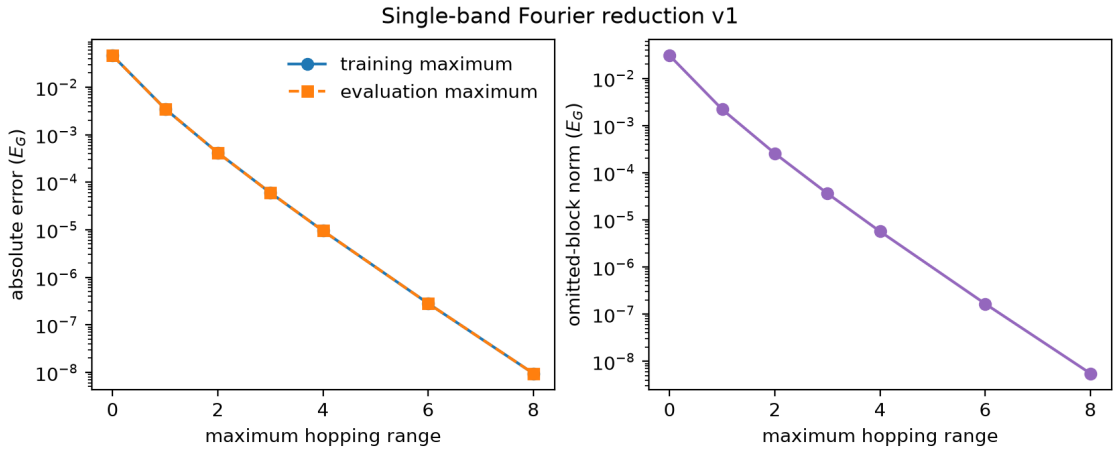


Figure G.1: Maintained single-band Fourier-reduction successor at $V_0/E_G = 0.5$. Training and disjoint evaluation maximum errors decrease across the frozen range inventory (left), while the omitted-block norm reports the corresponding locality tail (right). The figure is bound by the publication reference manifest.

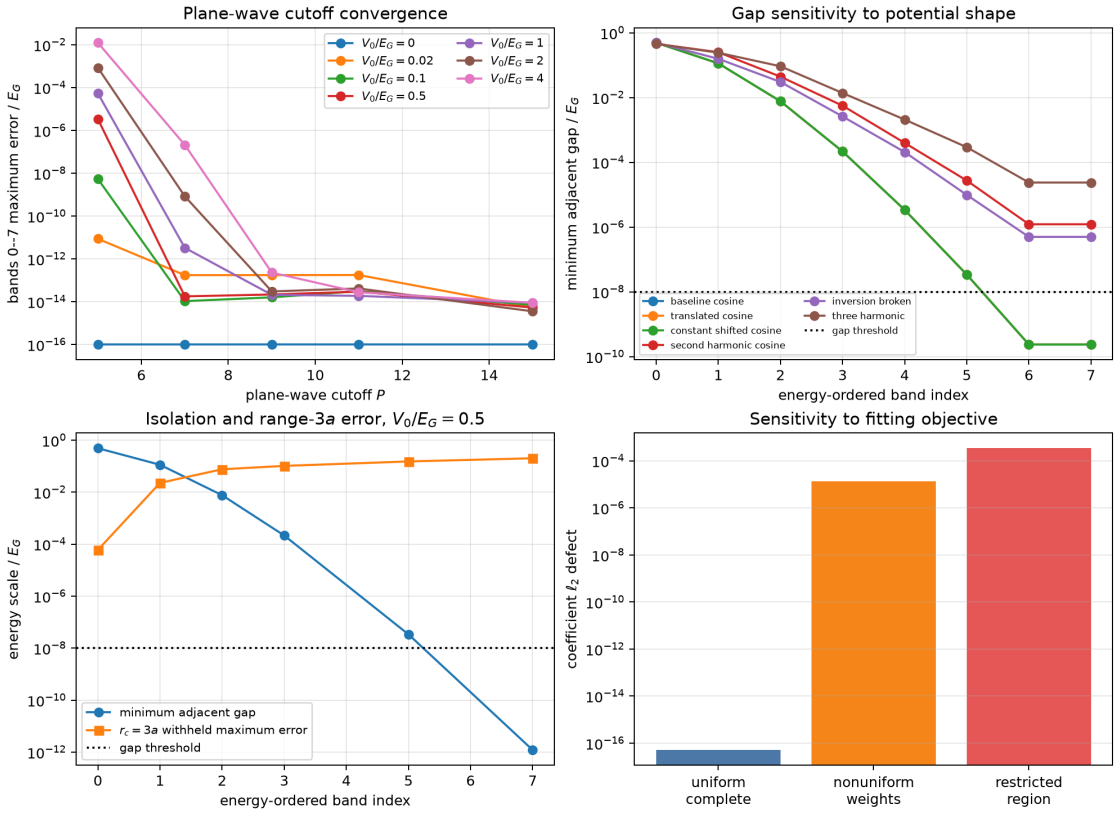


Figure G.2: Sensitivity of the reduction to cutoff, potential shape, band index, and fitting objective. Higher bands require larger plane-wave cutoffs (upper left), and their adjacent gaps depend strongly on the Fourier content of the potential (upper right). At fixed $V_0/E_G = 0.5$, diminishing isolation accompanies increasing range-$3a$ error (lower left). Direct and mediated routes cease to coincide when their reciprocal weights or fitting domains differ (lower right).

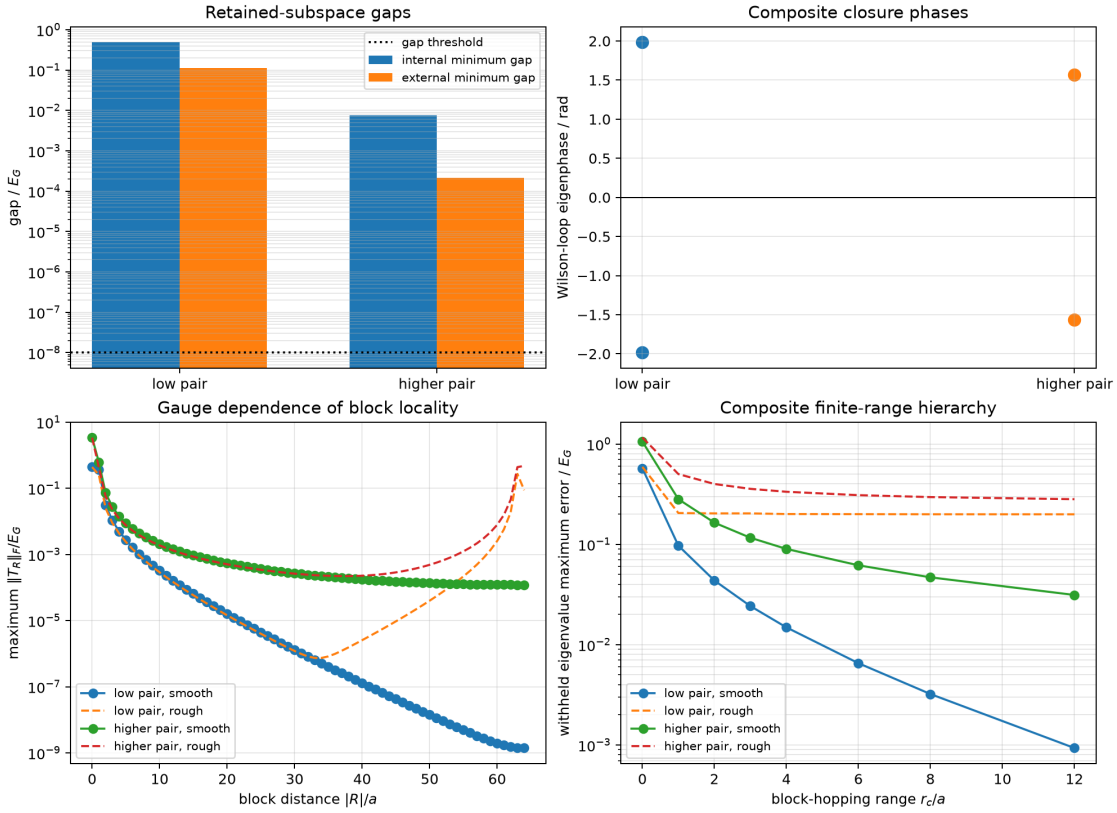


Figure G.3: Composite-band conditioning and finite-range reduction. Internal and external gaps distinguish the two retained band pairs (upper left), while Wilson-loop eigenphases determine their centers modulo a lattice period (upper right). Smooth polar frames produce more rapidly decaying hopping blocks (lower left) and smaller finite-range errors (lower right) than the spectrally equivalent rough gauges.

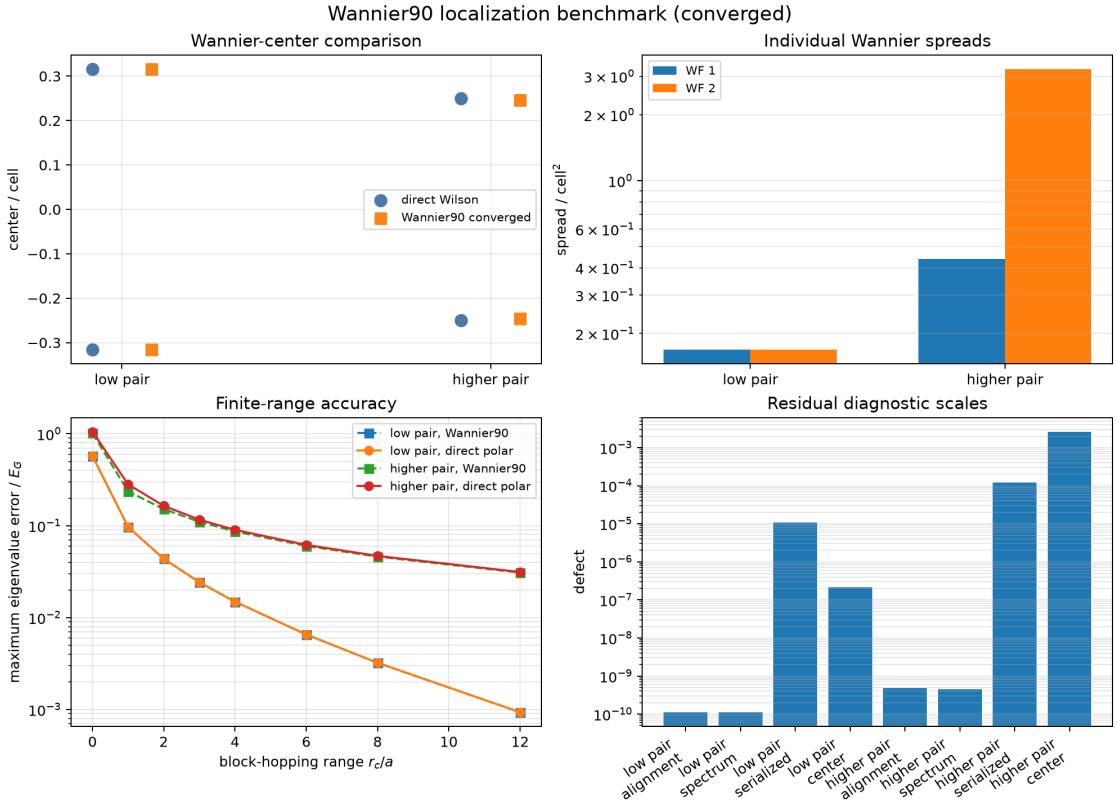


Figure G.4: Comparison between direct polar transport and converged, preconditioned Wannier90 localization. Centers are reduced modulo one lattice period (upper left), and individual Wannier spreads show the poorer locality of the higher pair (upper right). The two gauges give nearly identical finite-range errors after alignment (lower left); residual alignment, serialization, spectral, and center defects are shown separately (lower right).

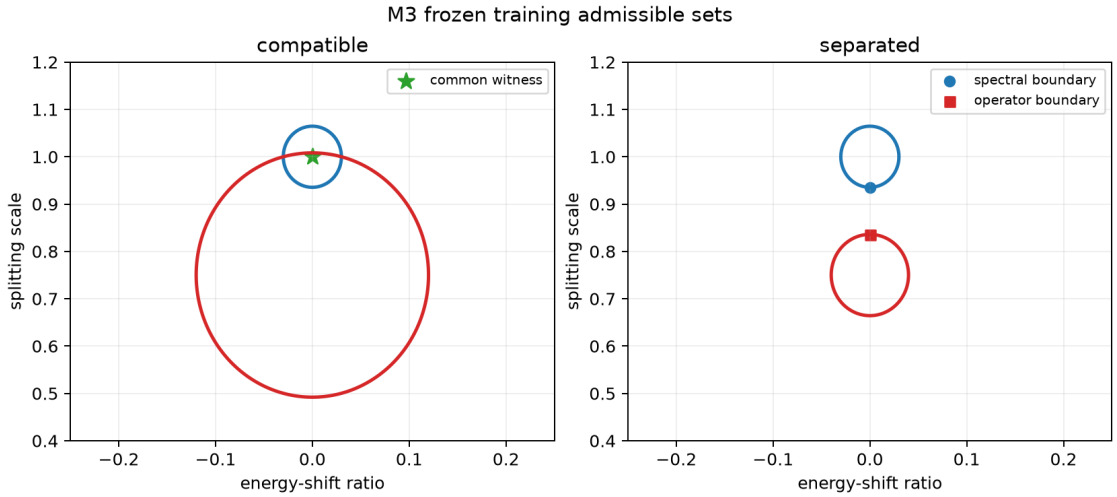


Figure G.5: Frozen M3 training admissible sets in the dimensionless (s, q) parameter plane. Blue curves bound the spectral-admissible region, and red curves bound the operator-admissible component associated with the feasible 0.3-radian global rotation. At left, the thresholds $(\tau_E, \tau_H) = (0.03, 0.33)$ admit the common witness $(0, 1)$, marked by the green star. At right, tightening the operator threshold to 0.31 separates the regions; the blue circle and red square mark the closest certified spectral and operator boundary points. The image is the checksummed retained artifact from the canonical synthetic M3 package, not a material result.

CHAPTER H

Two-Dimensional Wannier Reduction: From Separable to Coupled Potentials

Status calculated controlled synthetic numerical verification. Exact separable identities, independent plane-wave and real-space parent refinement, degenerate projectors, reciprocal-loop and topology diagnostics, scalar and rank-three composite Wannier hoppings, direct localization diagnostics, shell reduction, tensor masses, a frozen nonseparable-coupling sequence, and separate Qi–Wu–Zhang, Hofstadter, and Haldane Chern-band controls have been independently verified. The first Wannier90 preprocessing attempt failed and is retained; the corrected balanced-embedding attempt converged and was independently reconstructed. Six additional non-DFT cases retain nonmonotone mesh, cutoff, and embedding sensitivity. A 72-start pilot and an expanded 256-trajectory study are calculated and independently verified; their frozen mesh, convergence, and repeated-basin criteria fail. Post-hoc exact-equivalence controls bound classifier roundoff but do not retroactively satisfy the planned pre-execution check. No converged localization limit, material calculation, scientific validation, or uncertainty quantification is claimed.

H.1 Purpose

A two-dimensional periodic model is useful if it remains simple enough to verify while introducing features absent in one dimension. These include a Brillouin-zone mesh rather than a line, symmetry-related degeneracies, direction-dependent hopping shells, effective-mass tensors, gauge closure around reciprocal-space loops, and possible topological obstructions to Wannier localization.

The parent begins with a separable square-lattice cosine potential. Its bands and Bloch states factor into products of the one-dimensional problem, providing a strong reference. A controlled coupling then breaks separability without changing the lattice. The progression

is

$$\begin{array}{c}\text{verified separable parent}\\ \downarrow\\ \text{coupled two-dimensional parent}\\ \downarrow\\ \text{Wannier representation} \longrightarrow \text{finite-range tight binding.}\end{array} \tag{H.1}$$

The calculation addresses:

1. whether real-space and plane-wave Bloch fibers agree over a full two-dimensional reciprocal mesh;
2. whether the separable limit reproduces sums and products of the one-dimensional results;
3. how degeneracies affect band labels and retained-subspace selection;
4. how direct gauge transport and Wannier90 localization compare in two dimensions;
5. how hopping blocks decay by lattice vector and symmetry shell;
6. how finite-range TB models converge to isolated- and composite-band parent operators; and
7. how these conclusions change when separability, fourfold symmetry, or isotropy is broken in a controlled manner.

The principal negative result is that the expanded deterministic multi-initialization study does not meet its frozen combined convergence and repeated-basin criteria. Fine-mesh native convergence deteriorates, the $N = 31$ spread holdouts fail, and no best endpoint is reproduced under the declared density-aware symmetry quotient. These observations are restricted to the synthetic parent and fixed localization protocol.

H.2 Continuum parent family

Let the square lattice have primitive vectors

$$\mathbf{a}_1 = a\mathbf{e}_x, \qquad \mathbf{a}_2 = a\mathbf{e}_y, \tag{H.2}$$

and reciprocal scale

$$G = \frac{2\pi}{a}. \tag{H.3}$$

Define

$$V(x,y) = V_x \cos(Gx) + V_y \cos(Gy) + V_{xy} \cos(Gx)\cos(Gy). \tag{H.4}$$

The continuum Hamiltonian on $L^2(\mathbb{R}^2)$ is

$$\hat{H} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + V(x,y). \tag{H.5}$$

Use the dimensionless controls

$$\lambda_x = \frac{V_x}{E_G}, \qquad \lambda_y = \frac{V_y}{E_G}, \qquad \lambda_{xy} = \frac{V_{xy}}{E_G}, \qquad E_G = \frac{\hbar^2 G^2}{2m}. \tag{H.6}$$

The first Brillouin zone is

$$-\frac{G}{2} \le k_x, k_y < \frac{G}{2}. \tag{H.7}$$

For crystal momentum $\mathbf{k} = (k_x, k_y)$, write

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}), \tag{H.8}$$

where $u_{n\mathbf{k}}$ is periodic on the square cell. The cell-periodic fiber operator is

$$\hat{H}(\mathbf{k}) = \frac{1}{2m}\left(-i\hbar\nabla + \hbar\mathbf{k}\right)^2 + V(x,y). \tag{H.9}$$

H.3 Separable reference limit

When

$$V_{xy} = 0, \tag{H.10}$$

the Hamiltonian separates:

$$\hat{H}(\mathbf{k}) = \hat{H}_x(k_x) \otimes \hat{I}_y + \hat{I}_x \otimes \hat{H}_y(k_y). \tag{H.11}$$

Consequently, the exact product-form relations are

$$E_{n_x n_y}(k_x, k_y) = E^{(x)}_{n_x}(k_x) + E^{(y)}_{n_y}(k_y), \tag{H.12}$$

$$u_{n_x n_y, \mathbf{k}}(x,y) = u^{(x)}_{n_x,k_x}(x)\, u^{(y)}_{n_y,k_y}(y), \tag{H.13}$$

up to basis choices within degenerate product subspaces.

These identities make the separable model more than a qualitative warmup. The one-dimensional parent supplies expected two-dimensional eigenvalues, projectors, Wannier products, and axis-supported Kronecker-sum hopping structure. A two-dimensional construction that fails these checks in the separable limit has not yet isolated genuinely two-dimensional physics.

For the isotropic case $V_x = V_y$, product states related by $n_x \leftrightarrow n_y$ may become symmetry degenerate at selected $\mathbf{k}$. The stable comparison object is then the degenerate projector, not an arbitrary ordering of individual eigenvectors.

H.4 Plane-wave Bloch representation

Use reciprocal indices $(p, q) \in \mathbb{Z}^2$ and basis functions

$$\langle \mathbf{r} \mid p, q; \mathbf{k} \rangle = \frac{1}{a} e^{i[(k_x + pG)x + (k_y + qG)y]}. \tag{H.14}$$

The kinetic matrix is diagonal:

$$T_{pq,p'q'}(\mathbf{k}) = \frac{\hbar^2}{2m} \left[(k_x + pG)^2 + (k_y + qG)^2 \right] \delta_{pp'} \delta_{qq'}. \tag{H.15}$$

The potential couplings are

$$V_{pq,p'q'} = \frac{V_x}{2} \left(\delta_{p,p'+1} + \delta_{p,p'-1} \right) \delta_{qq'} \tag{H.16}$$

$$+ \frac{V_y}{2} \left(\delta_{q,q'+1} + \delta_{q,q'-1} \right) \delta_{pp'} \tag{H.17}$$

$$+ \frac{V_{xy}}{4} \sum_{s_x = \pm 1} \sum_{s_y = \pm 1} \delta_{p,p'+s_x} \delta_{q,q'+s_y}. \tag{H.18}$$

The separable terms move along one reciprocal axis at a time. The coupling V_{xy} connects diagonal reciprocal neighbors and therefore provides a visible algebraic marker of nonseparability.

Retain

$$-P_x \le p \le P_x, \qquad -P_y \le q \le P_y. \tag{H.19}$$

The represented dimension is

$$N_{\mathrm{PW}} = (2P_x + 1)(2P_y + 1). \tag{H.20}$$

Cutoff convergence should be tested independently in the two directions before an isotropic cutoff is assumed sufficient.

H.5 Real-space Bloch representation

Use an $N_x \times N_y$ grid on one unit cell with spacings

$$\Delta x = \frac{a}{N_x}, \qquad \Delta y = \frac{a}{N_y}. \tag{H.21}$$

The boundary identifications are

$$\psi(x + a, y) = e^{ik_x a} \psi(x, y), \tag{H.22}$$

$$\psi(x, y + a) = e^{ik_y a} \psi(x, y). \tag{H.23}$$

A centered second-order representation has the Kronecker-sum form

$$\mathbf{T}_h(\mathbf{k}) = \mathbf{T}_x(k_x) \otimes \mathbf{I}_y + \mathbf{I}_x \otimes \mathbf{T}_y(k_y), \tag{H.24}$$

where the boundary entries of $\mathbf{T}_x$ and $\mathbf{T}_y$ carry the corresponding Bloch phases. The represented Hamiltonian is

$$\mathbf{H}_h(\mathbf{k}) = \mathbf{T}_h(\mathbf{k}) + \operatorname{diag}\left(V(x_i, y_j)\right). \tag{H.25}$$

At $V_{xy} = 0$, this matrix should equal the Kronecker sum of the corresponding one-dimensional represented Hamiltonians, subject to one fixed vectorization order. Its eigenvalues should be sums of one-dimensional represented eigenvalues. This is an exact matrix-level check for the declared discretization, not merely a continuum expectation.

A two-dimensional discrete Bloch–Fourier transform transports the real-space matrix into reciprocal coordinates. When

$$N_x = 2P_x + 1, \qquad N_y = 2P_y + 1, \tag{H.26}$$

the transform is the Kronecker product of the one-dimensional transforms:

$$\mathbf{F}_{2\mathrm{D}}(\mathbf{k}) = \mathbf{F}_x(k_x) \otimes \mathbf{F}_y(k_y). \tag{H.27}$$

The transported matrix

$$\mathbf{A}_h^{\mathrm{FD}\to\mathrm{PW}}(\mathbf{k}) = \mathbf{F}_{2\mathrm{D}}(\mathbf{k})^{\dagger}\mathbf{H}_h(\mathbf{k})\mathbf{F}_{2\mathrm{D}}(\mathbf{k}) \tag{H.28}$$

can be compared with the plane-wave Galerkin matrix in a common basis. Low-reciprocal-sector and retained-band comparisons should accompany any full-space Frobenius norm because high-frequency grid modes can dominate the latter.

H.6 Symmetry and degeneracy diagnostics

For a real scalar potential,

$$E_n(-\mathbf{k}) = E_n(\mathbf{k}) \tag{H.29}$$

under consistent band or subspace matching. In the isotropic separable case, the square-lattice point symmetries imply additional relations among $\mathbf{k}$, rotated momenta, and reflected momenta. The usual high-symmetry points may be denoted

$$\Gamma = (0, 0), \qquad X = \left(\frac{G}{2}, 0\right), \qquad M = \left(\frac{G}{2}, \frac{G}{2}\right). \tag{H.30}$$

A plotted Γ–X–M–Γ path is useful for visualization but is not sufficient for Wannier localization or two-dimensional operator norms. The full reciprocal mesh and its neighbor relations remain required.

At degeneracies, individual eigenvectors are not uniquely defined. Comparisons should use the projector onto the complete degenerate or nearly degenerate cluster. Principal angles and overlap singular values determine whether two represented clusters span the same subspace before an internal gauge is selected.

Breaking $V_x = V_y$ while retaining $V_{xy} = 0$ supplies a controlled anisotropic case. Turning on V_{xy} supplies a controlled nonseparable case. These two changes should not be introduced simultaneously in the first comparison.

H.7 Effective-mass tensor

At an isolated band extremum $\mathbf{k}_0$, define the inverse effective-mass tensor

$$\left(m_n^{*-1}\right)_{ij} = \frac{1}{\hbar^2} \left.\frac{\partial^2 E_n(\mathbf{k})}{\partial k_i \partial k_j}\right|_{\mathbf{k}=\mathbf{k}_0} . \tag{H.31}$$

For the separable parent, the mixed derivative should vanish in the lattice axes, and the diagonal terms should agree with the corresponding one-dimensional curvatures. In the isotropic case, symmetry may force equal diagonal entries at selected extrema. These are expected structural checks, not calculated findings.

The reciprocal stencil, fit region, symmetry constraints, and units must be held fixed when plane-wave, real-space, Wannier, and TB representations are compared. A band-path curvature alone does not determine the full tensor.

H.8 Direct two-dimensional gauge construction

For an isolated band, neighboring overlaps are

$$M^{(\mathbf{k},\mathbf{b})} = \langle u_{\mathbf{k}} \mid u_{\mathbf{k}+\mathbf{b}} \rangle. \tag{H.32}$$

Parallel transport can be performed along one reciprocal direction on each mesh line, followed by transport between lines. Unlike the one-dimensional case, closing the gauge along one direction does not automatically close it along the other. Wilson loops around reciprocal cycles and elementary mesh plaquettes diagnose the accumulated holonomy.

For a composite M-band subspace, replace each scalar overlap by

$$\mathbf{M}_{mn}^{(\mathbf{k},\mathbf{b})} = \langle u_{m\mathbf{k}} \mid u_{n,\mathbf{k}+\mathbf{b}} \rangle. \tag{H.33}$$

Polar or singular-value decompositions supply neighboring unitary transport maps. Small singular values identify subspace mismatch or inadequate reciprocal resolution. A sequence of locally optimal rotations need not produce a globally periodic and localized frame; loop closure remains an independent condition.

The direct construction should retain every chosen spanning path, overlap, polar factor, loop holonomy, and closure transformation. Its purpose is not to replace a mature localization procedure, but to make the gauge problem visible before it is delegated.

Polar decomposition supplies the nearest unitary factor under the standard matrix-norm formulation, while its use in Wannier gauge transport and the need for global reciprocal-space consistency are established in the modern Wannier literature [46, 72].

H.9 Topology and the localization boundary

A globally smooth periodic frame for an isolated or composite two-dimensional subspace can be obstructed by nontrivial topology. The first Chern number may be written schematically as

$$C = \frac{1}{2\pi} \int_{\mathrm{BZ}} \operatorname{Tr} \boldsymbol{\Omega}(\mathbf{k}) \, \mathrm{d}^2 k, \tag{H.34}$$

where $\boldsymbol{\Omega}$ is the Berry-curvature matrix of the retained subspace. A nonzero C obstructs a complete set of exponentially localized Wannier functions spanning that subspace.

For periodic insulating subspaces, the relationship between vanishing Chern invariants, analytic quasi-Bloch functions, and exponentially localized Wannier functions is treated by Brouder et al. [13]. The retained scalar exercise checks only the expected trivial case and does not test the obstructed case.

The real scalar cosine family has spinless time-reversal symmetry and provides a topologically trivial starting point. The calculation checks that expectation through gauge closure and a discretized curvature diagnostic rather than inferring it from a visually smooth band plot. Degeneracies can make an individual-band Chern number undefined until a composite retained subspace is chosen.

Separate Chern-band models test the numerical obstruction diagnostics without identifying those models with the scalar parent. Topological obstruction is a stopping condition and interpretive lesson for the scalar reduction.

H.10 Wannier90 localization in two dimensions

Wannier90 consumes the two-dimensional model Bloch states through the same general interfaces used in Appendix G. The required data include:

1. eigenvalues on a uniform $N_{k_x} \times N_{k_y}$ mesh;
2. neighboring-state overlap matrices in both reciprocal directions;
3. optional trial-orbital projections;
4. the square or rectangular lattice convention;
5. retained band and Wannier counts; and
6. consistent reciprocal-boundary sewing.

These data correspond to the standard `.eig`, `.mmn`, optional `.amn`, and `.win` Wannier90 surfaces [91].

A formally three-dimensional lattice representation may use one reciprocal point in an inactive transverse direction. The auxiliary lattice vector and transverse orbital factor are then representation choices. Active-plane centers and spreads must be separated from artificial transverse contributions. The model remains two dimensional even though the localization interface uses a three-vector lattice convention.

For an isolated band, trial projections may be unnecessary when the direct parallel-transport gauge is already smooth and periodic. For a composite or entangled group, projections and possibly disentanglement windows become material choices [115]. Projection, disentanglement, gauge rotation, localization, and hopping truncation must remain separately identified.

The direct and Wannier90 routes should consume the same parent eigenvectors, mesh, subspace, and overlap matrices. After translations, orbital labels, and unitary freedom are aligned, compare:

1. retained projectors;
2. Wilson-loop or center phases;
3. active-plane Wannier centers and spreads;
4. real-space profiles and symmetry relations;
5. interpolated bands on withheld $\mathbf{k}$ points;
6. aligned real-space Hamiltonian blocks; and
7. mesh-refinement sensitivity under the frozen trial-projection family.

H.11 Two-dimensional Wannier Hamiltonian

Let $N_k = N_{k_x} N_{k_y}$ and choose one fundamental set $\mathcal{R}_{N_k}$ containing exactly N_k lattice-vector representatives

$$\mathbf{R} = r_x \mathbf{a}_1 + r_y \mathbf{a}_2 \tag{H.35}$$

modulo the two Born–von Karman superlattice vectors. For a retained M-band frame, define the represented real-space blocks

$$\mathbf{H}_{\mathrm{W}}(\mathbf{R}) = \frac{1}{N_k} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \mathbf{H}_{\mathrm{W}}(\mathbf{k}), \qquad \mathbf{R} \in \mathcal{R}_{N_k}. \tag{H.36}$$

The inverse discrete transform is

$$\mathbf{H}_{\mathrm{W}}(\mathbf{k}) = \sum_{\mathbf{R}\in\mathcal{R}_{N_k}} e^{i\mathbf{k}\cdot\mathbf{R}} \mathbf{H}_{\mathrm{W}}(\mathbf{R}). \tag{H.37}$$

The infinite-lattice interpretation is recovered only through a controlled reciprocal-mesh limit; periodic copies of one finite-transform coefficient must not be counted as independent hoppings.

In the separable isolated-product case, the two-dimensional Wannier functions should factor into one-dimensional Wannier functions up to gauge choices within degenerate product spaces. The Hamiltonian contains axis-directed contributions inherited from the one-dimensional factors. Any unexpected diagonal or mixed hopping should first be checked against gauge mixing, degeneracy, transform conventions, and aliasing.

The coupled term V_{xy} generally destroys product factorization and can produce genuine mixed-direction structure in the retained representation. Its controlled onset provides a direct test of whether the localization and hopping analysis detects nonseparability.

H.12 Two-dimensional tight-binding hierarchy

Choose a retained set

$$\mathcal{R}_c \subseteq \mathcal{R}_{N_k} \tag{H.38}$$

of distinct Born–von Karman representatives. The finite-range model is

$$\mathbf{H}_{\mathrm{TB}}^{(\mathcal{R}_c)}(\mathbf{k}) = \sum_{\mathbf{R}\in\mathcal{R}_c} e^{i\mathbf{k}\cdot\mathbf{R}} \mathbf{H}_{\mathrm{W}}(\mathbf{R}). \tag{H.39}$$

In two dimensions, “range” is not unique. Possible hierarchies include:

1. Euclidean-distance shells;

2. Manhattan-distance shells;
3. square-lattice point-group orbits; and
4. physically declared neighbor shells.

The hierarchy must be frozen before outcomes are inspected. A symmetry shell should be retained or omitted as a complete orbit when the model class is claimed to preserve that symmetry.

The omitted-block residual is

$$\Delta\mathbf{H}^{(\mathcal{R}_c)}(\mathbf{k}) = \sum_{\mathbf{R}\in\mathcal{R}_{N_k}\setminus\mathcal{R}_c} e^{i\mathbf{k}\cdot\mathbf{R}}\mathbf{H}_{\mathrm{W}}(\mathbf{R}). \tag{H.40}$$

For a complete uniform transform, Parseval's identity relates the full-mesh reciprocal Frobenius residual to the omitted real-space block norm:

$$\sum_{\mathbf{k}} \left\|\Delta\mathbf{H}^{(\mathcal{R}_c)}(\mathbf{k})\right\|_{\mathrm{F}}^2 = N_k \sum_{\mathbf{R}\in\mathcal{R}_{N_k}\setminus\mathcal{R}_c} \left\|\mathbf{H}_{\mathrm{W}}(\mathbf{R})\right\|_{\mathrm{F}}^2 . \tag{H.41}$$

Nonuniform reciprocal weights or incomplete transforms define a different objective.

H.13 Direct and mediated reduction routes

The mediated route constructs $\mathbf{H}_{\mathrm{W}}(\mathbf{R})$ and truncates or fits it within a declared shell hierarchy. The direct route fits the same finite-dimensional model class to parent band data over selected reciprocal points.

For one isolated band, a complete uniform mesh, equal weights, and an unconstrained set of retained lattice Fourier coefficients that are distinct modulo the Born–von Karman superlattice, direct least squares and mediated Fourier truncation are expected to agree. In a composite subspace, the direct spectral objective does not identify a unique matrix-valued operator because momentum-dependent unitary rotations preserve the eigenvalues. An aligned operator objective or additional orbital information is required.

The route defect is

$$D_{\mathrm{path}}^2 = \sum_{\mathbf{k}\in\mathcal{K}_{\mathrm{cmp}}} w_{\mathbf{k}} \left\|\mathbf{H}_{\mathrm{TB,dir}}(\mathbf{k}) - \mathbf{C}(\mathbf{k})^{\dagger}\mathbf{H}_{\mathrm{TB,med}}(\mathbf{k})\mathbf{C}(\mathbf{k})\right\|_{\mathrm{F}}^2 , \tag{H.42}$$

where $\mathbf{C}(\mathbf{k})$ is an admissible periodic alignment when the final orbital conventions differ. A constant cell-local matrix is sufficient only when the two gauges differ by a momentum-independent orbital rotation. A general momentum-dependent alignment can mix real-space cells and change the apparent hopping range, so its smoothness, periodicity, and

real-space action must be retained with the route comparison. The comparison should vary reciprocal weighting, withheld regions, shell constraints, gauge choices, and composite rank one at a time.

H.14 Staged method and execution status

The retained calculation executes the separable and parent-convergence stages, lowest-band reciprocal-loop and Chern diagnostics, symmetry-shell truncation, matched direct-versus-mediated reduction, and the frozen nonseparable-coupling continuation. Degeneracy is tested through a rank-two projector control. A rank-three extension adds a direct projected-gauge and localization comparison. Three separate topological models test nonzero Chern and Wilson-winding diagnostics without identifying any model with the scalar parent. The first Wannier90 attempt failed during preprocessing; the corrected balanced-embedding calculation then converged and supplies the external localization comparison.

H.14.1 Stage A: reproduce the separable product structure

Set $V_{xy} = 0$. For selected (V_x, V_y), compare two-dimensional plane-wave and real-space eigenvalues and projectors with sums and products of converged one-dimensional results. Verify Kronecker-sum identities at the represented matrix level and treat symmetry degeneracies as complete subspaces.

H.14.2 Stage B: converge the two parent representations

Refine (P_x, P_y) and $(\Delta x, \Delta y)$ independently over a full frozen reciprocal mesh. Compare low-sector transported operators, band energies, projectors, symmetry residuals, gaps, bandwidths, and effective-mass tensors. Freeze neither parent representation until the declared quantities are stable under both routes.

H.14.3 Stage C: construct direct two-dimensional gauges

For an isolated band, perform parallel transport along reciprocal lines, record loop holonomies, and impose periodic closure in both directions. For a composite subspace, use overlap polar factors and Wilson loops. Retain overlap singular values and stop if the selected subspace is not compatible across the mesh.

H.14.4 Stage D: compare with Wannier90

Provide Wannier90 with the same eigenvalues, overlap matrices, reciprocal mesh, and optional projections. Compare direct and Wannier90 centers, active-plane spreads, projectors, profiles, interpolated bands, and aligned hopping blocks. Repeat under mesh refinement. The closed benchmark freezes one trial-projection family and excludes cross-family projection sensitivity; such a comparison would require a separate controlled study.

H.14.5 Stage E: verify topology and symmetry

Evaluate reciprocal-loop closure and a discretized Berry-curvature or Chern diagnostic for the retained subspace. Verify the expected trivial starting case before interpreting localization. Check square-lattice symmetry in the isotropic model and its controlled reduction in the anisotropic model.

H.14.6 Stage F: truncate by two-dimensional hopping shell

Freeze a symmetry-respecting shell hierarchy. At each level report omitted block norms, reciprocal operator residuals, training and withheld band errors, gaps, bandwidths, and effective-mass-tensor errors. Compare Euclidean and physically declared shell orderings only as separately named model classes.

H.14.7 Stage G: compare direct and mediated reductions

Begin with an isolated band and complete uniform reciprocal data, where route agreement is expected. Then introduce composite bands, withheld reciprocal regions, nonuniform weights, symmetry-constrained shells, and independent gauge choices one at a time. Align final operators before forming the route defect.

H.14.8 Stage H: turn on nonseparable coupling

Increase λ_{xy} from zero through a frozen sequence while holding the remaining controls fixed. At each value repeat parent convergence, subspace, localization, hopping-decay, and route comparisons. Determine which observed changes are physical consequences of coupling and which arise from lost numerical convergence or a changed retained manifold.

H.15 Error accounting

The calculation retains separately:

1. one-dimensional reference error inherited by the separable check;
2. two-dimensional plane-wave cutoff error;
3. real-space grid and stencil error in each direction;
4. reciprocal-mesh and interpolation error;
5. symmetry and degeneracy-cluster matching error;
6. retained-subspace and overlap-conditioning error;
7. loop-closure, gauge, and topology diagnostics;
8. direct-versus-Wannier90 localization discrepancy;
9. hopping-shell truncation error;
10. direct-fit optimization and identifiability error;
11. effective-mass-tensor extraction error; and
12. nonseparable-coupling continuation error.

No scalar total error is defined. The experiment is an illustrative numerical verification problem, not scientific validation of a two-dimensional material or of silicon.

H.16 Calculated result

The retained input fixes $\lambda_x = \lambda_y = 0.5$ and varies $\lambda_{xy} = 0, 0.05, 0.15, 0.30$. The represented separable Kronecker-sum defect is exactly zero, the complete product-spectrum defect is $3.55 \times 10^{-14} E_G$, and the rank-two product-projector defect is 8.08×10^{-14}. A controlled internal 45° rotation reduces individual-state overlap to $1/\sqrt{2}$ while changing the projector by only 2.68×10^{-16}.

For the coupled parent at $\lambda_{xy} = 0.15$, the maximum lowest-four-band plane-wave cutoff error decreases from $5.25 \times 10^{-2} E_G$ at $P = 1$ to $6.27 \times 10^{-11} E_G$ at $P = 4$ relative to $P = 5$. The independent finite-difference sequence decreases monotonically to a $6.27 \times 10^{-3} E_G$ spectral error and a $5.34 \times 10^{-2} E_G$ transported low-sector operator defect at 25^2 points. These remain distinct parent errors rather than being combined.

Across the coupling sequence, the minimum isolation gap decreases from $0.4971E_G$ to $0.3251E_G$, while sewn neighboring overlaps remain above 0.978. The lattice Chern diagnostic is zero, the Wilson phases have no transverse winding, and a deterministic momentum-dependent phase attack changes the retained loop diagnostics by at most 1.14×10^{-15}. This verifies the expected trivial scalar family; it does not test a topological obstruction.

Mixed-direction hopping grows from a $3.00 \times 10^{-15}E_G$ numerical floor at the separable point to $1.94 \times 10^{-3}E_G$ at $\lambda_{xy} = 0.30$. Matched direct and mediated shell coefficients agree within $3.37 \times 10^{-16}E_G$. The complete finite transform reconstructs the 15×15 mesh to binary64 precision, but the strongest-coupling 31×31 withheld error remains $1.70 \times 10^{-6}E_G$; reciprocal-mesh interpolation and hopping truncation are therefore separate limits. The isotropic principal mass changes from $1.534m$ to $1.313m$. A separate anisotropic control gives $1.184m$ and $2.112m$ while retaining time reversal and reflection and deliberately breaking fourfold rotation by $7.46 \times 10^{-2}E_G$.

The independently implemented verifier reconstructs the parent matrices, separable spectra and projector, all retained mesh energies, reciprocal-boundary sewing, Chern sum, hoppings, shell fits, withheld errors, and mass tensors without importing the runner. It reports zero serialized-energy and hopping reconstruction defect at the retained binary64 precision. The complete reproduction package is retained under `calculations/research-monograph/periodic-2d/`.

H.16.1 Rank-three composite extension

At $\lambda_{xy} = 0.15$, the lowest three bands remain separated from the exterior by at least $3.48 \times 10^{-2}E_G$. Projection of centered Gaussian s, p_x, and p_y trials has minimum singular value 0.886, while the minimum sewn-neighbor singular value is 0.785. These controls establish a full-rank direct projected frame on the declared mesh; they do not establish a preferred material-orbital choice.

A deterministic rapidly varying internal $O(3)$ rotation supplies a rough gauge without changing the retained projector. Smooth and rough frames agree in mesh spectra within $1.22 \times 10^{-15}E_G$, in branch-aware Wilson eigenphase sets within 2.22×10^{-15}, and in zero total Chern number. They do not agree in localization: the rough gauge raises the total finite-supercell spread from $24.90a^2$ to $67.70a^2$. At squared-radius shell 18, the omitted hopping-block norm rises from $1.38 \times 10^{-2}E_G$ to $5.25 \times 10^{-1}E_G$. Thus subspace and topology diagnostics are invariant under the attack while represented hopping locality is not.

Independent reconstruction of the external comparison exposed a provisional direct-gauge error: multiplying the square trial-overlap matrix by its full inverse had returned the raw eigengauge. The corrected frame uses the polar factor $A(A^\dagger A)^{-1/2}$. The runner

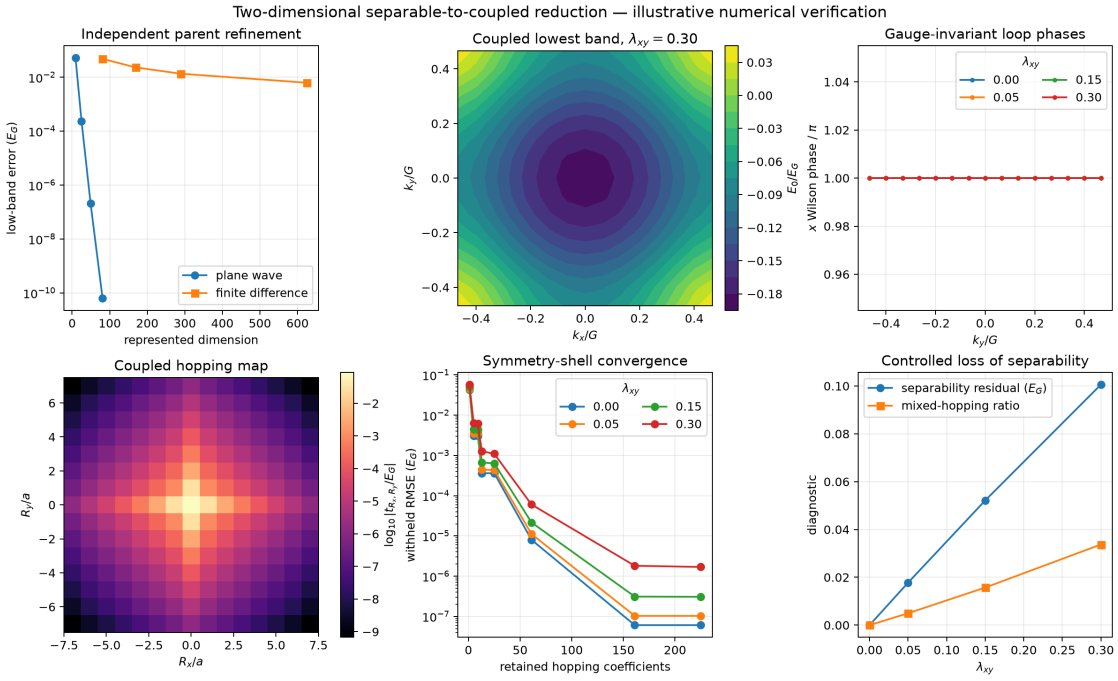


Figure H.1: Calculated two-dimensional controlled exercise. The panels retain parent-representation refinement, the full coupled lowest-band surface, gauge-invariant Wilson phases, the mixed-direction hopping map, frozen symmetry-shell convergence, and the controlled loss of separability. The plotted results are illustrative numerical verification, not material validation.

evaluates it by SVD, while the independent verifier uses a Hermitian eigendecomposition. The correction changes the gauge-coordinate diagnostics above but not the retained parent, projector, gap, or gauge-invariant topology conclusions.

The independent verifier reconstructs the trial projection, exterior gap, non-Abelian links, determinant Chern sum, Wilson spectra, hopping blocks, shell errors, density identities, and spreads without importing the runner. No independent Wannier90 result is inferred from this direct comparison.

H.16.2 Retained failure and corrected Wannier90 comparison

The first Wannier90 3.1.0 attempt stopped during preprocessing with `kmesh_get: something wrong, found too many nearest neighbours`. It exited with code 1 after 0.31 s and used 22,118,400 bytes maximum resident memory. No `.nnkp` was emitted; therefore the `.mmn` interface and localization stages were not started. The failure was retained without an automatic retry.

The $15 \times 15 \times 1$ reciprocal mesh had been embedded in an auxiliary unit cubic cell. Its inactive reciprocal increment is 15 times the active increments, so Wannier90's three-dimensional completeness search encounters too many shorter in-plane shells before reaching the inactive direction and exceeds its compiled 12-neighbor limit. This is an interface-embedding failure, not a spread- minimization result or evidence against localization.

The corrected run changed only the inactive direct-lattice length from 1 to 15, balancing the reciprocal increments while preserving the active parent, mesh, rank, projections, and localization controls. Preprocessing then found six axial neighbors in 0.28 s, and localization converged in 94 iterations and 0.47 s with maximum resident memory 27.3 MB.

The native Berry-link spreads are $0.2072a^2$, $0.2509a^2$, and $0.2513a^2$, totaling $0.7094a^2$. They are not directly substituted for the direct route's finite-supercell estimator. Reconstructing the Wannier90 frame on the same 128^2 grid gives spreads $0.2358a^2$, $2.9532a^2$, and $2.9469a^2$, totaling $6.1359a^2$, versus $24.5163a^2$ for the corrected direct projected frame reconstructed from that same external mesh. The separate centered-mesh direct route gives $24.8951a^2$. At squared-radius shell 50, the omitted hopping norm decreases from $5.90 \times 10^{-3} E_G$ to $2.72 \times 10^{-4} E_G$. The native/common-estimator discrepancy is retained as an estimator/discretization difference, not a localization gain. The frames span the same retained subspace within 2.65×10^{-10}; exact k-dependent alignment reduces their represented-operator defect to $3.57 \times 10^{-10} E_G$.

A bounded search over integer translations in $[-1, 1]^2$ for each orbital and one constant unitary leaves a per-orbital frame RMS defect of 1.188. Thus the optimized gauge is not merely a translated or constantly rotated copy of the direct projected frame. Native hopping blocks reconstruct the mesh within $3.06 \times 10^{-5} E_G$ in Frobenius norm and $2.23 \times 10^{-5} E_G$

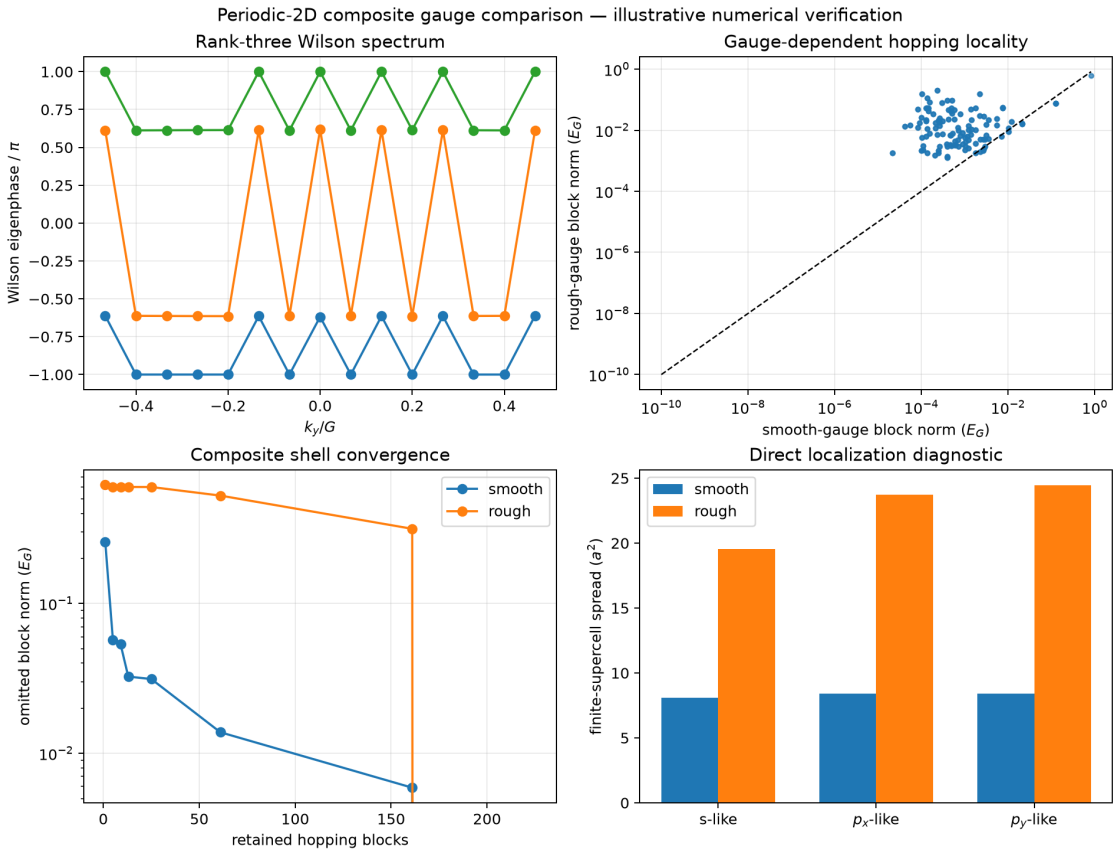


Figure H.2: Direct rank-three composite comparison. The Wilson spectrum and represented eigenvalues survive the internal gauge attack, whereas finite-supercell spreads and matrix-valued hopping locality worsen. The finite spreads are numerical diagnostics, not an infinite-mesh localization theorem.

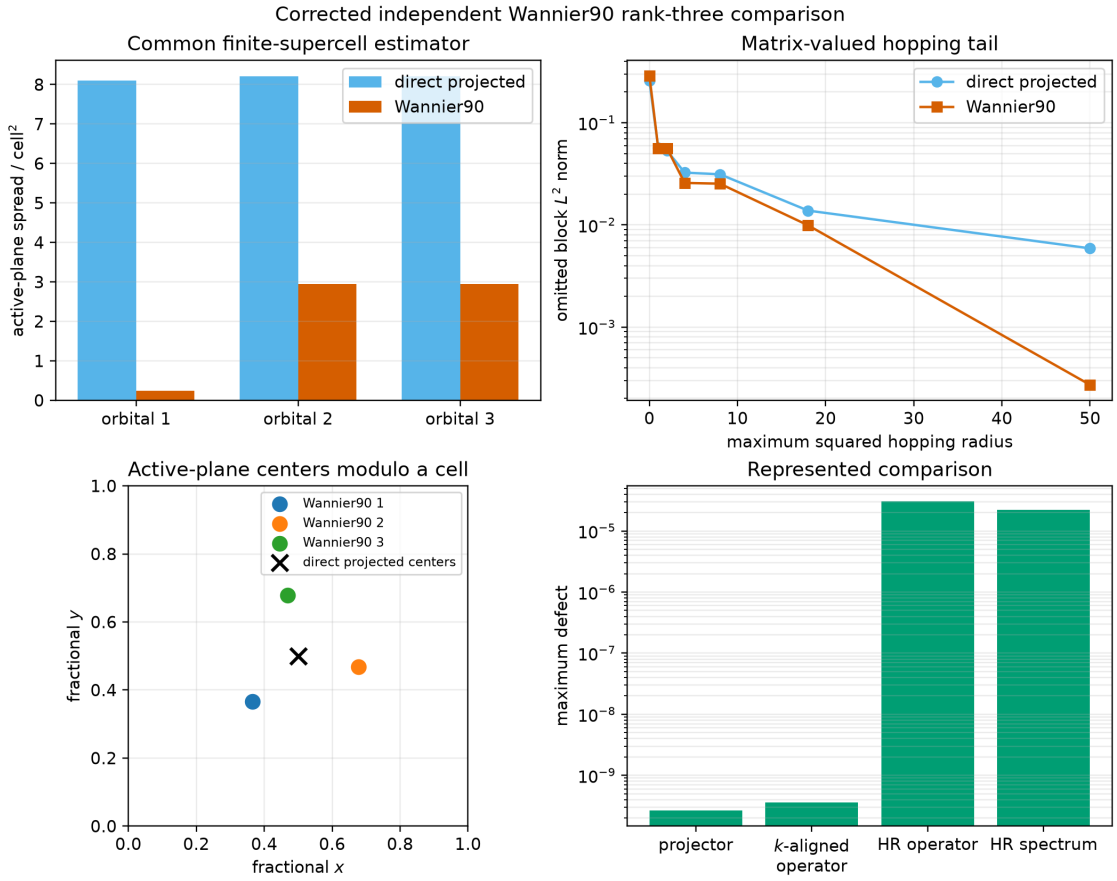


Figure H.3: Corrected independent Wannier90 rank-three comparison. On one common finite-supercell estimator, external spread minimization reduces active-plane spread and the long-range hopping tail. Projector and aligned-operator defects reflect printed U-matrix precision; the larger `_hr.dat` defects reflect its six-decimal serialization.

spectrally, retained separately as decimal serialization error. An independent verifier reconstructs the parent, polar frame, bounded alignment, native hoppings, common-grid densities, spreads, and shell tails without importing the extractor. A compact extracted numerical fixture retained with the result makes these reconstructions repository-portable; native-file identity, format, and execution-log checks remain a stronger mode requiring the external run.

H.16.3 Three distinct topological controls

The topological extension keeps three finite-dimensional Bloch models separate: the two-state Qi–Wu–Zhang model [94], the three-state flux-$1/3$ Hofstadter magnetic cell [49], and the two-state periodic-gauge Haldane model [43]. The magnetic-band Chern interpretation follows the Thouless–Kohmoto–Nightingale–den Nijs construction [123]. Each model has a declared topological parameter and a declared trivial control. None is a representation or approximation of the scalar continuum cosine operator, and their numerical errors and energy units are not combined.

For the implemented Qi–Wu–Zhang convention, gap closings occur at $m = -2, 0, 2$; the selected $m = -1$ case lies in the lower-band $C = -1$ interval, whereas $m = 3$ is analytically trivial. For Haldane, the selected masses are separated by the analytic boundary $|M| = 3\sqrt{3}|t_2 \sin\phi| \simeq 0.7794$: $M = 0$ is topological and $M = 1$ is trivial. At Hofstadter flux $1/3$, the Diophantine assignment gives $(-1, +2, -1)$ in the orientation used here. The finite $\Delta = 4$ atomic-superlattice case remains a declared numerical negative control; no analytic transition threshold is claimed for that value.

On each mesh $N = 21, 31, 51, 81$, normalized neighbor links define the gauge-invariant discretized plaquette Chern sums of Fukui, Hatsugai, and Suzuki [35] and horizontal Wilson phases of the type used for Wannier-center topology diagnostics [114]. At $N = 81$, the topological retained gaps are 2.000, 1.274, and 1.559 in the respective model units. Every topological lower band has Chern integer -1 and Wilson winding $+1$ on every mesh. The full-band Chern tuples are $(-1, +1)$ for Qi–Wu–Zhang and Haldane and $(-1, +2, -1)$ for Hofstadter. Every declared trivial control has zero retained Chern sum and zero winding.

A deterministic periodic phase attack changes a Chern sum by at most 2.3×10^{-16} and a Wilson phase by at most 1.5×10^{-15}. Reciprocal-seam projector defects remain below 1.9×10^{-15}. The independent verifier reconstructs every Bloch matrix and uses projector Bargmann products rather than the runner's normalized-link route.

The nonzero Chern integer and Wilson winding evaluate the numerical premises of the standard rank-one Wannier obstruction [13]. They do not prove the general theorem, establish topology for the scalar parent, or validate a material. The complete separate contracts and limitations are retained in `calculations/research-monograph/perio`

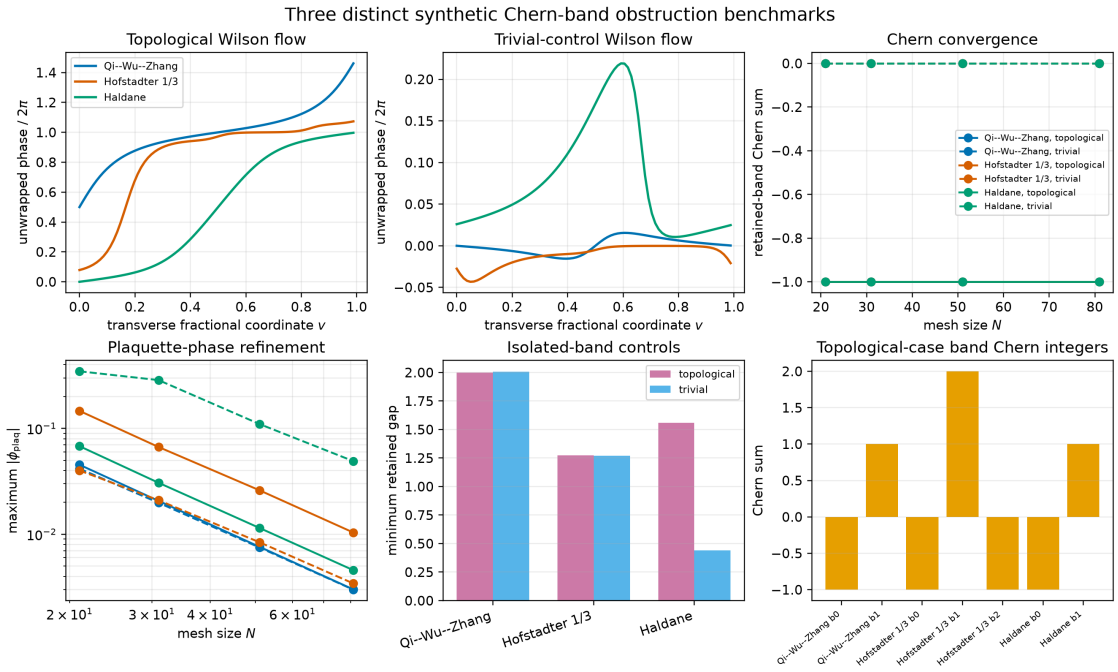


Figure H.4: Three separate synthetic Chern-band controls. Nonzero Chern sums and Wilson winding persist under mesh refinement in each topological case; the declared trivial controls remain at zero. Model-specific gaps and errors are not combined.

`dic-2d/topological-report.md`.

H.16.4 Bounded non-DFT Wannier90 sensitivity study

Six additional one-attempt cases vary reciprocal mesh, plane-wave cutoff, and inactive embedding around $(P, N, c) = (3, 15, 15)$. All complete within their retained resource limits, but the results are nonmonotone. The common finite-supercell spread is $4.6847a^2$, $6.1359a^2$, and $8.4973a^2$ at $N = 11, 15, 19$, respectively. At $P = 2$ and $P = 4$ it is $9.3161a^2$ and $10.3654a^2$, with radius-50 tails near $3.3 \times 10^{-2} E_G$ rather than the reference $2.72 \times 10^{-4} E_G$.

Changing only the inactive length to $c = 12$ closely reproduces the reference total spreads and tail, whereas $c = 18$ converges to a different basin. Thus the selected localized gauge is not demonstrated to be cutoff, mesh, or embedding independent. No less-localized case is discarded and no retry or alternative initialization is used to select a preferred outcome. A checksummed supplementary archive contains the native failure, reference, and study files. Persistent public deposit and DOI assignment remain later publication steps rather than scientific claims of this appendix.

The localization-functional literature already records multiple local minima and the importance of suitable initial gauges [72]. Wannier90 implements spread minimization on the unitary gauge manifold [78, 91]. Optimized projections, parallel-transport-based construction, and selected-columns approaches seek more robust or automated initialization [80, 16, 126]; related nonconvex initialization sensitivity is also discussed for the broader isolated and entangled localization problem [23]. The studies below do not compare those alternative algorithms. They test one fixed Wannier90 protocol and one frozen trial-projection family on one synthetic isolated rank-three subspace. Cross-family projection sensitivity is outside the closed benchmark. The studies make no priority, projection-robustness, global-optimization, or general-convergence claim.

H.16.5 Pilot deterministic multi-initialization study

A deterministic non-DFT follow-up applies eight recorded smooth reciprocal-periodic $U(3)$ initial-gauge perturbations to each of nine interfaces. The mesh sequence uses $N = 11, 15, 19, 23$ at $P = 4$ and $c = N$; the cutoff sequence uses $P = 2, 3, 4, 5$ at $N = c = 19$; and separate inactive-embedding controls use $c = 15, 19, 23$ at $P = 4, N = 19$. Every perturbation preserves the retained rank-three subspace. Native Wannier90 Berry-link spread owns optimizer and convergence classification; a distinct 512^2 common finite-supercell estimator is retained for represented within-configuration comparisons.

All 72 serial processes exit within the retained limits, but only 51 satisfy the native convergence criterion. The other 21 reach the 5000-iteration bound and remain nonconverged

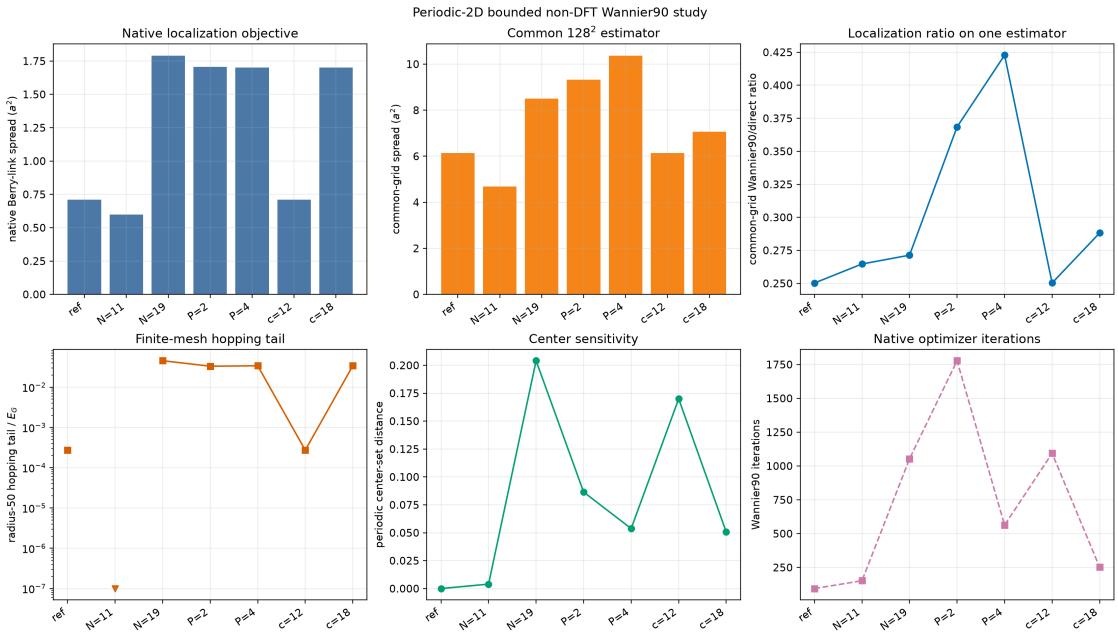


Figure H.5: Bounded non-DFT Wannier90 study. Native Berry-link and common-grid spreads occupy separate panels and are not compared as commensurate estimators. Localization ratios use only the common estimator. Hopping tails, center-set sensitivity, and optimizer iterations remain distinct. The nonmonotone records do not establish a converged localization limit.

evidence. At $N = 11, 15, 19, 23$, respectively, 7, 6, 5, and 2 of eight starts converge. Under the frozen spread-and-center classifier, every converged start occupies a separate observed basin, so the best-basin occupancy is one rather than the required two. The classification handles orbital permutation and periodic wrapping but not every spatial symmetry; it is a conservative observed-endpoint diagnostic, not proof of distinct mathematical local minima.

The best converged native spread changes from $0.796140901a^2$ at $N = 19$ to $0.863253246a^2$ at $N = 23$, a 7.77% difference that exceeds the frozen 1% criterion; the matched center-set distance is 0.187 cell. At fixed $N = c = 19$, the best spreads at $P = 4$ and $P = 5$ differ by only 6.24×10^{-7} relatively and their radius-50 tails by 3.54×10^{-4} relatively, but their center sets differ by 0.118 cell and the repeated-basin criterion still fails. Stable scalar objective values are therefore not promoted to a stable localized representation.

At $P = 4, N = 19$, the lowest converged native spreads for $c = 15, 19, 23$ all lie near $0.796141a^2$, while 8, 5, and 5 starts converge. The auxiliary embedding can thus preserve the lowest observed scalar objective while changing optimizer trajectories.

An offline reanalysis separates Ω_I from $\widetilde{\Omega} = \Omega_D + \Omega_{OD}$ [73, 72]. From $N = 19$ to $N = 23$, Ω_I changes by 8.57%, while the best converged $\widetilde{\Omega}$ changes by only 0.103%. The original total-spread failure is therefore dominated by the invariant subspace/discretization contribution, not by a comparably large change in the gauge-dependent objective. Center-set, repeated-best-basin, and multi-start convergence nevertheless still fail.

The reanalysis also minimizes center-set distance over all eight square-lattice D_4 operations. Only two non-best endpoints merge, at the $c = 23$ embedding point; best-basin occupancy remains one everywhere. Of the 21 nonconverged runs, exploratory terminal-trace diagnostics identify 11 still descending, nine near stationary without satisfying the convergence window, and one oscillatory or stalled. No endpoint is reclassified as converged.

For four representative endpoints, common-estimator spread changes by at most 1.32×10^{-6} relatively from 512^2 to 1024^2, with center changes at roundoff. Radius-18 tails replace radius 50 for all-mesh comparison because the smaller shell is supported by every declared mesh. The full negative-result-preserving protocol and independent native and offline verification are retained in `calculations/research-monograph/periodic-2d-optimizer-basin/`. The result does not support mesh-, cutoff-, embedding-, or initialization-independent localization, prove a global minimum, or supply a general convergence theorem.

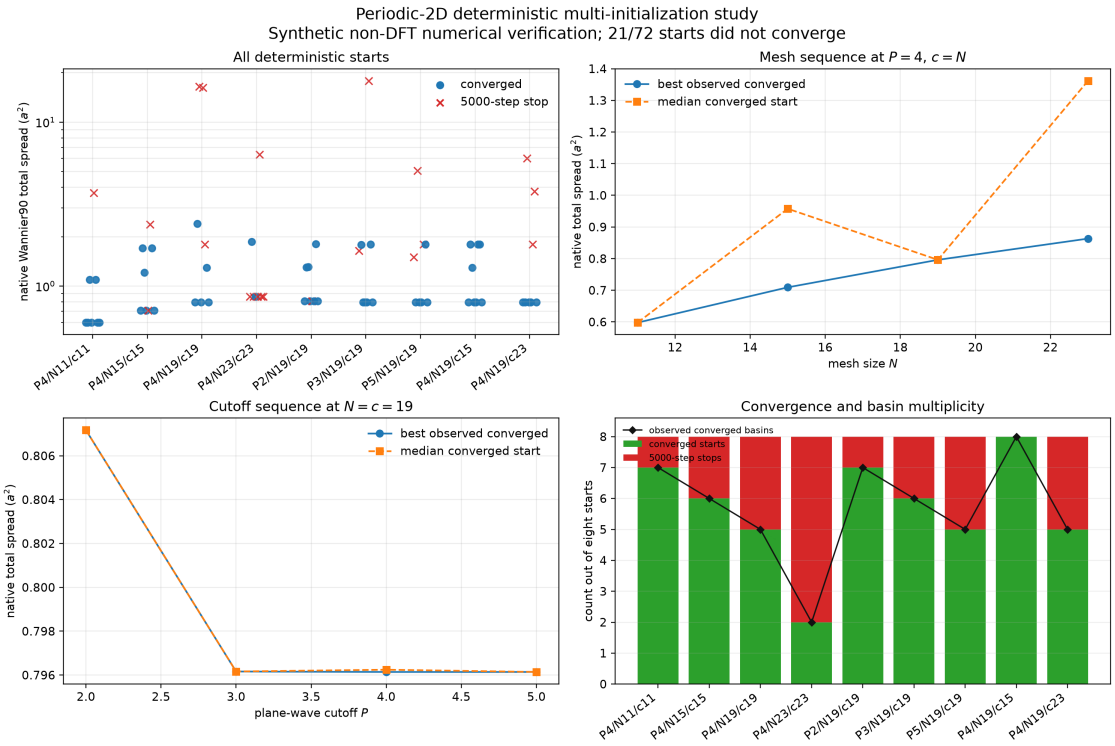


Figure H.6: Deterministic periodic-2D multi-initialization study. Blue points are native-converged endpoints and red crosses are 5000-step stops. Best and median converged-start sequences remain separate from the process-success count and observed-basin count.

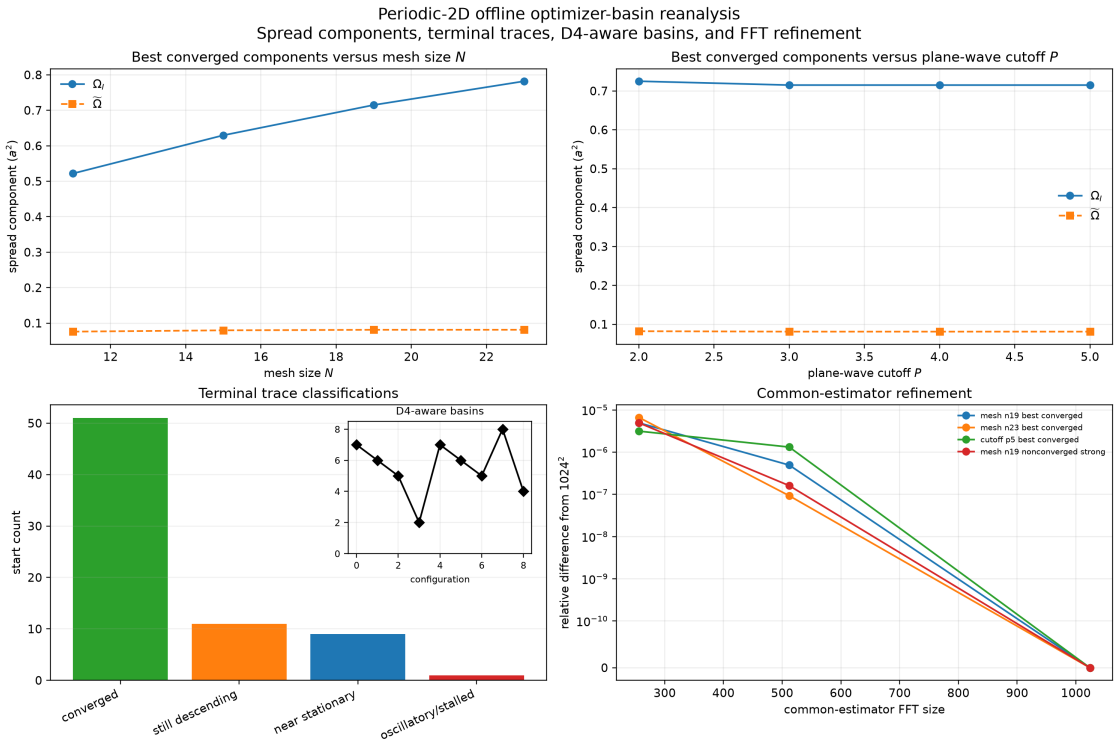


Figure H.7: Offline reanalysis of retained native files. Invariant and gauge-dependent spread components, terminal iteration traces, D_4-aware observed basins, and 256^2–1024^2 common-estimator refinement remain distinct diagnostics.

H.16.6 Expanded fixed-embedding and continuation study

The expanded follow-up decouples reciprocal mesh from the auxiliary embedding by comparing fixed $c = 31$ and balanced $c = N$ sequences at $N = 11, 15, 19, 23, 27, 31$. It applies identity plus 15 deterministic Halton-designed smooth periodic $U(3)$ starts to every interface, adds a $P = 3, 4, 5, 6$ cutoff sequence at $N = 23, c = 31$, and repeats the fixed and balanced $N = 23, P = 4$ interfaces with the preconditioner disabled. Every start lacking the native convergence statement after 5000 iterations is continued once from its exact retained checkpoint for at most 15000 additional iterations.

All 256 initial processes and all 120 required continuations exit successfully. Initially, 136 trajectories satisfy the native criterion. Continuation adds 60 converged endpoints, leaving 196 of 256 final endpoints converged and 60 nonconverged. Exploratory terminal traces classify the latter as 30 continuing descent, 25 near-stationary without five-step convergence, four oscillatory or stalled, and one stalled or nondescent. These labels do not replace the native criterion.

Final baseline convergence falls from 16 of 16 starts at $N = 11$ to 9 of 16 at fixed $c = 31, N = 27$ and 6 of 16 at $N = 31$. For the fixed $N = 27 \to 31$ pair, best Ω_I changes by 4.66% and best $\widetilde{\Omega}$ by 1.62%. The corresponding median $\widetilde{\Omega}$ change is 1.59%, and the median center-set distance is 0.0327 cell. A fit $a + b/N^2$ on $N = 15, 19, 23, 27$ misses the $N = 31$ best Ω_I and $\widetilde{\Omega}$ values by 3.69% and 2.54%. Only the radius-18 hopping-tail holdout meets its 1% target.

The $P = 5 \to 6$ cutoff pair satisfies its spread, center, hopping-tail, and converged-fraction thresholds. It still fails the basin requirement. The frozen classifier uses $\widetilde{\Omega}$, orbital permutation, periodic lattice translation, all eight D_4 operations, and matched active-plane density. Every converged endpoint remains a singleton observed basin. The nearest rejected density match is 2.23×10^{-5}, above the frozen 10^{-5} tolerance, so best-basin occupancy remains one rather than four. This is an observed numerical distinction, not proof of physically distinct minima.

The study design required exact self-symmetry and reconstruction-roundoff checks before execution, but no retained record shows that this check occurred at the specified time. This protocol deviation cannot be corrected retroactively. Post-hoc identity, orbital-permutation, all-eight-D_4, and integer-translation controls over the 16 group-best endpoints give maximum center and density mismatches of 5.56×10^{-17} cell and 6.69×10^{-16}, respectively. A threshold sensitivity analysis finds no matching pair through 2×10^{-5}, one non-best pair at 2.5×10^{-5}, and five pairs across four groups at 10^{-4}. The best-endpoint direct occupancy remains one at every tested threshold through 10^{-4}. These post-hoc checks bound numerical roundoff and show the reported best-basin failure is insensitive over the tested range; they do not restore compliance with the missed pre-execution step.

In the two matched $N = 23$ controls, enabling the preconditioner is associated with two and one additional native convergences, respectively: disabling it changes the counts

from 12 of 16 to 10 of 16 and 11 of 16 while leaving the best observed $\widetilde{\Omega}$ nearly unchanged. No general preconditioner advantage is inferred.

A post-hoc log-normal accelerated-failure-time regression [131] uses total optimizer iterations as the response and retains the 60 final nonconverged trajectories as right-censored observations. Configuration/optimizer categories are adjusted for fixed effects of the 16 deterministic starts through

$$\log T_i = \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta} + \sigma\epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0,1), \tag{H.43}$$

where T_i is the retained total iteration count or right-censoring limit. Relative to the fixed $c = 31$, $P = 4$, $N = 23$ preconditioned group, adjusted convergence-time ratios are 0.335 at fixed $N = 11$, 2.11 at fixed $N = 27$, 3.37 at fixed $N = 31$, and 8.19 at balanced $N = 27$. The fixed and balanced $N = 23$ preconditioner-off controls have ratios 2.49 and 1.50. The model therefore resolves iteration-scale variation hidden by final convergence counts and reinforces the concentration of slow or censored trajectories at finer meshes.

Every effective endpoint is reconstructed on a common 1024^2 grid, with radius-8 and radius-18 hopping tails. Independent verification reparses all native endpoints and hopping records and separately reconstructs common-grid density identities, centers, and spreads for the best and center-medoid endpoint of all 16 configuration/optimizer groups. The local supplementary archive `periodic-2d-standalone-optimizer-native-evidence-20260918T121659Z.tar.gz` is 397,083,715 bytes and has SHA-256 digest `ac02665dccc1480a9c5ec36997a5971c059c3612ea8def7f715a83922c5a02e8`. It retains all interfaces, initial trajectories, continuations, and the pre-resume snapshot. Public deposit remains a later publication step. The combined frozen criteria are not supported. This negative result is restricted to the synthetic parent and protocol; it neither predicts DFT behavior nor establishes general Wannier nonconvergence.

H.16.7 Separate topological phase sweeps

On a 51^2 mesh, separate sweeps recover the Qi–Wu–Zhang changes at $m = -2, 0, 2$ and bracket the Haldane changes around the analytic boundaries $M = \pm 0.7794$. The Hofstadter lower-band Chern integer changes from -1 to zero between $\Delta = 1.8$ and 2.0 and remains zero through the sampled continuation to $\Delta = 12$. This supplies a numerical basis for the declared $\Delta = 4$ control without claiming an exact analytic transition value.

H.17 Interpretation boundary

The separable model supplies an exact bridge from the one-dimensional warmup. The coupled model introduces genuinely two-dimensional structure without immediately in-

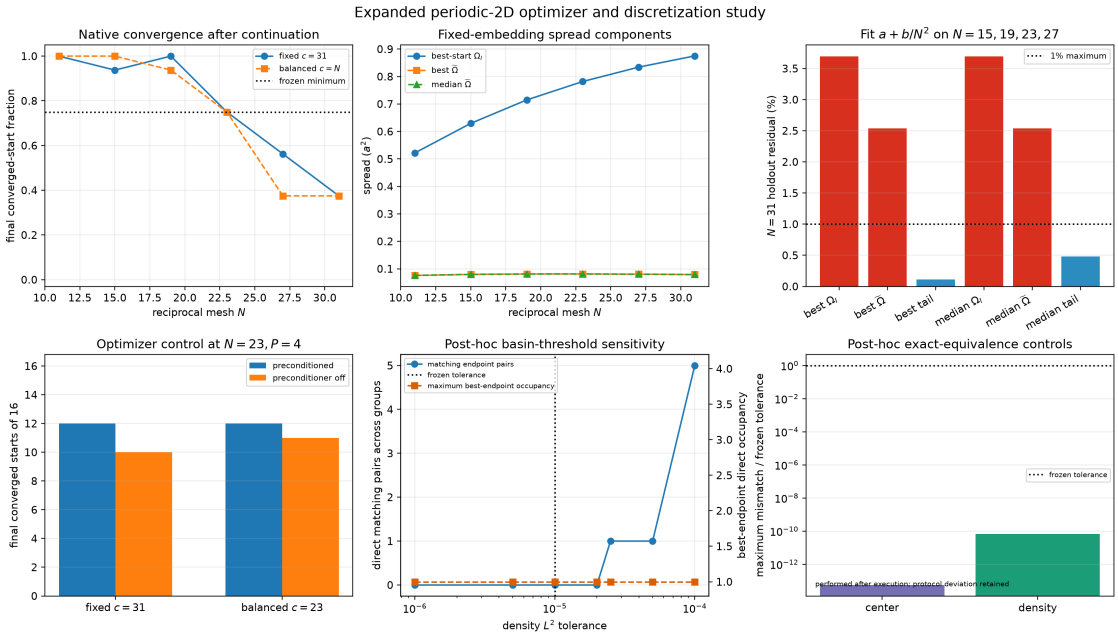


Figure H.8: Expanded optimizer and discretization study. Final convergence fractions deteriorate on fine meshes; invariant and gauge-dependent spreads remain separate; the $N = 31$ inverse-square holdout fails for both spread components; and the two matched preconditioner controls remain descriptive. Post-hoc threshold sensitivity and exact-equivalence controls retain the missed pre-execution check as a protocol deviation. All nonconverged trajectories and failed acceptance metrics remain in the evidence.

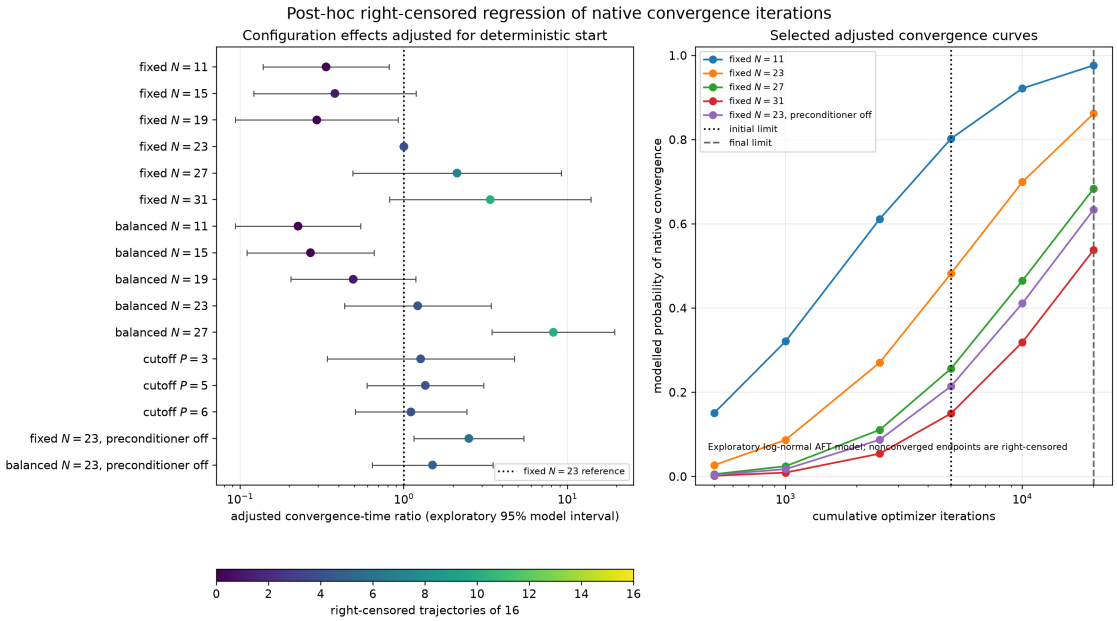


Figure H.9: Exploratory right-censored regression of native convergence iterations. The forest plot reports configuration time ratios relative to the fixed $N = 23$ group after adjustment for deterministic start; color records the number of right-censored trajectories. Selected fitted curves retain the 5000- and 20000-iteration limits. The displayed sandwich intervals are model diagnostics for this deterministic design, not population inference, physical uncertainty quantification, or causal effects.

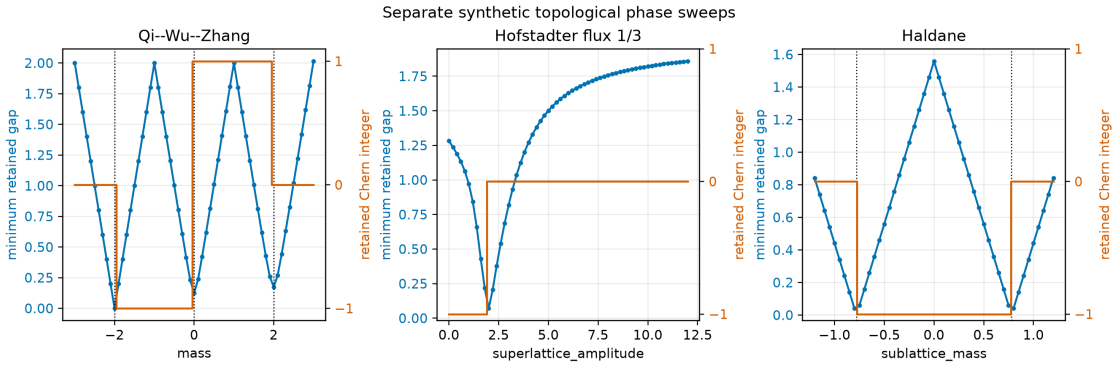


Figure H.10: Separate synthetic topological parameter sweeps. Dotted boundaries are analytic only for Qi–Wu–Zhang and Haldane. Gaps and energy units remain model specific, and every sample is independently reconstructed through projector Bargmann loops.

voking a material-specific parent. Direct gauge transport makes loop closure and topology visible; the corrected Wannier90 run supplies an independent localization route whose mesh, cutoff, and auxiliary-embedding sensitivity is now explicit. The shell hierarchy introduces directional and symmetry constraints absent in one dimension.

The completed scalar, composite, separately represented topological slices, phase sweeps, and bounded Wannier90 sensitivity cases establish numerical verification for their declared two-dimensional models under recorded conditions. They do not validate a two-dimensional material, determine an acceptable silicon Wannier subspace, show that the same hopping hierarchy is adequate for the diamond lattice, or turn the synthetic Wannier90 comparison into production or material evidence. Their role is to expose two-dimensional failure modes before the three-dimensional multiorbital bulk-silicon reduction is attempted.

CHAPTER I

Controlled Impurity-Extraction Benchmarks: Spin Spaces and One-Dimensional Defects

Status synthesis with six human-accepted calculated synthetic numerical-verification results and a human-accepted bounded primary-literature comparison. The spin-space and one-dimensional impurity calculations remain within bounded finite-model scopes; the proposed two-dimensional and material program is separated into Appendix J. This appendix does not claim an accepted production Kohn–Sham parent, Wannier operator, aligned material impurity operator, fitted material lattice or continuum model, crossover radius, calculated material binding energy, or scientific validation result.

I.1 Physical objective

The central objective is to determine when a finite operator extracted from a first-principles semiconductor calculation can be replaced by a constrained lattice or continuum model without losing the observables and operator structure required for its intended use. The problem is not to establish a universally exact effective Hamiltonian. It is to identify the least complex admissible model class that reproduces a declared parent problem within separately specified spectral, subspace, operator, and bound-state tolerances.

The parent description is a self-consistent Kohn–Sham calculation in the Born–Oppenheimer fixed-ion framework [50, 59]. In the initial program, exchange and correlation are represented by the PBE generalized-gradient approximation [89], and the electron–ion interaction is represented by compatible optimized norm-conserving Vanderbilt pseudopotentials from the approved PseudoDojo family [44, 106]. These choices define the parent model against which later reductions are evaluated. Disagreement between the parent calculation and experiment is not automatically a reduction error.

Prospective citation note—medium priority
Separate the Born–Oppenheimer attribution from the DFT and Kohn–Sham attributions. Check the fixed-ion statement against Born and Oppenheimer (1927), provisional key `bornOppenheimer1927` [10], and retain `hohenberg1964` [50] and `kohn1965` [59] only for the narrower claims they support.

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The Kohn–Sham band gap is likewise a property of the selected parent approximation and is not identified with the exact quasiparticle gap. The distinction between the Kohn–Sham eigenvalue gap and the fundamental gap is established by the derivative-discontinuity analysis of exact density-functional theory [90, 107].

I.2 Representation hierarchy

The proposed reduction is organized as the sequence

$$\hat{H}_{\mathrm{KS},s} \longrightarrow \hat{H}_{s}^{(P)} \longrightarrow \mathbf{H}_{\mathrm{W},s}^{(P)} \longrightarrow \Delta\mathbf{H}_{\mathrm{W},d}^{(P)} \longrightarrow \hat{H}_{\mathrm{model},d}, \tag{I.1}$$

where $s = b$ denotes the pristine host and $s = d$ a doped system. The arrows represent distinct operations:

1. selection of a retained Kohn–Sham subspace;
2. construction of a localized Wannier representation;
3. alignment of pristine and doped retained spaces;
4. construction of an impurity operator; and
5. approximation within a restricted lattice or continuum model class.

Projection, localization, alignment, truncation, and model fitting introduce different assumptions and must remain separately auditable. Wannier localization changes representation but does not by itself reduce matrix rank. For entangled bands, the retained subspace must be selected before the gauge is localized [72, 115].

I.3 Pristine silicon reference

The host is crystalline silicon in the diamond structure. A reproducible parent identity requires the lattice vectors, basis-site convention, reciprocal-coordinate convention, pseu-

dopotential files, plane-wave cutoff, k-point mesh, occupation treatment, self-consistency thresholds, relaxation criteria, spin treatment, spin–orbit setting, and producing-program version.

The initial bulk pilot targets a band-edge subspace suitable for comparison with a constrained orthogonal sp^3s^* Slater–Koster hierarchy. The principal spectral observables are the indirect Kohn–Sham gap, the conduction-valley position, and the longitudinal and transverse electron effective masses. Withheld-point band errors are required to separate interpolation or fitting performance from transferability outside the training set.

Spectral agreement is necessary but not sufficient. Two Hamiltonians can reproduce the same selected eigenvalues while differing substantially as operators. The bulk compatibility study must therefore evaluate both spectral observables and unitary-aligned operator residuals where a common operator representation has been established. The least complex acceptable tight-binding class is the first member of the declared hierarchy that satisfies all required tolerances; additional parameters are not justified solely because they further reduce the training error.

Prospective citation note—medium priority
Check the constrained orthogonal sp^3s^* silicon pilot and its Slater–Koster hierarchy against Slater–Koster (1954) and Vogl et al. (1983), bibliography keys `slater1954` [113] and `vogl1983` [127] (already present).

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

I.4 Aligned impurity operator

Let

$$\hat{H}_b^{(P)} : \mathcal{H}_b^{(P)} \longrightarrow \mathcal{H}_b^{(P)} \tag{I.2}$$

and

$$\hat{H}_d^{(P)} : \mathcal{H}_d^{(P)} \longrightarrow \mathcal{H}_d^{(P)} \tag{I.3}$$

be the retained pristine and doped Hamiltonians. Their matrices cannot be subtracted solely because they have equal dimensions. A unitary identification

$$\hat{U}_d : \mathcal{H}_b^{(P)} \longrightarrow \mathcal{H}_d^{(P)} \tag{I.4}$$

must be fixed using declared orbital, geometric, gauge, spin, and energy-reference conventions. This appendix chooses the pristine retained space as the reference and defines

$$\Delta\hat{H}_{d\to b}^{(P)} = \hat{U}_d^{\dagger}\hat{H}_d^{(P)}\hat{U}_d - \hat{H}_b^{(P)}. \tag{I.5}$$

An equivalent convention transports the pristine operator to the doped space:

$$\Delta\hat{H}^{(P)}_{b\to d} = \hat{H}^{(P)}_{d} - \hat{U}_d\hat{H}^{(P)}_{b}\hat{U}^{\dagger}_{d} = \hat{U}_d\Delta\hat{H}^{(P)}_{d\to b}\hat{U}^{\dagger}_{d}. \tag{I.6}$$

The direction must remain explicit. In the remainder of this appendix, $\Delta\hat{H}^{(P)}_{d}$ abbreviates the pristine-reference operator $\Delta\hat{H}^{(P)}_{d\to b}$.

This operator is the primary first-principles difference object. It should not be defined automatically as an impurity potential. It can contain local electrostatic structure, nonlocal pseudopotential contributions, changes in the self-consistent Hartree and exchange–correlation operators, lattice-relaxation effects, intervalley and interorbital coupling, and projection-induced nonlocality.

Equal total retained dimensions guarantee the existence of an abstract unitary identification, but not a physically justified one. A momentum-preserving identification additionally requires compatible retained dimensions in corresponding Bloch fibers. In a localized representation, the alignment must control site correspondence, orbital ordering, Wannier gauge, energy zero, lattice geometry, and supercell mapping.

I.5 Non-identifiability of an effective potential

Let $\hat{T}^{(P)}_{0}$ be a declared kinetic reference on $\mathcal{H}^{(P)}_{b}$. The difference

$$\hat{R}^{(P)}_{T_0,d} = \Delta\hat{H}^{(P)}_{d} - \Delta\hat{T}^{(P)}_{0,d} \tag{I.7}$$

is a reference-dependent residual operator. In the common case where the host kinetic reference is assumed unchanged by the impurity model,

$$\Delta\hat{T}^{(P)}_{0,d} = \hat{0}, \tag{I.8}$$

and the full aligned impurity operator is assigned to the residual. This assignment remains a model convention rather than a deduction that the operator is a local potential.

For any Hermitian operator $\hat{K}$ on the retained space,

$$\Delta\hat{H}^{(P)}_{d} = \left(\Delta\hat{T}^{(P)}_{0,d} + \hat{K}\right) + \left(\hat{R}^{(P)}_{T_0,d} - \hat{K}\right). \tag{I.9}$$

The aligned Hamiltonian difference alone does not uniquely determine a kinetic correction and a potential correction. Their separation becomes identifiable only after the kinetic structure and the admissible impurity-potential class have been fixed.

I.6 Admissible model classes

A native model class may act on a lattice, continuum, or other declared model space $\mathcal{X}_m$. Let

$$\mathcal{M}^{(m)}_{d,\text{native}} = \left\{ \Delta\hat{K}^{(m)}_d(\boldsymbol{\theta}) : \mathcal{X}_m \longrightarrow \mathcal{X}_m \;\middle|\; \boldsymbol{\theta} \in \Theta_m \right\}. \tag{I.10}$$

Before an operator norm is formed, each native candidate must be transported by a declared model-to-reference map into $\mathcal{L}(\mathcal{H}^{(P)}_b)$. Denote the resulting represented candidate by

$$\widetilde{\Delta\hat{H}}^{(m)}_d(\boldsymbol{\theta}) : \mathcal{H}^{(P)}_b \longrightarrow \mathcal{H}^{(P)}_b. \tag{I.11}$$

For a lattice class already expressed in the pristine reference coordinates, the transport may be the identity. A useful declared hierarchy may include:

1. a local isotropic screened-Coulomb potential;
2. an anisotropic effective-mass potential;
3. a multivalley local operator;
4. a finite-range nonlocal lattice operator;
5. orbital-block and neighbor-shell extensions; and
6. an explicitly nonlocal static downfolded operator.

These classes need not be nested merely because they appear in one list. Any claim of least complexity requires a separately declared ordering and complete acceptance rule. An energy-dependent downfolded operator $\Delta\hat{K}^{(m)}_d(E;\boldsymbol{\theta})$ requires an additional declared energy domain and an energy-resolved operator, action, or resolvent comparison. It is not governed by the static objective below merely by suppressing its energy argument.

For an energy-independent class in a finite-dimensional identified representation, suppose first that the minimum is attained and choose

$$\boldsymbol{\theta}^{\star}_{d,m} \in \operatorname*{arg\,min}_{\boldsymbol{\theta}\in\Theta_m} \left\| \Delta\hat{H}^{(P)}_d - \widetilde{\Delta\hat{H}}^{(m)}_d(\boldsymbol{\theta}) \right\|^2_{\text{HS}}. \tag{I.12}$$

The corresponding minimum operator separation is

$$d^{(m)}_d = \left\| \Delta\hat{H}^{(P)}_d - \widetilde{\Delta\hat{H}}^{(m)}_d(\boldsymbol{\theta}^{\star}_{d,m}) \right\|_{\text{HS}}. \tag{I.13}$$

If the minimum is not known to be attained, the model-class separation should instead be defined by the corresponding infimum over Θ_m.

The quantity $d_d^{(m)}$ measures the expressiveness of the model class after alignment. It is not merely an optimization diagnostic. If the minimum separation remains above the prescribed tolerance, the model class is structurally inadequate even when selected eigenvalues can be fitted.

An effective potential is therefore defined only relative to a declared class. If $\mathcal{M}_V$ is a class of admissible potential operators and the minimum is attained, then

$$\hat{V}^{\star}_{d,\mathcal{M}} \in \operatorname*{arg\,min}_{\hat{V}\in\mathcal{M}_V} \left\| \hat{R}^{(P)}_{T_0,d} - \hat{V} \right\|_{\mathrm{HS}}^2 . \tag{I.14}$$

The raw residual $\hat{R}^{(P)}_{T_0,d}$ is not itself assigned the name effective potential unless membership in, or approximation by, the declared potential class has been established.

I.7 Continuum effective-mass limit

Effective-mass and envelope-function theories derive continuum equations from perturbed periodic systems [70, 14, 15]. Their validity depends on retained-band structure, scale separation, valley content, interface or impurity length scales, and the treatment of boundary conditions. The effective Hamiltonian and admissible boundary conditions must be derived consistently; incorrect symmetrization can produce nonphysical solutions [30]. First-principles envelope-function constructions demonstrate that a continuum operator can, in principle, be derived from an atomistic parent rather than introduced solely by phenomenology [31].

The relevant literature separates limits that cannot be merged into one "continuum" sequence. Discrete-to-continuum convergence sends a numerical or physical lattice spacing to zero for a fixed limiting operator and can be posed in norm-resolvent form [81]. Band-edge homogenization instead sends a weak, slowly varying defect scale to zero and derives an effective-mass envelope from the curvature and Bloch state at a spectral edge [48, 26]. Finite-volume refinement enlarges the periodic domain at fixed local physics; exponential image effects are known in finite-volume bound-state settings, although their specific form must be re-established for the represented lattice problem rather than imported unchanged [61]. Finally, changing defect width or normalization changes the physical profile unless a scaling family is specified. Semiconductor donor theory adds a distinct central-cell and multivalley boundary: long-wavelength effective-mass behavior does not by itself resolve the atomically sharp core [58, 36]. Consequently, continuum-grid spacing, supercell size, band-edge scale, and defect-profile parameters must be refined independently. Agreement along a path that changes several of them simultaneously is not a continuum-limit result.

Let

$$\hat{J}_d : \mathcal{H}^{(P)}_{\mathrm{EMT},d} \longrightarrow \mathcal{H}^{(P)}_b \tag{I.15}$$

identify a finite-dimensional discretization of the continuum envelope-function space with a subspace of the pristine retained space. If $\hat{J}_d$ is an isometric embedding, a parameterized continuum impurity operator can be tested through

$$\boldsymbol{\theta}^{\star}_{\mathrm{EMT},d} \in \operatorname*{arg\,min}_{\boldsymbol{\theta}\in\Theta_{\mathrm{EMT}}} \left\| \Delta\hat{H}^{(P)}_d - \hat{J}_d \Delta\hat{H}_{\mathrm{EMT},d}(\boldsymbol{\theta}) \hat{J}^{\dagger}_d \right\|^2_{\mathrm{HS}}, \tag{I.16}$$

when the minimum is attained. This full-space loss treats the embedded continuum operator as zero on the orthogonal complement of Ran $\hat{J}_d$ and assigns all complement and cross-block content of the atomistic operator to the residual. If the intended comparison concerns only the embedded continuum sector, both operands must instead be compressed by $\hat{P}_J = \hat{J}_d \hat{J}^{\dagger}_d$. A non-isometric identification requires an explicit metric, dual map, or generalized comparison rather than this formula without qualification.

Recent multivalley analysis of Si/SiGe systems shows that an exact valley-sector reduction can produce band-limited envelopes and nonlocal potential operators, particularly when atomically sharp structure invalidates a conventional slowly varying local approximation [27]; the source equations and their project-specific limits are extracted in Appendix K. This provides a direct precedent for treating locality as a model-class hypothesis.

The atomistic-to-continuum crossover should consequently be expressed as a scale-dependent validation statement. A proposed crossover radius $r_{c,d}$ may be defined as the smallest tested radius beyond which both the declared exterior residual and the declared interior–exterior cross-coupling residual remain within their separate tolerances. Such a definition separates the existence of a long-range continuum regime from the unresolved short-range central-cell operator. If no tested radius passes, the result should remain explicitly unresolved or be recorded as no finite crossover over the tested domain.

I.8 Dopant branches and validation

For substitutional phosphorus, the neutral $\mathrm{P}^0_{\mathrm{Si}}$ supercell contains both the donor-core perturbation and an additional valence electron. The primary branch is scalar relativistic, collinearly spin polarized, and non-SOC so that the odd electron count is represented. The resulting neutral calculation must not be silently reinterpreted as an isolated ionized donor. Each finite supercell describes a periodic dopant array, so isolated-donor claims require supercell-size and shape convergence.

For substitutional boron, the neutral $\mathrm{B}^0_{\mathrm{Si}}$ supercell has one fewer valence electron than the matched pristine cell. Final acceptor and valence-manifold claims require a matching fully relativistic, noncollinear spinor treatment with spin–orbit coupling. Scalar-relativistic

calculations may remain useful for method development but must be labeled as a restricted branch.

Charged-defect corrections and dilute-limit interpretations require their own controlled assumptions and convergence evidence [33, 32]. These issues belong to the parent and finite-size validation layers and should not be absorbed into an operator-fitting residual without diagnosis.

Prospective citation note—medium priority
Check the phosphorus donor, boron acceptor, spin–orbit valence-manifold, and periodic-supercell qualifications against the appropriate donor/envelope, semiconductor, and defect-supercell sources. Candidate existing keys: `luttingerKohn1955` [70], `yuCardona2010` [133], and `freysoldt2014` [32].

Editorial status: prospective. Read the candidate sources and check the exact claim before adding or changing a citation.

The principal validation quantities should remain distinct:

- the global aligned operator error $\varepsilon_{H,d}$;
- the retained-subspace or projector error $\varepsilon_{\Pi,d}$;
- the binding-energy error $\Delta E_{b,d}$;
- the wavefunction or subspace fidelity F_d; and
- the continuum crossover radius $r_{c,d}$.

No single metric certifies the reduction. Spectral agreement does not imply operator agreement, a small global norm can conceal a localized central-cell failure, and a correct binding energy does not establish the transferability of the underlying impurity operator.

I.9 Controlled benchmark program

The impurity-reduction chain is exercised first on controlled models in which the relevant state-space maps and defect operators are known. The spin-space and one-dimensional exercises below are human accepted. Their purpose is to verify the mathematical construction and separate failure modes before a first-principles silicon impurity operator is interpreted.

The two completed headline warmups compare spinless and spinful state spaces and extract planted operators from matched one-dimensional pristine and defect supercells. The corresponding two-dimensional extension is specified separately in Appendix J. Each

stage retains applicable folding, null, planted-defect, alignment, finite-size, model-class, and observable controls.

I.9.1 Warmup 1: Spinless, spin-degenerate, collinear, and spinor spaces

Let $\mathcal{V}$ be a finite orbital space and let

$$\mathbf{H}_0 : \mathcal{V} \longrightarrow \mathcal{V} \tag{I.17}$$

be a spinless Hamiltonian. A nonmagnetic spinful lift acts on $\mathcal{V} \otimes \mathbb{C}^2$:

$$\mathbf{H}_{\text{deg}} = \mathbf{H}_0 \otimes \mathbf{I}_2. \tag{I.18}$$

The two matrices must not be subtracted directly because their state spaces and dimensions differ. The lift supplies the comparison map.

For a spin-independent impurity operator,

$$\Delta\mathbf{H}_{\text{deg}} = \Delta\mathbf{H}_0 \otimes \mathbf{I}_2, \tag{I.19}$$

and therefore

$$\|\Delta\mathbf{H}_{\text{deg}}\|_{\text{F}} = \sqrt{2}\,\|\Delta\mathbf{H}_0\|_{\text{F}}\,. \tag{I.20}$$

This identity provides an exact normalization check for the spin embedding.

The warmup should then introduce complexity in a fixed order:

1. the spinless operator $\mathbf{H}_0$;
2. the spin-degenerate lift $\mathbf{H}_0 \otimes \mathbf{I}_2$;
3. a collinear block operator
$$\mathbf{H}_{\text{col}} = \mathbf{H}_\uparrow \oplus \mathbf{H}_\downarrow; \tag{I.21}$$
4. a spinor operator with controlled spin mixing,
$$\mathbf{H}_{\text{spinor}} = \mathbf{H}_0 \otimes \mathbf{I}_2 + \sum_{j=x,y,z} \mathbf{B}_j \otimes \boldsymbol{\sigma}_j, \tag{I.22}$$
where the $\boldsymbol{\sigma}_j$ are Pauli matrices and each $\mathbf{B}_j = \mathbf{B}_j^\dagger$ is a declared Hermitian orbital operator, so that $\mathbf{H}_{\text{spinor}}$ is Hermitian.

The collinear and spinor stages should test spin-block ordering, spin-axis conventions, time-reversal behavior where applicable, and unitary changes of spin frame. A scalar model should recover the spin-degenerate case and fail explicitly when the planted operator contains unresolved spin splitting or spin mixing.

This sequence prepares the distinction between the primary collinear spin-polarized phosphorus branch and the fully relativistic noncollinear spinor branch required for final boron claims. It does not assert that the toy spin terms reproduce either dopant.

Calculated synthetic verification

The controlled calculation uses one authored four-orbital Hamiltonian and impurity operator, a Hermitian collinear splitting, and three Hermitian spin–orbital coefficients constructed from real antisymmetric generators. All operators carry explicit state-space, dimension, orbital-order, spin-order, spin-frame, unit, energy-reference, and geometry metadata. The complete input, protocol, result, independent verifier, report, and content identities are retained under `calculations/research-monograph/impurity-spin-spaces/`.

The spin-degenerate lift reproduces both spin-sector pullbacks exactly and has $\|\Delta\mathbf{H}_{\mathrm{deg}}\|_{\mathrm{F}} = 0.5213444159$, compared with $\|\Delta\mathbf{H}_{0}\|_{\mathrm{F}} = 0.3686461718$. The $\sqrt{2}$ scaling defect is 1.11×10^{-16}, while the root-dimension-normalized value is 0.1843230859 in both spaces. A raw spin-major versus orbital-major comparison gives a Frobenius defect of 0.5601356978, although their largest eigenvalue discrepancy is only 3.33×10^{-16}. Applying the declared permutation removes the matrix defect exactly.

Five nontrivial changes of spin frame produce raw matrix defects between 4.01×10^{-2} and 1.28×10^{-1} synthetic energy units. After the declared conjugation, the largest matrix defect is 1.77×10^{-16}; the largest spectral, Frobenius-norm, transformed time-reversal, and Hermiticity defects are respectively 2.22×10^{-16}, 1.11×10^{-16}, 6.36×10^{-17}, and 2.78×10^{-17}. The nonmagnetic lift and the constructed spin-mixing operator are time-reversal invariant to the retained arithmetic. The intentionally spin-split collinear operator has the predicted residual $2\sqrt{2}\|\mathbf{Z}\|_{\mathrm{F}} = 0.4659442027$.

The Frobenius-optimal spin-independent model recovers the degenerate lift exactly. Its relative residuals are 0.4079852850 for the planted collinear splitting and 0.2611910695 for the planted spin mixing; both absolute residuals agree with the independent Pauli-orthogonality formulas to 2.78×10^{-17}. Direct subtraction stops, without returning a nominal norm, for spinless versus spinful spaces, unaligned block orderings, different spin frames, and different energy references.

Figure I.1 summarizes the represented operators, model-class residuals, and covariance checks. These are human-accepted calculated synthetic numerical-verification results within the bounded finite-model exercise. They do not establish a material spin Hamiltonian, an explicit spin–orbit model, a phosphorus or boron impurity operator, scientific validation, or uncertainty quantification.

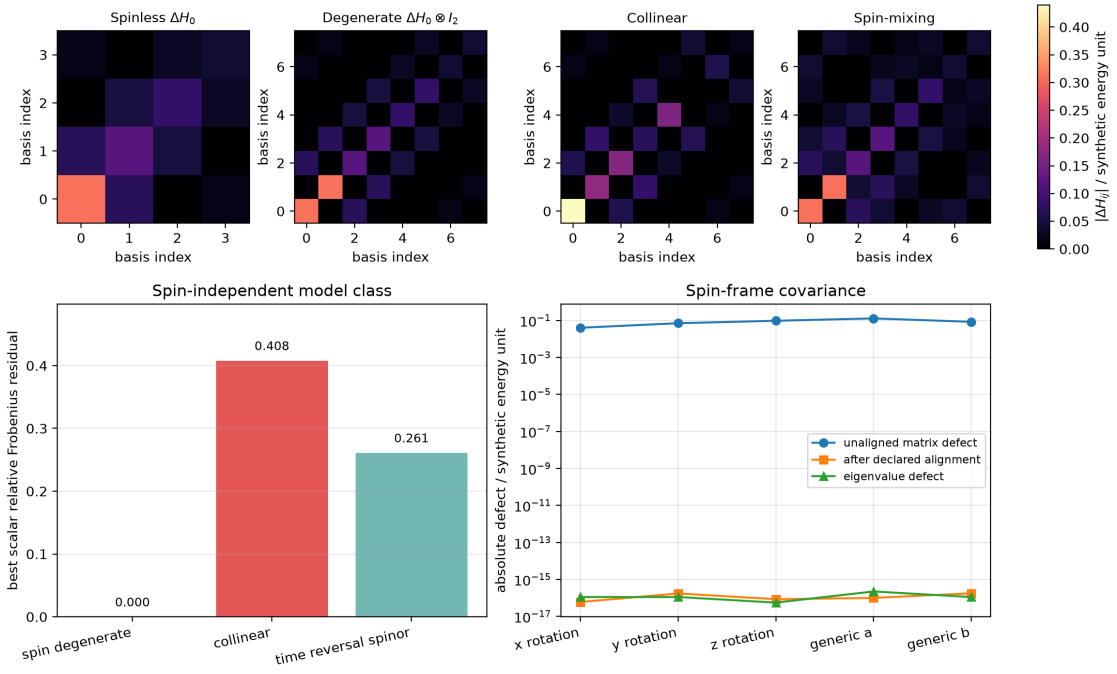


Figure I.1: Controlled spin-space embedding benchmark. The upper row shows the absolute represented matrices for the spinless, spin-degenerate, collinear, and spin-mixing impurity operators in their declared coordinates. The lower left compares the optimal spin-independent residual for the three spinful classes. The lower right contrasts raw spin-frame coordinate differences with aligned matrix and eigenvalue defects. Values at the roundoff floor are displayed logarithmically only to distinguish them from finite unaligned differences.

I.9.2 Warmup 2: One-dimensional matched pristine and defect supercells

Use the one-dimensional periodic parent developed in Appendix G. Let a supercell contain N_s primitive cells and have length

$$L = N_s a. \tag{I.23}$$

The reduced supercell crystal momentum is denoted K. Let

$$\mathbf{H}_{b,L}(K) : \mathcal{V}_{b,L}(K) \longrightarrow \mathcal{V}_{b,L}(K), \tag{I.24}$$

$$\mathbf{H}_{d,L}(K) : \mathcal{V}_{d,L}(K) \longrightarrow \mathcal{V}_{d,L}(K) \tag{I.25}$$

be matched pristine and defect supercell fibers. Both must use the same supercell length, boundary phase, discretization or orbital convention, spin space, units, and energy reference before alignment.

The pristine supercell must first reproduce primitive-cell band folding. For each K, the primitive momenta folded into the supercell fiber are

$$k_\alpha = K + \frac{2\pi\alpha}{L}, \qquad \alpha = 0, \dots, N_s - 1, \tag{I.26}$$

reduced modulo the primitive reciprocal lattice. Define a unitary folding map

$$\mathbf{F}_L(K) : \bigoplus_{\alpha=0}^{N_s-1} \mathcal{V}_b(k_\alpha) \longrightarrow \mathcal{V}_{b,L}(K). \tag{I.27}$$

It should satisfy

$$\mathbf{F}_L(K)^\dagger \mathbf{H}_{b,L}(K) \mathbf{F}_L(K) = \bigoplus_{\alpha=0}^{N_s-1} \mathbf{H}_b(k_\alpha) \tag{I.28}$$

for an exactly repeated pristine representation, up to the declared ordering and numerical approximation. This verifies primitive-to-supercell mapping before a defect is introduced.

The zero-defect control sets

$$\mathbf{H}_{d,L}(K) = \mathbf{H}_{b,L}(K) \tag{I.29}$$

before independent coordinate changes. After gauge, ordering, and energy alignment, the extracted impurity operator must be zero within the stated numerical tolerance.

Next introduce a known periodic supercell defect:

$$\mathbf{H}_{d,L}^{\text{known}}(K) = \mathbf{H}_{b,L}(K) + \mathbf{V}_{\text{known},L}(K). \tag{I.30}$$

The planted operator should progress through:

1. a one-site scalar shift;

2. an orbital-dependent onsite shift;
3. a modified nearest-neighbor hopping;
4. a finite-range nonlocal defect;
5. a collinear spin-dependent defect; and
6. a spin-mixing defect after the spinor warmup is verified.

The doped representation should then be independently transformed by a known unitary, orbital permutation, supercell translation, and scalar energy shift. The declared alignment procedure should recover the planted operator in the pristine reference coordinates.

The one-dimensional supercell warmup should be repeated over increasing N_s. Each finite L represents a periodic defect array. A localized isolated defect level is represented at finite L by a defect-derived supercell band $E_{d,\ell}(K; L)$ rather than by one K-independent bound-state energy. Its center, bandwidth, dispersion, and separation from the declared host band edge should be tracked together with central blocks, exterior blocks, cross couplings, and localization diagnostics as image separation increases.

Calculated synthetic verification

The controlled calculation reuses the accepted periodic one-dimensional parents: the lowest isolated-band hopping record and the smooth two-band `low_pair` blocks truncated to $|R| \leq 4a$. Their exact content identities, together with the complete input, protocol, result, independent verifier, report, and figure, are retained under `calculations/research-monograph/impurity-defect-1d/`. All defects and alignment maps are synthetic and known by construction.

For an $N_s = 16$ two-orbital supercell, three reduced supercell momenta give a largest primitive-folding operator defect of $9.69 \times 10^{-15} E_G$ and a largest eigenvalue defect of $7.77 \times 10^{-16} E_G$. The folding-map unitarity defect remains below 1.87×10^{-14}. Seven matched controls then plant null, scalar onsite, orbital onsite, range-one hopping, range-two hopping, collinear-spin, and spin-mixing operators. The raw defect representation is changed by a three-cell translation, an orbital swap, orbital rotation and phases, a site-dependent phase, a global spin-frame rotation where applicable, and a scalar reference shift of $0.137 E_G$. Metadata prevent direct subtraction. After applying the declared inverse maps, the largest planted-operator recovery defect is $1.32 \times 10^{-15} E_G$; the null recovery defect is $8.72 \times 10^{-16} E_G$. Omitting only the scalar-reference correction creates an exterior false offset of $0.750 E_G$ in the spinless controls.

A hierarchy frozen before evaluation selects the scalar onsite, orbital onsite, range-one, range-two, collinear, and general spinor classes exactly where those structures were

planted. The preceding relative residuals are respectively 0.474 for an orbital defect projected into the scalar class, unity for both nonlocal defects projected into insufficient-range classes, 0.278 for a collinear defect projected into the spin-independent class, and 0.184 for a spin-mixing defect projected into the collinear class. Six deliberately incompatible rank, spin, geometry, site, retained-subspace, and energy-reference cases stop without a nominal residual.

For a fixed Gaussian orbital defect, increasing N_s from 12 to 48 reduces the lowest defect-band width from $5.95 \times 10^{-7} E_G$ to the numerical floor. The center binding changes by $4.76 \times 10^{-12} E_G$, the two-state bound count is unchanged, and the lowest-state RMS radius stabilizes near $0.733776a$. The fixed core operator is unchanged across the sequence; its exterior norm, $2.07 \times 10^{-5} E_G$ for the declared radius-$2a$ split, remains separately reported rather than being inferred from the spectral convergence.

On a separate scalar $N_s = 128$ comparison grid, widening fixed-peak and fixed-integrated Gaussian profiles from $0.5a$ to $8a$ reduces the absolute lattice–parabolic binding discrepancy from $5.53 \times 10^{-3} E_G$ to $2.36 \times 10^{-5} E_G$ and from $8.84 \times 10^{-3} E_G$ to $2.65 \times 10^{-6} E_G$, respectively. The corresponding broad-profile fidelities are 0.999977 and 0.9999975. This is a trend on one finite, band-limited comparison grid, not a converged continuum-crossover result.

Two constructed residuals expose the independence of operator and observable metrics. A rank-one excited-sector residual of norm $0.500 E_G$ leaves the lowest binding and state unchanged to roundoff. A smaller $9.66 \times 10^{-3} E_G$ bound–continuum coupling changes the binding by $2.83 \times 10^{-3} E_G$ and lowers the state fidelity to 0.8536. Neither a large nor a small global operator norm is therefore an observable certificate without additional spectral information.

Figure I.2 summarizes these controls. This is a human-accepted calculated synthetic numerical-verification result within the bounded exercise. It establishes only software and numerical verification of the synthetic construction. It is not a silicon impurity calculation, material effective-mass validation, continuum-limit proof, or uncertainty quantification.

Calculated blind-alignment verification

A separate synthetic inverse problem withholds the authored translation, orbital, spin-frame, scalar energy shift, and planted defect from the inference action. The action instead receives the two represented Hamiltonians, an authored anchor cross-covariance, retained-subspace information, an exterior energy anchor, and spin labels. A truncated polar factor supplies the inferred map, and the scalar shift is estimated from the aligned exterior sector before subtraction. Oracle data enter only in post hoc evaluation. The complete input, protocol, result, independent verifier, report, and figures are retained under `calculations/research-monograph/impurity-defect-1d-blind-alignment/`.

In the 32-dimensional spinless and 64-dimensional spinor full-rank controls, the phase-

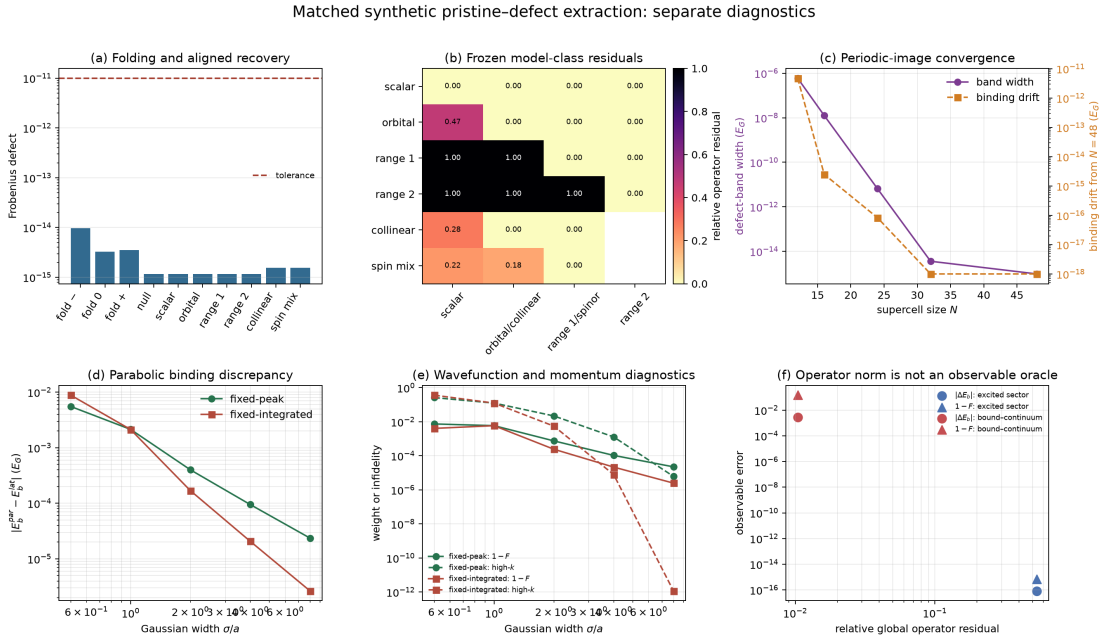


Figure I.2: Matched one-dimensional pristine–defect benchmark. Panel (a) reports primitive-folding and aligned planted-recovery defects against the frozen algebraic tolerance. Panel (b) gives classwise relative operator residuals for the frozen hierarchy. Panel (c) separates defect-band width from binding-center drift over the supercell sequence. Panels (d) and (e) retain lattice–parabolic binding, wavefunction, and high-momentum diagnostics for fixed-peak and fixed-integrated profile families. Panel (f) contrasts global operator residuals with binding and fidelity errors in two constructed spectral sectors.

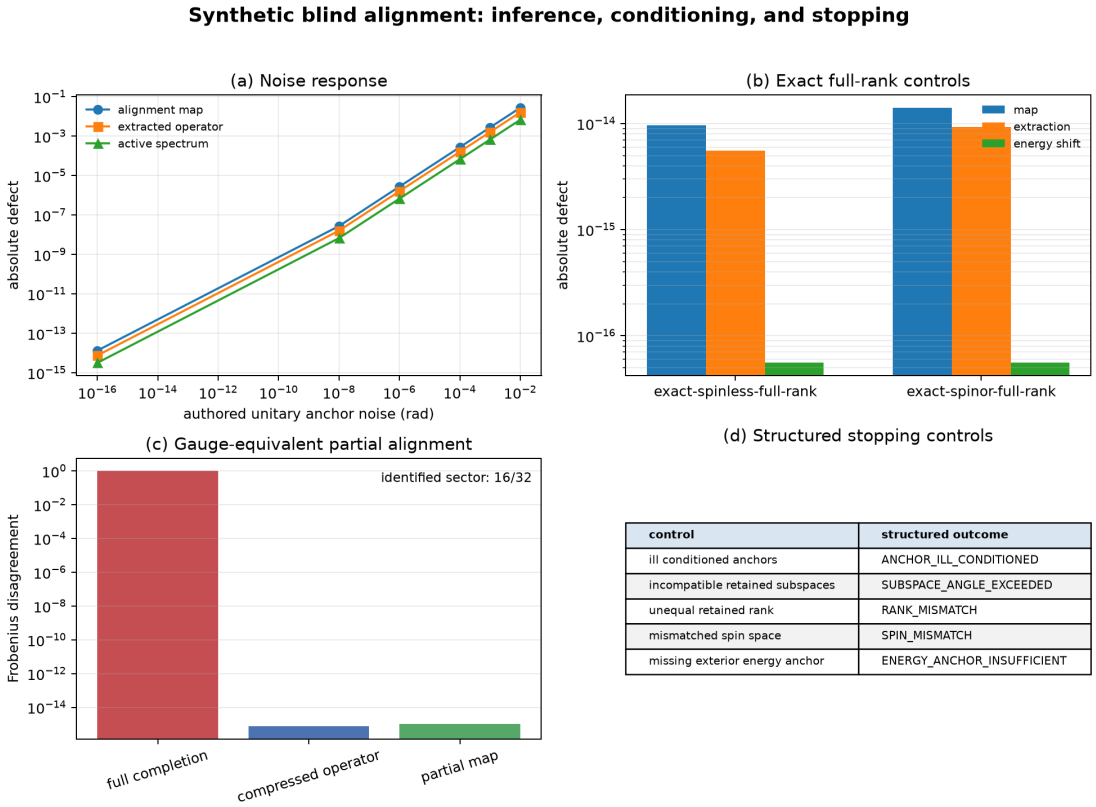
quotiented map defects are 9.68×10^{-15} and 1.41×10^{-14}, while the planted-operator recovery defects are $5.54 \times 10^{-15} E_G$ and $9.32 \times 10^{-15} E_G$. The scalar-shift error is $5.55 \times 10^{-17} E_G$ in both cases. Map, energy-reference, extraction, onsite-model, active-spectrum, and lowest-state errors remain separate.

A 16-of-32-dimensional undercomplete anchor identifies only a compressed sector. Two admissible unitary completions disagree by $9.92 \times 10^{-1} E_G$ as full extracted operators, but their compressed disagreement is $7.73 \times 10^{-16} E_G$ and their partial maps agree to 1.08×10^{-15}. The unobserved complement is therefore retained as a gauge freedom rather than assigned a forced map. A controlled unitary anchor perturbation produces graded downstream errors; at 10^{-2} rad the map, extraction, and active-spectrum defects are respectively 2.78×10^{-2}, $1.62 \times 10^{-2} E_G$, and $6.81 \times 10^{-3} E_G$. This is a deterministic sensitivity scan, not uncertainty quantification.

Ill-conditioned anchors, incompatible retained subspaces, unequal represented ranks, incompatible spin spaces, and a missing exterior energy anchor return structured stops without an extracted operator. An independent implementation reconstructs every retained nominal metric and stopping outcome without importing the runner. Figure I.3 summarizes the result. It is a human-accepted synthetic software and numerical verification result within this bounded exercise, not a material alignment or uniqueness claim.

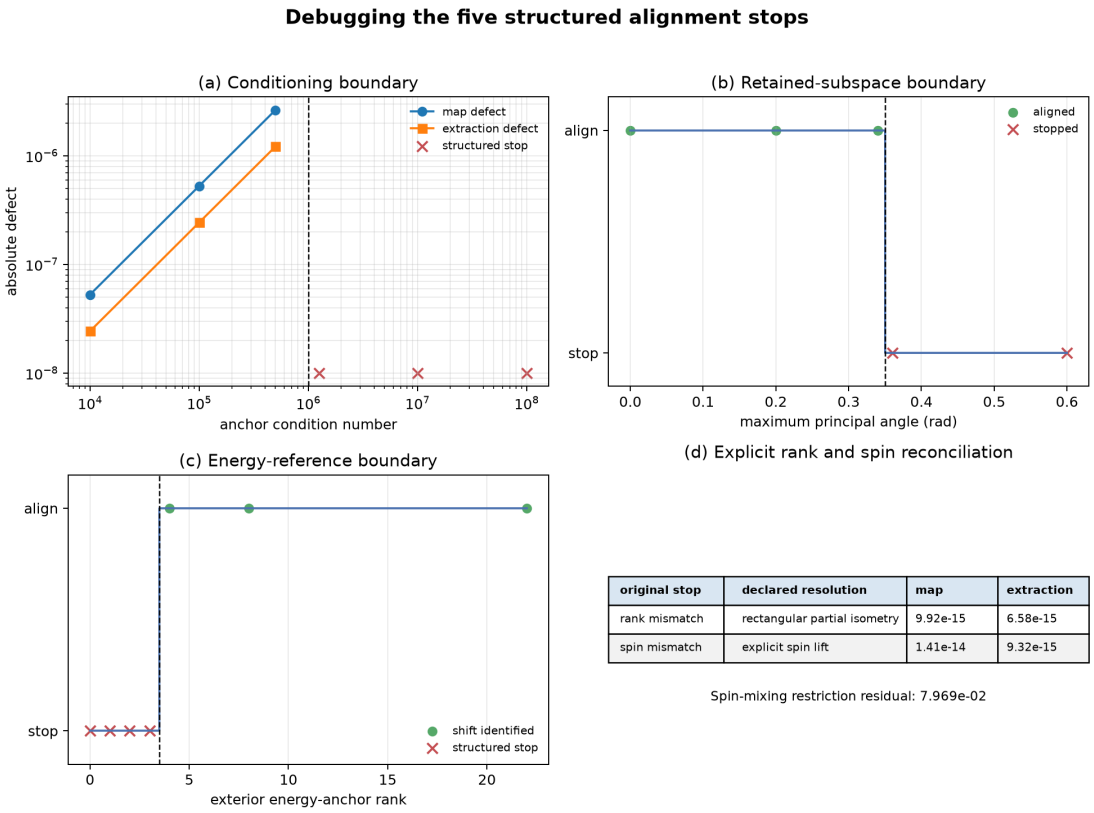
Neighboring diagnostics preserve those five stops while exposing their boundaries. With fixed additive anchor noise 10^{-10}, map and extraction errors grow to 2.64×10^{-6} and $1.22 \times 10^{-6} E_G$ at condition number 5.00×10^5; the next point, 1.25×10^6, stops against the frozen 10^6 limit. A one-direction principal-angle sweep passes at 0.34 rad and stops at 0.36 rad against the 0.35-rad rule. Exterior energy-anchor ranks zero through three stop, whereas rank four recovers the shift to $2.78 \times 10^{-17} E_G$. This boundary probe corrected an implementation defect: energy-anchor rank is now computed from singular values above the numerical-rank tolerance rather than by comparing a rounded floating-point trace with an integer threshold.

Unequal ranks remain invalid for full unitary comparison, but an explicitly declared 32×31 partial isometry recovers its common-sector map and zero operator to 9.92×10^{-15} and $6.58 \times 10^{-15} E_G$. Because the zero operator has a 31-dimensional lowest eigenspace, the individual lowest-state fidelity is left undefined and its eigenspace-projector defect, 7.28×10^{-15}, is reported instead. The spin mismatch similarly remains invalid directly; an explicit spin lift recovers the common spinor representation, while the spin-independent restriction residual $7.97 \times 10^{-2} E_G$ shows that the planted spin mixing cannot be restricted losslessly. Figure I.4 retains these diagnostics. The cutoffs are synthetic protocol settings rather than material tolerances.



control	structured outcome
ill conditioned anchors	ANCHOR_ILL_CONDITIONED
incompatible retained subspaces	SUBSPACE_ANGLE_EXCEEDED
unequal retained rank	RANK_MISMATCH
mismatched spin space	SPIN_MISMATCH
missing exterior energy anchor	ENERGY_ANCHOR_INSUFFICIENT

Figure I.3: Synthetic blind-alignment benchmark. Panel (a) separates alignment, extracted-operator, and active-spectrum responses to controlled unitary anchor perturbations. Panel (b) gives the exact full-rank controls. Panel (c) contrasts full-completion non-identifiability with agreement in the declared compressed sector. Panel (d) lists the five structured stopping controls.



original stop	declared resolution	map	extraction
rank mismatch	rectangular partial isometry	9.92e-15	6.58e-15
spin mismatch	explicit spin lift	1.41e-14	9.32e-15

Figure I.4: Debugging the five structured stops without weakening them. Panels (a)–(c) locate the conditioning, principal-angle, and exterior-energy-anchor boundaries. Panel (d) reports explicitly declared common-space resolutions for unequal rank and unequal spin space; these do not make the original raw comparisons admissible.

Calculated independent-route verification

A second synthetic follow-on compares two implementations after the same coordinate and energy-reference alignment has been declared. Route A assembles and subtracts finite-supercell matrices directly in site space. Route B does not call Route A: it evaluates the primitive 2×2 Bloch matrices, builds the 16-point folding map, and subtracts directly in folded-fiber space. The commutativity question is whether $F^\dagger V^{(A)} F = V^{(B)}$. Exact source identities, input, protocol, result, independent verifier, report, and figure are retained under `calculations/research-monograph/impurity-defect-1d-independent-route/`.

All seven null, local, nonlocal, collinear, and spin-mixing controls commute within the frozen 10^{-11} tolerance. The largest pristine-representation and route-noncommutativity defects are $9.58 \times 10^{-15} E_G$ and $9.61 \times 10^{-15} E_G$; the largest reconstructed eigenvalue discrepancy is $1.11 \times 10^{-15} E_G$. Direct site-space and direct fiber-space planted recovery remain below $1.56 \times 10^{-15} E_G$ and $9.70 \times 10^{-15} E_G$, respectively. An independent verifier reconstructs both routes without importing the runner.

Changing only the fiber-route hopping range from $4a$ to $3a$ produces operator noncommutativity $2.83 \times 10^{-2} E_G$ even though the reconstructed physical spectra still agree to $7.77 \times 10^{-16} E_G$. Thus spectral agreement does not certify the extracted operator when the pristine parent partition differs. A 15-fiber domain, nonuniform quadrature weights, and an unmatched alignment map stop before nominal extraction. All four altered contracts are paired with explicit compatible constructions: a common $|R| \leq 3$ parent, a common 15-cell/15-fiber domain, the dual $G^{-1} = W^{-1/2} F^\dagger$ and induced metric for weighted coordinates, and the relative unitary $R = T_A T_B^\dagger$ between alignment frames. Their route defects are at most $6.82 \times 10^{-15} E_G$. The common-range parent still differs from the nominal parent by $2.83 \times 10^{-2} E_G$; thus the original mixed-parent noncommutativity and the three stops remain valid guards because common contracts must be declared rather than inferred silently. Figure I.5 summarizes the comparison. This is a human-accepted bounded synthetic software and numerical-verification result, not material or continuum validation.

Calculated finite-rank analytical oracle

A third synthetic follow-on supplies a spectral oracle that does not diagonalize the full defect Hamiltonian. For the accepted two-orbital parent and local rank-one defect $V_g = -g|v\rangle\langle v|$, the matrix determinant lemma gives

$$1 - g G_N(E) = 0, \qquad G_N(E) = \frac{1}{N} \sum_j u^\dagger [H(k_j) - EI]^{-1} u. \tag{I.31}$$

The oracle evaluates only the primitive 2×2 Bloch-fiber inverses and resolves the unique root below the finite-host lower edge. This use of a host resolvent for a Wannier-basis

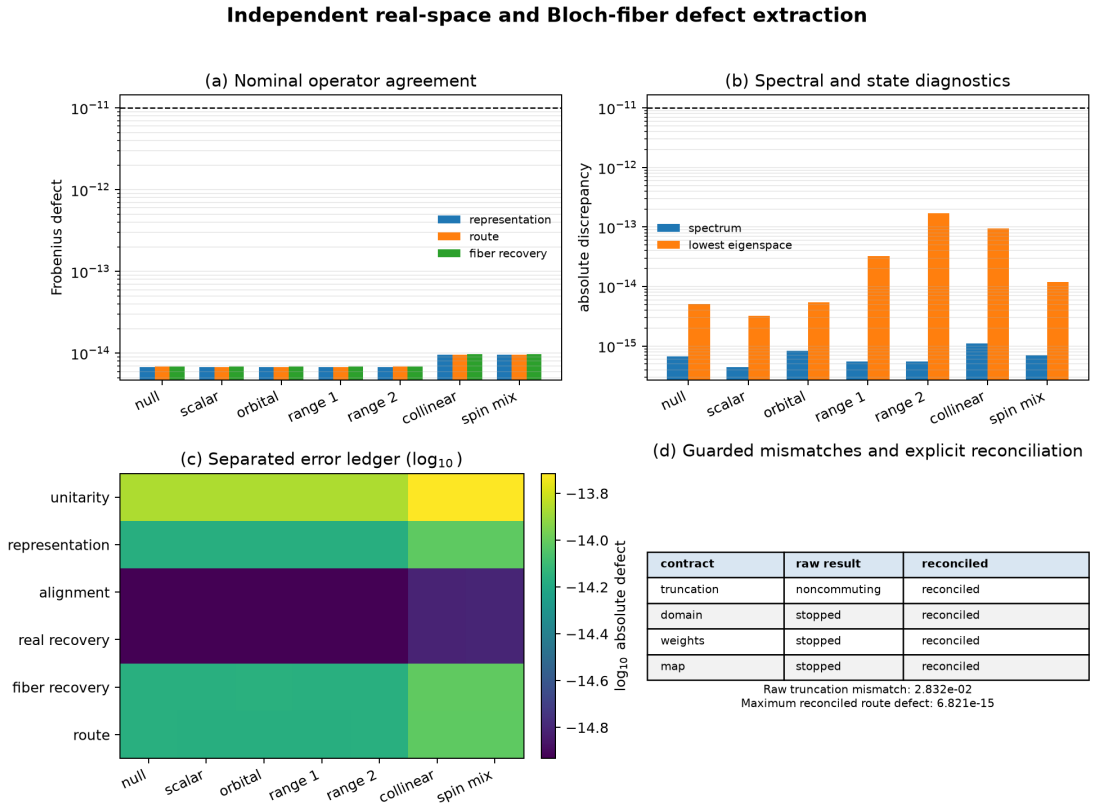


Figure I.5: Independent-route synthetic benchmark. Panel (a) compares representation, route, and planted-recovery errors. Panel (b) retains spectral and lowest-eigenspace discrepancies separately. Panel (c) shows the complete separated error ledger. Panel (d) distinguishes fail-closed mismatch outcomes from explicit common-parent, common-domain, dual-map, and relative-unitary reconciliations.

impurity follows the classical Koster–Slater construction [62]; the one-dimensional weak-binding and periodic-defect literature motivates the threshold and sign controls but does not supply the numerical result reported here [110, 87]. An independent route assembles and diagonalizes the full site-space defect operator. Inputs, source identities, root rules, retained results, an independent verifier, and the mini-paper are under `calculations/research-monograph/impurity-defect-1d-analytical-oracle/`.

For $N = 16, 24, 32, 48, 64$ and attractive magnitudes $g/E_G = 0.02, 0.08, 0.20, 0.50$, all 20 controls have exactly one state below the finite-host edge. The maximum oracle/numerical energy discrepancy is $6.11 \times 10^{-16} E_G$, the maximum secular residual is 8.16×10^{-13}, and the maximum rank-one projector defect is 6.14×10^{-13}. A zero-coupling threshold and a repulsive control have no state below the lower edge. A spin-degenerate rank-two lift agrees by projector to 1.55×10^{-14}, while a proposed rank-one comparison against that rank-two sector stops explicitly.

Figure I.6 retains the energy, finite-size, projector, threshold, no-state, and rank diagnostics separately. The supercell sequence is not an infinite-system or continuum extrapolation. This is a human-accepted bounded synthetic software and numerical-verification result, not material validation.

Calculated continuum refinement

The earlier smoothness scan changed Gaussian width on one band-limited grid and therefore did not identify a continuum limit. The fourth synthetic follow-on separates continuum-mesh, continuum-domain, lattice-supercell, represented lattice-scale, and profile-width axes. For lattice spacing a, the fixed-profile scaled parent is

$$E_a(q) = E_0 + \frac{E_{\mathrm{lat}}(aq) - E_0}{a^2}, \tag{I.32}$$

which preserves the band-edge curvature while changing the represented lattice scale. Inputs, criteria, source identities, retained results, and an independent site-space verifier are under `calculations/research-monograph/impurity-defect-1d-continuum-refinement/`.

The continuum-mesh, continuum-domain, and lattice-supercell support axes pass their frozen criteria. At fixed integrated profile with $\sigma = 2a_{\mathrm{ref}}$, the unrefined lattice fails five of six comparison criteria. From $a = 0.5a_{\mathrm{ref}}$ through the finest tested spacing $a = 0.125a_{\mathrm{ref}}$, all binding, bound-count, projector, compressed operator, cross-coupling, and Brillouin-edge criteria pass persistently. At the finest spacing the relative binding error is 4.67×10^{-5} and the projector defect is 2.69×10^{-4}.

Profile broadening at the original lattice spacing gives a different result. For the fixed-integrated family at $\sigma = 12a_{\mathrm{ref}}$, the relative binding error is 8.63×10^{-5} and the projector defect is 1.21×10^{-3}, but the compressed parent-dispersion residual remains $3.78 \times$

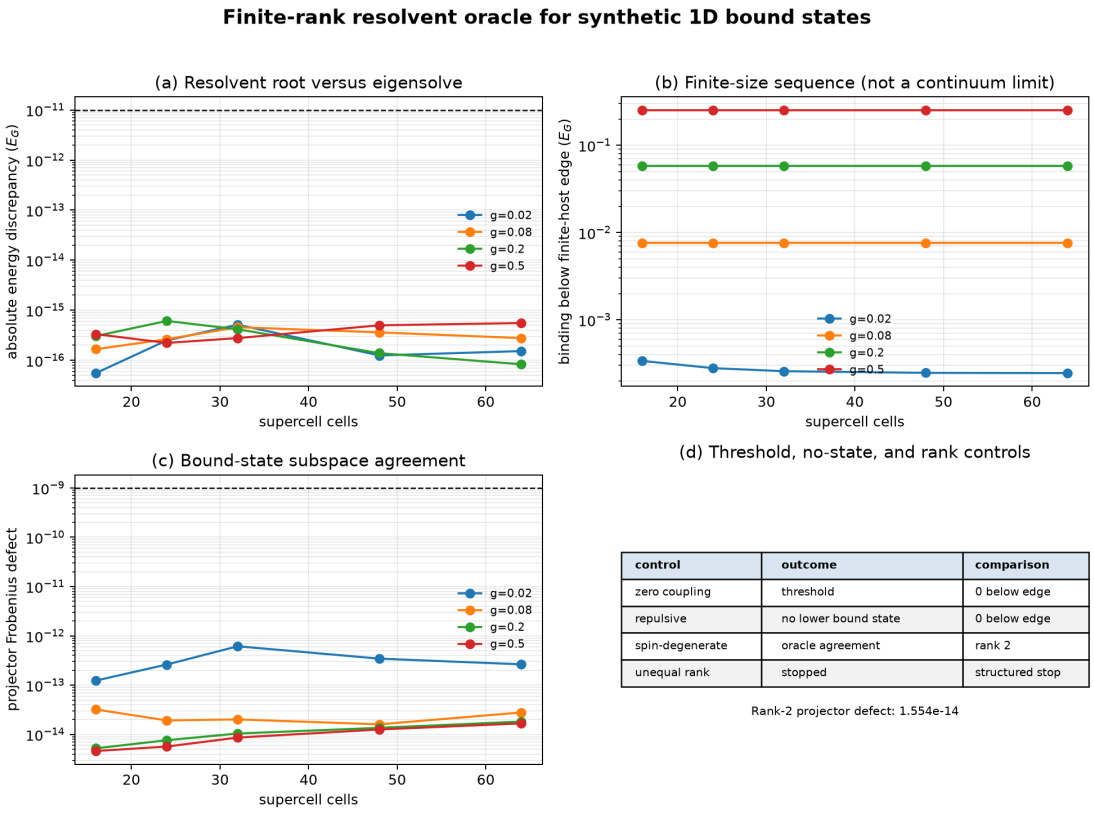


Figure I.6: Finite-rank resolvent oracle. Panel (a) compares secular roots with independent site-space eigenvalues. Panel (b) reports finite-size binding without a continuum claim. Panel (c) compares equal-rank bound-state projectors. Panel (d) retains threshold, repulsive no-lower-bound-state, spin-degenerate rank-two, and unequal-rank stopping controls.

$10^{-3}E_G$, above the frozen $10^{-3}E_G$ rule. The fixed-peak family additionally retains unequal below-edge counts, 41 versus 37, at that width. Neither family therefore has a profile-defined crossover over the tested width domain.

Figure I.7 distinguishes the successful bounded lattice-scale sequence from the negative width-crossover result. This is a human-accepted bounded synthetic numerical-verification result. It is not an asymptotic theorem, material validation, transferability, or uncertainty quantification.

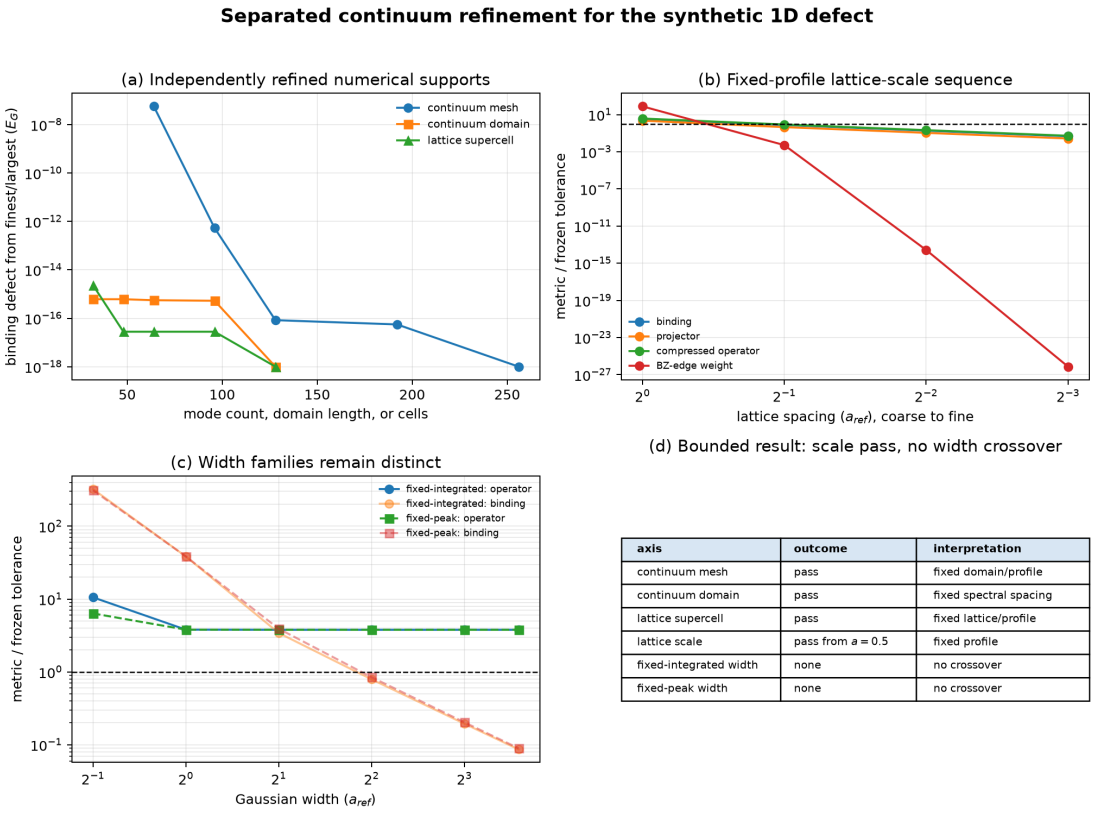
Literature position of the one-dimensional chain

The finite-rank and continuum ingredients are established separately in the literature. Host-resolvent impurity reasoning is classical in the Koster–Slater construction [62]; the retained review uses that source only at abstract level and does not attribute the benchmark's exact finite-matrix formula to unseen text. Weak-coupling existence results for continuum Schrödinger operators and periodic gaps do not supply the finite-supercell oracle reported here [110, 87]. Likewise, band-edge homogenization varies a weak, slowly varying defect, whereas discrete-to-continuum convergence varies an identified lattice spacing and finite-volume analysis varies the periodic domain [48, 26, 81, 61]. These distinct hypotheses motivate the separated axes; none proves the bounded crossover observed in the synthetic sequence.

Localized-basis defect operators also have substantial precedent. Berlijn, Volja, and Ku require normal-cell and impurity-supercell Hamiltonians to share a Wannier basis before impurity contributions are embedded in larger disordered configurations [5]. Lu and co-workers form and Wannier-interpolate electron–defect matrix elements from an aligned Kohn–Sham potential difference [69], while Corsetti and Mostofi use localized Wannier onsite elements as one route to the scalar potential alignment needed in silicon-vacancy supercells and analyze system-size effects separately [21]. These works establish localized defect differences, interpolation, energy alignment, and image-error precedents; they do not establish the hidden-map or model-class diagnostics of this exercise.

The nearest alignment precedent is the stitching analysis of Lihm and Park, which treats orbital pairing, spin orientation, phase, potential reference, and iterative Hamiltonian matching between independently generated MLWF models [66]. A 2026 preprint by Shi and co-workers provides close contemporary evidence: its clean and defect Wannier models use Hungarian assignment, a dense unitary Procrustes rotation, an explicit nonorthogonal cross-metric, and residual boundary checks before defect-patch embedding in a superconducting BdG calculation [109]. The preprint has a different physical objective and does not supply a uniqueness theorem, undercomplete-sector analysis, or the structured stops reported here. Thus the blind-alignment result is not positioned as the invention of Wannier gauge matching.

A second 2026 preprint uses an onsite-only Wannier substitution as a rapid defect-



axis	outcome	interpretation
continuum mesh	pass	fixed domain/profile
continuum domain	pass	fixed spectral spacing
lattice supercell	pass	fixed lattice/profile
lattice scale	pass from $a = 0.5$	fixed profile
fixed-integrated width	none	no crossover
fixed-peak width	none	no crossover

Figure I.7: Separated continuum refinement. Panel (a) retains independent mesh, domain, and lattice-supercell controls. Panel (b) shows the fixed-profile lattice-scale metrics relative to their frozen tolerances. Panel (c) shows that profile broadening improves spectral errors but cannot remove the fixed parent-dispersion residual. Panel (d) records the bounded scale pass and the absence of a profile-defined crossover.

screening class and reports limitations from omitted hopping, charge redistribution, and spin effects [52]. That objective is closer to one member of the benchmark hierarchy than to extraction of the full aligned operator. Classical and modern silicon-donor work further shows that multivalley and central-cell structure must be retained before a material effective-mass claim is made [58, 36].

The defensible contribution is therefore methodological integration: one reproducible, fail-closed synthetic chain retains compatibility, hidden-map identifiability, operator-model residuals, route commutativity, a finite-rank oracle, finite-volume diagnostics, and separated continuum axes. It is not a new impurity-resolvent, alignment, localization, or continuum theorem and is not material validation. The bounded search protocol, source-access register, claim matrix, and novelty disposition are retained under `docs/research/literature-reviews/impurity-defect-1d/`. The search is exhaustive enough only relative to its declared concepts, interfaces, dates, and stopping rule; it does not establish priority or comprehensive coverage.

One-dimensional follow-on sequence

The accepted known-map exercise remains the fixed ground-truth baseline. The five follow-ons are:

1. infer hidden translation, orbital, spin-frame, and scalar-reference maps from explicitly declared overlap or anchor data, with gauge equivalence, degeneracies, principal-angle stops, and noise treated as part of the alignment contract;

2. compare real-space extraction with an independently implemented Bloch-fiber or folding route and test their commutativity after alignment;

3. add an analytically solvable finite-rank defect whose bound-state energy and state follow from an independent resolvent equation;

4. refine continuum spacing, domain length, supercell size, and defect width independently before proposing a continuum crossover; and

5. position the definitions and observed failure modes against verified primary literature before making a novelty claim.

The first four items are the human-accepted blind-alignment, independent-route, analytical-oracle, and continuum-refinement results reported above. The fifth is the bounded literature synthesis just reported; it adds no numerical or material evidence. None is an active material calculation. Keeping the alignment inverse problem separate preserves attribution: failure to infer the map is not confused with failure of subtraction after a correct map is supplied.

I.10 Benchmark conclusion and two-dimensional handoff

The accepted spin-space and one-dimensional calculations establish the bounded synthetic prerequisites for extending aligned impurity extraction to two spatial dimensions. They verify spin embeddings, known-map and blind alignment, independent extraction routes, finite-rank spectral oracles, finite-volume diagnostics, model-class separation, and independently refined continuum comparisons under their declared contracts. They do not establish a two-dimensional defect result or transferability to a material calculation.

The proposed two-dimensional defect controls and subsequent material-transfer sequence are separated into Appendix J. That appendix consumes the accepted periodic parent from Appendix H and the spin and extraction contracts established here; it does not require these completed benchmarks to be repeated.

CHAPTER J

Two-Dimensional Defect Extraction and Material-Transfer Program

Status active controlled synthetic defect-extraction program and proposed material-transfer program. The periodic two-dimensional parent, spin-space contracts, and one-dimensional defect-extraction chain are completed prerequisites. The version-1 design is human-accepted, and the bounded synthetic Stage A null-and-folding calculation is human-accepted and independently reproduced. The bounded synthetic Stage B scalar-onsite multi-route calculation is complete, independently reproduced, and human-accepted at its exact bounded evidence boundary. The accepted-parent Stage C software contract is implemented, independently verified only with authored synthetic fixtures, and human-accepted at that exact execution-free boundary; it has not read an accepted parent. No accepted-parent directional or nonlocal two-dimensional defect result, material impurity operator, continuum crossover, scientific validation, or uncertainty- quantification result is claimed.

J.1 Purpose and prerequisite evidence

This appendix defines the next controlled extension of the impurity-extraction program. Appendix H supplies the accepted separable and coupled periodic parents. Appendix I supplies the accepted spin-space, alignment, extraction-route, finite-size, model-class, analytical-oracle, and continuum-refinement contracts. The controlled synthetic task is active. The accepted version-1 design and Stage A evidence are retained under `calculations/research-monograph/impurity-defect-2d/`; they fix staged synthetic inputs, represented-space maps, metrics, adverse controls, tolerances, stopping rules, independent-verification routes, authority, and resource bounds. The exact Stage A execution record binds the design, runner, scalar parent, canonical repository, output, resource envelope, and durable human decision; its separate acceptance record applies only to this bounded synthetic evidence. Stage B initially exposed an incompatibility among the chosen generic twist, componentwise $[0, 1)$ twist reduction, and bare-permutation full-Hamiltonian covariance. That single-route convention is retained as negative design

evidence. The adopted execution-free replacement constructs independent centred uniform-link and reduced seam matrices, relates them through one explicit gauge bridge, and executes A-then-B and B-then-A in fresh processes to expose hidden schedule state. Its authored toy retains all 2,304 blind candidates; clean schedules agree, an intentionally shared-cache mutation is discriminating, and an independent verifier reconstructs the toy result. A separately authorized calculation then applied the frozen two-route protocol to the accepted isotropic scalar parent. It retained 72 known-map route evaluations, 36 bridge records, and 2,304 blind candidate rows; runner criteria and independent Fourier/seam reconstruction pass. This exact calculated synthetic numerical-verification evidence is human-accepted, but it is not material validation. Accepted-parent calculations for Stages C–E remain unexecuted. Stage C has a human-accepted execution-free software contract and authored-fixture evidence, but it has not read an accepted parent. The program adds only genuinely two-dimensional defect structure; material transfer remains proposed and requires separate authorization and retained calculated evidence.

J.2 Two-dimensional specialization of the extraction contract

Chapter 22 owns the general definitions of a host, doped system, defect, perturbation operator, scalar onsite potential, finite-extent operator perturbation, compatibility, alignment, extraction, and locality. In particular, Eqs. (22.24) and (22.25) distinguish scalar onsite changes from bond or hopping changes, while Eqs. (22.28)– (22.33) define the minimum-image locality partition and its authored acceptance criterion. This appendix does not redefine those terms. It specializes them to the controlled two-dimensional campaign and retains the campaign's detailed controls and evidence history.

For this specialization, N_x and N_y are the positive primitive-cell counts along the two lattice directions, and (N_x, N_y) is the finite-periodic shape. The symbol $\boldsymbol{\phi} = (\phi_x, \phi_y)$ is the unreduced boundary-twist lift in turns. The scalar site basis has one basis state per ordered periodic site, so its represented dimension is $N_x N_y$. The campaign's bold symbols $\mathbf{H}_0$, $\Delta\mathbf{H}$, and $\mathbf{H}_{\text{def}}$ denote finite matrices for the periodic parent, the authored or extracted perturbation, and their compatible sum. They are campaign representations of the general operators in Chapter 22; they are not infinite-system operators or material potentials.

The controlled representation route constructs the parent and perturbation independently, checks shape, twist fiber, ordered basis, energy unit, and energy reference, and only then composes them. The extraction route receives already compatible and aligned parent and defect matrices and performs the signed subtraction defined by Eq. (22.19). It performs no implicit site mapping, gauge conversion, basis rotation, geometry transformation, or energy-zero inference. Locality is assessed with the minimum-image Chebyshev

partition of Eq. (22.27); a passing tolerance establishes only satisfaction of the declared finite represented criterion, not exact compact support or continuum localization.

The campaign adds independent representation routes, twist-gauge bridges, schedule-order controls, point-group covariance, blind site-map ambiguity, model-class recovery, and adverse inputs to the general extraction contract. The present synthetic software evidence verifies representation, compatible composition, exact planted-perturbation recovery, and sensitivity of the locality partition to core-crossing bonds. It does not derive a perturbation from DFT or Wannier90, establish supercell or twist convergence, validate a silicon impurity model, demonstrate transferability, or quantify uncertainty. Historical execution labels remain only where needed to identify immutable provenance records; the methodology itself is organized by representation, compatibility, extraction, locality, covariance, and model comparison.

J.3 Completed Stage A null and folding control

The human-accepted Stage A calculation used the separable scalar parent and the frozen Cartesian product of 6×6 and 8×8 cells with four boundary twists. All eight cases passed the declared finite-matrix criteria. The maximum folding-unitarity defect was 1.280×10^{-15}; the maximum folded off-block and primitive-block defects were both 2.584×10^{-16}; the maximum eigenvalue defect was $2.221 \times 10^{-16} E_G$; and the maximum recovered-null Frobenius defect was $2.179 \times 10^{-16} E_G$. These are calculated synthetic floating-point residuals under the frozen representation, not physical uncertainties.

An independently implemented Kronecker seam/Fourier route reconstructed every retained metric and reported a maximum retained-versus-reconstructed difference of 3.473×10^{-16}. The analytical seam controls separately reproduced 1, $\exp(+2\pi i\phi)$, and $\exp(-2\pi i\phi)$ for noncrossing, positive-crossing, and negative-crossing hops at $\phi = 0.37$ turns. Incorrect geometry, incorrect boundary phase, lost site correspondence, and unknown energy reference each produced the frozen structured stop before residual formation. This evidence verifies the Stage A folding, seam, null, and compatibility software contracts; it did not itself activate Stage B.

J.4 Completed Stage B scalar-onsite and symmetry control

The Stage B protocol fixes the integer D_4 matrices, the eight-site orbit of an off-axis scalar onsite plant, known-map extraction, transformed-twist full-Hamiltonian covariance, and

energy-reference adverse controls. Its adopted execution-free construction reconciles the finite-matrix twist gauges through independent centred uniform-link and reduced seam routes plus one explicit bridge. Its adversarial design analysis also records an identifiability limit: an off-diagonal host-only blind objective cannot determine an absolute translation in a homogeneous periodic parent. The blind route must therefore retain the complete symmetry-related ambiguity set and stop before extraction rather than select the planted site. The calculation reproduces that prediction: the Gamma case retains 512 tied maps per route and schedule, while the generic twist retains 64 tied translations. Every case stops with `DEFECT_2D.SITE_MAP_UNRESOLVED`, with null selected map and null extracted operator.

Across both routes and schedules, the largest known-map recovery maximum-entry defect is $2.238 \times 10^{-16} E_G$, the largest recovery Frobenius defect is $3.079 \times 10^{-16} E_G$, and the largest attacked bridge Frobenius residual is $3.537 \times 10^{-16} E_G$. Known-case and bridge schedule differences are exactly zero in the retained comparisons. The independent verifier reports a maximum reconstructed difference of 7.845×10^{-13}, below the frozen 10^{-10} criterion. Omitting the known $0.137 E_G$ shift produces a $1.096 E_G$ Frobenius residual, while the fixed-twist adverse controls produce maximum-entry residuals $9.979 \times 10^{-3} E_G$ and $4.065 \times 10^{-2} E_G$ in Routes A and B. These adverse outcomes remain separate from the passing correctly transformed cases.

This result verifies the bounded finite synthetic scalar-onsite, D_4, gauge- bridge, schedule, ambiguity, and adverse-control contracts only. It does not support a rerun, material transfer, or a directional accepted-parent result.

A subsequent execution-free Stage C software package uses authored bond coefficients without reading the accepted periodic parent. It distinguishes a directional nearest-neighbour plant from an isotropic class and a diagonal nonlocal plant from all shorter-range classes. Two independently constructed gauge routes, two fresh spawned schedule processes, all eight oriented D_4 operations at Gamma, five ordered model classes, locality shells, and four adverse controls are retained by the toy wire contract. Maintained tests and an independent implementation reproduce the toy behavior with maximum difference $1.388 \times 10^{-17} E_G$. This exact execution-free package is human-accepted at its bounded software-verification boundary. It is not an accepted-parent Stage C calculation or numerical evidence for a material.

The human-adopted accepted-parent Stage C design binds the accepted isotropic and anisotropic scalar parents. It separates isotropic D_4 covariance, anisotropic D_2 covariance, and an axis swap to the distinct $(\lambda_x, \lambda_y) = (0.7, 0.3)$ parent. A separately authorized execution-free implementation uses only an authored 61-hopping isotropic fixture and an authored 15×15 anisotropic energy grid. Across two fresh processes it retains the frozen 208 route evaluations, 104 gauge bridges, 1,040 ordered model fits, and 104 cross-schedule comparisons. The independent inverse-Fourier and QR verifier reconstructs

all records with maximum scalar difference $1.922 \times 10^{-15} E_G$. Invalid anisotropic D_4 invariance and omission of the separately swapped parent each produce $6.0 \times 10^{-3} E_G$, above their predeclared $10^{-3} E_G$ adverse floors. A synthetic run completed in 1.56 seconds with 99,041,280 bytes maximum resident set size and a 2,205,921-byte result, within the adopted envelope. Parent hopping reconstruction and truncation, represented-gauge error, alignment recovery, and defect-model residuals remain separate. This implementation has no accepted-parent input adapter and has read no accepted parent. It is human-accepted only at its execution-free software- contract boundary and supplies no calculated Stage C evidence or Stage D authority.

J.5 Matched pristine and defected-system supercells

Use the separable and coupled two-dimensional parents developed in Appendix H. Let the supercell contain $N_x \times N_y$ primitive cells with lattice vectors

$$\mathbf{L}_1 = N_x \mathbf{a}_1, \qquad \mathbf{L}_2 = N_y \mathbf{a}_2. \tag{J.1}$$

where

$\mathbf{L}_1$ **and** $\mathbf{L}_2$ the two supercell lattice vectors.
N_x **and** N_y the positive primitive-cell counts along the first and second lattice directions.
$\mathbf{a}_1$ **and** $\mathbf{a}_2$ the corresponding primitive lattice vectors.

The matched pristine and defected-system fibers are

$$\mathbf{H}_{b,\mathbf{L}}(\mathbf{K}), \qquad \mathbf{H}_{d,\mathbf{L}}(\mathbf{K}), \tag{J.2}$$

where

$\mathbf{H}_{b,\mathbf{L}}(\mathbf{K})$ the pristine finite matrix in the supercell fiber at momentum $\mathbf{K}$.
$\mathbf{H}_{d,\mathbf{L}}(\mathbf{K})$ the matched defected-system finite matrix in the same fiber.
b **and** d the pristine and defected-system branches.
$\mathbf{L}$ the two-dimensional supercell geometry.
$\mathbf{K}$ a crystal momentum in the reduced supercell Brillouin zone.

with $\mathbf{K}$ in the reduced supercell Brillouin zone.

The pristine control must reproduce two-dimensional folding from the primitive mesh into each supercell fiber. In the separable limit, the supercell construction should also reproduce products and sums of the verified one-dimensional results. Degenerate folded

clusters must be compared through their complete projectors rather than through arbitrary eigenvector labels.

The first planted perturbation should be a scalar onsite multiplication term at one declared supercell site. Accepted-parent calculations should add:

1. direction-dependent hopping changes;
2. a finite-range nonlocal defect;
3. anisotropic spatial profiles; and
4. a composite-orbital defect in the coupled two-dimensional parent.

Independent point-group-related placements should produce aligned operators related by the corresponding lattice symmetry. A failure of this covariance should be attributed among site mapping, supercell origin, gauge, degeneracy handling, and boundary phases before the defect model is interpreted.

The finite-size study must vary both area and shape. Square sequences probe increasing image separation under preserved symmetry, while rectangular sequences test anisotropic image interactions and alignment conventions. One apparently converged square supercell is not sufficient evidence for an isolated two-dimensional defect.

J.6 Energy-reference, gauge, ordering, and geometry stress tests

For a known aligned defected system, construct a raw candidate representation

$$\widetilde{\mathbf{H}}_d = \mathbf{G}\left(\mathbf{H}_b + \mathbf{V}_{\text{known}}\right)\mathbf{G}^{\dagger} + c\mathbf{I}, \tag{J.3}$$

where

$\widetilde{\mathbf{H}}_d$ the deliberately scrambled raw defected-system matrix.

$\mathbf{H}_b$ the known aligned pristine matrix.

$\mathbf{V}_{\text{known}}$ the independently authored planted perturbation matrix.

G and $\mathbf{G}^{\dagger}$ a known admissible coordinate transformation and its adjoint.

c the known scalar energy-reference shift.

I the identity matrix on the represented space.

where **G** is a known admissible coordinate transformation and c is a known scalar shift. The transformation sequence should include:

1. orbital phases;
2. orbital and site permutations;
3. unitary rotations within degenerate subspaces;
4. primitive or supercell translations;
5. admissible point-group transformations;
6. changes of localized gauge; and
7. a scalar energy-reference offset.

After applying the inverse declared alignment and energy-reference procedure, the extracted perturbation should equal $\mathbf{V}_{\text{known}}$ within the stated representation error.

The tests should also contain deliberately incompatible cases: unequal retained rank, mismatched spin space, incorrect supercell shape, lost site correspondence, or large principal angles between retained subspaces. These cases should stop the comparison. They should not be forced into a unitary fit merely because two final matrices can be made equal in dimension.

The energy-reference test is particularly important. Without correction, $c\mathbf{I}$ appears as a spatially extended false impurity term. The alignment procedure must remove only the declared reference offset and must not subtract a genuine localized onsite component.

J.7 Supercell-size and shape convergence

Let $\Delta\mathbf{H}_{\mathbf{L}}$ and $\Delta\mathbf{H}_{\mathbf{L}'}$ denote aligned impurity matrices for distinct two-dimensional supercell geometries $\mathbf{L}$ and $\mathbf{L}'$. Their matrix dimensions generally differ, so direct subtraction is undefined. The campaign must supply the explicit common-space maps $J_{\mathbf{L}\to c}$ and $J_{\mathbf{L}'\to c}$ and evaluate the core stability residual in Eq. (22.35). The maps must identify the same physical sites, orbitals, spin components, basis ordering, and localized gauge in the common represented space $\mathcal{S}_c$.

For each supercell separately, the exterior and cross-coupling diagnostics are those in Eq. (22.31). The radius r and core projector P_r must select the same physical neighborhood after both common-space maps. A global Frobenius norm can scale with represented volume or the number of repeated image contributions; it must not replace the mapped core, exterior, and cross-coupling diagnostics.

At finite supercell size, defect-derived energies must be tracked as functions of supercell momentum. Their bandwidth and $\mathbf{K}$ dependence diagnose image coupling and should not be collapsed into a single nominal bound-state energy. Wavefunction localization and

symmetry-resolved blocks should be tracked independently over the same size and shape sequence.

J.8 Recovery of known model classes

Let the planted impurity perturbation belong exactly to a declared class $\mathcal{M}^{(m_0)}$, where m_0 identifies that class. Fit the ordered hierarchy

$$\mathcal{M}^{(0)}, \mathcal{M}^{(1)}, \dots \tag{J.4}$$

where

$\mathcal{M}^{(m)}$ the model class at hierarchy level m.

m a nonnegative ordered model-complexity index.

$\dots$ continuation through every predeclared class in the finite hierarchy.

to the extracted operator. The experiment should verify that:

1. a scalar onsite class recovers a scalar onsite perturbation;
2. an orbital-dependent class is required for orbital splitting;
3. an onsite class fails for a planted hopping correction;
4. a local class leaves a nonzero residual for a planted nonlocal operator;
5. a spin-independent class fails for spin splitting or mixing; and
6. the first class containing the planted structure can recover it within the independently established extraction error.

The failure of a simpler class is a desired outcome when the planted defect lies outside it. This test determines whether the minimum operator separation acts as an expressiveness diagnostic rather than merely as an optimization score.

The model hierarchy, parameter bounds, weighting, and stopping rule must be frozen before the outcomes are inspected. An unnecessarily richer class may fit the planted operator but does not demonstrate that the selection rule can identify the least adequate class.

J.9 Smoothness-to-continuum crossover

Introduce a scalar defect-potential profile with tunable width σ. One fixed-peak family is

$$V_\sigma^{(\mathrm{peak})}(\mathbf{r}) = V_c \exp\left(-\frac{\|\mathbf{r}\|^2}{2\sigma^2}\right). \tag{J.5}$$

where

$V_\sigma^{(\mathrm{peak})}(\mathbf{r})$ the fixed-peak defect profile evaluated at position $\mathbf{r}$.
$\mathbf{r}$ a position relative to the declared defect center.
V_c the fixed peak energy amplitude.
σ the positive profile width, with units of length.
$\|\mathbf{r}\|$ the declared Euclidean radial distance.
exp the exponential function.

Its integrated strength changes with σ. A separate fixed-integrated family in spatial dimension D is

$$V_\sigma^{(\mathrm{int})}(\mathbf{r}) = \frac{g}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|\mathbf{r}\|^2}{2\sigma^2}\right), \tag{J.6}$$

where

$V_\sigma^{(\mathrm{int})}(\mathbf{r})$ the fixed-integrated defect profile evaluated at position $\mathbf{r}$.
g the fixed integrated profile strength, with units of energy times length to the power D.
D the positive integer spatial dimension; $D = 2$ in this campaign.
σ the positive profile width, with units of length.
$\mathbf{r}$ **and** $\|\mathbf{r}\|$ the relative position and its declared Euclidean norm.
π **and** exp the circle constant and exponential function.

where g is held fixed. The two protocols answer different questions and must not be mixed in one convergence sequence. Each profile is periodized only through a supercell large enough for image effects to be measured separately.

The ratio

$$\rho = \frac{\sigma}{a} \tag{J.7}$$

where

ρ the dimensionless defect-smoothness ratio.
σ the positive defect-profile width.
a the positive reference lattice spacing measured in the same length unit as σ.

is one control, not a complete continuum criterion. The study must also record the defect strength relative to the relevant bandwidth or gap, the bound-state or envelope length relative to a and the supercell, the retained-band and valley separations, and the lattice and continuum discretizations. Frozen sequences should range from atomically sharp profiles $\rho = O(1)$ to slowly varying profiles $\rho \gg 1$ under both normalization protocols.

For each ρ, compare:

1. the exact represented lattice perturbation;
2. a local lattice model;
3. a finite-range nonlocal lattice model;
4. a single-band effective-mass model where admissible; and
5. multiband or multivalley continuum classes when the retained parent requires them.

The study should measure operator residuals, band or valley mixing, bound-state energies, subspace fidelity, exterior residuals, and cross couplings. Expected improvement with increasing ρ is a hypothesis to test, not a result to assert in advance.

A proposed crossover radius or smoothness threshold is meaningful only when both exterior and cross-coupling criteria pass and remain stable under lattice, continuum-discretization, and supercell refinement. If no tested scale passes, the correct outcome is no established continuum crossover over the tested domain.

J.10 Operator error versus bound-state error

For a donor-like state below a declared conduction threshold, one possible binding convention is

$$E_b = E_{\text{edge}} - E_{\text{defect}} > 0. \tag{J.8}$$

where

E_b the positive donor-like binding energy under this convention.
E_{edge} the declared aligned conduction threshold.
E_{defect} the aligned donor-like defect-state energy below that threshold.
> 0 the sign requirement defining a bound donor-like state in this convention.

An acceptor-like branch requires its own signed convention relative to the aligned valence threshold. The parent and model must use the same declared band-edge or continuum reference. At finite supercell size, both the defect level and the relevant edge may depend

on supercell momentum, so the comparison must retain that dependence or state the rule used to extract an isolated-limit quantity.

For every planted or fitted model, retain separately

$$\varepsilon_H = \left\| \Delta\mathbf{H}_{\text{parent}} - \Delta\mathbf{H}_{\text{model}} \right\|_{\text{F}}, \tag{J.9}$$

$$\Delta E_b = E_{b,\text{model}} - E_{b,\text{parent}}, \tag{J.10}$$

$$F = \left| \langle \psi_{b,\text{parent}} \mid \psi_{b,\text{model}} \rangle \right|^2 \tag{J.11}$$

where

ε_H the Frobenius operator error between parent and model perturbations.

$\Delta\mathbf{H}_{\text{parent}}$ **and** $\Delta\mathbf{H}_{\text{model}}$ the aligned parent and reduced-model impurity matrices.

$\|\cdot\|_{\text{F}}$ the Frobenius matrix norm.

ΔE_b the signed model-minus-parent binding-energy error.

$E_{b,\text{model}}$ **and** $E_{b,\text{parent}}$ the model and parent binding energies under one convention.

F the squared fidelity of two isolated normalized bound states.

$\psi_{b,\text{parent}}$ **and** $\psi_{b,\text{model}}$ the aligned normalized parent and model bound states.

$\langle \cdot \mid \cdot \rangle$ **and** $|\cdot|$ the state-space inner product and complex modulus.

when individual normalized bound states are isolated, aligned, and present in both models. Before matching states, report the number of in-gap or bound states under the declared threshold convention. If the parent and model have unequal counts, the mismatch is itself a failed diagnostic and no arbitrary one-to-one fidelity should be forced. Degenerate or nearly degenerate clusters of equal rank require a projector or principal-angle fidelity instead of an arbitrary state overlap; unequal-rank clusters must retain both the rank mismatch and an appropriately defined subspace discrepancy.

The one-dimensional synthetic exercise in Appendix I realizes both contrasting cases: accurate binding with a large excited-sector operator residual, and a small global residual with a significant bound-state change. Those observations verify the metric separation for that constructed finite model only; corresponding two-dimensional and material behavior remains untested.

J.11 Null, covariance, and stopping conditions

Every campaign capability should include three outcomes:

Null control. With no planted defect, the aligned difference should vanish within the established representation and alignment errors.

Covariance control. Under an admissible coordinate change, the abstract impurity operator should be unchanged and its matrix should transform by the declared conjugation rule.

Stopping control. Incompatible rank, spin, geometry, subspace, or energy-reference data should prevent interpretation rather than produce a nominal residual.

These controls should be evaluated before any fitted model is called an impurity potential or effective-mass representation.

J.12 Required warmup order

The proposed sequence is:

1. verify the spinless-to-spinful embedding and Frobenius normalization;
2. verify primitive-to-pristine-supercell folding;
3. pass the zero-defect null extraction;
4. recover known one-dimensional planted defects under gauge and energy scrambling;
5. establish one-dimensional supercell-size convergence;
6. recover the declared one-dimensional model-class hierarchy;
7. scan one-dimensional defect smoothness toward a continuum regime;
8. reproduce the same controls in the separable two-dimensional parent;
9. introduce anisotropy and nonseparable two-dimensional coupling;
10. compare operator, bound-state, and subspace metrics; and
11. only then transfer the verified procedures to the matched pristine–doped silicon program.

The two-dimensional extraction should not be interpreted until the one-dimensional procedure recovers a known planted defect under alignment scrambling and supercell refinement. Likewise, first-principles execution would not be authorized or scientifically validated merely because the warmups pass. As recorded in Appendix I, the spin-space and one-dimensional prerequisites have human-accepted calculated synthetic evidence, including blind-alignment, independent-route, finite-rank-oracle, and continuum-refinement

follow-ons. The periodic two-dimensional folding parent is also accepted. Under the version-1 defect design, Stage A supplies human-accepted calculated and independently reproduced two-dimensional folding, seam, null-extraction, and incompatibility controls. Stage B supplies the bounded human-accepted scalar-onsite and symmetry result described above. Accepted-parent calculations for Stages C–E, new spin variants, and material transfer remain unexecuted and unauthorized.

J.13 Remaining warmup plot placeholder

The remaining placeholder is accepted-parent directional and nonlocal two-dimensional covariance: display extracted defect blocks for symmetry- related placements after alignment, together with the two-dimensional folding, null, supercell-shape, anisotropy, and metric controls. The completed one- dimensional Figure I.2 replaces the previous one-dimensional placeholders. Every future plot should identify the parent model, supercell, boundary condition, retained space, spin convention, planted defect, alignment map, energy reference, model class, normalization, and evidentiary status. A plot placeholder is not a calculated result.

J.14 Research questions

The resulting program is organized by four questions.

First, which retained subspace and localized representation preserve the declared pristine-silicon parent observables while remaining stable under changes in numerical resolution and gauge?

Second, what is the least complex constrained lattice-model class that satisfies both spectral and operator-level tolerances for the pristine host?

Third, which unitary identifications make pristine and doped retained operators comparable, and how sensitive is the aligned impurity operator to admissible choices of gauge, geometry, and energy reference?

Fourth, at what spatial scale, and within which anisotropic or multivalley model class, can the aligned phosphorus or boron impurity operator be replaced by an effective-mass description without losing the declared bound-state and operator observables?

These questions place the continuum approximation at the end of the reduction chain rather than assuming it at the outset. In the proposed material program, the aligned impurity operator would be the object extracted from first principles. A local screened potential, a nonlocal lattice operator, and a multivalley effective-mass model are competing admissible representations whose adequacy must be established by comparison.

CHAPTER K

Envelope Theory: Luttinger–Kohn, Burt, and Ermoneit

Status literature explanation and project interpretation. Luttinger and Kohn (1955), Burt (1992, 1999), and Ermoneit et al. (2026) have been read in full. This appendix explains their constructions in the context of the present reduction problem; it does not claim that these papers prove the project's modern Wannier, alignment, impurity-extraction, or continuum-convergence statements.

K.1 Why these papers matter to the present problem

The central problem of this monograph is to identify when a microscopic or first-principles semiconductor operator can be replaced by a smaller lattice or continuum operator without silently changing the state space, physical branch, or intended observable. Luttinger and Kohn's 1955 paper addresses an early and foundational version of that problem [70]. It begins with a one-electron Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{U}, \tag{K.1}$$

where $\hat{H}_0$ is periodic and $\hat{U}$ is an additional perturbing potential. The paper asks how the periodic problem can be replaced, near a selected band edge, by an equation for a slowly varying envelope.

Burt's 1992 and 1999 papers extend this question from a gentle perturbation of one bulk crystal to piecewise-periodic microstructures that may contain atomically abrupt interfaces [14, 15]. They distinguish an exact, invertible envelope-function representation from the approximate local differential equations conventionally called envelope or effective-mass models. Their central methodological shift is to derive exact equations first and to base subsequent approximations on the Fourier content of the envelope functions, rather than assuming that the material composition or microscopic perturbation itself varies slowly.

This is close to, but not identical with, the present project. Here the parent may be a

represented Kohn–Sham operator, the pristine and doped calculations may be produced independently, the extracted impurity term may be nonlocal or matrix valued, and a Wannier or other retained basis may intervene before a continuum limit is considered. Luttinger and Kohn therefore provide a model for how a controlled reduction can be organized, not a ready-made proof of the entire modern pipeline.

K.2 The periodic parent and the adapted basis

Let the Bloch eigenstates of the periodic parent satisfy

$$\hat{H}_0|\psi_{n\mathbf{k}}\rangle = \varepsilon_n(\mathbf{k})|\psi_{n\mathbf{k}}\rangle, \tag{K.2}$$

with position-space form

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r}), \tag{K.3}$$

where $u_{n\mathbf{k}}$ is lattice periodic. For a simple band with its minimum at $\mathbf{k}_0 = \mathbf{0}$, Luttinger and Kohn do not expand the perturbed state directly in ordinary Bloch states or in Wannier functions. They introduce the adapted family corresponding to their equation (II.4),

$$\chi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{0}}(\mathbf{r}). \tag{K.4}$$

They show that this family is complete and orthonormal when the underlying Bloch family is complete and orthonormal. The perturbed state can therefore be written as

$$\Psi(\mathbf{r}) = \sum_n \int_{\mathrm{BZ}} A_n(\mathbf{k})\chi_{n\mathbf{k}}(\mathbf{r})\,\mathrm{d}\mathbf{k}. \tag{K.5}$$

This choice is important conceptually. The reduced equation is not obtained only by differentiating a plotted band. A state-space representation is chosen first, the microscopic state is expanded in that representation, and only then are interband terms approximated or removed. In the language of this monograph, equation (K.4) supplies an analytical identification tied to one periodic parent. It is not a maximally localized Wannier gauge, an independently fitted alignment map, or a numerical pristine–doped basis match.

K.3 The gentle-potential approximation

The main physical assumption is that the fractional change of $U(\mathbf{r})$ over one unit cell is small. In reciprocal space, the matrix element of U between adapted states contains Fourier components shifted by reciprocal lattice vectors. Luttinger and Kohn retain the

component with zero reciprocal lattice vector when the perturbation is sufficiently smooth. In schematic notation this gives

$$\langle \chi_{n\mathbf{k}} | \hat{U} | \chi_{n'\mathbf{k}'} \rangle \approx \delta_{nn'}\, \widetilde{U}(\mathbf{k} - \mathbf{k}'). \tag{K.6}$$

The approximation suppresses large reciprocal-lattice momentum transfers and makes the leading perturbation band diagonal.

The paper repeatedly connects this assumption to scale separation. If a is a lattice spacing and a_i is the characteristic extent of an impurity state, the neglected contributions in the simple-band derivation are estimated at order $(a/a_i)^2$ under the stated assumptions. This is not a universal error bound for an arbitrary first-principles impurity operator. It is an order-of-magnitude estimate inside the paper's local-potential and slowly varying-envelope setting.

Luttinger and Kohn also identify the central-cell limitation explicitly. If the potential changes strongly inside the cell containing the impurity, the simple envelope equation is not valid there. They argue that an envelope description may remain valid sufficiently far from a short-range region. For the present project, this distinction motivates keeping a short-range central-cell operator separate from any proposed long-range screened continuum term rather than forcing one model to describe both regimes without evidence.

K.4 Removing interband coupling

In the adapted basis, the periodic Hamiltonian contains interband momentum matrix elements proportional to the wave vector measured from the band edge. Luttinger and Kohn treat those terms as small and introduce a canonical transformation

$$\mathbf{A} = e^{\mathbf{S}}\mathbf{B} \tag{K.7}$$

chosen to eliminate first-order interband coupling. Expanding the transformed Hamiltonian gives

$$e^{-\mathbf{S}}\mathbf{H}e^{\mathbf{S}} = \mathbf{H} + [\mathbf{H}, \mathbf{S}] + \frac{1}{2}[[\mathbf{H}, \mathbf{S}], \mathbf{S}] + \cdots . \tag{K.8}$$

The retained second-order terms reproduce the quadratic expansion of the selected band energy. This is a perturbative decoupling argument, not exact removal of the excluded bands for arbitrary energy, perturbation strength, or wave vector.

The distinction matters for the current research program. Modern retained-band selection, Wannier disentanglement, Feshbach downfolding, real-space truncation, and model fitting are different operations. The 1955 canonical transformation is evidence for the importance of accounting for excluded-band coupling; it does not make those later operations synonymous.

K.5 The simple-band envelope equation

After the smooth-potential and quadratic-band approximations, the transformed coefficient for a simple band satisfies the momentum-space equation given as (II.35) in the paper. Introducing the Fourier-transformed envelope

$$F_n(\mathbf{r}) = \int_{\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} B_n(\mathbf{k})\, \mathrm{d}\mathbf{k} \tag{K.9}$$

leads first to a short-range convolution form and then, when its kernel can be replaced by a delta distribution on the envelope scale, to the paper's equation (II.38):

$$[\varepsilon_n(-i\boldsymbol{\nabla}) + U(\mathbf{r})]\, F_n(\mathbf{r}) = E F_n(\mathbf{r}). \tag{K.10}$$

Here ε_n is understood only through its quadratic expansion about the selected extremum. Writing

$$\varepsilon_n(\mathbf{k}_0 + \mathbf{q}) = \varepsilon_n(\mathbf{k}_0) + \frac{\hbar^2}{2}\mathbf{q}^{\mathsf{T}}\mathbf{m}_n^{*-1}\mathbf{q} + O(\|\mathbf{q}\|^3) \tag{K.11}$$

and subtracting the constant edge energy yields the familiar anisotropic kinetic operator

$$-\frac{\hbar^2}{2}\boldsymbol{\nabla}\cdot\mathbf{m}_n^{*-1}\boldsymbol{\nabla}. \tag{K.12}$$

The leading microscopic wavefunction is the product stated in the paper's equation (II.40),

$$\Psi(\mathbf{r}) \approx F_n(\mathbf{r})\psi_{n\mathbf{0}}(\mathbf{r}). \tag{K.13}$$

Equation (K.13) displays the two-scale structure that the present monograph seeks to recover: a slowly varying envelope and a lattice-periodic microscopic factor. It is a leading approximation within a single-parent construction, not an exact identity between independently produced atomistic and continuum state spaces.

K.6 Band minima away from the zone center and multiple valleys

Section III shifts the adapted basis to a minimum at $\mathbf{k}_0 \neq \mathbf{0}$. The same derivation then uses band quantities evaluated at that minimum, and the envelope equation contains the quadratic expansion about $\mathbf{k}_0$. When symmetry produces several equivalent minima, the leading state has the schematic form

$$\Psi(\mathbf{r}) \approx \sum_{\nu=1}^{N_v} c_\nu F_\nu(\mathbf{r})\psi_{n\mathbf{k}_\nu}(\mathbf{r}). \tag{K.14}$$

The paper makes an important limitation explicit: the lowest-order effective-mass equations do not determine the correct linear combinations of degenerate valley solutions. Those combinations and their small splittings require corrections beyond the leading theory. For substitutional phosphorus in silicon, this prevents the existence of six equivalent conduction minima from being mistaken for a completely specified donor model. Valley ordering, phases, intervalley coupling, and the central-cell mechanism must still be identified.

K.7 Degenerate bands and coupled envelopes

For a band edge with an r-dimensional degeneracy, Section IV replaces the single envelope by components F_i. The quadratic band-edge information is a matrix-valued form with coefficients $D_{ij}^{\alpha\beta}$, and the impurity problem becomes a coupled system of the schematic form

$$\sum_{j=1}^{r}\left[-\sum_{\alpha,\beta} D_{ij}^{\alpha\beta}\partial_\alpha\partial_\beta + U(\mathbf{r})\delta_{ij}\right]F_j(\mathbf{r}) = EF_i(\mathbf{r}). \tag{K.15}$$

The exact coefficients and sign conventions must be taken from the declared band-edge basis and the paper's definitions. The structural conclusion is that degeneracy changes the effective equation from one scalar PDE to a coupled matrix differential equation.

This is directly relevant to the boron branch. A valence-band acceptor cannot be justified by copying the scalar donor equation and changing the sign of the charge. The retained valence manifold, its degeneracies, spin–orbit structure, and allowed couplings determine the continuum state space.

K.8 Spin–orbit coupling and the valence manifold

Section V incorporates spin–orbit coupling into the band-edge matrix elements. For a nondegenerate band, the formal structure of the effective-mass equation survives with modified momentum matrix elements and the spin–orbit-corrected band dispersion. For a degenerate valence edge, spin–orbit coupling partially lifts the degeneracy and yields coupled multiplets.

The paper emphasizes a scale criterion: a single spin–orbit-split band can be treated independently only when the energies induced by the impurity or magnetic field are small compared with the separation to the neighboring spin–orbit band. In its discussion of silicon acceptors, it warns that the binding-energy and spin–orbit scales can be comparable, so the simple single-band effective-mass theory need not be valid. The paper consequently

constructs a coupled six-component description before considering decoupled limits.

For this monograph, the lesson is methodological rather than numerical. Final boron claims require a spinor parent and a declared coupled valence-manifold continuum model. The historical numerical scale estimates in the 1955 paper are not adopted as calculated or current reference values for this project without separate verification.

K.9 Magnetic fields

Although magnetic-field calculations are outside the initial project scope, the paper's treatment is conceptually useful. It shows that the adapted-basis and canonical-transformation method extends to an external homogeneous field. At quadratic order, the effective equation follows the replacement

$$\mathbf{k} \longmapsto -i\boldsymbol{\nabla} - \frac{e}{\hbar c}\mathbf{A}, \tag{K.16}$$

with noncommuting products symmetrized. This part of the paper demonstrates that an effective operator must preserve the algebra required by the applied field; matching only the zero-field dispersion would not establish that property.

K.10 Burt's exact envelope-function representation

The equations in this and the following sections are explicit extractions from Burt's papers, rewritten only to match the notation of this monograph; the source equation numbers are stated at each step. Burt retains the Luttinger–Kohn form of the expansion but changes its scope and use [14, 15]. Choose one complete periodic basis $\{U_n(\mathbf{r})\}$ for the entire structure and write

$$\Psi(\mathbf{r}) = \sum_n F_n(\mathbf{r}) U_n(\mathbf{r}), \tag{K.17}$$

with every envelope F_n defined to contain only plane waves whose wave vectors lie in one chosen primitive reciprocal cell, usually a Brillouin zone. This is Burt's 1992 equation (3.1) and 1999 equation (2.4). If

$$U_n(\mathbf{r}) = \sum_{\mathbf{G}} U_{n\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}, \tag{K.18}$$

then the explicit envelope extraction in Burt's 1999 equation (2.5) is

$$F_n(\mathbf{r}) = \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\mathbf{G}} U^*_{n\mathbf{G}} \widetilde{\Psi}(\mathbf{k}+\mathbf{G}) e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{K.19}$$

For an orthonormal periodic basis, this is a unitary transformation at each in-zone wave vector. The expansion is therefore complete, unique, and invertible before it is truncated.

This definition differs from a conventional regional ansatz. In a conventional heterostructure treatment, one may choose different bulk band-edge functions in different layers and then match trial solutions at an interface. Burt instead uses the same periodic basis throughout, even where those functions do not diagonalize the local bulk Hamiltonian. The material change appears in the Hamiltonian matrix, including its off-diagonal entries, rather than through an unrecorded change of basis.

The band limit has two consequences that must be kept together. First, the exact F_n are smooth and continuous, with continuous derivatives, for a sufficiently well-behaved microscopic field. Second, one envelope by itself need not resemble the original wavefunction. Completeness belongs to the full sum in (K.17). Burt's 1999 delta-function example makes this point sharply: each individual band-limited envelope is broad and oscillatory, but the infinite collection reconstructs the delta distribution exactly.

K.11 Exact equations before reduction

Inserting equation (K.17) into the microscopic Schrödinger equation and applying the inverse basis transformation gives Burt's exact 1992 equation (3.8), which can be written

$$-\frac{\hbar^2}{2m}\nabla^2 F_n - \frac{i\hbar}{m}\sum_{n'} \mathbf{p}_{nn'}\cdot\boldsymbol{\nabla} F_{n'} + \sum_{n'}\int H_{nn'}(\mathbf{r},\mathbf{r}')F_{n'}(\mathbf{r}')\,\mathrm{d}\mathbf{r}' = EF_n(\mathbf{r}). \tag{K.20}$$

Here the periodic-basis kinetic and momentum matrix elements are contained in $H_{nn'}$ and $\mathbf{p}_{nn'}$ according to the paper's definitions, and the restricted delta kernel appearing in its equations (3.9)–(3.10) is

$$\Delta_{\mathrm{BZ}}(\mathbf{r}-\mathbf{r}') = \frac{1}{\Omega}\sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}. \tag{K.21}$$

The remaining kernel contains the transformed potential operator. It can be nonlocal even when the microscopic potential is local because the envelope Fourier domain is restricted to a Brillouin zone. Burt's alternative 1992 derivation exhibits equation (K.20) as a unitary change of representation from a plane-wave Hamiltonian. A truncated plane-wave Hamiltonian and its correspondingly transformed envelope equations therefore have identical eigenvalues before further approximation.

The word *exact* in this context refers to the representation transformation and the complete set of envelope equations. It does not mean that one dominant envelope, a local effective-mass equation, a finite band set, or a piecewise-constant coefficient model is exact. Those simplifications enter only later and must be assessed separately.

This distinction is particularly relevant to the present project. A nonlocal pseudopotential or represented impurity operator is not excluded by the exact envelope transformation.

But obtaining a local scalar continuum potential from such an operator is an additional model reduction, not a consequence of naming F_n an envelope.

K.12 Global slow variation rather than a gentle structure

Luttinger and Kohn focus on a perturbing potential that is gentle on the lattice scale. Burt's microstructure theory changes the approximation question. The microscopic composition may change abruptly; what matters for the simplified envelope equations is whether the relevant envelopes are *globally slowly varying*: their Fourier weight is concentrated near the selected in-zone wave vectors [14, 15].

This condition does not require every pointwise derivative to be small. Approximate envelopes may display a localized rapid change, kink, or interface feature while remaining dominated globally by small wave vectors. Conversely, a smooth-looking plotted function is not sufficient evidence: the Fourier support and the omitted high-wave-vector contributions must be checked.

Under the globally-slow condition, the nonlocal kernel in (K.20) may be approximated by a local or short-range operator. Burt's 1992 equation (5.1) then has the explicit local multiband form

$$-\frac{\hbar^2}{2m}\nabla^2 F_n - \frac{i\hbar}{m}\sum_{n'}\mathbf{p}_{nn'}\cdot\boldsymbol{\nabla}F_{n'} + \sum_{n'}H_{nn'}(\mathbf{r})F_{n'} = EF_n. \tag{K.22}$$

Burt separates bulk-profile terms, determined by the constituent bulk Hamiltonians and geometry, from interface terms that depend on how the microscopic potential changes. For the model analyzed in 1992, interface terms carry large reciprocal-wave-vector denominators and can be small for the low-lying state, but the paper does not claim that all interface terms are universally negligible. The appropriate test is made after the target state and operator have been identified.

For an impurity problem, the analogous project question is not simply whether the impurity atom is localized. It is whether the aligned microscopic state has envelope Fourier weight concentrated in the retained sectors and whether the terms discarded in localizing the envelope kernel are small on the declared comparison domain.

K.13 Dominant envelopes and effective-mass reduction

Burt obtains effective-mass equations by partitioning the envelopes into a small dominant set and a remote set, then eliminating the remote components perturbatively. This is a folding-down step performed after the exact envelope representation has been established.

For a dominant scalar conduction envelope, the resulting operator has the schematic form

$$-\frac{\hbar^2}{2}\boldsymbol{\nabla}\cdot\mathbf{m}^{-1}(E,\mathbf{r})\boldsymbol{\nabla}+E_c^{\mathrm{eff}}(\mathbf{r}), \tag{K.23}$$

where the energy and position dependence arise from the remote-band elimination and the transformed local Hamiltonian. This is the operator structure extracted from Burt's 1992 equations (6.9)–(6.10). The coefficients are derived rather than selected by choosing one of several formally Hermitian orderings.

The 1992 derivation keeps changes in the constituent zone-center eigenstates by retaining off-diagonal matrix elements of the zone-center Hamiltonian in one fixed basis. This explains why an effective-mass equation can remain useful even when the well and barrier Bloch functions are not approximately identical. It also identifies the approximation: interface terms, higher derivatives, remote envelopes, or nonlocal kernel contributions are neglected according to the chosen order.

Energy-dependent effective masses can arise naturally in this elimination. Burt distinguishes an energy-eigenvalue equation with energy-dependent coefficients from a time-dependent evolution equation and argues that the former is not forbidden by the superposition principle. Such a mass remains an approximation whose usefulness depends on the coupling structure and the neglected terms; it is not a universal prescription for every nonparabolic band.

K.14 Operator ordering and whole-matrix Hermiticity

Burt's 1999 tutorial derivation emphasizes that replacing constant bulk parameters by functions of position and symmetrizing every matrix element is not a reliable way to construct a multiband Hamiltonian [15]. The correct order of derivatives and coefficients follows from the order of momentum matrix elements in the remote-band elimination.

Burt's 1999 equations (4.4d)–(4.6) show the extraction directly. In the four-band example, eliminating the small conduction envelope gives

$$F_S \approx \frac{i\hbar p}{mE_g}\left(\partial_x F_X+\partial_y F_Y+\partial_z F_Z\right), \tag{K.24}$$

and the X-component equation becomes

$$\left(E_V-E-\frac{\hbar^2}{2m}\nabla^2\right)F_X+\partial_x\left[\frac{P^2}{E_g}\left(\partial_x F_X+\partial_y F_Y+\partial_z F_Z\right)\right]=0, \tag{K.25}$$

with cyclic equations for F_Y and F_Z. In $\mathbf{k}=-i\boldsymbol{\nabla}$ notation, Burt's equation (4.11) places an off-diagonal coefficient as $-k_i b(\mathbf{r})k_j$, with the order inherited from the two momentum matrix elements rather than from entrywise symmetrization.

For a coupled valence system, an individual off-diagonal differential operator need not be Hermitian by itself. Hermiticity is a property of the complete Hamiltonian matrix: the adjoint partner occurs in the transposed block. Different remote-band contributions to what is one bulk Luttinger parameter can acquire different operator orderings in an inhomogeneous structure. Consequently, a single fitted bulk parameter may need to be resolved into its microscopic contributions before an interface operator is defined.

The project analogue is immediate. Hermiticity of a complete represented impurity or continuum operator does not require every labeled physical component to be independently Hermitian before assembly. Conversely, entrywise symmetrization cannot substitute for a derivation from the chosen state-space map and eliminated sectors.

K.15 Interfaces, boundary conditions, and current

For the exact band-limited envelopes, continuity is built into the definition; it is not imposed by matching unrelated regional expansions. Integrating the local multiband equation gives the explicit interface quantity extracted from Burt's 1992 equations (7.1)–(7.2):

$$-\frac{\hbar^2}{2m}\,\partial_\perp F_n - \frac{i\hbar}{m}\sum_{n'} p^\perp_{nn'} F_{n'} \quad \text{is continuous across the interface,} \tag{K.26}$$

where $p^\perp$ is the momentum component normal to the interface. After a local multiband approximation is made, retaining the full second-order system gives continuity conditions for the envelopes and their derivatives in the setting analyzed by Burt. If remote envelopes or second derivatives are then eliminated, the reduced scalar equation (K.23) gives the familiar condition that the normal component of

$$\mathbf{m}^{-1}(E,\mathbf{r})\boldsymbol{\nabla} F \tag{K.27}$$

is continuous across an abrupt effective-mass step.

The corresponding kink in the approximate scalar envelope does not imply that the exact microscopic wavefunction has a literal derivative discontinuity. Burt's one-dimensional model shows a rapid but smooth change of derivative in the exact wavefunction and exact principal envelope; the piecewise-constant mass model replaces this soft interface region by a kink. The same derived boundary condition is consistent with the probability current of the reduced equation.

Thus boundary conditions belong to the approximation used. They must not be borrowed from one scalar effective-mass equation and applied unchanged to a multiband, nonlocal, or differently ordered operator.

K.16 Luttinger–Kohn and Wannier–Slater envelopes

Burt distinguishes two envelope representations. In the Luttinger–Kohn form, the periodic basis is independent of the in-zone wave vector:

$$\Psi(\mathbf{r}) = \sum_{n\mathbf{k}} F_n(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} U_n(\mathbf{r}). \tag{K.28}$$

In the Wannier–Slater form, the periodic basis is wave-vector dependent:

$$\Psi(\mathbf{r}) = \sum_{n\mathbf{k}} F_n^{\mathrm{WS}}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}). \tag{K.29}$$

These are Burt's 1992 equations (9.1)–(9.2). The latter envelopes are related to amplitudes in Wannier orbitals of a reference crystal by the paper's equation (9.10),

$$F_n^{\mathrm{WS}}(\mathbf{r}) = V_c^{1/2} \sum_{\mathbf{R}} \Delta_{\mathrm{BZ}}(\mathbf{r}-\mathbf{R}) \langle n\mathbf{R} \mid \Psi \rangle. \tag{K.30}$$

Exact equations can be derived for either representation. In Burt's 1992 analysis, the fixed-basis Luttinger–Kohn route gives a less cumbersome systematic effective-mass derivation for abrupt and substantial material changes, whereas the Wannier–Slater perturbation expansion generates additional interface terms and becomes more involved at consistent higher order.

Neither representation should be identified automatically with the modern construction of maximally localized Wannier functions. A Wannier90 calculation selects a retained subspace and gauge numerically; Burt's comparison concerns two analytical definitions of envelope coefficients relative to a reference periodic Hamiltonian. Any connection to the project's Wannier operator must therefore be made by an explicit map.

K.17 Out-of-zone solutions and representation completeness

The 1999 paper separates exact representation envelopes F_n from conventional trial envelopes that solve approximate local differential equations. The latter are not restricted to in-zone Fourier components. When one solves a derived multiband local system in piecewise-periodic regions, roots with wave vectors outside the reference Brillouin zone can appear.

The approximate one-dimensional system used for the explicit discussion is Burt's 1999 equation (5.11),

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2 F_n}{\mathrm{d}x^2} - \frac{i\hbar}{m}\sum_{n'} p_{nn'}\frac{\mathrm{d}F_{n'}}{\mathrm{d}x} + \sum_{n'} H_{nn'}(x) F_{n'} = E' F_n. \tag{K.31}$$

Burt argues that its out-of-zone solutions cannot be rejected solely because of their wave vector. In the extended local equation they can be replicas needed to satisfy the full set of interface conditions and reconstruct the microscopic state. In the one-dimensional example, using all such solutions gives an accurate wavefunction, and a reconstruction from one displaced reciprocal zone also succeeds. This does not make every out-of-zone root physically trustworthy: use of the extended equation requires the neglected interzone coupling and large-denominator terms to be controlled. The derivation and reconstruction, not the label "out of zone," determine admissibility.

For numerical reduction in this project, a large-wave-vector component should therefore be diagnosed rather than deleted by convention. It may indicate a necessary reconstruction component, a failure of the globally-slow assumption, an inadequate retained basis, or an artifact of extrapolating an approximate Hamiltonian beyond its validated domain.

K.18 Observables are not determined by the dominant envelope alone

Burt's 1999 discussion of interband dipole matrix elements supplies a broader warning for model reduction. Even when one envelope component dominates the norm and the product $F_n U_n$ closely approximates the wavefunction pointwise, that term alone need not approximate a matrix element of the position operator. Small envelope components and zone-boundary terms can determine the observable. For the one-dimensional dominant-component contribution, Burt's 1999 equation (6.12) isolates the zone-edge correction,

$$\begin{aligned} I_{cv} = {} & \pi i\, \widetilde{F}_c(\pi/a)^* \widetilde{F}_v(-\pi/a) \sum_G U^*_{cG} U_{v,G+2\pi/a} \\ & - \pi i\, \widetilde{F}_c(-\pi/a)^* \widetilde{F}_v(\pi/a) \sum_G U^*_{c,G+2\pi/a} U_{vG}. \end{aligned} \tag{K.32}$$

This term is often small for globally slow envelopes, but it is not identically zero merely because the periodic basis functions belong to different bands.

The lesson for the present work is that state or band reconstruction error does not automatically bound every observable error. Acceptance of a reduced impurity model must identify the observable, transform its operator consistently into the retained envelope space, and test whether discarded components enter at leading order for that observable.

K.19 Ermoneit's exact multivalley specialization

Ermoneit et al. specialize Burt–Foreman envelope theory to a decomposition of the Brillouin zone into disjoint valley sectors and apply it to valley splitting in strained Si/SiGe nanostructures [27]. The word *exact* again refers to the envelope representation and its transformed equation. Their microscopic model still assumes a periodic silicon potential V and represents the heterostructure, gate, and electric-field contributions through a chosen mesoscopic potential U; exactness is not a claim that this parent potential is an atomistically exact description of Si/SiGe.

Let $S_v = S(\mathbf{k}_v)$ denote the non-overlapping sector assigned to valley center $\mathbf{k}_v$. Their equations (2), (5), and (6) read

$$\Psi_\alpha(\mathbf{r}) = \sum_{n,v} u_{n\mathbf{k}_v}(\mathbf{r}) F_{nv\alpha}(\mathbf{r}), \tag{K.33}$$

$$F_{nv\alpha}(\mathbf{r}) = \sum_{\mathbf{K}\in S_v} e^{i\mathbf{K}\cdot\mathbf{r}} \sum_{\mathbf{G}} c_{n\mathbf{k}_v}(\mathbf{G})^* \widetilde{\Psi}_\alpha(\mathbf{K}+\mathbf{G}), \tag{K.34}$$

$$f_{nv\alpha}(\mathbf{r}) = e^{-i\mathbf{k}_v\cdot\mathbf{r}} F_{nv\alpha}(\mathbf{r}). \tag{K.35}$$

Here V_c is the finite crystal or computational normalization volume and Ω_p is one primitive-cell volume. The source uses the discrete Fourier convention

$$\Psi_\alpha(\mathbf{r}) = \sum_{\mathbf{q}} \widetilde{\Psi}_\alpha(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{r}}, \qquad \frac{1}{\Omega_p}\int_{\Omega_p} u_{n\mathbf{k}}(\mathbf{r})^* u_{n'\mathbf{k}}(\mathbf{r})\, \mathrm{d}^3 r = \delta_{nn'}. \tag{K.36}$$

Equation (9a) correspondingly gives

$$\sum_{n,v} \int_{V_c} F_{nv\alpha}(\mathbf{r})^* F_{nv\alpha'}(\mathbf{r})\, \mathrm{d}^3 r = \delta_{\alpha\alpha'}. \tag{K.37}$$

Exact completeness requires all bands and a collection of valley sectors whose disjoint union is the full first Brillouin zone. A later restriction to selected bands or valleys is a reduction, not part of the exact completeness statement. Unlike an informal sum of valley-centered trial functions, the disjoint support condition $\mathbf{K} \in S_v$ makes this decomposition unique. Equations (9b)–(11) make the associated projection explicit through

$$\Delta_v(\mathbf{r}-\mathbf{r}') = \frac{1}{V_c}\sum_{\mathbf{K}\in S_v} e^{i\mathbf{K}\cdot(\mathbf{r}-\mathbf{r}')}, \qquad \int_{V_c} \Delta_v(\mathbf{r}-\mathbf{r}') F_{nv'\alpha}(\mathbf{r}')\, \mathrm{d}^3 r' = \delta_{vv'} F_{nv\alpha}(\mathbf{r}). \tag{K.38}$$

Thus Δ_v is a sector-projected delta kernel, not the unrestricted Dirac delta distribution.

K.19.1 The project–multiply–project potential kernel

The source's equation (14) is the concrete nonlocal potential operator

$$U_{vv'}^{nn'}(\mathbf{r},\mathbf{r}') = \int_{V_c} \mathrm{d}^3x\, \Delta_v(\mathbf{r}-\mathbf{x}) u_{n\mathbf{k}_v}(\mathbf{x})^* U(\mathbf{x}) u_{n'\mathbf{k}_{v'}}(\mathbf{x}) \Delta_{v'}(\mathbf{x}-\mathbf{r}'). \tag{K.39}$$

Its structure is literally sector projection, multiplication by the microscopic Bloch-factor matrix element of U, and projection into the receiving sector. The transformed slowly varying kernel is

$$\mathsf{u}_{vv'}^{nn'}(\mathbf{r},\mathbf{r}') = e^{-i(\mathbf{k}_v\cdot\mathbf{r}-\mathbf{k}_{v'}\cdot\mathbf{r}')} U_{vv'}^{nn'}(\mathbf{r},\mathbf{r}'), \tag{K.40}$$

which is their equation (16). After perturbative interband decoupling, but without discarding intervalley coupling, their single-conduction-band equation (19) becomes

$$E_\alpha f_{v\alpha}(\mathbf{r}) = \left[-\frac{\hbar^2}{2}\boldsymbol{\nabla}\cdot\mathbf{m}_{cv}^{-1}\boldsymbol{\nabla} + E_{cv}\right] f_{v\alpha}(\mathbf{r}) + \sum_{v'}\int_{V_c} \mathrm{d}^3r'\, \mathsf{u}_{vv'}(\mathbf{r},\mathbf{r}') f_{v'\alpha}(\mathbf{r}'). \tag{K.41}$$

This equation is local in the effective kinetic term and nonlocal in the valley-projected potential. Interband elimination has occurred; intervalley coupling has not.

K.19.2 Energy-reference invariance

A constant energy-reference shift must change the diagonal energy but not the physical intervalley coupling. Ermoneit et al. prove this at the kernel level. Appendix B of the paper first derives the band-indexed identity

$$U_{vv'}^{nn'}(\mathbf{r},\mathbf{r}') \xrightarrow{U\mapsto U+U_0} U_{vv'}^{nn'}(\mathbf{r},\mathbf{r}') + U_0\,\delta_{nn'}\delta_{vv'}\Delta_v(\mathbf{r}-\mathbf{r}'). \tag{K.42}$$

After the phase transformation (K.40), their equation (26) is the corresponding single-band slowly varying kernel identity

$$\mathsf{u}_{vv'}(\mathbf{r},\mathbf{r}') \xrightarrow{U\mapsto U+U_0} \mathsf{u}_{vv'}(\mathbf{r},\mathbf{r}') + U_0\,\delta_{vv'} e^{-i\mathbf{k}_v\cdot(\mathbf{r}-\mathbf{r}')}\Delta_v(\mathbf{r}-\mathbf{r}'). \tag{K.43}$$

The phase-shifted projector acts as the identity on the corresponding slowly varying envelope. Because the sectors are disjoint, the additional constant is exactly diagonal in the valley label. Consequently,

$$\hat{H}_0 \mapsto \hat{H}_0 + U_0 I, \qquad \hat{H}_1 \mapsto \hat{H}_1, \tag{K.44}$$

where $\hat{H}_0$ and $\hat{H}_1$ are the intra- and intervalley blocks of their equations (22)–(24).

Replacing Δ_v by an unrestricted Dirac delta produces the conventional local model (21), but also removes the sector constraint. Under the same shift, the paper obtains the additional local off-diagonal contribution

$$\hat{H}_1^{\mathrm{loc}} \mapsto \hat{H}_1^{\mathrm{loc}} + U_0 \sum_{v'\neq v} e^{-i(\mathbf{k}_v-\mathbf{k}_{v'})\cdot\mathbf{r}} u_{\mathbf{k}_v}(\mathbf{r})^* u_{\mathbf{k}_{v'}}(\mathbf{r}). \tag{K.45}$$

This is not a physical valley-coupling response to a changed structure; it is an artifact of extending local envelopes beyond their defining valley sectors. For this problem, invariance under a constant energy shift is therefore a sharp operator-level diagnostic of the local approximation.

K.19.3 Intervalley coupling and valley splitting

For the two low-energy valleys at $\mathbf{k}_0^{\pm} = (0, 0, \pm k_0)^{\mathsf{T}}$, the unperturbed single-valley operators satisfy $\hat{H}_0^{-} = \hat{H}_0^{+*}$. The zeroth-order modes are chosen so that

$$f_-^{(0)} = f_+^{(0)*}, \qquad \int_{V_c} |f_+^{(0)}(\mathbf{r})|^2 \, \mathrm{d}^3 r = \int_{V_c} |f_-^{(0)}(\mathbf{r})|^2 \, \mathrm{d}^3 r = 1. \tag{K.46}$$

Degenerate perturbation theory reduces the first-order problem to

$$\begin{pmatrix} 0 & \Delta \\ \Delta^* & 0 \end{pmatrix} \begin{pmatrix} \eta_+ \\ \eta_- \end{pmatrix} = E_\alpha^{(1)} \begin{pmatrix} \eta_+ \\ \eta_- \end{pmatrix}, \qquad E_{\mathrm{vs}} = 2|\Delta|. \tag{K.47}$$

The first-order intervalley coupling evaluated with the exact sector-projected nonlocal kernel in their equation (30) is

$$\Delta = \int_{V_c} \mathrm{d}^3 r \int_{V_c} \mathrm{d}^3 r' \, f_+^{(0)}(\mathbf{r})^* \mathsf{u}_{+-}(\mathbf{r}, \mathbf{r}') f_-^{(0)}(\mathbf{r}'). \tag{K.48}$$

An exact algebraic rewriting of this first-order matrix element, using the convolution property of the sector projectors, gives their equation (31),

$$\Delta = \int_{V_c} \mathrm{d}^3 r \, e^{-2ik_0 z} f_+^{(0)}(\mathbf{r})^* u_+(\mathbf{r})^* U(\mathbf{r}) u_-(\mathbf{r}) f_-^{(0)}(\mathbf{r}). \tag{K.49}$$

Although this last expression looks like the conventional local-EFT formula, its envelopes remain sector projected. The support restriction, rather than the visual form of the final integral, is what preserves energy-reference invariance.

K.19.4 Spectrally filtered local approximation

The paper also gives an intermediate approximation that retains the inexpensive local single-valley solve and then restores the valley-sector support. For the one-dimensional reduction, define

$$\Delta_v^{(\mathrm{1D})}(z - z') = \frac{1}{L_z} \sum_{K_z \in S_v^{(z)}} e^{iK_z(z-z')}, \tag{K.50}$$

where L_z is the normalization length and $S_v^{(z)}$ is the relevant one-dimensional valley-sector interval. The paper's equation (39) is then

$$\widetilde{f}_{v\alpha}^{(0)}(z) = \int \mathrm{d}z'\, \Delta_v^{(1\mathrm{D})}(z-z') e^{-ik_v(z-z')} f_{\mathrm{loc},\alpha}^{(0)}(z'), \tag{K.51}$$

followed by normalization. Substitution of these projected envelopes into the coupling integral gives equation (40),

$$\Delta_{\mathrm{loc}}^{\mathrm{filtered}} = \sum_{n\in\mathbb{Z}} C_n^{(2)} \int \mathrm{d}z\, e^{-i(2k_0+nG_{0z})z} \widetilde{f}_+^{(0)}(z)^* U_\perp(z) \widetilde{f}_-^{(0)}(z). \tag{K.52}$$

By construction, this approximation is invariant under $U \mapsto U + U_0$. It is not identical to the nonlocal theory. The authors' reported numerical comparisons show close agreement for several quantum-well and wiggle-well cases, while their sharp Ge-spike example shows that filtering can overestimate the nonlocal result. These are literature-reported calculations for the stated Si/SiGe model, not calculated results or scientific validation for the present silicon-dopant project.

K.20 Mapping the papers onto the project reduction chain

The relation between the historical construction and the present project can be summarized as

$$\begin{aligned} \text{periodic parent } \hat{H}_0 &\longrightarrow \text{band-edge adapted basis} \\ &\longrightarrow \text{perturbative interband decoupling} \\ &\longrightarrow \text{envelope or coupled-envelope operator.} \end{aligned} \tag{K.53}$$

Burt inserts an exact representation layer before any local reduction:

$$\begin{aligned} \text{piecewise-periodic microscopic operator} &\longrightarrow \text{one fixed periodic basis} \\ &\longrightarrow \text{exact band-limited nonlocal envelopes} \\ &\longrightarrow \text{globally-slow Fourier diagnosis} \\ &\longrightarrow \text{local multiband equation} \\ &\longrightarrow \text{dominant-envelope effective operator.} \end{aligned} \tag{K.54}$$

Ermoneit et al. then make the multivalley support constraint and one diagnostic of its preservation explicit:

$$\begin{aligned}\text{disjoint valley sectors} &\longrightarrow \text{sector-projected delta kernels}\\ &\longrightarrow \text{project–multiply–project potential}\\ &\longrightarrow \text{nonlocal intervalley coupling}\\ &\longrightarrow \text{energy-reference-invariance test}\\ &\longrightarrow \text{optional filtered-local approximation}.\end{aligned} \tag{K.55}$$

The corresponding modern project chain is longer:

$$\begin{aligned}\text{pristine and doped KS parents} &\longrightarrow \text{compatible retained subspaces}\\ &\longrightarrow \text{gauge and energy alignment}\\ &\longrightarrow \text{aligned impurity operator}\\ &\longrightarrow \text{lattice reduction}\\ &\longrightarrow \text{continuum embedding and comparison}.\end{aligned} \tag{K.56}$$

Several differences must remain explicit.

1. **One microscopic problem versus independently produced calculations.** Luttinger and Kohn write $\hat{H} = \hat{H}_0 + \hat{U}$ on one microscopic state space, and Burt transforms one complete heterogeneous microscopic problem into a fixed envelope basis. The present project must first identify independently produced pristine and doped retained spaces before their operators can be subtracted.

2. **General exact kernel versus reduced impurity model.** Luttinger–Kohn begin with a multiplicative $U(\mathbf{r})$, while Burt's exact transformation can retain nonlocal and multicomponent structure. A local scalar impurity potential nevertheless emerges only after a separately justified localization and model-class reduction.

3. **Analytical periodic basis versus Wannier construction.** Luttinger–Kohn use band-edge periodic functions, Burt permits one fixed complete periodic basis or a wave-vector-dependent Wannier–Slater basis, and Ermoneit et al. choose valley sectors of a strained-silicon Bloch basis. None of these operations performs modern composite-band disentanglement, maximally localized gauge selection, or independent basis alignment.

4. **Asymptotic or Fourier scale separation versus finite computational evidence.** Luttinger–Kohn assume a gentle perturbation and small relevant wave vectors; Burt tests whether the envelopes are globally dominated by retained Fourier sectors; Ermoneit et al. enforce disjoint valley support and test constant-shift invariance. The project must additionally test finite supercells, reciprocal meshes, retained ranks, embeddings, and spatial domains before making a crossover claim.

5. **Reduction fidelity versus parent-model validity.** Agreement with a Luttinger–Kohn-type continuum equation can establish reduction fidelity to a selected Kohn–Sham parent. It does not by itself validate the parent against experiment.

K.21 Implications for phosphorus and boron in silicon

For phosphorus, Luttinger–Kohn support beginning with a multivalley envelope space associated with the equivalent silicon conduction minima. They also support separating the smooth long-range regime from the short-range region where the perturbation varies strongly within a cell. It does not determine the intervalley central-cell operator, the correct valley combinations, or the mapping from an independently generated doped calculation into the pristine band-edge basis.

For boron, Luttinger–Kohn support a coupled-envelope description of a degenerate valence edge and warn against a scalar single-band treatment when spin–orbit splitting is not large compared with the impurity scale. They do not select the modern parent calculation, spinor basis, continuum parameters, or validation observables required by this project's boron branch.

Burt's fixed-basis extraction adds a testable intermediate question for both dopants: whether the exact represented envelopes are concentrated in the retained reciprocal sectors strongly enough to justify the local multiband operator (K.22), and then whether a dominant-envelope elimination is accurate for the declared observable. Ermoneit et al. sharpen the phosphorus-side question by supplying explicit sector projectors and a constant-energy-shift invariance test for intervalley coupling. Their paper studies Si/SiGe conduction valleys, not substitutional phosphorus, boron, or a central-cell impurity operator; application to either dopant remains proposed work.

The two dopants therefore share the need for a declared microscopic-to-envelope map, but they do not share the same continuum state space. Valley multiplicity organizes the phosphorus donor problem; valence-band degeneracy and spin–orbit coupling organize the boron acceptor problem.

K.22 Error and evidence interpretation

The approximations discussed by Luttinger–Kohn, Burt, and Ermoneit et al. should be separated into the error categories used by this monograph.

Parent-model error. The periodic one-particle parent may disagree with the physical material. The effective-mass derivation does not remove this discrepancy.

Scale-separation and analytical reduction error. Neglected reciprocal- lattice Fourier components, localization of the exact nonlocal envelope kernel, omitted interface terms, spectral leakage between valley sectors, higher powers of wave vector, remote-envelope elimination, residual interband mixing, and short-range failure contribute to the difference between the microscopic parent and the envelope model.

Numerical error. Any finite representation of the parent bands, derivatives, envelopes, or coupled PDE adds discretization and solver errors not analyzed merely by the historical formal derivation.

Model-class error. A scalar, multivalley, or multiband continuum class may omit operator structure required by the aligned atomistic target.

The present reading establishes only what the cited papers state and how their assumptions relate to the declared project question. It is literature analysis, not a calculated result, numerical verification, scientific validation, or proof of the project's proposed continuum limit.

K.23 Summary of what the papers establish

Luttinger and Kohn go beyond the curvature mnemonic by expanding the perturbed state in a complete orthonormal family built from band-edge periodic functions and by using a canonical transformation to remove interband coupling through the retained quadratic order [70]. For a simple band and a potential that varies little across a unit cell, their momentum-space equation (II.35) becomes the coordinate-space envelope equation (II.38), and the leading microscopic state has the product form in (II.40). Their derivation treats minima away from the Brillouin-zone center and multiple equivalent valleys in Section III, degenerate bands through coupled envelope equations in Section IV, and spin–orbit coupling in Section V.

These are approximation statements. The relevant envelope or impurity scale must be large compared with the lattice spacing, the perturbation must be gentle away from any explicitly short-range region, and the band expansion is retained only through quadratic order. Lowest-order multivalley theory does not determine the correct valley combinations, and a single-band description is not justified when coupling to a nearby degenerate or spin–orbit-split manifold is comparable with the perturbation scale.

Burt makes the envelope transformation itself exact by equations (3.1) and (3.8) of the 1992 paper and equations (2.4)–(2.5) of the 1999 paper. The local multiband equation (1992, equation (5.1)), dominant-envelope elimination (1992, equations (6.3)–(6.10)), interface conditions (1992, equations (7.1)–(7.4)), operator ordering (1999, equations (4.4)–(4.11)),

and out-of-zone reconstruction (1999, equations (5.1)–(5.11)) are subsequent approximations or consequences with separately identifiable assumptions. The exact representation does not make a truncated local effective-mass model exact.

Ermoneit et al. make the multivalley decomposition unique through equations (5) and (9)–(11), retain the project–multiply–project kernel (14) in the single-band equation (19), and prove the constant-shift identity (26). Their first-order intervalley matrix element (30) gives $E_{\mathrm{VS}} = 2|\Delta|$, while equations (39)–(40) define a filtered-local approximation that restores the same constant-shift invariance. This is a concrete extension of Burt's exact representation logic, not evidence that an unprojected local multivalley model is exact.

Source review resolved Luttinger–Kohn (1955), key `luttingerKohn1955` [70], Burt (1992, 1999), keys `burt1992` [14] and `burt1999` [15], and Ermoneit et al. (2026), key `ermoneit2026` [27], have been read in full. Short Burt references: M. G. Burt, *J. Phys.: Condens. Matter* **4**, 6651–6690 (1992), doi:10.1088/0953-8984/4/32/003; and M. G. Burt, *J. Phys.: Condens. Matter* **11**, R53–R83 (1999), doi:10.1088/0953-8984/11/9/002. Ermoneit reference: L. Ermoneit, A. Thayil, T. Koprucki, and M. Kantner, *Phys. Rev. B* **113**, 245306 (2026), doi:10.1103/md2x-s44y. These sources do not establish modern Wannier disentanglement, independently generated pristine–doped alignment, or the project's proposed continuum crossover.

K.24 Questions carried forward

The papers leave the present project with five concrete questions:

1. What explicit identification maps an accepted retained Kohn–Sham space to the donor or acceptor envelope space?
2. Over what tested spatial domain is the aligned impurity operator gentle relative to the lattice scale?
3. Which intervalley or interband terms remain after the declared lattice reduction?
4. Which operator norm and observable tests establish the finite crossover from the atomistic model to the continuum class?
5. How do parent-model, numerical, and reduction discrepancies remain separate when the continuum model is compared with experiment?

Answering these questions requires the modern representation, alignment, and evidence program developed in the main chapters. The four papers supply the foundational representation, reduction pattern, and multivalley support diagnostic but not completed answers.

CHAPTER L

Candidate Contemporary Literature for Later Review

Status prospective literature register. The records below preserve the human-supplied candidate set, with identifier-level metadata corrections where primary publisher, repository, or official project pages were checked. Except where another appendix explicitly says that a paper was read, presence here means neither that the source has been read nor that its proposed manuscript claim has been accepted.

This appendix is a review queue, not a literature synthesis. A bibliography record allows a candidate citation to render; it does not establish that the source supports the proposed local claim. Before any citation is moved into a main chapter, the source must be read and checked against the exact sentence. Software websites are retained as secondary operational references rather than substitutes for archival papers. Placeholder identifiers are not entered in the bibliography.

The status phrases used below have the following meanings.

Metadata checked. The DOI or canonical record resolved and its core bibliographic fields were checked. The article itself remains unread unless stated elsewhere.

Official link checked. The official documentation, project, repository, or proceedings page resolved, but it may change over time.

Existing record. The bibliography already contained the source before this review queue was added.

Correction required. The supplied author, year, venue, title, or identifier did not match the resolved record.

Unresolved. The supplied record contains a placeholder or lacks enough verified metadata to create an authoritative bibliography entry.

L.1 Quantum ESPRESSO

1. **Quantum ESPRESSO toward the exascale.** Giannozzi et al. (2020) [38]. *Metadata checked.* Candidate use: supplement, rather than silently replace, the foundational suite papers when discussing accelerator and exascale development.

2. **One further step toward the exascale.** Carnimeo et al. (2023) [17]. *Metadata checked; correction required.* The resolved record has Ivan Carnimeo as first author and issue 20, not Giannozzi as first author and issue 21.

3. **Quantum ESPRESSO documentation.** Official documentation page [95]. *Official link checked.* Use for version-matched executable and input semantics; do not use it as the sole source for scientific-method claims.

L.2 Wannier90

1. **Wannier90 as a community code.** Pizzi et al. (2020) [91]. *Existing record; metadata checked.* Candidate use: modern software features and applications, with older method papers retained for the mathematical constructions they own.

2. **Wannier90 official website.** Project home page [130]. *Official link checked.* Retain as a secondary software reference, not as evidence for localization, disentanglement, or gauge claims.

L.3 PseudoDojo

1. **Actinide and super-heavy-element PseudoDojo extension.** Tantardini et al. (2024) [120]. *Metadata checked; correction required.* The resolved record is led by Christian Tantardini, not M. J. van Setten. It concerns 34 actinide and super-heavy-element pseudopotentials, so its relevance to silicon pseudopotential identity or convergence must be assessed narrowly rather than inferred from the shared validation framework.

2. **PseudoDojo repository.** Official GitHub repository [93]. *Official link checked.* Candidate use: software and validation-workflow provenance, with an immutable release or commit required for a reproduction claim.

L.4 Defect supercells and charged-defect alignment

1. **Claimed 2025 Freysoldt update.** *Correction required.* The supplied EPFL URL resolves to Komsa, Rantala, and Pasquarello (2012), not to a 2025 Freysoldt paper. The resolved paper is retained as a candidate [60]; no 2025 Freysoldt record has been added from the supplied metadata.

2. **Point-defect review.** Freysoldt et al. (2014) [32]. *Existing record.*

3. **Original FNV correction.** Freysoldt, Neugebauer, and Van de Walle (2009) [33]. *Existing record.*

L.5 Machine-learned tight-binding and DFT Hamiltonians

1. **Review of machine-learning interatomic potentials and Hamiltonians.** Li et al. (2025) [65]. *Metadata checked; source review pending.*

2. **Review of machine-learning tight-binding models.** Zou et al. (2026) [138]. *Metadata checked; correction required.* The resolved record is a 2026 article led by Jijie Zou, not a 2025 article led by Y. Wang.

3. **DeePTB.** Gu et al. (2024) [42]. *Metadata checked.* The official project article is retained separately [25] as a secondary software reference.

4. **DeepH.** Li et al. (2022) [64]. *Metadata checked; correction required.* The resolved paper is led by He Li and appears on pages 367–377, not under Yuan et al. on pages 300–311.

5. **DeepH-r preprint and journal version.** The 2024 arXiv version [134] and the 2026 journal version [135] describe the same research line. *Metadata checked; correction required.* The DOI resolves to *Physical Review Letters* **137**, 046401, not to *Physical Review B*. Prefer the journal version for the final citation unless a version-specific preprint statement is needed.

6. **Projected-density-of-states fitting for point defects.** Fried et al. (2025) [34]. *Metadata checked; title corrected.* Applicability to this project requires distinguishing its semi-empirical fitted Hamiltonian from the aligned operator-reduction problem posed here.

7. **Sparse tight-binding parameters for defects.** Schattauer et al. (2022) [102]. *Metadata checked; correction required.* The resolved record is led by Christoph Schattauer, not Y. Wang.

8. **Self-consistency training.** Zhang et al. (2024) [136]. *Official proceedings page checked.* The verified page range is 59329–59357, not 1–15.

9. **Claimed GPUTB article.** *Unresolved.* The supplied DOI contains the literal placeholder XXXXXX; no bibliography record has been created. A publisher record or stable preprint identifier is required before citation.

L.6 Verification, validation, and uncertainty quantification

1. **VECMAtk toolkit.** Groen et al. (2021) [41]. *Metadata checked.*

2. **Reliability and reproducibility in computational science.** Coveney, Groen, and Hoekstra (2021) [22]. *Metadata checked.* This is an editorial framing article for the theme issue; individual methodological claims may require the relevant article within that issue.

3. **Uncertainty-quantification patterns for multiscale models.** Ye et al. (2021) [132]. *Metadata checked; correction required.* The verified DOI is 10.1098/rsta.2020.0072, not the supplied placeholder Procedia DOI.

4. **Journal of Verification, Validation and Uncertainty Quantification.** *Not entered as a citation.* A third-party venue directory does not support a scientific claim. Cite a specific ASME article or the journal's official page only when the local statement actually concerns the venue.

5. **Discretization uncertainty.** Bect et al. (2021) [3]. *Official arXiv record checked.* The fuller title is *On the Quantification of Discretization Uncertainty: Comparison of Two Paradigms*; the associated conference DOI is recorded in the bibliography.

L.7 Effective mass and envelope theory

1. **Perturbative ab initio effective masses.** Rubel et al. (2021) [99]. *Metadata checked; source review pending.* Candidate use: computational extraction of effective masses,

not as a replacement for the state-space and nonlocal-envelope derivations reviewed in Appendix K.

2. **Claimed Oxford book chapter.** *Unresolved.* The supplied DOI contains the literal placeholder XXXX, and no chapter authors or verified chapter identifier were supplied. No bibliography record has been created.

L.8 Kohn–Sham and fundamental gaps

1. **Modern electronic-structure textbook.** Martin (2020) [71]. *Metadata checked; DOI corrected.* The verified book DOI is 10.1017/9781108555586, not 10 . 1017/97811088 60168. The exact gap discussion must still be checked before attaching the citation to a local claim.

2. **Paier et al. (2006).** The existing record [86] is *Screened Hybrid Density Functionals Applied to Solids*; it is not titled "Why does the DFT band gap underestimate the experimental band gap?" Its relevance is to screened hybrid-functional calculations and their solid-state results, not to the formal derivative-discontinuity statement owned by Perdew–Levy and Sham–Schlüter.

L.9 Review disposition

No candidate in this appendix should be inserted mechanically. Later review should record, for each proposed use, the exact manuscript sentence, the source passage that supports it, and whether the claim is about software capability, mathematical structure, numerical verification, scientific validation, or uncertainty quantification. Unresolved records require corrected primary metadata before source review begins.

CHAPTER M

Preliminary Silicon Work: Executions, Finite-Setting Evidence, and an Exploratory Wannier Decomposition

Status bounded synthesis of preliminary silicon work. The tutorial and finite-setting records cited here are retained repository evidence with their own statuses. The same-frame Wannier kinetic decomposition is a separately identified external, read-only prototype whose analysis package is not part of this manuscript branch. Nothing in this appendix establishes an accepted production silicon parent, converged Wannier representation, effective mass, scientific validation, uncertainty quantification, or authorization for new calculator execution.

M.1 Purpose and evidence boundary

The silicon program has progressed beyond an entirely prospective plan, but it has not crossed the production-material acceptance boundary stated in Chapter 27. This appendix gathers the preliminary work in one place so that successful execution observations, finite-setting numerical evidence, and exploratory represented-operator analysis are not mistaken for one homogeneous result.

The evidence is divided into four layers:

1. a legacy Quantum ESPRESSO 7.2 silicon tutorial reproduction and its associated band calculation;
2. a bounded Quantum ESPRESSO 7.5 rerun of the same tutorial-scale SCF input and a mechanical cross-version comparison;
3. a direct PBE/PseudoDojo cutoff–mesh bootstrap matrix with a read-only audit, provisional finite-setting selection, and an unresolved runtime warning boundary; and

4. a read-only analysis of a separate QE 7.5–Wannier90 3.1.0 silicon run, used to prototype a same-frame decomposition $H^W = T^W + R^W$, interpolation, local multiband expansion, and finite-mesh block truncation.

These layers do not share one physical parent. In particular, the tutorial uses a legacy LDA pseudopotential, the cutoff–mesh matrix uses the frozen PseudoDojo PBE norm-conserving artifact, and the exploratory Wannier source run uses the legacy ultrasoft `Si.pbe-n-van.UPF`. Their energies and operators must not be combined as though they were refinements of one calculation.

Table M.1: Status of the preliminary silicon evidence summarized here.

Layer	Supported statement	Unsupported promotion
QE 7.2 tutorial	The identified SCF reproduced the bundled printed energy and iteration count; a separate band run completed and retained 28 points with eight eigenvalues each.	Production convergence, modern band-path adequacy, a band gap, a valley, an effective mass, or pseudopotential suitability.
QE 7.5 tutorial comparison	The bounded rerun completed and the listed scalar and QEXSD observations were mechanically compared with QE 7.2.	General cross-version equivalence or validation of either physical parent.
PBE/PseudoDojo finite-setting matrix	The retained nine-case matrix passed its integrity audit; declared adjacent-setting criteria support a provisional C48/K8 selection.	Infinite-basis convergence, cutoff–mesh interaction control, accepted lattice geometry, effective-mass convergence, or downstream production eligibility.
Exploratory Wannier decomposition	The reported finite rank-four matrices obey the same-frame decomposition and bounded numerical checks for the identified external source bytes.	An accepted production Wannier operator, short-range tight-binding model, converged curvature, or silicon validation.

M.2 Source records

Table M.2 identifies the records from which this appendix is projected. Repository paths are relative to the repository root. The complete package in the final row remains external. A

bounded publication extraction containing the primary result, selected finite-mesh records and figures, and a copy of the external output manifest is retained locally; the omitted package payloads remain external.

Table M.2: Primary records for the preliminary silicon synthesis.

Evidence layer	Primary record or locator
QE 7.2 tutorial SCF	`calculations/bulk-silicon/qe-example01-si-scf-davidson/result.md`
QE 7.2 tutorial bands	`calculations/bulk-silicon/qe-example01-si-bands-davidson/result.md`
QE 7.5 smoke comparison	`calculations/bulk-silicon/qe-7.5-si-scf-smoke-comparison/execution-comparison.json`
Finite-setting audit	`calculations/bulk-silicon/production-convergence-preflight/direct-results-audit.md`
Provisional disposition	`calculations/bulk-silicon/production-convergence-preflight/direct-results-setting-disposition.json`
Warning classification	`calculations/bulk-silicon/production-convergence-preflight/ieee-warning-class-separated-nonmpi-c48-classification.md`
External Wannier prototype	Local bounded extraction: `calculations/ICMSEP2026/conference/paper_1/silicon-wannier-kinetic-preview/`; complete external package identifier `silicon-example11-hminus-t-readonly-analysis-20261004T193610Z`; `result.json` SHA-256 `846564b1d716044c9ed0ac4ddc94a1d6b7e81019ae2235979fcadab5033cfcf0`; copied output-manifest SHA-256 `5f509ab1cd4fa998f4c4002eba364982e94ee239237dd1adc6e1e29f1094dfb5`

M.3 Legacy tutorial baseline

The retained QE 7.2 Davidson SCF tutorial run exited with status zero, emitted `JOBDONE.`, and reported convergence after six iterations. Its printed total energy was

$$E_{\text{tutorial}} = -15.84452726\ \text{Ry}, \tag{M.1}$$

matching the bundled example output at printed precision. This is an accepted tutorial reproduction, not a production material result. Its primary record is `calculations/bulk-silicon/qe-example01-si-scf-davidson/result.md`.

The associated band calculation also exited with status zero and emitted the end-of-band-structure and completion markers. It retained 28 ordered reciprocal-space points with eight printed eigenvalues at each point. Of the 224 printed values, 166 were identical to the bundled legacy output at printed precision, and the largest observed printed difference was approximately 10^{-4} eV. No comparison tolerance was accepted, so that count is an observation rather than a numerical pass criterion. The coordinates form the legacy tutorial path; they are not an accepted modern silicon path or an effective-mass stencil.

The band result file retains older "awaiting human review" prose, while the later lifecycle summary records the tutorial band task as closed human-accepted. This appendix does not use that administrative difference to strengthen the scientific claim: under either description, the result remains bounded tutorial evidence only.

Both tutorial invocations emitted

```
IEEE_INVALID_FLAGIEEE_DIVIDE_BY_ZEROIEEE_OVERFLOW_FLAGIEEE_UNDERFLOW_FLA
G.
```

Successful termination does not classify those flags as harmless.

M.4 QE 7.5 smoke comparison

A separately authorized, isolated, single-process QE 7.5 invocation reused the same tutorial-scale SCF input and legacy pseudopotential identity. It also completed in six iterations with printed total energy -15.84452726 Ry. The parsed QEXSD energies differed by $-2.00030214614344 \times 10^{-10}$ Ha, or $-4.00060429228688 \times 10^{-10}$ Ry, between the QE 7.5 and QE 7.2 records. The largest listed absolute QEXSD eigenvalue difference was $2.3826225714618232 \times 10^{-11}$ Ha.

These values are a bounded compatibility observation for one input. Native charge-density and wavefunction files were not byte-identical, and byte identity was not required. The exercise also motivated fail-closed parser support for the exact observed QEXSD format versions `23.03.10` and `25.05.21`; it did not establish exhaustive compatibility with either upstream schema. The exact comparison and its limits are retained in `calculations/bulk-silicon/qe-7.5-si-scf-smoke-comparison/execution-comparison.json`. The same four-class IEEE report recurred.

M.5 PBE/PseudoDojo finite-setting matrix

The physically relevant bootstrap branch is distinct from the tutorial. It uses the frozen scalar-relativistic, non-SOC, non-spin-polarized PBE parent and the exact PseudoDojo

norm-conserving Si artifact required by the physical and numerical specifications. The direct campaign contains six cutoff labels `C30`, `C36`, `C42`, `C48`, `C54`, and `C60`, and four shifted-mesh labels `K6`, `K8`, `K10`, and `K12`. The logical `K8` case reuses `C48`. Nine unique cases therefore supplied nine SCFs and nine linked diagnostic NSCFs.

The later read-only audit recomputed the compact identities, checked all 18 input copies, matched 41 retained text-output and receipt identities, independently parsed all nine cases, and reconstructed the eight adjacent comparisons. Every invocation had one `JOBDONE.` marker and no retry. The audit explicitly did not reread or duplicate the dense restart and wavefunction payloads merely to restate compact observables.

The frozen adjacent-setting criteria were 10^{-5} Ry per atom for total energy, 0.05 kbar for pressure and each stress component, and 1 meV for the retained fixed-point band and gap probes. The first two cutoff comparisons failed at least the energy, pressure, and stress criteria. Comparisons C42→C48, C48→C54, and C54→C60 met all listed criteria. All three mesh comparisons, K6→K8, K8→K10, and K10→K12, also met the listed criteria at the printed resolution.

A later human disposition selected the smallest setting with passing lower and upper finite-setting guards:

$$E_{\mathrm{cut}}^{\psi} = 48\ \mathrm{Ry}, \qquad E_{\mathrm{cut}}^{\rho} = 192\ \mathrm{Ry}, \qquad \mathcal{K}_{\mathrm{SCF}} = 8 \times 8 \times 8\ \text{shifted}. \tag{M.2}$$

The selection is provisional. Sequential one-variable scans do not bound the cutoff–mesh interaction, and the diagnostic points do not locate the conduction valley or test curvature. No infinite-basis estimate follows.

M.5.1 What the finite-setting data do and do not show

The extracted per-atom energy changed over the cutoff sequence:

$$E_{\mathrm{C30}} = -8.461848470\ \mathrm{Ry}, \qquad E_{\mathrm{C60}} = -8.461975385\ \mathrm{Ry}. \tag{M.3}$$

The fixed-point probes were numerically much less sensitive over the upper cutoff sequence, but their limited coordinates and four-decimal electron-volt printing make them unsuitable substitutes for the later valley and effective-mass calculations. Pressure remained near 37.8 kbar because the provisional geometry was not the accepted zero-pressure PBE-relaxed lattice required by the physical specification. Thus the matrix supports finite-setting behavior at a fixed provisional geometry; it does not accept that geometry.

The authoritative synthesis is `calculations/bulk-silicon/production-convergence-preflight/direct-results-audit.md`. The selection and warning state are recorded separately in `direct-results-setting-disposition.json`; a digest or successful process exit is not a substitute for those semantic dispositions.

M.6 Runtime warning diagnosis and remaining block

Every SCF and diagnostic NSCF in the finite-setting matrix reported the same four IEEE classes. Subsequent separately authorized diagnostics localized more than the original text report alone could show:

1. one first instrumented trap occurred in the external MPI/PMIx/hwloc/OpenCL startup path;
2. after OpenCL discovery was disabled, a later floating-point division trapped below `ZHEGVX` during initial-wavefunction subspace diagonalization;
3. that later operation appeared with both Apple Accelerate and OpenBLAS;
4. a true non-MPI OpenBLAS build reproduced the matching OpenBLAS `ztrsv_CUN+388` operation; and
5. a divide-by-zero-only non-MPI build reproduced the exact instruction, classifying that matching first later operation as an IEEE divide-by-zero exception.

This localization does not recover the operands, explain the mathematical cause, establish whether the produced value affects retained observables, cover underflow, or identify every later invalid or overflow flag. The provisional setting therefore remains blocked from downstream protected execution pending an observable-effect and remaining-warning disposition. No new calculator execution is authorized by this appendix.

M.7 Exploratory same-frame Wannier kinetic decomposition

A separate local analysis consumed an externally retained QE 7.5–Wannier90 3.1.0 source run without executing either calculator. The source-run identifier is

`silicon-example11-hminus-t-20261004T171505Z`.

The analysis package identifies itself as `silicon-example11-hminus-t-readonly-analysis-20261004T193610Z`. Its `result.json` SHA-256 is

`846564b1d716044c9ed0ac4ddc94a1d6b7e81019ae2235979fcadab5033cfcf0`.

The complete package remains in a separate development worktree and is not part of the present manuscript branch. A bounded publication subset—the primary result, truncation record, truncation figure, external output manifest, tutorial source record, executed

Wannier90 input and diff, and local checksums—is retained under `calculations/ICMSEP2026/conference/paper_1/silicon-wannier-kinetic-preview/`. Its hashes bind bytes observed by the analysis, but the analysis did not consume a pre-existing authenticated production-lineage manifest. Its runtime used content-bound working-tree source over a base commit that alone is insufficient to reconstruct the implementation. The numerical values below are therefore tutorial-scale prototype observations, not promoted production evidence.

The numerical design is the official Wannier90 3.1.0 `examples/example11` silicon tutorial, not an arbitrary choice of exploratory settings. A post-hoc byte-identity check confirms that the source run's four original files match the retained upstream tutorial records. The executed `silicon.win` is a derived copy adding only `write_hr=true` and `write_u_matrices=true` output-artifact requests; it changes no tutorial numerical setting. The tutorial uses the ultrasoft PBE pseudopotential reference `Si.pbe-n-van.UPF`, a 25-Ry wavefunction cutoff, an unshifted $10 \times 10 \times 10$ SCF mesh, a $4 \times 4 \times 4$ Wannier mesh, 12 parent bands, four retained Wannier functions, and 6.5 eV frozen- and outer-window ceilings. This makes it an appropriate interface and analysis preview, but not a production design: it cannot be joined to the PseudoDojo C48/K8 branch by name, material, or software version alone. The tutorial's familiar smooth band diagrams test interpolation of the total represented Hamiltonian; their visual quality does not certify separate locality or short-range compressibility of the represented kinetic and non-kinetic components.

M.7.1 Same-frame construction

At each of the 64 retained reciprocal samples, the prototype composes the rectangular disentanglement matrix and square Wannier gauge as

$$W(\mathbf{k}) = U_{\text{dis}}(\mathbf{k})U(\mathbf{k}). \tag{M.4}$$

Let $C(\mathbf{k})$ contain the recovered pseudo-wavefunction plane-wave coefficients and let $\boldsymbol{\epsilon}(\mathbf{k})$ contain the parent band energies. The represented operators are

$$H^W(\mathbf{k}) = W(\mathbf{k})^\dagger \,\mathrm{diag}\big(\boldsymbol{\epsilon}(\mathbf{k})\big) W(\mathbf{k}), \tag{M.5}$$

$$T^W(\mathbf{k}) = W(\mathbf{k})^\dagger C(\mathbf{k})^\dagger \,\mathrm{diag}\left(\frac{|\mathbf{k}+\mathbf{G}|^2}{2}\right) C(\mathbf{k}) W(\mathbf{k}), \tag{M.6}$$

$$R^W(\mathbf{k}) = H^W(\mathbf{k}) - T^W(\mathbf{k}). \tag{M.7}$$

The kinetic diagonal is evaluated in Hartree before conversion to electron-volts. The remainder R^W is a finite represented non-kinetic remainder; it is not identified with a continuous local scalar potential. It may contain represented local, nonlocal-pseudopotential, Hartree, exchange–correlation, and other parent-Hamiltonian contributions.

The source-backed convention audit reports a maximum UNK Fourier round-trip defect of 2.16×10^{-15} and a maximum Parseval defect of 8.88×10^{-16}. The 25-Ry geometric support contains 537–568 plane waves over the mesh. The even $24 \times 24 \times 24$ FFT grid has signed-Nyquist ambiguity, so Nyquist-plane coordinates were excluded instead of assigned by preference; their largest observed band weight was 1.26×10^{-32} for this source run.

Because the pseudopotential is ultrasoft, the parent eigenproblem has a generalized overlap. The ordinary pseudo-coefficient norms, observed between 0.98598 and 1.02498, were not forced to one. The UNK payload does not contain the beta-projector augmentation required to reconstruct the overlap matrix independently. The generalized-overlap interpretation is therefore source-backed but not numerically reconstructed from UNK bytes alone.

M.7.2 Reported finite-representation diagnostics

The prototype reports reciprocal decomposition and Hermiticity defects at approximately 10^{-15} eV and Fourier round-trip defects below 7×10^{-15} eV. The independent comparison with native `hr.dat` differs by $1.2763350779 \times 10^{-5}$ eV in maximum Frobenius norm on the retained construction and by approximately 1.06×10^{-5} eV along the interpolated source-input path. That cross-route mismatch is not folded into the same-route algebraic closure.

For each Hamiltonian eigenstate $v_n(\mathbf{k})$, the reported expectation identity is

$$E_n(\mathbf{k}) = v_n^\dagger T^W v_n + v_n^\dagger R^W v_n, \tag{M.8}$$

with maximum closure defect 5.33×10^{-15} eV over 256 state samples. This does not define separate kinetic and remainder band structures: T^W and R^W are generally off diagonal in the Hamiltonian eigenbasis, and their individual expectations inside an exactly degenerate subspace depend on the chosen eigenvectors.

M.7.3 Local rank-three expansion at Γ

The reported represented Hamiltonian at Γ contains one lower state and an upper numerical triplet,

$$\operatorname{eig} H^W(\Gamma) = (-5.6982916462,\ 6.3868784657,\ 6.3868784667,\ 6.3868784673)\ \text{eV}. \tag{M.9}$$

The triplet splitting is 1.57×10^{-9} eV, below the declared 10^{-8} eV degeneracy tolerance. The analysis therefore uses a rank-three selected-space quadratic tensor rather than differentiating three independently ordered scalar bands through the degeneracy.

Analytic Cartesian derivatives of the Wigner–Seitz interpolation were checked by centered finite differences at seven step sizes. Reported gradient and Hessian convergence orders were approximately two for H^W, T^W, and R^W. The quadratic model's largest ordered-eigenvalue error over the tested directions and radii increased from 2.67×10^{-10} eV at the smallest reported offset to 4.29×10^{-5} eV at the largest. This checks the local expansion of the retained interpolation; it does not check reciprocal-mesh or parent-model convergence.

The remote-state quadratic term contains kinetic–kinetic, remainder–remainder, and kinetic–remainder products. The reported cross-term norm reaches 8.864 eV bohr^2. Separate additive "kinetic" and "remainder" effective masses are therefore not defined by this construction. The prototype deliberately reports directional multiband curvatures without converting them to silicon effective masses.

M.7.4 Finite-mesh block truncation

The 64 reciprocal matrices define exactly 64 centered lattice representatives. The prototype tested inversion-closed radius truncations and reconstructed only the same retained reciprocal mesh. Selected maximum relative matrix errors are summarized in Table M.3.

Table M.3: Reported tutorial-scale finite-mesh truncation errors. The final column gives the maximum ordered-eigenvalue error for $H^W/T^W/R^W$.

Blocks	Radius (bohr)	H^W	T^W	R^W	Error (eV)
1	0.000	0.594	0.376	0.206	$4.04/6.35/2.51$
13	7.212	0.0665	0.0688	0.0641	$0.558/1.467/0.942$
57	24.985	0.00196	0.00332	0.00421	$0.0207/0.0843/0.0636$
63	29.738	0.000594	0.00112	0.00127	$0.00426/0.0202/0.0159$
64	35.334	roundoff	roundoff	roundoff	roundoff

The data do not support an aggressive short-range truncation. Even 57 retained blocks permit a maximum kinetic eigenvalue error of about 84 meV. Thus a convincing total-band diagram and accurate complete finite reconstruction can coexist with componentwise truncation errors. This is a negative result about compression on one small periodic mesh, not an infinite-lattice localization conclusion and not proof that a separately fitted short-range model is impossible.

M.8 What has not yet been established

The preliminary work does not supply any of the following:

- an accepted zero-pressure PBE-relaxed lattice and production SCF parent;
- a resolved observable-effect disposition for the recurring IEEE flags;
- a cutoff–mesh interaction check or an infinite-setting error bound;
- authenticated common lineage connecting the provisional PseudoDojo parent to the exploratory ultrasoft Wannier source run;
- an accepted Wannier rank, projection family, windows, gauge policy, mesh-convergence result, or independent withheld interpolation test for the production branch;
- a converged conduction-valley coordinate, longitudinal or transverse electron effective mass, or compatible SOC/spinor valence analysis;
- an aligned sp^3s^* operator fit, model-class selection, or route comparison; or
- scientific validation or propagated uncertainty for a declared use.

The next scientifically meaningful sequence remains the one defined by the production program: close the warning and finite-setting interaction boundary, obtain and accept the zero-pressure lattice, construct the production SCF parent, run purpose-specific path and local-valley children, and only then freeze a production Wannier design. The exploratory decomposition is useful as a prototype of representation and diagnostic machinery, but it cannot bypass those gates.

References

[1] G. Angelone, P. Facchi, and M. Ligabò. "Classical Echoes of Quantum Boundary Conditions". In: *Journal of Physics A: Mathematical and Theoretical* 57 (2024), p. 425304. DOI: 10.1088/1751-8121/ad7428.

[2] A. Baldereschi and N. O. Lipari. "Spherical Model of Shallow Acceptor States in Semiconductors". In: *Physical Review B* 8 (1973), pp. 2697–2709. DOI: 10.1103/PhysRevB.8.2697.

[3] J. Bect, S. Zio, G. Perrin, C. Cannamela, and E. Vazquez. "On the Quantification of Discretization Uncertainty: Comparison of Two Paradigms". In: *14th World Congress in Computational Mechanics and ECCOMAS Congress 2020*. 2021. DOI: 10.23967/wccm-eccomas.2020.260. arXiv: 2103.14559.

[4] M. Benzi, P. Boito, and N. Razouk. "Decay Properties of Spectral Projectors with Applications to Electronic Structure". In: *SIAM Review* 55.1 (2013), pp. 3–64. DOI: 10.1137/100814019.

[5] T. Berlijn, D. Volja, and W. Ku. "Can Disorder Alone Destroy the e'_g Hole Pockets of Na_xCoO_2? A Wannier Function Based First-Principles Method for Disordered Systems". In: *Physical Review Letters* 106.7 (2011), p. 077005. DOI: 10.1103/PhysRevLett.106.077005.

[6] Y. Bertot and P. Castéran. *Interactive Theorem Proving and Program Development: Coq'Art: The Calculus of Inductive Constructions*. Texts in Theoretical Computer Science. An EATCS Series. Berlin and Heidelberg: Springer, 2004. DOI: 10.1007/978-3-662-07964-5.

[7] Å. Björck and G. H. Golub. "Numerical Methods for Computing Angles Between Linear Subspaces". In: *Mathematics of Computation* 27.123 (1973), pp. 579–594. DOI: 10.1090/S0025-5718-1973-0348991-3.

[8] P. E. Blöchl. "Projector Augmented-Wave Method". In: *Physical Review B* 50.24 (1994), pp. 17953–17979. DOI: 10.1103/PhysRevB.50.17953.

[9] G. Bonneau, J. Faraut, and G. Valent. "Self-adjoint Extensions of Operators and the Teaching of Quantum Mechanics". In: *American Journal of Physics* 69.3 (2001), pp. 322–331. DOI: 10.1119/1.1328351.

[10] M. Born and R. Oppenheimer. "Zur Quantentheorie der Molekeln". In: *Annalen der Physik* 389.20 (1927), pp. 457–484. DOI: 10.1002/andp.19273892002.

[11] T. B. Boykin, G. Klimeck, and F. Oyafuso. “Valence Band Effective-Mass Expressions in the $sp^3d^5s^*$ Empirical Tight-Binding Model Applied to a Si and Ge Parametrization”. In: *Physical Review B* 69 (2004), p. 115201. DOI: 10.1103/PhysRevB.69.115201.

[12] S. Bravyi, D. P. DiVincenzo, and D. Loss. “Schrieffer–Wolff Transformation for Quantum Many-Body Systems”. In: *Annals of Physics* 326.10 (2011), pp. 2793–2826. DOI: 10.1016/j.aop.2011.06.004.

[13] C. Brouder, G. Panati, M. Calandra, C. Mourougane, and N. Marzari. “Exponential Localization of Wannier Functions in Insulators”. In: *Physical Review Letters* 98.4 (2007), p. 046402. DOI: 10.1103/PhysRevLett.98.046402.

[14] M. G. Burt. “The Justification for Applying the Effective-Mass Approximation to Microstructures”. In: *Journal of Physics: Condensed Matter* 4.32 (1992), pp. 6651–6690. DOI: 10.1088/0953-8984/4/32/003.

[15] M. G. Burt. “Fundamentals of Envelope Function Theory for Electronic States and Photonic Modes in Nanostructures”. In: *Journal of Physics: Condensed Matter* 11.9 (1999), R53–R83. DOI: 10.1088/0953-8984/11/9/002.

[16] É. Cancès, A. Levitt, G. Panati, and G. Stoltz. “Robust Determination of Maximally Localized Wannier Functions”. In: *Physical Review B* 95.7 (2017), p. 075114. DOI: 10.1103/PhysRevB.95.075114.

[17] I. Carnimeo et al. “Quantum ESPRESSO: One Further Step toward the Exascale”. In: *Journal of Chemical Theory and Computation* 19.20 (2023), pp. 6992–7006. DOI: 10.1021/acs.jctc.3c00249.

[18] G. C. Cawley and N. L. C. Talbot. “On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation”. In: *Journal of Machine Learning Research* 11 (2010), pp. 2079–2107.

[19] J. R. Chelikowsky, N. Troullier, and Y. Saad. “Finite-Difference-Pseudopotential Method: Electronic Structure Calculations without a Basis”. In: *Physical Review Letters* 72.8 (1994), pp. 1240–1243. DOI: 10.1103/PhysRevLett.72.1240.

[20] J. des Cloizeaux. “Energy Bands and Projection Operators in a Crystal: Analytic and Asymptotic Properties”. In: *Physical Review* 135.3A (1964), A685–A697. DOI: 10.1103/PhysRev.135.A685.

[21] F. Corsetti and A. A. Mostofi. “System-Size Convergence of Point Defect Properties: The Case of the Silicon Vacancy”. In: *Physical Review B* 84.3 (2011), p. 035209. DOI: 10.1103/PhysRevB.84.035209.

[22] P. V. Coveney, D. Groen, and A. G. Hoekstra. “Reliability and Reproducibility in Computational Science: Implementing Validation, Verification and Uncertainty Quantification In Silico”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 379.2197 (2021), p. 20200409. DOI: 10.1098/rsta.2020.0409.

[23] A. Damle and L. Lin. “Disentanglement via Entanglement: A Unified Method for Wannier Localization”. In: *Multiscale Modeling & Simulation* 16.3 (2018), pp. 1392–1410. DOI: 10.1137/17M1129696.

[24] C. Davis and W. M. Kahan. “The Rotation of Eigenvectors by a Perturbation. III”. In: *SIAM Journal on Numerical Analysis* 7.1 (1970), pp. 1–46. DOI: 10.1137/0707001.

[25] DeepModeling. *DeePTB: Efficient Deep Learning-Based Tight-Binding Approach*. 2024.

[26] V. Duchêne, I. Vukićević, and M. I. Weinstein. “Homogenized Description of Defect Modes in Periodic Structures with Localized Defects”. In: *Communications in Mathematical Sciences* 13.3 (2015), pp. 777–823. DOI: 10.4310/CMS.2015.v13.n3.a9.

[27] L. Ermoneit, A. Thayil, T. Koprucki, and M. Kantner. “Exact Multivalley Envelope Function Theory of Valley Splitting in Si/SiGe Nanostructures”. In: *Physical Review B* 113 (2026), p. 245306. DOI: 10.1103/md2x-s44y.

[28] H. Feshbach. “Unified Theory of Nuclear Reactions”. In: *Annals of Physics* 5.4 (1958), pp. 357–390. DOI: 10.1016/0003-4916(58)90007-1.

[29] H. Feshbach. “A Unified Theory of Nuclear Reactions. II”. In: *Annals of Physics* 19.2 (1962), pp. 287–313. DOI: 10.1016/0003-4916(62)90221-X.

[30] B. A. Foreman. “Effective-Mass Hamiltonian and Boundary Conditions for the Valence Bands of Semiconductor Microstructures”. In: *Physical Review B* 48.7 (1993), pp. 4964–4967. DOI: 10.1103/PhysRevB.48.4964.

[31] B. A. Foreman. “First-Principles Envelope-Function Theory for Lattice-Matched Semiconductor Heterostructures”. In: *Physical Review B* 72 (2005), p. 165345. DOI: 10.1103/PhysRevB.72.165345.

[32] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti, and C. G. Van de Walle. “First-Principles Calculations for Point Defects in Solids”. In: *Reviews of Modern Physics* 86.1 (2014), pp. 253–305. DOI: 10.1103/RevModPhys.86.253.

[33] C. Freysoldt, J. Neugebauer, and C. G. Van de Walle. “Fully Ab Initio Finite-Size Corrections for Charged-Defect Supercell Calculations”. In: *Physical Review Letters* 102.1 (2009), p. 016402. DOI: 10.1103/PhysRevLett.102.016402.

[34] H. P. Fried, D. Barragan-Yani, F. Libisch, and L. Wirtz. “A Machine Learning Approach to Predict Tight-Binding Parameters for Point Defects via the Projected Density of States”. In: *npj Computational Materials* 11.1 (2025), p. 176. DOI: 10.1038/s41524-025-01634-1.

[35] T. Fukui, Y. Hatsugai, and H. Suzuki. “Chern Numbers in Discretized Brillouin Zone: Efficient Method of Computing (Spin) Hall Conductances”. In: *Journal of the Physical Society of Japan* 74.6 (2005), pp. 1674–1677. DOI: 10.1143/JPSJ.74.1674.

[36] J. K. Gamble, N. T. Jacobson, E. Nielsen, A. D. Baczewski, J. E. Moussa, I. Montaño, and R. P. Muller. “Multivalley Effective Mass Theory Simulation of Donors in Silicon”. In: *Physical Review B* 91.23 (2015), p. 235318. DOI: 10.1103/PhysRevB.91.235318.

[37] P. Giannozzi et al. “Advanced Capabilities for Materials Modelling with Quantum ESPRESSO”. In: *Journal of Physics: Condensed Matter* 29.46 (2017), p. 465901. DOI: 10.1088/1361-648X/aa8f79.

[38] P. Giannozzi et al. “Quantum ESPRESSO toward the Exascale”. In: *The Journal of Chemical Physics* 152.15 (2020), p. 154105. DOI: 10.1063/5.0005082.

[39] P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, et al. “QUANTUM ESPRESSO: A Modular and Open-Source Software Project for Quantum Simulations of Materials”. In: *Journal of Physics: Condensed Matter* 21.39 (2009), p. 395502. DOI: 10.1088/0953-8984/21/39/395502.

[40] G. H. Golub and C. F. Van Loan. *Matrix Computations*. 4th ed. Johns Hopkins Studies in the Mathematical Sciences. Baltimore, MD: Johns Hopkins University Press, 2013. ISBN: 978-1-4214-0794-4.

[41] D. Groen et al. “VECMAtk: A Scalable Verification, Validation and Uncertainty Quantification Toolkit for Scientific Simulations”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 379.2197 (2021), p. 20200221. DOI: 10.1098/rsta.2020.0221.

[42] Q. Gu, Z. Zhouyin, S. K. Pandey, P. Zhang, L. Zhang, and W. E. “Deep Learning Tight-Binding Approach for Large-Scale Electronic Simulations at Finite Temperatures with Ab Initio Accuracy”. In: *Nature Communications* 15.1 (2024), p. 6772. DOI: 10.1038/s41467-024-51006-4.

[43] F. D. M. Haldane. “Model for a Quantum Hall Effect without Landau Levels: Condensed-Matter Realization of the “Parity Anomaly””. In: *Physical Review Letters* 61.18 (1988), pp. 2015–2018. DOI: 10.1103/PhysRevLett.61.2015.

[44] D. R. Hamann. “Optimized Norm-Conserving Vanderbilt Pseudopotentials”. In: *Physical Review B* 88 (2013), p. 085117. DOI: 10.1103/PhysRevB.88.085117.

[45] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, et al. “Array Programming with NumPy”. In: *Nature* 585 (2020), pp. 357–362. DOI: 10.1038/s41586-020-2649-2.

[46] N. J. Higham. “Computing the Polar Decomposition—with Applications”. In: *SIAM Journal on Scientific and Statistical Computing* 7.4 (1986), pp. 1160–1174. DOI: 10.1137/0907079.

[47] D. Hilbert. “Mathematical Problems”. In: *Bulletin of the American Mathematical Society* 8.10 (1902). Translated by Mary Winston Newson, pp. 437–479. DOI: 10.1090/S0002-9904-1902-00923-3.

[48] M. A. Hoefer and M. I. Weinstein. “Defect Modes and Homogenization of Periodic Schrödinger Operators”. In: *SIAM Journal on Mathematical Analysis* 43.2 (2011), pp. 971–996. DOI: 10.1137/100807302.

[49] D. R. Hofstadter. “Energy Levels and Wave Functions of Bloch Electrons in Rational and Irrational Magnetic Fields”. In: *Physical Review B* 14.6 (1976), pp. 2239–2249. DOI: 10.1103/PhysRevB.14.2239.

[50] P. Hohenberg and W. Kohn. “Inhomogeneous Electron Gas”. In: *Physical Review* 136 (1964), B864–B871. DOI: 10.1103/PhysRev.136.B864.

[51] R. A. Horn and C. R. Johnson. *Matrix Analysis*. 2nd ed. Cambridge: Cambridge University Press, 2013. DOI: 10.1017/CBO9781139020411.

[52] H. Kang and L. Muechler. *Predicting Defect States: A Quick Screening Protocol for Substitutional Point Defect Engineering*. Preprint, version 1, 8 June 2026. 2026. DOI: 10.48550/arXiv.2606.10139. arXiv: 2606.10139 [cond-mat.mtrl-sci].

[53] T. Kato. *Perturbation Theory for Linear Operators*. 2nd ed. Classics in Mathematics. Berlin and Heidelberg: Springer, 1995. DOI: 10.1007/978-3-642-66282-9.

[54] R. D. King-Smith and D. Vanderbilt. “Theory of Polarization of Crystalline Solids”. In: *Physical Review B* 47 (1993), pp. 1651–1654. DOI: 10.1103/PhysRevB.47.1651.

[55] C. Kittel. *Introduction to Solid State Physics*. 8th ed. Hoboken, New Jersey: John Wiley & Sons, 2004. ISBN: 978-0-471-41526-8.

[56] G. Klimeck, R. C. Bowen, T. B. Boykin, C. Salazar-Lazaro, T. A. Cwik, and A. Stoica. “Si Tight-Binding Parameters from Genetic Algorithm Fitting”. In: *Superlattices and Microstructures* 27.2–3 (2000), pp. 77–88. DOI: 10.1006/spmi.1999.0797.

[57] W. Kohn. “Analytic Properties of Bloch Waves and Wannier Functions”. In: *Physical Review* 115 (1959), pp. 809–821. DOI: 10.1103/PhysRev.115.809.

[58] W. Kohn and J. M. Luttinger. “Theory of Donor States in Silicon”. In: *Physical Review* 98 (1955), pp. 915–922. DOI: 10.1103/PhysRev.98.915.

[59] W. Kohn and L. J. Sham. “Self-Consistent Equations Including Exchange and Correlation Effects”. In: *Physical Review* 140 (1965), A1133–A1138. DOI: 10.1103/PhysRev.140.A1133.

[60] H.-P. Komsa, T. T. Rantala, and A. Pasquarello. “Finite-Size Supercell Correction Schemes for Charged Defect Calculations”. In: *Physical Review B* 86.4 (2012), p. 045112. DOI: 10.1103/PhysRevB.86.045112.

[61] S. König, D. Lee, and H.-W. Hammer. “Volume Dependence of Bound States with Angular Momentum”. In: *Physical Review Letters* 107.11 (2011), p. 112001. DOI: 10.1103/PhysRevLett.107.112001.

[62] G. F. Koster and J. C. Slater. “Wave Functions for Impurity Levels”. In: *Physical Review* 95.5 (1954), pp. 1167–1176. DOI: 10.1103/PhysRev.95.1167.

[63] P. Kuchment. “An Overview of Periodic Elliptic Operators”. In: *Bulletin of the American Mathematical Society* 53.3 (2016), pp. 343–414. DOI: 10.1090/bull/1528.

[64] H. Li, Z. Wang, N. Zou, M. Ye, R. Xu, X. Gong, W. Duan, and Y. Xu. “Deep-Learning Density Functional Theory Hamiltonian for Efficient Ab Initio Electronic-Structure Calculation”. In: *Nature Computational Science* 2.6 (2022), pp. 367–377. DOI: 10.1038/s43588-022-00265-6.

[65] Y. Li, X. Zhang, M. Liu, and L. Shen. “A Critical Review of Machine Learning Interatomic Potentials and Hamiltonian”. In: *Journal of Materials Informatics* 5.4 (2025). DOI: 10.20517/jmi.2025.17.

[66] J.-M. Lihm and C.-H. Park. “Reliable Methods for Seamless Stitching of Tight-Binding Models Based on Maximally Localized Wannier Functions”. In: *Physical Review B* 99.12 (2019), p. 125117. DOI: 10.1103/PhysRevB.99.125117.

[67] P.-O. Löwdin. “Studies in Perturbation Theory. IV. Solution of Eigenvalue Problem by Projection Operator Formalism”. In: *Journal of Mathematical Physics* 3.5 (1962), pp. 969–982. DOI: 10.1063/1.1724312.

[68] P.-O. Löwdin. “Partitioning Technique, Perturbation Theory, and Rational Approximations”. In: *International Journal of Quantum Chemistry* 21.1 (1982), pp. 69–92. DOI: 10.1002/qua.560210105.

[69] I.-T. Lu, J. Park, J.-J. Zhou, and M. Bernardi. “Ab Initio Electron-Defect Interactions Using Wannier Functions”. In: *npj Computational Materials* 6 (2020), p. 17. DOI: 10.1038/s41524-020-0284-y.

[70] J. M. Luttinger and W. Kohn. “Motion of Electrons and Holes in Perturbed Periodic Fields”. In: *Physical Review* 97.4 (1955), pp. 869–883. DOI: 10.1103/PhysRev.97.869.

[71] R. M. Martin. *Electronic Structure: Basic Theory and Practical Methods*. 2nd ed. Cambridge University Press, 2020. ISBN: 9781108429900. DOI: 10.1017/9781108555586.

[72] N. Marzari, A. A. Mostofi, J. R. Yates, I. Souza, and D. Vanderbilt. “Maximally Localized Wannier Functions: Theory and Applications”. In: *Reviews of Modern Physics* 84 (2012), pp. 1419–1475. DOI: 10.1103/RevModPhys.84.1419.

[73] N. Marzari and D. Vanderbilt. “Maximally Localized Generalized Wannier Functions for Composite Energy Bands”. In: *Physical Review B* 56.20 (1997), pp. 12847–12865. DOI: 10.1103/PhysRevB.56.12847.

[74] N. W. McLachlan. *Theory and Application of Mathieu Functions*. Oxford: Clarendon Press, 1947.

[75] M. Milanese, J. Norton, H. Piet-Lahanier, and É. Walter, eds. *Bounding Approaches to System Identification*. New York: Plenum Press, 1996. DOI: 10.1007/978-1-4757-9545-5.

[76] H. J. Monkhorst and J. D. Pack. “Special Points for Brillouin-Zone Integrations”. In: *Physical Review B* 13 (1976), pp. 5188–5192. DOI: 10.1103/PhysRevB.13.5188.

[77] R. E. Moore, R. B. Kearfott, and M. J. Cloud. *Introduction to Interval Analysis*. Philadelphia: Society for Industrial and Applied Mathematics, 2009. DOI: 10.1137/1.9780898717716.

[78] A. A. Mostofi, J. R. Yates, Y.-S. Lee, I. Souza, D. Vanderbilt, and N. Marzari. “wannier90: A Tool for Obtaining Maximally-Localised Wannier Functions”. In: *Computer Physics Communications* 178.9 (2008), pp. 685–699. DOI: 10.1016/j.cpc.2007.11.016.

[79] L. de Moura and S. Ullrich. “The Lean 4 Theorem Prover and Programming Language”. In: *Automated Deduction—CADE 28*. Vol. 12699. Lecture Notes in Computer Science. Cham: Springer, 2021, pp. 625–635. DOI: 10.1007/978-3-030-79876-5_37.

[80] J. I. Mustafa, S. Coh, M. L. Cohen, and S. G. Louie. “Automated Construction of Maximally Localized Wannier Functions: Optimized Projection Functions Method”. In: *Physical Review B* 92.16 (2015), p. 165134. DOI: 10.1103/PhysRevB.92.165134.

[81] S. Nakamura and Y. Tadano. “On a Continuum Limit of Discrete Schrödinger Operators on Square Lattices”. In: *Journal of Spectral Theory* 11.1 (2021), pp. 355–367. DOI: 10.4171/JST/343.

[82] National Research Council. *Assessing the Reliability of Complex Models: Mathematical and Statistical Foundations of Verification, Validation, and Uncertainty Quantification*. Washington, DC: The National Academies Press, 2012. DOI: 10.17226/13395.

[83] A. Neumaier. “Complete Search in Continuous Global Optimization and Constraint Satisfaction”. In: *Acta Numerica* 13 (2004), pp. 271–369. DOI: 10.1017/S0962492904000194.

[84] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, 2010.

[85] T. Nipkow, L. C. Paulson, and M. Wenzel. *Isabelle/HOL: A Proof Assistant for Higher-Order Logic*. Vol. 2283. Lecture Notes in Computer Science. Berlin and Heidelberg: Springer, 2002. DOI: 10.1007/3-540-45949-9.

[86] J. Paier, M. Marsman, K. Hummer, G. Kresse, I. C. Gerber, and J. G. Ángyán. “Screened Hybrid Density Functionals Applied to Solids”. In: *The Journal of Chemical Physics* 124.15 (2006), p. 154709. DOI: 10.1063/1.2187006.

[87] A. Parzygnat, K. K. Y. Lee, Y. Avniel, and S. G. Johnson. “Sufficient Conditions for Two-Dimensional Localization by Arbitrarily Weak Defects in Periodic Potentials with Band Gaps”. In: *Physical Review B* 81.15 (2010), p. 155324. DOI: 10.1103/PhysRevB.81.155324.

[88] M. C. Payne, M. P. Teter, D. C. Allan, T. A. Arias, and J. D. Joannopoulos. “Iterative Minimization Techniques for Ab Initio Total-Energy Calculations: Molecular Dynamics and Conjugate Gradients”. In: *Reviews of Modern Physics* 64.4 (1992), pp. 1045–1097. DOI: 10.1103/RevModPhys.64.1045.

[89] J. P. Perdew, K. Burke, and M. Ernzerhof. “Generalized Gradient Approximation Made Simple”. In: *Physical Review Letters* 77 (1996), pp. 3865–3868. DOI: 10.1103/PhysRevLett.77.3865.

[90] J. P. Perdew and M. Levy. “Physical Content of the Exact Kohn–Sham Orbital Energies: Band Gaps and Derivative Discontinuities”. In: *Physical Review Letters* 51.20 (1983), pp. 1884–1887. DOI: 10.1103/PhysRevLett.51.1884.

[91] G. Pizzi, V. Vitale, R. Arita, S. Blügel, F. Freimuth, G. Géranton, M. Gibertini, D. Gresch, C. Johnson, T. Koretsune, et al. “Wannier90 as a Community Code: New Features and Applications”. In: *Journal of Physics: Condensed Matter* 32.16 (2020), p. 165902. DOI: 10.1088/1361-648X/ab51ff.

[92] E. Prodan and W. Kohn. “Nearsightedness of Electronic Matter”. In: *Proceedings of the National Academy of Sciences* 102.33 (2005), pp. 11635–11638. DOI: 10.1073/pnas.0505436102.

[93] PseudoDojo developers. *PseudoDojo: Python Package to Generate and Validate Pseudopotentials*.

[94] X.-L. Qi, Y.-S. Wu, and S.-C. Zhang. “Topological Quantization of the Spin Hall Effect in Two-Dimensional Paramagnetic Semiconductors”. In: *Physical Review B* 74.8 (2006), p. 085308. DOI: 10.1103/PhysRevB.74.085308.

[95] Quantum ESPRESSO Foundation. *Documentation for Quantum ESPRESSO*.

[96] Quantum ESPRESSO Foundation. *pw.x: Input Description*. Official online input reference; record the consulted software/documentation version and access date at editorial freeze. Quantum ESPRESSO Foundation.

[97] M. Reed and B. Simon. *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*. New York: Academic Press, 1975.

[98] C. Rostgaard. *The Projector Augmented-Wave Method*. 2009. DOI: 10.48550/arXiv.0910.1921. arXiv: 0910.1921 [cond-mat.mtrl-sci].

[99] O. Rubel, F. Tran, X. Rocquefelte, and P. Blaha. “Perturbation Approach to Ab Initio Effective Mass Calculations”. In: *Computer Physics Communications* 261 (2021), p. 107648. DOI: 10.1016/j.cpc.2020.107648.

[100] W. Rudin. *Principles of Mathematical Analysis*. 3rd ed. New York: McGraw-Hill, 1976.

[101] G. K. Sandve, A. Nekrutenko, J. Taylor, and E. Hovig. “Ten Simple Rules for Reproducible Computational Research”. In: *PLOS Computational Biology* 9.10 (2013), e1003285. DOI: 10.1371/journal.pcbi.1003285.

[102] C. Schattauer, M. Todorović, K. Ghosh, P. Rinke, and F. Libisch. “Machine Learning Sparse Tight-Binding Parameters for Defects”. In: *npj Computational Materials* 8.1 (2022), p. 116. DOI: 10.1038/s41524-022-00791-x.

[103] K. Schmüdgen. *Unbounded Self-adjoint Operators on Hilbert Space*. Vol. 265. Graduate Texts in Mathematics. Dordrecht: Springer, 2012. DOI: 10.1007/978-94-007-4753-1.

[104] P. H. Schönemann. “A Generalized Solution of the Orthogonal Procrustes Problem”. In: *Psychometrika* 31.1 (1966), pp. 1–10. DOI: 10.1007/BF02289451.

[105] M. Schwade, S. Zhang, F. Vonhoff, F. P. Delgado, and D. A. Egger. “Physics-Informed Hamiltonian Learning for Large-Scale Optoelectronic Property Prediction”. In: *Nature Communications* 17.1 (2026), p. 2652. DOI: 10.1038/s41467-026-70865-7.

[106] M. J. van Setten, M. Giantomassi, E. Bousquet, M. J. Verstraete, D. R. Hamann, X. Gonze, and G.-M. Rignanese. “The PseudoDojo: Training and Grading a 85 Element Optimized Norm-Conserving Pseudopotential Table”. In: *Computer Physics Communications* 226 (2018), pp. 39–54. DOI: 10.1016/j.cpc.2018.01.012.

[107] L. J. Sham and M. Schlüter. “Density-Functional Theory of the Energy Gap”. In: *Physical Review Letters* 51.20 (1983), pp. 1888–1891. DOI: 10.1103/PhysRevLett.51.1888.

[108] R. Shankar. *Principles of Quantum Mechanics*. 2nd ed. New York: Plenum Press, 1994. DOI: 10.1007/978-1-4757-0576-8.

[109] H. Shi, Y. Xie, T. Zhang, W. Chu, and X.-G. Gong. *First-Principles Quantum-Spectral Framework for Elementary Vortex Pinning in Superconductors*. Preprint, version 2, 4 August 2026. 2026. DOI: 10.48550/arXiv.2606.25862. arXiv: 2606.25862 [cond-mat.supr-con].

[110] B. Simon. “The Bound State of Weakly Coupled Schrödinger Operators in One and Two Dimensions”. In: *Annals of Physics* 97.2 (1976), pp. 279–288. DOI: 10.1016/0003-4916(76)90038-5.

[111] B. Simon. *Trace Ideals and Their Applications*. 2nd ed. Vol. 120. Mathematical Surveys and Monographs. Providence, RI: American Mathematical Society, 2005. DOI: 10.1090/surv/120.

[112] S. H. Simon. *The Oxford Solid State Basics*. Oxford: Oxford University Press, 2013. ISBN: 978-0-19-968076-4.

[113] J. C. Slater and G. F. Koster. “Simplified LCAO Method for the Periodic Potential Problem”. In: *Physical Review* 94 (1954), pp. 1498–1524. DOI: 10.1103/PhysRev.94.1498.

[114] A. A. Soluyanov and D. Vanderbilt. “Computing Topological Invariants without Inversion Symmetry”. In: *Physical Review B* 83.23 (2011), p. 235401. DOI: 10.1103/PhysRevB.83.235401.

[115] I. Souza, N. Marzari, and D. Vanderbilt. “Maximally Localized Wannier Functions for Entangled Energy Bands”. In: *Physical Review B* 65 (2001), p. 035109. DOI: 10.1103/PhysRevB.65.035109.

[116] F. Starrost, S. Bornholdt, C. Solterbeck, and W. Schattke. “Band-Structure Parameters by Genetic Algorithm”. In: *Physical Review B* 53 (1996). Erratum: Physical Review B 54, 17226 (1996), pp. 12549–12552. DOI: 10.1103/PhysRevB.53.12549.

[117] M. D. Stiles. “Generalized Slater–Koster Method for Fitting Band Structures”. In: *Physical Review B* 55 (1997), pp. 4168–4173. DOI: 10.1103/PhysRevB.55.4168.

[118] J. C. Strikwerda. *Finite Difference Schemes and Partial Differential Equations*. 2nd ed. Philadelphia, PA: Society for Industrial and Applied Mathematics, 2004. DOI: 10.1137/1.9780898717938.

[119] S. M. Sze and M.-K. Lee. *Semiconductor Devices: Physics and Technology*. 3rd ed. Hoboken, New Jersey: John Wiley & Sons, 2012. ISBN: 978-0-470-53794-7.

[120] C. Tantardini et al. "Generating and Grading 34 Optimised Norm-Conserving Vanderbilt Pseudopotentials for Actinides and Super-Heavy Elements in the PseudoDojo". In: *Computer Physics Communications* 295 (2024), p. 109002. DOI: 10.1016/j.cpc.2023.109002.

[121] G. Teschl. *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*. 2nd ed. Vol. 157. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2014. DOI: 10.1090/gsm/157.

[122] The mathlib Community. "The Lean Mathematical Library". In: *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*. New York, NY, USA: Association for Computing Machinery, 2020, pp. 367–381. DOI: 10.1145/3372885.3373824.

[123] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs. "Quantized Hall Conductance in a Two-Dimensional Periodic Potential". In: *Physical Review Letters* 49.6 (1982), pp. 405–408. DOI: 10.1103/PhysRevLett.49.405.

[124] L. N. Trefethen. *Spectral Methods in MATLAB*. Vol. 10. Software, Environments, and Tools. Philadelphia, PA: Society for Industrial and Applied Mathematics, 2000. DOI: 10.1137/1.9780898719598.

[125] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, et al. "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python". In: *Nature Methods* 17 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

[126] V. Vitale, G. Pizzi, A. Marrazzo, J. R. Yates, N. Marzari, and A. A. Mostofi. "Automated High-Throughput Wannierisation". In: *npj Computational Materials* 6 (2020), p. 66. DOI: 10.1038/s41524-020-0312-y.

[127] P. Vogl, H. P. Hjalmarson, and J. D. Dow. "A Semi-Empirical Tight-Binding Theory of the Electronic Structure of Semiconductors". In: *Journal of Physics and Chemistry of Solids* 44.5 (1983), pp. 365–378. DOI: 10.1016/0022-3697(83)90064-1.

[128] Z. Wang, S. Ye, H. Wang, J. He, Q. Huang, and S. Chang. "Machine Learning Method for Tight-Binding Hamiltonian Parameterization from Ab-Initio Band Structure". In: *npj Computational Materials* 7.1 (2021), p. 11. DOI: 10.1038/s41524-020-00490-5.

[129] G. H. Wannier. “The Structure of Electronic Excitation Levels in Insulating Crystals”. In: *Physical Review* 52 (1937), pp. 191–197. DOI: 10.1103/PhysRev.52.191.

[130] Wannier90 developers. *Wannier90*.

[131] L. J. Wei. “The Accelerated Failure Time Model: A Useful Alternative to the Cox Regression Model in Survival Analysis”. In: *Statistics in Medicine* 11.14–15 (1992), pp. 1871–1879. DOI: 10.1002/sim.4780111409.

[132] D. Ye et al. “Uncertainty Quantification Patterns for Multiscale Models”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 379.2197 (2021), p. 20200072. DOI: 10.1098/rsta.2020.0072.

[133] P. Y. Yu and M. Cardona. *Fundamentals of Semiconductors: Physics and Materials Properties*. 4th ed. Berlin and Heidelberg: Springer, 2010. DOI: 10.1007/978-3-642-00710-1.

[134] Z. Yuan et al. *Deep Learning Density Functional Theory Hamiltonian in Real Space*. 2024. DOI: 10.48550/arXiv.2407.14379. arXiv: 2407.14379 [physics.comp-ph].

[135] Z. Yuan et al. “Deep-Learning Density Functional Theory Hamiltonian in Real Space”. In: *Physical Review Letters* 137.4 (2026), p. 046401. DOI: 10.1103/mbhs-vlby.

[136] H. Zhang, C. Liu, Z. Wang, X. Wei, S. Liu, N. Zheng, B. Shao, and T.-Y. Liu. “Self-Consistency Training for Density-Functional-Theory Hamiltonian Prediction”. In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. Proceedings of Machine Learning Research. PMLR, 2024, pp. 59329–59357.

[137] L. Zhang, B. Onat, G. Dusson, A. McSloy, G. Anand, R. J. Maurer, C. Ortner, and J. R. Kermode. “Equivariant Analytical Mapping of First Principles Hamiltonians to Accurate and Transferable Materials Models”. In: *npj Computational Materials* 8.1 (2022), p. 158. DOI: 10.1038/s41524-022-00843-2.

[138] J. Zou et al. “Review of Machine Learning Tight-Binding Models: Route to Accurate and Scalable Electronic Simulations”. In: *Chinese Physics B* 35.1 (2026), p. 017101. DOI: 10.1088/1674-1056/ae15ef.

Index

abrupt interface
  envelope theory, 418
acceptor, 210
  coupled valence manifold, 415
adjoint, 16
admissibility
  operator, 203
  spectral, 203
admissible set, **162**
  controlled result, 331
  operator, 168
  spectral, 168
aligned subtraction, 259
alignment
  absent by construction in Luttinger–Kohn, 426
  constrained family, 167
  global unitary, 124
  impurity extraction, 212
  independent constructions, 198
  retained spaces, 159, 289
  tight-binding anchored, 261
alignment map
  pristine to doped, 219
approximation
  evidence boundary, 161
array ordering, 4
artificial intelligence
  research assistance, i
audience, ii

band gap
  candidate literature, 435
  sampled isolation, 94
band isolation, 106
band truncation, 413
bandwidth
  scalar hopping model, 112
basis, **9**
  ordered, 12
basis invariant, 19
basis transformation, 17, 158
Bloch boundary condition, **76**
Bloch fiber, **70**
Bloch phase
  quotient seam, 80
Bloch representation, 194
Bloch state, 194
  band-edge basis, 412
Bloch–Wannier bridge, 121
block Hamiltonian
  periodic parent, 119
block hopping
  complete transform, 128
  gauge dependence, **118**
bond term, 221
Born rule, **43**
Born–von Karman representative, 107
boron
  Luttinger–Kohn interpretation, 428
  spin–orbit branch, 227
  substitutional, 210
  transferability, 240
boundary condition
  Burt envelope theory, 420
boundary twist, 80

Brillouin zone
  one-dimensional, 69
Brillouin-zone restriction, 416
Brillouin-zone sampling, 191
bulk model
  unresolved question, 206
bulk observable, 192
bulk program
  dependency sequence, 206
Burt envelope-function theory, **411**
  exact representation, 416
  summary, 429

candidate family, 163
canonical transformation
  Luttinger–Kohn, 413
central hypothesis, 154
central-cell correction, 412
characteristic equation, 28
  differential equation, 39
charge-density cutoff, 191
charged defect
  candidate literature, 433
citation
  prospective literature register, 431
claim
  evidence discipline, 184
  evidence requirement, 141
codomain, 15
commutativity
  reduction paths, 265
compatibility
  before arithmetic, 7
  certified incompatibility, 264
  host and dopant, 209
  impurity extraction, 216
compatibility certificate, **162**
composite band
  external gap, 120
  frame and alignment, **118**
composition
  reduction paths, 241
compressed operator, **91**
compression, **285**, 287
  common additive, 287
  operator, 92
computational quantity, **2**
continuum embedding, 229
convention, **273**
convergence
  bulk parent, 191
  finite-setting, 143, 180
coordinate convention, 279
coordinate dependent, 19
coordinate representation, **38**
coordinate system
  metadata, 4
coordinate transformation, 17
  Laplacian, 77
coordinate vector, 12
cosine potential
  pedagogical parent, 103
  plane-wave and finite-difference representations, 82
crossover
  continuum, 263
crossover radius, 229
curvature
  reduced momentum, 112

data leakage, 182
defect, **212**
defect perturbation, 212
degenerate band
  coupled envelope equations, 415
derived quantity, 180
determinant
  geometric meaning, 28
diagnostic example, 153
diamond structure, 187

differential operator
  characteristic polynomial, 41
dimension, 12
  physical, 3
dipole matrix element
  envelope theory, 422
Dirac notation, **36**
direct integral, 70
Dirichlet boundary, 299
Dirichlet boundary condition, 39, 52
Dirichlet Laplacian, 44
discrete Fourier transform
  band to hopping, 107
discrete Laplacian
  Dirichlet interval, 45
discretization
  comparison, 301
disentanglement, 195
  distinction, 98
domain, 15
  operator, **37**, 285
dominant envelope, 418
donor, 209
dopant, **212**
  research questions, 211
doped system, **212**
downfolding, 262, 287

effective Hamiltonian, 152
  retained-operator distinction, 95
effective mass
  candidate literature, 434
effective-mass equation
  derivation, 411
effective-mass kink, 420
effective-mass model
  continuum limit, 377
  continuum reduction, **229**
  Luttinger–Kohn, 411
effective-mass operator
  position dependent, 418
effective-mass tensor, 347
  band curvature, 414
eigendecomposition, 27
eigenpair residual, 32
eigensolver, 32
eigenspace, **25**
eigensystem, **25**
  ordering, 31
eigenvalue, 26
  degenerate, 30
eigenvector, 26
  phase freedom, 31
ellipse
  admissible set, 169
energy reference, 3, 160
  impurity extraction, 217
  sensitivity, 200
energy zero
  intervalley invariance, 424
energy-reference invariance, 423
envelope function
  band-limited definition, 416
  representation versus trial solution, 421
  simple band, 414
envelope-function equation
  exact nonlocal form, 417
envelope-function theory, **411**
Ermoneit multivalley theory, 423
  summary, 429
error
  accounting, 205
  amplification, 180
  categories, iv
  category separation, 143
  Luttinger–Kohn approximation, 428
evidence
  classes, **140**

common witness, 178
controlled synthetic benchmarks, 179
current boundary, **243**
impurity extraction benchmark, 249
legacy tutorial, 244
literature analysis, 428
M3 constrained admissible sets, 248
model adequacy, **177**
PBE bootstrap, 245
periodic reduction benchmarks, 247
separation certificate, 178
silicon Wannier prototype, 246
status vocabulary, 278
tutorial versus production, 193
vocabulary, iv
excluded-space reduction, 262
expectation value, 36
experiments
periodic reduction, **316**
external program, 189

failure, 184
figure
contract, 253
finite difference, 291
Bloch-periodic, **76**
mixed derivative, 79
one-dimensional Bloch fiber, 105
particle in a box, 45
two-dimensional box, 56
finite representation
contract, 19
finite-extent perturbation, **213**
finite-range model, **102**
folding down
Burt envelope theory, 418
Fourier mode
finite-difference symbol, 86
fractional coordinate, 64

frame
retained, 258
frame defect, 122
Frobenius norm, **280**
bulk residual, 309

gauge, 158
composite frame, 120
localization, 197
sensitivity, 200
gauge alignment
composite band, **118**
gauge attack
rank two, 124
gentle perturbation, 411
Fourier content, 412
globally slowly varying envelope, 418
Gram metric, **65**
grid
endpoint-excluded periodic, 79

Hamiltonian
additive decomposition, 38
definition, **36**
real-space, 197
reciprocal-space, 197
harmonic oscillator
finite constructions, **297**
Hermitian matrix, **25**, 26
Hermiticity, 161
whole multiband Hamiltonian, 419
Hilbert space, 36
Hilbert's sixth problem, 156
Hilbert–Schmidt geometry, **280**
Hilbert–Schmidt inner product, 280
holonomy
closure, 123
hopping coefficient, **102**
hopping truncation, 108
host system, **212**

identifiability
continuum, 264
effective potential, 375
particle in a box, 290
identification map, 159
identity map, 16
impurity model
lattice, **228**
impurity operator
aligned, 374
definition, **213**
extraction, **212**
model classes, **372**
motivation, 153
two-dimensional program, **398**
indexing convention, 4
infinite square well, 39
infinite-dimensional qualification, 283
inner product, 14
interband coupling, 413
Isabelle, 267
isolated band
Fourier reduction, **102**
retained space, 94

k-point mesh, 191
kinetic energy
scalar Laplacian distinction, 82
Kohn–Sham operator, 186
Kronecker product, **51**
definition, 52
Kronecker sum, 54
two-dimensional Hamiltonian, 56

laboratory
isolated-band Fourier reduction, 115
nonorthogonal finite difference, 87
periodic geometry, 74
ladder operator, 298
Laplacian
nonorthogonal periodic cell, **76**
one-dimensional Dirichlet, 44
lattice
embedded plane, **62**
lattice constant
equilibrium procedure, 192
Lean, 267
learning
aligned target, 231
continuum crossover, 234
joint identification and reduction, 236
parameter prediction, 232
residual model, 235
least squares
direct hopping fit, 109
limitation
current, 250
linear combination, 12
linear independence, 12
linear map, 15
literature review
candidate sources, 431
locality
diagnostic role, 173
gauge dependence, 128, 260
partition-resolved norm, 222
loss
operator, 165
spectral, 165
Luttinger–Kohn model, **411**
comparison with Wannier–Slater, 421
summary, 429

M3 controlled calculation, 174, 331
machine learning
Hamiltonian literature, 433
magnetic field
effective-mass equation, 416

mathematical operator, 273
matrix
  column meaning, 15
matrix representation, **9**, 15
measurement
  energy, 43
  position, 43
mechanized proof
  status, **267**
metadata, **2**
metric tensor
  direct lattice, 65
minimal model, 228
minimum-image distance, 222
model adequacy, **149**
  simultaneous requirements, 163
model class
  admissible, 154
  bulk, **201**
  impurity, 376
model expressiveness, 233
model hierarchy
  bulk, 202
  impurity, 228
model-class expressivity, 163
multiband model, 229
  Luttinger–Kohn, 415
multivalley model, 229
  Luttinger–Kohn, 414

negative evidence, 145, 184
negative finding, 252
negative result, 241
noncommutativity
  reduction, 204
nonlocal operator
  Burt envelope representation, 417
  Ermoneit multivalley theory, 424
norm, 14, 277
  unitary invariant, 260
normalized operator loss, 166
normalized spectral loss, 165
notation, **273**
  alignment, 275
  Bloch and Wannier, 274
nuisance parameter, 167
number
  quantity distinction, 3

objective
  aligned operator, 203
  composite, 235
observable
  envelope reconstruction, 422
optimized norm-conserving Vanderbilt pseudopotential, 190
onsite block, 221
onsite potential, **213**
operator
  definition, **36**
  finite-dimensional foundations, **258**
operator comparison, **158**
  transport, 199
operator difference
  aligned pristine–doped, 282
  direction, 219
  impurity, 212
  represented, 160
operator ordering
  Burt envelope theory, 419
operator residual, 161
operator space, **285**
operator-to-observable bound, 263
orthogonality, 14
orthonormal basis, 14
orthonormal frame
  embedded plane, 72
out-of-zone solution, 421
overlap metric
  PAW, 149

parallel transport, 199
  polar, 122
parameter domain, 163
parent model
  bulk silicon, **189**
  Kohn–Sham, 186
Parseval identity
  finite hopping transform, 109
particle in a box, **290**
  one-dimensional, **39**
  three-dimensional, **51**
  two-dimensional, **51**
path dependence, 204
  bulk reduction, 310
projector augmented-wave (PAW) method, 149
pedagogy, ii
periodic geometry, **62**
periodic image, 213
periodic parent, 411
periodic potential, 153
periodic-image contamination, 224
perturbation
  operator, **213**
phosphorus
  demonstration, 240
  Luttinger–Kohn interpretation, 428
  neutral branch, 227
  substitutional, 209
physical model, 273
PhysKit
  Bloch-periodic Laplacian, 84
  embedded lattice, 73
plane-wave cutoff, 191
plane-wave representation
  one-dimensional fiber, 104
position-space representation, **38**
primitive vector, 64
principal angle, 199
  retained spaces, 122
probability current
  interface, 420
probability density, 43
Procrustes alignment
  pointwise, 124
  rank deficiency, 126
projected Hamiltonian
  gauge covariance, 120
projection
  ambient plane, 71
  band subspace, 195
  common, 281
  orthogonal, **91**
projector defect, 122
proof
  evidence boundary, 268
  status vocabulary, 256
proof package
  dependency graph, 256
proof program
  architecture, **255**
provenance, 146, 183
  bulk parent, 193
pseudoinverse, 71
pseudopotential
  candidate literature, 432
  identity, 190
pullback
  identification, 259

quadratic loss, 169
Quantum ESPRESSO, 189
  candidate literature, 432
quantum state, **36**
quasimomentum sector, 70

rank-two model, 164
Rayleigh quotient, 30
reciprocal lattice

dual basis, 68
reciprocal mesh
centered uniform, 107
reciprocal sewing, 123
recovery residual, 225
rectangular box, 52
reduced coordinates, **62**
reduced model, **149**
reduced momentum, 69
reduction
constrained problem, 201
Luttinger–Kohn mapping, 426
mathematical structure, **262**
reduction hierarchy, 187
reduction route
bulk comparison, **305**
direct, 202, 308
Wannier-mediated, 202, 308
reference
finite represented, 144
representation
bulk selection, **206**
bulk silicon, **194**
represented data, 5
represented operator
impurity extraction, 214
software foundation, 244
represented-operator RMS residual, 166
reproducibility, 146, 183
research sequence, **269**
residual, 277
inequivalent constructions, 293
result
scientific boundary, 249
result table
contract, 252
retained subspace, **91**
impurity extraction, 214
target, 195
Rocq, 267
route comparison
Fourier mediated, 109

scalar, 10
scholarly output, 270
scientific branch
bulk silicon, 189
scientific rigor, i
self-adjoint matrix, 26
separation certificate, 171
separation of variables, 55
serialization
embedded lattice, 73
silicon
bulk program, **186**
bulk research questions, 188
conduction valleys, 414
doped program, **209**
finite-setting evidence, 439
preliminary evidence, **436**
pristine host, 187
Quantum ESPRESSO tutorial, 438
Wannier kinetic decomposition, 441
singular value
neighboring frames, 122
slab
distinction from embedded plane, 72
software artifact, 273
span, 12
sparse matrix
Kronecker construction, 58
spectral filtering
valley envelope, 425
spectral gap
external, 120
spectral projector, 30
spectral reconstruction, 93, 152
spectral restriction, 286
spectral RMS error, 165

spectral theorem, **25**
spectrum, 26
  operator information, 159
spin mixing, 221
spin–orbit coupling, 210
  Luttinger–Kohn, 415
state space
  Luttinger–Kohn basis, 412
stop condition, 145, 216
structured learning, **231**
subspace diagnostic, 199
supercell
  matched pristine, 209
  matched pristine and doped, 213
supercell convergence
  impurity operator, 224

tensor-product basis, **51**
threshold sensitivity, 173
tight-binding model
  orthogonal sp3s-star, 202
  two-dimensional, 350
topology
  Wannier localization, 348
training data, 182
  admissible set, 173
  bulk model, 203
  finite transform, 144
training mesh
  complete Fourier, 111
transferability
  dopant, **240**
transformation
  PAW, 149
truncation
  distinction, 98

uncertainty quantification, 181
  candidate literature, 434
unit
  computational quantity, **2**
unitary matrix, 16
units, 279

validation
  dopant, 378
  scientific, **140**, 181
validation data, 182
valley
  linear combination, 414
valley splitting
  nonlocal coupling, 425
valley-sector projector, 423
vector, 10
  finite representation, **9**
vector space, 10
verification, **140**
  candidate literature, 434
  impurity extraction, 225
  independent reconstruction, 145
  numerical, 178
  software, 177

Wannier function, 196
Wannier localization, 197
Wannier representation
  validation, 198
Wannier transformation, 196
  composite frame, 121
Wannier–Slater envelope, 421
Wannier90
  candidate literature, 432
warmup
  impurity extraction, **372**
  two-dimensional, **342**
  two-dimensional defect extraction, **398**
withheld evaluation
  role separation, 144
withheld mesh

multiband alignment, 129
staggered reciprocal, 111
withheld validation, 203
witness
admissible set, 173
common admissible, 171